FANUC MANUAL GUIDE *i*

For Machining Center System

OPERATOR'S MANUAL

B-63874EN-2/04

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In this manual we have tried as much as possible to describe all the various matters.
However, we cannot describe all the matters which must not be done, or which cannot be done, because there are so many possibilities.
Therefore, matters which are not especially described as possible in this manual should be regarded as "impossible".

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SAFETY PRECAUTIONS

When using machines incorporating FANUC MANUAL GUIDE *i*, be sure to observe the descriptions herein.

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DEFINITION OF WARNING, CAUTION, AND NOTE

This manual includes safety precautions for protecting the user and preventing damage to the machine. Precautions are classified into Warning and Caution according to the degree of the risk or the severity of damage.
Also, supplementary information is described as Note.
Read the Warning, Caution, and Note thoroughly before attempting to use the machine.

> ⚠ **WARNING**
> Applied when there is a danger of the user being injured or when there is a damage of both the user being injured and the equipment being damaged if the approved procedure is not observed.

> ⚠ **CAUTION**
> Applied when there is a danger of the equipment being damaged, if the approved procedure is not observed.

> **NOTE**
> The Note is used to indicate supplementary information other than Warning and Caution.

- Read this manual carefully, and store it in a safe place.

GENERAL WARNINGS AND CAUTIONS

To ensure safety while using a machine featuring the MANUAL GUIDE *i* function, observe the following precautions:

> ⚠ **WARNING**
> 1 Confirm, on the screen, that the data has been entered correctly before proceeding to the next operation. Attempting operation with incorrect data may cause the tool to strike the workpiece or machine, possibly breaking the tool or machine or injuring the operator.
> 2 Before starting the machine using the tool compensation function, carefully determine the direction of compensation and the compensation value, and ensure that the tool will not strike the workpiece or machine. Otherwise the tool or machine may be damaged or the operator may be injured.

⚠ WARNING

3 When using constant surface speed control, set the maximum rotating speed of the spindle to a value that is allowed for the workpiece and workpiece hold unit. Otherwise, the workpiece or hold unit may be removed by centrifugal force to damage the tool machine or injure the operator.
4 Set all necessary parameters and data items before starting MANUAL GUIDE *i* operations. Note that if the cutting conditions are not suitable for the workpiece, the tool may be damaged or the operator may be injured.
5 After creating a machining program using MANUAL GUIDE *i* functions, do no run the machine on that program immediately. Instead, confirm every step of the resultant program, and make sure that the tool path and machining operation are correct and that the tool will not strike the workpiece or machine. Before starting production machining, run the machine with no workpiece attached to the machine to make sure that the tool will not strike a workpiece or the machine. If the tool strikes the machine and/or work-piece, the tool and/or machine may be damaged, and even injuries the operator.

⚠ CAUTION

After pressing the power-on button, do not touch any keys on the keyboard until the initial screen appears. Some keys are used for maintenance or special operations such that pressing such a key may cause an unexpected operation.

OVERVIEW OF THIS MANUAL

This manual describes the functions of MANUAL GUIDE *i* for the Series 0*i*-MODEL C, 0*i*-MODEL D, Series 16*i*/18*i*/21*i*-MODEL B, or Series 30*i*/31*i*/32*i*-MODEL A.

For other functions, refer to the operator's manual for the Series 0*i*-MODEL C, 0*i*-MODEL D, Series 16*i*/18*i*/21*i*-MODEL B, Series 30*i*/31*i*/32*i*-MODEL A, or Series 30*i*/31*i*/32*i*-MODEL B.
The specifications and usage of MANUAL GUIDE *i* may vary according to the specifications of the operator's panel of a machine tool. Be sure to read the manual provided by the machine tool builder.

The functions of the CNC machine tool system are determined not only by the CNC, but by the combination of the machine tool, the power magnetic circuit in the machine tool, the servo system, the CNC, and the operator's panel.

It is impossible to cover all possible combinations of all functions, programming methods, and operations in a single manual.

This manual explains only the MANUAL GUIDE *i* operations provided for the CNC. For individual CNC machine tools, refer to applicable manuals from the machine tool builders.

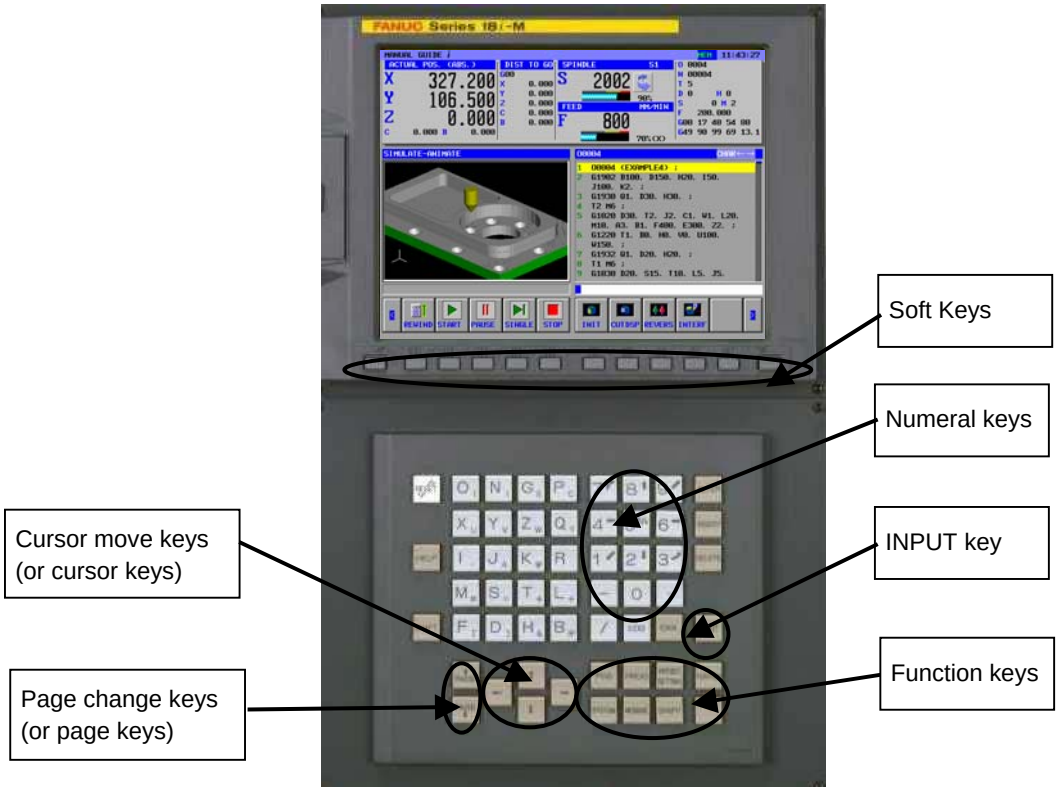
This manual explains as many detailed functions as possible. However, it is not possible to describe all of the items which cannot be done or which the operator must not do. Therefore, please assume that functions other than those described in this manual cannot be performed.

Detailed information and special conditions are explained in notes. The readers may encounter new technical terms in the notes not previously defined or described. In this case, read this manual through first, then review the details.

SYMBOLS USED

In this manual, the following conventions are used for keys.

(1) Function keys are represented by enclosing their names between the special characters < and >:
Example) <PROGRM>, <OFSET>
(2) The numbers to be input by numerical keys are underlined.
Example) 12.345
(3) Similarly to functions keys, the input (INPUT) and editing keys are represented by enclosing their names between the special characters < and >:
Example) 12.345 <INPUT>, <ALTER>
(4) Soft-keys are enclosed in brackets [ ]:
Example) [LIST], [LINE]
(5) The cursor move keys are indicated by the following symbols :
Example) <↑>, <↓>, <←>, <→>
(6) The page change keys are indicated by the following symbols :
Example) <↑ PAGE>, <PAGE ↓>



FORMAT OF PROGRAMS TO BE USED

The ISO code format, which is in wide use for CNC machine tools, is used on machining programs used with MANUAL GUIDE *i*. In addition, G4-digit machining cycles are used to implement advanced machining operations.
G4-digit machining cycles can be input and edited easily using a menu window, which enables machining data to be entered interactively.
When a machining program is being edited, an address at which the cursor is placed is explained at the bottom of the screen.

1. Word

The term “word” refers to a minimum unit in NC machining programs. It is represented using a combination of an address (one of letters A to Z), some of digits 0 to 9, an arithmetic symbol (+ or -), and/or a decimal point.
The “address” represents the meaning of numeric data, such as a machine movement axis

2. Cursor

The term “cursor” used with machining programs refers to a display portion where a string of characters is shown against a yellow background. This portion is referred to as “cursor-selected display portion”. This is the place where an actual editing action is made during machining program editing.
Either of the following two items can be cursor-selected.

<1> 1 block

When the cursor is shifted to the beginning of the next block by pressing the cursor move key <→>, the entire block is selected.
When the cursor is placed at the address of the beginning of a certain block, pressing the cursor move key <←> causes the entire block to be selected.
When a block is already selected, pressing the cursor move key <↑> or <↓> causes the block, respectively, following or preceding the block of interest to be selected.

<2> 1 word

When a whole block is selected, pressing the cursor move key <←> or <→> causes a word to be selected. Each time the cursor move key <←> or <→> is pressed, a new word is selected.
When a word in a certain block is selected, pressing the cursor move key <↑> or <↓> causes the same-number address in the block, respectively, following or preceding the block of interest to be selected.

NOTE

1 On a screen where numeric data, such as offset data or cycle machining data, is input directly, the term “cursor” refers to a data portion enclosed in a blue frame.
2 A data portion enclosed in a blue frame is referred to as “cursor-selected data item”. Once numeric data is keyed in with numeric keys, pressing the <INPUT> key causes the numeric data to be input as the data item of interest.
3 Pressing the cursor move key <←>, <↑>, <↓>, or <→> causes a new data item to be selected.

PREFACE

This document is configured as follows:

Explanations about the document contents

SAFETY PRECAUTIONS

Describes the rules you should observe so that you can safely use machines incorporating FANUC MANUAL GUIDE *i*.

PREFACE

Explains how this document is configured. It also lists applicable CNC units and related documents.

Part I, "WHAT IS MANUAL GUIDE *i*?"

Describes the features of MANUAL GUIDE *i*.

Part II, "BASIC SCREEN AND OPERATION"

Describes the configuration of the MANUAL GUIDE *i* screen. It also explains all basic operations, from creating machining programs in ISO code format to running machine tools using the programs.

Part III, "CONVENIENT FUNCTIONS"

Explains convenient functions that will help create and run programs.

Part IV, "MACHINING CYCLE"

Explains how to program machining cycles offered by MANUAL GUIDE *i*.

Part V, "OTHER FUNCTIONS"

Describes tool management and other functions usable with MANUAL GUIDE *i*.

APPENDIX

Provides information (such as examples of creating and running programs, alarms, and parameter setting) which machine operators can reference when operating MANUAL GUIDE *i*.

APPENDIX (FOR MACHINE TOOL BUILDER)

Provides information (such as MANUAL GUIDE *i* startup procedure and customization) which machine tool builders will need in using MANUAL GUIDE *i*.

Applicable models

This manual can be used with the following models.
The abbreviated names may be used.

- FANUC Series 30*i*/31*i*/32*i*-MODEL A

Product name	Abbreviation	
FANUC Series 30*i*-MODEL A	30*i* –A	Series 30*i*
FANUC Series 31*i*-MODEL A	31*i* –A	Series 31*i*
FANUC Series 31*i*-MODEL A5	31*i* –A5	
FANUC Series 32*i*-MODEL A	32*i* –A	Series 32*i*

- FANUC Series 30*i*/31*i*/32*i*-MODEL B

Product name	Abbreviation	
FANUC Series 30*i*-MODEL B	30*i* –B	Series 30*i*
FANUC Series 31*i*-MODEL B	31*i* –B	Series 31*i*
FANUC Series 31*i*-MODEL B5	31*i* –B5	
FANUC Series 32*i*-MODEL B	32*i* –B	Series 32*i*

NOTE
1 For an explanatory purpose, the following descriptions may be used according to the types of path control used:
  • T series: For the lathe system
  • M series: For the machining center system
2 Unless otherwise stated, the model names Series 30*i*, Series 31*i*, and Series 32*i* used in this document sometimes collectively refer to the models mentioned above except for item 3 below.
3 Some functions described in this manual may not be applied to some products. For details, refer to the DESCRIPTIONS (B-63942EN) related to applied CNC.
4 MANUAL GUIDE i is not available in Series 30i-MODEL A or MODEL B with 5 paths or over.

- FANUC Series 16*i*/18*i*/160*i*/180*i*/160*i*s/180*i*s-MODEL B

Product name	Abbreviation	
FANUC Series 16*i*-TB	16*i*-TB	Series 16*i*
FANUC Series 16*i*-MB	16*i*-MB	
FANUC Series 160*i*-TB	160*i*-TB	
FANUC Series 160*i*-MB	160*i*-MB	
FANUC Series 160*i*s-TB	160*i*s-TB	
FANUC Series 160*i*s-MB	160*i*s-MB	
FANUC Series 18*i*-TB	18*i*-TB	Series 18*i*
FANUC Series 18*i*-MB5	18*i*-MB5	
FANUC Series 18*i*-MB	18*i*-MB	
FANUC Series 180*i*-TB	180*i*-TB	
FANUC Series 180*i*-MB5	180*i*-MB5	
FANUC Series 180*i*-MB	180*i*-MB	
FANUC Series 180*i*s-TB	180*i*s-TB	
FANUC Series 180*i*s-MB5	180*i*s-MB5	
FANUC Series 180*i*s-MB	180*i*s-MB	
FANUC Series 21*i*-TB	21*i*-TB	Series 21*i*
FANUC Series 21*i*-MB	21*i*-MB	
FANUC Series 210*i*-TB	210*i*-TB	
FANUC Series 210*i*-MB	210*i*-MB	
FANUC Series 210*i*s-TB	210*i*s-TB	
FANUC Series 210*i*s-MB	210*i*s-MB	

NOTE
1 For an explanatory purpose, the following descriptions may be used according to the types of path control used:
  • T series: For the lathe system
  • M series: For the machining center system
2 Unless otherwise stated, the model names Series 16*i*, Series 18*i*, and Series 21*i* used in this document sometimes collectively refer to the models mentioned above except for item 3 below.
3 Some functions described in this manual may not be applied to some products. For details, refer to the DESCRIPTIONS (B-63532EN) related to applied CNC.

- FANUC Series 0*i*–MODEL C

Product name	Abbreviation	
FANUC Series 0*i*-TC	0*i*-TC	Series 0*i*-C
FANUC Series 0*i*-TTC	0*i*-TTC	
FANUC Series 0*i*-MC	0*i*-MC	

- FANUC Series 0*i*–MODEL D

Product name	Abbreviation	
FANUC Series 0*i*-TD	0*i*-TD	Series 0*i*-D
FANUC Series 0*i*-MD	0*i*-MD	

NOTE
1 For an explanatory purpose, the following descriptions may be used according to the types of path control used:
  • T series: For the lathe system
  • M series: For the machining center system
2 Unless otherwise stated, the model names Series 0*i* used in this document sometimes collectively refer to the models mentioned above except for item 3 below.
3 Some functions described in this manual may not be applied to some products. For details, refer to the DESCRIPTIONS (B-64112EN, B-64302EN) related to applied CNC.

Related manuals

- Manuals related to MANUAL GUIDE *i*

The table below lists manuals related to MANUAL GUIDE *i*.

Manual name	Specification Number	
OPERATOR'S MANUAL (Common to Lathe System/Machining Center System)	B-63874EN	*
OPERATOR'S MANUAL (For Machining Center System)	B-63874EN-2	
OPERATOR'S MANUAL (SET-UP GUIDANCE FUNCTIONS)	B-63874EN-1	

(*) In the table, this manual is marked with an asterisk (*).

- Manuals related to FANUC Series 30*i*/31*i*/32*i*-MODEL A

Manual name	Specification Number
DESCRIPTIONS	B-63942EN
CONNECTION MANUAL (HARDWARE)	B-63943EN
CONNECTION MANUAL (FUNCTION)	B-63943EN-1
OPERATOR'S MANUAL (Common to Lathe System/Machining Center System)	B-63944EN
OPERATOR'S MANUAL (For Lathe System)	B-63944EN-1
OPERATOR'S MANUAL (For Machining Center System)	B-63944EN-2
MAINTENANCE MANUAL	B-63945EN
PARAMETER MANUAL	B-65950EN

- Manuals related to FANUC Series 30*i*/31*i*/32*i*-MODEL B

Manual name	Specification Number
DESCRIPTIONS	B-64482EN
CONNECTION MANUAL (HARDWARE)	B-64483EN
CONNECTION MANUAL (FUNCTION)	B-64483EN-1
OPERATOR'S MANUAL (Common to Lathe System/Machining Center System)	B-64484EN
OPERATOR'S MANUAL (For Lathe System)	B-64484EN-1
OPERATOR'S MANUAL (For Machining Center System)	B-64484EN-2
MAINTENANCE MANUAL	B-64485EN
PARAMETER MANUAL	B-64490EN

- Manuals related to FANUC Series 16*i*/18*i*/160*i*/180*i*/160*i*s/180*i*s-MODEL B

Manual name	Specification Number
DESCRIPTIONS	B-63522EN
CONNECTION MANUAL (HARDWARE)	B-63523EN
CONNECTION MANUAL (FUNCTION)	B-63523EN-1
OPERATOR'S MANUAL (For Lathe System)	B-63524EN
OPERATOR'S MANUAL (For Machining Center System)	B-63534EN
MAINTENANCE MANUAL	B-63525EN
PARAMETER MANUAL	B-63530EN

- Manuals related to FANUC Series 0*i*–MODEL C

Manual name	Specification Number
DESCRIPTIONS	B-64112EN
CONNECTION MANUAL (HARDWARE)	B-64113EN
CONNECTION MANUAL (FUNCTION)	B-64113EN-1
Series 0*i*-TC OPERATOR'S MANUAL	B-64114EN
Series 0*i*-MC OPERATOR'S MANUAL	B-64124EN
Series 0*i*-TTC OPERATOR'S MANUAL	B-64284EN
MAINTENANCE MANUAL	B-64115EN
PARAMETER MANUAL	B-64120EN

- Manuals related to FANUC Series 0*i*–MODEL D

Manual name	Specification Number
DESCRIPTIONS	B-64302EN
CONNECTION MANUAL (HARDWARE)	B-64303EN
CONNECTION MANUAL (FUNCTION)	B-64303EN-1
OPERATOR'S MANUAL (Common to Lathe System/Machining Center System)	B-64304EN
OPERATOR'S MANUAL (For Lathe System)	B-64304EN-1
OPERATOR'S MANUAL (For Machining Center System)	B-64304EN-2
MAINTENANCE MANUAL	B-64305EN
PARAMETER MANUAL	B-64310EN
START-UP MANUAL	B-64304EN-3

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ADDITIONAL INFORMATION

I. WHAT IS MANUAL GUIDE *i*?

1 WHAT IS MANUAL GUIDE *i*?

1.1 WHAT IS MANUAL GUIDE *i*?

MANUAL GUIDE *i* guides you on how to operate CNC units installed on machine tools such as lathes and machining centers. It features a single screen that can be used to create machining programs, check them using animation, set up machining, and perform actual machining.
With MANUAL GUIDE *i*, the ISO code format, which is in wide use for CNC-based machine tools, is used for machining programs and machining cycles are available to implement advanced machining operations.

1.2 MAIN FEATURES OF MANUAL GUIDE *i*

By using MANUAL GUIDE *i*, the operator can carry out routine machining easily.

(1) Integrated operation screen that enables almost all routine machining operations
A single integrated operation screen enables routine machining operations including machining program input/editing, animated simulation based machining program checks, production machining, MDI operations, and manual operations with JOG and HANDLE.

(2) Machining programs in ISO code format
Using ISO code machining programs, which are in wide use, enables the operator to specify simple operations with simple commands, such as those for straight lines and arcs, and complicated machining operations with machining cycles easily.

(3) High affinity with CAD/CAM
ISO code machining programs created using CAD/CAM can be used without modification. Adding advanced machining cycles to these machining programs makes them perfect machining programs. They can be checked easily, using animated simulation.

(4) Advanced machining program editing
Using advanced editing functions, such as substring search and cut/paste via the clipboard, enables easy editing of machining programs.

(5) Advanced machining using machining cycles
Advanced machining cycles are available which cover various types of machining including milling and turning. With these machining cycles, it is possible to perform complex machining by creating and running programs easily.

(6) Fixed format program menu-driven simple program entry
Register a series of frequently used machining operations previously as a menu, and select necessary machining operations from the menu when creating a machining program. This method can eliminate the trouble of entering similar machining operations repeatedly.

(7) M code menu
It is possible to input M codes easily by referencing explanations displayed in an M code menu. Machine tool builders can create the explanations easily.

(8) Realistic animated simulation
Machining programs can be checked easily, using an animated simulation method that can realistically show what the surface machined with a specific type of tool tip is like. In addition, you

can check a simulated workpiece as if you were seeing at a real workpiece because the animated simulation method uses solid models for all operations, from milling to turning, for the workpiece.

(9) Advanced set-up guidance

It is possible to set up machining operations and check the precision of machined workpieces easily, using an advanced set-up guidance function that can handle all measurements, from tool offset measurement to the measurement of workpieces in machine tools.

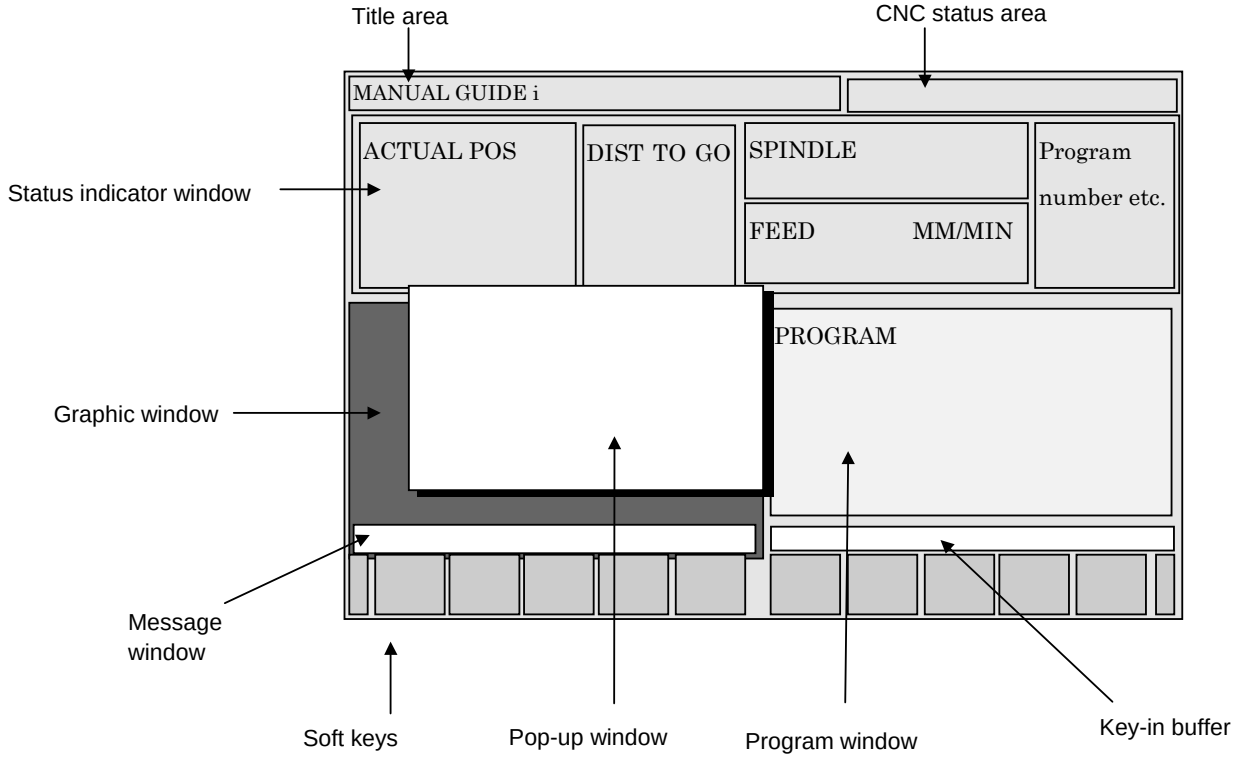
(10) Supporting a wide variety of machine types including lathes and machining centers

MANUAL GUIDE *i* supports lathes 1 to 3, vertical and horizontal machining centers, and tilting head-equipped machining centers with 3 + 2 axes. It also supports compound machine tools that are capable of turning and milling.

II. BASIC SCREEN AND OPERATION

1 BASIC SCREEN

In MANUAL GUIDE *i*, basically, only one screen called the All-in-one Screen is used for all the operations from trial machining to actual machining.



(1) Title area:
The title of MANUAL GUIDE *i* is always displayed.

(2) CNC status area:
The following CNC statuses are displayed.
- Mode
- Alarm status
- Reset or emergency stop status
- Actual time

(3) Status indicator window:
The following information about CNC is displayed.
- Actual machine position
- Remaining moving distance of the actual block
- Actual speed and load meter (for the axis with the maximum load)

NOTE
1 It is possible to display the maximum 8 axes.
2 It is possible to display Actual speed in Feed per revolution. (Refer to the bit 0 (FDS) of parameter No.14703.)

- Spindle rotating speed and spindle load meter
- Program number and process number
- Command values during automatic operation (M,S,T,F)

- Machine status icons
  (a) Direction of spindle rotation:
   (Normal/vertical)  (Reverse/vertical)

   (Normal/horizontal)  (Reverse/horizontal)

  (b) Constant surface speed control mode:
   (Enabled/vertical)  (Enabled/horizontal)

  (c) Handle mode:
   (HNDL mode)

(4) Graphic window
The following graphical drawing is displayed as needed.
- Animated drawing with a solid model (machining simulation)
- Tool path drawing

(5) Program window
The machining program is displayed.

(6) Pop-up window
The following supplemental screens are displayed as needed.
- M-code menu
- Fixed format statement menu
- Set data (workpiece coordinate system, tool offset, fixed format statement registration)

NOTE
Tool offset pairs are the maximum 9990pairrs.

- Program list
- Data input window for machining cycles

(7) Key-in buffer
Comments on data and input numerical data are displayed as needed.

(8) Message window
The following messages are displayed as needed.
- Meaning of the word which a cursor is allocated (Guidance Message)

NOTE
It is possible to display two Guidance Message of M/T mode in the CNC for compound lathe. (Refer to the bit 2 (GDM) of parameter No.14703.)

- Meaning of Machine Cycle which is executing
- Content of Warning and Alarm

(9) Soft keys

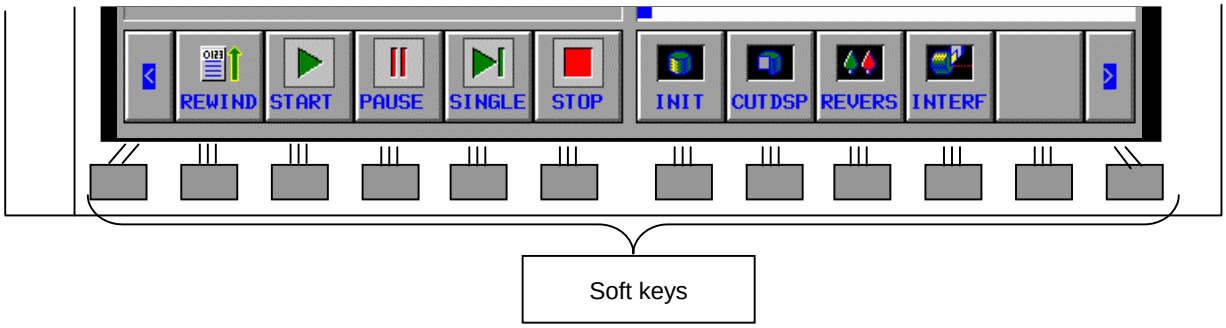
The comments on the following soft-keys are displayed as needed.

- Editing operation menu
- Machining cycle menu
- Pop-up window menu

Remark)

The soft keys described in this manual are specified to 12 keys placed under the screen, LCD, as shown in the following example.
The meaning of each soft key is various by the displayed content on the screen, and will be displayed on the relevant part for each key.

Example of soft keys)



2 EDITING PROGRAMS

2.1 PROGRAM WINDOW AND EDITING

MANUAL GUIDE *i* uses a program window to input and edit machining programs (in ISO code format). The program window is operated using the following soft keys, which are displayed by pressing the leftmost soft key [<] or rightmost soft key [>] several times.

REWIND				CHSIZE	LINENO				SETING

[REWIND] : This soft key takes you to the beginning of the program.
[CHSIZE] : This soft key zooms the program window to full screen.
[LINENO] : This soft key selects whether to display the line number at the beginning of each block. The line number is not memorized in the machining program.
[SETING] : This soft key opens the setting screen.

The scroll bar displayed on the right edge of the program window indicates the approximate position of the cursor throughout the machining program.
The cursor is moved around in the program window, using the cursor keys <←>, <↑>, <↓>, and <→>. Pressing cursor key <→> moves the cursor right in a block. To move the cursor to the next block, you need to select the block entirely once and then select an individual address. Pressing cursor key <←> has the same effect as for cursor key <→> except for the direction in which the cursor moves.

The program is edited using the following soft keys, which are displayed by pressing the leftmost soft key [<] or rightmost soft key [>] several times.

NEWPRG	OPEN	SRCH↑	SRCH↓	O SRCH	COPY	CUT	DELETE	KEYPST	PASTE

Remark)
Basically, soft keys are placed in form of 1 line. By pressing the rightmost soft key [>], the soft keys shifted to right direction by 10 will appear. By pressing the leftmost soft key [<], the left side soft keys shifted by 10 will appear.

NOTE
Data server device and Memory card device are not available in Series 0*i*-D.

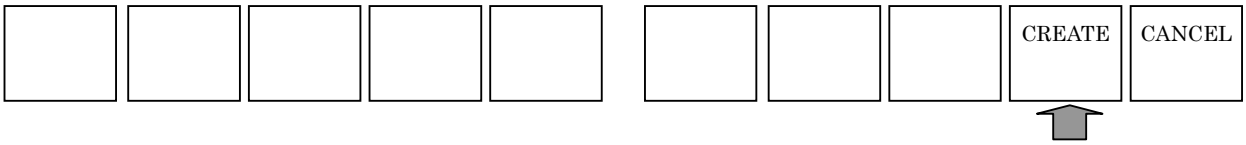
2.2 CREATING MACHINING PROGRAMS

NEWPRG	O LIST	SRCH↑	SRCH↓	O SRCH	COPY	CUT	DELETE	KEYPST	PASTE

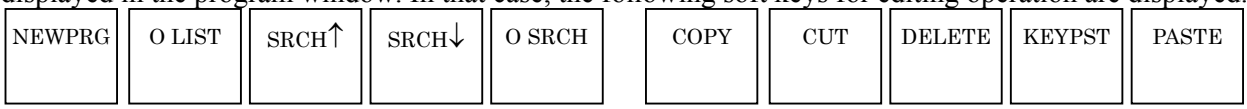
When you press the soft key [NEWPRG], the program creation window as shown below appears.

CREATE NEW PROGRAM
NEW PROGRAM NO.
KEY IN NEW PROGRAM NUMBER.

For the program creation window, the following soft keys are displayed.

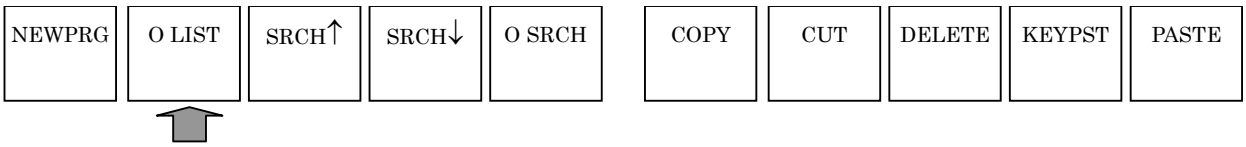


In this window, enter a program number, using numeric keys, and then press soft key [CREATE].
Pressing soft key [CREATE], new program with entered program number only is created, and it is displayed in the program window. In that case, the following soft keys for editing operation are displayed.



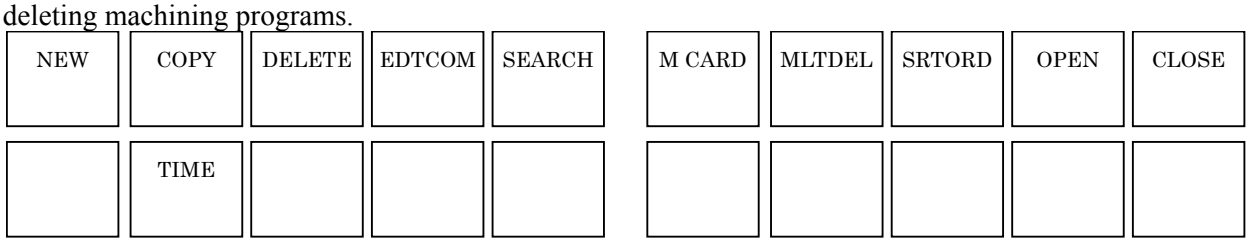
In addition to the above operation, you can create new part program by entering new program number after the address "O", and pressing <INSERT> key.

2.3 EDITING IN A PROGRAM LIST



Pressing soft key [O LIST] displays a window that lists registered machining programs.
By pressing the cursor key <←> or <→>, a sort type (sort by number, sort by date and time of updating, or sort by size) can be selected.

The following soft keys are displayed in the window. They can be used for editing such as copying or deleting machining programs.



[NEW] : This soft key displays the program creation window, in which you can create machining programs.

[COPY] : This soft key displays the program copy window. After a program number is entered using numeric keys, pressing soft key [COPY] copies the specified program.

[DELETE] : This soft key displays a message for asking you whether you really want to delete a specified program. Pressing soft key [YES] deletes the specified program. Pressing soft key [NO] cancels the request to delete the specified program.

[EDTCOM] : This soft key displays the comment (program name) edit window. If you move the cursor, using cursor key <←> or <→>, and enter a character, using MDI keys, the character appears at the left of the cursor. Using the <CAN> key can delete the character at the left. Pressing soft key [ALTER] causes the comment to be changed.

[SEARCH] : This soft key displays the program search window. After entering a desired program number in the window, using numeric keys, press soft key [SEARCH].

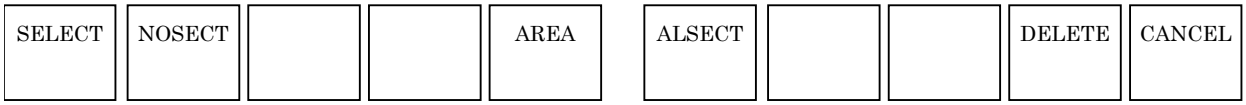
[M CARD] : This soft key enables input/output to and from the memory card. See Chapter 2, "MEMORY CARD INPUT/OUTPUT FUNCTION", in Part III for details of memory card input/output.

[MLTDEL] : This soft key deletes more than one program at a time. For details, see Subsection 2.3.1, "Simultaneous Deletion of Multiple Programs".

[SRTORD] : This soft key switches the sort order for displaying programs between ascending order and descending order.
[OPEN] : After placing the cursor on a program you want to edit, using the cursor key <←> or <↓>, pressing this soft key selects the program and closes the list window. Then, the soft keys described at the top of this subchapter are displayed again.
[CLOSE] : This soft key closes the program list window.
[TIME] : This soft key starts the machining time display function. For details, see Section 3.7, "DISPLAYING MACHINING TIME".

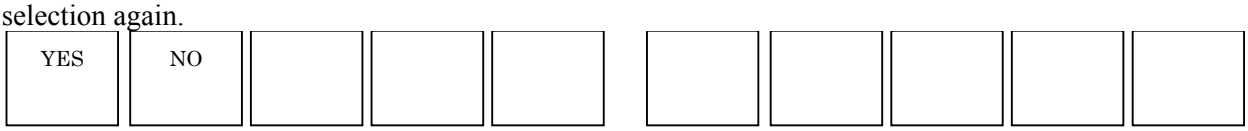
2.3.1 Simultaneous Deletion of Multiple Programs

In the O list window, pressing the soft key [MLTDEL] displays a window for selecting a program to be deleted.



[SELECT] : This soft key selects a program to be deleted. Place the cursor on the program to be selected, and press the soft key [SELECT]. When the program is selected, the color of its background changes to yellow.
[NOSECT] : This soft key deselects a program. Place the cursor on the program to be deselected, and press the soft key [NOSECT]. When the program is deselected, the color of its background changes from yellow back to white.
[AREA] : This soft key selects a range. Place the cursor on the first program in a sequence of programs to be selected, press the soft key [AREA], and then move the cursor to the last program in the sequence of programs. The color of the specified range changes to purple. After this, press the soft key [SELECT] to assert the range. The color of the background of the selected programs changes from purple to yellow.
[ALSECT] : This soft key can select all programs. Pressing the key selects all programs in the folder of interest. The color of the background of the selected programs changes to yellow.
[DELETE] : This soft key deletes selected programs.
[CANCEL] : This soft key deselects all programs in the folder of interest at a time. The color of the background of the deselected programs changes from yellow back to white.

Pressing the soft key [DELETE] displays a confirmation message in the message section of a window for selecting programs to be deleted. The message asks whether you really want to delete the selected programs. It also displays [YES] and [NO] as soft keys. Pressing the soft key [YES] deletes the programs. Pressing the soft key [NO] cancels the request to delete them and displays the soft keys for range selection again.



2.3.2 Managing Program List Folder (for Series 30*i*/31*i*/32*i* Only)

Described herein is the specification of folder management in the program list window.

2.3.2.1 What is displayed in the program list?

(1) Name
Corresponds to a conventional program number.
(2) Comment
If a comment is too large to fit the display area, the portion of the comment that cannot be displayed is represented with ellipses (…).
(3) Program update time (year, month, day, hour, minute)

(4) Program size (in character units)
(5) [RETURN UPPER FOLDER]
(6) Program attribute (whether the program of interest can be edited)
If a program cannot be edited, the letter "R" is displayed at the right of the program name.

2.3.2.2 Operations added to the program list window

(1) Changing a program/folder name
Pressing the soft key [RENAME] displays the [ALTER PROGRAM NAME OR FOLDER NAME] window.

(2) Detailed program information
Pressing the soft key [DETAIL] displays the [PROGRAM PROPERTY] window, which contains the following information.
<1> Program name (consisting of up to 32 characters) (Note)
<2> Comment (consisting of up to 48 characters) (Note)
<3> Program update time (year, month, day, hour, minute, second)
<4> Program size (in character units)
<5> Program attribute (whether the program of interest can be edited)

NOTE
For the Series 30*i*/31*i*/32*i*, lowercase letters can be used in program names and comments.

(3) Changing the program/folder attribute (disabling editing)
Pressing the soft key [WRPROT] changes the attribute cyclically.
The letter "R" is displayed on the right side of programs that cannot be edited.

(4) Selecting devices
Pressing the soft key [DEVICE] displays the [SELECT DEVICE] window.

Once a device has been selected, pressing the soft key [SELECT] displays a list of programs in the selected device.
Refer to also Section 11.1, "SELECTING DEVICES", in Part III in the "Series 30*i*/31*i*/32*i* Operator's Manual" (B-63944EN, or B-64484EN).

(5) Creating programs
Specify which to create, programs or folders, by checking the corresponding check box.

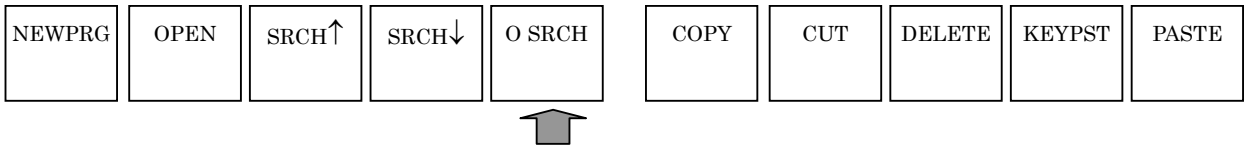
2.3.2.3 Supporting data server operations

(1) MANUAL GUIDE *i* supports program editing and program operations on the data server.
Switching devices in the program list window enables programs on the data server to be handled in the same manner as for programs in program memory.
The differences are listed below. They meet the same specification as for operations on the NC screen.

Operation	Program memory	Data server
Creating programs	Supported	Not supported
Copying programs	Supported	Not supported if they are already selected
Deleting programs	Supported	Not supported if they are already selected
Renaming programs	Supported	Not supported if they are already selected
Memory card input/output	Supported	Not supported

(2) The other functions (such as setting and FTP-based transfer) on the data server are performed on the NC screen (program list window).
(3) The specification of ATA flash card that can be used by MANUAL GUIDE *i* and has a size that is bigger than 2Gbyte is followings.
① Made by TDK
② The size is 4Gbyte

2.4 SEARCHING FOR A MACHINING PROGRAM TO BE EDITED



After entering a desired program number, using numeric keys, pressing soft key [O SRCH] can select the program.
Pressing soft key [O SRCH] without entering a program number causes the next program to be selected.
Pressing the O key on the MDI panel, the soft keys changes to the above one, so enter the program number to be searched and press soft key [O SRCH].
And as another way, you can select the program in the program list by pressing soft key [OPEN] after placing the cursor on the program number to be selected.
When the part program is selected, the content of the part program is displayed in the program window, and the above soft keys will be displayed.

2.5 PROGRAM NUMBER O8-DIGIT FUNCTION

It is possible to use program number O8-digit on Series 16*i* / 18*i*/ 21*i*-MODEL B and Series 30*i* / 31*i*/ 32*i*. On Series 16*i* / 18*i*/ 21*i*-MODEL B, the option “Program number O8-digit” is necessary. On Series 30*i* / 31*i*/ 32*i*, this function is available in case parameter No.11304#3 is set to 1.

2.6 BASIC EDITING OPERATIONS OF PART PROGRAM

Since MANUAL GUIDE *i* uses ISO-code form part program, editing of 1 word, minimum unit of the program and made from address and numerical data, are available by using <INSERT>, <ALTER> and <DELETE> keys, which are placed on the MDI panel.

2.6.1 Entering a Word (<INSERT> key)

Operation
(1) Select the word, before which you want to insert the new word, by placing the cursor. Otherwise, select the block, before which you want to insert the new word, by placing the cursor.
(2) Enter a new word by MDI keys. Plural words can be entered at once.
(3) Press <INSERT> key.

NOTE
In the case a bit 2 (C4E) of parameter No.14852 is set to 0, when a cursor is placed on the word in the cycle machining block and an operator is going to insert a new word, the warning to confirm an operator whether it can be done or not will be displayed.

2.6.2 Modifying a Word (<ALTER> key)

Operation
(1) Select the word to be modified by placing the cursor on the word.
(2) Enter new word by MDI keys. Plural words can be entered at once.
(3) Press <ALTER> key.

2.6.3 Modifying the Numeric Value of a Word (<ALTER> key)

Operation
(1) Select the word to be modified by placing the cursor on the word.
(2) Enter new numeric data only by MDI keys. Plural words can not be entered.
(3) Press <ALTER> key.

2.6.4 Deleting a Word (<DELETE> key)

Operation
(1) Select the word to be deleted by placing the cursor on the word.
(2) Press <DELETE> key.

NOTE
1 As the deleting operation in the CNC program screen, no prompting message for deleting a word is displayed.
2 Program number, Oxxxx, and end of program, %, cannot be deleted.

2.6.5 Modifying a Block (<ALTER> key)

Operation
(1) Select the block to be modified by placing the cursor on the block.
(2) Enter a new word by MDI keys. Plural words can be entered at once.
(3) Press <ALTER> key.

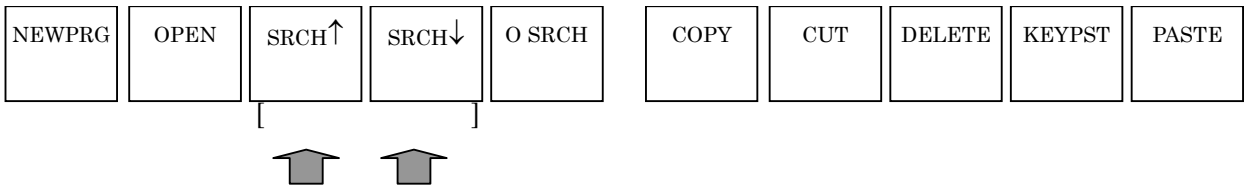
2.6.6 Deleting a Block (<DELETE> key)

Operation
(1) Select the block to be deleted by placing the cursor on the block.
(2) Press <DELETE> key.

2.6.7 Changing the Program Number (ALTER key)

Operation
(1) Select the program number word, Oxxxx, by placing the cursor on the word.
(2) Enter the address “O” and new program number by MDI keys.
(3) Press <ALTER> key.

2.7 SEARCH (FORWARD AND BACKWARD)



After a character string is entered using MDI keys, pressing soft key [SRCH↑] (backward search) or [SRCH↓] (forward search) searches for the specified character string. A substring can also be searched for.
If the character string is found, key-in buffer is cleared, but the string is displayed in the [ ] under soft keys [SRCH↑] and [SRCH↓].
This displayed string can be used for the next time searching operation. Only pressing soft key [SRCH↑] or [SRCH↓] without entering same string again, you can search the same string over and over.

2.7.1 Simultaneous String Search Function of Multi Programs

Simultaneous string search function of multi programs is available on multi-part program editing screen or multi-path concurrent display screen.

2.7.1.1 Mode to use this function

This function is available on only following mode.

(1) EDIT mode
(2) MEM mode (the background edit)

2.7.1.2 Program window where search is executed

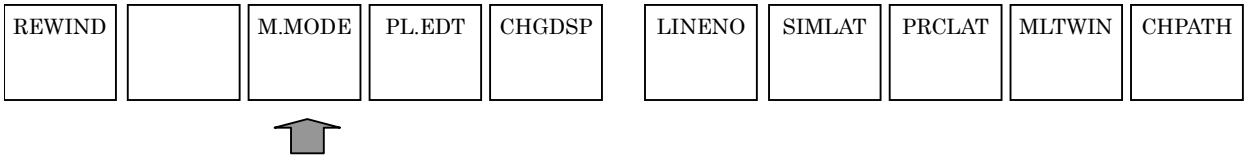
The search of this function is executed only to the following program windows.

(1) Program window of mode to use this function on multi-path concurrent display screen
(2) Program window on multi-part program editing screen

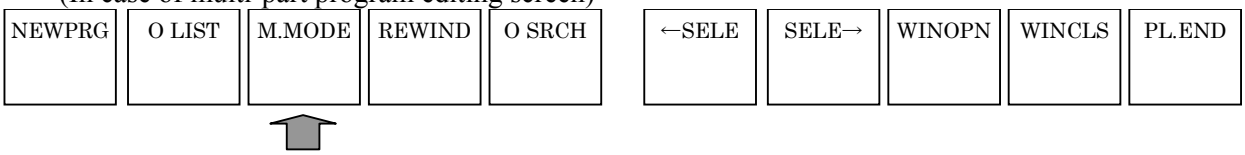
2.7.1.3 Changing search mode

By changing to multi-search mode with following operation, simultaneous string search function become available.

(1) In order to change to multi-search mode, press soft key [M.MODE].
(In case of multi-path concurrent display screen)



(In case of multi-part program editing screen)



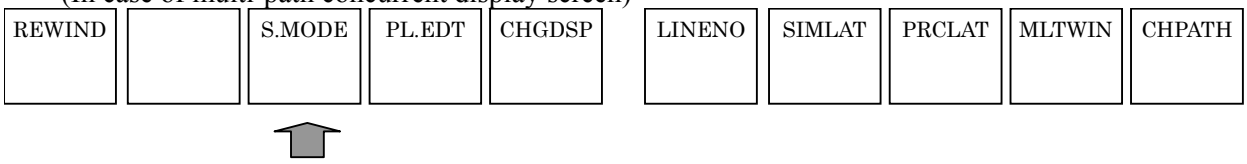
(2) The icon of the soft key [↑ search] / [↓ search] is changed as follows, when becoming multi-search mode.
(Icon on multi-search mode)



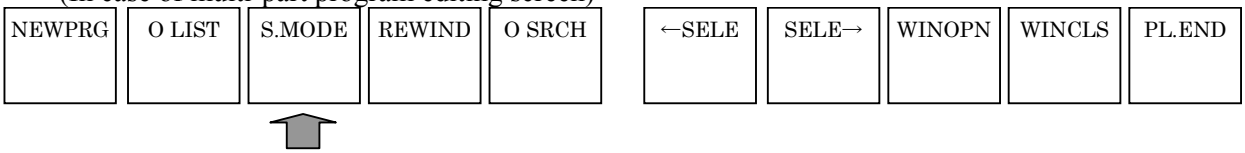
(Icon on single-search mode(conventional))



(3) In order to return to single-search mode, press soft key [S.MODE].
(In case of multi-path concurrent display screen)

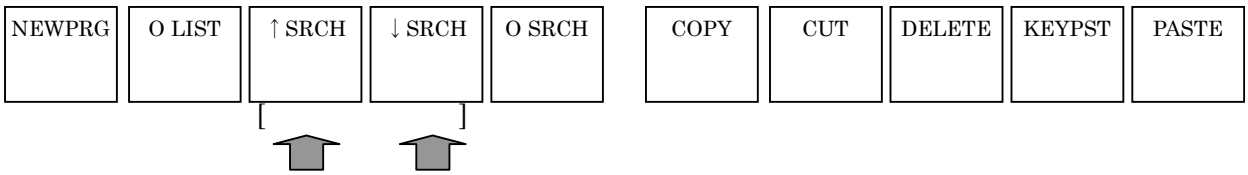


(In case of multi-part program editing screen)

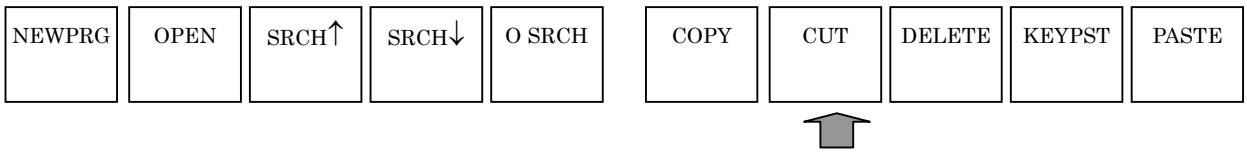


2.7.1.4 Execute simultaneous string search of multi programs

(1) When soft key [↑ search] / [↓ search] is pressed in the state of the multi-search mode, search of the character string is executed to programs of all windows which satisfy the requirement for search.



2.8 CUT

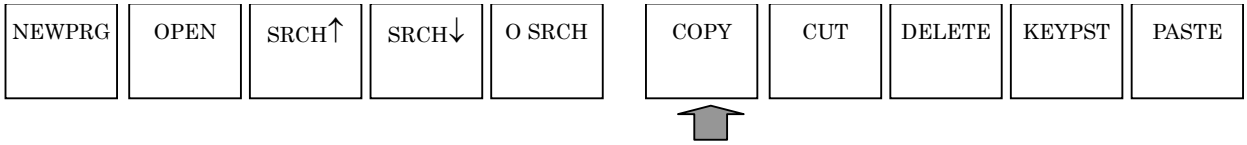


Pressing soft key [CUT] displays a message that prompts you to select a range of data to be cut. First select the cut range (by displaying it in yellow), using cursor keys, and then press soft key [CUT]. The data cut off from the specified range is stored on the clipboard. The contents of the clipboard can be pasted to other places in the program and to other programs.
Clip board is the memory, in which the content of part program can be stored temporary. By pressing soft key [CUT] or [COPY], the content of the cursor selected is stored in it, then the old content is cleared. The content of the clip board is cleared at the power off of CNC.
A clip board size can be selected using bits 4 (CLP1) and 5 (CLP2) of parameter No. 14701.

Bit 4(CLP1)= 0, bit 5(CLP2)= 0: Sets a clip board size of 1024 bytes.
Bit 4(CLP1)= 1, bit 5(CLP2)= 0: Sets a clip board size of 2048 bytes.
Bit 4(CLP1)= 0, bit 5(CLP2)= 1: Sets a clip board size of 4096 bytes.
Bit 4(CLP1)= 1, bit 5(CLP2)= 1: Sets a clip board size of 8192 bytes.

NOTE
The contents of the clipboard are preserved until the power is shut off or other data is stored on the clipboard. So, the clipboard contents can be used any number of times.

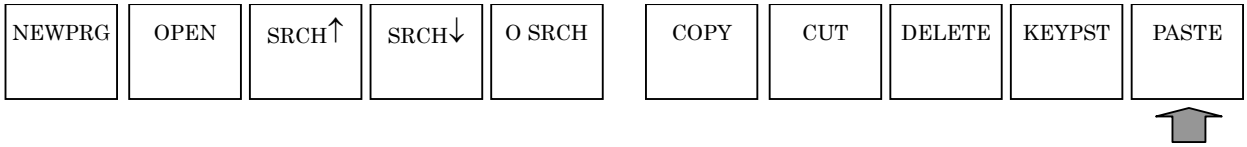
2.9 COPY



Pressing soft key [COPY] displays a message that prompts you to select a range of data to be copied. First select the copy range (by displaying it in yellow), using cursor keys, and then press soft key [COPY]. The data copied from the specified range is stored on the clipboard. The contents of the clipboard can be pasted to other places in the program and to other programs.

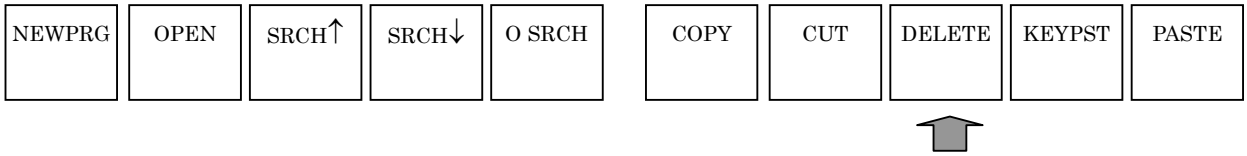
NOTE
Once a range of data to be copied has been selected, the cursor is placed back to the first block in the selected range if parameter CEB (No.14705#1) = 0 or the last block if CEB (No.14705#1) = 1.

2.10 PASTE



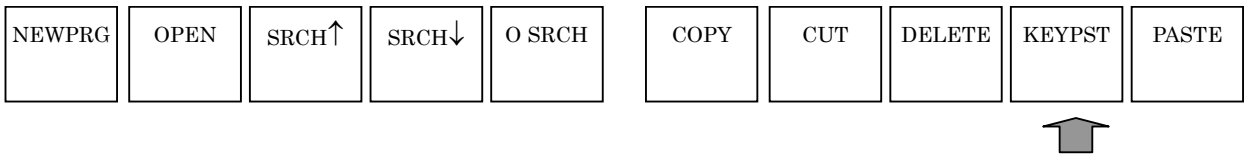
Pressing soft key [PASTE] pastes the contents of the clipboard to the place that immediately follows the current cursor position.
The clipboard contents are preserved.

2.11 DELETE



Pressing soft key [DELETE] displays a message that prompts you to select a range of data to be deleted. First select the delete range (by displaying it in yellow), using cursor keys, and then press soft key [SELECT]. A message appears which asks whether you really want to delete the selected data. Pressing soft key [YES] deletes the data.
The deleted data is not stored on the clipboard. The clipboard preserves the previous contents.

2.12 KEY-IN PASTE



Pressing soft key [KEYPST] copies the contents of a range selected (displayed in yellow) using the cursor to the key-in buffer.
Using the cursor keys <←> and <→> can move the cursor around in the key-in buffer, so a character entered using an MDI key can be placed immediately before the cursor. Pressing the <CAN> key deletes the character at the left of the cursor.
The character strings in the key-in buffer can be used in the same manner as for ordinary editing. Pressing the <ALTER> key, for example, replaces the cursor-selected current contents in the program with the current contents of the key-in buffer. Pressing the <INSERT> key inserts the key-in buffer contents to the place that immediately follows the cursor-selected contents in the program.
Pressing <CAN> after <SHIFT> clears the contents of the key-in buffer.

NOTE
Using key-in paste enables a very long comment program and part of a custom macro program block to be changed easily.

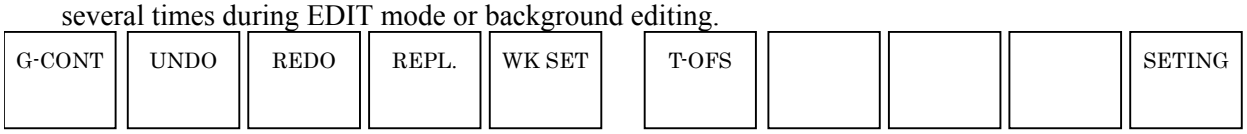
2.13 REPLACING WORD

It is possible to search a word that includes specified characters and replace it to other specified word.

2.13.1 Replacing a Word

The operations for replacing a word are as follows.

1) The following soft keys are displayed by pressing the leftmost soft key [<] or rightmost soft key [>] several times during EDIT mode or background editing.



2) If [REPL.] soft key is pressed, “REPLACE WORD” window will be displayed.
3) Input characters in “SRCH. CHAR. STR.” for search and press <INPUT> key.
4) Input word in “RPLC. CHAR. STR.” for replacement and press <INPUT> key.

5) Press [SRCH ↑] or [SRCH ↓] soft key and the cursor will move on a word that includes the characters specified in “SRCH. CHAR. STR.”.
6) Press [REPL.] soft key and the word will be replaced to the word that specified in “RPLC. CHAR. STR.”. The next word will be searched automatically.
7) Press [REPL.] soft key again, if the operator wants to replace the word which is selected by the cursor. Press [SRCH ↑] or [SRCH ↓] soft key, if the operator wants to search the next word without replacing.

2.13.2 Replace All Words

After the items “SRCH. CHAR. STR.” and “RPLC. CHAR. STR.” are filled and [ALL] soft key is pressed, the message “ARE YOU SURE TO EXECUTE REPLACE ALL?” will be displayed
If [YES] soft key is pressed during the message is displayed, all words that include characters of “SRCH. CHAR. STR.” will be replaced to the word of “RPLC. CHAR. STR.”.
After all words are replaced, the cursor moves on the last word that is replaced.

2.13.3 Replaced Words

The result of replaced word format is same as altering word by [ALTER] soft key.
The following table indicates examples of replaced words.

Examples of Replaced Word

Pattern and result of replacement	Search char.	Replace char.	Ex. of hit word	Result of replacement
Search character: Whole word Replace character: Free Result: Hit word is replaced to replace character.	X10.	Z20.	X10.	Z20.
Search character: A part of a word Replace character: Free Result: Hit word is replaced to replace character.	X1	Z20.	X10.	Z20.
Search character: Address Replace character: Free Result: Hit word is replaced to replace character.	X	Z20.	X10.	Z20.
Search character: Free Replace character: Value Result: Only the part of value of hit word is replaced to replace character.	X10.	20.	X10.	X20.

Pattern and result of replacement	Search char.	Replace char.	Ex. of hit word	Result of replacement
Search character: Free Replace character: Plural words Result: Hit word is replaced to replace character.	X10.	X20. Z30.	X10.	X20. Z30. ;
Search character: Plural words Replace character: Free Result: Hit word is replaced to replace character.	G0 X10.	G1 Z20.	G0 X10.	G1 Z20. X10. ;
Search character: Character in comment Replace character: Free Result: Whole hit word is replaced replace character.	COMMENT	Z20.	(COMMENT)	Z20.
Search character: Macro sentence Replace character: Free Result: Hit word is replaced to replace character.	BPRNT	DPRNT	BPRNT	DPRNT
Search character: EOB Replace character: Free Result: Hit word is replaced to replace character.	;	Z20.;	;	Z20. ;

2.13.4 Note

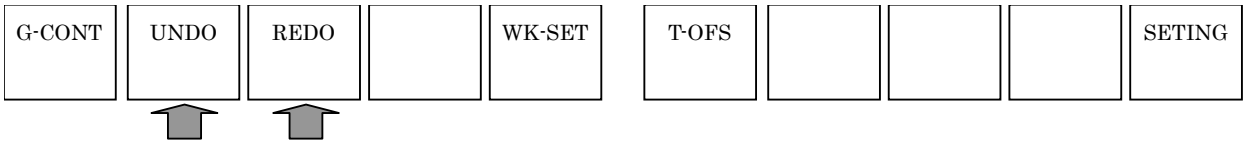
Please take notice of the followings.

1) O number and program name can't be replaced.
2) If the format of a word is incorrect after the operation of replacing, the operation becomes invalid.
3) If the number of characters of a block exceeds the limitation (200 characters) after the operation of replacing, the operation becomes invalid.
4) Replacing word function isn't executed when the items "SRCH. CHAR. STR." and "RPLC. CHAR. STR." are empty.
5) The cursor don't move during replacing all words.
6) If the mode is changed during replacing all words, it will be stop halfway. The cursor will move on the last word that is replaced.
7) If the screen changing by pressing function key or changing the path is executed during replacing all words, it will be executed after replacing all words.
8) "REPLACE WORD" window will be displayed at the lower right of the program window while the program window is enlarged.

9) Replacing word function can't be used during all path display / edit screen is displayed. The [REPL.] soft key isn't displayed.
10) The corresponding software version of the each function described in this part is as follows.

BX24/BX26/BX28/BX29 (Series 0*i*-C)	: After version 03
BX32 (Series 0*i*-D)	: After version 01
BX00/BX01 (Series 16*i*/18*i*/21*i*-B)	: After version 21
BY70 (Series 30*i*/31*i*/32*i*-A)	: After version 26
BX71 (Series 30*i*/31*i*/32*i*-B)	: After version 01

2.14 UNDO, REDO



Pressing soft key [UNDO] during editing in the MDI mode, EDIT mode, or MEM mode can cancel (undo) a program editing operation using the MANUAL GUIDE *i*. Pressing soft key [REDO] can cancel (redo) the cancellation of an operation. This function can be used, for example, when a block has been deleted by mistake.

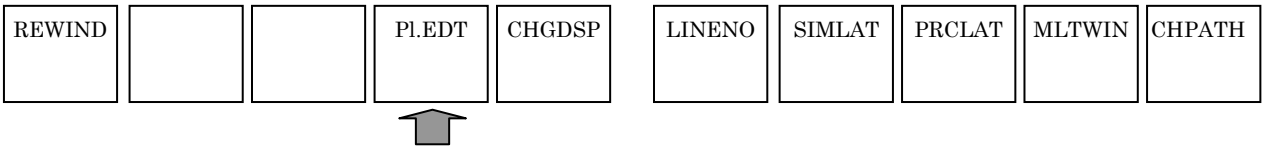
NOTE
1 Before the undo function or redo function can be used with MANUAL GUIDE *i*, the size of a buffer for storing operations must be set in parameter No. 14861. When 0 is set in the parameter, a 5-KB buffer is allocated.
2 The undo and redo functions can be used for editing operations in a selected program. This means that the creation of a new program and the deletion of a program cannot be undone, and the editing of the key-in buffer cannot be undone.
3 When the screen display is switched or an operation for opening a program is performed, the buffer is cleared.

2.15 MULTI PART PROGRAM EDITING FUNCTION

Plural programs are displayed on one screen, and it is possible to edit them Simultaneously.

2.15.1 Start of Multi Part Program Editing Function

In the base screen of EDIT mode or the background edit screen of MEM mode, by pressing [Pl.EDT] soft key, multi part program editing screen is displayed.



2.15.2 Operation of Display Program Selection

(1) When screen of multi part program editing function of programs is displayed, please select the program for display by following methods.

<1> Selection of program on [O LIST] window
<2> Selection of program by pressing [O SRCH] soft key

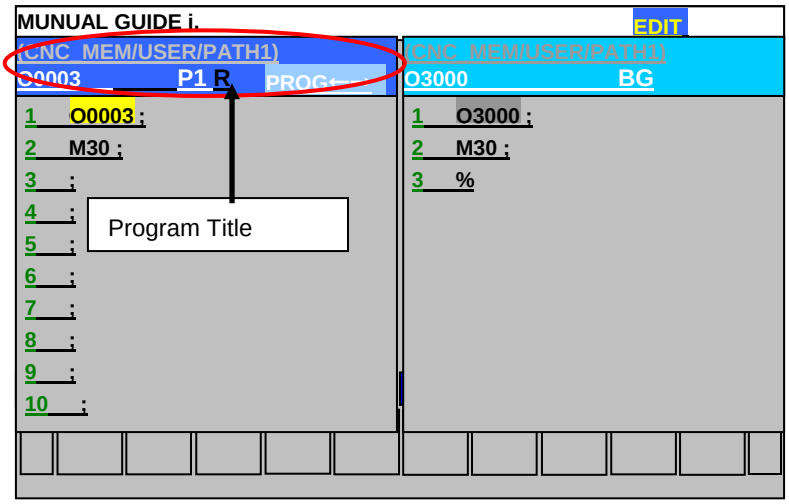
(2) On target of editing window, the selected program is displayed. However, only the following programs are displayed on read only mode.

<1> Main program on selected path (the background edit screen of MEM mode)
<2> Main program on other path

2.15.3 Display of Multi Part Program Editing Screen

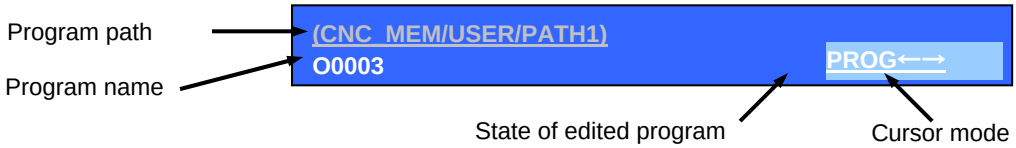
On multi part program editing function screen, plural programs are displayed on one screen, and it is possible to edit them simultaneously.

(Example of Multi part program editing function screen(10.4inch))



The title of the program is displayed when entering multi part program editing function .

(1) Program window title being edited
Program window title being edited is displayed in blue.



<1> Program path
The program path where the displayed program exists is displayed. When the number of characters of program path exceeds the width of the program window title, it is omitted by "...".

<2> Program name
Name of the displayed program is displayed. When the number of characters of program name exceeds the width of the program window title, it is omitted by "...".

<3> State of edited program
It is an State of the displayed program in each program window. The displayed State is as follows.

P1 : The main program selected by 1 path
P2 : The main program selected by 2 path
P3 : The main program selected by 3 path
BG : Program that was not selected in each path
R : read only program

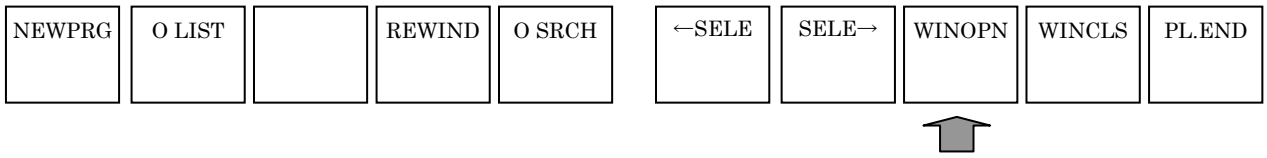
(2) Program window title not being edited
Program window title not being edited is displayed in light blue.

Moreover, the cursor of the program being edited displayed in yellow, the cursor of the program not being displayed in gray.

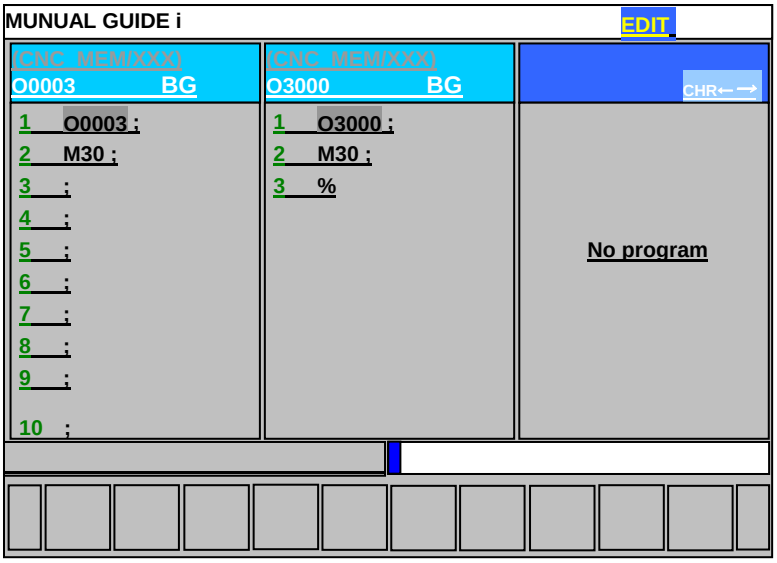
2.15.4 Opening and Shutting of Program Window

The program window is opened or shut. The program windows can be opened up to three in maximums.

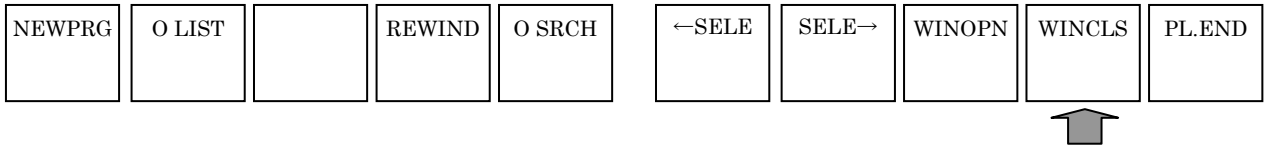
(1) By pressing [WINOPN] soft key, the blank window is added. At this time, the blank window become target of editing.



(Example of Screen where three programs are displayed simultaneously.)



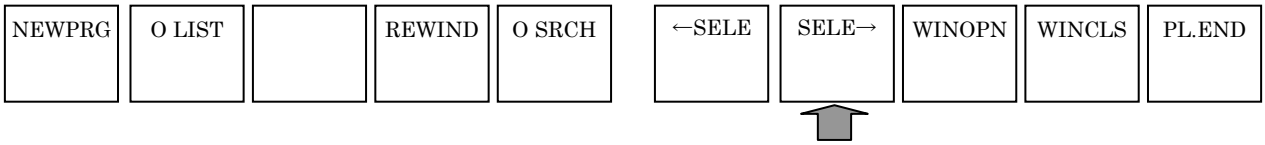
(2) When three program window exist, the program window is shut by pressing [WINCLS] soft key. The shutting program window is the program window that is target of editing.



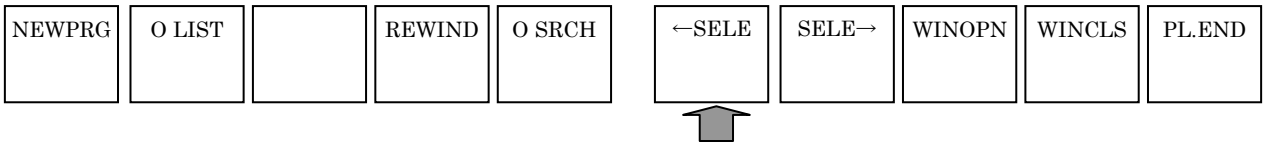
2.15.5 Switching Target of Editing Program

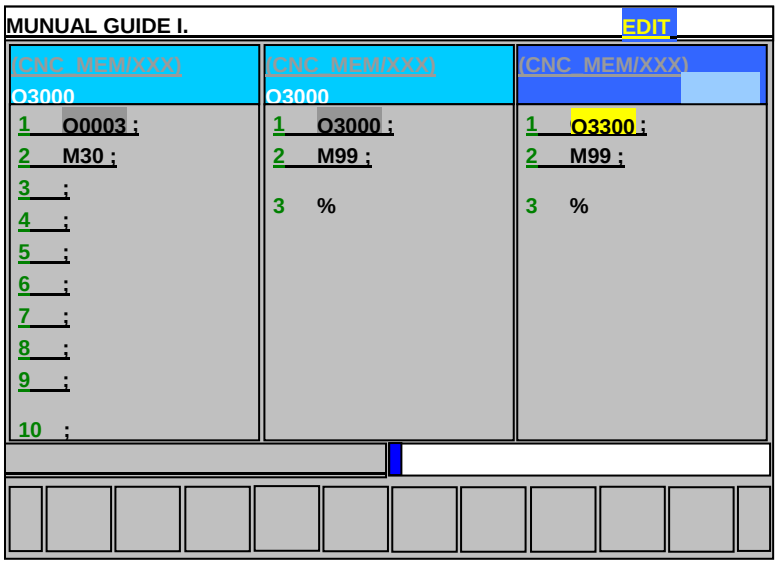
Operator can switch target of editing program.(*1)

(1) By pressing [SELE→] soft key or <SHIFT> key + cursor move key <→>,the target of editing program is switched in order of <1> → <2> → <3> → <1> on the figure below.



(2) By pressing [←SELE→] soft key or <SHIFT> key + cursor move key <←>,the target of editing program is switched in order of <1> → <3> → <2> →<1> on the figure below.

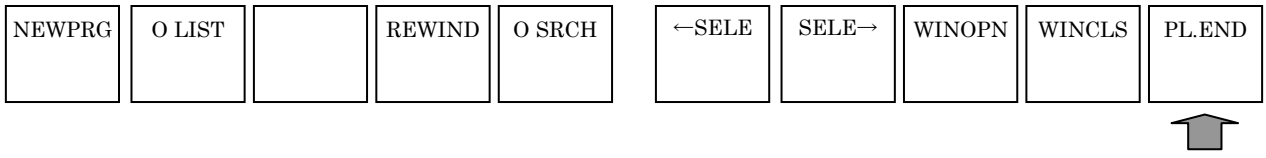




(*1) The main program being selected in the foreground was not switched.

2.15.6 End of Multi Part Program Editing Function of Programs

By pressing [PL.END] soft key, state of multi part program editing function of programs is released.



2.15.7 Operation of Switching To CNC Screen

State of multi part program editing function of programs is not released, when switching to the CNC screen from the multi part program editing function screen.

2.15.8 Operation of Changing Path

The state of multi part program editing function is independent in each path. Therefore, when the path is switched, if the path after switched is multi part program editing function, the multi part program editing function screen is displayed.

(Example) When switching from path to path 2, and returning it to path again.
Multi part program editing function screen in path 1 is displayed.



Path 1 **Path 2** **Path 1**

2.15.9 Operation of Changing Mode

When the mode is changed, state of multi part program editing function is released.

2.15.10 About the Maintenance of Displayed Program Information

The program that had been selected before the release is displayed, when the multi part program editing function is released once, and the multi part program editing function is started again.

2.15.11 Restrictions

(1) The following functions cannot be used while the multi part program editing function is used. Please release the state of multi part program editing function when you want to use their functions.

<1> Enlargement of displayed program
<2> Display of preview of Arbitrary figure
<3> Simultaneous editing of multipath program
<4> Display of machining simulation
<5> Process list editing function
<6> Replacement function

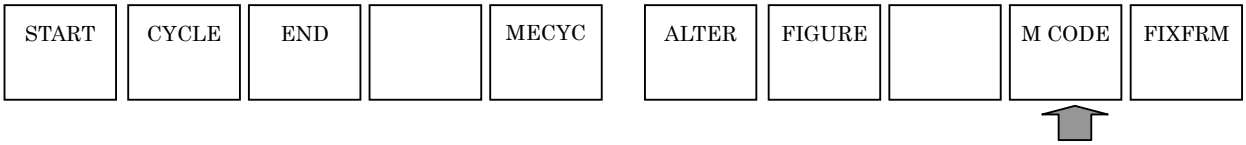
(2) The following operations cannot be done to the program that has already been displayed on multi part program editing function screen.

<1> Overwrite copy of program
<2> Delete of program
<3> Comment editing
<4> Deletion of Multiple Programs
<5> Opening program
<6> Alter program name of folder name

(3) The program displayed on the multi part program editing function screen cannot be edited on the CNC screen. However, the main program can be edited.

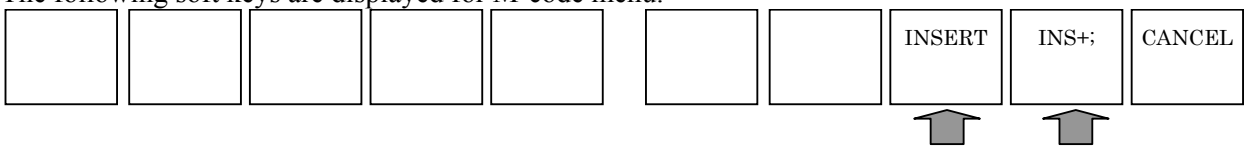
2.16 M-CODE MENU

Pressing the leftmost soft key [<] or rightmost soft key [>] several times displays soft key [M CODE] as follows:



Pressing soft key [M CODE] displays the M code menu.

The following soft keys are displayed for M-code menu.



Select an M code group, using the cursor keys <←> and <→>, and then select an M code from the M code group, using the cursor keys <↑> and <↓>.
Pressing soft key [INSERT] inserts the selected M code to the place that follows immediately the cursor position in the program. Pressing soft key [INS+;] inserts an EOB to the place that follows immediately the M code at the same time.
Entering M-code continuously and pressing soft key [INS+;] at last, plural M-code can be entered in one block.

If an M-code is selected in the program window using the cursor, pressing the soft key [ALTER] displays the M-code menu, which enables another M-code to be selected.

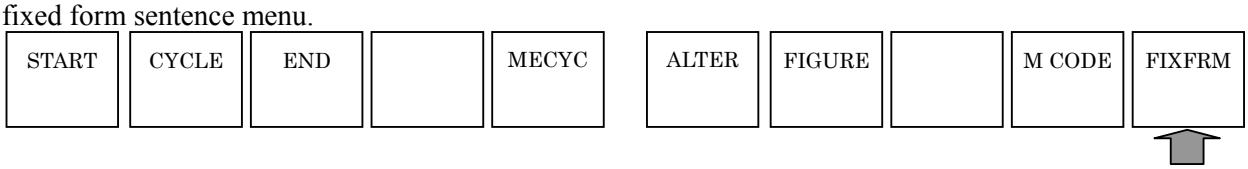
NOTE
1 In many cases, M codes in the M code menu are set up to a machine tool by the machine tool builder. So, the M code menu varies from one machine tool to another.
2 If the bit 4 (MCD) of parameter No. 14850 is set to 1, M code menu is disabled.

2.17 FIXED FORM SENTENCE INSERTION

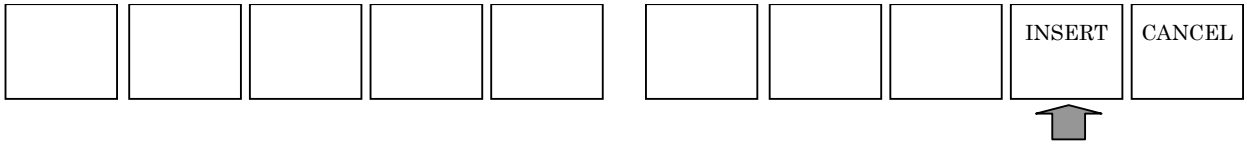
2.17.1 Fixed Form Sentence

Pressing the leftmost soft key [<] or rightmost soft key [>] several times displays soft key [FIXFRM] for milling or turning.
However, there is a case that either of them is displayed depending on the machine construction, so into details, refer to the document made by machine tool builder.

If soft key [FIXFRM] is displayed together with a milling soft key menu, pressing it displays a milling fixed form sentence menu.



The following soft keys are displayed for the fixed form sentence menu.



Select a fixed form sentence group, using the cursor keys <←> and <→>, and then select a fixed form sentence from the fixed form sentence group, using the cursor keys <↑> and <↓>.
Pressing soft key [INSERT] inserts the selected fixed form sentence to the place that follows immediately the cursor position in the program.
Using this function makes it easy to enter machining program patterns (fixed form sentences) that may be used frequently.

As the following example, you can register the fixed form sentence, in which undefined data is replaced to the special character such as "?".

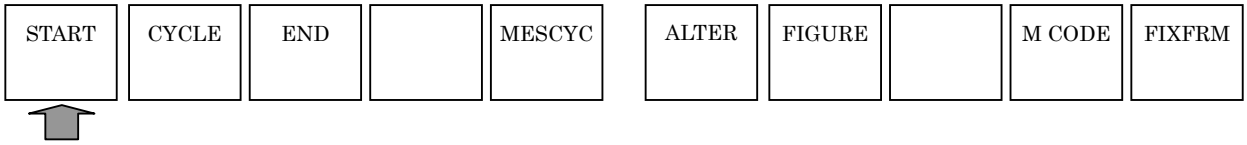
```
1. TOOL CHANGE
     G28 G91 X0. Y0. ;
     G28 Z0. ;
     T? ;
     M03 S? ;
```

When the fixed form sentence, in which this special character is included, is inserted to the part program, warning to urge an operator to enter the defined data.
Entering the data that is used actually and pressing <ALTER> can make the correct part program.

This special character is displayed in red in the program window.
This special character and displaying color can be modified by setting corresponding parameters.
Set ASCII code in decimal value of the special character to the parameter No.14860. If 0 is set to this parameter, "?" is used as the special character.
Set the color code of the displaying color to the parameter No.14480 in form of "AABBCC". AA is red color value, BB is green color value and CC is blue color value. If 0 is set to this parameter, the character is displayed in red.

NOTE
1 In many cases, fixed form sentences in the fixed form sentence menu are set up to a machine tool by the machine tool builder. So, the fixed form sentence menu varies from one machine tool to another.
Operators can make changes and additions to the menu. For details, see the descriptions about [SETING].
2 Start and end command fixed form sentences can be called from the fixed form sentence group separately, using, respectively, [START] and [END].

2.17.2 Start Command



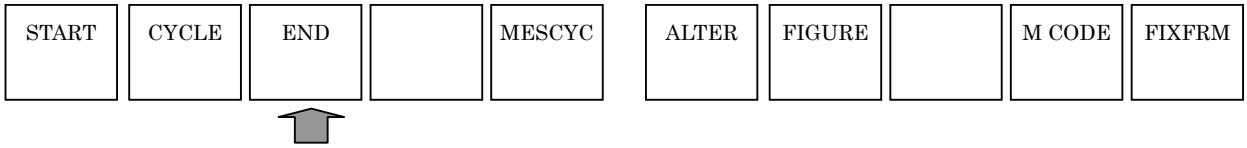
Pressing soft key [START] displays the start command fixed form sentence menu.

Select a fixed form sentence, using the cursor keys <↑> and <↓>.
Pressing soft key [INSERT] inserts the selected fixed form sentence to the place that follows immediately the current cursor position in the program.

This operation makes it easy to enter machining program patterns (fixed form sentences) that are used frequently at machining program start.

NOTE
In many cases, fixed form sentences in the fixed form sentence menu are set up to a machine tool by the machine tool builder. So, the fixed form sentence menu varies from one machine tool to another.
Operators can make changes and additions to the menu. For details, see the descriptions about [SETING].

2.17.3 End Command



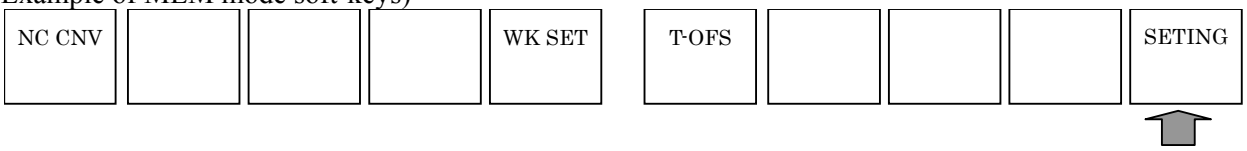
Pressing soft key [END] displays the end command fixed form sentence menu.

Select a fixed form sentence, using the cursor keys <↑> and <↓>.
Pressing soft key [INSERT] inserts the selected fixed form sentence to the place that follows immediately the current cursor position in the program.
This operation makes it easy to enter machining program patterns (fixed form sentences) that are used frequently at machining program start.

2.18 REGISTERING FIXED FORM SENTENCES

Soft key [SETING] to open the setting window can be displayed on all mode such as MEM, MDI, and manual mode.
Pressing the leftmost soft key [<] or rightmost soft key [>] several times displays the soft-keys including [SETING]

Example of MEM mode soft-keys)



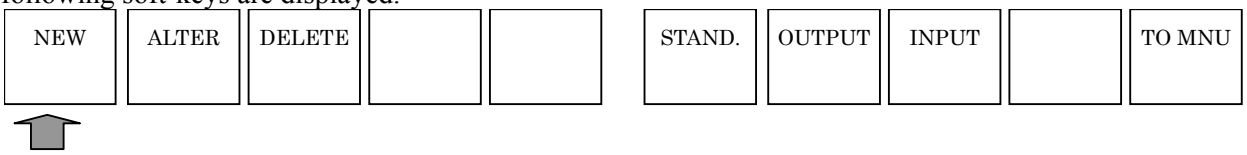
Pressing soft key [SETING] displays the window of setting data.
After selecting the menu of “REGISTER FIXED FORM SENTENCE FOR MILLING” by placing the cursor, pressing soft key [SELECT] displays the window of data setting.

With "FIXED FORM SENTENCE FOR MILLING," which is called by soft key [FIXFRM] displayed together with the milling menu, you can modify the contents of a selected fixed form sentence or add a new sentence.
Selecting “FIXED FORM SENTENCE FOR MILLING” displays the following screen.

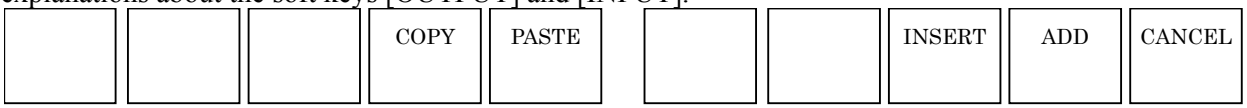
NOTE
1 The fixed form sentence menu displayed in the tab of “FORM1” has same contents with the one displayed in the “START” menu. Into detail, refer to the Subsection 2.14.2, “Start Command”.
2 The fixed form sentence menu displayed in the tab of “FORM5” has same contents with the one displayed in the “END” menu. Into detail, refer to the Subsection 2.14.3, “End Command”.
3 When the bit 3 (PRT) of parameter No. 14850 is ‘1’, prohibiting registering fixed form sentences by memory protection key will be enabled.

2.18.1 Registering a New Fixed Form Sentence

When the REGISTER FIXED FORM SENTENCE MILLING window is displayed on a screen, the following soft-keys are displayed.



By pressing soft key [NEW], a window for registering a new fixed form sentence appears. At the same time, the following soft keys appear:
See Section 2.3, “MEMORY CARD INPUT/OUTPUT OF FIXED FORM SENTENCES”, in Part III, for explanations about the soft keys [OUTPUT] and [INPUT].

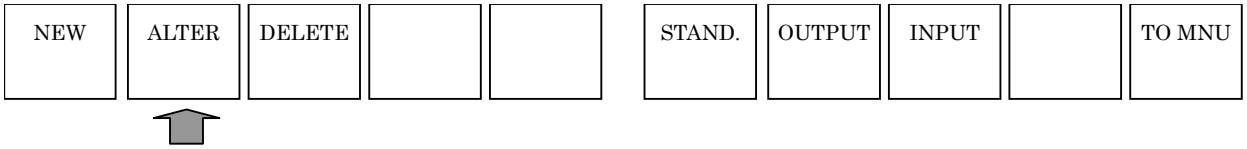


Position the cursor at the "REGISTERED NAME" item, from the MDI keyboard enter the name of the fixed form sentence you want to register, then press the <INPUT> key to input the name.
Then, position the cursor at the "REGISTERED SENTENCE" item, enter a fixed form sentence from the MDI keyboard, press the <INPUT> key, then press soft key [INSERT] or [ADD]. This can register the new fixed form sentence.

[COPY] : Select all registered names or fixed form sentences and copy them to the clipboard.
[PASTE] : Paste the clip board contents. You can copy a part of a machining program being edited to the clipboard in advance, then you can use it later to, for example, copy it in "REGISTERED SENTENCE."
[INSERT] : Add the name of the new fixed form sentence to be registered to a location immediately before the item positioned by the cursor when [NEW] is pressed. The menu numbers of the subsequent fixed form sentences are all incremented by one.
[ADD] : Add the name of the new fixed form sentence to be registered to the end of the menu already registered.
[CANCEL] : Cancel a registration operation.

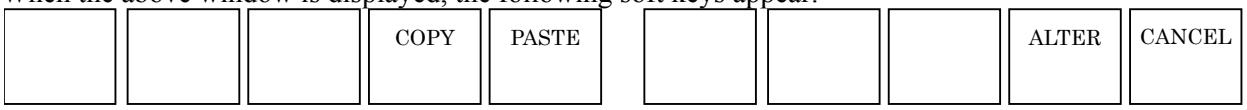
NOTE
1 About the number of fixed form sentences per tab and the maximum characters per fixed form sentence, the following settings can be selected.
<1> The number of fixed form sentences per tab is 10 and the maximum characters per fixed form sentence is 128.
<2> The number of fixed form sentences per tab is 5 and the maximum characters per fixed form sentence is 256.
2 The characters per fixed form sentence increase to 256 when bit 4 (CFF) of parameter No.14852 = 1.
3 Please power on again when the above parameter is changed. And after that, fixed form sentences will be initialized on starting the machine again.

2.18.2 Modifying a Fixed Form Sentence



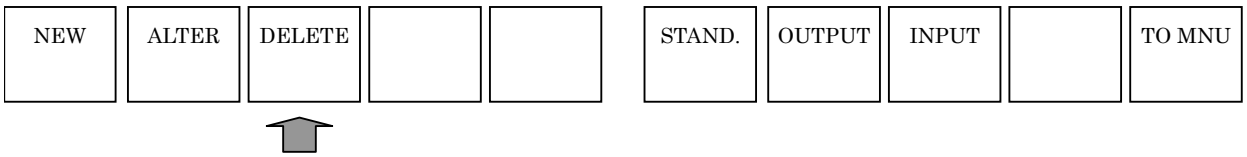
Position the cursor to the name of the fixed form sentence you want to modify, and press soft key [ALTER]. A window for modifying a fixed form sentence then appears.

When the above window is displayed, the following soft keys appear:



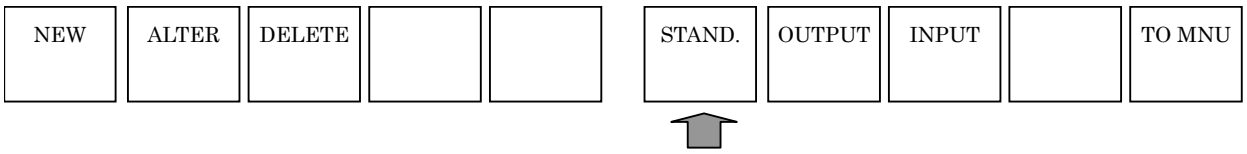
In the "REGISTERED NAME" and "REGISTERED SENTENCE" items, the contents of the selected fixed form sentence are displayed.
Select an item by using the cursor keys <↑> and <↓>, select the part you want to modify by using the cursor keys <←> and <→>, then enter a new character string from the MDI keyboard. This character string is inserted immediately before the cursor position.
With the <CAN> key, you can delete the previous character string one character at a time.
In the same way as in registration, soft keys [COPY] and [PASTE] can also be used.
When you have modified "REGISTERED NAME" and "REGISTERED SENTENCE," you must press the <INPUT> key at the end of the modification.
Pressing soft key [ALTER] replaces the original fixed form sentence with the new fixed form sentence.

2.18.3 Deleting a Fixed Form Sentence



Position the cursor at the name of the fixed form sentence you want to delete, and press soft key [DELETE]. Then a message for confirming a deletion operation is displayed. If you press soft key [YES], the fixed form sentence is deleted.
Pressing soft key [NO] cancels the deletion operation.

2.18.4 Initialization to Standard Fixed Form Sentences



The machine tool builder can factory-set particular fixed form sentences as standard fixed form sentences. For details, refer to the relevant manual issued by the machine tool builder.
In this case, the fixed form sentences are stored as initial data in a memory area in which data cannot be deleted.
You can use soft key [STAND.] to restore the initial state factory-set by the machine tool builder.

NOTE
1 When fixed form sentences are initialized, the sentences that have been entered or modified so far are all deleted; so, care is necessary.
2 Also when the machine tool builder reads standard fixed form sentences into memory, initialization using [STAND.] is required first.

When soft key [STAND.] is pressed, a message for confirming initialization is displayed. By pressing soft key [YES], initialization is performed.
Selecting soft key [NO] cancels the initialization.

2.18.5 Extension of Fixed Form Sentences (For only Series 30*i*/31*i*/32*i*-B, Series 0*i*-D)

The maximum number of the following items of fixed form sentences is extended.

Item	Maximum number
The number of tabs	7
The number of fixed form sentences per tab	10
The number of characters per fixed form sentence	256

(*) Fixed form sentences in the “INSERT ENDING COMMAND FOR TURNING” window are the same as 7th tab from the left in “REGISTER FIXED FORM SENTENCE FOR TURNING” window.
(*) Fixed form sentences in the “INSERT ENDING COMMAND FOR MILLING” window are the same as 7th tab from the left in “REGISTER FIXED FORM SENTENCE FOR MILLING” window.

2.18.5.1 Parameter

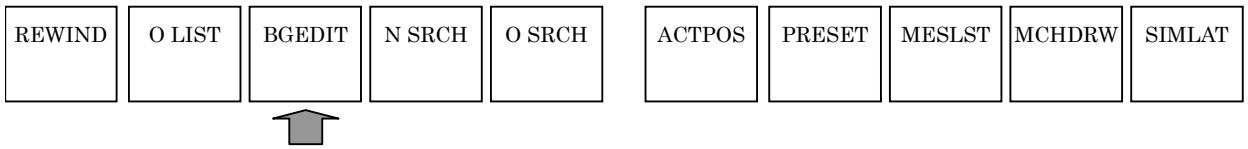
When the parameter No.14791#4 is set to 1, this function is available.

NOTE
Please power on again when the above parameter is changed. And after that, fixed form sentences will be initialized on starting the machine again.

2.19 BACKGROUND EDITING

During actual machining on the machine, contents of the other part program can be edited.

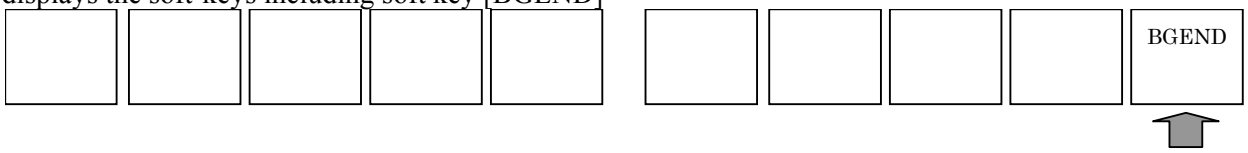
2.19.1 Starting Background Editing



In the MEM mode, pressing the soft key [BGEDIT] switches the screen display to the background editing screen and displays the program list screen.
Move the cursor to a machining program to be edited, with the cursor keys, then select the machining program by pressing soft key [OPEN].

2.19.2 Ending Background Editing

During background editing, pressing the leftmost soft key [<] or rightmost soft key [>] several times displays the soft-keys including soft key [BGEND]



Pressing the [BGEND] soft key ends the background editing screen and returns the screen display to the automatic operation screen.

NOTE
Switching the CNC modes during background editing will result in the background editing being terminated.

2.19.3 Operations During Background Editing

Operations during background editing are basically the same as editing operations in the EDIT mode. Note, however, that the use of some operations is restricted.

2.20 NOTES ON CREATING PROGRAMS

The notes that should be observed when creating a program are described below. Read the notes before creating a program.

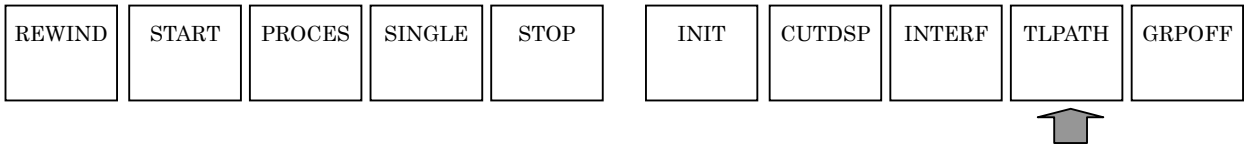
2.20.1 General Notes on Machining Programs

(1) Use ISO code form (G code commands) basically for a machining program. Enter a command for simple motion directly with a G code, and use cycle machining for complicated machining.
(2) Enter directly a tool change command involving the calling of the next tool by using a code such as a T code and M code or by using a subprogram call prepared by the machine tool builder, referring to the relevant operator's manual of the machine tool builder.
(3) Enter directly a tool length compensation command used with a machining center by using a G code such as G43.
(4) For changing tools or tool length compensation, a similar command is repeatedly entered. So, such a command can be entered easily by registering a fixed form sentence beforehand.
(5) MANUAL GUIDE *i* cannot handle any program having a block consisting of over 199 characters. Selecting a program having a block consisting of 200 characters or more (including space characters) will result in the message "BLOCK EXCEED MAXIMUM SIZE." appearing, thus disabling further editing.

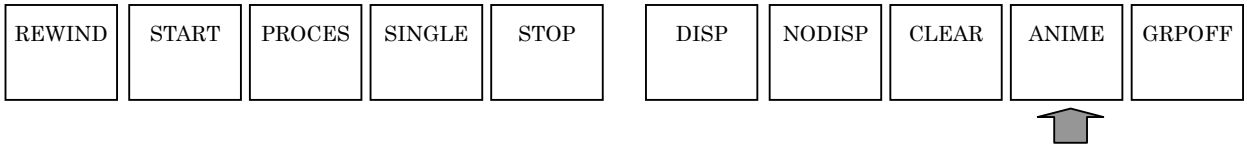
3 MACHINING SIMULATION

Select the MEM mode on the machine operator's panel.

When you press soft key [SIMLAT], the SIMULATE-ANIMATE screen (animation drawing) appears with the following soft keys:



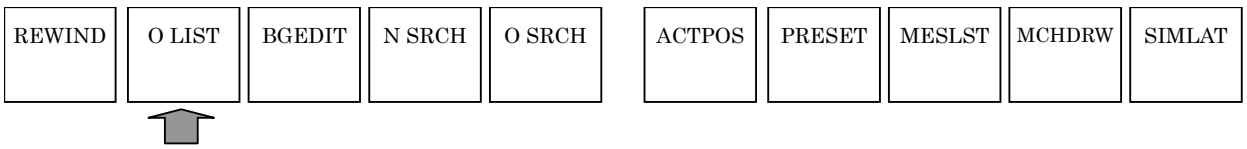
When you press soft key [TLPATH], the SIMULATE-TOOL PATH screen (tool path drawing) appears with the following soft keys:



Pressing soft key [ANIME] selects the mode for machining simulation (animation drawing) again.

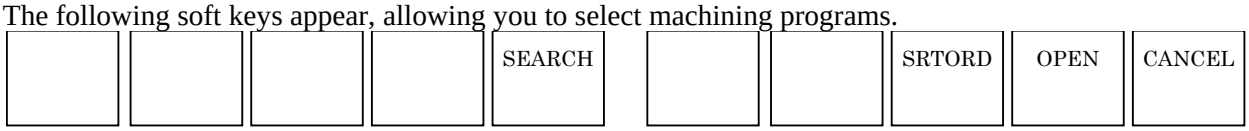
In machining simulation (animated tool path), pressing the soft key [GRPOFF] resumes the MEM mode window.

3.1 EDITING WITH THE PROGRAM LIST



By pressing soft key [O LIST], a window showing a list of registered machining programs is displayed.

The following soft keys appear, allowing you to select machining programs.



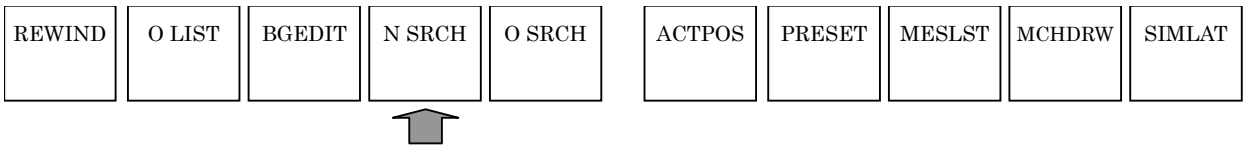
[SEARCH] : A window for searching programs appears. Enter the number of the program you want to search for with numeric keys, then press soft key [SEARCH]. The search window can be closed using soft key [CANCEL].

[OPEN] : After performing the above search operation or moving the cursor to a desired program number by using the cursor keys <↑> and <↓>, press soft key [OPEN]. Then the program is selected.

[CANCEL] : Close the list window.

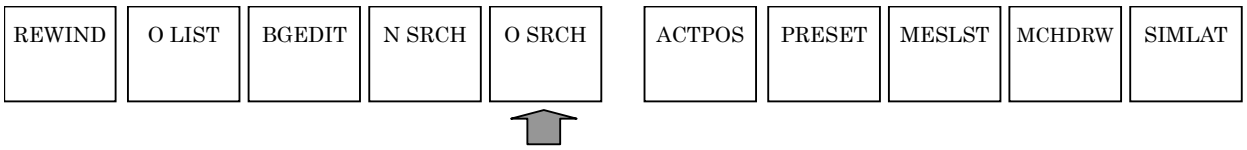
[SRTORD] : This soft key switches the sort order for displaying programs between ascending order and descending order.

3.2 SEARCHING FOR A SEQUENCE NUMBER IN A PROGRAM



When you enter the sequence number you want to search for by using numeric keys then press soft key [N SRCH], you can search for the block having the sequence number.
When you press soft key [N SRCH] without entering any data, the sequence number (N) is searched for sequentially.
If the sequence number is not found, a warning is displayed. It can be released by pressing a key.

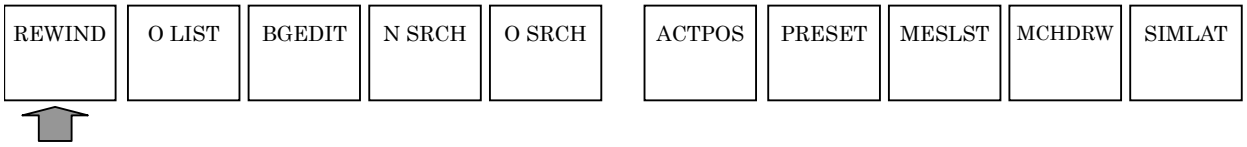
3.3 SELECTING A PROGRAM FOR MACHINING SIMULATION



When you enter the program number you want to select by using numeric keys then press soft key [O SRCH], the program with the program number is selected and displayed in the program window.
By pressing soft key [O SRCH] without entering any key, the next program is selected.

NOTE
If the bit 3 (OSR) of parameter No.3202 is set to “1”, pressing soft key [O SRCH] without entering any data, the next program is not selected.

3.4 REWINDING A PROGRAM



By pressing soft key [REWIND], you can return to the beginning of a selected program.

3.5 SETTING OF DATA FOR MACHINING SIMULATION

When machining simulation (animation drawing) can be performed, a blank figure and tool figure must be set. Such animation data must be set in the DRAWING DEFINITION block, which is to be entered in a machining program.
To display the window for entering DRAWING DEFINITION data, press [START] in the soft key menu for editing machining programs.

(1) Defining materials
Select the [BLANK] tab.
(2) Defining tools
Select the [TOOL] tab.
(3) Selecting a spindle
Select the [SEL. SPIND.] tab.

DRAWING DEFINITION			
	Blank form block	G1902	Rectangular solid
		G1900	Column
		G1906	Column (around X)
		G1901	Column with a hole
		G1907	Column with a hole (around X)
		G1903	Polygon
		G1904	Polygon with a hole
	Free contour cylinder blank figure block	G1970	Start point
		G1971	Line
		G1972	Arc (CW)
		G1973	Arc (CCW)
		G1974	Chamfering
		G1975	Corner rounding
		G1976	End
	Free contour cylinder blank figure block (around X)	G1970	Start point
		G1971	Line
		G1972	Arc (CW)
		G1973	Arc (CCW)
		G1974	Chamfering
		G1975	Corner rounding
		G1976	End
	Tool definition block	G1921	Drill (turning, milling)
		G1931	Counter sink tool (milling)
		G1932	Flat end mill (milling)
		G1933	Ball end mill (milling)
		G1922	Tap (turning, milling)
		G1923	Reamer (turning, milling)
		G1924	Boring tool (turning, milling)
		G1930	Face mill (milling)
	Spindle selection block	G1998	Spindle selection

NOTE
1 A tool figure during animation drawing is drawn by a tool definition block.
2 A blank is initialized by a blank figure block.
3 When you use ‘Tool Data Base Function’, you do not need to specify ‘Tool Definition Block’.
4 Displaying a column around X, column with a hole, or any other blank figure requires setting up parameter ML2 (No.27003#2) = 1.

(1) Input item of blank form block (rectangular solid: G1902)

WORK		
Data item		Meaning
B	WIDTH	Width of the rectangular solid blank. Length in the X-axis direction (positive value)
D	DEPTH	Depth of the rectangular solid blank. Length in the Y-axis direction (positive value)
H	HEIGHT	Height of the rectangular solid blank. Length in the Z-axis direction (positive value)
I	WORK ORIGIN X	Distance in the X-axis direction from the lower left corner of the rectangular solid to workpiece origin (positive or negative value)

WORK		
Data item		Meaning
J	WORK ORIGIN Y	Distance in the Y-axis direction from the lower left corner of the rectangular solid to workpiece origin (positive or negative value)
K	WORK ORIGIN Z	Distance in the Z-axis direction from the lower left corner of the rectangular solid to workpiece origin (positive or negative value)

(2) Input item of blank form block (column: G1900)

WORK		
Data item		Meaning
D	DIAMETER	Diameter of the column blank (positive value)
L	LENGTH	Length of the column blank (positive value)
K	WORK ORIGIN Z	Cutting allowance of the end face of the blank (Z-axis distance between the end face and workpiece origin) (positive value)
W	WORK ORIGIN Z (SPN 2)	Cutting allowance of the back end face of the blank (Z-axis distance between the back end face and workpiece origin) (positive value)

NOTE
The input item of “WORK ORIGIN Z(SPN 2) (W)“ is prepared for a machine with a sub spindle, and is displayed when the bit 1 (SUB) of parameter No.14702 = 1.

(3) Input item of blank form block (column(around X): G1906)

This blank is used for the machining simulation whose workpiece rotates around X-axis in machining centers.
Data setting is the same as the above column.

NOTE
1 There is the following limitations when the simulation whose workpiece rotates around X axis is executed.
- This function is available in only machining centers.
- Turning cycle can’t be simulated.
- Polar coordinate interpolation can’t be simulated.
- Under simulation as workpiece rotates workpiece around X-axis, coordinate is displayed in X, Y, -Z.

2 The following parameter setting is necessary.
- No.27003#2(ML2)=1, 27003#1(ML1)=0 and 27003#0(ML0)=0
- No.14717 : The axis number of work rotation axis

(4) Input item of blank form block (column with a hole: G1901)

WORK		
Data item		Meaning
D	DIAMETER	Diameter of the column blank (positive value)
E	INNER DIAMETER	Inner diameter of the column blank (positive value)
L	LENGTH	Length of the column blank (positive value)
K	WORK ORIGIN Z	Cutting allowance of the end face of the blank (Z-axis distance between the end face and workpiece origin) (positive value)
W	WORK ORIGIN Z (SPN 2)	Cutting allowance of the back end face of the blank (Z-axis distance between the back end face and workpiece origin) (positive value)

NOTE
The input item of “WORK ORIGIN Z (SPN 2) (W)” is prepared for a machine with a sub spindle, and is displayed when the bit 1 (SUB) of parameter No.14702 = 1.

(5) Input item of blank form block (column with a hole (around X): G1907)

This blank is used for the machining simulation whose workpiece rotates around X-axis in machining centers.
Data setting is the same as the above column with a hole.

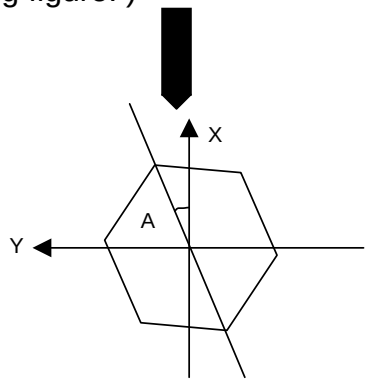
NOTE
1 There is the following limitations when the simulation whose workpiece rotates around X axis is executed.
- This function is available in only machining centers.
- Turning cycle can’t be simulated.
- Polar coordinate interpolation can’t be simulated.
- Under simulation as workpiece rotates workpiece around X-axis, coordinate is displayed in X, Y, -Z.

2 The following parameter setting is necessary.
- No.27003#2(ML2)=1, 27003#1(ML1)=0 and 27003#0(ML0)=0
- No.14717 : The axis number of work rotation axis

(6) Input item of blank form block (polygon: G1903)

WORK		
Data item		**Meaning**
R	NUMBER OF CORNER	The number of corner This must be a integer, larger than 2 and smaller than 100.
D	DIAMETER	Diameter of the polygon blank (positive value)
L	LENGTH	Length of the polygon blank (positive value)
A	ANGLE OF ATTACHMENT	The angle between a corner and X axis
K	WORK ORIGIN Z	Cutting allowance of the end face of the blank (Z-axis distance between the end face and workpiece origin) (positive value)
W	WORK ORIGIN Z (SPN 2)	Cutting allowance of the back end face of the blank (Z-axis distance between the back end face and workpiece origin) (positive value)

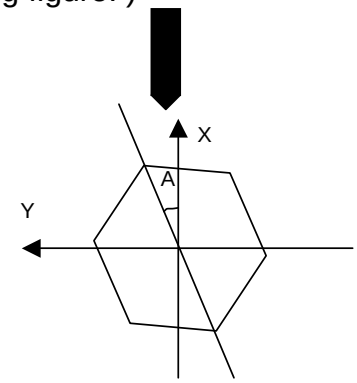
NOTE
1 The input item of “WORK ORIGIN Z(SPN 2) (W)“ is prepared for a machine with a sub spindle, and is displayed when the bit 1 (SUB) of parameter No.14702 = 1.
2 Only equilateral polygon can be displayed.
3 Angle of attachment is the angle that is formed by X-axis and one comer of blank figure.(Refer to the following figure.)



(7) Input item of blank form block (polygon with a hole: G1904)

WORK		
Data item		**Meaning**
R	NUMBER OF CORNER	The number of corner This must be a integer, larger than 2 and smaller than 100.
D	DIAMETER	Diameter of the polygon blank (positive value)
E	INNER DIAMETER	Inner diameter of the polygon blank (positive value)
L	LENGTH	Length of the polygon blank (positive value)
A	ANGLE OF ATTACHMENT	The angle between a corner and X axis
K	WORK ORIGIN Z	Cutting allowance of the end face of the blank (Z-axis distance between the end face and workpiece origin) (positive value)
W	WORK ORIGIN Z (SPN 2)	Cutting allowance of the back end face of the blank (Z-axis distance between the back end face and workpiece origin) (positive value)

NOTE
1 The input item of “WORK ORIGIN Z(SPN 2) (W)“ is prepared for a machine with a sub spindle, and is displayed when the bit 1 (SUB) of parameter No.14702 = 1.
2 Only equilateral polygon can be displayed.
3 Angle of attachment is the angle that is formed by X-axis and one comer of blank figure.(Refer to the following figure.)



(8) Input item of free contour cylinder blank figure block (start point: G1970)

ELEMENT (INPUT DATA)		
Data item		**Meaning**
DX	START POINT DX	X coordinate of the start point of an arbitrary figure (positive value)
Z	START POINT Z	Z coordinate of the start point of an arbitrary figure
W	WORK ORIGIN Z (SPN 2)	Cutting allowance of a blank rear end face (distance between the rear end face and workpiece origin in the Z-axis direction) (positive value) **(Note 3)**

NOTE
1 'INPUT DATA' means the items, which are displayed on the input data window in editing or altering.
This applies also for G codes G1971 to G1975.
2 Figure input is performed based on the +X-side cross section of the ZX plane of a round rod.
3 The input item of "WORK ORIGIN Z (SPN 2) (W)" is prepared for a machine with a sub spindle, and is displayed when the bit 1 (SUB) of parameter No.14702 = 1.

ELEMENT (OUTPUT DATA)		
Data item		**Meaning**
H	START POINT DX	X coordinate of the start point (input value)
V	START POINT Z	Z coordinate of the start point (input value)
W	WORK ORIGIN Z (SPN 2)	Cutting allowance of a blank rear end face (distance between the rear end face and workpiece origin in the Z-axis direction) (positive value) (input value)

NOTE
4 'OUTPUT DATA' means the items, which are displayed on the program window as creating program. It can be referenced only for program display purposes.
This applies also for G codes G1971 to G1975.

(9) Input item of free contour cylinder blank figure (line: G1971)

ELEMENT (INPUT DATA)		
Data item		**Meaning**
D	LINE DIRECTION	Select a line direction from the displayed soft key menu.
DX	END POINT DX	X coordinate of a line end point Remark) This item may not be displayed, depending on the input line direction value.
Z	END POINT Z	Z coordinate of a line end point Remark) This item may not be displayed, depending on the input line direction value.
A	ANGLE	Line angle Remark) This item may not be displayed, depending on the input line direction value.
L	LAST CONNECTION	[TANGNT] : Contacts the immediately preceding figure. [NOTHING] : Does not contact the immediately preceding figure. (initial value)
M	NEXT CONNECTION	[TANGNT] : Contacts the next figure. [NOTHING] : Does not contact the next figure. (initial value)

ELEMENT (OUTPUT DATA)		
Data item		Meaning
H	END POINT X	X coordinate of the end point of a straight line (calculation result)
V	END POINT Z	Z coordinate of the end point of a straight line (calculation result)
K	LINE DIRECTION	The direction of a straight line is selected from a menu indicated on a soft key. (input value)
C*	END POINT DX	X coordinate of the end point of a straight line (input value)
D*	END POINT Z	Z coordinate of the end point of a straight line (input value)
A*	ANGLE	Straight-line angle (input value)
L	LAST CONNECTION	[0] : Not in contact with the immediately preceding figure [1] : In contact with the immediately preceding figure (input value)
M	NEXT CONNECTION	[0] : Not in contact with the immediately preceding figure [1] : In contact with the immediately preceding figure (input value)
S	SELECT FIG. INFO.	In the case of plural intersection or contact, the operator sets a candidate. (input value)

(10) Input item of free contour cylinder blank figure (arc(CW): G1972) Input item of free contour cylinder blank figure (arc(CCW): G1973)

ELEMENT (INPUT DATA)		
Data item		Meaning
DX	END POINT DX	X coordinate of an arc end point
Z	END POINT Z	Z coordinate of an arc end point
R	RADIUS	Radius of an arc
CDX	CENTER POINT CDX	X coordinate of an arc center
CZ	CENTER POINT CZ	Z coordinate of an arc center
L	LAST CONNECTION	[TANGNT] : Contacts the immediately preceding figure. [NOTHING] : Does not contact the immediately preceding figure. (initial value)
M	NEXT CONNECTION	[TANGNT] : Contacts the next figure. [NOTHING] : Does not contact the next figure. (initial value)

ELEMENT (OUTPUT DATA)		
Data item		Meaning
H	END POINT X	X coordinate of an arc end point (calculation result)
V	END POINT Z	Z coordinate of an arc end point (calculation result)
R	RADIUS	Arc radius (calculation result)
I	CENTER POINT X	X coordinate of an arc center (calculation result)
J	CENTER POINT Z	Z coordinate of an arc center (calculation result)
C*	END POINT X	X coordinate of an arc end point (input value)
D*	END POINT Z	Z coordinate of an arc end point (input value)
E*	RADIUS	Arc radius (input value)
P*	CENTER POINT CDX	X coordinate of an arc center (input value)
Q*	CENTER POINT CZ	Z coordinate of an arc center (input value)
L	LAST CONNECTION	[0] : Not in contact with the immediately preceding figure [1] : In contact with the immediately preceding figure (input value)
M	NEXT CONNECTION	[0] : Not in contact with the immediately preceding figure [1] : In contact with the immediately preceding figure (input value)
S	SELECT FIG. INFO.	In the case of plural intersection or contact, the operator sets a candidate. (input value)

(11) Input item of free contour cylinder blank figure (chamfering: G1974)

ELEMENT (INPUT DATA)		
Data item		Meaning
C	CHAMFER	Chamfer (radius value, positive value)

ELEMENT (OUTPUT DATA)		
Data item		Meaning
H	END POINT X	X coordinate of an arc end point (calculation result)
V	END POINT Z	Z coordinate of an arc end point (calculation result)
C	CHAMFER	Amount of chamfering (radius value, positive value) (input value)

(12) Input item of free contour cylinder blank figure (corner rounding: G1975)

ELEMENT (INPUT DATA)		
Data item		Meaning
R	CORNER RADIUS	Corner R radius (radius value, positive value)

ELEMENT (OUTPUT DATA)		
Data item		Meaning
H	END POINT X	X coordinate of an arc end point (calculation result)
V	END POINT Z	Z coordinate of an arc end point (calculation result)
R	RADIUS	Arc radius (calculation result)
I	CENTER POINT X	X coordinate of an arc center (calculation result)
J	CENTER POINT Z	Z coordinate of an arc center (calculation result)
K	ROTATION DIRECTION	[2] : clockwise [3] : counterclockwise

(13) Input item of free contour cylinder blank figure (end: G1976)

This block is output at the end of a series of arbitrary figure blocks.

(14) Input item of free contour cylinder blank figure (around X) (start point: G1790)
Input item of free contour cylinder blank figure (around X) (line: G1791)
Input item of free contour cylinder blank figure (around X) (arc(CW): G1792)
Input item of free contour cylinder blank figure (around X) (arc(CCW): G1793)
Input item of free contour cylinder blank figure (around X) (chamfering: G1794)
Input item of free contour cylinder blank figure (around X) (corner rounding: G1795)
Input item of free contour cylinder blank figure (around X) (end: G1796)

This blank is used for the machining simulation whose workpiece rotates around X-axis in machining centers.
Data setting is the same as the above free contour cylinder blank figure.

NOTE

1 There is the following limitations when the simulation whose workpiece rotates around X axis is executed.
- This function is available in only machining centers.
- Turning cycle can't be simulated.
- Polar coordinate interpolation can't be simulated.
- Under simulation as workpiece rotates workpiece around X-axis, coordinate is displayed in X, Y, -Z.

2 The following parameter setting is necessary.
- No.27003#2(ML2)=1, 27003#1(ML1)=0 and 27003#0(ML0)=0
- No.14717 : The axis number of work rotation axis

NOTE

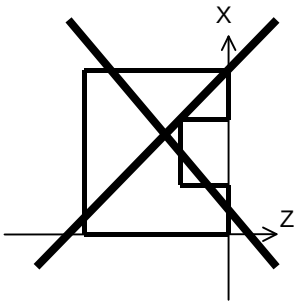
1 The arc commands (G1972 and G1973) and corner rounding command (G1975) are changed to linear elements with several blocks, and then displayed. Depending on the figure, it may require a longer time before being completely displayed.

2 Input the end point of a free contour cylinder blank figure such that it matches with the start point.

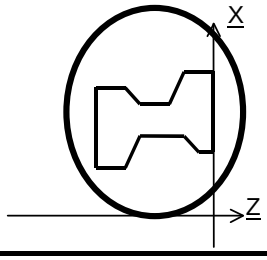
3 A series of arbitrary figures must be enclosed with G1970 (start point) and G1976 (arbitrary figure end).

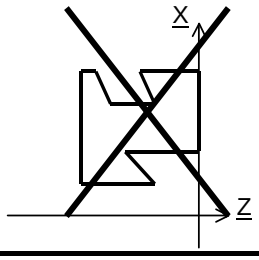
4 To edit an arbitrary figure, place the cursor on the block of G1970 (start point), then press [ALTER].

5 A figure for an end face portion cannot be specified. An end face portion must contain vertical lines only.



6 A figure in the Z direction from an end face portion must be a monotonously increasing or decreasing figure.





(15) Input item of tool definition block (drill: G1921)

TOOL		
Data item		Meaning
Q	SETTING	General-purpose tool: See the descriptions about G1910.
D	TOOL DIAMETER	Drill diameter (positive value)
B	NOSE ANGLE	Drill nose angle (positive value)
H	TIP LENGTH	Drill length (positive value)

(16) Input item of tool definition block (counter sink tool: G1931)

TOOL		
Data item		Meaning
Q	SETTING	General-purpose tool: See the descriptions about G1910.
D	BOTTOM DIAMETER	Diameter of the end of the counter sink tool end (positive value)
E	CUTER DIAMETER	Diameter of the counter sink tool (positive value)
H	TIP LENGTH	Length of the cutting portion of the counter sink tool (tool axis direction, positive value)
B	CUTTER LENGTH	Cutter length of the counter sink tool (tool axis direction, positive value)
L	SHANK LENGTH	Entire length of the counter sink tool (positive value)
W	SHANK DIAMETER	Diameter of the shank of the counter sink tool (positive value)

(17) Input item of tool definition block (flat end mill: G1932)

TOOL		
Data item		Meaning
Q	SETTING	General-purpose tool: See the descriptions about G1910.
D	TOOL DIAMETER	End mill diameter (positive value)
H	TIP LENGTH	End mill tool length (positive value)

(18) Input item of tool definition block (ball end mill: G1933)

TOOL		
Data item		Meaning
Q	SETTING	General-purpose tool: See the descriptions about G1910.
D	TOOL DIAMETER	End mill diameter (positive value)
H	TIP LENGTH	End mill tool length (positive value)

(19) Input item of tool definition block (tap: G1922)

TOOL		
Data item		Meaning
Q	SETTING	General-purpose tool: See the descriptions about G1910.
D	TOOL DIAMETER	Tap diameter (positive value)
H	TIP LENGTH	Tap length (positive value)

(20) Input item of tool definition block (reamer: G1923)

TOOL		
Data item		Meaning
Q	SETTING	General-purpose tool: See the descriptions about G1910.
D	TOOL DIAMETER	Reamer diameter (positive value)
H	TIP LENGTH	Reamer length (positive value)

(21) Input item of tool definition block (boring tool: G1924)

TOOL		
Data item		Meaning
Q	SETTING	General-purpose tool: See the descriptions about G1910.
D	TOOL DIAMETER	Boring tool diameter (positive value)
H	TIP LENGTH	Boring tool length (positive value)

(22) Input item of tool definition block (face mill: G1930)

TOOL		
Data item		Meaning
Q	SETTING	General-purpose tool: See the descriptions about G1910.
D	TOOL DIAMETER	Face mill diameter (positive value)
H	TIP LENGTH	Face mill length (positive value)

(23) Input item of spindle selection block: G1998

SEL. SPIND.		
Data item		Meaning
S	SPINDLE NUMBER	Spindle number of a sub spindle (positive number) Remark) Enter 2 when the sub spindle has the spindle number 2. Enter 3 when the sub spindle has the spindle number 3.

NOTE
The menu above is prepared for a machine with a sub spindle, and is displayed when bit 1 (SUB) of parameter No.14702 = 1.

3.6 MACHINING SIMULATION DISPLAY

3.6.1 Machining Simulation (Tool Path) (for Series 0*i*-C/16*i*/18*i*/21*i*)

The path of the tool in a machining program can be drawn without performing actual machining operation on the machine (machining simulation). In the case of Series 0*i*-D/30*i*/31*i*/32*i* see Subsection 3.6.3, “Machining Simulation (Tool Path Drawing) (for Series 0*i*-D/ 30*i*/31*i*/32*i*)”.

NOTE
1 To perform machining simulation, select the MEM mode with the mode switch on the machine operator’s panel.
2 If a coordinate system change is specified in the program during simulation, simulation is performed without changing the coordinate system.

NOTE

3 In the drawing during machining, when the workpiece specifying block is executed, form of a workpiece is drawn in form of wire frame. Drawing color of the workpiece can be specified by the parameter No.14773.
4 When a simulation window is opened, a blank figure drawing at the last drawing is displayed. However, if bit 4 (WDK) of parameter No. 27310 is set to 1, no blank figure is displayed when a simulation window is opened.
5 Tool path drawing with a multi-path system is performed only at the selected path for displaying. And a simulation window is initialized if the selected path for displaying is changed. So if the selected path for displaying is changed in displaying a free contour cylinder blank figure, the drawing is initialized for a column figure which encloses it.
6 Path drawing is performed using values in the workpiece coordinate system. Coordinates that allow for tool compensation (cutter compensation, tool length compensation, geometry compensation, and wear compensation), tool tip control, or the like are not used for drawing.
7 Each control modes of the high-speed and high-precision functions (commands for advanced preview control, AI advanced preview control, AI contour control, AI nano contour control, high-precision contour control, AI high-precision contour control, and AI nano high-precision contour control) cannot be turned ON in machining simulation.
8 In simulation of a program having a sequence of contiguous small blocks, drawing requires longer processing time, possibly causing machining time to become longer than actual machining time.

To stop the machining simulation (tool path drawing), press [GRPOFF].

When machining simulation (tool path drawing) is selected, the soft keys shown below appear. Pressing the leftmost soft key [<] or the rightmost soft key [>] changes the page of the soft key display to the second, third, or fourth page.

(1) 1st page soft-keys

REWIND	START	PROCES	SINGLE	STOP	DISP	NODISP	CLEAR	ANIME	GRPOFF

(2) 2nd page soft-keys

LARGE	SMALL	AUTO		ROTATE	←MOVE	MOVE→	↑MOVE	↓MOVE	CENTER

(3) 3rd page soft-keys

				WK SET	T-OFS				SETING

(4) 4th page soft-keys

REWIND	O LIST	CHGDSP		O SRCH	ACTPOS		MESLST		

3.6.1.1 Starting and stopping simulation

REWIND	START	PAUSE	SINGLE	STOP	DISP	NODISP	CLEAR	ANIME	GRPOFF

On the 1st page soft key, you can perform operations related to execution in machining simulation (tool path).
For [SETING], see Chapter 10, "SETTING DATA," in Part II.

[REWIND] : Return to the beginning of the machining program selected for machining simulation.
[START] : Start machining simulation for the currently selected machining program.
[PAUSE] : Stop machining simulation temporarily.
[SINGLE] : Cause a single-block stop when machining simulation is performed in the continuous mode. When machining simulation is in the stopped state, this soft key starts machining simulation in the single-block mode.
[STOP] : End machining simulation.
[NODISP] : From the block immediately after this soft key is pressed, suppress tool path drawing. Tool path drawing is not performed until soft key [DISP] is pressed next.
[DISP] : From the block immediately after this soft key is pressed, start drawing of the tool path.
Remark) Only necessary tool path portions can be drawn by using soft keys [DISP] and [NODISP].
[CLEAR] : Erase the tool path drawn so far. Tool path drawing immediately after this soft key is pressed is displayed.

NOTE
By setting bit 0 (PRC) of parameter No. 27310 to 1, soft key [PAUSE] can be changed to soft key [PROCES] (for temporarily stopping machining simulation at a block specifying M01;).

3.6.1.2 Scaling, movement, and other operations

LARGE	SMALL	AUTO		ROTATE	←MOVE	MOVE→	↑MOVE	↓MOVE	CENTER

2nd page soft keys allow you to perform scaling and movement for machining simulation (tool path drawing) as well as selection of a drawing coordinate system. These operations are the same as those in drawing during machining. For details, see Subsection 4.3.3, “Scaling, Movement, and Other Operations in Drawing during Machining”.

NOTE
1 Operations such as scaling, movement, and selection of a drawing coordinate system must be performed before machining simulation (tool path drawing) is started.
2 If a coordinate system change is specified in the program during simulation, simulation is performed without changing the coordinate system.
3 Using the following parameter can change a reference position used in enlarging or reducing drawn figures.
Parameter GCN (No.27310#1)
The reference position is the center of the:
Figure if GCN = 1
Window if GCN = 0

3.6.2 Machining Simulation (Animated Drawing) (FOR Series 0*i*-C /16*i*/18*i*/21*i*)

Animated simulation of a machining operation by a machining program can be performed without performing actual machining operation on the machine. In the case of Series 0*i*-D/30*i*/31*i*/32*i* see Subsection 3.6.4, "Machining Simulation (Animated) (for Series 0*i*-D/Series 30*i*/31*i*/32*i*)".

NOTE
1 To perform machining simulation, select the MEM mode with the mode switch on the machine operator's panel.
2 If a coordinate system change is specified in a program under simulation, whether to perform simulation with the same coordinate system or a changed coordinate system can be chosen using bit 1 (ACD) of parameter No. 27311. For details, see Subsection 3.6.7, "Changing Workpiece Coordinate During Machining Simulation (Animation, Tool Path Drawing)".
3 When a program that is composed by repeated minute line segments like a program outputted by CAD/CAM is executed, sometimes the speed of simulation much slows down. In this case, please set tool motion unit to parameter No. 27323 by dot. When tool moves over the value of this parameter on screen, the screen is updated and the speed will be accelerated. But, if the value of the parameter is too large, the resolution will decrease.
4 A tool tip position in animated simulation has coordinates of values in the workpiece coordinate system. Coordinates that allow for tool compensation (cutter compensation, tool length compensation, geometry compensation, and wear compensation), tool tip control, or the like are not used for drawing.
5 Each control modes of the high-speed and high-precision functions (commands for advanced preview control, AI advanced preview control, AI contour control, AI nano contour control, high-precision contour control, AI high-precision contour control, and AI nano high-precision contour control) cannot be turned ON in machining simulation.
6 In simulation of a program having a sequence of contiguous small blocks, drawing requires longer processing time, possibly causing machining time to become longer than actual machining time.

To close the machining simulation (animated drawing) window, press [GRPOFF].
When machining simulation (animated drawing) is selected, the soft keys shown below appear. Pressing the leftmost soft key [<] or the rightmost soft key [>] changes the page of the soft key display to the second, third, or fourth page.

(1) 1st page soft-keys

REWIND	START	PROCES	SINGLE	STOP		INIT	CUT DSP	INTERF	TLPATH	GRPOFF

(2) 2nd page soft-keys :

LARGE	SMALL	AUTO	REVERS	ROTATE		←MOVE	MOVE→	↑MOVE	↓MOVE	CENTER

(3) 3rd page soft-keys :

				WK SET		T-OFS				SETING

(4) 4th page soft-keys :

REWIND	O LIST	CHGDSP	N SRCH	O SRCH	ACTPOS	PRESET	MESLST		

3.6.2.1 Starting and stopping machining simulation

REWIND	START	PROCES	SINGLE	STOP	INIT	CUT DSP	INTERF	TLPATH	GRPOFF

On the 1st page soft keys, you can perform operations related to execution in machining simulation (animated). The operations of [REWIND], [START], [PAUSE], [SINGLE], and [STOP] are the same as the operations in machining simulation (tool path). For details, see Subsection 3.6.1, “Machining Simulation (Tool Path) (for Series 0*i*-C/16*i*/18*i*/21*i*)”.

[INIT] : Initialize the machined blank used for animation.
[CUT DSP] : Lets you switch between 1/4 workpiece and entire periphery for rod blanks and drilled rod blanks. Before starting animated simulation, be sure to switch if necessary.
[INTERF] : Lets you select whether to make an interference check during animated simulation. If an interference check is enabled, a warning is displayed when the tool tip collides with a workpiece during rapid traverse, and the portion collided with the tool tip is displayed in the same color as that of the tool.

NOTE
With bit 0 (ITF) of parameter No. 27311, you can select continued operation (ITF (No. 27311#0) = 0) or temporary stop (ITF (No. 27311#0) = 1) if tool interferes with the workpiece during animation.

3.6.2.2 Scaling, movement, and other operations

LARGE	SMALL	AUTO	REVERS	ROTATE	←MOVE	MOVE→	↑MOVE	↓MOVE	CENTER

The soft keys on the 2nd page allow you to perform scaling and movement for machining simulation (animated drawing) as well as selection of a drawing coordinate system. These operations are the same as those in drawing during machining. For details, see Subsection 4.3.3, “Scaling, Movement, and Other Operations in Drawing during Machining”.

[REVERS] : Change the viewpoint of the blank in animation to the exactly opposite position. You can use this soft key when, during machining with a sub-spindle or machining with the C-axis rotated, for example, you want to see animation from the opposite side.

NOTE
1 Although soft key [REVERS] may be used even during machining simulation, switching will sometimes be delayed depending on the CNC status.
2 Operations such as scaling, movement, and selection of a drawing coordinate system must be performed before machining simulation (tool path) is started.

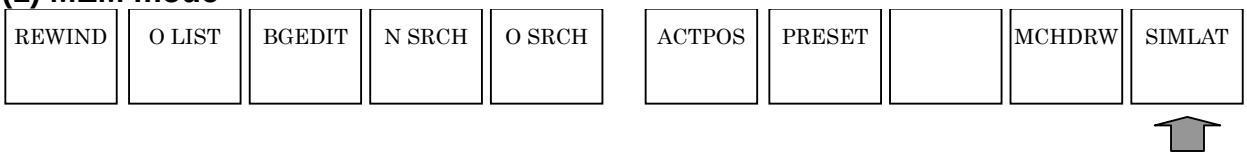
3.6.3 Machining Simulation (Tool Path Drawing) (for Series 0*i*-D/ 30*i*/31*i*/32*i*)

During machining, the tool path of another program can be drawn.
With Series 0*i*-D/30*i* MANUAL GUIDE *i*, the terms related to operation and drawing are defined as follows:

Automatic operation	Operation performed for actual machining
Drawing during machining	Tool path drawing during automatic operation
Background operation	Virtual operation performed for simulation. This operation can be performed simultaneously with automatic operation.
Machining simulation	Machining simulation in background operation (generic term for tool path drawing and animated simulation)
Tool path drawing	Machining simulation for drawing the tool path in background operation
Animated drawing	Machining simulation for animated simulation of background operation

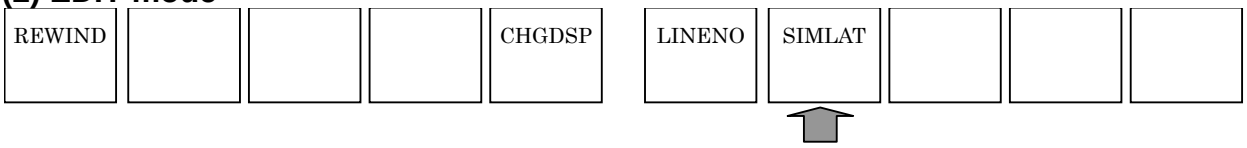
With FANUC Series 0*i*-D/30*i* MANUAL GUIDE *i*, machining simulation can be performed in the MEM mode and EDIT mode. When each mode is selected, the machining simulation screen can be displayed using the operation described below.

(1) MEM mode



Pressing soft key [SIMLAT] displays the machining simulation (animated) screen and the soft keys for machining simulation.

(2) EDIT mode



Pressing soft key [SIMLAT] displays the machining simulation (animated) screen and the soft keys for machining simulation.

NOTE

1 A program subject to simulation is placed in the background editing selection state. So, if background editing is in progress when the soft key [SIMLAT] is pressed, the simulation screen cannot be displayed. (The warning message "CAN NOT OPERATE ON BG EDIT" is displayed.) Terminate background editing then press the [SIMLAT].
2 A program selected when the soft key [SIMLAT] is pressed is the program being selected in the foreground.
3 When machining simulation is terminated by pressing the soft key [GRPOFF], the program selected for drawing is closed, and the program selected in the foreground is placed in the foreground selection state.
4 If the screen display is switched to another screen during drawing, machining simulation is terminated.
5 If the machining simulation screen is displayed again after the machining simulation screen is terminated, the results of the previous simulation are erased.
6 Animated simulation assumes that the tool tip is placed at the current position. So, if the tool is inclined with the tool inclination axis, drawing is performed at a position different from the actual tool position until the next move command is specified for the tool tip. Coordinates that allow for tool compensation (cutter compensation, tool length compensation, geometry compensation, and wear compensation), tool tip control, or the like are not used for drawing.
7 Each control modes of the high-speed and high-precision functions (commands for AI contour control I, AI contour control II) cannot be turned ON in machining simulation.
8 In simulation of a program having a sequence of contiguous small blocks, drawing requires longer processing time, possibly causing machining time to become longer than actual machining time.

To close the machining simulation (tool path drawing) window, press [GRPOFF].

When machining simulation (tool path drawing) is selected, the soft keys shown below appear. Pressing the leftmost soft key [<] or the rightmost soft key [>] changes the page of the soft key display to the second, third, or fourth page.

(1) 1st page soft-keys :

REWIND	START	PAUSE	SINGLE	STOP	DISP	NODISP	CLEAR	ANIME	GRPOFF

(2) 2nd page soft-keys :

LARGE	SMALL	AUTO		ROTATE	←MOVE	MOVE→	↑MOVE	↓MOVE	CENTER

(3) 3rd page soft-keys :

				WK SET	T-OFS				SETING

(4) 4th page soft-keys

REWIND	O LIST	↑ SRCH	↓ SRCH	O SRCH	ACTPOS	PRESET	MESLST		

NOTE
1 The data displayed on the machining simulation screen such as the current position and remaining travel distance is not automatic operation state data but is machining simulation data.
2 The machining simulation screen does not display spindle information and feed information but displays the next travel distance only.
3 On the machining simulation screen, the remaining travel distance is 0 at all times.
4 To stop machining simulation, press the soft key [STOP] soft key. If <RESET> key is pressed, foreground operation is reset.
5 If a blank registration command is executed in tool path drawing during machining simulation, the blank figure is displayed using a line (wire frame). Use parameter No. 14773 to specify the display color of the blank figure.
6 When the simulation window is displayed, the blank figure data previously displayed is used for drawing. However, when bit 4 (WDK) of parameter No. 27310 is set to 1, no blank figure is displayed if the simulation window is open.
7 In machining simulation performed with a multipath system, the tool path of the displayed path only is drawn. If the displayed path is switched during path drawing, the drawing screen is initialized. So, if the displayed path is switched when an arbitrary figure blank is displayed, the blank is initialized to a round rod figure enclosing the arbitrary figure.
8 If the animated simulation option is not selected, a blank figure command is ignored, and no blank figure (wire frame) is displayed.
9 When simulation is started from a midway point, the modal state of the NC is not always executed up to the midway point. So, if simulation is started from a midway point, correct drawing operation may not be performed.

If you want to prevent the macro executor program from running during machining simulation, change its processing according to the state of system variable #3010.

#3010 can take the values listed below.

System variable	Value	Executing State
#3010	1	Normal condition (Other than the following status)
	-1	Executing machining simulation (animated or tool path drawing)

3.6.3.1 Starting and stopping machining simulation

REWIND	START	PAUSE	SINGLE	STOP		DISP	NODISP	CLEAR	ANIME	GRPOFF

On the 2nd page soft key, you can perform operations related to execution in machining simulation (tool path).

[REWIND] : Return to the beginning of the machining program selected for machining simulation.
[START] : Start machining simulation for the currently selected machining program.
[PAUSE] : Stop machining simulation temporarily.
[SINGLE] : Cause a single-block stop when machining simulation is performed in the continuous mode. When machining simulation is in the stopped state, this soft key starts machining simulation in the single-block mode.
[STOP] : End machining simulation.

[NODISP] : From the block immediately after this soft key is pressed, suppress tool path drawing. Tool path drawing is not performed until soft key [DISP] is pressed next.
[DISP] : From the block immediately after this soft key is pressed, start drawing of the tool path.
Remark) Only necessary tool path portions can be drawn by using soft keys [DISP] and [NODISP].
[CLEAR] : Erase the tool path drawn so far. Tool path drawing immediately after this soft key is pressed is displayed.

NOTE
By setting bit 0 (PRC) of parameter No. 27310 to 1, soft key [PAUSE] can be changed to soft key [PROCES] (for temporarily stopping machining simulation at a block specifying M01;).

3.6.3.2 Scaling, movement, and other operations

The operations are the same as those operations performed with Series 0*i* -C/Series 16*i*/18*i*/21*i*-MODEL B.

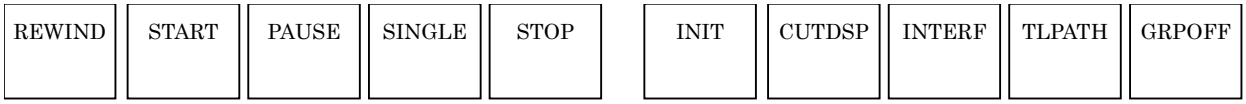
3.6.4 Machining Simulation (Animated Drawing) (for Series 0*i*-D/ 30*i*/31*i*/32*i*)

During machining, animated simulation can be performed for another program.
The terms related to operation and drawing for tool path drawing described in Subsection 3.6.3, "Machining Simulation (Tool Path Drawing) (for Series 0*i*-D/30*i*/31*i*/32*i*)" are applicable. See Subsection 3.6.3.
With Series 0*i*-D/30*i* MANUAL GUIDE *i*, machining simulation (animated) can be performed in the MEM mode and EDIT mode. The operation for displaying the machining simulation screen in each mode is the same as for tool path drawing. See Section 9.4.

If you want to prevent the macro executor program from running during machining simulation, change its processing according to the state of system variable #3010.
#3010 can take the values listed below.

System variable	Value	Executing State
#3010	0	Normal condition (Other than the following status)
	-1	Executing machining simulation (animated or tool path drawing)

3.6.4.1 Starting and stopping machining simulation



On the 1st page soft keys, you can perform operations related to execution in machining simulation (animated drawing). The operations of [REWIND], [START], [PAUSE], [SINGLE], and [STOP] are the same as the operations in machining simulation (tool path drawing). For details, see Subsection 3.6.3, "Machining Simulation (Tool Path Drawing) (for Series 30*i*/31*i*/32*i*)".

[INIT] : Initialize the machined blank used for animation.
[CUTDSP] : Lets you switch between 1/4 workpiece and entire periphery for rod blanks and drilled rod blanks. Before starting animated simulation, be sure to switch if necessary.
[INTERF] : Lets you select whether to make an interference check during animated simulation. If an interference check is enabled, a warning is displayed when the tool tip collideswith a workpiece during rapid traverse, and the portion collided with the tool tip is displayed in the same color as that of the tool.

NOTE
With bit 0 (ITF) of parameter No. 27311, you can select continued operation (ITF (No.27311#0) = 0) or temporary stop (ITF (No.27311#0) = 1) if tool interferes with the workpiece during animation.

3.6.4.2 Scaling, movement, and other operations

The operations are the same as those operations performed with Series 0*i*-C/16*i*/18*i*/21*i*-MODEL B.

3.6.5 Data Handled during Machining Simulation (for Series 0*i*-D/ 30*i*/31*i*/32*i*)

During machining simulation (background operation), data is handled as indicated below.

(1)	Parameter	The same parameters are used for machining simulation and automatic operation.
(2)	Tool compensation value Workpiece origin offset value Extended workpiece origin offset value Macro variable Fixture offset data	Data used for machining simulation differs from data used for automatic operation.
(3)	Tool life management data Tool number offset data 3-dimensional error compensation data	These data items are not used for machining simulation.

(1) Parameter
The same parameters as used for automatic operation are used for machining simulation. However, no parameter can be rewritten in background operation. (If an attempt is made to rewrite a parameter by programmable parameter input (G10L52), machining simulation stops (in case that bit 3 (G10) of parameter No.27310 is set to 0) or the G10 command is ignored (in case that bit 3 (G10) of parameter No.27310 is set to 1).)

(2) Tool compensation value, workpiece origin offset value, extended workpiece origin offset value, macro variable, etc.
Data used for background operation differs from data used for automatic operation. When background operation is started, the data for automatic operation is copied to produce data for background operation. Afterwards, the data for background operation and the data for automatic operation are handled separately from each other. So, even when data is rewritten using G10, for example, background operation and automatic operation do not affect each other. Note, however, that data rewritten in background operation is erased without being reflected in the actual data (foreground data).

(3) Tool life management data, tool number offset data, 3-dimensional error compensation data
These data items are not used for background operation. (If an attempt is made to rewrite data by using G10, machining simulation stops (in case that bit 3 (G10) of parameter No.27310 is set to 0) or the G10 command is ignored (in case that bit 3 (G10) of parameter No.27310 is set to 1).)

3.6.6 Functions Operating Differently between Machining Simulation and Automatic Operation (for Series 0*i*-D/ 30*i*/31*i*/32*i*)

The functions listed below are major functions that operate in background operation and automatic operation differently from each other. There are additional functions that operate differently.

(1) Custom macro
  (a) Interface signal
  #1000 to #1035 are always assumed to be 0 at all times.
  (b) Message output
  A message output with #3006 is not displayed but is ignored.
  (c) Clock
  #3001 and #3002 are ignored. So, note that if a command as indicated below is specified, for example, drawing does not proceed:
  #3001=1;
  WHILE[#3001 LE100]DO1;
  END1;
  (d) Mirror image
  #3007 is assumed to be 0 at all times.
  (e) State during program restart operation
  #3008 is assumed to be 0 at all times.
  (f) External output command
  BPRNT, DPRNT, POPEN, and PCLOS are ignored.

(2) Functions ignored
  (a) G04 (Dwell)
  (b) G20, G21 (Inch/metric conversion)
  (c) Auxiliary function (M, S, T, B)
  (d) G22,G23 (Stored stroke limit on/off)
  (e) G10.6 (Tool retraction data setting)
  (f) G10.9 (Programmable diameter/radius switching)
  (g) G81.1 (Chopping)
  (h) G25/G26 (Spindle speed fluctuation detection on/off)

(3) Functions that operate differently in part
  (a) G28 (automatic reference position return) is drawn up to a midway point.
  (b) G29 (automatic return from the reference position) is drawn from a midway point.
  (c) G27 (reference position return check) does not make a reference position return check.
  (d) Stored stroke limit checks are not made.
  (e) G31 (skip function), and G31.1, G31.2, and G31.3 (multi-step skip) are drawn up to a specified position, regardless of the skip signal.
  (f) G60 (single direction positioning) is drawn directly up to a specified position at all times even when the positioning direction is reverse.

3.6.6.1 Functions that cannot be used for machining simulation

(1) Functions that operate differently in background drawing
When the functions below are specified, the operations described below are performed.
G02.2/G03.2 : Involute interpolation
  Circular interpolation is performed.
G06.1 : Spline interpolation
  Linear interpolation is performed.
When the functions below are specified, the operations described below are performed.
G02.1/G03.1 : Circular threading B
  Circular interpolation is performed. No rotation axis can be drawn.

G02.3/G03.3 : Exponential interpolation
Linear interpolation is performed for a straight line only.
G07 : Hypothetical axis interpolation
Circular interpolation is performed.

(2) Functions that disable background drawing
When the functions below are specified, an alarm (PS1080 (DUPLICATE DEVICE SUB PROGRAM CALL)) is output, and background drawing stops.
G10 : Data setting Part of the function can be used.
M198 : External sub program call

(3) It is not possible to execute machining simulation, if the P-CODE variables #30000 and up are used in the execution macro function of macro executor function.

3.6.6.2 Functions that can be used for machining simulation (M series)

G00 : Positioning
G01 : Linear interpolation
G02/G03 : Circular interpolation
However, helical interpolation with the instruction axis other than basic three axes cannot be drawn.
G17/G18/G19 : Plane selection
G33 : Threading (Drawn as linear interpolation)
G40/G41/G42 : Cutter compensation / Cancel
G52 : Local coordinate system
G53 : Machine coordinate system select
G54 to G59 : Workpiece coordinate system select
G54.1 : Extended workpiece coordinate system select
G65 : Macro call
G68/G69 : Coordinate system rotation, 3-dimensional coordinate conversion / Cancel
G90/G91 : Absolute/incremental programming
G92 : Workpiece coordinate system change
G92.1 : Workpiece coordinate system preset
G94 : Feed per minute
G95 : Feed per revolution
G96/G97 : Constant surface speed control / Cancel
M98 : Sub program call
G07.1 : Cylindrical interpolation
G12.1 : Polar coordinate interpolation

3.6.7 Changing Workpiece Coordinate During Machining Simulation (Animation, Tool Path Drawing)

By setting parameter No.27311#1(ACD) to 1, when work coordinate system(G92(M system), G54-G59) is changed in the part program during machining simulation, animation or tool path drawing, the drawing is performed on the changed work coordinate system.
In machining simulation, work coordinate system(G92(M system), G54-G59) when G1900-G1907 (drawing definition of blank form block) or G1998(Spindle definition block for drawing) is specified is drawing coordinate system. The origin of the drawing coordinate system is the origin specified by G1900-G1907.
In the path drawing during machining, the drawing is performed without changing the coordinate system.

NOTE
1 Even if the coordinate system is changed in machining simulation, the axis drawing of coordinate system is not changed.
2 Drawing of machining simulation is performed on the premise that the top point of tool (or the center point of tool diameter) is on a commanded point at the workpiece coordinate system. So, there are some case where machining simulation is different from the movement of real machining before a position command after slanting a tool.
3 Drawing is not performed in a coordinate system where tool compensation (such as cutter compensation, tool length compensation, geometry compensation and wear compensation), tool nose control, and so forth are considered.
4 The rotary center of rotary axis(Cs axis) is on the origin of drawing coordinate specified by drawing definition of blank form block. In case of column blank, rotary axis rotates about center of the diameter. In case of rectangular solid blank, rotary axis rotates about the axis specified in parameter No.27334.
5 If 3-dimentional coordinate conversion command is performed in slanting a tool, the slanted angle of angular axis is made ineffective and a tool is drawn on the slant of the coordinate system specified by 3-dimentional coordinate conversion command.
6 If drawing of machining simulation is performed from an intermediate point of the program, it is performed as the continuous-state workpiece coordinate system is same as one of the last simulation.
7 Machining simulation is not available in case that C is not “0” on G12.1.

3.6.8 Machining Simulation Screen on 15 inch LCD (Tool Path) (for Series 30*i*/31*i*/32*i*)

In 15-inch LCD, tool path can be drawn by full screen display.
Software BY70 (Series 30/31/32i-A) after version 26 and Software BX71 (Series 30i/31i/32i-B) after version 01 are available.

3.6.8.1 Operations at Machining simulation screen

(1) Press [CHSIZE] soft key on the standard mode of the tool path screen.
(2) The full screen mode of tool path is displayed. When the machining simulation stops and [CHSIZE] soft key is pressed, the tool path is erased.
(3) When the [CHSIZE] soft key is pressed on the full screen mode of the tool path screen, the standard mode of the tool path screen is displayed.
(4) When the [CHSIZE] soft key is pressed during executing simulation, a warning message ‘EXECUTING OPERATION’ is displayed.

3.6.8.2 Operations at drawing during machining screen

(1) In case of the operation at drawing during machining screen, it is same with the case of ‘3.6.8.1 Operation at machining simulation screen ’.

(2) When the [CHSIZE] soft key is pressed during drawing tool path, the screen is displayed as follows.
  i) The size of the drawing window changes and drawn tool paths are erased.
  ii) Tool paths executed after pressing the [CHSIZE] soft key are drawn in the screen.

3.7 MACHINING SIMULATION SPEED CHANGE FUNCTION

It is possible to change execution speed of the machining simulation.

3.7.1 Operation on Machining Simulation Screen

1) In the machining simulation screen, it is possible to change the execution speed of the machining simulation by the soft keys [SPEED +] and [SPEED -].
2) Up or down cursor keys allow to change the execution speed during the machining simulation execution.
3) The yellow indicator displayed on top of the simulation window represents the actual level of simulation speed.

NOTE
Set the execution speed of machining simulation by the following parameters.

27335 **Number of speed steps for the machining simulation speed level**

[Data type] Byte
[Path attribute] Common to all paths
[Valid data range] 0-9
[Standard value] 0

The number of speed steps for the machining simulation speed level can be specified between 2 and 9.
Example) If you want to have 3 levels of simulation speed , then set parameter No.27335 to 3.

NOTE
If 0, 1, or a negative value is set in parameter No.27335, then the function is disabled. If a value bigger than nine is set in parameter No.27335, then it is equivalent to set parameter No.27335 to 9.

27336 **Waiting time between blocks at lowest speed of simulation**

[Data type] Byte
[Path attribute] Common to all paths
[Unit of data] 8msec
[Valid data range] 0-127
[Standard value] 0

Waiting time between each block at the lowest speed of simulation. A value from 1 (8msec) to 127 (1016msec) can be specified as the waiting time between blocks at the lowest speed

NOTE
When 0 or a negative value is set in parameter No.27336 then it is equivalent to set parameter No.27336 to 64(512msc).

If the machining simulation is executed at the highest speed, then the waiting time between blocks becomes “0”. It is the same execution speed as if the function is disabled.
If the machining simulation is executed at the lowest speed, then the waiting time becomes blocks specified by the parameter No.27336.
If the machining simulation is executed in an interval between the minimum and the maximum speed, then the waiting time is adjusted according to the selected speed step.

NOTE
If the setting value of the parameter No.27335 is bigger than the value of the parameter No.27336, then the number of speed steps for the machining simulation speed level is equivalent to be the same value with the setting in No.27336.

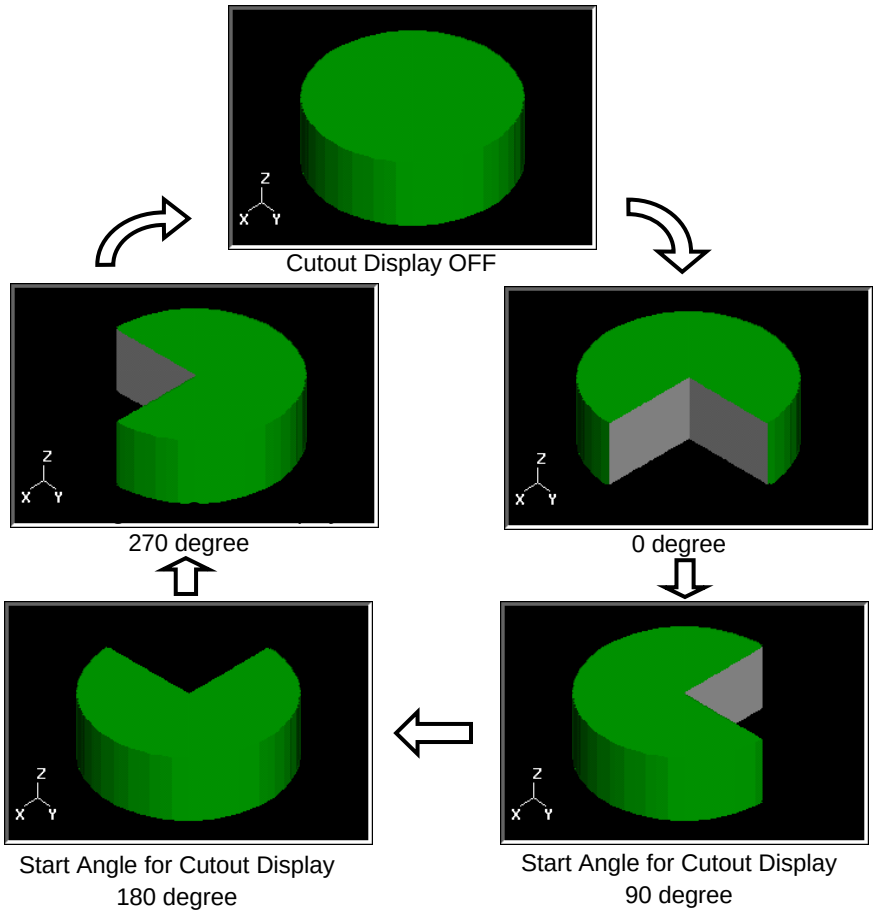
3.8 BLANK CUTOUT DISPLAY IN ANIMATED SIMULATION

The beginning angle of the blank cutout display can be changed in machining animated simulation. This function is enabled when the parameter No.27312#7 is set to 1.

3.8.1 Operation

By pressing [CUTDSP] soft key the beginning angle of the blank cutout display can be changed according to following order.

0 degree -> 90 degree -> 180 degree -> 270 degree->OFF



Beginning angle of the blank cutout and the display position is kept until CNC switching off.

3.8.2 Available Blank Figures for Blank Cutout Display

This function can be used with following blank figures.

Blank Figure	G code
CYLINDER BLANK FIGURE (around Z-axis)	G1900
CYLINDER BLANK FIGURE (around X-axis)	G1906
CYLINDER BLANK FIGURE (around Y-axis)	G1756
HOLLOW CYLINDER BLANK FIGURE (around Z-axis)	G1901
HOLLOW CYLINDER BLANK FIGURE (around X-axis)	G1907
HOLLOW CYLINDER BLANK FIGURE (around Y-axis)	G1757
FREE CONTOUR CYLINDER BLANK FIGURE (around Z-axis)	G1970
FREE CONTOUR CYLINDER BLANK FIGURE (around X-axis)	G1790
FREE CONTOUR CYLINDER BLANK FIGURE (around Y-axis)	G1780
POLYGON BLANK FIGURE (around Z-axis)	G1903
POLYGON BLANK FIGURE (around Y-axis)	G1758
HOLLOW POLYGON BLANK FIGURE (around Z-axis)	G1904
HOLLOW POLYGON BLANK FIGURE (around Y-axis)	G1759

3.9 DISPLAYING MACHINING TIME

During machining simulation, the logical machining time of each block is calculated from feedrate and distance for movement. And the result is displayed.

3.9.1 Format of Machining Time Data

Machining time data is inserted in program and it is conserved.
The place when the data are inserted is in the comment that is next to O number.
The format is "**,T_,A_**". "**,T_**" is cutting time data and "**,A_**" is non-cutting time data.

3.9.2 Operation for Inserting Machining Time

The following soft keys are displayed on MEM mode after soft key [SIMLAT] is pushed.

REWIND	START	PAUSE	SINGLE	STOP	INIT	CUTDSP	INTERF	TLPATH	GRPOFF
LARGE	SMALL	AUTO	REVERS	ROTATE	←MOVE	MOVE→	↑ MOVE	↓ MOVE	CENTER
				WK SET	T-OFS				
REWIND	O LIST			O SRCH	ACTPOS			INSERT	

After soft key [START] is pushed and machining simulation is started, the machining time is calculated and it is kept in the memory temporarily. After that, the color of soft key [INSERT] is changed. If soft key [INSERT] is pushed in this state, the message "MACHINE TIME DATA IS INSERTED. OK?" is displayed. Then, if soft key [YES] is selected, the machining time data are inserted to the end of comment.

NOTE
1 Don’t operate the machine during inserting machining time data.
2 When the machining time isn’t kept in the memory, the machining time can’t be inserted.
3 When the program has been protected, the machining time can’t be inserted.
4 The machining time can’t be inserted during operation.
5 All paths must be on MEM mode to insert machining time data. When at least one path isn’t on MEM mode, the machining time can’t be inserted.
6 Even if all path is on MEM mode, if at least one path is on background editing, machining time data can’t be inserted.
7 When the machining time data are inserted on one path, the machining time data are inserted on other paths too. But when machining simulation for each path is executed, the machining time data are inserted on only the path in which the machining simulation is executed.

3.9.3 Display Machining Time

The following soft keys are displayed after soft key [O LIST] is pushed.

NEW	COPY	DELETE	EDTCOM	SEARCH	M CARD	ALLDEL	SRTORD	OPEN	CLOSE
	TIME								

If soft key [TIME] is pushed, the indication of modified date and program size are changed to machining time as follows.
If soft key [TIME] is pushed again, modified date is displayed again.

NOTE
1 The execution time of M98, M99, and custom macro call M codes is not included in the machining time.
2 The execution time of T and S codes used as arguments in custom macro calls is not included in the machining time either.
3 The execution time of M00, M01, and M30 is added to the machining time.
4 The parameters about displaying machining time are from 27390 to 27392.
5 The execution time of commands only used for drawing is added to the machining time.

3.10 LARGE WINDOW DISPLAY OF MACHINING SIMULATION (ANIMATED DRAWING) (For Series 30*i*/31*i*/32*i*–B only)

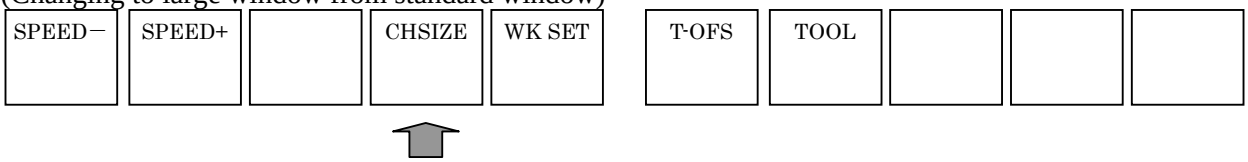
Large window display of machining simulation (animated drawing) is available.

NOTE
1 One of the following hardware is necessary to use this function.
  (1) CNC with 15”color LCD or 10”color LCD B
  (2) CNC with Personal computer function with Windows XP
  (3) CNC is connected to Personal computer with CNC display screen function.
2 Setting parameter No.27314#7=1 is necessary to use this function, in addition to satisfying 1.

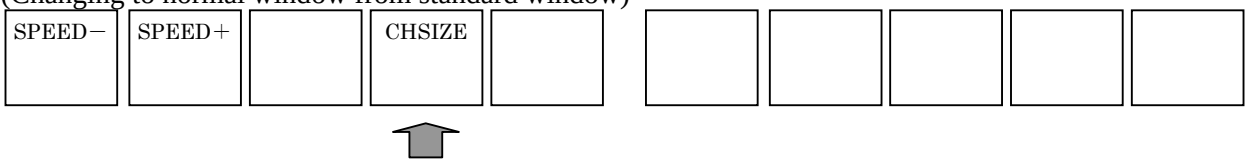
3.10.1 Changing Window Size of Machining Simulation (Animated Drawing)

In order to change window size of machining simulation (animated drawing), press soft key [CHSIZE]. At this time, the scale magnification for drawing is not changed.

(Changing to large window from standard window)



(Changing to normal window from standard window)



When the machining simulation stops and soft key [CHSIZE] is pressed, the drawing result is erased.
When the soft key [CHSIZE] is pressed during executing simulation, a warning message ‘EXECUTING OPERATION’ is displayed.

3.10.2 Restrictions on Large Window Display

(1) Soft key [O LIST]、[INSERT]、[WK SET]、[T-OFS]、[TOOL] are not available.
(2) The following information is not displayed.
  - Distance to go
  - Next distance
  - Spindle information
  - Feed information
  - Program

3.11 2 WINDOWS MACHINING SIMULATION (For only Series 30*i*/31*i*/32*i*–B)

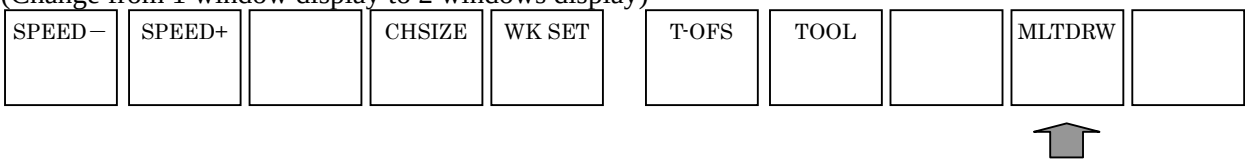
2 windows display of machining simulation is available.

NOTE
1 One of the following hardware is necessary to use this function.
(1) CNC with 15”color LCD
(2) CNC with Personal computer function with Windows XP
(3) CNC is connected to Personal computer with CNC display screen function.
2 Setting parameter No.27314#3=1 is necessary to use this function, in addition to satisfying 1.

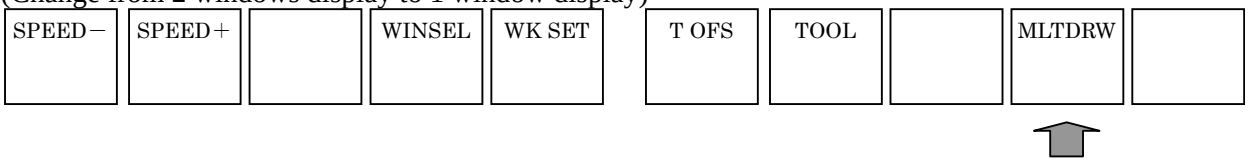
3.11.1 Changing Window of Machining Simulation

In order to change the number of windows of machining simulation, press soft key [MLTDRW].

(Change from 1 window display to 2 windows display)



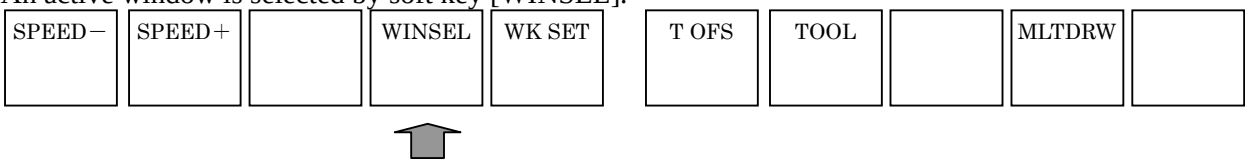
(Change from 2 windows display to 1 window display)



When the machining simulation stops and soft key [MLTDRW] is pressed, the drawing result is erased.
When the soft key [MLTDRW] is pressed during executing machining simulation, a warning message ‘EXECUTING OPERATION’ is displayed.

3.11.2 Selecting Active Window

An active window is selected by soft key [WINSEL].



3.11.3 Restrictions on 2 Windows Display

(1) Soft key [INSERT], [CHSIZE] are not available.
(2) The program is not displayed.

3.12 NOTES

3.12.1 Notes on Machining Simulation (for Series 0*i*-C/16*i*/18*i*/21*i*)

During the execution of machining simulation, the system is automatically placed in the machine locked state. During the execution of drawing, no machine control signals are output by auxiliary and other functions, but control signals such as "OP," "STL," "SPL," "RST," and "AL" may be output. During the execution of drawing, the drawing-in-progress signal "CKGRP," shown below, is output. If the control signals that may be output during the execution of drawing affect machine control, the PMC Ladder program must be modified so that these signals are ignored, by using the drawing-in-progress signal.

	#7	#6	#5	#4	#3	#2	#1	#0
F62			CKGRP					

CKGRP Indicates that a dynamic graphics display is being drawn for check purposes.

In addition, if you want to prevent the macro executor program from running during machining simulation, change its processing according to the state of system variable #3010.
#3010 can take the values listed below.

System variable	Value	Executing State
#3010	0	Normal condition (other than the following status)
	1	Executing automatic operation (including drawing during machining)
	4	Executing machining simulation (animated or tool path drawing)

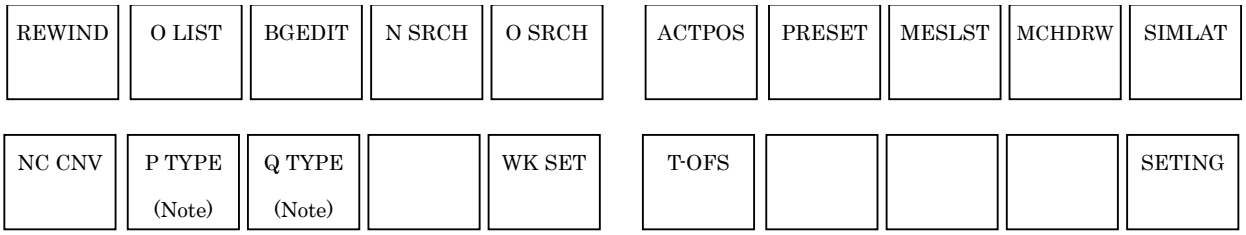
When machining simulation for a program containing a 3- dimensional coordinate conversion command is being performed, increasing the dry run speed (for example, selecting a rapid traverse rate) can speed the execution.

3.12.2 Tool size in Machining Simulation

Tool in animated drawing is drawn within about quarter of the graphic window. Therefore, if the scale magnification for machining simulation is large, a part of the tool might not be displayed.

4 PROGRAM OPERATION

When the MEM mode is selected with the machine operator's panel, the soft keys shown below appear on the MANUAL GUIDE *i* screen. Pressing the leftmost soft key [<] or the rightmost soft key [>] changes the page of the soft key display to the second page.



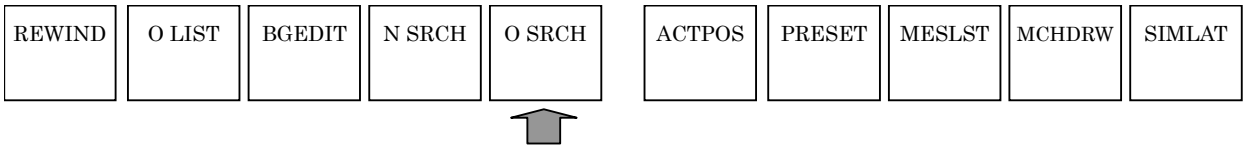
For soft key [SETING], see Section 2.15, "REGISTERING FIXED FORM SENTENCES".

For soft key [WK SET], see Chapter 7, "SETTING THE WORKPIECE COORDINATE DATA".

For soft key [T-OFS], see Chapter 8, "SETTINGS RELATED TO TOOLS".

NOTE
When the optional function of PROGRAM RE-START, the soft keys [P TYPE] and [Q TYPE] are displayed on the screen.

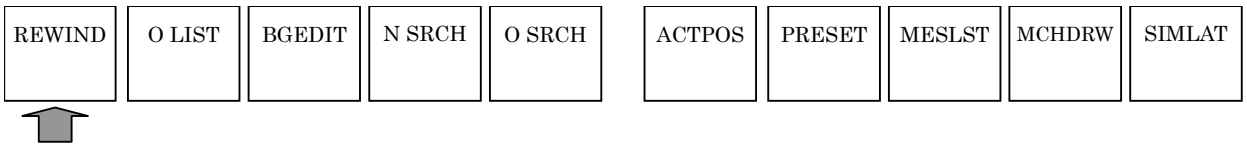
4.1 SELECTING A PROGRAM FOR OPERATION



When you enter the program number you want to select by using numeric keys then press soft key [O SRCH], the program with the program number is selected and displayed in the program window.
By pressing soft key [O SRCH] without entering any key, the next program is selected.

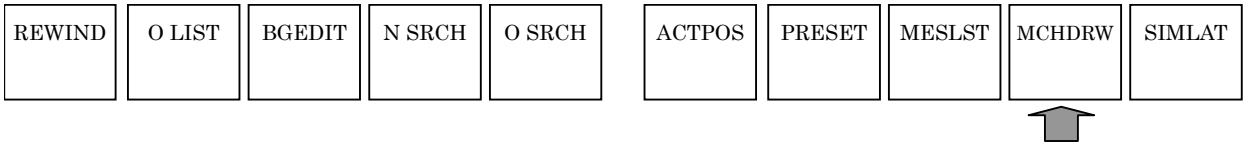
NOTE
If the bit 3 (OSR) of parameter No.3202 is set to "1", pressing soft key [O SRCH] without entering any data, the next program is not selected.

4.2 REWINDING A PROGRAM



By pressing soft key [REWIND], you can return to the beginning of a selected program.

4.3 DISPLAYING THE DRAWING-DURING-MACHINING



By pressing soft key [MCHDRW], the drawing-during-machining window can be displayed. Press soft key [GRPOFF] to close the drawing-during-machining window.

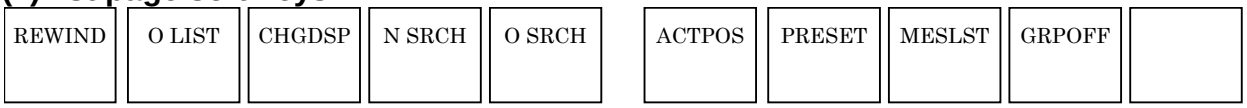
While a machining operation is being performed on the machine, the tool path can be drawn. This function is available also during machine lock and dry run operation.

NOTE

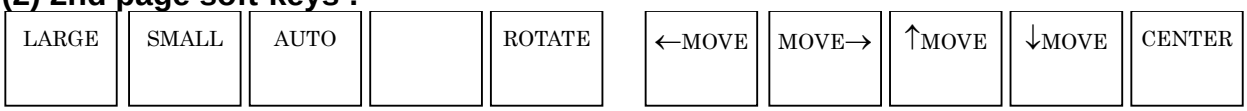
1 To perform drawing during machining, open the window for drawing during machining by pressing soft key [GRP ON] before starting machining operation.
2 If a coordinate system change is specified in the program during drawing, drawing is performed without changing the coordinate system.
3 In the drawing during machining, when the workpiece specifying block is executed, form of a workpiece is drawn in form of wire frame. Drawing color of the workpiece can be specified by the parameter No.14773.
4 When a simulation window is opened, a blank figure drawing at the last drawing is displayed. However, if bit 4 (WOK) of parameter No. 27310 is set to 1, no blank figure is displayed when a simulation window is opened.
5 Tool path drawing with a multi-path system is performed only at the selected path for displaying. And a simulation window is initialized if the selected path for displaying is changed. So if the selected path for displaying is changed in displaying a free blank figure, the drawing is initialized for a column figure which encloses it.
6 Path drawing is performed using values in the workpiece coordinate system. Coordinates that allow for tool compensation (cutter compensation, tool length compensation, geometry compensation, and wear compensation), tool tip control, or the like are not used for drawing.

When drawing during machining (tool path) is selected, the soft keys shown below appear. Pressing the leftmost soft key [<] or the rightmost soft key [>] changes the page of the soft key display to the second or third page.

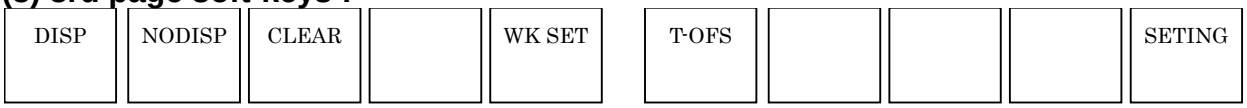
(1) 1st page soft-keys :



(2) 2nd page soft-keys :



(3) 3rd page soft-keys :



4.3.1 Program Selection Operation and Other Operations in Drawing during Machining

REWIND	O LIST	CHGDSP	N SRCH	O SRCH	ACTPOS	PRESET	MESLST	GRPOFF	

1st page Soft keys are used for operations such as selection of a program. These key operations are the same as the operations in the MEM mode, so see the sections listed below.
To close the window for drawing during machining and stop the mode for drawing during machining, press soft key [GRPOFF].

Section 4.1, “SELECTING A PROGRAM FOR OPERATION”
Section 4.2, “REWINDING A PROGRAM”
Section 4.4, “CHANGING THE CURRENT POSITION DISPLAY”
Section 4.5, “PRESETTING RELATIVE COORDINATES”
Section 4.6, “NEXT-BLOCK DISPLAY FUNCTION”

4.3.2 Selecting Whether to Display the Tool Path or Not in Drawing during Machining

DISP	NODISP	CLEAR		WK SET	T-OFS				SETING

2nd page soft keys mainly allow you to select whether to display the tool path or not.
For soft key [SETING], see Section 2.15, "REGISTERING FIXED FORM SENTENCES".

[NODISP] : As soon as this soft key is pressed, drawing of the tool path is stopped. The tool path is not drawn until soft key [DISP] is pressed next.
[DISP] : As soon as this soft key is pressed, drawing of the tool path is started.
Remark) Only necessary tool path portions can be drawn by using soft keys [DISP] and [NODISP].
[CLEAR] : Erase the tool path drawn so far. Immediately after this soft key is pressed, tool path drawing is displayed.

4.3.3 Scaling, Movement, and Other Operations in Drawing during Machining

LARGE	SMALL	AUTO		ROTATE	←MOVE	MOVE→	↑MOVE	↓MOVE	CENTER

3rd soft keys allow you to perform scaling and movement of a tool path drawing as well as selection of a drawing coordinate system.

NOTE
The operations explained below must be performed before tool path drawing is started.

(1) Scaling and movement

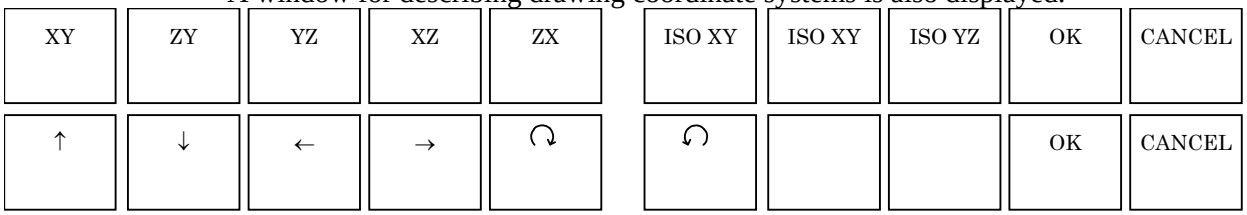
[LARGE] : Increase the drawing magnification.
[SMALL] : Decrease the drawing magnification
[AUTO] : When the blank form block for animation has been input to the machining program, perform automatic scaling so that the blank figure is fit in the window.

NOTE
If a blank form for animation has been entered, automatic scaling is performed when the corresponding block is executed for the first time.

[←MOVE] : Move the viewpoint leftward. As a result, the tool path drawn moves rightward.
[MOVE→] : Move the viewpoint leftward. As a result, the tool path drawn moves leftward.
[↑MOVE] : Move the viewpoint upward. As a result, the tool path drawn moves downward.
[↓MOVE] : Move the viewpoint downward. As a result, the tool path drawn moves upward.
[CENTER] : Set the center of the tool path to the center of the window.

(2) Selecting a drawing coordinate system

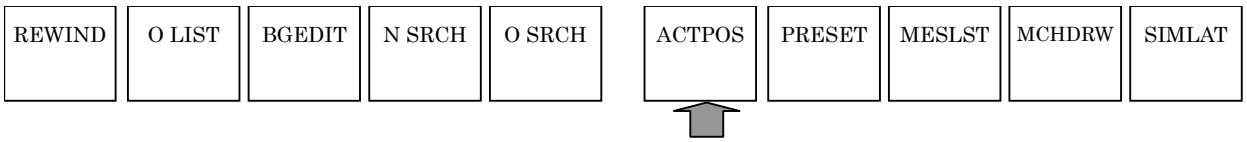
[ROTATE] : Display the soft keys shown below for selecting the drawing coordinate system. A window for describing drawing coordinate systems is also displayed.



[XY] : Select the XY plane.
[ZY] : Select the ZY plane.
[YZ] : Select the YZ plane.
[XZ] : Select the XZ plane.
[ZX] : Select the ZX plane.
[ISO XY] : Select an equiangular coordinate system with the positive direction of the Z-axis facing up.
[ISO XY] : Select an equiangular coordinate system with the positive direction of the Z-axis facing up. But, the view point is opposed to the above one.
[ISO YZ] : Select an equiangular coordinate system with the positive direction of the X-axis facing up.
[OK] : Perform drawing in the coordinate system selected by one of the above soft keys.
[CANCEL] : Cancel the coordinate system selected by one of the above soft keys and use the original coordinate system for drawing.
[↑] : Make a counterclockwise rotation by selecting the right direction viewed when you face the screen as the center axis.
[↓] : Make a clockwise rotation by selecting the right direction viewed when you face the screen as the center axis.
[←] : Make a counterclockwise rotation by selecting the upward direction viewed when you face the screen as the center axis.
[→] : Make a clockwise rotation by selecting the upward direction viewed when you face the screen as the center axis.
[↷] : Make a clockwise rotation by selecting the direction toward you viewed when you face the screen as the center axis.
[↶] : Make a counterclockwise rotation by selecting the direction toward you viewed when you face the screen as the center axis.

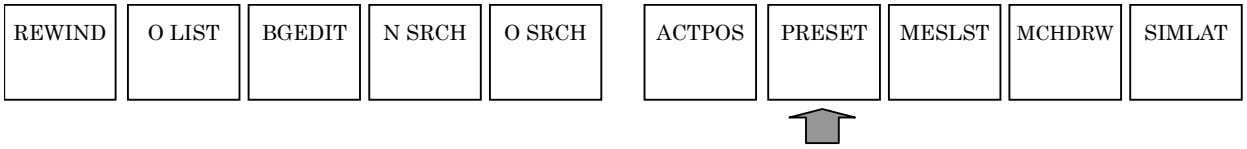
NOTE
A rotation is made according to the unit specified in parameter No. 14716.

4.4 CHANGING THE CURRENT POSITION DISPLAY



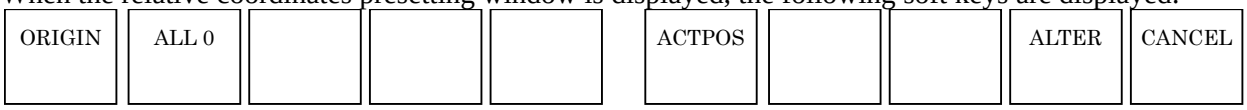
By pressing soft key [ACTPOS], the current position display in the status display window in the upper part of the screen changes in the sequence "ACTUAL POS. (ABS)" → "ACTUAL POS. (RELATIVE)" → "ACTUAL POS. (MACHINE)"
If the manual handle interrupt option function is attached, the position display changes in the sequence "ACTUAL POS. (ABS)" → "ACTUAL POS. (RELATIVE)" → "ACTUAL POS. (MACHINE)" → “HANDLE INTER.(INPUT)” → “HANDLE INTER.(OUTPUT)”

4.5 PRESETTING RELATIVE COORDINATES



By pressing soft key [PRESET], a relative coordinates presetting window appears, allowing you to preset relative coordinates.

When the relative coordinates presetting window is displayed, the following soft keys are displayed:



[ORIGIN] : Set the axis selected with the cursor keys to "0." At this time, no relative coordinates are preset.
[ALL 0] : Set all axes to "0." At this time, no relative coordinates are preset.
Remark) You can set a desired coordinate value for an axis selected by the cursor by entering the value you want to preset by using numeric keys then pressing the <INPUT> key. At this time, as described above, the relative coordinates are not yet preset.
[ACTPOS] : Change the current position display in the status display window in the upper part of the screen in the sequence "ACTUAL POS. (ABS)" → "ACTUAL POS. (RELATIVE)" → "ACTUAL POS. (MACHINE)."If the manual handle interrupt option function is attached, the position display changes in the sequence "ACTUAL POS. (ABS)" → "ACTUAL POS. (RELATIVE)" → "ACTUAL POS. (MACHINE)" → “HANDLE INTER.(INPUT)” → “HANDLE INTER.(OUTPUT)”
[ALTER] : Preset the relative coordinates to coordinate values set by the above operation. This soft key also closes the relative coordinates presetting window.
[CANCEL] : Cancel presetting of coordinates and just close the window.

4.6 NEXT-BLOCK DISPLAY FUNCTION

In the MEM mode or MDI mode, the travel distance data of the block to be executed next is displayed in status indicator window.

In DRAWING DURING MACHINING screen in MEM mode, the display can be chose between spindle/feedrate information and distance data of the block to be executed next by a soft key [CHGDSP]

NOTE
1 n Series 0*i*-C/16*i*/18*i*/21*i*, set a parameter No.3106#2 to 1 in order to display distance data of the block to be executed next for SINGLE BLOCK OPERATION.
2 In Series 0*i*-D/30*i*/31*i*/32*i*, set a parameter No.3106#2 to 0. The distance data of the block to be executed next for SINGLE BLOCK OPERATION is displayed as follows.
- While the block to be executed next is buffered, the distance data of the block is displayed.
- While the block to be executed next is not buffered, the distance data of the current block is displayed.

(1) In case of machining simulation nor path drawing during actual machining is not executed
The travel distance data of the next block is automatically displayed.
(2) In case of machining simulation or path drawing during actual machining is executed
Usually, the travel distance of the next block is not displayed.

Pressing soft key [CHGDSP] erases the display of spindle and actual feedrate information and displays the travel distance data of the next block. Pressing soft key [CHGDSP] again returns the screen to the display of spindle and actual feedrate information.

4.7 FEEDRATE DISPLAY

The actual feedrate is displayed during machining operation.

NOTE
In MANUAL GUIDE *i* screen, the digits in the decimal part of actual feedrate are not displayed in feed per revolution mode, even if the following parameter is set. Therefore, please set the following parameter to be 0.

No.3135 Number of decimal places in actual feedrate display
(Series 0*i*-D/30*i*/31*i*/32*i* only)

4.8 DISPLAYING 3-DIMENSIONAL INTERFERENCE CHECK STATUS (FOR Series 30*i*/31*i*/32*i* ONLY)

The 3-dimensional interference check status is displayed at the bottom left corner of the MANUAL GUIDE *i* screen.

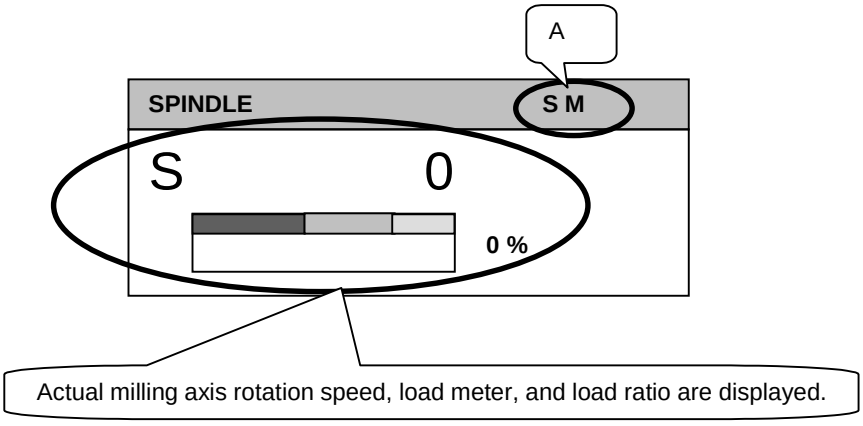
4.9 DISPLAYING THE ROTATION SPEED OF A SERVO MOTOR-DRIVEN MILLING AXIS

If an arbitrary servo motor-driven axis is used as a milling axis, its actual rotation speed can be displayed. This function is unavailable on the Series 0*i*-C.

(1) The “SPINDLE” is displayed as explained below according to the setting of the parameter MSD (No.14836#1), which is intended to specify whether to enable this function, and the setting of the milling axis rotation speed display switching signal SDPC<Gn038#5>.
<1> If MSD (No.14836#1) = 1 and SDPC<Gn038#5> = ON:
- Actual rotation sped, load meter, and load ratio (%) of the milling axis

<2> If MSD (No.14836#1) = 0 or SDPC<Gn038#5> = OFF:
- Actual rotation speed, load meter, and load ratio (%) of the spindle motor axis (hereafter called the spindle)

(2) The display of the portion A in the following figure varies as described below according to which is displayed, the actual rotation speed of the spindle or that of the milling axis.
<1> If the actual rotation speed of the spindle is displayed, the spindle number is also displayed.
<2> If the actual rotation speed of the milling axis is displayed, a character specified in the milling axis suffix parameter (No.14868) is also displayed.



Milling axis actual rotation speed display box

NOTE
This function is unavailable on the Series 0*i*-C.

4.9.1 Conditions for Using This Function

Using this function requires the following settings.

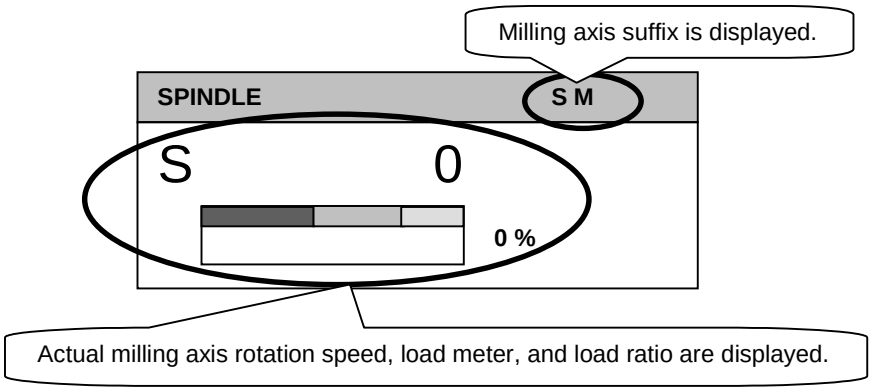
(1) Parameter
MSD (No.14836#1) = 1

(2) Signal
Actual milling axis rotation speed display switching signal SDPC<Gn038#5> = ON
This signal is provided for each path separately. For multipath applications, turn on the signal for a path for which the milling axis is specified.
- Path 1: G0038#5
- Path 2: G1038#5
- Path 3: G2038#5

4.9.2 About the “SPINDLE” Display Block

4.9.2.1 Path 1 display window (10.4-inch screen)

(1) If this function is enabled and the milling axis rotation speed display switching signal is ON (parameter MSD (No.14836#1) = 1 and signal SDPC<Gn038#5> = 1):
In the path 1 display window (10.4-inch screen), the “SPINDLE” display block shows the actual rotation speed, load meter, load ratio (%), and suffix of the milling axis as shown below.
The following information is hidden.
- Spindle number
- Constant surface speed control icon
- Load meter
- Load ratio



Milling axis actual rotation speed display box

(a) Displaying the actual rotation speed of the milling axis when more than one spindle is available
Turning on the milling axis rotation speed display switching signal SDPC <Gn038#5> to request to display the actual rotation speed of the milling axis when the actual rotation speed of spindle No. n is currently being displayed causes the actual rotation speed of the milling axis to appear and overwrite the actual rotation speed of the spindle as shown above.

(2) If this function is enabled and the milling axis rotation speed display switching signal is OFF (parameter MSD (No.14836#1) = 1 and signal SDPC <Gn038#5> = 0):
(a) If a spindle number is selected (signals PC3SLC <Gn026#0> and PC2SLC <Gn028#7>):
In the path 1 display window (10.4-inch screen):
The “SPINDLE” display block shows the following information about the selected spindle.
- Actual rotation speed
- Load meter
- Load ratio (%)
- Spindle number

(b) If no spindle number is selected (signals PC3SLC <Gn026#0> = 0 and PC2SLC <Gn028#7> = 0):
In the path 1 display window (10.4-inch screen), the “SPINDLE” display block shows the following information about spindle 1.
- Actual rotation speed
- Load meter
- Load ratio (%)
- Spindle number (that is, 1)

(3) If this function is disabled and the milling axis rotation speed display switching signal is ON (parameter MSD (No.14836#1) = 0 and signal SDPC <Gn038#5> = 1):
The same information as stated in item (2) is displayed.

(4) If this function is disabled and the milling axis rotation speed display switching signal is OFF (parameter MSD (No.14836#1) = 0 and signal SDPC <Gn038#5> = 0):
The same information as stated in item (2) is displayed.

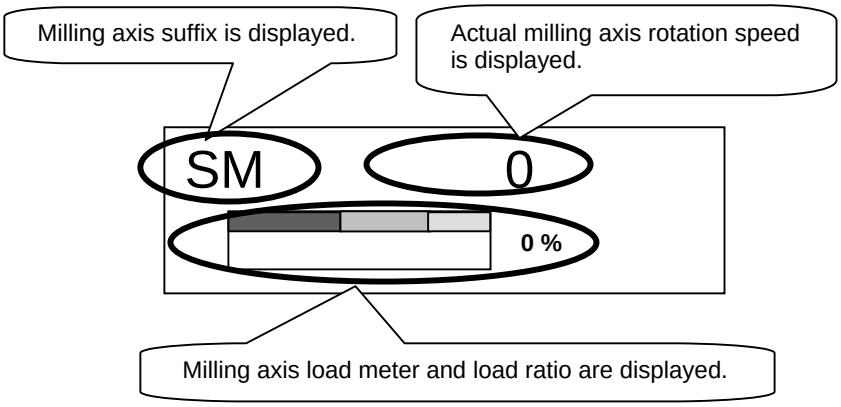
4.9.2.2 Path 1 display window (15-inch screen)

(1) If this function is enabled (parameter MSD (No.14836#1) = 1):
In the path 1 display window (15-inch screen), the following information is displayed at the same time.
- Actual spindle and milling axis rotation speed
- Load meter

- Load ratio (%)
- Expanded spindle name (or spindle suffix)
- Milling axis suffix

The following information is hidden.

- Constant surface speed control icon
- Load meter
- Load ratio



Milling axis information in the "SPINDLE" display (enlarged view)

4.9.2.3 Simultaneous multipath display window (10.4-inch screen)

(1) If this function is enabled and the milling axis rotation speed display switching signal is ON (parameter MSD (No.14836#1) = 1 and signal SDPC <Gn038#5> = 1):
In the simultaneous multipath display window (10.4-inch screen), the "SPINDLE" display block of each path shows the actual rotation speed of the milling axis.
The spindle number is hidden, and the milling axis suffix is shown.

(2) If this function is enabled and the milling axis rotation speed display switching signal is OFF (parameter MSD (No.14836#1) = 1 and signal SDPC <Gn038#5> = 0):
(a) If a spindle number is selected (signals PC3SLC <Gn026#0> and PC2SLC <Gn028#7>):
In the simultaneous multipath display window (10.4-inch screen), the "SPINDLE" display block shows the following information about the selected spindle.
- Actual rotation speed
- Load meter
- Load ratio (%)
- Spindle number

(b) If no spindle number is selected (signal PC3SLC <Gn026#0> = 0 and signal PC2SLC <Gn028#7> = 0):
In the simultaneous multipath display window (10.4-inch screen), the "SPINDLE" display block shows the following information about spindle 1.
- Actual rotation speed
- Load meter
- Load ratio (%)
- Spindle number (that is, 1)

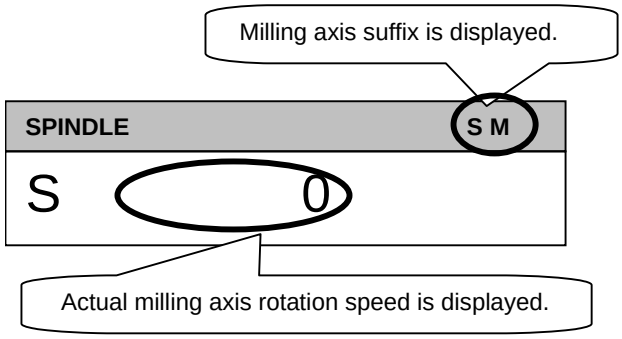
(3) If this function is disabled and the milling axis rotation speed display switching signal is ON (parameter MSD (No.14836#1) = 0 and signal SDPC <Gn038#5> = 1):
The same information as stated in item (2) is displayed.

(4) If this function is disabled and the milling axis rotation speed display switching signal is OFF (parameter MSD (No.14836#1) = 0 and signal SDPC <Gn038#5> = 0):
The same information as stated in item (2) is displayed.

4.9.2.4 Simultaneous multipath display window (15-inch screen)

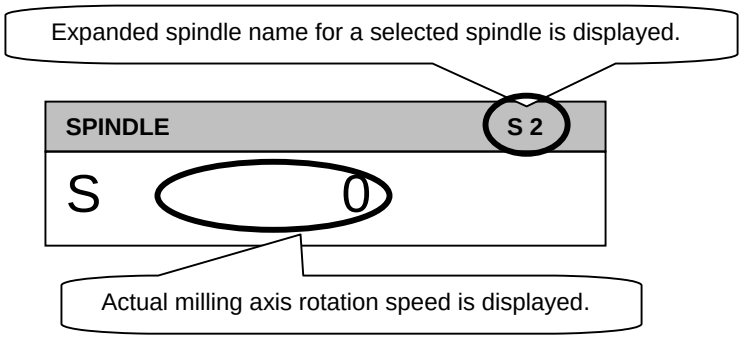
(1) If this function is enabled and the milling axis rotation speed display switching signal is ON (parameter MSD (No.14836#1) = 1 and signal SDPC <Gn038#5> = 1):
In the simultaneous multipath display window (15-inch screen), the "SPINDLE" display block for each path shows the actual rotation speed and suffix of the milling axis in the place shown below.

(a) Two-path environment



"SPINDLE" display block (enlarged) in the simultaneous multipath display window

(2) If this function is enabled and the milling axis rotation speed display switching signal is OFF (parameter MSD (No.14836#1) = 1 and signal SDPC <Gn038#5>= 0):
(a) If a spindle number is selected (signals PC3SLC <Gn026#0> and PC2SLC <Gn028#7>):
In the simultaneous multipath display window (15-inch screen), the "SPINDLE" display block shows the following information about the spindle selected for each path.
- Actual rotation speed
- Expanded spindle name (or spindle suffix)



"SPINDLE" display block in the simultaneous multipath display window

(b) If no spindle number is selected (signals PC3SLC <Gn026#0> = 0 and PC2SLC <Gn028#7> = 0):
In the simultaneous multipath display window (15-inch screen), the "SPINDLE" display block shows the following information about spindle 1 for each path.
- Actual rotation speed
- Expanded spindle name (or spindle suffix)

(3) If this function is disabled and the milling axis rotation speed display switching signal is ON (parameter MSD (No.14836#1) = 0 and signal SDPC <Gn038#5> = 1):
The same information as stated in item (2) is displayed.

(4) If this function is disabled and the milling axis rotation speed display switching signal is OFF (parameter MSD (No.14836#1) = 0 and signal SDPC <Gn038#5> = 0):
The same information as stated in item (2) is displayed.

4.9.2.5 Machining simulation and NC statement conversion windows (10.4-inch screen)

The "SPINDLE" block shows the following information no matter whether the milling axis display function is enabled.

- The actual rotation speed, load meter, and load ratio are displayed always as "0".
- The spindle number and milling axis suffixes are hidden.

4.9.2.6 Machining simulation and NC statement conversion windows (15-inch screen)

(1) If this function is enabled (parameter MSD (No.14836#1) = 1):
The "SPINDLE" display block shows the following information always as "0".
- Actual spindle and milling axis rotation speed
- Load meter
- Load ratio

(2) If this function is disabled (parameter MSD (No.14836#1) = 0):
The "SPINDLE" display block shows the following information always as "0".
- Actual spindle rotation speed
- Load meter
- Load ratio

4.9.3 "CURRENT POSITION" Display Block

Whether to show the current position of the milling axis in the following target windows depends on the parameter NDPx (No.3115#0) and the signal NPOSn <Gn198>.

(To show)
For the target axis, set up parameter NDPx (No.3115#0) = 0 and turn off the signal NPOSn <Gn198>.

(To hide)
For the target axis, set up parameter NDPx (No.3115#0) = 1 or turn on the signal NPOSn <Gn198>.

(1) Target windows (common to both 10.4- and 15-inch screens)
(a) One-path display window
(b) Simultaneous multipath display window (non-EDIT modes)
(c) Drawing-during-machining window
(d) Machining simulation window
(e) NC statement conversion window
(f) Re-threading window

4.9.4 "FEED" Display Block

Whether to show the load meter and load ratio (%) of the milling axis in the following target window depends on the parameter NDPx (No.3115#0) and the signal NPOSn <Gn198>.

(To show) For the target axis, set up parameter NDPx (No.3115#0) = 0 and turn off the signal NPOSn <Gn198>.

(To hide) For the target axis, set up parameter NDPx (No.3115#0) = 1 or turn on the signal NPOSn <Gn198>.

(1) Target windows (15-inch screen only)
  (a) One-path display window
  (b) Drawing-during-machining window
  (c) Machining simulation window
  (d) NC statement conversion window

4.9.5 Details of Actual Rotation Speed Shown in the SPINDLE Display Block

The SPINDLE block shows the actual rotation speed of the milling axis as explained below.

<1> The actual rotation speed to be displayed is obtained by multiplying the servo motor rotation speed by the following gear ratio.

$$\text{Gear ratio} = \frac{\text{Number of gear teeth on servo motor axis (parameter (No.1896))}}{\text{Number of gear teeth on milling axis (parameter (No.1897))}}$$

(The above expression applies to the Series 16*i*/18*i*/21*i*. For the Series 30*i*/31*i*/32*i*, the parameter must be changed as stated in NOTE below.)

<2> The displayed actual rotation speed is up to 8 digits.

<3> The displayed actual rotation speed varies depending on what path, spindle, or servo axis is selected.

<4> If a linear motor-based control axis is specified as a milling axis by mistake, the displayed actual rotation speed is "0".

NOTE

The parameter used in calculating a gear ratio differs between the Series 0*i*-C/16*i*/18*i*/21*i* and Series 0*i*-D/30*i*/31*i*/32*i*.

The parameter to be referenced for each machine model is as follows:

- Number of gear teeth on the servo motor axis

  Series 0*i*-C/16*i*/18*i*/21*i* : Parameter No.1896

  Series 0*i*-D/30*i*/31*i*/32*i* : Parameter No.1898

- Number of gear teeth on the milling axis

  Series 0*i*-C/16*i*/18*i*/21*i* : Parameter No.1897

  Series 0*i*-D/30*i*/31*i*/32*i* : Parameter No.1899

4.9.6 Details of Load Ratio (%) Shown in the SPINDLE Display Block

The SPINDLE block shows the load ratio of the milling axis as explained below.

(1) The load ratio to be displayed is obtained using the following expression.

$$\text{Load ratio} = \frac{\text{<1> - <3>}}{\text{<2> - <3>}} \times 100$$

<1> : Digital form of servo motor load current

<2> : Digital form of rated current (parameter (No.2086))

<3> : Digital form of steady-state load current (parameter (No.14815))

NOTE
The parameter (No.14815) varies according to the servo axis number used for the milling axis.
The parameter to be referenced for each axis is as follows:
(Common to both the Series 0*i*/16*i*/18*i*/21*i* and Series 30*i*/31*i*/32*i*)
Axis 1: Parameter No.14815
Axis 2: Parameter No.14816
Axis 3: Parameter No.14817
Axis 4: Parameter No.14818
Axis 5: Parameter No.14819
Axis 6: Parameter No.14820
Axis 7: Parameter No.14821
Axis 8: Parameter No.14822

(2) The displayed load ratio is up to 3 digits.
(3) The displayed load ratio varies depending on what path or servo motor is selected.

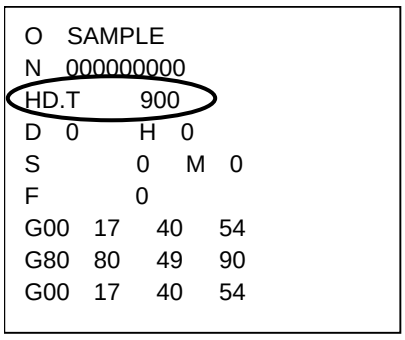
4.10 DISPLAYING THE SPINDLE-POSITION AND STANDBY-POSITION TOOL NUMBERS (FOR Series 30*i*/31*i*/32*i* ONLY)

On the NC screen, setting up parameter PCT (No.3108#2) = 1 or DHN (No.11320#0) = 1 enables HD.T (spindle tool number) and NX.T (standby-position tool number) to be displayed.
HD.T and NX.T can be displayed also on the MANUAL GUIDE *i* screen.

NOTE
For details of this function, refer to the applicable “Connection Manual: Function” (B-63943EN-1) and “PMC Programming Manual” (B-63983EN).

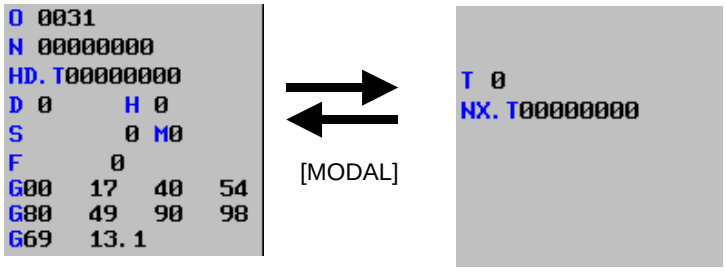
4.10.1 Displaying the Spindle-Position Tool Number

On the MANUAL GUIDE *i* screen, the modal display block shows HD.T.



4.10.2 Displaying the Standby-Position Tool Number

Pressing the soft key [<] or [>] several times causes the soft key [MODAL] to appear.
Pressing the soft key [MODAL] changes the modal information display block as shown below, thus displaying the standby-position tool number.
Pressing the soft key [MODAL] once again resumes the previous display.



Page 1 **Page 2**

4.11 G54.4 MODAL DISPLAY

G54.4 is displayed in the modal information when the workpiece setting error compensation is effective.

4.11.1 Detail

(1) Where screen effective invalidity of workpiece setting error compensation mode can be confirmed.
a) Modal information of base screen
b) Modal information of DRAWING-DURING-MACHINING window (Usual display)

(2) When the workpiece setting error compensation mode is effective.
The workpiece setting error compensation mode becomes effective if commanded in G54.4 P1-7. G54.4 is displayed in the modal information.

(3) When the workpiece setting error compensation mode is ineffective.
The workpiece setting error compensation mode becomes ineffective if commanded in G54.4 P0. G54.4 displayed in the modal information becomes non-display.

4.11.2 Note

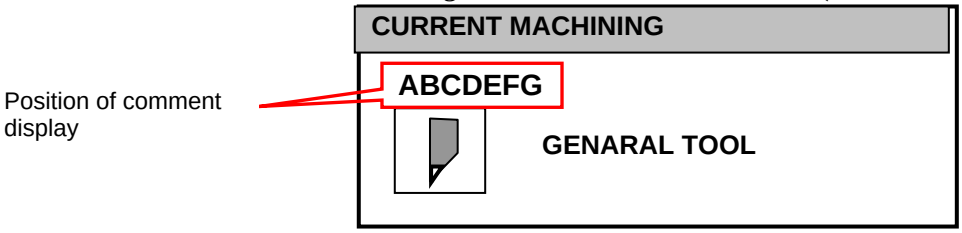
(1) G54.4 is not displayed in the modal information while machining simulation.

(2) At the following screens are not display G54.4 on the modal information part.
a) All path screen display.
b) DRAWING-DURING-MACHINING window. (Usual display)
c) Machining simulation screen.(Animation)
d) Machining simulation screen. (Tool path)

4.12 COMMENT DISPLAY DURING MACHINING

By using G code (G2900) for comment display, it is possible to display comments on the screen during automatic operations. When G2900 is executed during automatic operation, then the character string of specified comment is displayed at the current machining portion on the base screen.

(Example)
Displayed comment on the base screen during execution G code "**G2900 P1 (ABCDEFG)**"



4.12.1 Format of Comment Display G code G2900

The format of comment display G code G2900 is as follows.

G2900 Pp(xxxxx) ;

Address	Meaning
P	Display / Deletion setting P0 : The comment is deleted from the screen P1 : The comment is displayed on the screen
(xxxxx)	Comment The character string in round parentheses is displayed on the screen. The string can be formed by the following set of characters: • Only a special alphabet/figure/sign • A special sign; The following signs except % () ?,@#=*+[]& /. SP (Note) Note) SP means space. • The number of characters which can be displayed is 37 characters or less. (Do not depend on the size of the indicator)

(Examples)

(1) G2900 P1 (ABC 12345)
"ABC 12345" is displayed.

(2) G2900 P0
The comment is deleted.

4.12.2 Notes

The notes of G2900 are as follows.

(1) In the case of specifying number for address P excluding 0 or 1, and specifying no comment, nether comment display nor deletion are done.
(2) When there is no character string in the specified comment, the comment is erased from the screen.
(3) The name of machining cycle is not displayed during the comment display.
(4) If the G code G2900 is executed in machining simulation or machining drawing, then the character string is not displayed.
(5) In the NC program conversion, comment display G code G2900 is output as it is into the converted program.
(6) The comment remains on the base screen when returning from machining simulation screen or machining drawing screen after switching to them. However, the comment is erased on the base screen when returning from machining simulation screen after switching to them in Series 30*i*/31*i*/32*i*-A or Series 0*i*-D.
(7) If the mode switch is done in the state of STOP or the state of HOLD then the comment remains being displayed.
(8) When the G code G2900 is executed during the automatic operation in the RMT mode or the subprogram is called from M198, then the comment is not displayed.
(9) If the parameter No.14700#0 is set to 1, then the comment cannot be displayed until switching to the NC screen once and returning back to MANUAL GUIDE *i*.

4.13 OPERATION RESTART CONFIRMATION FUNCIOTN AFTER EDITING PROGRAM (FOR Series 30*i*/31*i*/32*i*)

A confirmation message, to restart the machining, is displayed when the machining operation is restarted after program editing in EDIT mode during single block stop.
Only when the restart is allowed by soft-key, the machining can be restarted.
This function is enabled when the parameter No.10330#7 is set to 1.

4.13.1 Operation Restart Confirmation Screen

After changing EDIT mode during operation stop status or operation hold status and the amount of time specified in parameter No.10333 is passed, if the start button is pushed, a message and soft-keys for confirmation of machining restart will be displayed. The detail of the soft-keys to confirm machining restart is as follows.

(1) [YES] soft-key
Machining restart is allowed. If start button is pushed again, the machining will be restarted.
(2) [No] soft-key
Operation restart isn't allowed. If the start button is pushed again, then the message for confirmation of machining restart will be displayed again.

4.13.2 Editing Function during Single Block Stop

All editing functions of MANUAL GUIDE *i* are available in EDIT mode during single block stop.

4.13.3 Operation Restart Confirmation in Multi Path System

The confirmation message is displayed irrespective of the selected path. For example, though the path 1 is selected, if the machining in the path 3 is restarted, then the confirmation message will be displayed.

NOTE
If the confirmation is necessary in more than one path at the same time, the soft -keys [YES] or [NO] operation one time is effective to all paths.

4.13.4 Parameter

Set the following parameter to use this function.

	#7	#6	#5	#4	#3	#2	#1	#0
10330	RSC							

[Path attribute] Common to all paths

RSC If the machining is restarted after program editing during operation stop status or operation hold status,

0: the machining is restarted. And program editing during operation stop status or operation hold status in MANUAL GUIDE *i* screen is prohibited.

1: The message and soft-keys for confirmation which machining will be restarted or not are displayed. While the soft-keys are displayed, if [YES] soft-key is pushed, machining will be restarted after the cycle operation is started again. If [NO] soft-key is pushed, the operation will be invalid. And program editing during operation stop status or operation hold status in MANUAL GUIDE *i* screen is allowed.

NOTE
If the amount of time that is specified in parameter No.10333 is passed after switching to EDIT mode during operation stop status or operation hold status, then CNC system will judge that the program is being edited.

10333	Amount of time that has to pass to judge that the program is being edited in OPERATION RESTART CONFIRMATION FUNCIOTN AFTER EDITING PROGRAM

[Path attribute] Common to all paths

[Unit of data] msec

[Valid data range] 0-32767

When parameter RSC (10330#7) = 1, if the time that is set in this parameter is passed after the mode is changed to EDIT during stop status or hold status, then the CNC system will judge that the program is being edited.

If this parameter is set to 0, the time is set to 1000msec.

5 EDITING AND OPERATION MDI PROGRAMS

When the MDI mode is selected with the machine operator's panel, the soft keys shown below appear on the screen of MANUAL GUIDE *i*. Pressing the leftmost soft key [<] or the rightmost soft key [>] changes the page of the soft key display to the second and third pages sequentially.

REWIND				LINENO	ACTPOS	PRESET			

REWIND		SRCH↑	SRCH↓	CHSIZE	COPY	CUT	DELETE	KEYPST	PASTE

	UNDO	REDO		WK SET	T-OFS				SETING

For soft keys [ACTPOS] and [PRESET] on the first page, see the following sections:

Section 4.4, "CHANGING THE CURRENT POSITION DISPLAY"
Section 4.5, "PRESETTING RELATIVE COORDINATES"

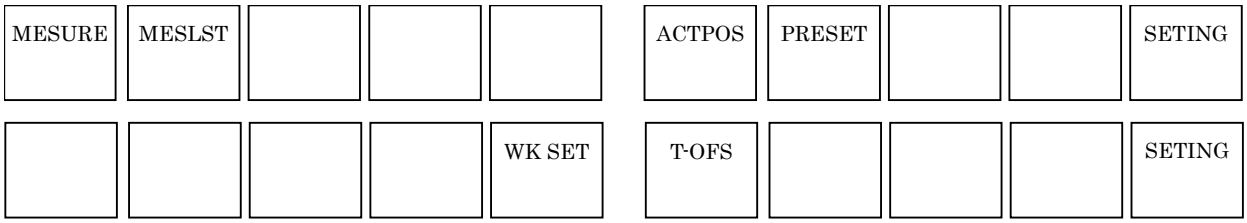
The soft keys on the second and third pages are used for editing machining programs entered by MDI. For details on these soft keys, see the following sections:

Section 2.1, "PROGRAM WINDOW AND EIDITNG"
Section 2.6, "SEARCH (FORWARD AND BACKWARD)"
Section 2.7, "CUT"
Section 2.8, "COPY"
Section 2.9, "PASTE"
Section 2.10, "DELETE"
Section 2.11, "KEY-IN PASTE"
Section 2.12, "UNDO, REDO"

For soft key [SETING], see Section 2.15, "REGISTERING FIXED FORM SENTENCES".

6 MANUAL OPERATION

When the handle or jog mode is selected with the machine operator's panel, the following soft keys appear on the screen of MANUAL GUIDE *i*:



NOTE
Soft keys [MESURE] and [MESLST] are displayed only when the Set-up guidance optional function is attached.

By pressing soft key [MESURE], a window for set-up guidance functions that are performed in the manual mode is displayed. By pressing soft key [MESLST], a window showing a list of measurement results is displayed. For details of these windows, see "Set-up Guidance" in Part IV.
For soft keys [ACTPOS] and [PRESET], see the following sections:

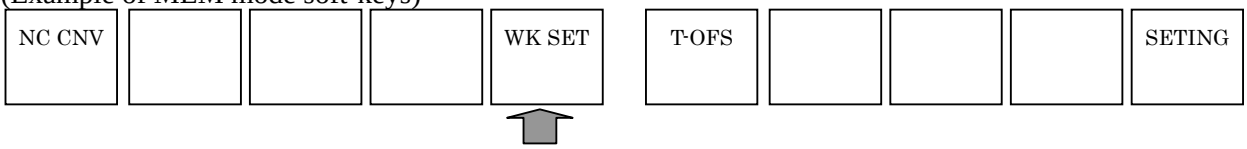
Section 4.4, "CHANGING THE CURRENT POSITION DISPLAY"
Section 4.5, "PRESETTING RELATIVE COORDINATES"

For soft key [SETING], see Section 2.15, "REGISTERING FIXED FORM SENTENCES".

7 SETTING THE WORKPIECE COORDINATE DATA

Soft key [WK SET] to open the workpiece coordinate data window can be displayed on all mode such as MEM, EDIT and manual mode.
Pressing the leftmost soft key [<] or rightmost soft key [>] several times displays the soft-keys including soft key [WK SET]

(Example of MEM mode soft-keys)

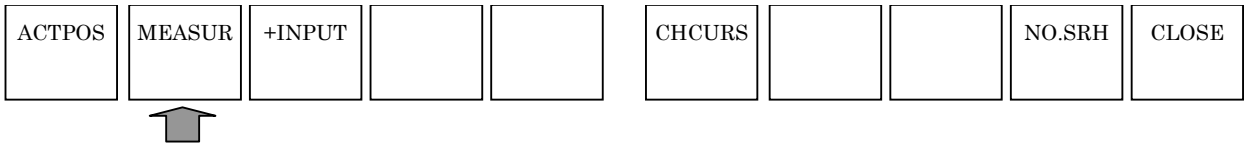


Pressing soft key [WK SET] displays the workpiece coordinate data window.

For machining centers, a window for setting the workpiece origin offset is displayed.

The data items to be set and displayed are common to the corresponding data items of the CNC. So, for details, refer to the operator's manual of the CNC.

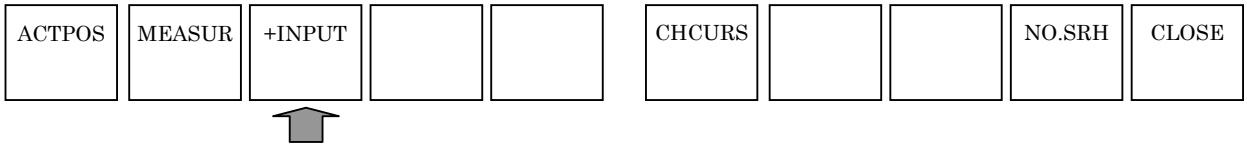
7.1 SOFT KEY [MEASUR]



By pressing soft key [MEASUR], the calculations below can be made.

(1) Workpiece origin offset window for M series
Current machine coordinate value - Target value of workpiece coordinate
(2) Workpiece origin offset window for T series
Current absolute coordinate value + Current setting - Target value of workpiece coordinate
(3) Workpiece coordinate system shift amount with the T series
Current setting - Current value of absolute coordinate + Target value of workpiece coordinate

7.2 SOFT KEY [+INPUT]



By pressing the soft key [+INPUT], "current value + offset value" can be calculated.

7.3 CALCULATING METHOD OF MEASURING OF SETTING WORKPIECE COORDINATE DATA (for only Series 0*i*-D)

7.3.1 Outline

Calculating method of measuring of workpiece origin offset value can be selected by parameter No.27482#4 in workpiece coordinate data window.

(1) No.27482#4=0 (specifications of MANUAL GUIDE *i*)

Method of calculation	
Lathe	Machining
Eq.5	Eq.4

(2) No.27482#4=1 (specifications of NC screen)

Parameter		Method of calculation	
No.11278#0	No.1015#6	Lathe	Machining
0	0	Eq.2	Eq.1
	1	Eq.2	Eq.2
1	0	Eq.2	Eq.4
	1	Eq.3	Eq.3

(Equation)

No.	Equation
1	Result = Current absolute coordinate value + Workpiece origin offset value on the modal workpiece coordinate + External workpiece origin offset value - Target value of workpiece coordinate
2	(External workpiece origin offset value) Result = Current absolute coordinate value + External workpiece origin offset value - Target value of workpiece coordinate (Workpiece origin offset value) Result = Current absolute coordinate value + Workpiece origin offset value on the modal workpiece coordinate - Target value of workpiece coordinate
3	(External workpiece origin offset value) Result = Current machine coordinate value - Workpiece origin offset value on the modal workpiece coordinate - Target value of workpiece coordinate (Workpiece origin offset value) Result = Current machine coordinate value - External workpiece origin offset value - Target value of workpiece coordinate
4	Result = Current machine coordinate value - Target value of workpiece coordinate
5	(External workpiece origin offset value) Result = Current absolute coordinate value + External workpiece origin offset value - Target value of workpiece coordinate (Workpiece origin offset value) Result = Current absolute coordinate value + Workpiece origin offset value on the selected workpiece coordinate by cursor - Target value of workpiece coordinate

7.3.2 Parameter Setting

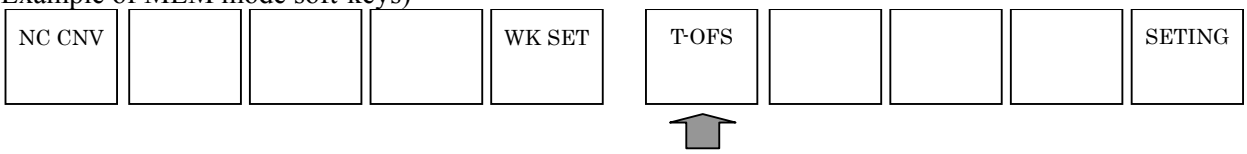
Please set parameter CMW(No.27482#4) to 1.

8 SETTINGS RELATED TO TOOLS

8.1 SETTING TOOL OFFSET DATA

[T-OFS] to open the tool offset data window can be displayed on all mode such as MEM, EDIT and manual mode.
Pressing the leftmost soft key [<] or rightmost soft key [>] several times displays the soft-keys including [T-OFS]

Example of MEM mode soft-keys)



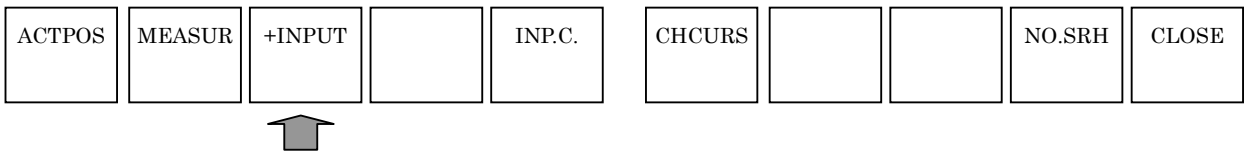
Pressing [T-OFS] displays the tool offset data window.

(1) M: TOOL OFFSET (TOOL LENGTH COMP. / CUTTER COMPENSATION)
(2) M: TOOL TYPE OFFSET (TOOL LENGTH COMP. / CUTTER COMPENSATION)

Data for each mode can be selected with a corresponding tab.

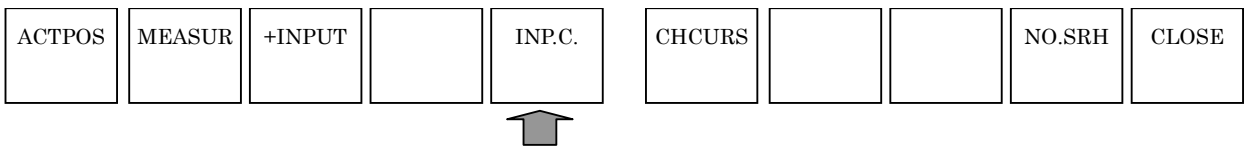
The data to be set and displayed in (1) is common to the corresponding data in the CNC. For details, refer to the operator's manual of the CNC.
For data items (2) that are related to the tool management function, see Chapter 2, "TOOL MANAGEMENT FUNCTION" in the Part VI.

8.1.1 Soft Key [+INPUT]



By pressing the soft key [+INPUT] soft key, "Current value + Offset value" can be calculated.

8.1.2 Soft Key [INP.C.]



By pressing the soft key [INP.C.] soft key, "Relative coordinate value" can be entered to the offset value directly.

8.2 SPECIFYING TOOL FIGURE DIMENSION DATA

8.2.1 Overview

(1) In the tool offset window, it is possible to key in tool data (such as tool diameter, type, name, mounting, and figure data) necessary when animation is used to draw figures or perform cycle machining.
(2) The tool diameter and the counter sink tool small diameter (tool nose radius) are input as a tool offset value (CUTTER COMPENSATION (machining center)).
(3) Selecting the TOOL DATA tab causes the TOOL FIGURE DIMENSION DATA SETTING window to appear. Only up to 300 tools can be specified in this window.
(4) Using data specified on the TOOL DATA tab during cycle machining requires specifying its number, using a T or D code.
(5) Cycle machining on machining centers is performed by referencing the tool information having the same number as a cutter compensation number specified using a D code.
(6) A check is made on the relationship between specified tool types and cycles. When a drilling cycle is executed, for example, an alarm is issued if a counter sink tool is selected. No check is made unless a tool type has been input.
(7) A tool figure is drawn during machining simulation if a T or D code is executed.
(8) The term “tool database function” herein refers to an operation in which cycle machining and animation are performed by inputting or referencing tool figure dimension data.

NOTE
1 This function is enabled if parameter TOL (No.14850#0) = 0.
2 Tool figures can be drawn by specifying either a tool information number explained above or a G code. When issuing a G code-based drawing command after tool information is specified, separate them by at least 2 blocks. Otherwise, it is likely that G code-based tool drawing may not take place.

8.2.2 Setting of Tool Type

By placing cursor on an item for selecting kind of tool, the following soft keys appear. When appropriate soft key is pushed, a type of tool is selected and its icon is displayed. Also tool name is displayed on the right of the icon.

(1) Soft keys for selecting kind of tool for machining centers

DRILL	CHAMFR	F END	B END	TAP	CHCURS	INIT		NO.SRH	CLOSE
REAMER	BORING	F MILL							CLOSE

8.2.3 Editing of Tool Name

To edit tool name, place cursor on tool name, change mode into character, input alphabets or numerals, and push <INPUT>.
This function is useful to distinguish similar tools.

8.2.4 Setting of Tool Set

When a cursor is placed on tool setting number, a guidance window is automatically displayed in the right of the screen. By inputting tool setting number and pushing <INPUT>, tool setting can be set.

8.2.5 Setting of Tool Data

By placing cursor on items of tool data, guidance window is automatically displayed. By inputting tool data value and pushing <INPUT>, tool data can be set. Then three-whole and one-decimal number can be inputted when unit is [deg.]. When [inch] or [mm], eight-figure number can be inputted. But numbers that is inputted to 7 decimal places are rounded off to 6 decimal places.
The names of item and the numbers of item depend on kind of tool. See below about the detail. In this table, tools that don't have to set tool data are omitted.
Tool data isn't necessary to execute milling cycle. So milling cycle can be executed though tool data isn't set.

Tool form data for milling

KIND OF TOOL	DRILL	CHAMFER
Data	Nose angle (*)	Cutter diameter (*)

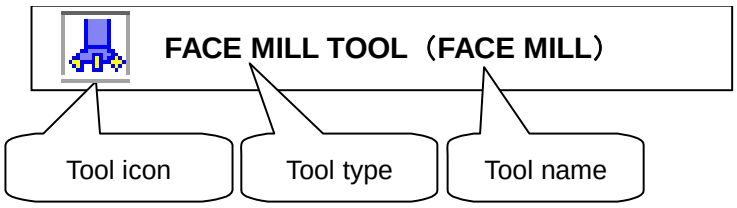
(*) : Cycle can be executed if not set

8.2.6 Initializing of Tool Data

Tool data can be initialized by soft key [INIT]. When soft key [INIT] is pushed, a message for confirming initialization is displayed. By pressing soft key [YES], initialization is performed.
But items to be initialized are only items placed in tool data tab and [INIT] has no connection with offset value.

8.2.7 Displaying Tool Icons

If tool figure dimension data can be specified, placing the cursor on a tool offset command (for machining centers having a tool management function, D code) causes the tool icon, tool type, and tool name corresponding to the number to be displayed in the message window.



NOTE
1 Enabling this function requires setting the parameter TLJ (No.14705#7) to 1.

8.2.8 Shortcut to the Tool Figure Dimension Data Setting Window

If tool figure dimension data can be specified, placing the cursor on a tool offset command and then pressing the soft key [ALTER] or the <INPUT> key displays the tool database window. In this window, the cursor moves to the data having a tool number corresponding to the cutter compensation number (machining centers).

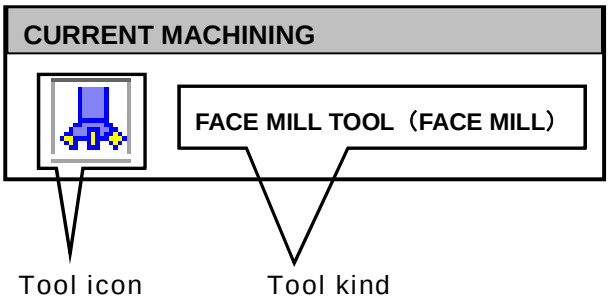
NOTE
1 Enabling this function requires setting the parameter TLJ (No.14705#7) to 1.

8.3 TOOL INFORMATION DISPLAY

The tool information can be always displayed referring to the modal information. The modal information is T code in case of the lathe, and D code in case of the machining center.
This function is enabled when the parameter No.27480#3 is set to 1.

8.3.1 Displaying Screen

The icon and the tool kind which corresponds to D codes are displayed on the base screen.



NOTE
Tool information is not displayed on the machining simulation screen, the machining drawing screen, and the NC program conversion screen.

8.3.2 Tool Information Display

(1) The icon and the tool kind of the tool, which correspond to the modal information of D code for the milling system, are displayed.
(2) Icon and tool kind, set in the tool database, are displayed.
(3) If the modal value for D code is 0, then the icon and the tool kind are not displayed.
(4) If in the tool database there is no tool specified, corresponding to the active modal D code, then the icon and the tool kind are not displayed.
(5) If the tool database is not enabled then the icon and the tool kind are not displayed.
(6) This function shows the tool information while the base screen is displayed in the modes MDI, MEM, RMT, HND, JOG, TJOG, THAND, and REF (Excluding the EDIT mode). Tool information is displayed also when the CNC is not machining.

8.3.3 Notes when Commanding Offset Number by D Code

In following cases there is the possibility that it is displayed a different tool than the tool actually selected.

(1) If T code and D code are commanded in different blocks, then the tool information corresponding to the last commanded D code (Excluding D0) will be displayed until the D code corresponding to the most recent T code is commanded.
(2) When D code is not commanded, as for example a drill tool, then the tool information corresponding to the last commanded D code (Excluding D0) will be displayed. The tool information is not updated until a different D code is commanded. (The tool information is not displayed when D code has been not commanded yet.)

Therefore, it is advised to command a D code soon after the T code is commanded in order to have congruous tool information displayed in respect to the selected tool.

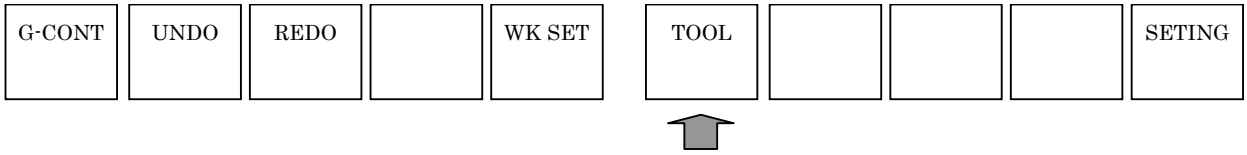
8.4 COMBINING A TOOL DATABASE WITH THE TOOL MANAGEMENT FUNCTION

If the tool management option is available, the tool management function data table can be displayed on the MANUAL GUIDE *i* screen. In this case, a tool database setting window is added to the tool management function table.

NOTE
1 To use this function, you require tool management function options.
2 To use this function the parameter TOL (No.14705#5) must be set to 1.

8.4.1 Tool Management Data Window

Pressing the soft key [TOOL] causes the tool management data window to appear.
In this window, it is possible to specify tool management data, magazine information, tool offset, and tool information.



8.4.2 Tool Management Data Tab

8.4.2.1 Tool management data

On the tool management data tab, it is possible to specify a compensation number, tool life information, spindle rotation speed, feedrate, customization information, and tool information.

The following data can be specified.

(1) NO. (TOOL MANAGEMENT DATA NO)
A tool management data number is displayed. It cannot be changed; it can only be displayed.
(2) TYPE-NO
A tool type number (T code) is displayed.
(3) MG (Magazine number)
A magazine number is displayed. It cannot be changed; it can only be displayed.
(4) POT
A pot number is displayed. It cannot be changed; it can only be displayed.
(5) T-INFO
The tool information consists of the 6 different pieces of information listed below, as viewed from the right.
The 2 leftmost bits are not in use.
Tool management data : Valid (R)/invalid (-)
Tool life count type : Time (T)/count (C)
Tool type : Large-diameter (B)/regular-diameter (N) tool
Data access : Locked (L)/unlocked (U)
Whether to include the tool as a search target when its life status is ”DISABLE”
: Yes (-)/no (S)
Tool life count period : 1 second (0)/8 millisecond (1)

(6) L-COUNT
The too life count of the tool (how many times or how long has it been used) is displayed.
(7) MAX-LIFE
The maximum life value/time of the tool is displayed.
(8) NOTICE-L
The life expectancy value/time of the tool is displayed.
(9) L-STATE
The current life status of the tool is displayed. It can be one of the following:
DISABLE, ENABLE, OVER, DAMAGE

8.4.2.2 Spindle rotation speed and feedrate

Pressing the page change key <↑PAGE> or <PAGE↓> several times displays a window for setting the spindle rotation speed and feedrate.

The following data can be specified.

(1) S (SPINDLE SPEED)
The rotation speed of the spindle is displayed.
(2) F (FEED RATE)
The feedrate is displayed.

8.4.2.3 Tool offset

Pressing the page change key <↑PAGE> or <PAGE↓> several times displays a window for specifying a tool offset.

The following data can be specified.

(1) H (TOOL LENGTH COMPENSATION NUMBER), D (CUTTER COMPENSATION NUMBER)
The tool length compensation number and cutter compensation number are displayed.
(2) GEOM, WEAR
The compensation value corresponding to a compensation number specified with H or D is displayed.

Pressing the page change key <↑PAGE> or <PAGE↓> several times displays a window for specifying tool information.
For detailed descriptions of data, see chapters explaining the tool database.

8.4.2.4 Customization data

Pressing the page change key <↑PAGE> or <PAGE↓> several times displays a window for specifying customization data.

The following data can be specified.
(1) CUSTOM 0
This is customization information in bit format.
Either 1 or 0 can be keyed in for individual bits separately.
(2) CUSTOM 1 to 4
This is customization information.
(3) CUSTOM 5 to 20
This is customization information. It is displayed only if an option for adding customization data to the tool management function is available.

8.4.3 Customizing Tool Management Data Displays

In the tool management data window, it is possible to specify whether to show or hide items (such as type number, tool information, and tool life counter) and to change the position where they are displayed.

8.4.4 Magazine Management Table Tab

The magazine data tab displays the magazine, spindle, tool magazine data number and tool type number corresponding to the pot in the standby position.

(1) POT
The pot number is displayed.
(2) NO.
The tool management data number is displayed.
(3) TYPE- NO.
The tool type number corresponding to the tool type number is displayed.
(4) SPDL
The tool management data number and tool type number for the spindle position are displayed.
(5) WAIT
The tool management data number and tool type number for the standby position are displayed.

8.4.5 Tool Offset Tab

The tool offset tab displays a window for specifying tool offset.

8.4.6 Tool Information Tab

The tool information tab displays a window for specifying tool figure dimension data.

8.4.7 Outputting Tool Management Data to the Memory Card

The following procedure can output tool management data.

<1> On the machine operator's panel, select the edit mode to display a window for tool management.
<2> Press the soft key [TOOL] to display the tool management function window, and then press the leftmost soft key [<] or the rightmost soft key [>] several times. The soft keys shown below appear.

						OUTPUT	INPUT		CLOSE

<3> Press the soft key [OUTPUT]. The following array of soft keys appears, enabling data to be selected for output.

TOOL	MAG	STATUS	CSTM		DSPCT	P_NAME	POINT		CANCEL

[TOOL] : Outputs tool management information and tool database data.
[MAG] : Outputs magazine management table data.
[STATUS] : Outputs tool life status name data.
[CSTM] : Outputs customization data name data.

Displaying the following 3 items requires installing the tool management expansion option.

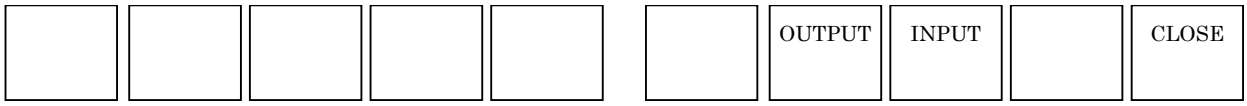
[DSPCT] : Outputs window customization data.
[P_NAME] : Outputs spindle and standby position name data.
[POINT] : Outputs customization data decimal point position data.

<4> Pressing the soft key corresponding to the tool management data to be output displays a confirmation message. Pressing the soft key [YES] directs the tool management data to the memory card.

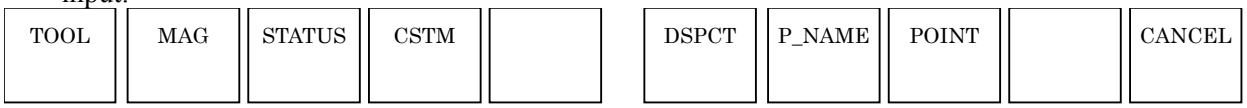
8.4.8 Inputting Tool Management Data from the Memory Card

The following procedure can input tool management data.

<1> On the machine operator's panel, select the edit mode to display a window for tool management.
<2> Press the soft key [TOOL] to display the tool management function window, and then press the leftmost soft key [<] or the rightmost soft key [>] several times. The soft keys shown below appear.



<3> Press the soft key [INPUT]. The following array of soft keys appear, enabling data to be selected for input.



[TOOL] : Inputs tool management information and tool database data.
[MAG] : Inputs magazine management table data.
[STATUS] : Inputs tool life status name data.
[CSTM] : Inputs customization data name data.

Displaying the following 3 items requires installing the tool management expansion option.

[DSPCT] : Inputs window customization data.
[P_NAME] : Inputs spindle and standby position name data.
[POINT] : Inputs customization data decimal point position data.

<4> Pressing the soft key corresponding to the tool management data to be input displays a confirmation message. Pressing the soft key [EXEC] causes the tool management data to be input to the memory card.

III. CONVENIENT FUNCTIONS

1 CONTOUR PROGRAMMING

Contour programming allows arbitrary figures (contour programs) consisting of circles and straight lines, rather than those figures handled in cycle machining, to be input.
Pressing the soft key [G-CONT] displays a window similar to one used with arbitrary figure data described in the previous chapter. This window is used to perform all operations, from inputting figures to writing them to a machining program, in the same manner.
In this case, the created machining program consists of G01, G02, and G03 codes in ISO code format.

NOTE
See Chapter 5, “Inputting Arbitrary Figures”, of Part IV for explanations about contour program input data.

(1) Inputting insertion character strings

Arbitrary character strings can be input when inputting any figure block to contour programs.
The 「INS.CODE」 tab is added to the window for inputting contour program figure blocks as explained below.

Insertion character string		
Data item		**Meaning**
F*	LAST CODE	Character string (up to 16 characters) to be input at the beginning (right after an EOB) of each figure block
B*	NEXT CODE	Character string (up to 16 characters) to be input at the end (right before an EOB) of each figure block

The 「INS.CODE」 described above can be used to input G codes and feedrates into figure blocks.

NOTE
The character string that can be input is only a word. It can include neither parentheses nor space characters.

(2) Editing contour program figure blocks

Once a contour program figure block has been input, it can be re-edited by performing figure calculation in the same manner as when it was entered.
For this reason, each figure block has the input figure data written as a comment.

The first figure block of a contour program has a G code (G1200, G1300, G1450, G1500, or G1600) signifying a start point as a comment. Place the cursor on that block and press the soft key [ALTER].
This operation displays a window for editing contour programs. In this window, the figures can be edited in the same manner as arbitrary figures.

(3) Select style of contour program of ZC plane

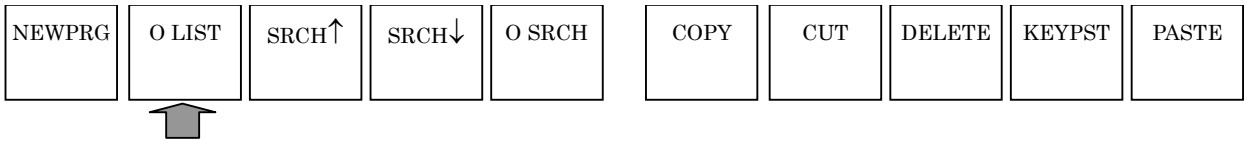
Select style of contour program of ZC plane by parameter No.27096#0.
(Parameter No.27096#0)

0 : The program for cylindrical interpolation on G18 plane is generated in Contour programming window of ZC plane.
1 : The program for cylindrical interpolation on G19 plane is generated in Contour programming window of ZC plane.

2 MEMORY CARD INPUT/OUTPUT FUNCTION

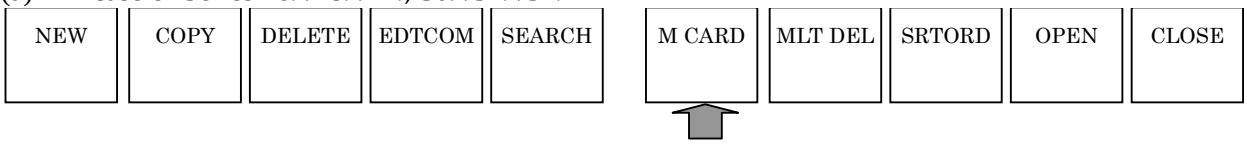
2.1 MEMORY CARD INPUT/OUTPUT OF PROGRAM

2.1.1 Memory Card Input/Output Screen of Program

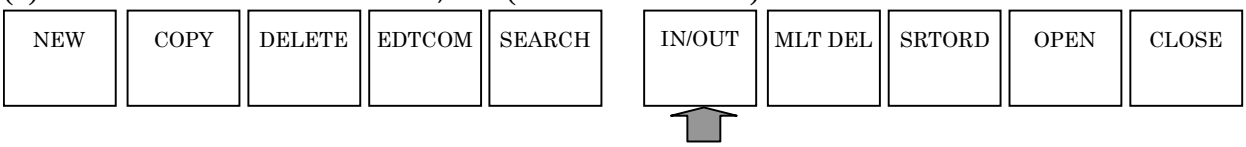


Select EDIT mode on the machine operator's panel. Pressing soft key [O LIST] displays the program list window, programs registered in the CNC. Following soft-keys are displayed.

(a) In case of Series 16*i* /18*i* /21*i*, 30*i* /31*i* /32*i*-A

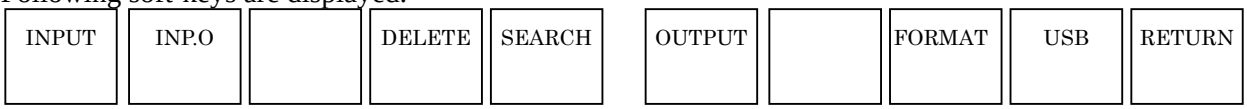


(b) In case of Series 30*i* /31*i* /32*i*-B, 0*i*-D(with USB interface)



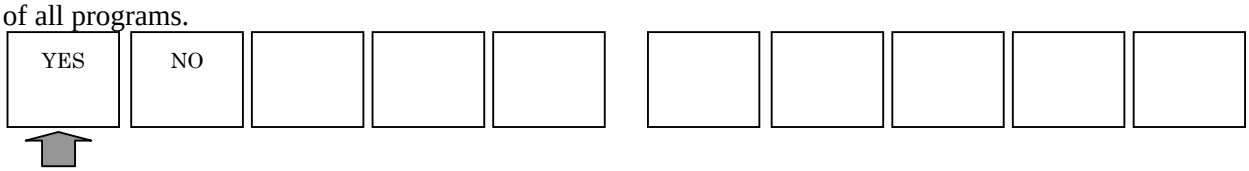
Pressing the soft key [M CARD] or the soft key [IN/OUT] on the program list screen displays the [INPUT/OUTPUT PROGRAM BY MEMORY CARD] screen.

Following soft-keys are displayed.



[INPUT] : Inputs a program from the memory card.
[INP.O] : Inputs a program from the memory card (by changing the O number).
[DELETE] : Deletes files on the memory card.
[SEARCH] : Searches for a file on the memory card.
[OUTPUT] : Displays the screen for output to the memory card.
[FORMAT] : Formats the memory card. (Note)
[USB] : Displays the [INPUT/OUTPUT PROGRAM BY USB MEMORY] screen. (Note)
[RETURN] : Returns the screen display to the program list screen.

In order to delete a file in the memory card, place the cursor on the file name in the above window and press soft key [DELETE]. This soft key displays a message for checking if selected file may be deleted. Pressing soft key [YES] deletes the file in the memory card. Pressing soft key [NO] cancels the deletion of all programs.



In case of initializing the memory card such as deleting all files in the memory card, press soft key [FORMAT]. This soft key displays a message for checking if the memory card may be initialized. Pressing soft key [YES] initializes the memory card and all files in the memory card are deleted. Pressing soft key [NO] cancels the deletion of all programs.

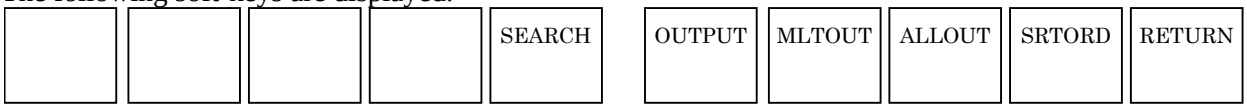
NOTE
1 When the CNC screen display function is in use, neither of the following items is displayed.
- Soft key [FORMAT]
- Whether the memory area is in use or not

2 Soft key [USB] is displayed for Series 30*i* /31*i* /32*i*-B and Series 0*i*-TD /0*i*-MD only. And, in case of Series 0*i*-TD /0*i*-MD, It is necessary to meet the following requirements.
- Using the following main boards with USB interface
  <1> main board B2 (A20B-8200-0842)
  <2> main board B3 (A20B-8200-0843)
- Using the following softwares
  <1> 19.0th edition or more of NC basic software (D4F1/D6F1)
  <2> 1.0th edition or more of USB software (659B)
  <3> 12.0th edition or more of MANUAL GUIDE i software (BX32)
- The parameter No.14854#0=1

2.1.2 Memory Card Output Operation for Program

Pressing the soft key [OUTPUT] on the INPUT/OUTPUT PROGRAM BY MEMORY CARD screen displays the OUTPUT PROGRAM TO MEMORY CARD screen.

The following soft-keys are displayed.

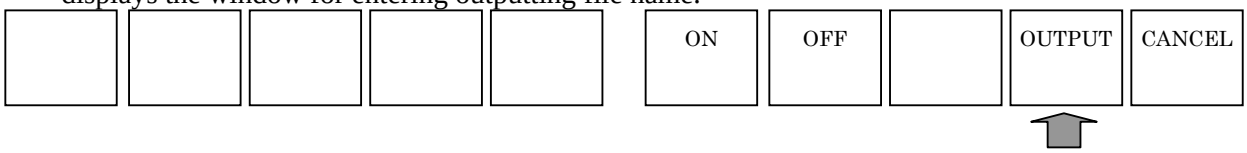


[SEARCH] : Searches for a program.
[OUTPUT] : Output a selected program to the memory card.
[MLTOUT] : Output multi programs to the memory card.
[ALLOUT] : Output all programs to the memory card.
[SRTORD] : Switches the sort order for displaying a program list between ascending order and descending order.
[RETURN] : Returns the screen display to the [INPUT/OUTPUT PROGRAM BY MEMORY CARD] screen.

Pressing soft key [SEARCH] displays the following window of program searching.
Enter the program number to be searched, then press soft key [SEARCH].

(1) Output single part program

Select the part program to be output by placing the cursor on it. Pressing soft key [OUTPUT] displays the window for entering outputting file name.



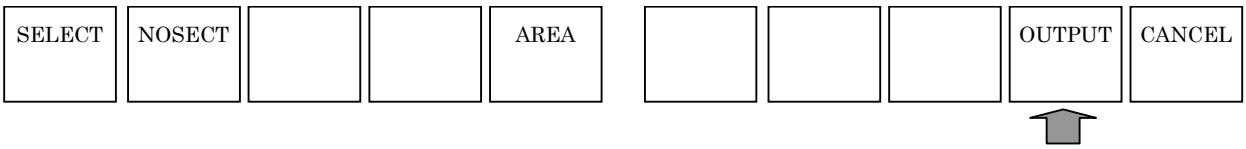
If the program number can be used as the output file name, press soft key [OUTPUT] without entering file name.

In case that the output file name should be changed, enter the file name to OUTPUT FILE NAME and press soft key [OUTPUT].

In order to output the selected program together with subprogram called from the program, press soft key [ON] for OUTPUT WITH SUBPROGRAM item. If not, press soft key [OFF].

(2) Output multi part programs

Pressing soft key [MLTOUT] displays the screen for selecting a program to be output



[SELECT] : This soft key selects a program to be deleted. Place the cursor on the program to be selected, and press the soft key [SELECT]. When the program is selected, the color of its background changes to yellow.

[NOSECT] : This soft key deselects a program. Place the cursor on the program to be deselected, and press the soft key [NOSECT]. When the program is deselected, the color of its background changes from yellow back to white.

[AREA] : This soft key selects a range. Place the cursor on the first program in a sequence of programs to be selected, press the soft key [AREA], and then move the cursor to the last program in the sequence of programs. The color of the specified range changes to purple. After this, press the soft key [SELECT] to assert the range. The color of the background of the selected programs changes from purple to yellow.

[OUTPUT] : This soft key outputs selected programs to memory card.

[CANCEL] : This soft key deselects all programs in the folder of interest at a time and returns the screen display to the [OUTPUT PROGRAM TOUSB MEMORY] screen. The color of the background of the deselected programs changes from yellow back to white.

Pressing soft key [OUTPUT] displays the window for entering outputting file name.

If the file name “PROGRAM ALL” can be used as it is, press soft key [OUTPUT] without entering file name and selected part programs stored in the CNC, the currently selected path when multi-path lathe, are output to the memory card with this name.

In case that the output file name should be changed, enter the file name to OUTPUT FILE NAME and press soft key [OUTPUT].

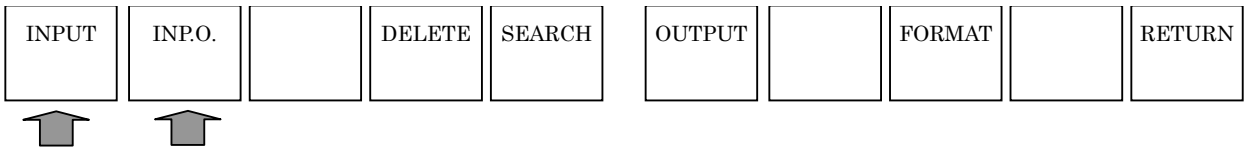
(3) Output all part programs

Pressing soft key [ALLOUT] displays the window for entering outputting file name.

If the file name “PROGRAM ALL” can be used as it is, press soft key [OUTPUT] without entering file name and all part programs stored in the CNC, the currently selected path when multi-path lathe, are output to the memory card with this name.

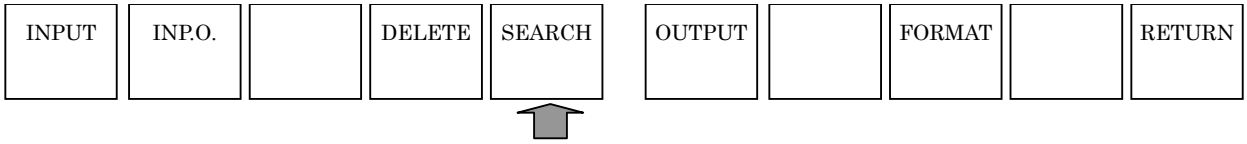
In case that the output file name should be changed, enter the file name to OUTPUT FILE NAME and press soft key [OUTPUT].

2.1.3 Memory Card Input Operation for Program



Select the file to be input by placing the cursor on it on the program input/output window.
Pressing soft key [INPUT] begins to read the part program in the file from memory card to CNC.
Pressing soft key [INP.O.] displays the window for changing the reading program number, so enter the new program number and press soft key [INPUT].

When the file, in which all part program was output with file name of “PROGRAM ALL”, is read to CNC with changing the program number by soft key [INP.O.], the program number of the 1st program is changed to the new number.



In order to search the file to be input to CNC, press soft key [SEARCH] and the file searching window is displayed.

Enter the file name to be searched, and press soft key [SEARCH], then the file is searched if the file is stored in the memory card.

2.1.4 File Format Allowed for Memory Card Input/Output

Only text files can be input to and output from the memory card. The file format described below must be observed.

(1) A file must start with "%" and "LF".
(2) A file must end with "%".
(3) For input, data read operation is skipped after the first "%" is detected until an "LF" is detected.
(4) A block must end not with a semicolon (;) but with an "LF". ("LF" is 0A of ASCII code.)
(5) When a file containing lowercase alphabetic characters, kana characters, and some special characters (such as $, ¥, and !) is input, those characteristics are ignored.
(6) ASCII code is used as input/output code, regardless of the setting parameter (ISO/EIA).
(7) Whether to output an "LF" only or an "LF, CR, CR" as an EOB can be chosen using bit 3 (NCR) of parameter No. 0100.
(8) Characters usable for a file name
Alphabetic characters : A to Z
Numeric characters : 0 to 9
Special characters : $ & # % ' ( ) - @ ^ { } ~ ` ! _

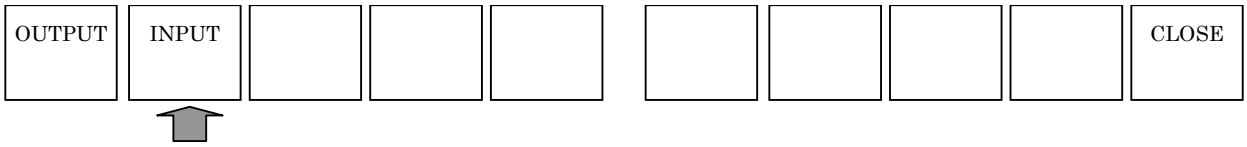
2.1.5 Notes

For the Series 30*i*/31*i*/32*i*, the current folder is treated as an input/output target.

2.2 MEMORY CARD INPUT/OUTPUT OF TOOL DATA

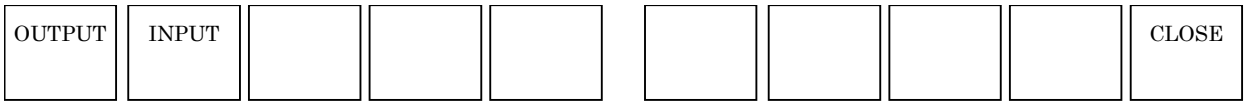
2.2.1 Memory Card Input/Output Screen of Tool Data

During displaying TOOL DATA window, the following soft-keys are displayed by pressing the leftmost soft key [<] or rightmost soft key [>] several times.



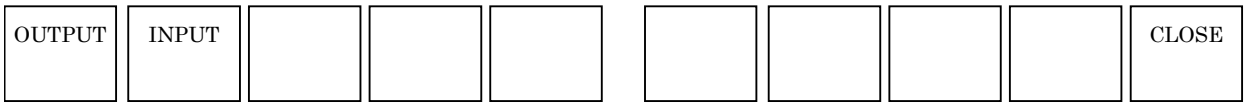
In order to input or output the tool data between memory card, select EDIT mode on the machine operator's panel.
Insert the memory card into the memory card slot on the LCD/MDI panel.

2.2.2 Memory Card Output Operation for Tool Data



Pressing soft key [OUTPUT] displays the following window for entering the output file name.
Pressing soft key [OUTPUT] without entering the file name outputs the tool data with the file name "TOOLDAB.DAT".
In order to change the file name, enter the file name and press soft key [OUTPUT].

2.2.3 Memory Card Input Operation for Tool Data



Pressing soft key [INPUT] displays the following window of f the file list store in the memory card.
Select the file in which tool data are stored and to be read to CNC by placing the cursor by the cursor key.
Pressing soft key [INPUT] begins reading the tool data to CNC form the memory card.

2.2.4 Data Format

The following format can be input or output.

(1) Machining Center

G1980 P_ K_ T_ S_ A_ ;

P : Offset Number (1 to 999)
K : Kind of Tool
T : Name of Tool
S : Setting
A : Tool Data

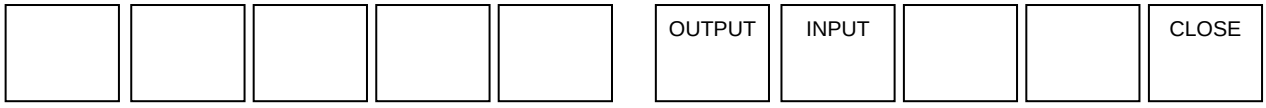
2.3 MEMORY CARD INPUT / OUTPUT OF OFFSET DATA (FOR Series 16*i*/18*i*/21*i*/30*i*/31*i*/32*i*)

Tool Offset data can be input or output from/to memory card.
This function is available when the parameter No.27480#4 is set to 0.

2.3.1 Output Operation

Offset data can be output by the following operation.

(1) Offset window
After changing EDIT mode, push [T-OFS] soft-key to open the tool offset window. The following soft-keys will be displayed by pressing the [<] or [>] soft-key several times.



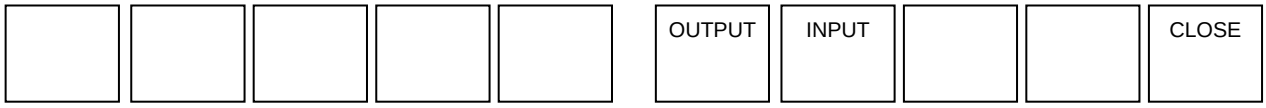
If [OUTPUT] is pushed, then the window for entering outputting file name will be displayed.

(a) In case of Series 16*i* /18*i* /21*i*
Because the outputting file name can't be changed, push [OUTPUT] soft-key without pushing other button. The outputting file name is followings.

TOOLOFST.DAT (When 1 path is selected)
TOOLOFST.P-2 (When 2 path is selected)
TOOLOFST.P-3 (When 3 path is selected)

(b) In case of Series 30*i* /31*i* /32*i*
If [OUTPUT] is pushed without changing file name, tool offset data with default file name will be output. Enter file name and push [OUTPUT] in case the operator wants to change outputting file name.

(2) Tool management function integrated in MANUAL GUIDE *i*
After changing EDIT mode, push [TOOL] soft-key to open the tool management function integrated in MANUAL GUIDE *i*. The following soft-keys will be displayed if the [<] or [>] soft-key is pushed several times after the offset tab is selected.

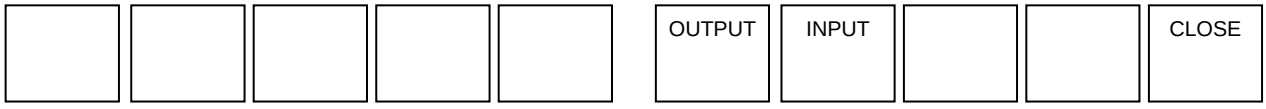


From now the operation is same as the case of offset window.

2.3.2 Input Operation

Offset data can be input by the following operation.

(1) Offset window
After changing EDIT mode, push [T-OFS] soft-key to open the tool offset window. The following soft-keys will be displayed by pressing the [<] or [>] soft-key several times.



If [INPUT] is pushed, then the file list in memory card will be displayed.

(a) In case of Series 16*i* /18*i* /21*i*
Because one of the following file name is displayed in the file list, push [INPUT] soft-key without pushing other button.

TOOLOFST.DAT (When 1 path is selected)
TOOLOFST.P-2 (When 2 path is selected)
TOOLOFST.P-3 (When 3 path is selected)

(b) In case of Series 30*i* /31*i* /32*i*
All files in the memory card are displayed in the file list. Select the offset file by cursor and press [INPUT].

(2) Tool management function integrated in MANUAL GUIDE *i*
The operation is same as the case of offset window.

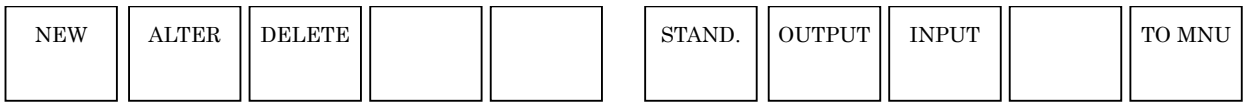
2.3.3 Notes

If the operator reset the CNC or set the CNC in emergency stop during inputting or outputting operations, the inputting or outputting will be interrupted. Then, input or output again.

2.4 MEMORY CARD INPUT/OUTPUT OF FIXED FORM SENTENCES

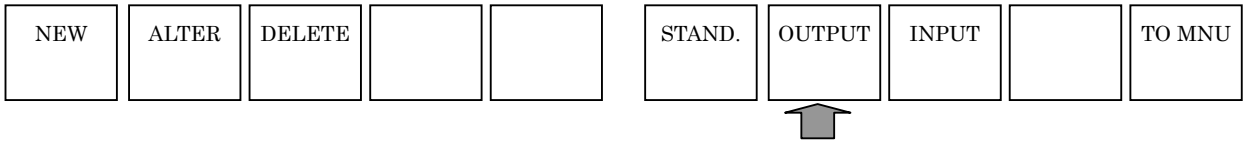
2.4.1 Memory Card Input/Output Screen of Fixed Form Sentences

The following soft key is displayed after soft key [SETING] is pushed and “REGISTER FIXED FORM SENTENCES FOR MILLING” is selected.



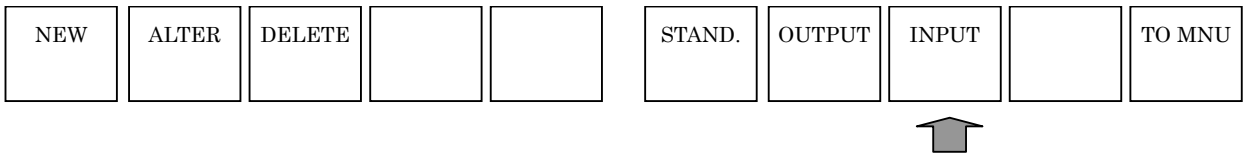
Select EDIT mode on the machine operator’s panel.
Insert the memory card into the memory card slot on the LCD/MDI panel.

2.4.2 Output Operation for Fixed Form Sentences



Pressing soft key [OUTPUT] displays the following window for entering the output file name.
Pressing soft key [OUTPUT] without entering the file name outputs the fixed form sentences with the file name “FIXFRM.DAT” (Output fixed form sentences for milling in a lump.)
In order to change the file name, enter the file name and press soft key [OUTPUT].

2.4.3 Input Operation for Fixed Form Sentences



Pressing [INPUT] displays the following window of the file list store in the memory card.
Select the file in which fixed form sentences are stored and to be read to CNC by placing the cursor by the cursor key.
Pressing [INPUT] begins reading the tool data to CNC form the memory card.

2.4.4 Format of Fixed Form Sentences

The following format can be input or output for fixed form sentences.
<Mode>=_,<Tab>=_,<Title>=_,<Code>=_

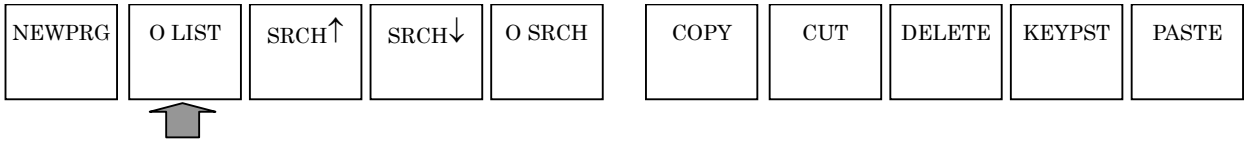
Mode : Setting the mode of Fixed form sentences (Milling or Turning).
1 : Fixed Form Sentence for turning mode
2 : Fixed Form Sentence for milling mode
3 : Fixed Form Sentence for milling and turning mode
Tab : The number of tab(1 - 5)
Title : Setting the name of Fixed Form Sentence
Code : Setting Fixed Form Sentence

4 words, <Mode>=, <Tab>=, <Title>=, <Code>=, can’t be input as the name of fixed form sentence or fixed form sentence.

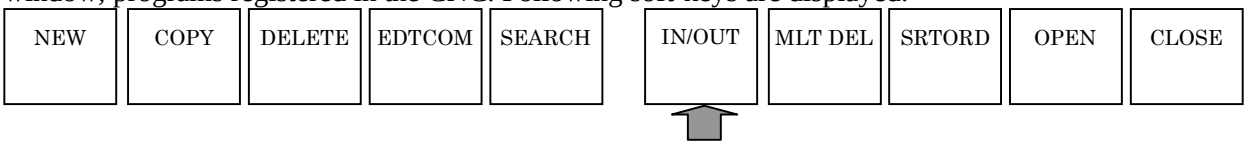
3 USB MEMORY INPUT/OUTPUT FUNCTION (FOR Series 0*i*–D,30*i*/31*i*/32*i*–B)

3.1 USB MEMORY INPUT/OUTPUT OF PROGRAM

3.1.1 USB Memory Input/Output Screen of Program

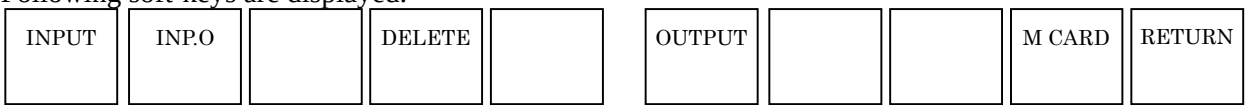


Select EDIT mode on the machine operator's panel. Pressing soft key [O LIST] displays the program list window, programs registered in the CNC. Following soft-keys are displayed.



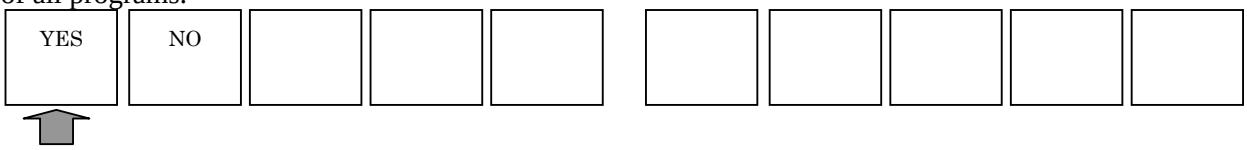
Pressing the soft key [IN/OUT] on the program list screen displays the [INPUT/OUTPUT PROGRAM BY USB MEMORY] screen.(Note)

Following soft-keys are displayed.



[INPUT] : Inputs a program from the USB memory.
[INP.O] : Inputs a program from the USB memory (by changing the O number).
[DELETE] : Deletes files on the USB memory.
[OUTPUT] : Displays the screen for output to the USB memory.
[M CARD] : Displays the [INPUT/OUTPUT PROGRAM BY MEMORY CARD] screen.
[RETURN] : Returns the screen display to the program list screen.

In order to delete a file in the USB memory, place the cursor on the file name in the above window and press soft key [DELETE]. This soft key displays a message for checking if selected file may be deleted. Pressing soft key [YES] deletes the file in the USB memory. Pressing soft key [NO] cancels the deletion of all programs.

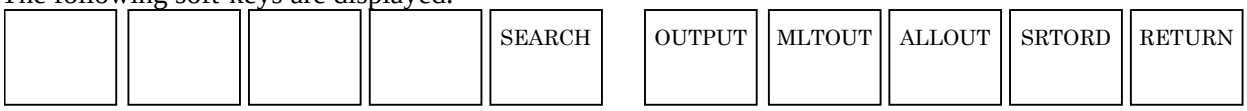


NOTE
1 When both the USB memory and memory card are inserted, pressing the soft key [IN/OUT] displays the [INPUT/OUTPUT PROGRAM BY MEMORY CARD] screen.
2 The item whether the memory area is in use or not is displayed on USB memory.

3.1.2 USB Memory Output Operation for Program

Pressing the soft key [OUTPUT] on the INPUT/OUTPUT PROGRAM BY MEMORY CARD screen displays the OUTPUT PROGRAM TO MEMORY CARD screen.

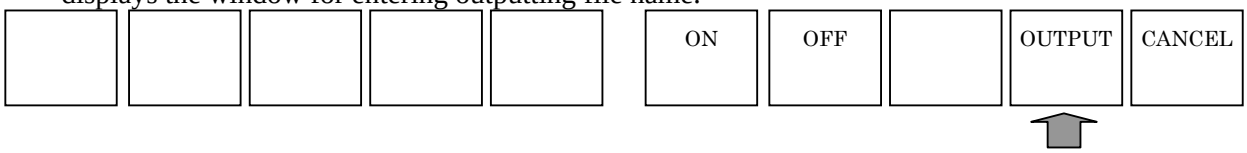
The following soft-keys are displayed.



[SEARCH] : Searches for a program.
[OUTPUT] : Output a selected program to the USB memory.
[MLTOUT] : Output multi programs to the USB memory.
[ALLOUT] : Output all programs to the USB memory.
[SRTORD] : Switches the sort order for displaying a program list between ascending order and descending order.
[RETURN] : Returns the screen display to the [INPUT/OUTPUT PROGRAM BY MEMORY CARD] screen.

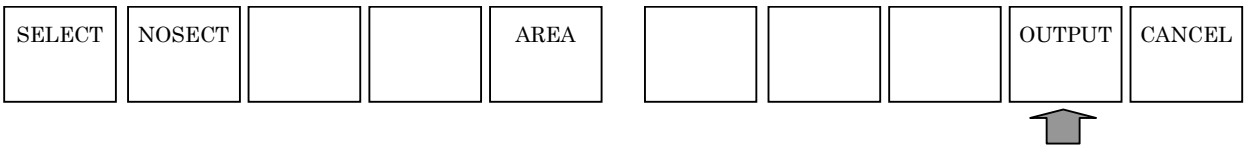
Pressing soft key [SEARCH] displays the following window of program searching.
Enter the program number to be searched, then press soft key [SEARCH].

(1) Output single part program
Select the part program to be output by placing the cursor on it. Pressing soft key [OUTPUT] displays the window for entering outputting file name.



If the program number can be used as the output file name, press soft key [OUTPUT] without entering file name.
In case that the output file name should be changed, enter the file name to OUTPUT FILE NAME and press soft key [OUTPUT].
In order to output the selected program together with subprogram called from the program, press soft key [ON] for OUTPUT WITH SUBPROGRAM item. If not, press soft key [OFF].

(2) Output multi part programs ram
Pressing soft key [MLTOUT] displays the screen for selecting a program to be output



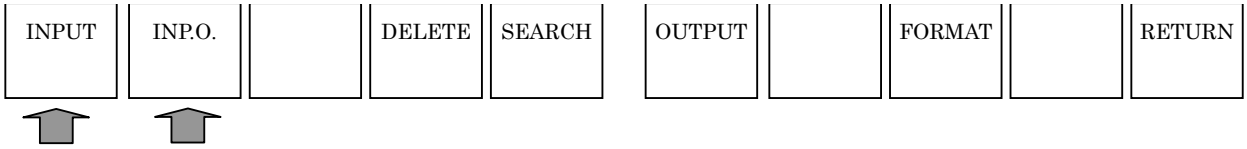
[SELECT] : This soft key selects a program to be deleted. Place the cursor on the program to be selected, and press the soft key [SELECT]. When the program is selected, the color of its background changes to yellow.
[NOSECT] : This soft key deselects a program. Place the cursor on the program to be deselected, and press the soft key [NOSECT]. When the program is deselected, the color of its background changes from yellow back to white.
[AREA] : This soft key selects a range. Place the cursor on the first program in a sequence of programs to be selected, press the soft key [AREA], and then move the cursor to the last program in the sequence of programs. The color of the specified range changes to purple. After this, press the soft key [SELECT] to assert the range. The color of the background of the selected programs changes from purple to yellow.

[OUTPUT] : This soft key outputs selected programs to USB memory.
[CANCEL] : This soft key deselects all programs in the folder of interest at a time and returns the screen display to the [OUTPUT PROGRAM TO USB MEMORY] screen. The color of the background of the deselected programs changes from yellow back to white.

Pressing soft key [OUTPUT] displays the window for entering outputting file name.
If the file name “PROGRAM ALL” can be used as it is, press soft key [OUTPUT] without entering file name and selected part programs stored in the CNC, the currently selected path when multi-path lathe, are output to the USB memory with this name.
In case that the output file name should be changed, enter the file name to OUTPUT FILE NAME and press soft key [OUTPUT].

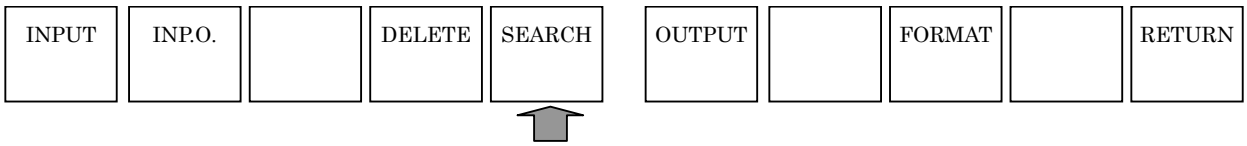
(3) Output all part programs
Pressing soft key [ALLOUT] displays the window for entering outputting file name.
If the file name “PROGRAM ALL” can be used as it is, press soft key [OUTPUT] without entering file name and all part programs stored in the CNC, the currently selected path when multi-path lathe, are output to the USB memory with this name.
In case that the output file name should be changed, enter the file name to OUTPUT FILE NAME and press soft key [OUTPUT].

3.1.3 USB Memory Input Operation for Program



Select the file to be input by placing the cursor on it on the program input/output window.
Pressing soft key [INPUT] begins to read the part program in the file from USB memory to CNC.
Pressing soft key [INP.O.] displays the window for changing the reading program number, so enter the new program number and press soft key [INPUT].

When the file, in which all part program was output with file name of “PROGRAM ALL”, is read to CNC with changing the program number by soft key [INP.O.], the program number of the 1st program is changed to the new number.



In order to search the file to be input to CNC, press soft key [SEARCH] and the file searching window is displayed.

Enter the file name to be searched, and press soft key [SEARCH], then the file is searched if the file is stored in the USB memory.

3.1.4 File Format Allowed for USB Memory Input/Output

Only text files can be input to and output from the memory card. The file format described below must be observed.

(1) A file must start with "%" and "LF".
(2) A file must end with "%".
(3) For input, data read operation is skipped after the first "%" is detected until an "LF" is detected.
(4) A block must end not with a semicolon (;) but with an "LF". ("LF" is 0A of ASCII code.)

(5) When a file containing lowercase alphabetic characters, kana characters, and some special characters (such as $, ¥, and !) is input, those characteristics are ignored.
(6) ASCII code is used as input/output code, regardless of the setting parameter (ISO/EIA).
(7) Whether to output an "LF" only or an "LF, CR, CR" as an EOB can be chosen using bit 3 (NCR) of parameter No. 0100.
(8) Characters usable for a file name
Alphabetic characters : A to Z , a to z
Numeric characters : 0 to 9
Special characters : $ & # % ' ( ) - @ ^ { } ~ ` ! _

3.1.5 Notes

The current folder is treated as an input/output target.

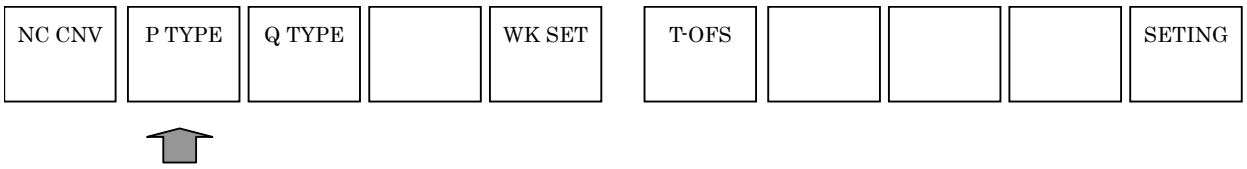
4 PROGRAM RESTART FUNCTION

When a tool is broken, or machining is to be restarted after holidays, for example, the block number or sequence number of a block from which machining is to be restarted can be specified using this function to enable machining to be restarted from the block. This function can also be used as a high-speed program check function.
Two methods of restarting are available: P type and Q type.

NOTE
To use this function, the order of movement to a program restart position needs to be set in parameter No. 7310.

4.1 RESTART METHOD

4.1.1 Soft-key [P TYPE]



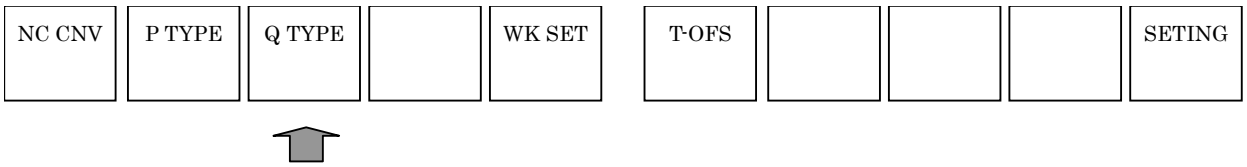
(1) When the program restart signal SRN <G006#0> turns to 0 :
Pressing the soft key [P TYPE] has no effect. (Nothing occurs.)

(2) When the program restart signal SRN <G006#0> turns to 1 :
<1> Enter one of the following in the key-in buffer:
(a) Sequence number
(b) N sequence number
(c) B block number
(d) Nxxxyyyyy (xxx: Number of repetition (3-digit)
yyyyy: Sequence number (5-digit))
(e) xxxyyyyy (xxx: Number of repetition (3-digit)
yyyyy: Sequence number (5-digit))

(*) For items (d) or (e), enter an 8-digit number. For sequence number 3 for two times, for example, enter N00200003.
<2> Press the soft key [P TYPE].
If a value entered into the key-in buffer satisfies none of the items (a) through (e), an error is indicated.
<3> The program restart screen is displayed.

NOTE
If soft key [P TYPE] is pressed without performing an automatic operation even once after the power is turned on, an emergency stop is reset, or alarms (PS094 to PS097) is reset, an alarm (PS097) is issued.

4.1.2 Soft-key [Q TYPE]



(1) When the program restart signal SRN <G006#0> turns to 0 :
Pressing the soft key [Q TYPE] has no effect. (Nothing occurs.)

(2) When the program restart signal SRN <G006#0> turns to 1 :
<1> Enter one of the following in the key-in buffer:
(a) Sequence number
(b) N sequence number
(c) B block number
(d) Nxxxyyyyy (xxx: Number of repetition (3-digit)
yyyyy: Sequence number (5-digit))
(e) xxxyyyyy (xxx: Number of repetition (3-digit)
yyyyy: Sequence number (5-digit))

(*) For items (d) or (e), enter an 8-digit number. For sequence number 3 for two times, for example, enter N00200003.
<2> Press the soft key [Q TYPE].
If a value entered into the key-in buffer satisfies none of the items (a) through (e), an error is indicated.
<3> The program restart screen is displayed.

4.2 PROGRAM RESTART IN MACHINING CYCLE (For only Series30*i*/31*i*/32*i*-A /B /0*i*-D)

Program restart can be executed on the halfway of the following machining cycles.

G code	Cycle name
G1000 - G1006	Hole machining
G1020 - G1021	Facing
G1030 - G1033 G1060 - G1071	Contouring
G1040 - G1045	Pocketing
G1050 - G1059	Grooving
G1080 - G1083	Embossing
G1100 - G1105	Hole machining (Work rotation)
G1110 - G1115	Hole machining (Tool rotation)
G1120 - G1129	Turning
G1130 - G1138	Turning grooving
G1140 - G1142	Threading
G1160 - G1168	Residual machining

(Note on using)
<1> This function is not available when a machining cycle is first block of program.
<2> B block number can be endtered in the key-in buffer.

5 CALCULATOR FUNCTION

When numeric data is input, expressions for arithmetic operations, trigonometric functions, square root calculations, and so forth can be input for calculation.

5.1 APPLICATIONS

The fixed-point format calculation function can be used for cycle input, arbitrary figure input, contour program input, setting of various data items (basis setting, measurement condition setting, calibration setting), relative coordinate preset input, and the key-in buffer(*).

(*) If the calculation result becomes an integer during key-in buffer input, no decimal point is used with it.

5.2 CALCULATION METHODS

Arithmetic operations (addition, subtraction, multiplication, and division)
Arithmetic operations are performed using the key operations described below. The result of a calculation is displayed at the cursor position for input data.

(1) Addition : 100.+200. [INPUT]
(2) Subtraction : 100.-200. [INPUT]
(3) Multiplication : 100.*200. [INPUT]
(4) Division : 100./200. [INPUT]

Example of input

```
BOTTOM THICKNESS      T= 20
CUT DEPTH OF AXIS     J=3.
BOTTOM FINISH AMOUNT  H=0.2
SIDE FINISH AMOUNT    K=0.3
FEED RATE             F=100
CLEARANCE OF AXIS     C=200 - 180
```

5.3 TRIGONOMETRIC FUNCTIONS (SINE, COSINE, TANGENT, ARCSINE, ARCCOSINE, ARCTANGENT)

Trigonometric function calculations are made using the key operations described below. The result of a calculation is displayed at the cursor position for input data.

(1) Sin : SIN(45) [INPUT]
(2) Cosine : COS(45) [INPUT]
(3) Tangent : TAN(45) [INPUT]
(4) Arcsine : ASIN(0.5) [INPUT]
(5) Arccosine : ACOS(0.5) [INPUT]
(6) Arctangent : ATAN(20,2) [INPUT]
(Note that for an arctangent calculation, a special format using two arguments is required. Enter data according to the format ATAN(a,b). arctan(a/b) is calculated.)

For a calculation, () is required at all times.

5.4 SQUARE ROOT

A square root calculation is made using the key operations described below. The result of a calculation is displayed at the cursor position for input data.

(1) Square root : SQRT(45) [INPUT]

For a calculation, () is required at all times.

5.5 EXPONENTIAL FUNCTIONS

Exponential function calculations are made using the key operations described below. The result of a calculation is displayed at the cursor position for input data.

(1) Exponential function 1 (An exponential function of e = 2.718... can be calculated.) :
EXP(4) [INPUT]
(2) Exponential function 2 ("a" raised to the power of "b" can be calculated.) :
PWR(4,3) [INPUT]

(Note that for a calculation of exponential function 2, a special format using two arguments is required. Enter data according to the format PWR(a,b). "a" raised to the power of "b" is calculated.)

For a calculation, () is required at all times.

5.6 LOGARITHMIC FUNCTIONS (COMMON LOGARITHM, NATURAL LOGARITHM)

Logarithmic function calculations are made using the key operations described below. The result of a calculation is displayed at the cursor position for input data.

(1) Common logarithm : LOG(45) [INPUT]
(2) Natural logarithm : LN(45) [INPUT]

For a calculation, () is required at all times.

5.7 ABSOLUTE VALUE

An absolute value calculation is made using the key operations described below. The result of a calculation is displayed at the cursor position for input data.

(1) Absolute value : ABS(-45) [INPUT]

For a calculation, () is required at all times.

5.8 ROUNDING

Rounding operations are performed using the key operations described below. The result of a calculation is displayed at the cursor position for input data.

(1) Rounding 1 (rounding off to an integer) :
RND(1.234) [INPUT]

(2) Rounding 2 (rounding off "a" to the decimal places specified by "b") :
RND2(1.267,0.01) [INPUT]

(Note that for a calculation of rounding 2, a special format using two arguments is required. Enter data according to the format RND(a,b). The value of "a" is rounded off to the decimal places specified by "b". As "b", do not specify a value other than 1, 0.1, 0.01, and so forth.)

For a calculation, () is required at all times.

5.9 DISCARDING

This operation discards all decimal places. A discarding operation is performed using the key operations described below. The result of a calculation is displayed at the cursor position for input data.

(1) Discarding : FIX(1.234) [INPUT]

For a calculation, () is required at all times.

5.10 CIRCLE RATIO

A circle ratio calculation is made using the key operations described below. The circle ratio 3.14... is indicated.

(1) Circle ratio : PAI [INPUT]

(*) In the case of key-in buffer input, no decimal point is assigned when an integer is produced as the result of a calculation. In cases other than key-in buffer input, the respective input formats are to be followed.

6 SHORTCUT KEY OPERATIONS

On MANUAL GUIDE *i*, almost all the operations excepting numerical data entering are done by soft-keys. However, if you are well experienced in those operations, you can operate more quickly by using other key instead of the assigned soft-key. This other key operation is called shortcut key.

Pressing <HELP> key on the MDI keyboard displays the window in which explanations of shortcut key are displayed. Into details, refer to Chapter 6, “HELP SCREEN”.

NOTE
If the small MDI key board is attached to the CNC, shortcut key operations cannot be used.

6.1 SHORTCUTS FOR VARIOUS CONFIRMATION OPERATIONS

Soft key	Shortcut key
[YES]	[INPUT]
[NO]	[CAN]

6.2 SHORTCUTS FOR RANGE SELECTION

Soft key	Shortcut key
[SELECT]	[INPUT]
[CANCEL]	[CAN]

6.3 SHORTCUTS FOR COPY OPERATION

Soft key	Shortcut key
[COPY]	[INPUT]
[CANCEL]	[CAN]

6.4 SHORTCUTS FOR CUT OPERATION

Soft key	Shortcut key
[CUT]	[INPUT]
[CANCEL]	[CAN]

6.5 SHORTCUTS FOR THE BASE SCREEN SOFT KEYS

Soft key	Shortcut key
[<]	[-]+[INPUT]
[SF1]	[1]+[INPUT]
[SF2]	[2]+[INPUT]
[SF3]	[3]+[INPUT]
[SF4]	[4]+[INPUT]

Soft key	Shortcut key
[SF5]	[5]+[INPUT]
[SF6]	[6]+[INPUT]
[SF7]	[7]+[INPUT]
[SF8]	[8]+[INPUT]
[SF9]	[9]+[INPUT]
[SF10]	[0]+[INPUT]
[>]	[.]+[INPUT]

* By setting bit 1 (SFN) of parameter No. 14703 to 1, a number for shortcut operation can be displayed under each soft key.

6.6 SHORTCUT FOR STARTING THE CYCLE CHANGE SCREEN

Soft key	Shortcut key
[ALTER]	[INPUT]

6.7 SHORTCUTS FOR THE MENU SELECTION SCREEN

Soft key	Shortcut key
[SELECT]	[INPUT] or numeral +[INPUT]
[CANCEL]	[CAN]

6.8 SHORTCUTS FOR THE REGULAR PROGRAM INSERTION SCREEN

Soft key	Shortcut key
[INSERT]	[INPUT] or numeral +[INPUT]
[CLOSE]	[CAN]

6.9 SHORTCUTS FOR THE M CODE INSERTION SCREEN

Soft key	Shortcut key
[INSERT]	[INPUT]
[CLOSE]	[CAN]

6.10 SHORTCUTS FOR THE PROGRAM LIST SCREEN

Soft key	Shortcut key
[OPEN]	[INPUT] or [9]
[CLOSE]	[CAN] or [0]
[DELETE]	[DELETE] or [3]
[EDTCOM]	[ALTER] or [4]
[NEW]	[1]
[COPY]	[2]
[SEARCH]	[5]
[M CARD]	[6]
[ALLDEL]	[7]

Soft key	Shortcut key
[SRTORD]	[8]

6.11 INVALIDATION OF SHORTCUT KEY OPERATIONS

Shortcut key operation by a numeric key can be invalidated on the program list window.
This function is enabled when the parameter No.14705#6 is set to 1.

6.11.1 Operation of Shortcut Key

(1) Invalid shortcut key operation
Shortcut key by a numeric key becomes invalid. In this case, <INPUT>, <INSERT>, <ALTER>, and <DELETE> keys on the MDI keyboard can be used.

(2) Screen affected by the invalidation of shortcut key operations
The shortcut key operation becomes invalid, when the following windows are displayed
- Program list window
- Memory card list window

This function does not affect other screens. It does not affect the sub-windows opened from the Program list and Memory card list windows, too.

6.12 SHORTCUTS FOR THE PROGRAM CREATION SCREEN

Soft key	Shortcut key
[CREATE]	[INPUT]
[CANCEL]	[EOB]

6.13 SHORTCUTS FOR THE COMMENT EDITING SCREEN

Soft key	Shortcut key
[CREATE]	[INPUT]
[CANCEL]	[EOB]

6.14 SHORTCUTS FOR THE SEARCH SCREEN

Soft key	Shortcut key
[SEARCH]	[INPUT]
[CANCEL]	[EOB]

6.15 SHORTCUTS FOR THE CYCLE INPUT SCREEN

Soft key	Shortcut key
[INSERT]	[INSERT]
[ALTER]	[ALTER]
[CANCEL]	[EOB]
[SF1]	[1]
[SF2]	[2]
[SF3]	[3]

Soft key	Shortcut key
[SF4]	[4]
[SF5]	[5]
[SF6]	[6]
[SF7]	[7]
[SF8]	[8]
[SF9]	[9]
[SF0]	[0]
[>]	[.]+[INPUT]

6.16 SHORTCUT FOR THE WORKPIECE COORDINATE SYSTEM SETTING SCREEN

Soft key	Shortcut key
[CLOSE]	[EOB]

6.17 SHORTCUT FOR THE TOOL OFFSET SETTING SCREEN

Soft key	Shortcut key
[CLOSE]	[EOB]

6.18 SHORTCUTS FOR THE REGULAR PROGRAM REGISTRATION SCREEN

Soft key	Shortcut key
[CLOSE]	[INPUT]
[NEW]	[INSERT]
[DELETE]	[DELETE]
[ALTER]	[ALTER]

6.19 SHORTCUTS FOR THE CREATION SCREEN FOR REGULAR PROGRAM REGISTRATION

Soft key	Shortcut key
[INSERT]	[INSERT]
[ADD]	[ALTER]
[CANCEL]	[EOB]

6.20 SHORTCUTS FOR THE ALTER SCREEN FOR REGULAR PROGRAM REGISTRATION

Soft key	Shortcut key
[ALTER]	[ALTER]
[CANCEL]	[EOB]

6.21 SHORTCUTS FOR THE PRESET SCREEN

Soft key	Shortcut key
[ALTER]	[ALTER]
[CANCEL]	[EOB]

6.22 SHORTCUT FOR THE MEASUREMENT RESULT SCREEN

Soft key	Shortcut key
[CLOSE]	[CAN]

6.23 SHORTCUT FOR THE MANUAL MEASUREMENT SCREEN

Soft key	Shortcut key
[CLOSE]	[EOB]

6.24 SHORTCUT FOR VARIOUS SETTING SCREENS

Soft key	Shortcut key
[CLOSE]	[EOB]

6.25 SHORTCUTS FOR THE FREE FIGURE MAIN SCREEN

Soft key	Shortcut key
[CREATE]	[INPUT]
[CANCEL]	[CAN]
[DELETE]	[DELETE]
[ALTER]	[ALTER]
[<]	[-]
[SF1]	[1]
[SF2]	[2]
[SF3]	[3]
[SF4]	[4]
[SF5]	[5]
[SF6]	[6]
[SF7]	[7]
[SF8]	[8]
[SF9]	[9]
[SF10]	[0]
[>]	[.]

6.26 SHORTCUTS FOR THE FREE FIGURE INPUT SCREEN

Soft key	Shortcut key
[OK]	[INSERT]
[CANCEL]	[EOB]

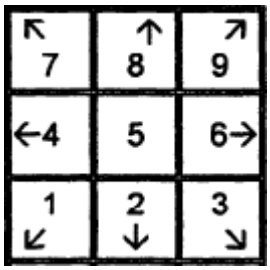
6.27 SHORTCUTS FOR THE FREE FIGURE CREATION SCREEN

Soft key	Shortcut key
[OK]	[INSERT]
[CANCEL]	[EOB]

7 INPUT OF DIRECTION ALONG THE LAYOUT OF NUMERIC KEYS

The following directions can be inputted along the layout of numeric keys of MDI unit.
This function is enabled when the parameter No.14836#4 is set to 1.

(1) Direction for line in line-insert window of arbitrary figure input.
(2) Direction for division of machining range in range selection window of arbitrary figure input.
(3) Direction for line in line-insert window of free contour cylinder blank figure input
(4) Direction for line in line-insert window of contour program input



Numeric keys of MDI unit

8 HELP SCREEN

Pressing the <HELP> key on the MDI keyboard displays the HELP window, in which explanations for shortcut key operation are displayed.
In the window, “CONTENTS” and “TOPIC” tabs are displayed.

Moving the cursor key <↑> or <↓>, place the cursor to the item of shortcut key to display the explanation.
Pressing the cursor key <→> displays the tab “TOPIC” tab and explanation of the selected shortcut key.
Pressing the cursor key <←> returns to “CONTENTS” tab.
Pressing [CLOSE] closes the HELP window.

9 OPERATION NAVIGATING FUNCTION

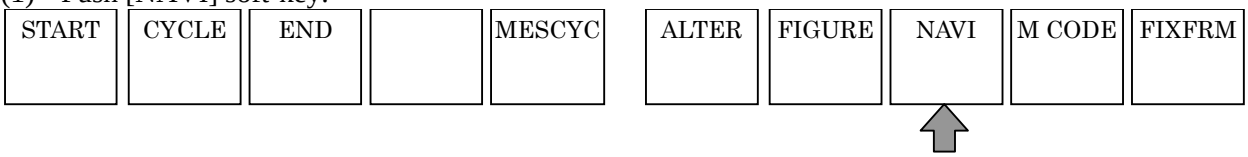
An unaccustomed operator can program MANUAL GUIDE *i* program in correct order because windows for next operation are displayed automatically.

9.1 START METHOD FOR OPERATION NAVIGATING

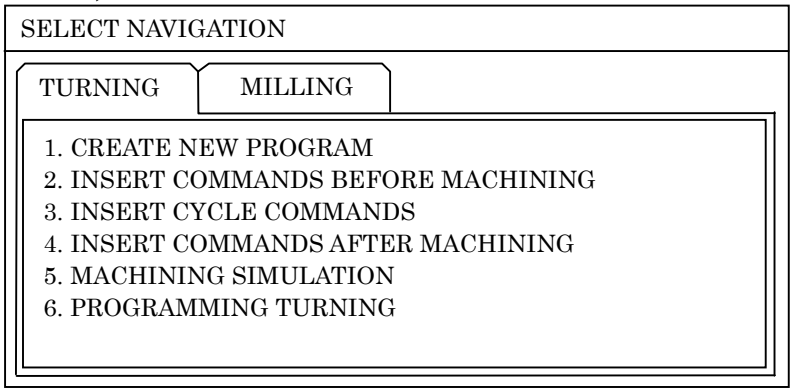
Operation navigating starts by one of following methods.
Refer to the manual published by the machine tool builder about which method is effective.

9.1.1 First Method of Start (“SELECT NAVIGATION” window is displayed)

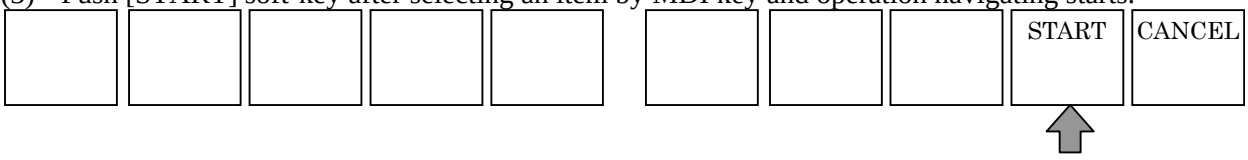
(1) Push [NAVI] soft-key.



(2) “SELECT NAVIGATION” window is displayed.
(Example of window)

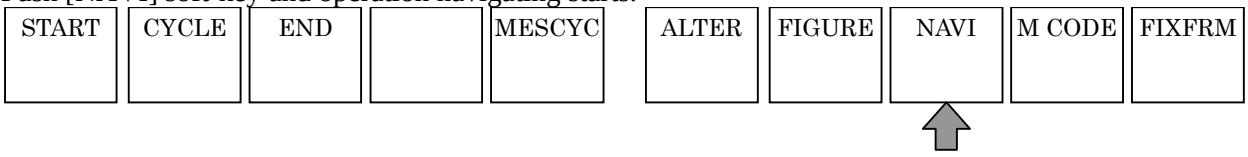


(3) Push [START] soft-key after selecting an item by MDI key and operation navigating starts.



9.1.2 Second Method of Start (“SELECT NAVIGATION” window isn’t displayed)

Push [NAVI] soft-key and operation navigating starts.



9.2 OPERATION WHILE NAVIGATION IS EFFECTIVE

When a window is closed by pushing soft-key during operation navigating, the window for next operation is displayed automatically.
And the icon of compass is displayed in the edge of left lower of MANUAL GUIDE *i* screen during operation navigating.

NOTE
1 Operate by soft-keys during operation navigating. When a window is closed by pushing other keys than soft-keys, the window for next operation is not displayed.
2 When a window isn't closed because a warning occurs by incorrect input data or other causes, the window for next operation is not displayed.
3 It is necessary to set parameter No.14853#6 to 1 to use this function.
4 Operation navigating is available in only EDIT mode or MEM mode (background editing).
The soft key [NAVI] is not displayed when one of the other modes is selected.

9.3 THE METHOD TO FINISH OPERATION NAVIGATING

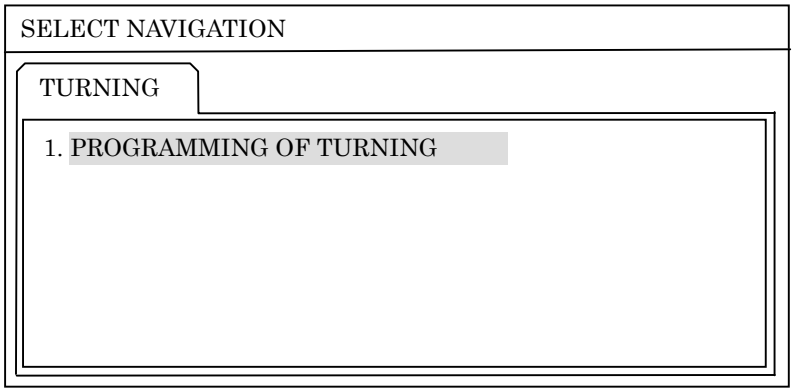
Each of the following operations is done, operation navigating is finished.

(1) Completing operation navigating
(2) Changing to the NC screen
(3) Changing mode
(4) Changing selected path
(5) Cutting power supply

9.4 OPERATION WHEN STANDARD OPERATION NAVIGATING DATA ARE USED

Some windows are automatically displayed along a basic flow of programming when the standard operation navigating data are used.

< Operation in case of lathe >
(1) Push soft-key [NAVI] and the following window is displayed.
[Displayed window]

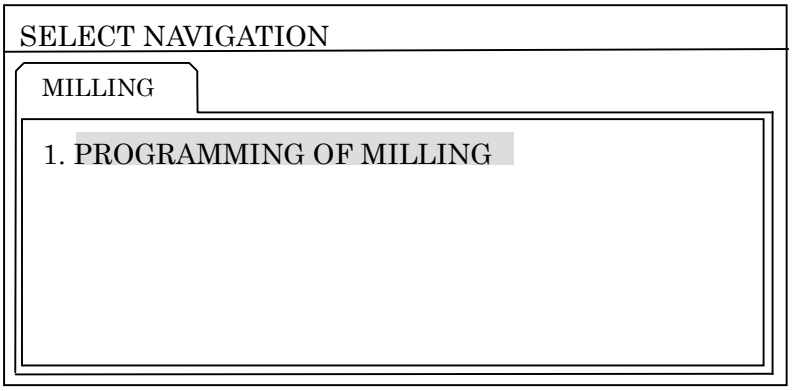


(2) Order of displaying window
After the cursor is placed on “1. PROGRAMMING OF TURNING” and soft-key [START] is pushed, windows are displayed in order of followings.

(a) CREATE NEW PROGRAM OR NEW FOLDER
(b) INSERT STARTING COMMAND FOR MILLING (The menu for blanks.)
(c) INSERT STARTING COMMAND FOR MILLING (The menu for fixed form sentences.)
(d) INSERT MILLING CYCLE
(e) INSERT MILLING FIGURE
(f) Returning to (4). After this, the procedure is repeated.
(If [CANCEL] soft-key is pushed, the repeat is ended.)

< Operation in case of machining center >

(1) Push soft-key [NAVI] and the following window is displayed.
[Displayed window]



(2) Order of displaying window
After the cursor is placed on “1. PROGRAMMING OF MILLING” and [START] soft-key is pushed, windows are displayed in order of followings.

(a) CREATE NEW PROGRAM OR NEW FOLDER
(b) INSERT STARTING COMMAND FOR MILLING (The menu for blanks.)
(c) INSERT STARTING COMMAND FOR MILLING
(The menu for fixed form sentences.)
(d) INSERT MILLING CYCLE
(e) INSERT MILLING FIGURE
(f) Returning to (4). After this, the procedure is repeated.
(If [CANCEL] soft-key is pushed, the repeat is ended.)

10 ADVANCED GUIDANCE FUNCTION (for only Series 30*i*/31*i*/32*i*-B)

The following functions are included in MANUAL GUIDE *i* advanced guidance function.

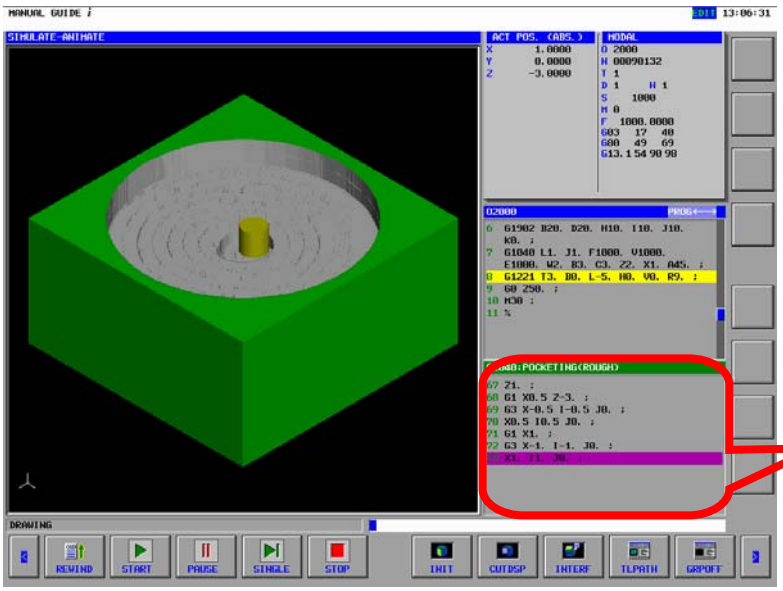
(1) Decomposed display
(2) Input data check by simulation
(3) Help-window function
(4) Cooperation with animated software with PANEL *i*

The following option is necessary to use MANUAL GUIDE *i* advanced guidance function.

- MANUAL GUIDE *i* advanced guidance function : S774

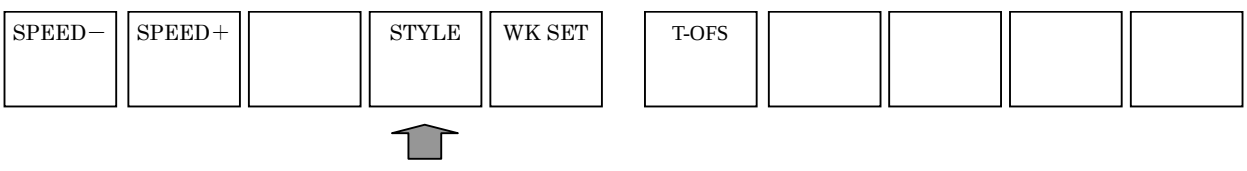
10.1 DECOMPOSED CYCLE DISPLAY

On machining simulation screen, this function can display the program which is decomposed to move commands during machining simulation. By this function, the operator can check details of the movement of machining cycle easily.



10.1.1 Method of Displaying the Decomposed Cycle Display Screen

Press soft key [STYLE] on machining simulation screen.



10.1.2 Condition

It is necessary for this function to satisfy the following conditions.

(1) Parameter No.27313#0 = 0
(2) Foreground operation is not executed.
(3) A new program can be created.
(4) Capacity of the unused program memory is enough.

10.1.3 Restriction

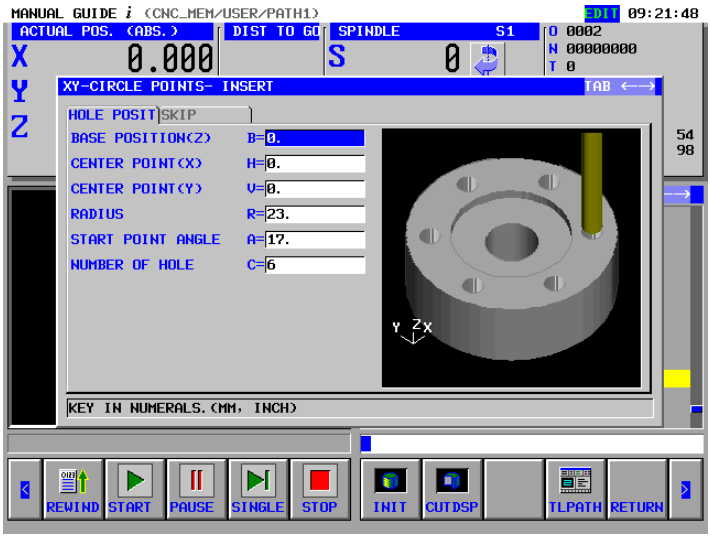
(1) When the decomposed cycle display is executed during foreground operation, warning is displayed. However, foreground operation and machining simulation are not interrupted.

(2) If a new program can not be created on machining simulation of the decomposed cycle display, warning is displayed, when machining cycle is executed.

(3) If capacity of the unused program memory is not enough on machining simulation of the decomposed cycle display, warning is displayed, when machining cycle is executed.

10.2 INPUT DATA CHECK BY SIMULATION

The operation of a cycle that is being edited can be checked by machining simulation (animation / tool path drawing) while cycle data input screen is displayed.



10.2.1 Condition

It is possible to use this function on the following conditions.

(1) MANUAL GUDIE *i* screen can be operated on only first path. Concretely, it is the case that one of following conditions is met.
<1> "MANUAL GUIDE *i* Multi Path function" is invalid.
<2> Parameter No.14864 = 1 (MANUAL GUIDE *i* is valid on only first path.), parameter No.27401#0 - #1 = 0 and parameter No.27402#0 - #1 = 0.(Turret 2 and 3 are unavailable.)
(2) A program can be newly made.

NOTE
In case a program can't be made (ex. The number of registered program is maximum), a warning is generated when this function is starting up.

(3) Available program memory size > program size of the machining simulation execution range

NOTE
If available program memory size is not enough, a warning is generated when this function is starting up.

(4) Edit mode is selected.
(5) While multi part program editing is not used.
(6) Program window size is quarter of screen.
(7) Parameter No.27313#2 = 0

10.2.2 Operational Procedure

(1) Push [CHECK] soft-key while cycle data input screen is displayed. Then, after a message "PREPARING" is displayed on the part of guidance, soft-keys for machining simulation ([REWIND], [START] and so on) are displayed.

(2) Next, push [START] soft-key. Machining simulation is started and input data can be checked. And, the following function for machining simulation can be used in this window.
<1> Rewind (Note : This is rewind function for execution pointer of machining simulation. Editing pointer is not rewinded.)
<2> Machining simulation start, stop and pause
<3> Cut display of workpiece
<4> Rotation, scaling and movement of drawing coordinate

10.2.3 Available Screen

(1) Machining cycle data input screen
<1> Machining type
<2> Fixed form figure
(2) Measurement cycle data input screen
(3) Coordinate system conversion cycle
(4) Tilted working plan indexing
(5) Mirror
(6) Rotation
(7) Scaling
(8) Figure copy

10.2.4 Execution Range of Input Data Check by Simulation

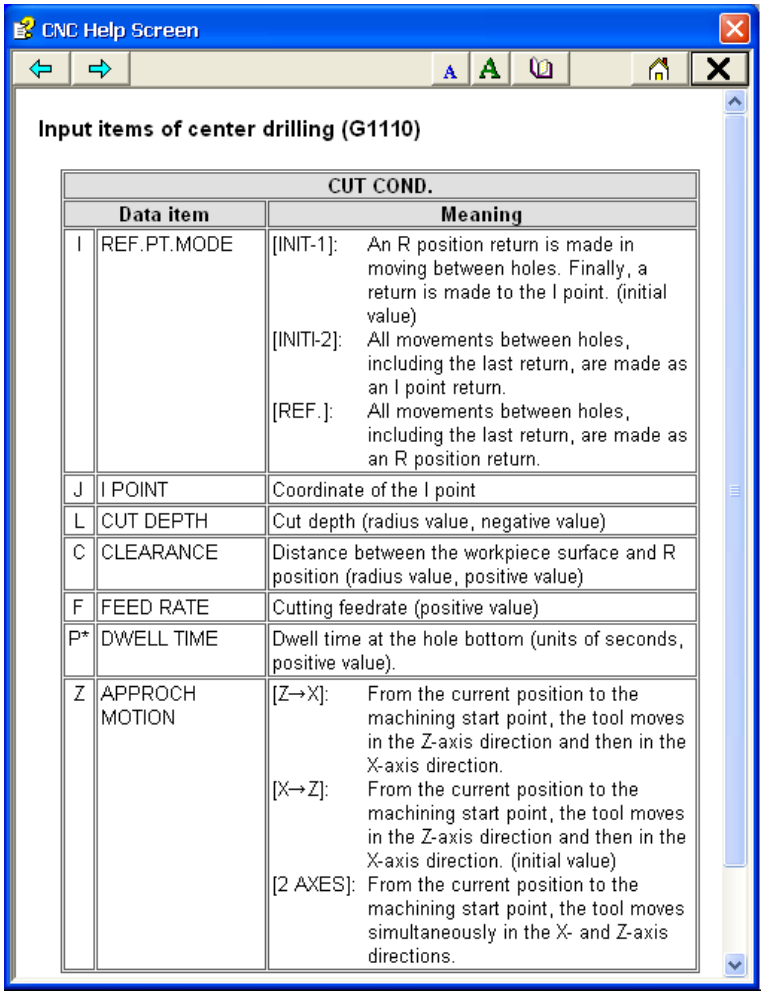
Execution range of input data check by simulation is as following table.

The editing cycle			Execution range
Classification 1	**Classification 2**	**Classification 3**	
Machining cycle	Command of machining type	Possible to execute without figure command	Top of program - The editing cycle
		Impossible to execute without figure command	Top of program - Figure command corresponding the editing cycle
	Command of figure (Fixed form figure)		Top of program - The editing cycle
Measurement cycle			Top of program - The editing cycle
Coordinate system conversion cycle			Top of program - Cancel command of the mode
Tilted working plane indexing			
Mirror			
Scaling			
Rotation			
Figure copy			

10.3 HELP WINDOW ACCORDING TO EACH SCREEN

10.3.1 Outline

Help window according to each screen is displayed by pressing <HELP> key while displaying the cycle input window.



10.3.2 Condition

It is possible to use this function on the following conditions.

(1) Available display unit
PANEL *i*

(2) Necessary parameter setting
Please set parameter HLP(No.27313#3) to 1.

(3) Necessary software/data
Please install the following items in PANEL *i*.
- CNC screen display function (software : A02B-0207-K775) which supports this function.
- Displayed data in help window.

(4) Available languages
- Japanese
- English
- German

10.3.3 Restriction

(1) Even if <HELP> key is pushed while displaying input window of cycle which is not supported, Help window is not displayed.

(2) Even if <HELP> key is pushed while selecting the language which is not supported, Help window is not displayed.

10.4 COOPERATION WITH ANIMATED SOFTWARE THAT IS OPERATED WITH PANEL *i*

10.4.1 Outline

It is possible to change from MANUAL GUIDE *i* screen to the screen of commercial animated software by pressing the soft-key.

10.4.2 Condition

It is possible to use this function on the following conditions.

(1) Available display unit
PANEL *i*

(2) Necessary parameter
Please set the following parameters.
Parameter to make this function effective
- Parameter No.27313#4 = 1
- Parameter No.14489

(3) Necessary software/data

Please install the following items in PANEL *i*.

- CNC screen display function (software : A02B-0207-K775) which supports this function.

(4) Necessary setting of CNC screen display function

Please set the execution file of activating application to the following registries.

< Registry >

Preservation of setting of CNC screen display function	Registry key
Common	HKEY_CURRENT_USER¥Software¥FANUC¥CNCScrn¥MGiAnimeLink¥
Every connection destination (Node)	HKEY_CURRENT_USER¥Software¥FANUC¥CNCScrn¥NodeX¥MGiAnimeLink¥

（NodeX : Node0～Node7）

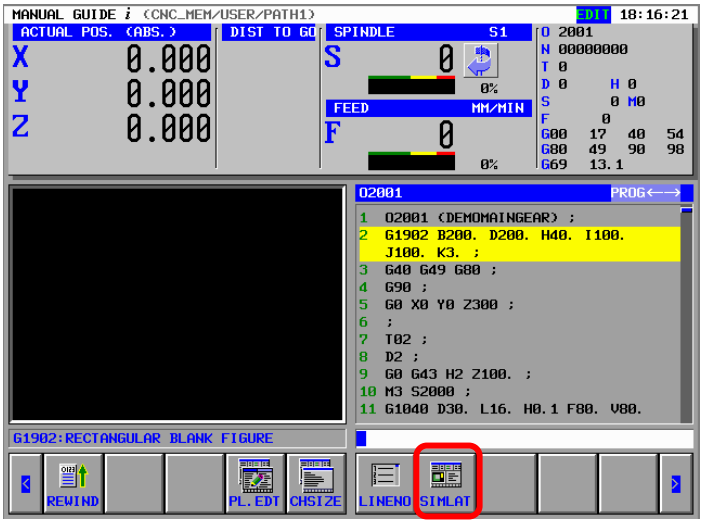
< Set item >

Name	Type	Data
ProgramName	REG_SZ	Execution file of application

In the case that the argument is specified for the execution file of the application, "%d" in the character string is replaced as the node number of CNC. When you want to represent "%" in itself, please describe as "%%".

10.4.3 Operation

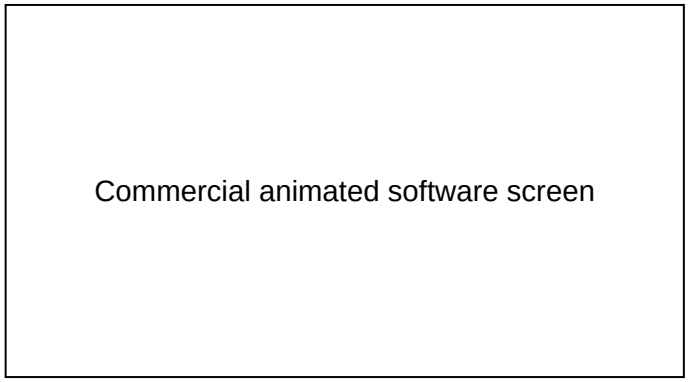
Commercial animated software is displayed by pressing [SIMLAT] soft-key. A machining simulation screen of MANUAL GUIDE *i* is not displayed.



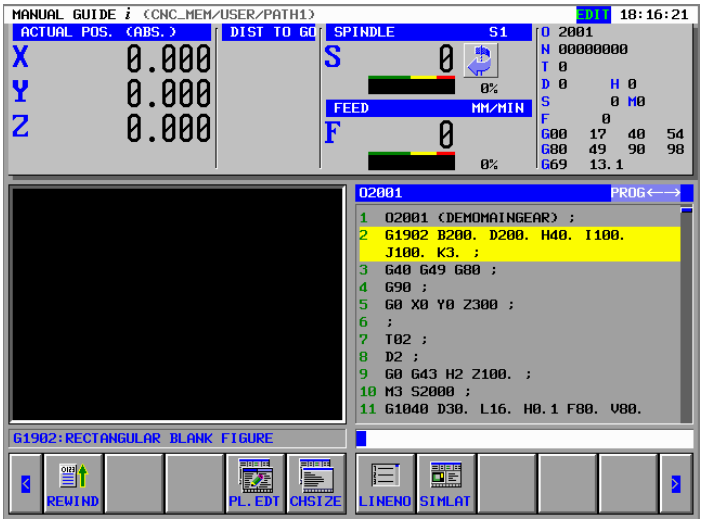
By pressing [SIMLAT] soft-key

Commercial animated software screen

MANUAL GUIDE *i* screen is displayed by closing commercial animated software.



By closing commercial animated software



10.4.4 Restriction

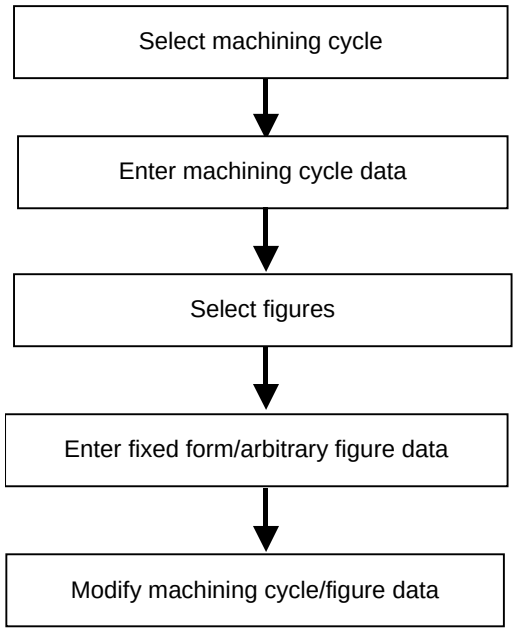
(1) Commercial animated software has to access the tape storage memory of CNC, and acquire to the machining program, because MANUAL GUIDE *i* does not output the machining program to commercial animated software side.

(2) When the screen of commercial animated software shuts, MANUAL GUIDE *i* screen is displayed.

IV. MACHINING CYCLE

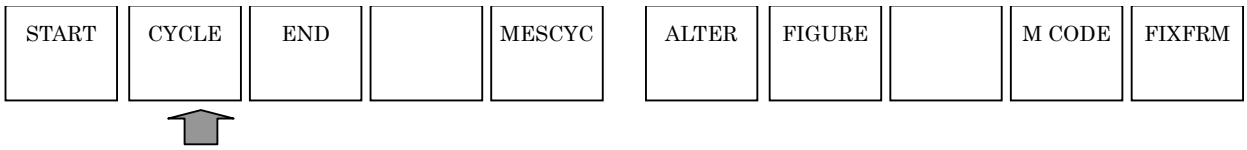
1 OVERVIEW

The following shows the flow of programming a machining cycle by using MANUAL GUIDE *i*:

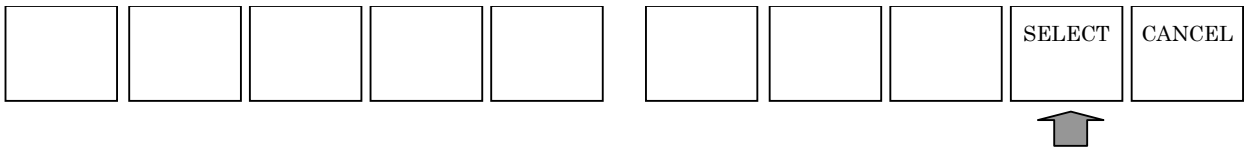


1.1 SELECTING A MACHINING CYCLE TYPE

Press the leftmost soft key [<] or rightmost soft key [>] several times to display the machining cycle soft key menu shown below.
Soft keys for milling are provided.



Press soft key [CYCLE] to display the machining cycle menu. The following soft keys are displayed:



Select a machining cycle using the cursor keys.
(Select a machining cycle group, using the cursor keys <←> and <→>, and then select a machining cycle type from the machining cycle group, using the cursor keys <↑> and <↓>.
Pressing soft key[SELECT] displays the data entry window for the selected machining cycle type.
(After a menu number is inputted, it is possible that machining cycle is selected by pressing <INPUT> key.)

NOTE
The scroll bar displayed on the right edge of the machining cycle menu window indicates the approximate position of the cursor throughout the machining cycle menu.
If the scroll bar marker is on the middle of the scroll bar, therefore, it is likely that part of the machining cycle menu is hidden behind the window. If this is the case, pressing the cursor key <↓> makes the hidden part visible.

1.2 ENTERING MACHINING CYCLE DATA

The machining cycle data entry window is divided into two sections, one section for cutting conditions and the other for detailed data.



Pressing the cursor key <←> or <→> switches between the two sections and changes the displayed tab. The character in the selected tab is displayed in blue.

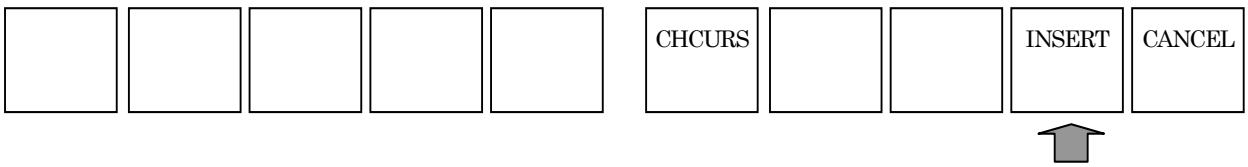
(1) Selecting an input item
Pressing the cursor key <↑> or <↓> can place the cursor on a desired data input item.

Example of the machining cycle data entry window

```
BOTTOM THICKNESS      T= 20
CUT DEPTH OF AXIS     J=3.
BOTTOM FINISH AMOUNT  H=0.2
SIDE FINISH AMOUNT    K=0.3
FEED RATE             F=100
CLEARANCE OF AXIS     C=0.2
```

(2) Entering data
There are two types of data input items. One type includes those entered as numbers, and the other type, those selected from a menu displayed using a soft key. For the first type, the message "KEY IN NUMERALS." is displayed in the lower section of the window. For the second type, the message "SELECT SOFT KEY" is displayed.
Necessary data can be entered by referencing a comment for the data input item and a guide chart displayed in the window.

(3) Inserting entered data into the program
The following soft keys are displayed for the machining cycle data window.



When completing the entry of necessary data, press soft key [INSERT] to insert the entered data (machining cycle block) into the program.

NOTE

1 Among the data item displayed in the cutting condition window, there are data should be danger if they are set automatically such as cutting amount or feedrate. These data should be entered by an operator always. Other data are set automatically.

2 Data items displayed in the detail window are usually set automatically by duplicating the value entered last time. So, check these automatically set value and modify them if necessary.

3 Executing machining cycle requires a machining cycle block and a figure block, which will be explained later.
Once a machining cycle block has been entered, therefore, be sure to enter a figure block in succession.

4 A data item with an asterisk displayed at its right end has a default value. You need to enter no data if you accept the default value.

5 The [CHCURS] soft key is displayed in data entry windows for machining cycle, figure, and contour programs.
Pressing this soft key can select whether the cursor keys <←> and <→> are to be used for tab switching or cursor movement within the entered data. If tab switching is selected, "Tab ←→" is displayed in the upper right section of the window. If cursor movement is selected, "Character ←→" is displayed.

6 Initial values automatically set in the data entry window
In the data entry window for a blank figure block or tool definition block, previously entered data is automatically set as initial values.
Therefore, the operator must enter data only once for the first time; in the second and subsequent operations, the entered data is then set as initial values.
Because figure data depends on the drawing data, initial values are not set automatically in the figure data entry window.

7 Automatically copied tool definition block
When the tool database function is not used, a tool definition block in the drawing definition menu must be specified before a milling cycle or turning cycle is specified.
Data of the tool definition block can then be copied and automatically set as initial values of tool condition data for each cycle.
The operator therefore needs to enter only a tool definition block; the operator need not enter tool condition data in subsequent cycle menus.
The following tool definition block data is copied:
- Milling tool
  (1) Tool diameter (D)

1.3 SELECTING FIGURES

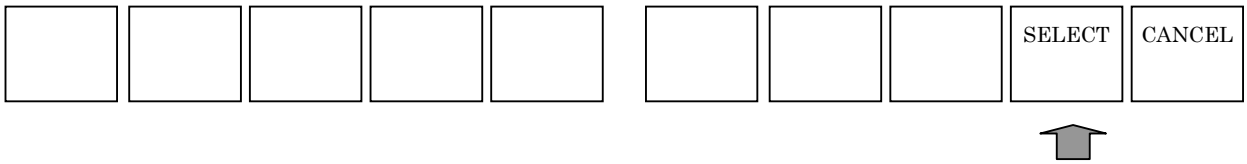
When a machining cycle block is entered, a machining cycle figure menu is displayed.
The menu shows only figures that can be used for the previously entered machining cycle.

The cycle figure menu window is divided into two sections, one section for selecting figure and the other for subprogram.

(Example of tab for pocketing)



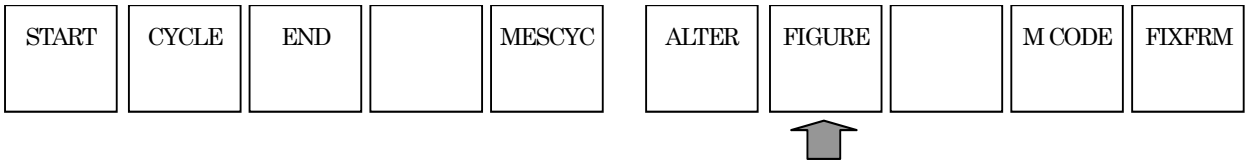
(1) Selecting a figure from the figure selection window
When the figure selection window is selected, the characters of the tab are displayed in blue. Select a desired figure by moving the cursor with the cursor keys <↑> and <↓>.



Pressing soft key [SELECT] displays a data entry window for the selected figure. (Figures for machining cycles are classified into the following two major types.)
Fixed form figure :
Fixed form figure is predefined figure such as square and circle, and can be specified by entering minimum data such as side length or circle radius.
Free form figure :
Free form figure is specified by entering figures by line or arc along the figures written on a blue print one by one.

(2) Selecting a figure from the subprogram window
Pressing the cursor key <→> displays the subprogram window with its tab characters displayed in blue.
The subprogram window shows a list of the subprogram numbers and names of figure blocks registered as subprograms in advance.
Select a subprogram to be used by moving the cursor to the subprogram with the cursor keys <↑> and <↓>.
Pressing soft key [SELECT] create the block for calling the selected subprogram as “M98 Pxxxx ;” in the actual part program.
(It is specified by program number to be displayed in the subprogram menu or not.
For milling machining cycle figure, the minimum and maximum program number of the programs should be used as the subprogram are set in the parameter No.14722 (MFIGSNO) and No.14723 (MFIGENO).
Subprogram whose number is within these range is displayed in the subprogram menu for milling cycle figure.)

(3) Entering another figure block in succession
To enter an additional figure block after entering machining cycle and figure block data, press [FIGURE] among the displayed soft keys shown below to display a figure menu, then select a necessary figure.
Different from the menu displayed for entering machining cycle data, the figure menu displayed here shows all menu items.



NOTE
The scroll bar displayed on the right edge of the figure menu window indicates the approximate position of the cursor throughout the figure menu.
If the scroll bar marker is on the middle of the scroll bar, therefore, it is likely that part of the figure menu is hidden behind the window. If this is the case, pressing the cursor key <↓> makes the hidden part visible.

1.4 ENTERING FIXED FORM FIGURE DATA

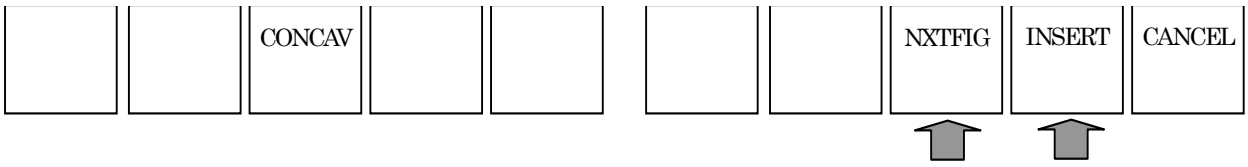
Selecting the fixed form figure displays the data entry window for machining cycle fixed form data entry window.

(1) Selecting a tab
A data entry window for hole positions, fixed form figures (milling) may be extended over two pages as necessary.
Pressing the cursor key <←> or <→> switches between the two pages. It also changes the displayed tab.

(2) Selecting an input item
Pressing the cursor key <↑> or <↓> can place the cursor on a desired data input item.

(3) Entering data
There are two types of data input items. One type includes those entered as numbers, and the other type, those selected from a menu displayed using a soft key. For the first type, the message "KEY IN NUMERALS." is displayed in the lower section of the window. For the second type, the message "SELECT SOFT KEY" is displayed.
Necessary data can be entered by referencing a comment for the data input item and a guide chart displayed in the window.

(4) Inserting the entered data into the program
The following soft keys are displayed for the fixed form figure data entry windows. For some data items, soft keys for selection are displayed.



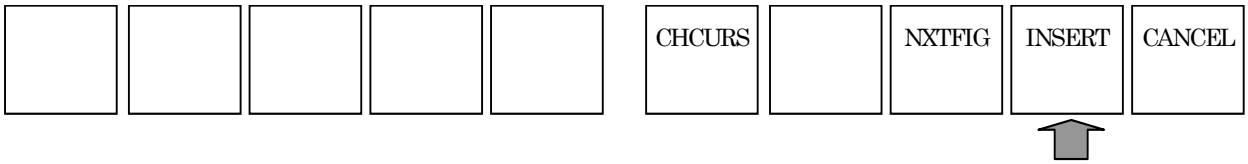
When completing the entry of necessary data, press soft key [INSERT].
Soft key [INSERT] inserts the entered figure block into the program and displays the program window.

(5) Entering another figure in succession
When another figure is to be entered in succession, press soft key [NXTFIG]. The entered figure block is then inserted to the program, and the figure selection menu is displayed again.

NOTE
1 More than two figures can be entered in succession for a single machining cycle type.
Machining cycle is executed for each of the specified figures sequentially.
2 An ordinary ISO code block can be entered between machining cycle and figure blocks. If this is done, no machining operation takes place in the machining cycle block. Instead, the ISO code block entered before the figure block is executed. Machining cycle is executed only in the figure block after the ISO code block has been executed.

(6) Outputting entered data as a subprogram
When the entered figure data is to be output as a subprogram, follow the steps below.
(Before a fixed form figure data can be output as a subprogram, bit 1 (SBP) of parameter No. 14851 must be set to 1.)

<1> After completing the entry of necessary data, press soft key [INSERT].



<2> A screen for selecting fixed form figure data as a subprogram is displayed as shown below.
When creating fixed form figure data as a subprogram, select "CREATE AS SUB PROGRAM".

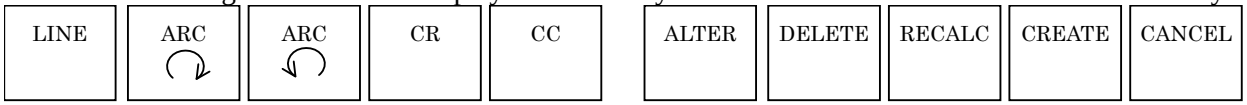
METHOD OF FREE FIGURE CREATION
○ INSERT IN CURRENT PROGRAM
◎ CREATE AS SUB PROGRAM
SELECT CREATIVE METHOD.

<3> If a comment is entered, the entered comment is added to the M98P**** block of the main program and the subprogram at the time of program output.
<4> Pressing the [OK] soft key outputs a created fixed form figure program as a subprogram to the machining program.

1.5 ENTERING ARBITRARY FIGURE DATA

For machining cycle, an arbitrary figure consisting of circles and straight lines can be entered by performing automatic calculation on entered data to obtain the end point of each figure even if no end point is specified on the drawing.

(1) Entering start points
Be sure to enter a start point for the beginning of an arbitrary figure.
After entering necessary data, press soft key [OK].
(2) Selecting a figure
A menu for figure selection is displayed on soft keys as shown below. Select whatever is necessary.



(3) Entering figure data
When a figure is selected, a figure data entry window appears. In this window, enter the figure data specified on the drawing.
There are two types of data input items. One type includes those entered as numbers, and the other type, those selected from a menu displayed using a soft key. For the first type, the message "KEY IN NUMERALS." is displayed in the lower section of the window. For the second type, the message "SELECT SOFT KEY" is displayed.
After entering the necessary data, press soft key [OK].

NOTE
As figures are entered, they are drawn in the figure entry window.
In the upper section of the window, symbols for entered figures are displayed sequentially, starting at the left. The cursor key <←> or <→> can be used to select an entered figure; the lines that correspond to the selected figure will be displayed in yellow.

(4) Changing figure data
Place the cursor on the figure that contains the figure data you want to change, and press soft key [ALTER]. The figure data entry window appears. In this window, place the cursor on the data input item you want to change, and enter new data.
If you want to delete data, press the <CAN> and <INPUT> keys in the stated sequence.
Pressing soft key [OK] closes the figure data entry window. Press soft key [RECALC] to perform calculation for all entered figures, and check drawn figures to see if the intended changes have been made.

(5) Inserting new figures
After placing the cursor on the figure that precedes immediately the place where your want to add a figure, select the figure to be added from the soft key menu.
In the figure data entry window that appears for the new figure, enter figure data necessary for the figure, and press soft key [OK].
Press soft key [RECALC] to perform calculation for all entered figures, and check drawn figures to see if the intended additions have been made.

(6) Deleting unnecessary figures
After placing the cursor on the figure you want to delete, press soft key [DELETE].
Press soft key [RECALC] to perform calculation for all entered figures, and check drawn figures to see if the intended deletions have been made.

(7) Changing figures
If you want to change the type of an entered figure, delete it and enter a new figure.

(8) Writing entered arbitrary figures to machining programs
After all necessary figures have been entered, they can be written to machining programs.
There are two methods for writing figures to machining programs. The first method writes directly to the currently selected machining program. The second method writes to a newly created subprogram.
After necessary data has been entered, pressing soft key [CREATE] displays a window for selecting one of the creation methods. Select whichever creation method you want, using the cursor keys <↑> and <↓> .
To write to the machining program that has been selected, simply press soft key [OK].
To create a subprogram, enter a new subprogram number to the subprogram number item, and then press the <INPUT> key. You can enter character strings as a comment if necessary. The comment is displayed as a program name in the list window.
Pressing soft key [OK] now creates a subprogram and writes figures to the subprogram.

NOTE
See Chapter 5, "ENTERING ARBITRARY FIGURES," for descriptions about data for arbitrary figures for machining cycle.

1.6 CHANGING MACHINING CYCLE AND FIGURE DATA

To change the contents of a machining cycle and fixed form figure, place the cursor on the related block, and press soft key [ALTER].
To change the contents of an arbitrary figure and contour program, place the cursor at the first of the related blocks, and press soft key [ALTER].
In any case, a data entry window for these items appears. After making the necessary data changes, press soft key [CREATE] in the same manner as for new entry. New figures are written to the machining program.

1.7 NOTES ON CREATING PROGRAMS

This section provides notes on creating programs. Read the notes before creating a program.

(1) Programmed commands required before machining cycle input
  (a) When using a tool database with a CNC of the M series, be sure to enter a D command (offset number).
  (b) Be sure to enter a spindle rotation command. At the same time, specify whether to enable or disable constant surface speed control.
  (c) No feedrate command needs to be specified, because a feedrate command is separately entered as a machining cycle command. However, specify a command for feed per revolution or feed per minute as required. For example, operator know-how on executing milling in pocketing by feed per revolution can be utilized.
  (d) Enter an M code command for switching between the spindle and C-axis, and a command for C-axis reference position return as required.
  (e) Enter M codes such as for coolant ON/OFF as required.
  (f) Upon completion of machining cycle, the tool always returns to the position before the start of the machining cycle. By setting bit 7 (ESC) of parameter No. 27002 to 1, the tool can be prevented from returning to the position before the start of the machining cycle.
  (g) For the machining cycle mentioned above, a similar command is repeatedly entered. So, such a command can be entered easily by registering a fixed form sentence beforehand.

(2) Machining cycle
  (a) Specify an I point coordinate not as a distance from point R but as a coordinate value.
  (b) Figure data should be entered as a subprogram for utilization in roughing, finishing, and chamfering. This eliminates the need to enter figure data each time.
  (c) In machining of a projected portion (island figure), efficient machining can be achieved by using pocketing with an island instead of using contouring. In this case, specify a blank as the outer wall of a pocket.
  (d) In contouring (side facing), enter a start point figure and end point figure so that both figures contact each other. This means that by starting not from a corner of a figure but from an intermediate point on a straight line, undercutting due to approaching the figure and retraction can be eliminated.

2 MILLING

MANUAL GUIDE *i* supports the following types of milling.

(1) Hole machining

Hole machining (with the tool rotated)			
	Machining command block M series (Note 1)	G1000	Center drilling
		G1001	Drilling
		G1002	Tapping
		G1003	Reaming
		G1004	Boring
		G1005	Fine boring
		G1006	Back boring

Figure for hole machining			
	Hole position block (XY plane)	G1210	Random points
		G1211	Linear points (same interval)
		G1212	Linear points (different interval)
		G1213	Grid points
		G1214	Rectangle points
		G1215	Circle points
		G1216	Arc points (same interval)
		G1217	Arc points (different interval)
	Hole position block (YZ plane)	G1310	Random points
		G1311	Linear points (same interval)
		G1312	Linear points (different interval)
		G1313	Grid points
		G1314	Rectangle points
		G1315	Circle points
		G1316	Arc points (same interval)
		G1317	Arc points (different interval)
	Hole position block (B-axis hole, cylindrical surface(YB),)	G1777	Arc points
		G1778	Random points

NOTE
1 Hole machining cycles for milling machines (M series) are provided. When using a milling machine, set bit 1 (MC1) of parameter No. 27000 to 0.

(2) Facing

Facing			
	Machining command block	G1020	Roughing
		G1021	Finishing

Figure for facing			
	Fixed-figure block (XY plane)	G1220	Rectangle
		G1221	Circle
		G1222	Track
		G1225	Polygon
	Arbitrary-figure block (XY plane)	G1200	Start point
		G1201	Straight line
		G1202	Arc (CW)
		G1203	Arc (CCW)
		G1204	Chamfering
		G1205	Corner rounding
		G1206	End
	Fixed-figure block (YZ plane)	G1320	Rectangle
		G1321	Circle
		G1322	Track
		G1325	Polygon
	Arbitrary-figure block (YZ plane)	G1300	Start point
		G1301	Straight line
		G1302	Arc (CW)
		G1303	Arc (CCW)
		G1304	Chamfering
		G1305	Corner rounding
		G1306	End

(3) Contouring

Contouring			
	Machining command block	G1060	Outer Wall Roughing
		G1061	Outer Wall Bottom finishing
		G1062	Outer Wall Side finishing
		G1063	Outer Wall Chamfering
		G1064	Inner Wall Roughing
		G1065	Inner Wall Bottom finishing
		G1066	Inner Wall Side finishing
		G1067	Inner Wall Chamfering
		G1068	Partial Roughing
		G1069	Partial Bottom finishing
		G1070	Partial Side finishing
		G1071	Partial Chamfering

Figure for contouring			
	Fixed-figure block (XY plane)	G1220	Rectangle
		G1221	Circle
		G1222	Track
		G1225	Polygon
	Arbitrary-figure block (XY plane)	G1200	Start point
		G1201	Straight line
		G1202	Arc (CW)
		G1203	Arc (CCW)
		G1204	Chamfering
		G1205	Corner rounding
		G1206	End
	Fixed-figure block (YZ plane)	G1320	Rectangle
		G1321	Circle
		G1322	Track
		G1325	Polygon
	Arbitrary-figure block (YZ plane)	G1300	Start point
		G1301	Straight line
		G1302	Arc (CW)
		G1303	Arc (CCW)
		G1304	Chamfering
		G1305	Corner rounding
		G1306	End

(4) Emboss machining

Emboss machining			
	Machining command block	G1080	Roughing
		G1081	Bottom finishing
		G1082	Side finishing
		G1083	Chamfering

Figure for emboss machining			
	Arbitrary-figure block (XY plane)	G1200	Start point
		G1201	Straight line
		G1202	Arc (CW)
		G1203	Arc (CCW)
		G1204	Chamfering
		G1205	Corner rounding
		G1206	End
	Arbitrary-figure block (YZ plane)	G1300	Start point
		G1301	Straight line
		G1302	Arc (CW)
		G1303	Arc (CCW)
		G1304	Chamfering
		G1305	Corner rounding
		G1306	End

(5) Pocketing

Pocketing			
	Machining command block	G1040	Roughing
		G1041	Bottom finishing
		G1042	Side finishing
		G1043	Chamfering

Figure for pocketing			
	Fixed-figure block (XY plane)	G1220	Rectangle
		G1221	Circle
		G1222	Track
		G1225	Polygon
	Arbitrary-figure block (XY plane)	G1200	Start point
		G1201	Straight line
		G1202	Arc (CW)
		G1203	Arc (CCW)
		G1204	Chamfering
		G1205	Corner rounding
		G1206	End
	Fixed-figure block (YZ plane)	G1320	Rectangle
		G1321	Circle
		G1322	Track
		G1325	Polygon
	Arbitrary-figure block (YZ plane)	G1300	Start point
		G1301	Straight line
		G1302	Arc (CW)
		G1303	Arc (CCW)
		G1304	Chamfering
		G1305	Corner rounding
		G1306	End

(6) Grooving

Grooving			
	Machining command block	G1050	Roughing
		G1051	Bottom finishing
		G1052	Side finishing
		G1053	Chamfering
		G1054	Roughing (cylindrical surface, B/Y axis)

Figure for grooving			
	Fixed-figure block (XY plane)	G1220	Rectangle
		G1221	Circle
		G1222	Track
		G1223	Radial groove
		G1224	Straight line
		G1225	Polygon
	Arbitrary-figure block (XY plane)	G1200	Start point
		G1201	Straight line
		G1202	Arc (CW)
		G1203	Arc (CCW)
		G1204	Chamfering
		G1205	Corner rounding
		G1206	End
	Fixed-figure block (YZ plane)	G1320	Rectangle
		G1321	Circle
		G1322	Track
		G1323	Radial groove
		G1325	Polygon
	Arbitrary-figure block (YZ plane)	G1300	Start point
		G1301	Straight line
		G1302	Arc (CW)
		G1303	Arc (CCW)
		G1304	Chamfering
		G1305	Corner rounding
		G1306	End
	Fixed-figure block (Cylindrical surface, B/Y axis)	G1775	B axis groove
		G1776	X axis groove

Notes on the CNC for milling

1 As hole machining enabled when a CNC for milling is used, MANUAL GUIDE *i* provides hole machining (tool rotation: M series) only.
2 In the input data item of “CUT ANGLE” of Pocketing Rough (G1040) and Bottom Finish (G1041), when the machine is 2 axes of Maximum simultaneously controlled axes, please sure not to set the data.
(If the data is set, the alarm (PS0015) occurred during execution of Cycles.)
3 On the CNC for milling, when cycle machining is specified, a switchover to G90 (absolute command) occurs internally. Therefore, G91 must be specified if incremental commands are used after cycle machining.

NOTE

1 When MANUAL GUIDE *i* is used to enter cycle machining data, combinations of a machining type block and a figure block must be entered in succession. More than one figure block can be entered for one machining command block. For arbitrary figures, however, more than one set of arbitrary figures can be entered if one set is assumed to consist of up to 90 figures enclosed between start and end points.

2 A data item whose address is indicated with [*] in the following tables is automatically set with a typical value unless a value is entered for it. You need to enter no data if you accept the typical value. On the screen, [*] is displayed in the right end of the data item.

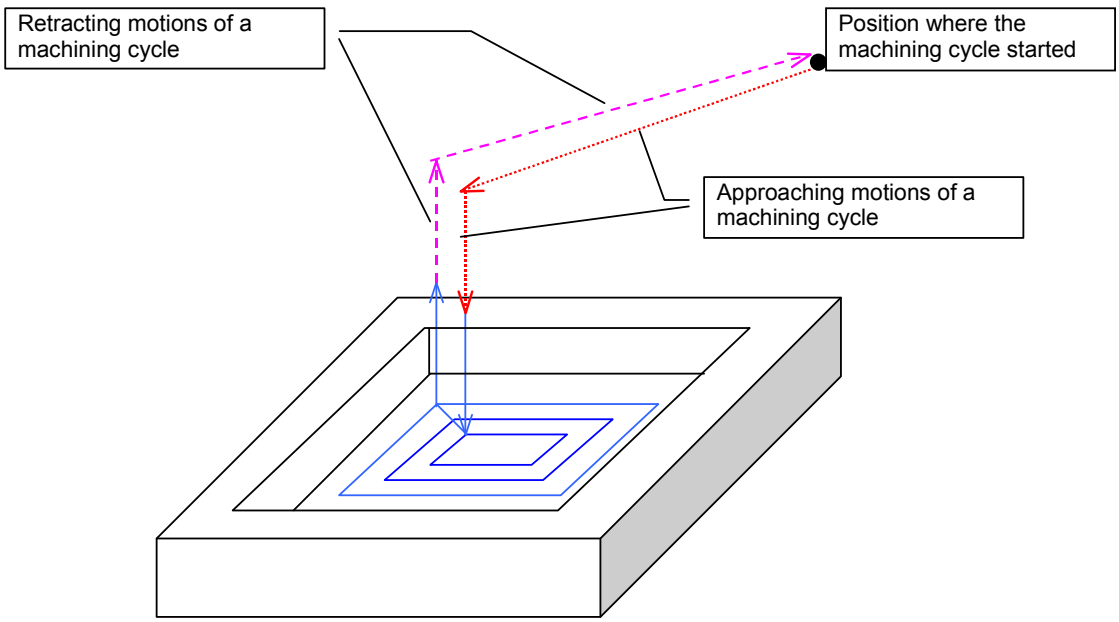
3 In some of the input data items, the system sets the last inputted data as an initial data. As to these input data items, the mark of “(COPY)” was described.

4 In the input data item of “APPROCH MOTION”, when the machine is 2 axes of Maximum simultaneously controlled axes, please sure not to specify the [3 AXES] soft-key. (If [3 AXES] is specified, the alarm (PS0015) occurred during execution of Cycles.)

5 When a cycle ends, the tool always returns to the position where the cycle started. (If bit 7 (ESC) of parameter No. 27002 is set to 1, it is possible not to cause the tool to return to the start position when a cycle ends.)

Remarks) Cycle retract motions

In case of bit 7 (ESC) of parameter No.27002 = 0, Retracting motions indicated as broken lines in the following drawing will be outputted. The order of motion axis will be opposite to the approached motions



In case of carrying out plural figures machining by one G4 digit cycle, such like drilling, this retracting motions will be available at the last figure machining.

2.1 HOLE MACHINING

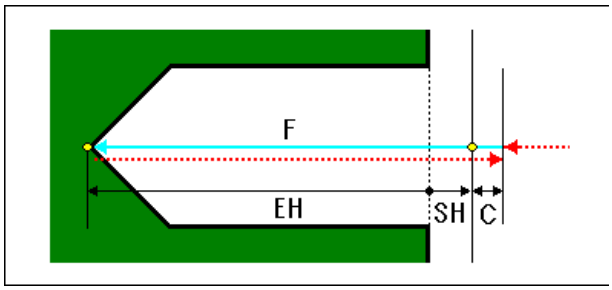
2.1.1 Machining Command (Tool Rotation: M Series)

(1) Input items of center drilling (G1000)

CUT COND.		
Data item		**Meaning**
W	MACHINING TYPE	[NORMAL] : No dwelling is performed. (initial value) [DWELL] : Dwelling is performed.
I	REF. PT. MODE	[INIT-1]: An R position return is made in moving between holes. Finally, a return is made to the I point. (initial value) [INIT-2]: All movements between holes, including the last return, are made as an I point return. [REF.] : All movements between holes, including the last return, are made as an R position return.
J	I POINT	Coordinate of the I point (COPY)
L	CUT DEPTH	Cut depth (radius value, negative value)
C	CLEARANCE	Distance between the workpiece surface and R position (radius value, positive value) (COPY)
F	FEED RATE	Cutting feedrate (positive value) (COPY)
P*	DWELL TIME	Dwell time at the hole bottom. If omitted, 0 is assumed. (units of seconds, positive value) (COPY)
Z	APPROCH MOTION (Note)	[2 AXES] : When moving from the current position to the machining start point, the tool first moves in the machining plane in two-axis synchronous operation and then moves along the tool axis. (initial value) [3 AXES] : The tool moves from the current position to the machining start point in three-axis synchronous operation.

NOTE
When the option for expanding the number of simultaneously controlled axes is not provided, this input item is not displayed. Similarly, the input item is not displayed for hole machining cycles G1001 to G1006.

(a) Tool path



<1> Move the tool to the position "cutting start position + clearance (C)" in rapid traverse.
<2> Move the tool to the cutting end position at the cutting feedrate (F).
<3> Move the tool to the position "cutting start position + clearance (C)" in rapid traverse.

(2) Input items of drilling (G1001)

CUT COND.		
Data item		**Meaning**
W	MACHINING TYPE	[NORMAL] : One cut with no dwelling performed (initial value) [DWELL] : One cut with dwelling performed [PECK] : Peck drilling (Note 1) [H SPED] : High-speed peck drilling (Note 2)
H	GO PATH AMOUNT SET	[SETTING]: GO PATH AMOUNT is entered on the DETAIL tab. [NOTHING]: GO PATH AMOUNT is not specified. Remark) This item is displayed when bit 0 (HL0) of parameter No. 27020 is set to 1. When “SETTING” is specified, GO PATH AMOUNT is displayed on the DETAIL tab.
Q	PECKING CUT DEPTH	Depth of cut made by one cut (radius value, positive value) (COPY)
I	REF. PT. MODE	See the descriptions Center drilling: about G1000.
J	I POINT	Coordinate of the I point (COPY)
L	CUT DEPTH	Cut depth (radius value, negative value)
K	GO PAST AMOUNT	Length of the incomplete hole portion at the tool tip. If omitted, 0 is assumed. (Radius value, positive value) (COPY) Remark) This item is not displayed when bit 0 (HL0) of parameter No. 27020 is set to 1.
C	CLEARANCE	Distance between the workpiece surface and R position (radius value, positive value) (COPY)
F	FEED RATE	Cutting feedrate (positive value) (COPY)
P*	DWELL TIME	Dwell time at the hole bottom. If omitted, 0 is assumed. (units of seconds, positive value) (COPY)
Z	APPROCH MOTION	See the descriptions about Center drilling: G1000.

NOTE
1 In the case of ‘MACHINING TYPE’ = ‘PECK’, the system refers to the parameter No.5115 as the return amount. Therefore, please set No.5115 to suitable value before machining.
2 In the case of ‘MACHINING TYPE’ = ‘H SPED’, the system refers to the parameter No.5114 as the return amount. Therefore, please set No.5114 to suitable value before machining.

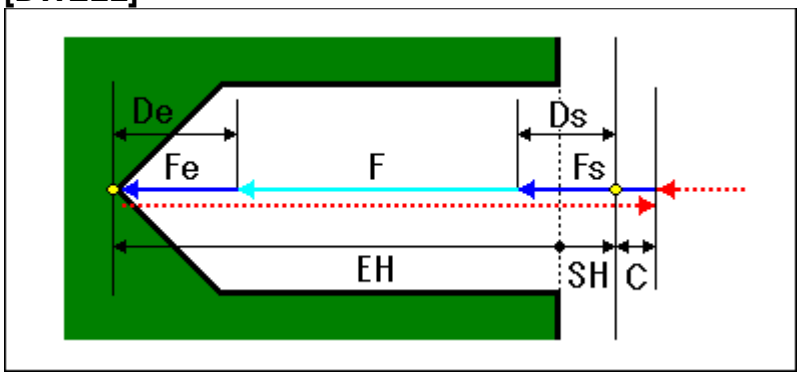
If bit 0 (HL0) of parameter No. 27020 is set to 1, when “SETTING” is specified for GO PATH AMOUNT SET on the CUT COND. tab, the tool diameter, tool angle, and GO PATH AMOUNT are displayed on the DETAIL tab.

DETAIL		
Data item		**Meaning**
A*	START CUT DEPTH	Depth of cut at the start time
S*	START FEED RATE	Feedrate at the start time
D*	END CUT DEPTH	Depth of cut at the end time In a high-speed peck drilling cycle, when the depth of cut and feedrate at the start time or at the end time are entered, the depth of cut at the end time is cut by just one cut operation without pecking.
E*	END FEED RATE	Feedrate at the end time
U	TOOL DIAMETER	Tool diameter
V	NOSE ANGLE	Tool angle
K	GO PAST AMOUNT	When the tool diameter and tool angle are entered then soft key [CALC] is pressed, an automatically calculated value is assigned to this item.

(a) Tool path

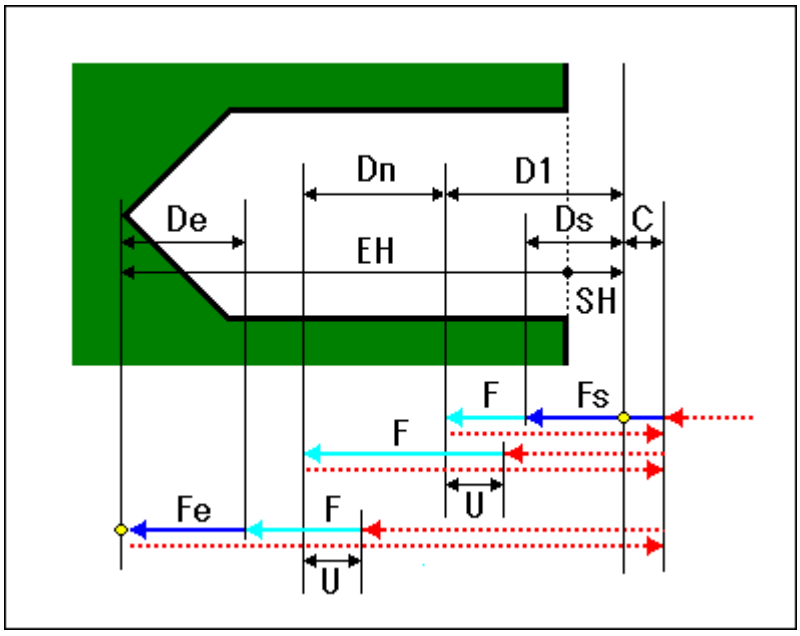
A drilling tool path can be selected from the following:

(i) [NORMAL]/[DWELL]



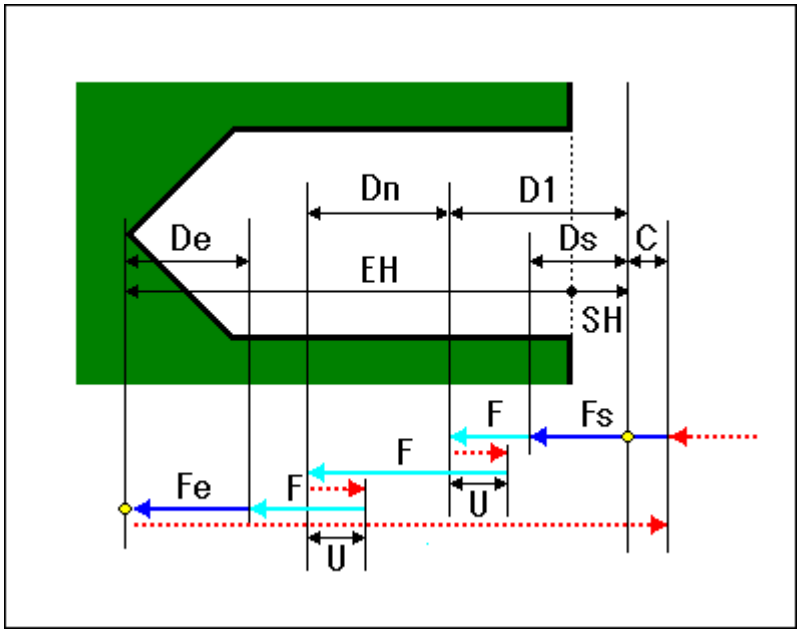
<1> Move the tool to the position "cutting start position + clearance (C)" in rapid traverse.
<2> Move the tool to the cutting end position at the cutting feedrate (F).
<3> Move the tool to the position "cutting start position + clearance (C)" in rapid traverse.

(ii) [PECK]



<1> Move the tool to the position "cutting start position + clearance (C)" in rapid traverse.
<2> Move the tool to the position "cutting start position - primary cut depth (D1)" at the cutting feedrate (F).
<3> Move the tool to the position "cutting start position + clearance (C)" in rapid traverse.
<4> Move the tool to the position "previous cutting end position + return clearance (U)" in rapid traverse.
<5> Move the tool to the position "previous cutting end position - compensation cut depth (Dn)" at the cutting feedrate (F).
<6> Repeat steps <3> to <5> until the last cutting end position is reached.
<7> Move the tool to the position "cutting start position + clearance (C)" in rapid traverse.

(iii) [H SPED]



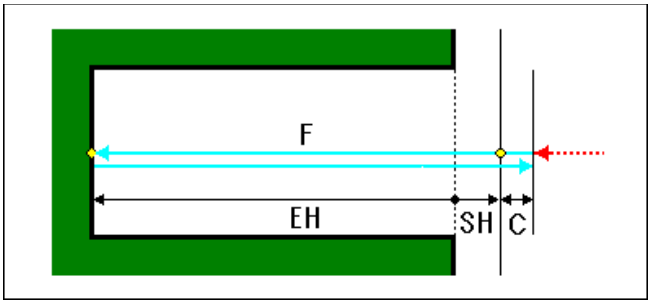
<1> Move the tool to the position "cutting start position + clearance (C)" in rapid traverse.
<2> Move the tool to the position "cutting start position - primary cut depth (D1)" at the cutting feedrate (F).
<3> Move the tool to the position "current position + return clearance (U)" in rapid traverse.
<4> Move the tool to cut to the position "previous cutting end position - compensation cut depth (Dn)" at the cutting feedrate (F).
<5> Repeat steps <3> and <4> until the cutting end position is reached.
<6> Move the tool to the position "cutting start position + clearance (C)" in rapid traverse.

(3) Input items of tapping (G1002)

CUT COND.		
Data item		**Meaning**
W	MACHINING TYPE	[NORMAL] : CW tapping (initial value) [REVERS] : CCW tapping
R	TAP TYPE	[FLOAT] : Float tapping [RIGID] : Rigid tapping (NOTE) Remark) FLOAT is not displayed when the rigid tapping option is provided and bit 0 (G84) of parameter No. 5200 is set to 1. RIGID is not displayed when the rigid tapping option is not provided. The initial value is FLOAT if the rigid tapping option is not provided; otherwise, the initial value is RIGID.
D	THREAD LEAD	Tapping tool lead (radius value, positive value) (COPY)
I	REF. PT. MODE	See the descriptions about Center drilling: G1000.
J	I POINT	Coordinate of the I point (COPY)
L	CUT DEPTH	Cut depth (radius value, negative value)
C	CLEARANCE	Distance between the workpiece surface and R position (radius value, positive value) (COPY)
P*	DWELL TIME	Dwell time at the hole bottom. If omitted, 0 is assumed. (units of seconds, positive value) (COPY)
Z	APPROCH MOTION	See the descriptions about Center drilling: G1000.
S	SPINDLE SPEED	Spindle speed (min^{-1})

NOTE
1 When you use rigid tapping mode M code command (bit 0 (G84) of parameter No.5200 = 0), the system refers to No.5210 or No.5212 as the value of M code. Therefore, please set No.5210 or No.5212 to suitable value before machining.

(a) Tool path



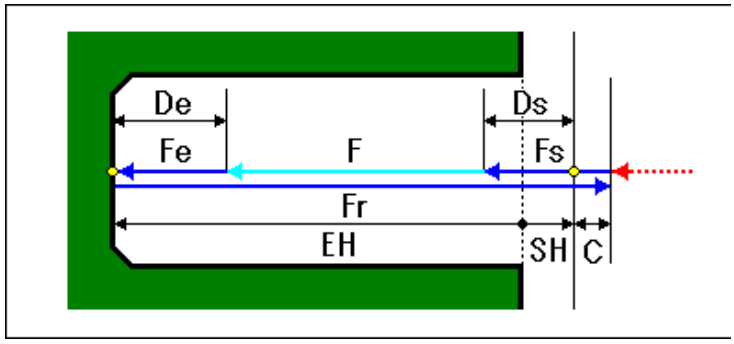
<1> Move the tool to the position "cutting start position + clearance (C)" in rapid traverse.
<2> Move the tool to the cutting end position at the cutting feedrate (F).
<3> Stop the spindle.
<4> Rotate the spindle in reverse.
<5> Move the tool to the position "cutting start position + clearance (C)" at the cutting feedrate (F).
<6> Cause the spindle to start rotating normally.

(4) Input items of reaming (G1003)

CUT COND.		
Data item		**Meaning**
W	MACHINING TYPE	[CUT] : The tool retracts from the hole bottom in cutting feed. (initial value) [RAPID] : The tool retracts from the hole bottom in rapid traverse. [DWELL] : After dwelling at the hole bottom, the tool retracts in cutting feed.
I	REF. PT. MODE	See the descriptions about Center drilling: G1000.
J	I POINT	Coordinate of the I point (COPY)
L	CUT DEPTH	Cut depth (radius value, negative value)
C	CLEARANCE	Distance between the workpiece surface and R position (radius value, positive value) (COPY)
F	FEED RATE	Cutting feedrate (positive value) (COPY)
P*	DWELL TIME	Dwell time at the hole bottom. If omitted, 0 is assumed. (units of seconds, positive value) (COPY)
Z	APPROCH MOTION	See the descriptions about Center drilling: G1000.

DETAIL		
Data item		**Meaning**
A*	START CUT DEPTH	Depth of cut at the start time when the workpiece is cut by changing the feedrate at the end of cutting
S*	START FEED RATE	Feedrate at the start time when the workpiece is cut by changing the feedrate at the end of cutting
D*	END CUT DEPTH	Depth of cut applicable when the workpiece is cut by changing the feedrate at the end of cutting
E*	END FEED RATE	Feedrate applicable when the workpiece is cut by changing the feedrate at the end of cutting

(a) Tool path



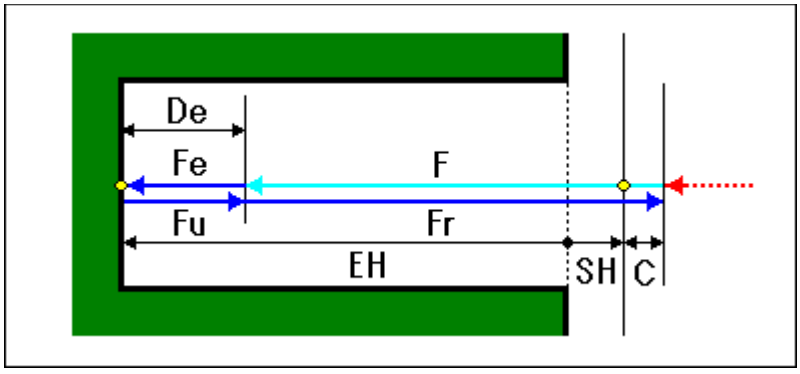
<1> Move the tool to the position "cutting start position + clearance (C)" in rapid traverse.
<2> Move the tool to the cutting end position at the cutting feedrate (F).
<3> Move the tool to the position "cutting start position + clearance (C)" at the returning feedrate (Fr).

(5) Input items of boring (G1004)

CUT COND.		
Data item		**Meaning**
W	MACHINING TYPE	[CUT] : The tool retracts from the hole bottom in cutting feed. (initial value) [RAPID] : The tool retracts from the hole bottom in rapid traverse. [DWELL] : After dwelling at the hole bottom, the tool retracts in cutting feed.
I	REF. PT. MODE	See the descriptions about Center drilling: G1000.
J	I POINT	Coordinate of the I point (COPY)
L	CUT DEPTH	Cut depth (radius value, negative value)
C	CLEARANCE	Distance between the workpiece surface and R position (radius value, positive value) (COPY)
F	FEED RATE	Cutting feedrate (positive value) (COPY)
P*	DWELL TIME	Dwell time at the hole bottom. If omitted, 0 is assumed. (units of seconds, positive value) (COPY)
Z	APPROCH MOTION	See the descriptions about Center drilling: G1000.

DETAIL		
Data item		**Meaning**
D*	END CUT DEPTH	Depth of cut applicable when the workpiece is cut by changing the feedrate at the end of cutting
E*	END FEED RATE	Feedrate applicable when the workpiece is cut by changing the feedrate at the end of cutting

(a) Tool path



<1> Move the tool to the position "cutting start position + clearance (C)" in rapid traverse.

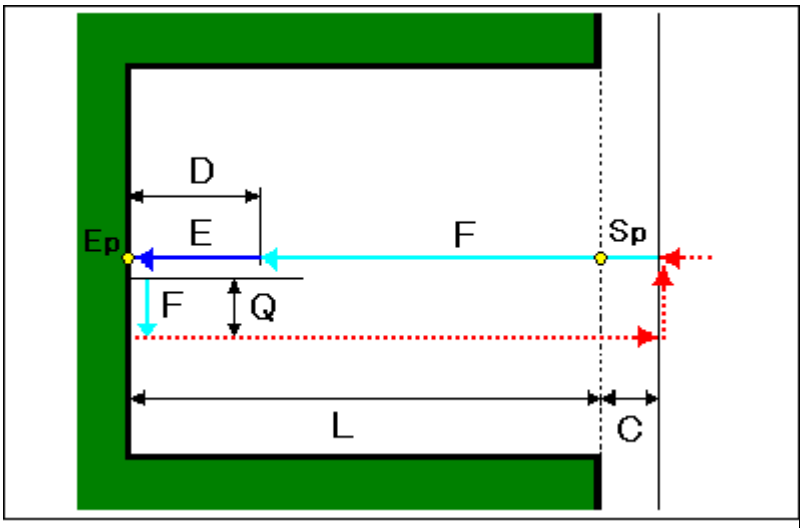
<2> Move the tool to the cutting end position at the cutting feedrate (F).
<3> Move the tool to the position "cutting start position + clearance (C)" at the returning feedrate (Fr).

(6) Input items of fine boring (G1005)

CUT COND.		
Data item		**Meaning**
Q	SHIFT AMOUNT	Shift amount (radius value) at the hole bottom after spindle orientation (COPY)
I	REF.PT.MODE	See the descriptions about Center drilling: G1000.
J	I POINT	Coordinate of the I point (COPY)
L	CUT DEPTH	Cut depth (radius value, negative value)
C	CLEARANCE	Distance between the workpiece surface and R position (radius value, positive value) (COPY)
F	FEED RATE	Cutting federate (positive value) (COPY)
P*	DWELL TIME	Dwell time at the hole bottom. If omitted, 0 is assumed. (units of seconds, positive value) (COPY)
Z	APPROCH MOTION	See the descriptions about Center drilling: G1000.

DETAIL		
Data item		**Meaning**
D*	END CUT DEPTH	Depth of cut applicable when the workpiece is cut by changing the feedrate at the end of cutting
E*	END FEED RATE	Feedrate applicable when the workpiece is cut by changing the feedrate at the end of cutting

(a) Tool path



When the end feed rate (E) is entered, tool path is as follows.
<1> Move the tool to the position "cutting start position (Sp) + clearance (C)" in rapid traverse.
<2> Move the tool to the position "cutting end position (Ep) + end cut depth (D)" at the cutting feed rate (F).
<3> Move the tool to the cutting end position (Ep) at the end feed rate (E).
(Dwell is performed when dwell time is set.)
<4> Stop the spindle.
<5> Spindle orientation is performed.
<6> Move the tool to the position "current position + shift amount (Q)" at the feed rate (F).
<7> Move the tool to the position "cutting start position (Sp) + clearance (C)" in rapid traverse.
<8> Move the tool to the position " current position - clearance (C)" in rapid traverse.

<9> Rotate the spindle in the positive direction.
<10> Move the next hole position and repeat the same motion.

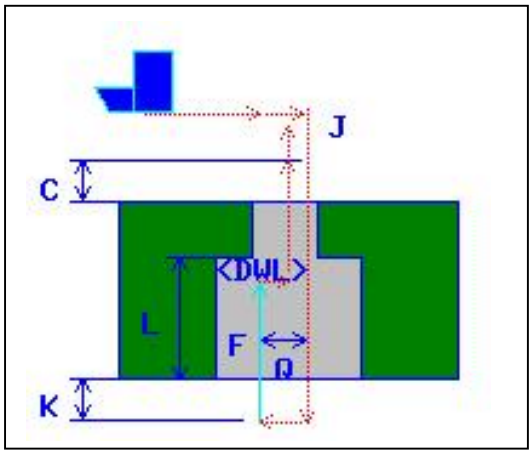
When the end feed rate (E) is empty, tool path is as follows.
<1> Move the tool to the position "cutting start position (Sp) + clearance (C)" in rapid traverse.
<2> Move the tool to the cutting end position (Ep) at the cutting feed rate (F).
(Dwell is performed when dwell time is set.)
<3> Stop the spindle.
<4> Spindle orientation is performed.
<5> Move the tool to the position "current position + shift amount (Q)" in rapid traverse.
<6> Move the tool to the position "cutting start position (Sp) + clearance (C)" in rapid traverse.
<7> Move the tool to the position " current position - clearance (C)" in rapid traverse.
<8> Rotate the spindle in the positive direction.
<9> Move the next hole position and repeat the same motion.

NOTE
Direction of tool rotation is only positive direction.

(7) Input items of back boring (G1006)

CUT COND.		
Data item		**Meaning**
Q	SHIFT AMOUNT	Shift amount (radius value) at the hole bottom after spindle orientation (COPY)
M	CUT DEPTH	Cut depth (radius value, negative value)
L	DIST. FROM BOTTOM	Distance (radius value) at the hole bottom in the lifting direction
I	REF.PT.MODE	See the descriptions about Center drilling: G1000.
J	I POINT	Coordinate of the I point (COPY)
K*	GO PAST AMOUNT	Go-past amount at the hole bottom. If omitted, 0 is assumed. (radius value, positive value) (COPY)
C	CLEARANCE	Distance between the workpiece surface and R position (radius value, positive value) (COPY)
F	FEED RATE	Cutting feedrate (positive value) (COPY)
P*	DWELL TIME	Dwell time at the hole bottom after lift machining. If omitted, 0 is assumed. (units of seconds, positive value) (COPY)
Z	APPROCH MOTION	See the descriptions about Center drilling: G1000.

(a) Tool path



<1> Move the tool to the position "cutting start position + clearance (C)" in rapid traverse.
<2> The tool is shifted away from the tool tip.

<3> The tool moves to the bottom of the hole (R point) by rapid traverse.
<4> The tool returns by a shift amount toward the tool tip.
<5> The spindle is turned in the normal direction to cut in to the position "K + L" along the tool axis at the feedrate (F) specified for cut-in.
<6> The spindle is stopped.
<7> After being shifted away from the tool tip, the tool is pulled out from the hole.

2.1.2 Figure (XY Plane)

(1) Input items of random points (G1210)

HOLE POS-1		
Data item		Meaning
B	BASE POSITION	Z coordinate of the workpiece surface
H	POINT-1 (X)	X coordinate of the first hole
V	POINT-1 (Y)	Y coordinate of the first hole
A*	POINT-2 (X)	X coordinate of the second hole
C*	POINT-2 (Y)	Y coordinate of the second hole
D*	POINT-3 (X)	X coordinate of the third hole
E*	POINT-3 (Y)	Y coordinate of the third hole
F*	POINT-4 (X)	X coordinate of the fourth hole
I*	POINT-4 (Y)	Y coordinate of the fourth hole

HOLE POS-2		
Data item		Meaning
J*	POINT-5 (X)	X coordinate of the fifth hole
K*	POINT-5 (Y)	Y coordinate of the fifth hole
M*	POINT-6 (X)	X coordinate of the sixth hole
P*	POINT-6 (Y)	Y coordinate of the sixth hole
Q*	POINT-7 (X)	X coordinate of the seventh hole
R*	POINT-7 (Y)	Y coordinate of the seventh hole
S*	POINT-8 (X)	X coordinate of the eighth hole
T*	POINT-8 (Y)	Y coordinate of the eighth hole

NOTE
It is unnecessary to enter values for every hole position. If entered, however, both X and Y coordinates in a pair must be entered for a hole position.

(2) Input items of linear points (same interval) (G1211)

HOLE POSIT		
Data item		**Meaning**
B	BASE POSITION	Z coordinate of the workpiece surface
H	START POINT (X)	X coordinate of the start point (first hole) of a straight line
V	START POINT (Y)	Y coordinate of the start point (first hole) of a straight line
A	ANGLE	Angle of a straight line from the X-axis (initial value = 0)
D	HOLE POS.TYPE	[LENGTH] : Specified with the distance between the first and last holes, and the number of holes. [PITCH] : Specified with the distance between two adjacent holes, and the number of holes.
E	LENGTH / PITCH	Length : Distance between the first and last holes (if [LENGTH] is selected for item D) Pitch : Distance between two adjacent holes (if [PITCH] is selected for item D)
C	NUMBER OF HOLE	Number of holes

SKIP		
Data item		**Meaning**
F*	OMITTING POINT 1	Point where no hole is to be made (1)
I*	OMITTING POINT 2	Point where no hole is to be made (2)
J*	OMITTING POINT 3	Point where no hole is to be made (3)
K*	OMITTING POINT 4	Point where no hole is to be made (4)

NOTE
It is unnecessary to enter a value for any item of an omitting point.
Similarly, this applies to G1213 to G1216 described later.

(3) Input items of linear points (different interval) (G1212)

HOLE POS-1		
Data item		**Meaning**
B	BASE POSITION	Z coordinate of the workpiece surface
H	START POINT (X)	X coordinate of the start point (first hole) of a straight line
V	START POINT (Y)	Y coordinate of the start point (first hole) of a straight line
A	ANGLE	Angle of a straight line from the X-axis (initial value = 0)

HOLE POS-2		
Data item		**Meaning**
C	PITCH WIDTH-1	Distance between the first and second holes (positive or negative value)
D	PITCH WIDTH-2	Distance between the second and third holes (positive or negative value)
E*	PITCH WIDTH-3	Distance between the third and fourth holes (positive or negative value)
F*	PITCH WIDTH-4	Distance between the fourth and fifth holes (positive or negative value)
I*	PITCH WIDTH-5	Distance between the fifth and sixth holes (positive or negative value)
J*	PITCH WIDTH-6	Distance between the sixth and seventh holes (positive or negative value)
K*	PITCH WIDTH-7	Distance between the seventh and eighth holes (positive or negative value)
M*	PITCH WIDTH-8	Distance between the eighth and ninth holes (positive or negative value)
P*	PITCH WIDTH-9	Distance between the ninth and tenth holes (positive or negative value)
Q*	PITCH WIDTH-10	Distance between the tenth and eleventh holes (positive or negative value)

NOTE
It is unnecessary to enter a value for every pitch width data item.

(4) Input items of grid points (G1213)

HOLE POSIT		
Data item		Meaning
B	BASE POSITION	Z coordinate of the workpiece surface
H	START POINT (X)	X coordinate of the start point (first hole) of a straight line
V	START POINT (Y)	Y coordinate of the start point (first hole) of a straight line
U	LENGTH FOR X AXIS	Length of the first side of a grid (positive value)
W	LENGTH FOR Y AXIS	Length of the second side of a grid (positive value)
I	NUMBER FOR X AXIS	Number of holes on the first side of a grid (positive value)
J	NUMBER FOR Y AXIS	Number of holes on the second side of a grid (positive value)
K	ANGLE FOR X AXIS	Angle of the first side of a grid from the X-axis (initial value = 0)
M	ANGLE FOR Y AXIS	Angle of the second side of a grid from the X-axis (initial value = 90)

SKIP		
Data item		Meaning
A*	OMITTING POINT 1	Point where no hole is to be made (1)
C*	OMITTING POINT 2	Point where no hole is to be made (2)
D*	OMITTING POINT 3	Point where no hole is to be made (3)
E*	OMITTING POINT 4	Point where no hole is to be made (4)

(5) Input items of rectangle points (G1214)

HOLE POSIT		
Data item		Meaning
B	BASE POSITION	Z coordinate of the workpiece surface
H	START POINT (X)	X coordinate of the start point (first hole) of a straight line
V	START POINT (Y)	Y coordinate of the start point (first hole) of a straight line
U	LENGTH FOR X AXIS	Length of the first side of a rectangle (positive value)
W	LENGTH FOR Y AXIS	Length of the second side of a rectangle (positive value)
I	NUMBER FOR X AXIS	Number of holes on the first side of a rectangle (positive value)
J	NUMBER FOR Y AXIS	Number of holes on the second side of a rectangle (positive value)
K	ANGLE FOR X AXIS	Angle of the first side of a rectangle from the X-axis (initial value = 0)
M	ANGLE FOR Y AXIS	Angle of the second side of a rectangle from the X-axis (initial value = 90)

SKIP		
Data item		Meaning
A*	OMITTING POINT 1	Point where no hole is to be made (1)
C*	OMITTING POINT 2	Point where no hole is to be made (2)
D*	OMITTING POINT 3	Point where no hole is to be made (3)
E*	OMITTING POINT 4	Point where no hole is to be made (4)

(6) Input items of circle points (G1215)

HOLE POINTS		
Data item		Meaning
B	BASE POSITION	Z coordinate of the workpiece surface
H	CENTER POINT (X)	X coordinate of the center of a circle
V	CENTER POINT (Y)	Y coordinate of the center of a circle
R	RADIUS	Radius of a circle (positive value)
A	START POINT ANGLE	Central angle of the first hole from the X-axis (positive or negative value) (initial value = 0)
C	NUMBER OF HOLE	Number of holes to be made (positive value)

SKIP		
Data item		Meaning
D*	OMITTING POINT 1	Point where no hole is to be made (1)
E*	OMITTING POINT 2	Point where no hole is to be made (2)
F*	OMITTING POINT 3	Point where no hole is to be made (3)
I*	OMITTING POINT 4	Point where no hole is to be made (4)

(7) Input items of arc points (same interval) (G1216)

HOLE POSIT		
Data item		Meaning
B	BASE POSITION	Z coordinate of the workpiece surface
H	CENTER POINT (X)	X coordinate of the center of an arc
V	CENTER POINT (Y)	Y coordinate of the center of an arc
R	RADIUS	Radius of an arc (positive value)
A	START POINT ANGLE	Central angle of the first hole from the X-axis (positive or negative value) (initial value = 0)
C	PITCH ANGLE	Central angle between two adjacent holes (positive or negative value)
D	NUMBER OF HOLE	Number of holes to be made (positive value)

SKIP		
Data item		Meaning
E*	OMITTING POINT 1	Point where no hole is to be made (1)
F*	OMITTING POINT 2	Point where no hole is to be made (2)
I*	OMITTING POINT 3	Point where no hole is to be made (3)
J*	OMITTING POINT 4	Point where no hole is to be made (4)

(8) Input items of arc points (different interval) (G1217)

HOLE POS-1		
Data item		Meaning
B	BASE POSITION	Z coordinate of the workpiece surface
H	CENTER POINT (X)	X coordinate of the center of an arc
V	CENTER POINT (Y)	Y coordinate of the center of an arc
R	RADIUS	Radius of an arc (positive value)
A	START POINT ANGLE	Central angle of the first hole from the X-axis (positive or negative value) (initial value = 0)

HOLE POS-2		
Data item		Meaning
C*	PITCH ANGLE-1	Central angle between the first and second holes (positive or negative value)
D*	PITCH ANGLE-2	Central angle between the second and third holes (positive or negative value)
E*	PITCH ANGLE-3	Central angle between the third and fourth holes (positive or negative value)
F*	PITCH ANGLE-4	Central angle between the fourth and fifth holes (positive or negative value)
I*	PITCH ANGLE-5	Central angle between the fifth and sixth holes (positive or negative value)
J*	PITCH ANGLE-6	Central angle between the sixth and seventh holes (positive or negative value)
K*	PITCH ANGLE-7	Central angle between the seventh and eighth holes (positive or negative value)
M*	PITCH ANGLE-8	Central angle between the eighth and ninth holes (positive or negative value)
P*	PITCH ANGLE-9	Central angle between the ninth and tenth holes (positive or negative value)
Q*	PITCH ANGLE-10	Central angle between the tenth and eleventh holes (positive or negative value)

2.1.3 Figure (YZ Plane)

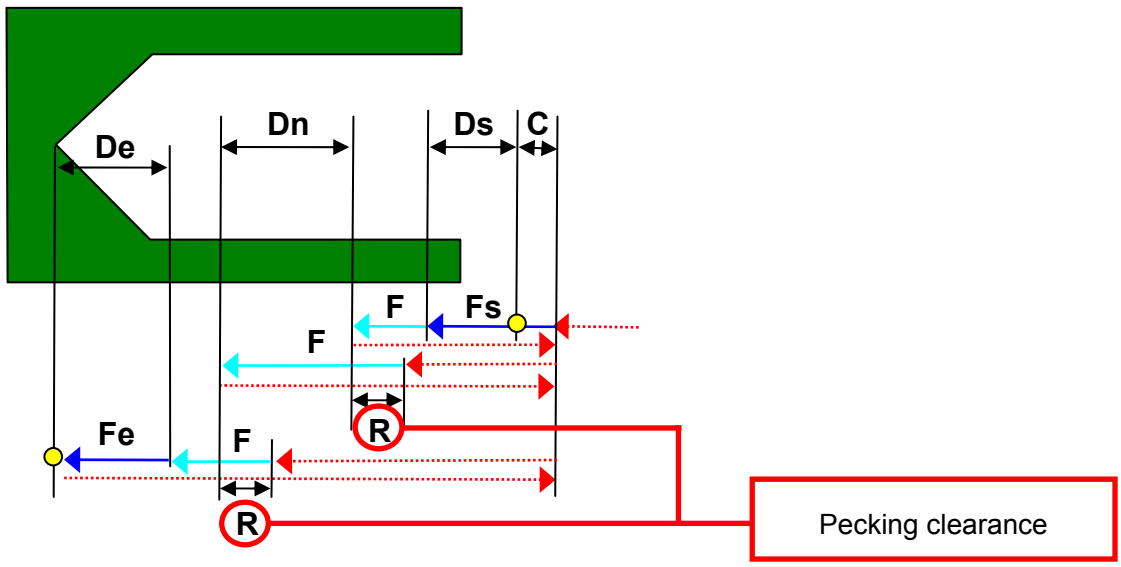
YZ plane

The same hole position block types as for the XY plane explained in the previous subsection are available for the YZ plane. They are provided with the following G codes.
The data to be set for the YZ plane is the same as for the XY plane except that the XY plane is changed to the YZ plane and that the direction in which the tool moves to cut is changed from the Z-axis to the X-axis.

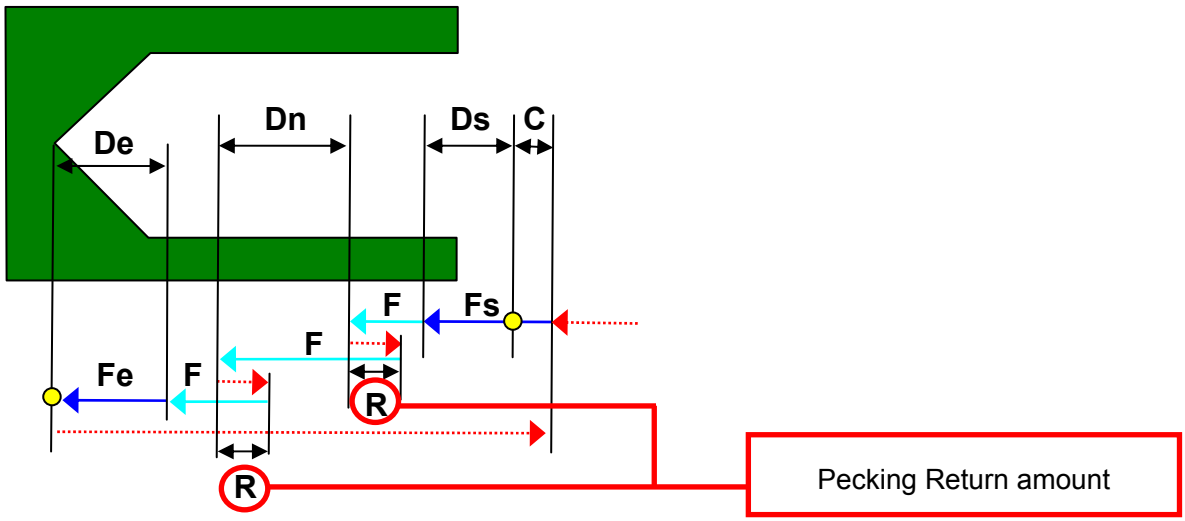
Random Points	:	G1310
Linear Points (Same Interval)	:	G1311
Linear Points (Different Interval)	:	G1312
Grid Points	:	G1313
Rectangle Points	:	G1314
Circle Points	:	G1315
Arc Points (Same Interval)	:	G1316
Arc Points (Different Interval)	:	G1317

2.1.4 Specifying Pecking Clearance / Return Amount of Drill Cycle for Machining Center in Input Window (for only Series 30*i*/31*i*/32*i*-A/B, 0*i*-MD)

(1) G1001 : Drill cycle (tool rotation)
In case of peck drill, pecking clearance R can be specified in input window instead of parameter No.5115.



In case of high-speed peck drill, pecking return amount R can be specified in input window instead of No.5114.



2.1.4.1 Parameter setting to use this function

Set the parameter No.27195#4 to 1.

2.1.4.2 Format of drill cycle

(1) G1001 : Drill cycle (tool rotation) input item
The following item is added. Refer to MANUAL GUIDE *i* operator's manual (B-63874EN) about other items.

Detail		
Data item		Meaning
R	PECKING CLEARANCE / PECKING RETURN AMT.	(In case "PECK DRILL" is inputted in "MACHINING TYPE") Pecking clearance When this item is blank, pecking clearance is value of parameter No.5115. (In case "HIGH SPEED" is inputted in "MACHINING TYPE") Pecking return amount When this item is blank, pecking return amount is value of parameter No.5114.

2.1.5 C axis Hole on End Face (Random Points) - Cartesian Coordinates Input Cycle

C axis hole on end face (random points) can be commanded in virtual cartesian coordinate system using the following command.

C axis hole on end face (random points) - cartesian coordinates input : G1574

2.1.5.1 Input items of C axis hole on end face (random points) - cartesian coordinates (G1574)

HOLE POS-1		
Data item		Meaning
B	BASE POSITION	Z coordinate of the workpiece surface - virtual cartesian coordinate
H	X AXIS POS.1(RAD.)	X coordinate of the first hole (radius value) - virtual cartesian coordinate
V	C AXIS POS.1	C coordinate of the first hole - virtual cartesian coordinate
A	X AXIS POS.2(RAD.)	X coordinate of the second hole (radius value) - virtual cartesian coordinate
C	C AXIS POS.2	C coordinate of the second hole - virtual cartesian coordinate
D	X AXIS POS.3(RAD.)	X coordinate of the third hole (radius value) - virtual cartesian coordinate
E	C AXIS POS.3	C coordinate of the third hole - virtual cartesian coordinate
F	X AXIS POS.4(RAD.)	X coordinate of the fourth hole (radius value) - virtual cartesian coordinate
I	C AXIS POS.4	C coordinate of the fourth hole - virtual cartesian coordinate

HOLE POS-2		
Data item		Meaning
J	X AXIS POS.5(RAD.)	X coordinate of the fifth hole (radius value) - virtual cartesian coordinate
K	C AXIS POS.5	C coordinate of the fifth hole - virtual cartesian coordinate
M	X AXIS POS.6(RAD.)	X coordinate of the sixth hole (radius value) - virtual cartesian coordinate
P	C AXIS POS.6	C coordinate of the sixth hole - virtual cartesian coordinate
Q	X AXIS POS.7(RAD.)	X coordinate of the seventh hole (radius value) - virtual cartesian coordinate
R	C AXIS POS.7	C coordinate of the seventh hole - virtual cartesian coordinate
S	X AXIS POS.8(RAD.)	X coordinate of the eighth hole (radius value) - virtual cartesian coordinate
T	C AXIS POS.8	C coordinate of the eighth hole - virtual cartesian coordinate

2.2 FACING

2.2.1 Machining Command

(1) Input items of ROUGH (G1020)

TOOL COND.		
Data item		Meaning
D	TOOL DIAMETER	Face mill diameter

NOTE
1 Tab ‘TOOL COND.’ is enabled when bit 0 (TLG) of parameter No.27002 = 1.
2 The operator ordinarily sets the above data on the tab of ‘TOOL DATA’ in Tool Offset window. Therefore, it is not necessary to display the tab ‘TOOL COND.’

CUT COND.		
Data item		Meaning
T*	THICKNESS	Face mill diameter
H*	FINISH AMOUNT	Finishing allowance in facing
L	CUT DEPTH OF RADIUS	Depth of cut in the tool radius direction to the next cutting path
J	CUT DEPTH OF AXIS	Depth of cut in the tool axis direction per cutting operation
F	FEED RATE – RADIUS	Feedrate applicable when cutting is performed in the tool radius direction
E	FEED RATE – AXIS	Feedrate applicable when cutting is performed in the tool axis direction

NOTE
If the bit 0 (FC1) of parameter No.27030 = 1, only the menu of perpendicular directions to “CUTTING DIRECTION” can have been displayed during in case of selecting “CUT SHIFT DIRECTION”. Thus, the extra selection can be removed.

DETAIL		
Data item		**Meaning**
I	INITIAL FEED OVERRIDE	Feedrate override value for the first cutting. The initial value is 100 (1 to 200, positive value).
W	CUTTING METHOD	[SINGLE] : Cutting in the tool radius direction is always performed in the same direction. [ZIGZAG] : Cutting in the tool radius direction is performed back and forth. (COPY)
P	PATH MOVE METHOD	[PULL] : Retracts to point R before moving to the start point of the next cutting path (in the tool axis direction). [KEEP] : Moves to the start point of the next cutting path directly without retracting to point R. Remark1) This data item is indicated only when [ZIGZAG] is selected for CUTTING METHOD. Remark2) This data item is enabled when the bit 0 (FC0) of parameter No.27030 = 0. Remark3) In case that the movement is done by 1 axis, tool is kept even if [PULL] is specified. Remark4) If [KEEP] is specified, an end point of previous cutting motion is adjusted in order to enable a tool to move with 1 axis to next cutting start point (COPY)
V	PATH MOVE FEED RATE	Feedrate applicable when the tool moves to the start point of the next cutting path. In case that the feedrate is set to 0, the tool moves at a rapid traverse rate. Remark1) This data item is indicated only when [ZIGZAG] is selected for CUTTING METHOD. Remark2) This data item is enabled when the bit 0 (FC0) of parameter No.27030 = 0. (COPY)
C	CLEARANCE OF AXIS	Distance between the surface of a blank being machined and a cutting start point (point R) in the tool axis direction (radius value) Remark) By referring to the parameter No.27009 (minimum clamp value), the system sets the data as Initial value.
M	CLERANCE OF RADIUS	Distance between the end of a blank being machined and the end of the tool placed at the retract position (radius value) Remark) By referring to the parameter No.27009 (minimum clamp value), the system sets the data as Initial value.

DETAIL		
Data item		Meaning
A	CUTTING DIRECTION	[RIGHT] : Performs cutting rightward as indicated in the illustration. When both directions are selected, cutting for the first cutting path is performed rightward. [LEFT] : Performs cutting leftward as indicated in the illustration. When both directions are selected, cutting for the first cutting path is performed leftward. [UP] : Performs cutting upward as indicated in the illustration. When both directions are selected, cutting for the first cutting path is performed upward. [DOWN] : Performs cutting downward as indicated in the illustration. When both directions are selected, cutting for the first cutting path is performed downward. Remark) The actual cutting direction is determined by the coordinate axis indicated in the illustration. (COPY)
B	CUT DEPTH DIRECTION	[RIGHT] : Performs cutting while shifting the cutting path rightward as indicated in the illustration. [LEFT] : Performs cutting while shifting the cutting path leftward as indicated in the illustration. [UP] : Performs cutting while shifting the cutting path upward as indicated in the illustration. [DOWN] : Performs cutting while shifting the cutting path downward as indicated in the illustration. Remark) The actual cutting direction is determined by the coordinate axis indicated in the illustration. (COPY)
Z	APPROCH MOTION	[2 AXES] : When moving from the current position to the machining start point, the tool first moves in the machining plane in two-axis synchronous operation and then moves along the tool axis. (initial value) [3 AXES] : The tool moves from the current position to the machining start point in three-axis synchronous operation.

NOTE
When the option for expanding the number of simultaneously controlled axes is not provided, this input item is not displayed. Similarly, the input item is not displayed for finishing cycle G1021.

(2) Input items of FINISH (G1021)

TOOL COND.		
Data item		Meaning
D	TOOL DIAMETER	See the descriptions about Roughing: G1020.

CUT COND.		
Data item		Meaning
L	CUT DEPTH OF RADIUS	See the descriptions about Roughing: G1020.
F	FEED RATE – RADIUS	-
E	FEED RATE – AXIS	-

DETAIL		
Data item		Meaning
W	CUTTING METHOD	See the descriptions about Roughing: G1020.
P	PATH MOVE METHOD	-
V	PATH MOVE FEED RATE	-

DETAIL		
Data item		Meaning
C	CLEARANCE OF AXIS	-
M	CLERANCE OF RADIUS	-
A	CUTTING DIRECTION	-
B	CUT DEPTH DIRECTION	-
Z	APPROCH MOTION	-

2.2.2 Fixed Form Figure Blocks (XY Plane)

(1) Input items of SQUARE (XY plane) (G1220)

POS./SIZE		
Data item		Meaning
T	FIGURE TYPE	[FACE] : Used as a figure for facing [CONVEX] : Used as an outer figure for contouring [CONCAVE]: Used as an inner figure for contouring or as a figure for pocketing [GROOVE] : Used as a figure for grooving Remark) When facing is selected as machining type, be sure to select [FACE].
B	BASE POSITION	Z coordinate of the final surface for facing (in the tool axis direction)
H	CENTER POINT (X)	X coordinate of the center position of a rectangular figure
V	CENTER POINT (Y)	Y coordinate of the center position of a rectangular figure
U	LEBGTH FOR X AXIS	Length of the side in the X-axis direction (radius value, positive value)
W	LENGTH FOR Y AXIS	Length of the side in the Y-axis direction (radius value, positive value)
R*	CORNER RADIUS	Radius for corner rounding (positive value)
A*	ANGLE	Inclination angle of a rectangular figure relative to the X-axis (positive or negative value)

(2) Input items of CIRCLE (XY plane) (G1221)

POS./SIZE		
Data item		Meaning
T	FIGURE TYPE	See the descriptions about SQUARE: G1220.
B	BASE POSITION	Z coordinate of the final surface for facing (in the tool axis direction)
H	CENTER POINT (X)	X coordinate of the center position of a circular figure
V	CENTER POINT (Y)	Y coordinate of the center position of a circular figure
R	RADIUS	Radius of a circular figure (radius value, positive value)

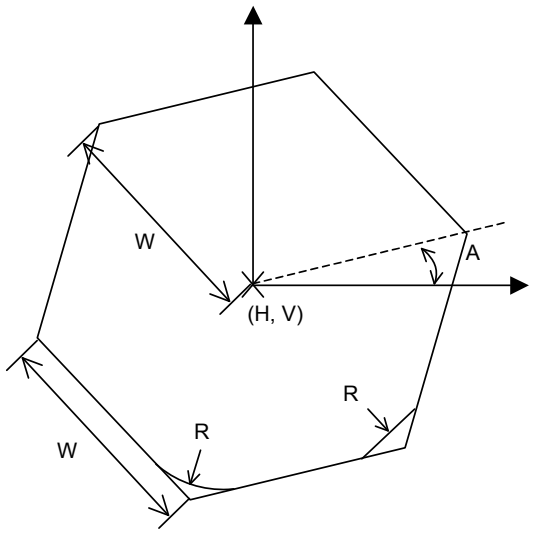
(3) Input items of TRACK (XY plane) (G1222)

POS./SIZE		
Data item		Meaning
T	FIGURE TYPE	See the descriptions about SQUARE: G1220.
B	BASE POSITION	Z coordinate of the final surface for facing (in the tool axis direction)
H	CENTER POINT (X)	X coordinate of the center position of the left semicircle
V	CENTER POINT (Y)	Y coordinate of the center position of the left semicircle
U	BETWEEN DISTANCE	Distance between the centers of the right and left semicircles (radius value, positive value)
R	RADIUS	Radius of the left and right semicircles (radius value, positive value)
A*	ANGLE	Inclination angle of a track figure relative to the X-axis. The blank is regarded as 0 degrees. (positive or negative value)

(4) Input items of POLYGON (XY plane) (G1225)

This figure can be specified when bit 2 (P2) of parameter No. 27095 is set to 1.

POS./SIZE		
Data item		Meaning
T	FIGURE TYPE	See the descriptions about SQUARE: G1220.
B	BASE POSITION	
L	HEIGHT/DEPTH	Not displayed when facing is performed.
H	CENTER POINT(X)	
V	CENTER POINT(X)	
E	NUMBER OF ANGLE	Positive integer ranging from 3 to 99
U	INPUT TYPE OF SIZE	U1:Polygon radius,U2:Side length, U3: Width across flat
W	POLYGON RADIUS / SIDE LENGTH	
A	ANGLE	Inclination angle of a straight line connecting a vertex and the center with respect to the 1st axis
D	GROOVE WIDTH	Specified only when this item is used for grooving.
C	CORNER TYPE	C1:None、C2:Chamfering、C3:Corner rounding
R	CORNER SIZE	
Z	FACE POSITION	Z1:+ end face、Z2:－ end face (*1)
Y	ROTATION AXIS NAME	(*2)



(*1) “FACE POSITION” is displayed on a tab added when bit 4 (MC4) of parameter No. 27000 is set to 1.

(*2) “ROTATION AXIS NAME” is displayed on a tab added only for the polar coordinate plane (XC) when bit 4 (MC4) of parameter No. 27000 is set to 1.

2.2.3 Fixed Form Figure (YZ Plane)

The same fixed-figure block types as for the XY plane explained in the previous subsection are available for the YZ plane. They are provided with the following G codes.
The data to be set for the YZ planes is the same as for the XY plane except that the XY plane is changed to the YZ plane and that the direction in which the tool moves to cut is changed to the X-axis (YZ plane).

Square : G1320 (YZ plane)
Circle : G1321 (YZ plane)
Track : G1322 (YZ plane)
Polygon : G1325 (YZ plane)

2.2.4 Arbitrary Figure (XY Plane)

When an arbitrary figure for facing is input, data such as a figure type and machining reference position is specified in the start point block. Other data items to be input such as a straight line and arc are exactly the same as for arbitrary figures of other machining types.
So, this subsection describes only the start point block of an arbitrary figure for facing.
For other arbitrary figures, see Chapter 5, "DETAILED DESCRIPTIONS ABOUT ENTERING ARBITRARY FIGURES".

(1) Input items of START POINT (XY plane) (G1200)

ELEMENT		
Data item		Meaning
T	FIGURE TYPE	See the descriptions about SQUARE: G1220.
X	START POINT X	X coordinate of the start point of an arbitrary figure
Y	START POINT Y	Y coordinate of the start point of an arbitrary figure
Z	BASE POSITION	Z coordinate of the final surface for facing (in the tool axis direction)

2.2.5 Arbitrary Figure (YZ Plane)

The same arbitrary-figure block types as for the XY plane explained in the previous subsection are available for the YZ plane. Their start points are specified with the following G codes.
The data to be set for the YZ planes is the same as for the XY plane except that the XY plane is changed to the YZ plane.

Start point : G1300 (YZ plane)

2.3 CONTOURING

2.3.1 Machining Command

(1) Input items of OUTER WALL (ROUGH) (G1060)
Input items of INNER WALL (ROUGH) (G1054)
Input items of PARTIAL (ROUGH) (G1068)

TOOL COND.		
Data item		Meaning
D	TOOL DIAMETER	End mill diameter

NOTE
1 Tab ‘TOOL COND.’ is enabled when bit 0 (TLG) of parameter No. 27002 = 1.
2 The operator ordinarily sets the above data on the tab of ‘TOOL DATA’ in Tool Offset window. Therefore, it is not necessary to display the tab ‘TOOL COND.’

CUT COND		
Data item		Meaning
T	BOTTOM THICKNESS (*2)	Cutting allowance of the bottom in side face machining (radius value, positive value)
S	SIDE THICKNESS (*2)	Cutting allowance of the side face (radius value, positive value)
L	CUT DEPTH OF RADIUS (*1)	Depth of cut per side face machining operation (in the tool radius direction) (radius value, positive value)

CUT COND		
Data item		Meaning
J	CUT DEPTH OF AXIS (*1)	Depth of cut in the tool axis direction per cutting operation (radius value, positive value) The default is (bottom surplus thickness - bottom finishing allowance).
K*	SIDE FINISH AMOUNT (*1)	Finishing allowance on the side face. The blank is regarded as 0. (radius value, positive value)
H*	BOTTOM FINISHI AMT. (*1)	Finishing allowance at the bottom in side face machining. The blank is regarded as 0. (radius value, positive value)

(*1) When bit 5 (RCF) of parameter No. 27002 is set to 1, the displayed order is changed by exchanging the input items "CUT DEPTH OF RADIUS" and "CUT DEPTH OF AXIS" and the input items "SIDE FINISH AMOUNT" and "BOTTOM FINISH AMT." so that the items related to workpiece dimensions are displayed collectively.

(*2) When bit 0 (P0) of parameter No. 27095 is set to 1, the "BOTTOM THICKNESS" and "SIDE THICKNESS" items can be entered on the figure input screen. In this case, these input items are not displayed on the cycle input screen.
The advantages of displaying these input items on the figure input screen instead of the cycle input screen are as follows:

- "BOTTOM THICKNESS" and "SIDE THICKNESS" are closely related to "BASE POSITION" and "HEIGHT/DEPTH" on the figure input screen. These items can be entered in the same block.
- A single machining cycle can contain more than one figure block. Therefore, when "BOTTOM THICKNESS" and "SIDE THICKNESS" are entered in a figure block, thickness values can be set separately for each machining profile.

NOTE

When a setting is made to use the figure input screen for entering "BOTTOM THICKNESS" and "SIDE THICKNESS", it is impossible to directly use a machining program that was created with a setting to enter "BOTTOM THICKNESS" and "SIDE THICKNESS" on the cycle input screen. The program needs to partly be modified.

Depending on the setting, the following alarms are issued:

1 When the bottom thickness and side thickness are entered on the figure input screen (bit 0 (P0) of parameter No. 27095 is set to 1)

<1> A machining type block contains a bottom thickness or side thickness item

→ Alarm (PS3043(Series 16*i*/18*i*/21*i*),
PS3543(Series 30*i*/31*i*/32*i*) ”WRONG THICKNESS”)

<2> In a machining profile block, a bottom thickness or side thickness is required, but its argument is not present.

→ Alarm (PS3016(Series 16i/18i/21i),
PS3516(Series 30i/31i/32i) ”NO NECESSARY ADDRESS”)

NOTE

2 When the bottom thickness and side thickness are entered on the cycle input screen (bit 0 (P0) of parameter No. 27095 is set to 0)

<1> A machining profile block contains a bottom thickness or side thickness item.

→ Alarm (PS3043(Series 16*i*/18*i*/21*i*), PS3543(Series 30*i*/31*i*/32*i*) ”WRONG THICKNESS”)

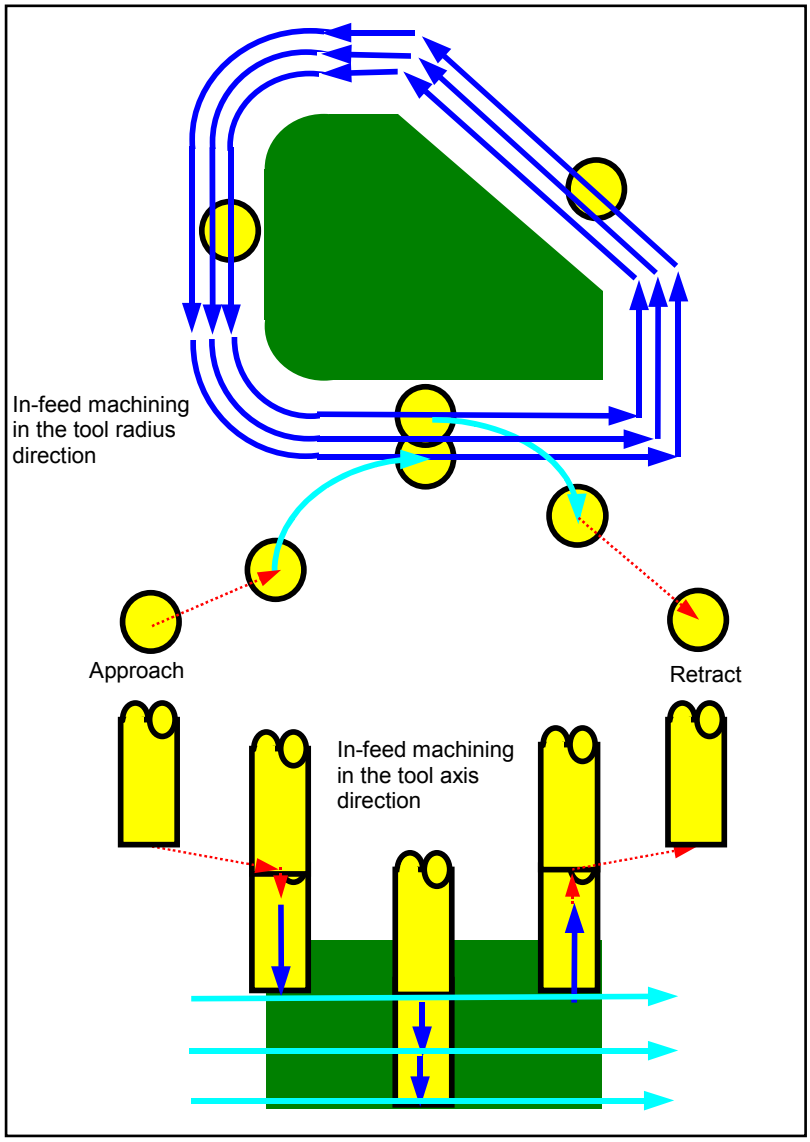
* For machining type blocks in which the bottom and side thickness items are permitted to be omitted, an alarm is not issued even when the bottom or side thickness item is missing. If the bottom and side thickness items are required (part of facing and contouring), the alarm “NO NECESSARY ADDRESS” is issued.

CUT COND		
Data item		**Meaning**
F	FEED RATE- SING.CUT	Feedrate applicable when only the one-side cutter portion of an end mill is used for cutting. This feedrate is used for cutting in retract operation and on the side face other than initial cutting.
V	FEED RATE- BOTH CUT	Feedrate applicable when the entire front side of an end mill is used for cutting. This feedrate is used for initial cutting.
E	FEED RATE – AXIS	Feedrate applicable when cutting is performed in the tool axis direction toward the bottom of a side face being machined

DETAIL		
Data item		**Meaning**
M	INITIAL FEED OVERRIDE	Feedrate override value for the first cutting. The initial value is 100 (1 to 200, positive value).
W	UP CUT/DOWN CUT	[UP CUT] : Performs machining in up-cut mode, assuming that the tool is rotating clockwise. [DWNCUT] : Performs machining in down-cut mode, assuming that the tool is rotating clockwise. (COPY)
C	CLEARANCE OF AXIS	Distance between the surface of a blank being machined and a cutting start point (point R) in the tool axis direction (radius value, positive value) Remark) By referring to the parameter No.27009 (minimum clamp value), the system sets the data as Initial value.
P	APPROACH TYPE	[ARC] : Approaches a side face along an arc. [TANGEN]: Approaches a side face along the straight line tangent to the first figure in side face cutting. [VERTIC] : Approaches a side face along the straight line normal to the first figure in side face cutting. (COPY)
R	APPROACH RAD./DIST.	Radius when [ARC] is specified. Straight line length when [TANGEN] or [VERTIC] is specified. (radius value, positive value) Remark) By referring to the parameter No.27010 (minimum clamp value), the system sets the data as Initial value.
A*	APPROACH ANGLE	Center angle of the arc when [ARC] is specified. The default is 90 degrees. (positive value) Remark) This data item is indicated only when [ARC] is selected for APPROACH TYPE. (COPY)

DETAIL		
Data item		Meaning
Q	ESCAPE TYPE	[ARC] : Retracts from a side face along an arc. [TANGEN]: Retracts from a side face along the straight line tangent to the last figure in side face cutting. [VERTIC] : Retracts from a side face along the straight line normal to the last figure in side face cutting. (COPY)
X	ESCAPE RAD./DIST.	Radius when [ARC] is specified. Straight line length when [TANGEN] or [VERTIC] is specified. (radius value, positive value) Remark) By referring to the parameter No.27010 (minimum clamp value), the system sets the data as Initial value.
Y*	ESCAPE ANGLE	Center angle of the arc when [ARC] is specified. The default is 90 degrees. (positive value) Remark) This data item is indicated only when [ARC] is selected for ESCAPE TYPE. (COPY)
Z	APPROCH MOTION	[2 AXES] : When moving from the current position to the machining start point, the tool first moves in the machining plane in two-axis synchronous operation and then moves along the tool axis. (initial value) [3 AXES] : The tool moves from the current position to the machining start point in three-axis synchronous operation. Remark) This input item is not displayed when the option for expanding the number of simultaneously controlled axes is not provided.

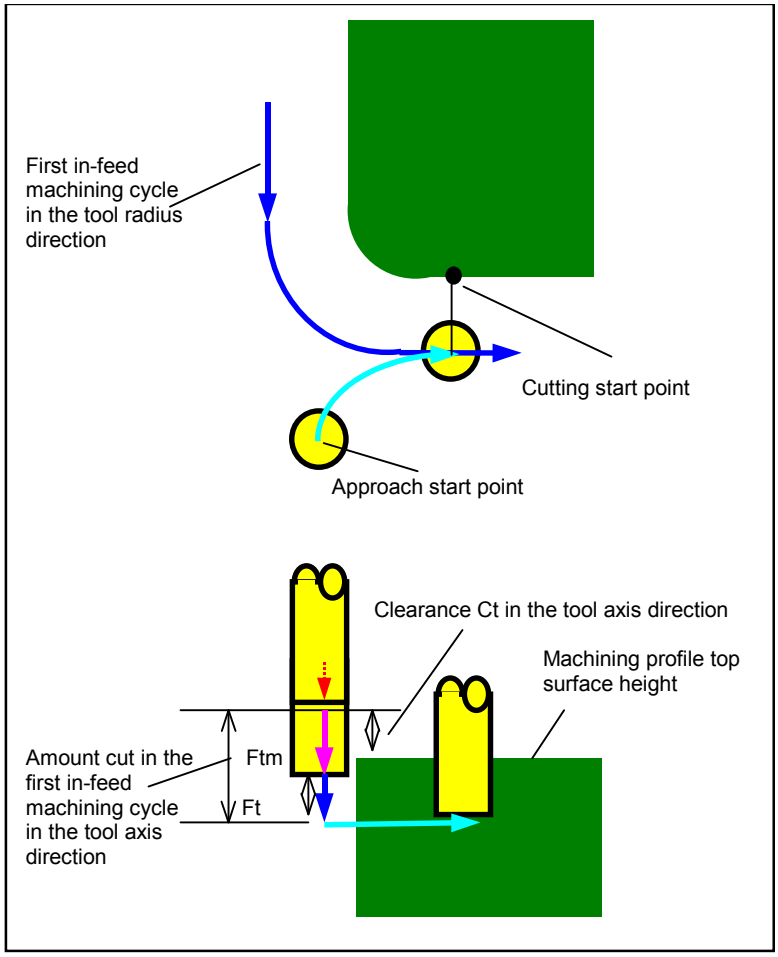
(a) Tool path



The side-face contour of a machining profile is cut off.
The following tool path is created.

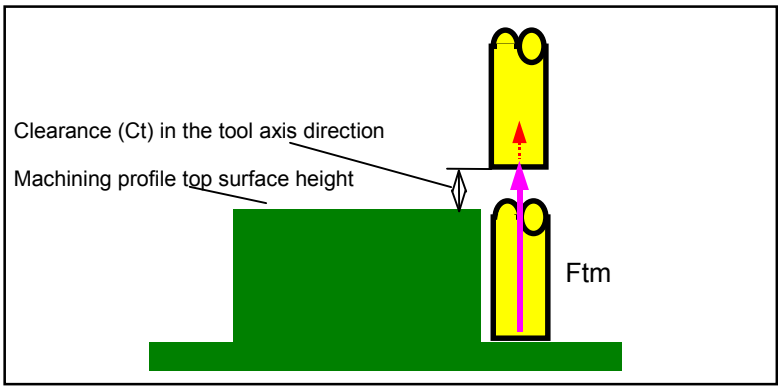
<1> The tool moves to above the approach start point.
<2> The tool moves to the height of the cutting surface.
<3> The tool cuts along the side-face contour of the machining profile.
The tool performs in-feed machining in the tool radius direction until the cutting allowance in the tool radius direction is removed.
<4> Steps <2> and <3> are repeated until the cutting allowance in the tool axis direction is removed.
<5> The tool retracts.

(i) Approach



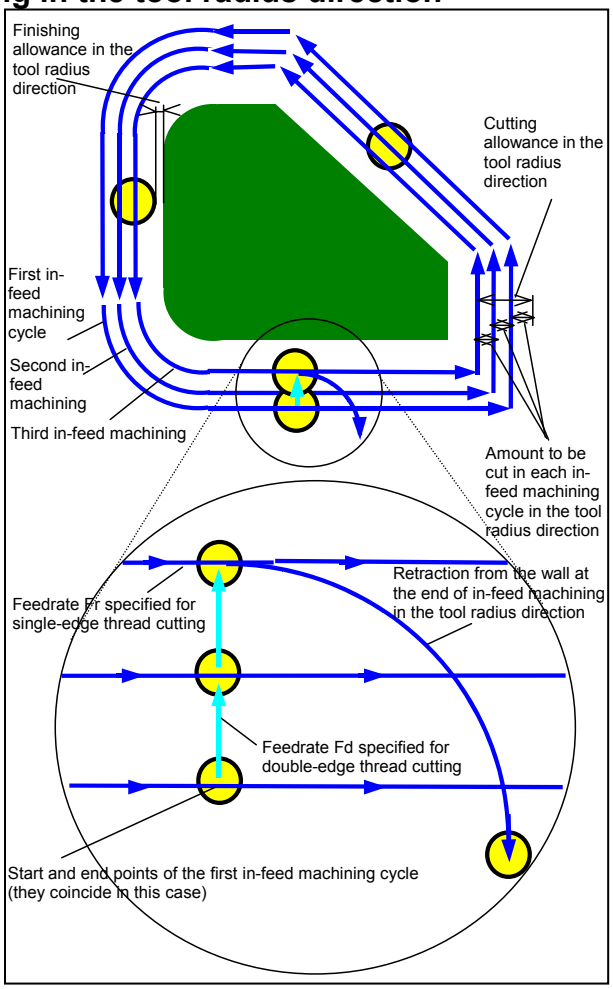
<1> The tool moves to the position "machining profile top surface height + clearance (Ct) in the tool axis direction" by rapid traverse.
<2> The tool moves to the position "amount to be cut in the first in-feed machining cycle in the tool axis direction - clearance (Ct) in the tool axis direction" at the feedrate (Ftm) specified for movement in the tool axis direction.
<3> The tool cut in to the position "amount to be cut in the first in-feed machining cycle in the tool axis direction" at the feedrate (Ft) specified for cutting in the tool axis direction.
<4> The tool approaches, in the tool radius direction, the start point of the first in-feed machining cycle in the tool radius direction.

(ii) Retraction



<1> The tool moves from the approach end point to the position "machining profile top surface height + clearance (Ct) in the tool axis direction" at the feedrate (Ftm) specified for movement in the tool axis direction.

(iii) In-feed machining in the tool radius direction



<1> The tool moves to cut along the contour from the first in-feed machining cycle start point to the end point at the feedrate (Fs) specified for single-edge thread cutting.
<2> The tool approaches, using the following method.

When the in-feed machining start point coincides with the in-feed machining end point:
The tool directly approaches the next in-feed machining start point in the normal direction at the feedrate (Fd) specified for double-edge thread cutting.

When the in-feed machining start point does not coincide with the in-feed machining end point:
The tool approaches the second in-feed machining start point.

<3> The tool moves to cut along the contour of the machining profile at the feedrate (Fs) specified for single-edge thread cutting.
<4> Steps <2> and <3> are repeated until the cutting allowance (cutting allowance in the tool radius direction - finishing allowance) is removed.
<5> The tool retracts.

(2) Input items of OUTER WALL (BOTTOM FINISH) (G1061)
Input items of INNER WALL (BOTTOM FINISH) (G1065)
Input items of PARTIAL (BOTTOM FINISH) (G1069)

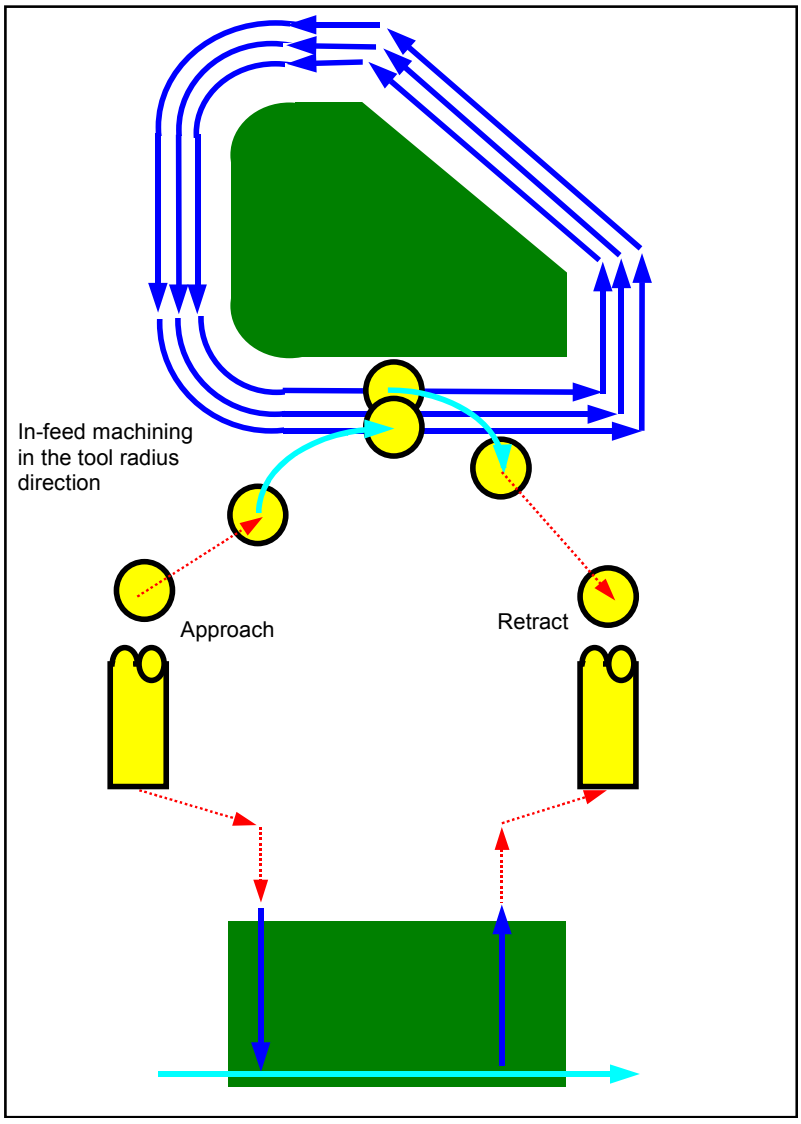
TOOL COND.		
Data item		Meaning
D	TOOL DIAMETER	See the descriptions about Roughing: G1060.

CUT COND		
Data item		Meaning
T	BOTTOM THICKNESS	See the descriptions about Roughing: G1060.
S	SIDE THICKNESS	-
K*	SIDE FINISH AMOUNT	-
H*	BOTTOM FINISHI AMT.	-
L	CUT DEPTH OF RADIUS	-
F	FEED RATE- SING.CUT	-
V	FEED RATE- BOTH CUT	-
E	FEED RATE- AXIS	-

DETAIL		
Data item		Meaning
W	UP CUT/DOWN CUT	See the descriptions about Roughing: G1060.
C	CLEARANCE OF AXIS	-
P	APPROACH TYPE	-
R	APPROACH RAD./DIST.	-
A*	APPROACH ANGLE	-
Q	ESCAPE TYPE	-
X	ESCAPE RAD./DIST.	-
Y*	ESCAPE ANGLE	-
Z	APPROCH MOTION	-

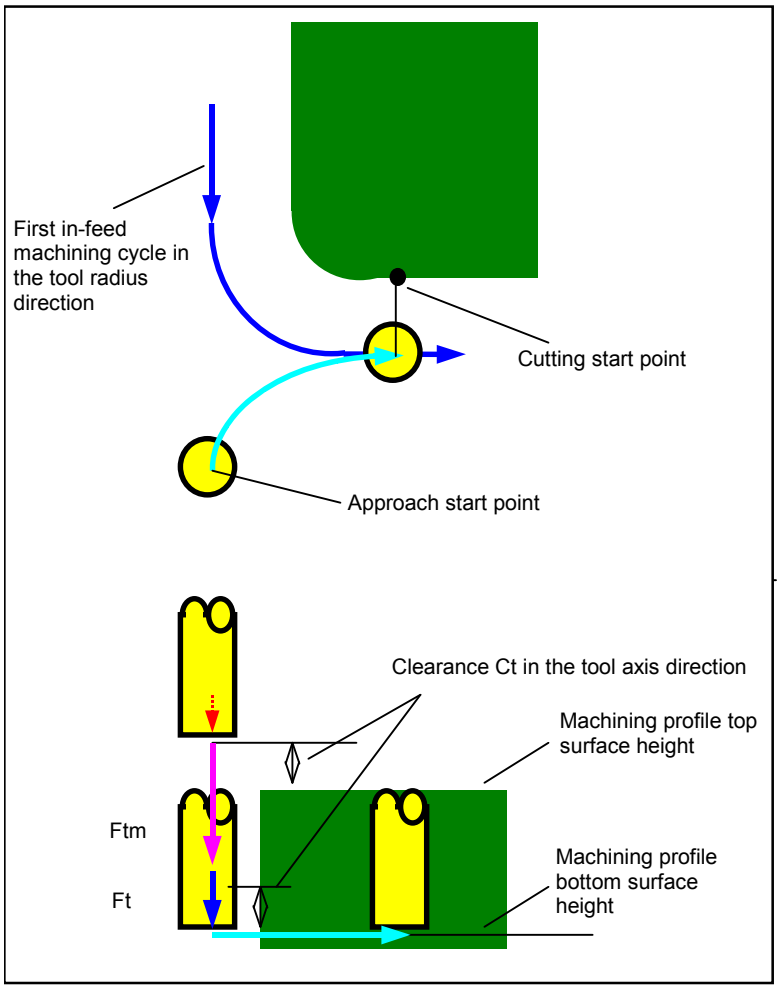
(a) Tool path

The bottom surface of the side-face contour of the machining profile is finished.
The following tool path is created.



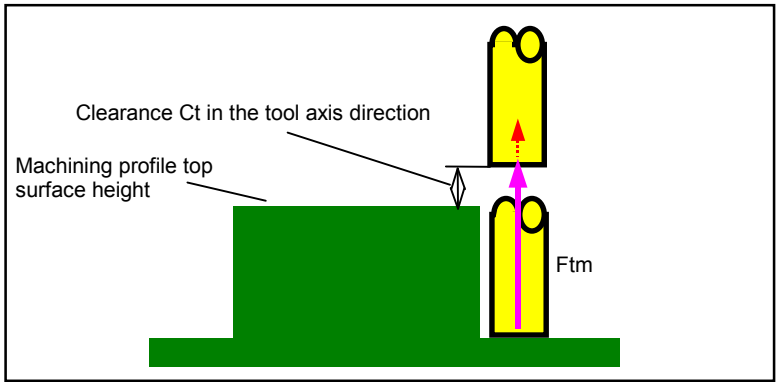
<1> The tool approaches the approach start point of the machining profile.
<2> The tool moves to the bottom surface height of the machining profile.
<3> The tool moves to cut along the side-face contour of the machining profile.
In-feed machining in the tool radius direction is performed until the cutting allowance in the tool radius direction is removed.
<4> The tool retracts.

(i) Approach



<1> The tool moves to the position "machining profile top surface height + clearance (Ct) in the tool axis direction" by rapid traverse.
<2> The tool moves to the position "machining profile bottom surface + cutting allowance (Vt) in the tool axis direction + clearance (Ct) in the tool axis direction" at the feedrate (Ftm) specified for movement in the tool axis direction.
<3> The tool moves to the machining profile bottom surface at the feedrate (Ft) specified for cutting in the tool axis direction.
<4> The tool approaches, in the tool radius direction, the in-feed machining start point in the tool radius direction.

(ii) Retraction



<1> The tool moves from the approach end point to the position "machining profile top height + clearance (Ct) in the tool axis direction" at the feedrate (Ftm) specified for movement in the tool axis direction.

- In-feed machining in the tool radius direction
  This movement is the same as for contouring (roughing).
  See descriptions about contouring (roughing) for details.

(3) Input items of OUTER WALL (SIDE FINISH) (G1062)
Input items of INNER WALL (SIDE FINISH) (G1066)
Input items of PARTIAL (SIDE FINISH) (G1070)

TOOL COND.		
Data item		**Meaning**
I	INPUT TYPE	[INPUT] : Inputs a cutter compensation value directly. [REF.] : Inputs a cutter compensation number to read a cutter compensation value by that number.
D	TOOL DIAMETER	See the descriptions about Roughing: G1060.
M	CUTTER COMP.NO.	Cutter compensation number of an end mill (positive value) Remark) This item is indicated only when [REF.] is selected for INPUT TYPE.

NOTE
Tab ‘TOOL COND.’ is not enabled when bit 3 (CN3) of parameter No. 27040 = 1.

CUT COND.		
Data item		**Meaning**
S	SIDE THICKNESS	See the descriptions about Roughing: G1060.
K*	SIDE FINISH AMOUNT	-
B	NUMBER OF FINISHING	Number of cuts for finishing (positive value) Remark) Depth of each cut = (side surplus thickness)/(number of finishing cuts)
F	FEED RATE- SING.CUT	Feedrate applicable when only the one-side cutter portion of an end mill is used for cutting. This feedrate is used for cutting in retract operation and on the side face other than initial cutting.
V	FEED RATE- BOTH.CUT	-
E	FEED RATE- AXIS	-

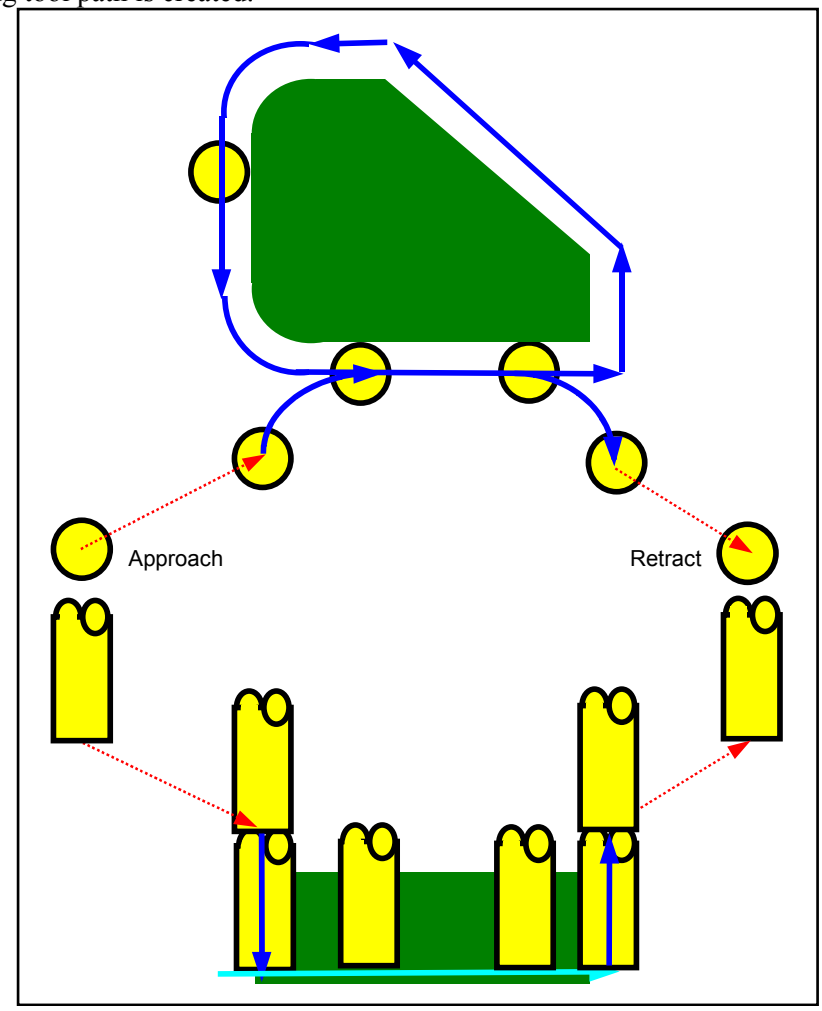
NOTE
When bit 4 (FSP) of parameter No. 27002 is set to 1 to make it possible to set the feedrate for side face finishing for each element, a cutting feedrate that is usually set to "FEED RATE-SINGLE.CUT" specified in the machining type block is set to “SIDE FIN. FEEDRATE” if it is specified.

DETAIL		
Data item		**Meaning**
W	UP CUT/DOWN CUT	See the descriptions about Roughing: G1060.
C	CLEARANCE OF AXIS	-
P	APPROACH TYPE	-
R	APPROACH RAD./DIST.	-
A*	APPROACH ANGLE	-
Q	ESCAPE TYPE	-
X	ESCAPE RAD./DIST.	-

DETAIL		
Data item		Meaning
Y*	ESCAPE ANGLE	-
Z	APPROCH MOTION	-

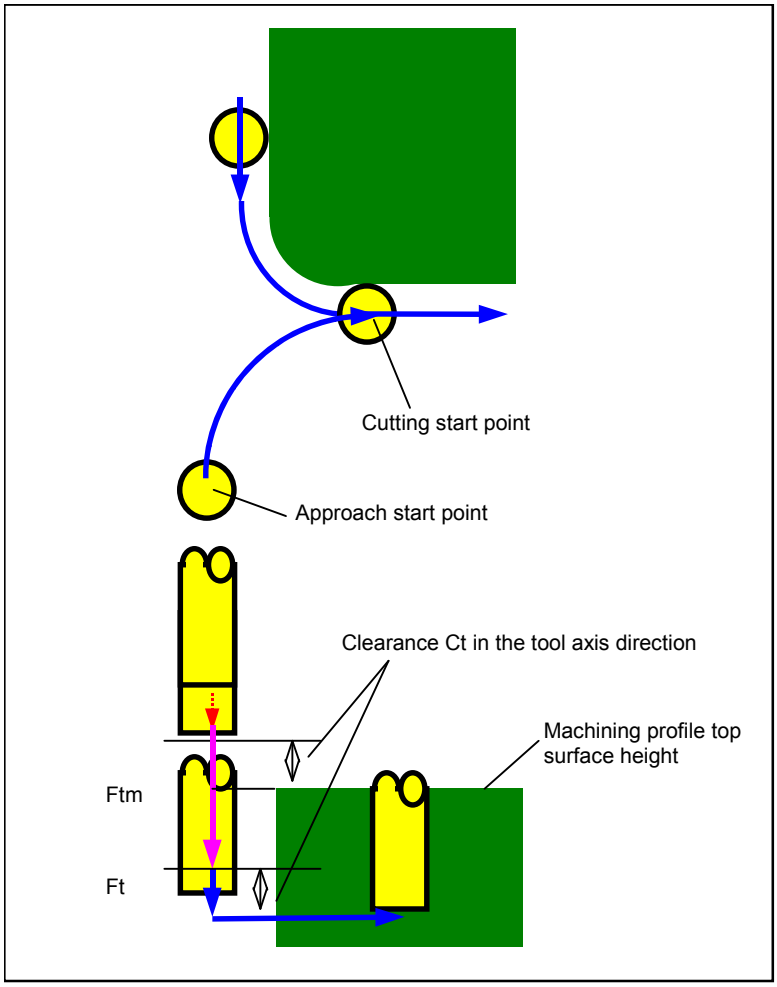
(a) Tool path

The side-face contour of the machining profile is finished.
The following tool path is created.



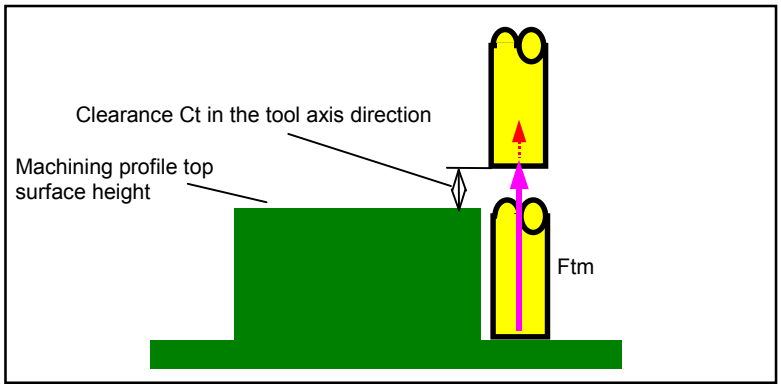
<1> The tool approaches a point above the approach start point.
<2> The tool moves to the bottom surface of the machining profile.
<3> The tool moves to cut along the side-face contour of the machining profile.
In-feed machining is performed for the cutting allowance (Vt) in the tool radius direction a specified number of finishing cycles.
<4> The tool retracts.

(i) Approach



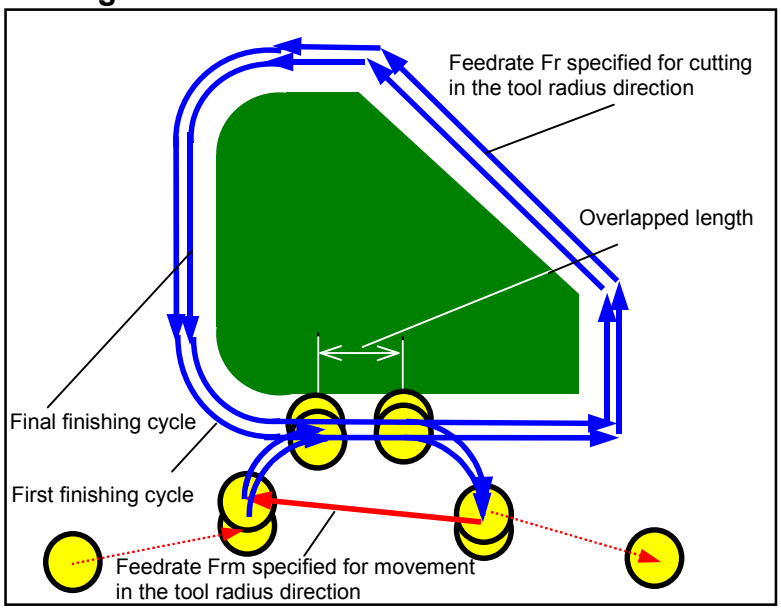
<1> The tool moves to the position "machining profile top surface height + clearance (Ct) in the tool axis direction" by rapid traverse.
<2> The tool moves to the position "machining profile bottom surface + finishing allowance (Tt) in the tool axis direction + clearance (Ct) in the tool axis direction" at the feedrate (Ftm) specified for movement in the tool axis direction.
<3> The tool moves to the position "machining profile bottom surface + finishing allowance (Tt) in the tool axis direction" at the feedrate (Ft) specified for cutting in the machining profile.
<4> The tool approaches, in the tool radius direction, the in-feed machining start point in the tool radius direction.

(ii) Operation



<1> The tool moves from the approach end point to the position "machining profile top surface height + clearance (Ct) in the tool axis direction" at the feedrate (Ftm) specified for movement in the tool axis direction.

(iii) In-feed machining in the tool radius direction



<1> The tool approaches the cutting start point in the tool radius direction at the feedrate (Fr) specified for cutting in the tool radius direction.
<2> The tool moves to cut along the contour from the first in-feed machining cycle start point to the first in-feed machining cycle end point at the feedrate (Fr) specified for cutting in the tool radius direction.
<3> The tool retracts from the cutting end point in the tool radius direction at the feedrate (Fr) specified for cutting in the tool radius direction.
<4> The tool approaches the next in-feed machining cycle start point, using the specified [movement method for in-feed machining].
<5> Steps <2> to <4> are repeated as many times as the required number of finishing cycles.

(4) Input items of OUTER WALL (CHAMFER) (G1063)
Input items of INNER WALL (CHAMFER) (G1067)
Input items of PARTIAL (CHAMFER) (G1071)

TOOL COND.		
Data item		Meaning
K	TOOL SMALL DIAMETER	Diameter of the tip of a chamfering tool (positive value)

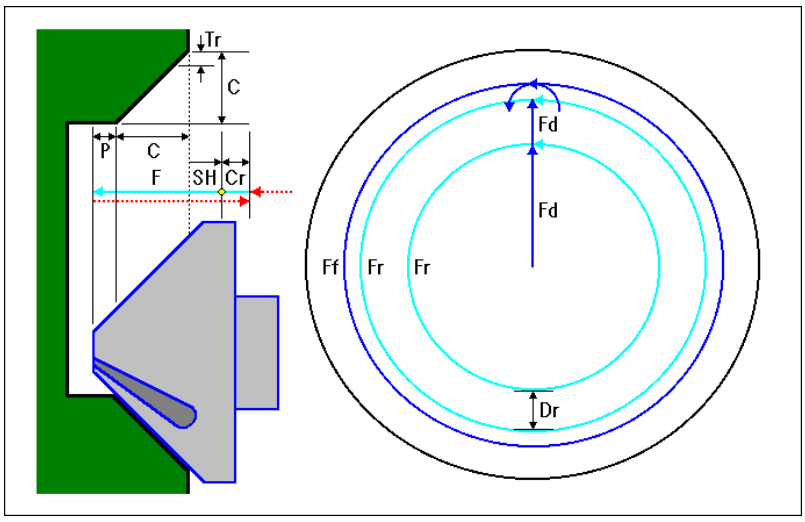
NOTE
1 Tab ‘TOOL COND.’ is enabled when bit 0 (TLG) of parameter No. 27002 = 1.
2 The operator ordinarily sets the above data on the tab of ‘TOOL DATA’ in Tool Offset window. Therefore, it is not necessary to display the tab ‘TOOL COND.’

CUT COND.		
Data item		Meaning
S	CHAMFER	Chamfer length (radius value, positive value)
H	EJECTION STROKE	Distance between the tip of a chamfering tool and an actually cut position in the tool axis direction (radius value, positive value)

CUT COND.		
Data item		Meaning
F	FEED RATE- RADIUS	Feedrate applicable when cutting is performed in the tool radius direction
E	FEED RATE- AXIS	Feedrate applicable when cutting is performed in the tool axis direction

DETAIL		
Data item		Meaning
W	UP CUT/DOWN CUT	See the descriptions about Roughing: G1060.
C	CLEARANCE OF AXIS	-
P	APPROACH TYPE	-
R	APPROACH RAD./DIST.	-
A*	APPROACH ANGLE	-
Q	ESCAPE TYPE	-
X	ESCAPE RAD./DIST.	-
Y*	ESCAPE ANGLE	-
Z	APPROCH MOTION	-

(a) Tool path



<1> The tool moves to the position "cutting start position + clearance (Cr)" by rapid traverse.
<2> The tool moves to the position "amount to be chamfered (C) + ejection stroke (P)" at the feedrate (F) specified for cutting.
<3> The tool cuts in to the portion to be chamfered by the cut depth (Dr) in the tool radius direction at the feedrate (Fd) specified for both-edge thread cutting.
<4> The tool cuts along the contour of the hole machining profile at the feedrate (Fr) specified for single-edge thread cutting.
<5> Steps <3> and <4> are repeated until only the finishing allowance (Tr) in the tool radius direction is left uncut.
<6> The tool turns around the finishing allowance (Tr) in the tool radius direction to perform finishing at the feedrate (Ff) specified for finishing.
<7> The tool moves to the position "cut-in start position + clearance (Cr)" by rapid traverse.

2.3.2 Fixed Form Figure (XY Plane)

(1) Input items of SQUARE (XY plane) (G1220)

POS./SIZE		
Data item		Meaning
T	FIGURE TYPE	[FACE] : Used as a figure for facing [CONVEX] : Used as an outer figure for contouring [CONCAVE]: Used as an inner figure for contouring or as a figure for pocketing [GROOVE] : Used as a figure for grooving Remark) When contouring is selected as machining type, select [CONVEX] if an outer figure is to be machined; if an inner figure is to be machined, select [CONCAVE].
B	BASE POSITION	Z coordinate of the top surface or the bottom of the side face of a part to be subject to contouring (in the tool axis direction)
L	HEIGHT/DEPTH	When the top surface of a workpiece is selected as the BASE POSITION, specify the distance to the bottom of the side face by using a negative value (radius value). → Depth When the bottom of a side face is selected as the BASE POSITION, specify the distance to the top surface of the workpiece by using a positive value (radius value). → Height
H	CENTER POINT (X)	X coordinate of the center position of a rectangular figure
V	CENTER POINT (Y)	Y coordinate of the center position of a rectangular figure
U	LENGTH FOR X AXIS	Length of the side in the X-axis direction (radius value, positive value)
W	LENGTH FOR Y AXIS	Length of the side in the Y-axis direction (radius value, positive value)
R*	CORNER RADIUS	Radius for corner rounding (positive value)
A*	ANGLE	Inclination angle of a rectangular figure relative to the X-axis (positive or negative value)

When bit 0 (P0) of parameter No. 27095 is set to 1, the "BOTTOM THICKNESS" and "SIDE THICKNESS" displayed on the cycle input screen are displayed on the THICKNESS tab of the figure input screen.

(2) Input items of CIRCLE (XY plane) (G1221)

POS./SIZE		
Data item		Meaning
T	FIGURE TYPE	See the descriptions about SQUARE: G1220.
B	BASE POSITION	-
L	HEIGHT/DEPTH	-
H	CENTER POINT (X)	X coordinate of the center position of a circular figure
V	CENTER POINT (Y)	Y coordinate of the center position of a circular figure
R	RADIUS	Radius of a circular figure (radius value, positive value)

When bit 0 (P0) of parameter No. 27095 is set to 1, the “BOTTOM THICKNESS” and “SIDE THICKNESS” displayed on the cycle input screen are displayed on the THICKNESS tab of the figure input screen.

(3) Input items of TRACK (XY plane) (G1222)

POS./SIZE		
Data item		Meaning
T	FIGURE TYPE	See the descriptions about SQUARE: G1220.
B	BASE POSITION	-
L	HEIGHT/DEPTH	-
H	CENTER POINT (X)	X coordinate of the center position of the left semicircle
V	CENTER POINT (Y)	Y coordinate of the center position of the left semicircle
U	BETWEEN DISTANCE	Distance between the centers of the right and left semicircles (radius value, positive value)
R	RADIUS	Radius of the left and right semicircles (radius value, positive value)
A*	ANGLE	Inclination angle of a track figure relative to the X-axis (positive or negative value)

(4) Input items of POLYGON (XY plane) (G1225)

See Subsection 4.2.2, "Fixed Form Figure Blocks (XY Plane)", in "FACING".
This code cannot be specified for part machining.

2.3.3 Fixed Form Figure (YZ Plane)

The same fixed-figure block types as for the XY plane explained in the previous subsection are available for the YZ plane. They are provided with the following G codes.
The data to be set for the YZ planes is the same as for the XY plane except that the XY plane is changed to the YZ plane and that the direction in which the tool moves to cut is changed to the X-axis (YZ plane).

Square : G1320 (YZ plane)
Circle : G1321 (YZ plane)
Track : G1322 (YZ plane)
Polygon : G1325 (YZ plane)

2.3.4 Arbitrary Figure (XY Plane)

When an arbitrary figure for contouring is input, data such as a figure type and machining reference position is specified in the start point block. Other data items to be input such as a straight line and arc are exactly the same as for arbitrary figures of other machining types.
So, this subsection describes only the start point block of an arbitrary figure for contouring.
For other arbitrary figures, see Chapter 5, "DETAILED DESCRIPTIONS ABOUT ENTERING ARBITRARY FIGURES".

(1) Input items of START POINT (XY plane) (G1200)

ELEMENT		
Data item		Meaning
T	FIGURE TYPE	[FACE] : Used as a figure for facing [CONVEX] : Used as an outer figure for contouring [CONCAVE]: Used as an inner figure for contouring and emboss machining or as a figure for pocketing [GROOVE] : Used as a figure for grooving [OPEN] : Used when only one side is cut during contouring Remark) When contouring is specified as a machining type, one of [CONVEX], [CONCAVE], and [OPEN] must be selected.
X	START POINT X	X coordinate of the start point of an arbitrary figure
Y	START POINT Y	Y coordinate of the start point of an arbitrary figure

ELEMENT		
Data item		Meaning
Z	BASE POSITION	Z coordinate of the top surface or the bottom of the side face of a part to be subject to contouring (tool axis direction)
L	HEIGHT/DEPTH	When the top surface of a workpiece is selected as the BASE POSITION, specify the distance to the bottom of the side face by using a negative value (radius value). → Depth When the bottom of a side face is selected as the BASE POSITION, specify the distance to the top surface of the workpiece by using a positive value (radius value). → Height
P	FIGURE ATTRIBUTE	[RIGHT] : The right side of an entered figure as viewed with respect to the direction of movement is cut. (initial value) [LEFT] : The left side of an entered figure as viewed with respect to the direction of movement is cut. (initial value) Remark) These items are displayed only when [OPEN] is selected as a figure type.

(a) Side face finishing feedrate for each figure element

When bit 4 (FSP) of parameter No. 27002 is set to 1, the feedrate for side face finishing can be set for each element.

When the side face finishing feedrate can be set, the input item for the side face finishing feedrate is displayed on the "ATTRIBUTE" tab for each of the figure elements that are straight line, arc, chamfer, and corner rounding elements.

"SIDE FIN. FEEDRATE" is used for every cut when finishing is performed as many times as specified.

(i) Outputting the side face finishing feedrate to a program

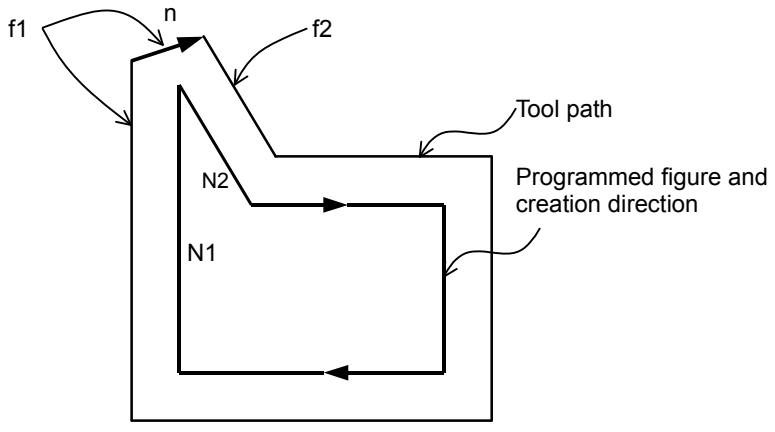
When a figure element is output to a program, the value entered for "SIDE FIN. FEEDRATE" is output following address F in the format shown below. When "SIDE FIN. FEEDRATE" is a space, address F is not output.

(Example of straight lines on an XY plane)
G1201 H-10. V-20. K5. C-10. L0. M0. F50. ;

When 0 or a space is set for "SIDE FIN. FEEDRATE" for an element, the element is cut at "FEED RATE-SING.CUT" set in the machining type block.

(ii) Feedrate applied to a block added during cutting

When an acute-angled figure as shown in the figure below is programmed, a new block may be added as the tool path for the acute-angle corner during cutting. In such a case, the feedrate specified for the previous element when the figure was created is applied to the added block. In the following figure, for example, when f1 is specified as "SIDE FIN. FEEDRATE" for block N1 of a programmed figure, the feedrate of the added block, n, is assumed also to be f1.



(iii) Handling of the side face finishing feedrate at the time of figure copy

When a figure is copied, a space is set for “SIDE FIN. FEEDRATE” of all part elements of a figure generated by the copy operation. When a value needs to be specified, use soft key [ALTER] on the arbitrary figure input screen to enter the value.
A figure copy replaces rounded corners with arc elements, and chamfers with straight line elements. Also in such a case, a space is set for “SIDE FIN. FEEDRATE”.

(iv) Handling of the side face finishing feedrate at the time of NC program conversion

NC program conversion outputs the value specified for “SIDE FIN. FEEDRATE” following address F to the cutting block generated as a result of the conversion.
If 0 or a space is set for “SIDE FIN. FEEDRATE”, the value of "FEED RATE-SING.CUT" specified in the machining type block is output.

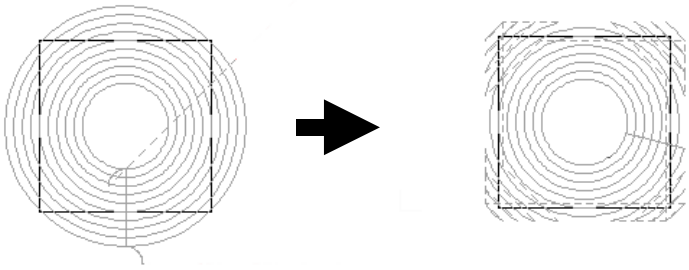
2.3.5 Arbitrary Figure (YZ Plane)

The same arbitrary-figure block types as for the XY plane explained in the previous subsection are available for the YZ plane. Their start points are specified with the following G codes.
The data to be set for the YZ planes is the same as for the XY plane except that the XY plane is changed to the YZ plane and that the direction in which the tool moves to cut is changed to the X-axis (YZ planes).

Start point : G1300 (YZ plane)

2.4 EMBOSS MACHINING

In the case of the contouring, the tool cuts along the side-face contour of the machining profile and performs in-feed machining in the tool radius direction. These tool passes sometimes generate many air-cut movement as the following left figure.
So, the machining called “EMBOSS MACHINING CYCLE” that can reduce the air-cut passes as the following right figure is prepared.



NOTE
1 This function is available when bit 7 (MC7) of parameter No.27000 = 1.
2 The parameters for the machining conditions for embossing cycles are same with the parameters for pocketing cycles. (Ex. No.27066)

2.4.1 Machining Command

(1) Input items of ROUGH (G1080)

TOOL COND.		
Data item		Meaning
D	TOOL DIAMETER	End mill diameter

NOTE
1 Tab 'TOOL COND.' is enabled when bit 0 (TLG) of parameter No. 27002 = 1.
2 The operator ordinarily sets the above data on the tab of 'TOOL DATA' in Tool Offset window. Therefore, it is not necessary to display the tab 'TOOL COND.'

CUT COND.		
Data item		**Meaning**
T*	BOTTOM THICKNESS (*1)	Cutting allowance of the bottom in side face machining. The blank is regarded as 0. (radius value, positive value)
S*	SIDE THICKNESS (*1)	Cutting allowance of the side face. The blank is regarded as 0. (radius value, positive value) Remark) When both the side surplus thickness and bottom surplus thickness are omitted, an entire pocket area is cut.
L (*2)	CUT DEPTH OF RADIUS	Depth of cut on the side face (tool radius direction) per cutting operation (radius value, positive value)
J (*2)	CUT DEPTH OF AXIS	Depth of cut in the tool axis direction per cutting operation (radius value, positive value)
K* (*2)	SIDE FINISH AMOUNT	Finishing allowance on the side face. The blank is regarded as 0. (radius value, positive value)
H* (*2)	BOTTOM FINISHI AMT.	Finishing allowance at the bottom in side face machining. The blank is regarded as 0. (radius value, positive value)
F	FEED RATE- SING.CUT	Feedrate applicable when only the one-side cutter portion of an end mill is used for cutting. This feedrate is used for cutting in retract operation and on the side face other than initial cutting.
V	FEED RATE- BOTH.CUT	Feedrate applicable when the entire front side of an end mill is used for cutting. This feedrate is used for initial cutting.
E	FEED RATE- AXIS	Feedrate applicable when cutting is performed in the tool axis direction toward the bottom of a side face being machined

(*1) These items can be entered on the figure input screen. For details, see Subsection 4.3.1, "Machining Command", in Section 4.3, "CONTOURING".
(*2) When bit 5 (RCF) of parameter No. 27002 is set to 1, the displayed order is changed by exchanging the input items "CUT DEPTH OF RADIUS" and "CUT DEPTH OF AXIS" and the input items "SIDE FINISH AMOUNT" and "BOTTOM FINISH AMT." so that the items related to workpiece dimensions are displayed collectively.

DETAIL		
Data item		**Meaning**
W	UP CUT/DOWN CUT	[UP CUT] : Performs machining in up-cut mode, assuming that the tool is rotating clockwise. [DWNCUT] : Performs machining in down-cut mode, assuming that the tool is rotating clockwise. (COPY)
B	CLEARANCE OF RADIUS	Distance between the side face and a tool retract position in the tool radius direction (radius value, positive value) Remark1) When one pocket cutting operation is completed, the tool performs a retract operation in the tool axis direction from the side face of the pocket by this clearance amount. Remark2) By referring to the parameter No.27009 (minimum clamp value), the system sets the data as Initial value.
C	CLEARANCE OF AXIS	Distance between the surface of a blank being machined and a cutting start point (point R) in the tool axis direction (radius value, positive value) Remark) By referring to the parameter No.27009 (minimum clamp value), the system sets the data as Initial value.

DETAIL		
Data item		Meaning
Z	APPROCH MOTION	[2 AXES] : When moving from the current position to the machining start point, the tool first moves in the machining plane in two-axis synchronous operation and then moves along the tool axis. (initial value) [3 AXES] : The tool moves from the current position to the machining start point in three-axis synchronous operation. Remark) This input item is not displayed when the option for expanding the number of simultaneously controlled axes is not provided.

(2) Input items of BOTTOM FINISH (G1081)

TOOL COND.		
Data item		Meaning
D	TOOL DIAMETER	See the descriptions about Roughing: G1080.

NOTE
1 Tab 'TOOL COND.' is enabled when bit 0 (TLG) of parameter No. 27002 = 1.
2 The operator ordinarily sets the above data on the tab of 'TOOL DATA' in Tool Offset window. Therefore, it is not necessary to display the tab 'TOOL COND.'

CUT COND.		
Data item		Meaning
T*	BOTTOM THICKNESS	See the descriptions about Roughing: G1080.
K*	SIDE FINISH AMOUNT	-
H*	BOTTOM FINISHI AMT.	-
L	CUT DEPTH OF RADIUS	-
F	FEED RATE- SING.CUT	-
V	FEED RATE- BOTH.CUT	-
E	FEED RATE- AXIS	-

DETAIL		
Data item		Meaning
W	UP CUT/DOWN CUT	See the descriptions about Roughing: G1080.
B	CLEARANCE OF RADIUS	-
C	CLEARANCE OF AXIS	-
Z	APPROCH MOTION	-

(3) Input items of SIDE FINISH (G1082)

TOOL COND.		
Data item		Meaning
I	INPUT	[INPUT] : Inputs a cutter compensation value directly. [REF.] : Inputs a cutter compensation number to read a cutter compensation value by that number.
D	TOOL DIAMETER	End mill diameter (positive value) Remark) This item is indicated only when [INPUT] is selected for INPUT.
M	CUTTER COMP.NO.	Cutter compensation number of an end mill (positive value) Remark) This item is indicated only when [REF.] is selected for INPUT.

CUT COND.		
Data item		Meaning
K*	SIDE FINISH AMOUNT	See the descriptions about Roughing: G1080.
H*	BOTTOM FINISHI AMT.	-
F	FEED RATE- SING.CUT	-

CUT COND.		
Data item		Meaning
V	FEED RATE- BOTH.CUT	-
E	FEED RATE- AXIS	-

NOTE
When bit 4 (FSP) of parameter No. 27002 is set to 1 to make it possible to set the feedrate for side face finishing for each element, a cutting feedrate that is usually set to "FEED RATE-SINGLE.CUT" specified in the machining type block is set to “SIDE FIN. FEEDRATE” if it is specified.

DETAIL		
Data item		Meaning
W	UP CUT/DOWN CUT	See the descriptions about Roughing: G1080.
C	CLEARANCE OF AXIS	-
P	APPROACH TYPE	[ARC] : Approaches a side face along an arc. [TANGEN]: Approaches a side face along the straight line tangent to the first figure in side face cutting. [VERTIC] : Approaches a side face along the straight line normal to the first figure in side face cutting. (COPY)
R	APPROACH RAD./DIST.	Radius when [ARC] is specified. Straight line length when [TANGEN] or [VERTIC] is specified. (radius value, positive value) Remark) By referring to the parameter No.27010 (minimum clamp value), the system sets the data as Initial value.
A*	APPROACH ANGLE	Center angle of the arc when [ARC] is specified. The default is 90 degrees. (positive value) Remark) This data item is indicated only when [ARC] is selected for APPROACH TYPE. (COPY)
Q	ESCAPE TYPE	[ARC] : Retracts from a side face along an arc. [TANGEN]: Retracts from a side face along the straight line tangent to the last figure in side face cutting. [VERTIC] : Retracts from a side face along the straight line normal to the last figure in side face cutting. (COPY)
X	ESCAPE RAD./DIST.	Radius when [ARC] is specified. Straight line length when [TANGEN] or [VERTIC] is specified. (radius value, positive value) Remark) By referring to the parameter No.27010 (minimum clamp value), the system sets the data as Initial value.
Y*	ESCAPE ANGLE	Center angle of the arc when [ARC] is specified. The default is 90 degrees. (positive value) Remark) This data item is indicated only when [ARC] is selected for ESCAPE TYPE. (COPY)
Z	APPROCH MOTION	[2 AXES]: When moving from the current position to the machining start point, the tool first moves in the machining plane in two-axis synchronous operation and then moves along the tool axis. (initial value) [3 AXES]: The tool moves from the current position to the machining start point in three-axis synchronous operation.

(4) Input items of CHAMFER (G1083)

TOOL COND.		
Data item		Meaning
K	TOOL SMALL DIAMETER	Diameter of the tip of a chamfering tool (positive value)

CUT COND.		
Data item		Meaning
S	CHAMFER AMOUNT	Chamfer length (radius value, positive value)
H	EJECTION STROKE	Distance between the tip of a chamfering tool and an actually cut position in the tool axis direction (radius value, positive value)
F	FEED RATE - RADIUS	Feedrate applicable when cutting is performed in the tool radius direction
E	FEED RATE - AXIS	Feedrate applicable when cutting is performed in the tool axis direction

DETAIL		
Data item		Meaning
W	UP CUT/DOWN CUT	See the descriptions about Roughing: G1080.
C	CLEARANCE OF AXIS	-
P	APPROACH TYPE	See the descriptions about Side finishing: G1082.
R	APPROACH RAD./DIST.	-
A*	APPROACH ANGLE	-
Q	ESCAPE TYPE	-
X	ESCAPE RAD./DIST.	-
Y*	ESCAPE ANGLE	-
Z	APPROCH MOTION	-

2.4.2 Arbitrary Figure (XY Plane)

When an arbitrary figure for emboss machining is input, data such as a figure type and machining reference position is specified in the start point block. Other data items to be input such as a straight line and arc are exactly the same as for arbitrary figures of other machining types.
So, this subsection describes only the start point block of an arbitrary figure for pocketing.
For other arbitrary figures, see Chapter 5, "DETAILED DESCRIPTIONS ABOUT ENTERING ARBITRARY FIGURES".

(1) Input items of START POINT (XY plane) (G1200)

ELEMENT		
Data item		Meaning
T	FIGURE TYPE	[FACE] : Used as a figure for facing [CONVEX] : Used as an outer figure for contouring [CONCAVE]: Used as an inner figure for contouring and emboss machining or as a figure for pocketing [GROOVE] : Used as a figure for grooving Remark) When emboss machining is selected as machining type, be sure to select [CONCAVE].
X	START POINT X	X coordinate of the start point of an arbitrary figure
Y	START POINT Y	Y coordinate of the start point of an arbitrary figure
B	BASE POSITION	Z coordinate of the top surface of a workpiece subject to pocketing or the bottom of a pocket (in the tool axis direction)

ELEMENT		
Data item		Meaning
L	HEIGHT/DEPTH	When the top surface of a workpiece is selected as the BASE POSITION, specify the distance to the bottom of the side face by using a negative value (radius value). → Depth When the bottom of the side face is selected as the BASE POSITION, specify the distance to the top surface of the pocket by using a positive value (radius value). → Height

(a) Side face finishing feedrate for each figure element

When bit 4 (FSP) of parameter No. 27002 is set to 1, the feedrate for side face finishing can be set for each element.
When the side face finishing feedrate can be set, the input item for the side face finishing feedrate is displayed on the "ATTRIBUTE" tab for each of the figure elements that are straight line, arc, chamfer, and corner rounding elements.
For details, see Subsection 4.3.4, "Arbitrary Figure (XY Plane)", in Section 4.3, "CONTOURING".

2.4.3 Arbitrary Figure (YZ plane)

The same arbitrary-figure block types as for the XY plane explained in the previous subsection are available for the YZ plane. Their start points are specified with the following G codes.
The data to be set for the YZ planes is the same as for the XY plane except that the XY plane is changed to the YZ plane and that the direction in which the tool moves to cut is changed to the X-axis (YZ planes).

Start point : G1300 (YZ plane)

2.5 POCKETING

2.5.1 Machining Command

(1) Input items of ROUGH (G1040)

TOOL COND.		
Data item		Meaning
D	TOOL DIAMETER	End mill diameter

NOTE
1 Tab 'TOOL COND.' is enabled when bit 0 (TLG) of parameter No. 27002 = 1.
2 The operator ordinarily sets the above data on the tab of 'TOOL DATA' in Tool Offset window. Therefore, it is not necessary to display the tab 'TOOL COND.'

CUT COND.		
Data item		Meaning
T*	BOTTOM THICKNESS (*1)	Cutting allowance of the bottom in side face machining. The blank is regarded as 0. (radius value, positive value)
S*	SIDE THICKNESS (*1)	Cutting allowance of the side face. The blank is regarded as 0. (radius value, positive value) Remark) When both the side surplus thickness and bottom surplus thickness are omitted, an entire pocket area is cut.
L	CUT DEPTH OF RADIUS (*2)	Depth of cut on the side face (tool radius direction) per cutting operation (radius value, positive value)

CUT COND.		
Data item		Meaning
J	CUT DEPTH OF AXIS (*2)	Depth of cut in the tool axis direction per cutting operation (radius value, positive value)
K*	SIDE FINISH AMOUNT (*2)	Finishing allowance on the side face. The blank is regarded as 0. (radius value, positive value)
H*	BOTTOM FINISHI AMT. (*2)	Finishing allowance at the bottom in side face machining. The blank is regarded as 0. (radius value, positive value)

(*1) These items can be entered on the figure input screen. For details, see Subsection 4.3.1, "Machining Command", in Section 4.3, "CONTOURING".
(*2) When bit 5 (RCF) of parameter No. 27002 is set to 1, the displayed order is changed by exchanging the input items "CUT DEPTH OF RADIUS" and "CUT DEPTH OF AXIS" and the input items "SIDE FINISH AMOUNT" and "BOTTOM FINISH AMT." so that the items related to workpiece dimensions are displayed collectively.

CUT COND.		
Data item		Meaning
F	FEED RATE- SING.CUT	Feedrate applicable when only the one-side cutter portion of an end mill is used for cutting. This feedrate is used for cutting in retract operation and on the side face other than initial cutting.
V	FEED RATE- BOTH.CUT	Feedrate applicable when the entire front side of an end mill is used for cutting. This feedrate is used for initial cutting.
E	FEED RATE- AXIS	Feedrate applicable when cutting is performed in the tool axis direction toward the bottom of a side face being machined

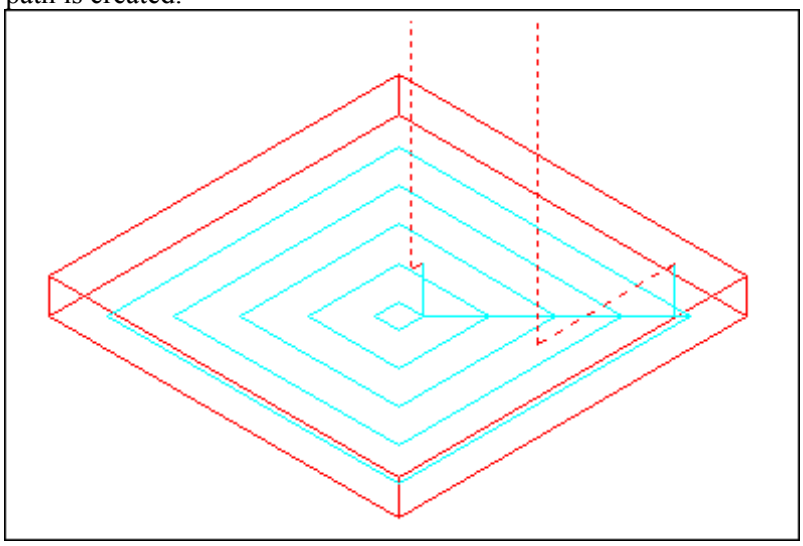
NOTE
When a non-zero value is input for just one of the data items, side surplus thickness and bottom surplus thickness, cutting is performed for the input surplus thickness, and for the omitted surplus thickness, cutting operation is not performed on the assumption that the omitted surplus thickness is 0.
When non-zero values are input for both the side surplus thickness and bottom surplus thickness, cutting is performed for each surplus thickness.

DETAIL		
Data item		Meaning
W	UP CUT/DOWN CUT	[UP CUT] : Performs machining in up-cut mode, assuming that the tool is rotating clockwise. [DWNCUT] : Performs machining in down-cut mode, assuming that the tool is rotating clockwise. (COPY)
B	CLEARANCE OF RADIUS	Distance between the wall of a pocket and a tool retract position in the tool radius direction (radius value, positive value) Remark1) When one pocket cutting operation is completed, the tool performs a retract operation in the tool axis direction from the side face of the pocket by this clearance amount. Remark2) By referring to the parameter No.27009 (minimum clamp value), the system sets the data as Initial value.
C	CLEARANCE OF AXIS	Distance between the surface of a blank being machined and a cutting start point (point R) in the tool axis direction (radius value, positive value) Remark) By referring to the parameter No.27009 (minimum clamp value), the system sets the data as Initial value.

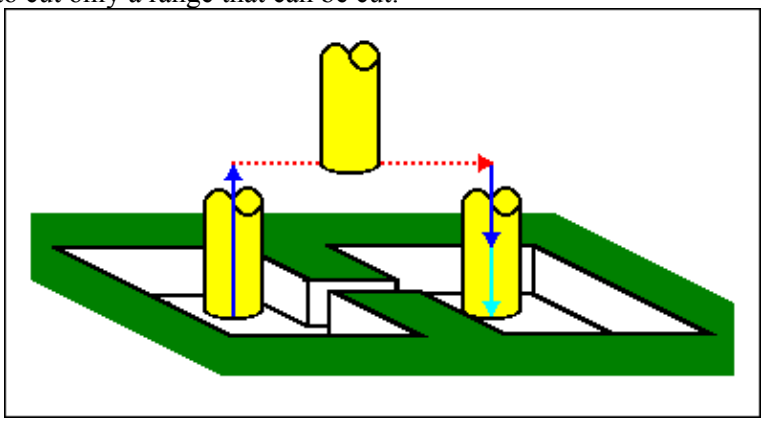
DETAIL		
Data item		Meaning
Z	APPROCH MOTION	[2 AXES] : When moving from the current position to the machining start point, the tool first moves in the machining plane in two-axis synchronous operation and then moves along the tool axis. (initial value) [3 AXES] : The tool moves from the current position to the machining start point in three-axis synchronous operation. Remark) This input item is not displayed when the option for expanding the number of simultaneously controlled axes is not provided.
X	CUT DEPTH METHOD	[STRGHT]: Plunging straightly to the pocket area [HELICAL]: Plunging helically to the pocket area In the following plane, [HELICAL] can be used. (1) XY-plane (2) YZ-plane In the following plane, [HELICAL] cannot be used, because helical interpolation function cannot be used. (1) ZC-plane (cylindrical surface) (C-axis is parallel to Y-axis) (2) XC-plane (polar coordinate interpolation plane) (C-axis is parallel to Y-axis) (3) XA-plane (cylindrical surface) (C-axis is parallel to Y-axis)
A*	CUT ANGLE	Angle at which the tool cuts into the pocket obliquely or helically. The blank is regarded as 0 degrees. (in one-degree increments, positive value) (COPY) Remark) By performing cutting also using the side face of an end mill, the load on the tool can be reduced.
R	HELICAL RADIUS	The radius of helical circle when a tool plunges helically
M	HELICAL FEED RATE	The feed rate when a tool plunges helically
P*	START PT.(1ST AXIS)	1st-axis coordinate of the cutting start point of pocketing. When omitting this item, also omit the 2nd-axis coordinate. In this case, the coordinates of the start point are determined automatically. Remark1) The 1st axis is the X-axis on the XY plane, the Y-axis on the YZ plane. Remark2) This item is enable when the bit 7 (PR7) of parameter No. 27060 = 1.
Q*	START PT.(2ND AXIS)	2nd-axis coordinate of the cutting start point of pocketing. When this item is omitted, the coordinate is determined automatically. Remark1) The 2nd axis is the Y-axis on the XY plane, the Z-axis on the YZ plane. Remark2) This item is enable when the bit 7 (PR7) of parameter No. 27060 = 1.

(a) Tool path

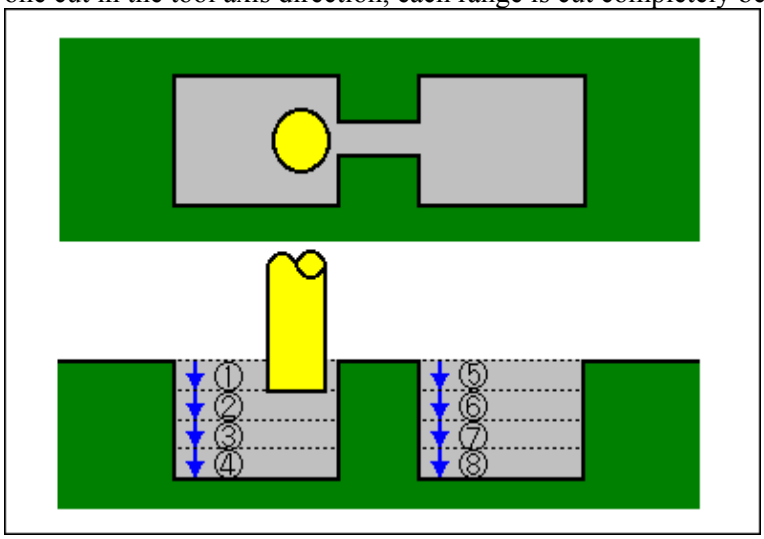
The inside of a pocket machining profile is cut off in a spiral manner.
The following tool path is created.



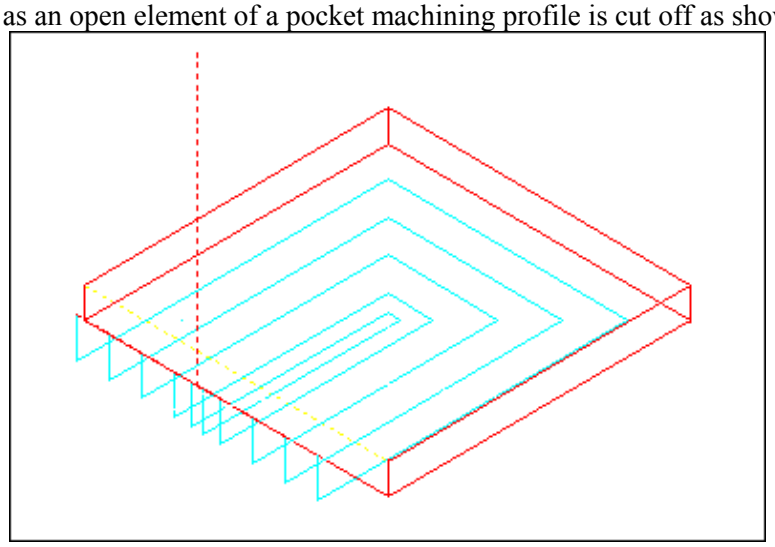
More than one island machining profile and more than one cavity machining profile can be defined for a pocket machining profile. The island machining profiles are left uncut.
The cavity machining profiles are detoured so that they will not be cut.
The tool path is created in such a way that an anticipated interference with pocket machining profiles or island machining profiles can be avoided.
The created tool path is effective because retraction in the tool axis direction is avoided as much as possible.
For this tool path, in-feed machining in the tool axis direction is possible.
Only specified cutting allowances can be cut off.
The direction of cutting can be either up-cut or down-cut. The direction of cutting is controlled automatically around islands.
It is possible to cut in from the inside and outside of the machining profile.
If there is a portion left uncut at a corner, it is possible to automatically discriminate the uncut portion and cut it off.
It is possible to cut in at an arbitrary angle in the tool axis direction.
It is possible to specify an arbitrary position as a cutting start point.
It is possible to determine a cutting start point automatically.
It is possible to select a movement method for tool movement.
It is possible to control automatically a cut-in depth on the island machining profile top surface automatically.
In the following pocket machining profile, which has a pocket through which the tool can pass, the tool is lifted automatically to cut only a range that can be cut.



If there is more than one cut in the tool axis direction, each range is cut completely before another.



A contour specified as an open element of a pocket machining profile is cut off as shown below.



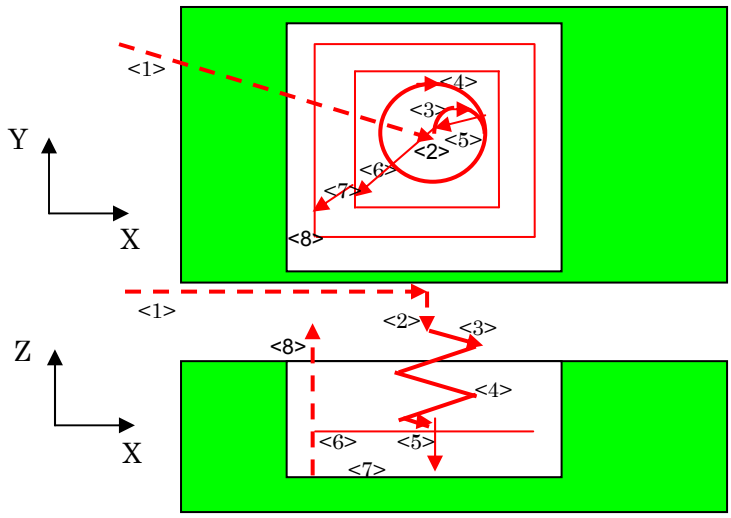
In case “HELICAL” is specified in “CUT DEPTH METHOD”, machining motion is as follows.

<1> Tool moves rapidly the position “Start point of the 1st axis (P), Start point of the 2nd axis (Q)” to the tool radius direction (If the start points (P,Q) are not specified, start points decided automatically are used).
<2> Tool moves rapidly the position “Base position (B) + clearance (C)” to the tool axis direction.
* In case the cutting motion is after the 2nd time, after the <2> motion, tool moves the position “Base position(B) + Clearance(C) + (Cutting depth to the tool axis direction(J) x (Cutting times -1))”
<3> In case of approaching helically with the cutting feed rate (M), tool moves “Start point of the 1st axis(P) + Radius of helical(R)” to the tool radius direction, and moves by the “Depth of cutting in 1 helical motion/4” to the tool axis direction.
<4> Tool plunges helically with the cutting feed rate (M) to the position “Base position (B) – (“Cutting depth to the tool axis direction (J)” x (Cutting times – 1))” with cutting angle (A) and helical radius (R).
<5> Tool moves the position “Start point of the 1st axis (P), Start point of the 2nd axis (Q)” to the tool radius direction with the feed rate (V).
<6> Tool moves the initial start point of the pocket area to the tool radius direction with the feed rate (V).
<7> Tool cuts the pocket area to the tool radius direction at the feed rate (F,V) with the cutting depth of the tool radius direction (L).

<8> After the completion of machining the pocket area, tool escapes the position “Base position (B) + Clearance (C)” to the tool axis direction with the feed rate (E).
<9> Repeating the step <1> to <8> until the position “Base position (B) + High/Depth (L)”

NOTE
In order to use for specifying helical plunge, the optional function “Helical interpolation” is needed.

Example) In case of XP-plane



(2) Input items of BOTTOM FINISH (G1041)

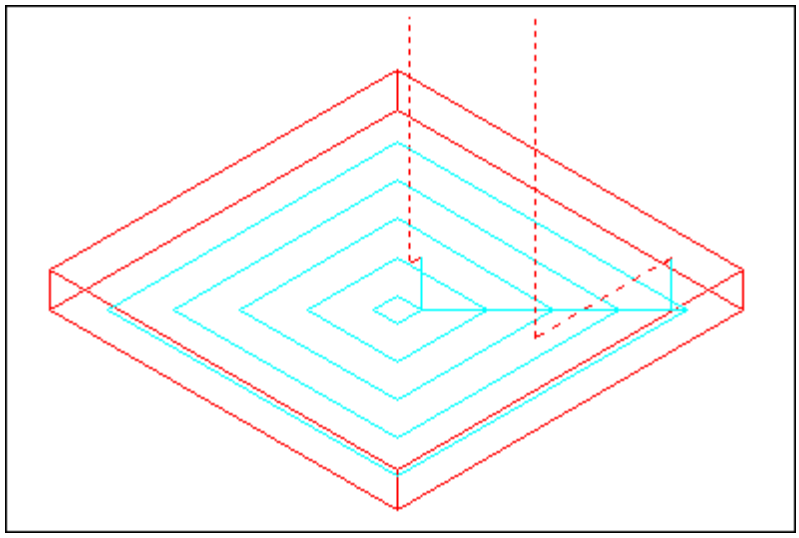
TOOL COND.		
Data item		Meaning
D	TOOL DIAMETER	See the descriptions about Roughing: G1040.

CUT COND.		
Data item		Meaning
T*	BOTTOM THICKNESS	See the descriptions about Roughing: G1040.
K*	SIDE FINISH AMOUNT	-
H*	BOTTOM FINISHI AMT.	-
L	CUT DEPTH OF RADIUS	-
F	FEED RATE- SING.CUT	-
V	FEED RATE- BOTH.CUT	-
E	FEED RATE- AXIS	-

DETAIL		
Data item		Meaning
W	UP CUT/DOWN CUT	See the descriptions about Roughing: G1040.
B	CLEARANCE OF RADIUS	-
C	CLEARANCE OF AXIS	-
Z	APPROCH MOTION	-

DETAIL		
Data item		Meaning
X	CUT DEPTH METHOD	[STRGHT]: Plunging straightly to the pocket area [HELICAL]: Plunging helically to the pocket area In the following plane, [HELICAL] can be used. (1) XY-plane (2) YZ-plane In the following plane, [HELICAL] cannot be used, because helical interpolation function cannot be used. (1) ZC-plane (cylindrical surface) (C-axis is parallel to Y-axis) (2) XC-plane (polar coordinate interpolation plane) (C-axis is parallel to Y-axis) (3) XA-plane (cylindrical surface) (C-axis is parallel to Y-axis)
R	HELICAL RADIUS	The radius of helical circle when a tool plunges helically
M	HELICAL FEED RATE	The feed rate when a tool plunges helically
A*	CUT ANGLE	Angle at which the tool cuts into the pocket obliquely or helically. The blank is regarded as 0 degrees.
P*	START PT.(1ST AXIS)	-
Q*	START PT.(2ND AXIS)	-

(a) Tool path



The bottom surface of a pocket machining profile is finished in a spiral manner.
The tool path for it is the same as for pocketing (roughing).
See descriptions about pocketing (roughing) for details.
No in-feed machining in the tool axis direction is performed, though.
The top surface of island machining profiles is not cut off either.

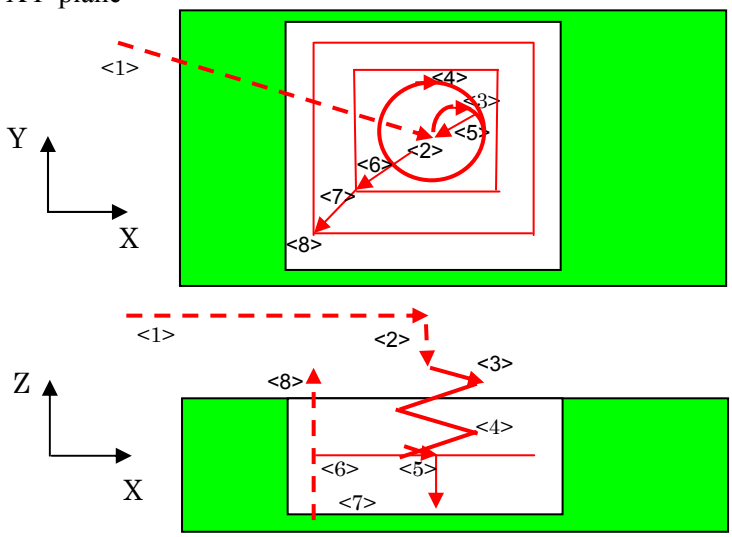
In case “HELICAL” is specified in “CUT DEPTH METHOD”, machining motion is as follows.

<1> Tool moves rapidly the position “Start point of the 1st axis (P), Start point of the 2nd axis (Q)” to the tool radius direction (If the start points (P,Q) are not specified, start points decided automatically are used).
<2> Tool moves rapidly the position “Base position (B) + clearance (C)” to the tool axis direction.
<3> In case of approaching helically with the cutting feed rate (M), tool moves “Start point of the 1st axis (P) + Radius of helical (R)” to the tool radius direction, and moves by the “Depth of cutting in 1 helical motion/4” to the tool axis direction.

<4> Tool plunges helically at cutting feed rate (M) to the position "Base position (B) – Bottom thickness (T)" with cutting angle (A) and helical radius (R).
<5> Tool moves the position "Start point of the 1st axis (P), Start point of the 2nd axis (Q)" to the tool radius direction at the feed rate (V).
<6> Tool moves the initial start point of the pocket area to the tool radius direction at the feed rate (V).
<7> Tool cuts the pocket area to the tool radius direction at the feed rate (F,V) with the cutting depth of the tool radius direction (L).
<8> After the completion of machining the pocket area, tool escapes the position "Base position (B) + Clearance (C)" to the tool axis direction with the feed rate (E).
<9> Repeating the step <1> to <8> until the position "Base position (B) + High/Depth (L)"

NOTE
In order to use for specifying helical plunge, the optional function "Helical interpolation" is needed.

Example) In case of XY-plane



(3) Input items of SIDE FINISH (G1042)

TOOL COND.		
Data item		**Meaning**
I	INPUT	[INPUT] : Inputs a cutter compensation value directly. [REF.] : Inputs a cutter compensation number to read a cutter compensation value by that number.
D	TOOL DIAMETER	End mill diameter (positive value) Remark) This item is indicated only when [INPUT] is selected for INPUT.
M	CUTTER COMP.NO.	Cutter compensation number of an end mill (positive value) Remark) This item is indicated only when [REF.] is selected for INPUT.

NOTE
Tab 'TOOL COND.' is not enabled when bit 3 (PF3) of parameter No. 27061 = 1.

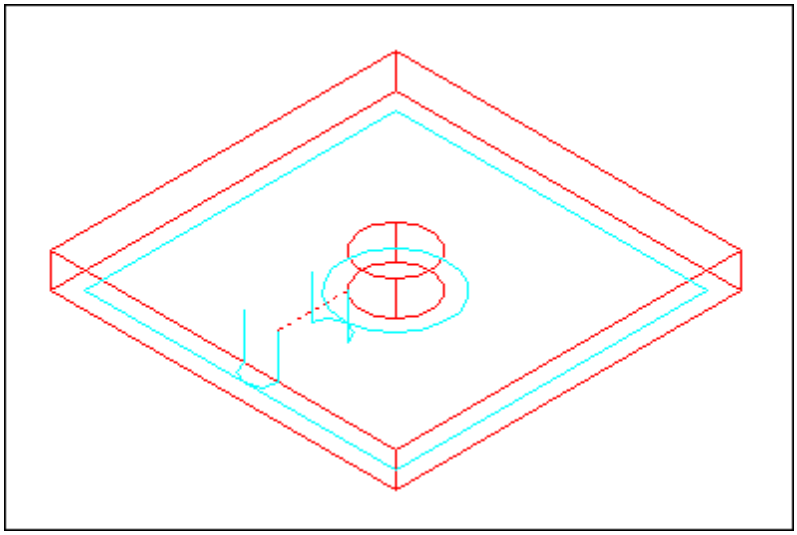
CUT COND.		
Data item		**Meaning**
K*	SIDE FINISH AMOUNT	See the descriptions about Roughing: G1040.
H*	BOTTOM FINISHI AMT.	-
F	FEED RATE- SING.CUT	-

CUT COND.		
Data item		Meaning
V	FEED RATE- BOTH.CUT	-
E	FEED RATE- AXIS	-

NOTE
When bit 4 (FSP) of parameter No. 27002 is set to 1 to make it possible to set the feedrate for side face finishing for each element, a cutting feedrate that is usually set to "FEED RATE-SINGLE.CUT" specified in the machining type block is set to “SIDE FIN. FEEDRATE” if it is specified.

DETAIL		
Data item		Meaning
W	UP CUT/DOWN CUT	See the descriptions about Roughing: G1040.
C	CLEARANCE OF AXIS	-
P	APPROACH TYPE	[ARC] : Approaches a side face along an arc. [TANGEN]: Approaches a side face along the straight line tangent to the first figure in side face cutting. [VERTIC] : Approaches a side face along the straight line normal to the first figure in side face cutting. (COPY)
R	APPROACH RAD./DIST.	Radius when [ARC] is specified. Straight line length when [TANGEN] or [VERTIC] is specified. (radius value, positive value) Remark) By referring to the parameter No.27010 (minimum clamp value), the system sets the data as Initial value.
A*	APPROACH ANGLE	Center angle of the arc when [ARC] is specified. The default is 90 degrees. (positive value) Remark) This data item is indicated only when [ARC] is selected for APPROACH TYPE. (COPY)
Q	ESCAPE TYPE	[ARC] : Retracts from a side face along an arc. [TANGEN]: Retracts from a side face along the straight line tangent to the last figure in side face cutting. [VERTIC] : Retracts from a side face along the straight line normal to the last figure in side face cutting. (COPY)
X	ESCAPE RAD./DIST.	Radius when [ARC] is specified. Straight line length when [TANGEN] or [VERTIC] is specified. (radius value, positive value) Remark) By referring to the parameter No.27010 (minimum clamp value), the system sets the data as Initial value.
Y*	ESCAPE ANGLE	Center angle of the arc when [ARC] is specified. The default is 90 degrees. (positive value) Remark) This data item is indicated only when [ARC] is selected for ESCAPE TYPE. (COPY)
Z	APPROCH MOTION	[2 AXES] : When moving from the current position to the machining start point, the tool first moves in the machining plane in two-axis synchronous operation and then moves along the tool axis. (initial value) [3 AXES] : The tool moves from the current position to the machining start point in three-axis synchronous operation.

(a) Tool path



The side-face contour of pocket and island machining profiles is finished. The tool path for it is the same as for contouring (side-face finishing).
See descriptions about contouring (side-face finishing) for details.

The specifications for the following points differ in part, though.
No in-feed machining is performed in the tool radius direction or tool axis direction.
Even if it is anticipated that the tool may interfere with a pocket or island machining profile during finishing, no tool path that can avoid interference is created.

(4) Input items of CHAMFER (G1043)

TOOL COND.		
Data item		Meaning
K	TOOL SMALL DIAMETER	Diameter of the tip of a chamfering tool (positive value)

CUT COND.		
Data item		Meaning
S	CHAMFER AMOUNT	Chamfer length (radius value, positive value)
H	EJECTION STROKE	Distance between the tip of a chamfering tool and an actually cut position in the tool axis direction (radius value, positive value)
F	FEED RATE - RADIUS	Feedrate applicable when cutting is performed in the tool radius direction
E	FEED RATE - AXIS	Feedrate applicable when cutting is performed in the tool axis direction

DETAIL		
Data item		Meaning
W	UP CUT/DOWN CUT	See the descriptions about Roughing: G1040.
C	CLEARANCE OF AXIS	-
P	APPROACH TYPE	See the descriptions about Side finishing: G1042.
R	APPROACH RAD./DIST.	-
A*	APPROACH ANGLE	-
Q	ESCAPE TYPE	-
X	ESCAPE RAD./DIST.	-
Y*	ESCAPE ANGLE	-
Z	APPROCH MOTION	-

(a) Tool path

The top surface of a wall of a pocket is chamfered. The tool path for it is the same as for contouring (chamfering).
See descriptions about contouring (chamfering) for details.

2.5.2 IMPROVEMENT OF CHECKING CUTTING CONDITION FOR POCKETING (ROUGH) AND EMBOSS MACHINING (ROUGH)

Pocketing (rough) and emboss machining (rough) have the following limitation.

(Limitation)
When the cutting condition (SIDE THICKNESS - SIDE FINISH AMOUNT) ≤ CUT DEPTH OF RADIUS is given, CLERANCE OF RADIUS is ignored. Then, the tool performs a retract operation in the tool axis direction when the tool is touching the side face of the pocket.

If the above cutting condition is given, the alarm "WRONG CUTTING DEPTH" can be generated when cycle machining is executed.

2.5.2.1 PARAMETER

Setting the following parameter to 1 is necessary to use this function.

	#7	#6	#5	#4	#3	#2	#1	#0
27061	CDR							

[Data format] bit

CDR When Pocketing (rough) and emboss machining (rough) is executed, if the cutting condition (SIDE THICKNESS - SIDE FINISH AMOUNT) ≤ CUT DEPTH OF RADIUS is given,
1: An alarm is generated.
0: An alarm isn't generated.

2.5.3 Fixed Form Figure (XY Plane)

(1) Input items of SQUARE (XY plane) (G1220)

POS./SIZE		
Data item		Meaning
T	FIGURE TYPE	[FACE] : Used as a figure for facing [CONVEX] : Used as an outer figure for contouring [CONCAVE]: Used as an inner figure for contouring or as a figure for pocketing [GROOVE] : Used as a figure for grooving Remark) When pocketing is selected as machining type, be sure to select [CONCAVE].
B	BASE POSITION	Z coordinate of the top surface of a workpiece subject to pocketing or the bottom of a pocket (in the tool axis direction)
L	HEIGHT/DEPTH	When the top surface of a workpiece is selected as the BASE POSITION, specify the distance to the bottom of a pocket by using a negative value (radius value). → Depth When the bottom of a pocket is selected as the BASE POSITION, specify the distance to the top surface of the pocket by using a positive value (radius value). → Height
H	CENTER POINT (X)	X coordinate of the center position of a rectangular figure

POS./SIZE		
Data item		Meaning
V	CENTER POINT (Y)	Y coordinate of the center position of a rectangular figure
U	LENGTH FOR X AXIS	Length of the side in the X-axis direction (radius value, positive value)
W	LENGTH FOR Y AXIS	Length of the side in the Y-axis direction (radius value, positive value)
R*	CORNER RADIUS	Radius for corner rounding (positive value)
A*	ANGLE	Inclination angle of a rectangular figure relative to the X-axis (positive or negative value)

When bit 0 (P0) of parameter No. 27095 is set to 1, the “BOTTOM THICKNESS” and “SIDE THICKNESS” displayed on the cycle input screen are displayed on the THICKNESS tab of the figure input screen.

(2) Input items of CIRCLE (XY plane) (G1221)

POS./SIZE		
Data item		Meaning
T	FIGURE TYPE	See the descriptions about SQUARE: G1220.
B	BASE POSITION	-
L	HEIGHT/DEPTH	-
H	CENTER POINT (X)	X coordinate of the center position of a circular figure
V	CENTER POINT (Y)	Y coordinate of the center position of a circular figure
R	RADIUS	Radius of a circular figure (radius value, positive value)

(3) Input items of TRACK (XY plane) (G1222)

POS./SIZE		
Data item		Meaning
T	FIGURE TYPE	See the descriptions about SQUARE: G1220.
B	BASE POSITION	-
L	HEIGHT/DEPTH	-
H	CENTER POINT (X)	X coordinate of the center position of the left semicircle
V	CENTER POINT (Y)	Y coordinate of the center position of the left semicircle
U	BETWEEN DISTANCE	Distance between the centers of the right and left semicircles (radius value, positive value)
R	RADIUS	Radius of the left and right semicircles (radius value, positive value)
A*	ANGLE	Inclination angle of a track figure relative to the X-axis (positive or negative value)

(4) Input items of POLYGON (XY plane) (G1225)

See Subsection 4.2.2, "Fixed Form Figure Blocks (XY Plane)", in "FACING".

2.5.4 Fixed Form Figure (YZ Plane)

The same fixed-figure block types as for the XY plane explained in the previous subsection are available for the YZ plane. They are provided with the following G codes.
The data to be set for the YZ planes is the same as for the XY plane except that the XY plane is changed to the YZ plane and that the direction in which the tool moves to cut is changed to the X-axis (YZ plane).

Square : G1320 (YZ plane)
Circle : G1321 (YZ plane)
Track : G1322 (YZ plane)
Polygon : G1325 (YZ plane)

2.5.5 Arbitrary Figure (XY Plane)

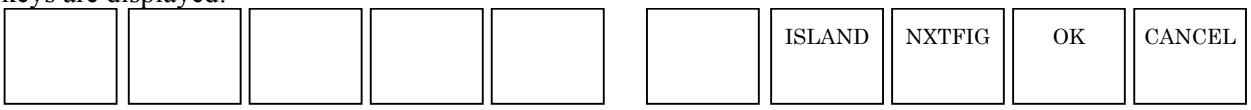
When an arbitrary figure for pocketing is input, data such as a figure type and machining reference position is specified in the start point block. Other data items to be input such as a straight line and arc are exactly the same as for arbitrary figures of other machining types.
So, this subsection describes only the start point block of an arbitrary figure for pocketing.
For other arbitrary figures, see Chapter 5, "DETAILED DESCRIPTIONS ABOUT ENTERING ARBITRARY FIGURES".

(1) Input items of START POINT (XY plane) (G1200)

ELEMENT		
Data item		Meaning
X	START POINT X	X coordinate of the start point of an arbitrary figure
Y	START POINT Y	Y coordinate of the start point of an arbitrary figure
T	FIGURE TYPE	See the descriptions about SQUARE: G1220.
B	BASE POSITION	-
L	HEIGHT/DEPTH	-

(a) Input of Island

After entering the outer wall figure for pocketing, press soft key [CREATE]. Then, the following soft keys are displayed:



If there is a island, push the soft-key [ISLAND] in order to input the island figure. The START POINT screen is displayed.
As to the input data item [FIGURE TYPE] is set “CONVEX” automatically. (When a outer wall, it is set “CONCAV”.)
After that, input the island figure as the same of a outer wall figure.

NOTE
As to the soft-key [NXTFIG], it is used in the case of inputting the other pocket figure one after another.

Side face finishing feedrate for each figure element

When bit 4 (FSP) of parameter No. 27002 is set to 1, the feedrate for side face finishing can be set for each element.
When the side face finishing feedrate can be set, the input item for the side face finishing feedrate is displayed on the “ATTRIBUTE” tab for each of the figure elements that are straight line, arc, chamfer, and corner rounding elements.
For details, see Subsection 4.3.4, "Arbitrary Figure (XY Plane)", in Section 4.3, "CONTOURING".

2.5.6 Arbitrary Figure (YZ Plane)

The same arbitrary-figure block types as for the XY plane explained in the previous subsection are available for the YZ plane. Their start points are specified with the following G codes.
The data to be set for the YZ planes is the same as for the XY plane except that the XY plane is changed to the YZ plane and that the direction in which the tool moves to cut is changed to the X-axis (YZ planes).

Start point : G1300 (YZ plane)

2.5.7 Pocket Residual Machining Cycle (for only Series 30*i*/31*i*/32*i*-B, 0*i*-D)

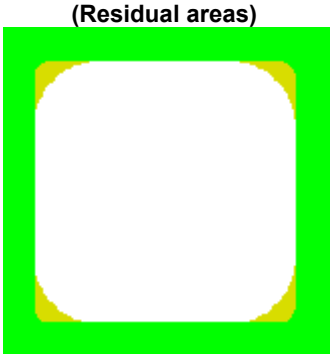
2.5.7.1 Outline

It is possible to calculate plural residual areas automatically and machine only residual areas in pocket roughing cycle.

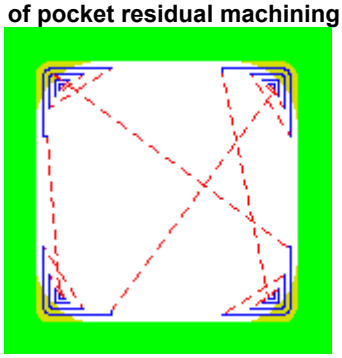
(Example program)

```
D1 ;
G1040 L10. F500. V500. E500. W2. B1. C1. Z2. X1. ;   ← Pocket roughing cycle
G1220 T3. B0. L-20. H50. V50. U80. W80. R5. ;        ← Pocket figure ..... Same figure
D2 ;                                                                       :
G1044 N1. L2. F500. V500. E500. W2. B1. C1. Z2. ;    ← Pocket residual machining cycle
G1220 T3. B0. L-20. H50. V50. U80. W80. R5. ;        ← Pocket figure .....
```

(Residual areas)



(Tool path of pocket residual machining cycle)



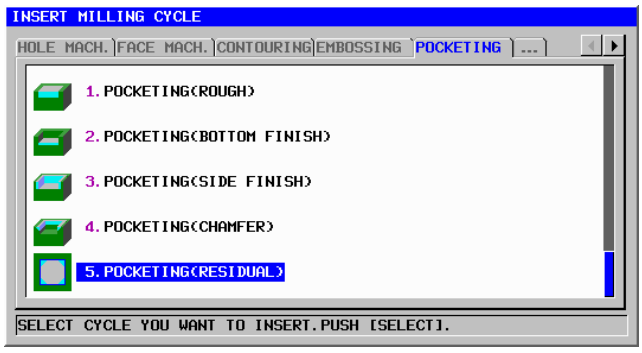
2.5.7.2 Condition

This function is enabled when the parameter No.27096#5 is set to 0.

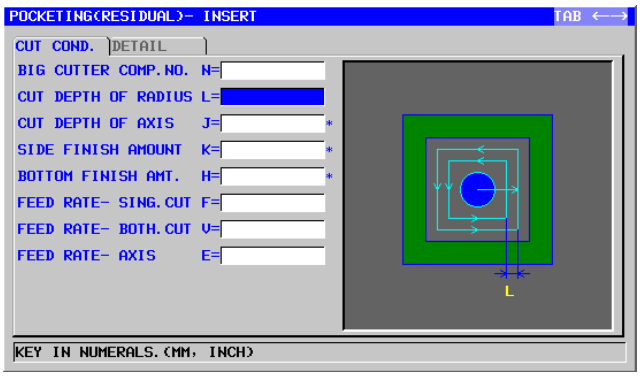
2.5.7.3 Operation

Method of program creation

When “POCKETING(RESIDUAL)” is selected at the [POCKETING] tab of milling cycle menu screen, the creation screen of the residual machining cycle is displayed.



It is possible to create the program of residual machining cycle by inputting data according to the guidance window.



(Example of program)

G1044 Nn Ll Jj Kk Hh Ff Vv Ee Ww Bb Cc Zz;

2.5.7.4 Format of pocket residual machining cycle command

Pocket residual machining cycle is commanded by the following format.

G1044 Dd Nn Pp Ll Jj Kk Hh Ff Vv Ee Ww Bb Cc Zz;

TOOL COND.		
Data item		**Meaning**
D	TOOL DIAMETER	End mill diameter used for residual machining. Specify only in case of No.27002#0=1.

CUT COND.		
Data item		**Meaning**
N	LARGE CUTTER COMP.NO.	Cutter compensation number of the end mill that had been used for pocket roughing in previous process of residual machining. Specify only in case of No.27002#0=0.
P	LARGE TOOL DIAMETER	Diameter of the end mill that had been used for pocket roughing in previous process of residual machining. Specify only in case of No.27002#0=1.
L	CUT DEPTH OF RADIUS	Depth of cut on the side face (tool radius direction) per cutting operation
J*	CUT DEPTH OF AXIS	Depth of cut in the tool axis direction per cutting operation
K*	SIDE FINISH AMOUNT	Finishing allowance on the side face. The blank is regarded as 0.
H*	BOTTOM FINISHI AMT.	Finishing allowance at the bottom in side face machining. The blank is regarded as 0.
F	FEED RATE- SING.CUT	Feedrate applicable when only the one-side cutter portion of an end mill is used for cutting.
V	FEED RATE- BOTH.CUT	Feedrate applicable when the entire front side of an end mill is used for cutting.
E	FEED RATE- AXIS	Feedrate applicable when cutting is performed in the tool axis direction toward the bottom of a side face being machined

DETAIL		
Data item		Meaning
W	UP CUT/DOWN CUT	[UP CUT] : Performs machining in up-cut mode, assuming that the tool is rotating clockwise. [DWNCUT] : Performs machining in down-cut mode, assuming that the tool is rotating clockwise.
B	CLEARANCE OF RADIUS	Distance between the wall of a pocket and a tool retract position in the tool radius direction
C	CLEARANCE OF AXIS	Distance between the surface of a blank being machined and a cutting start point (point R) in the tool axis direction
Z	APPROCH MOTION	[2 AXES] : When moving from the current position to the machining start point, the tool first moves in the machining plane in two-axis synchronous operation and then moves along the tool axis. (initial value) [3 AXES] : tool moves from the current position to the machining start point in three-axis synchronous operation. Remark) This input item is not displayed when the option for expanding the number of simultaneously controlled axes is not provided.

G1044 is commanded together with figure blocks as follows.

(Example program)

```
G1044 N1. L2. F500. V500. E500. W2. B1. C1. Z2. ;   ← Pocket residual machining cycle
G1220 T3. B0. L-20. H50. V50. U80. W80. R5. ;       ← Figure block
```

The maximum number of definable island figure

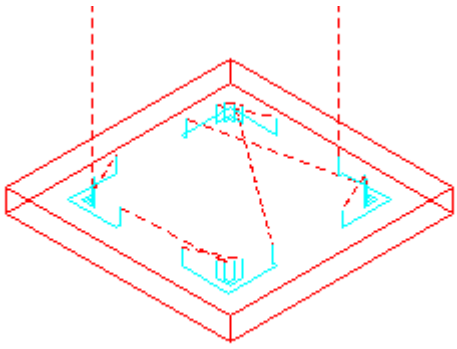
The maximum number of definable island figure is 16.

Plane to calculate residual area

Residual areas are calculated based on the bottom surface of a pocket figure.
Residual areas on higher than bottom surface cannot be calculated.

2.5.7.5 Tool path

Residual areas are machined in a spiral manner.
The following tool path is created.



NOTE
If tool cannot machine from the side, change tool for smaller diameter or drill prepared holes for approach in the tool axis direction.

Island figures inside residual area

It is possible to select the behavior of island figures inside residual area in the following parameter.

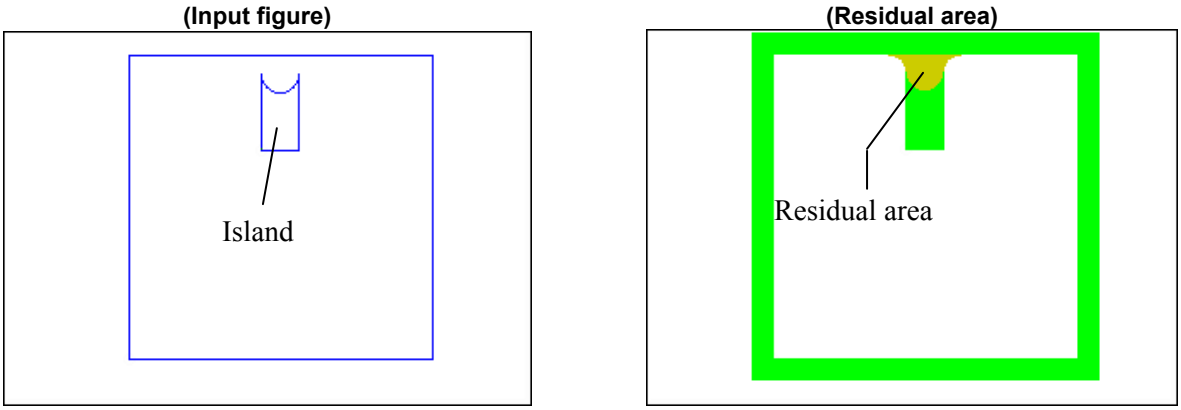
(Parameter No.27061#6=0)
Island figures inside residual area are invalidation. (Island figures are machined)

(Parameter No.27061#6=1)
Island figures inside residual area are left.
Maybe time of tool path creation will be long.

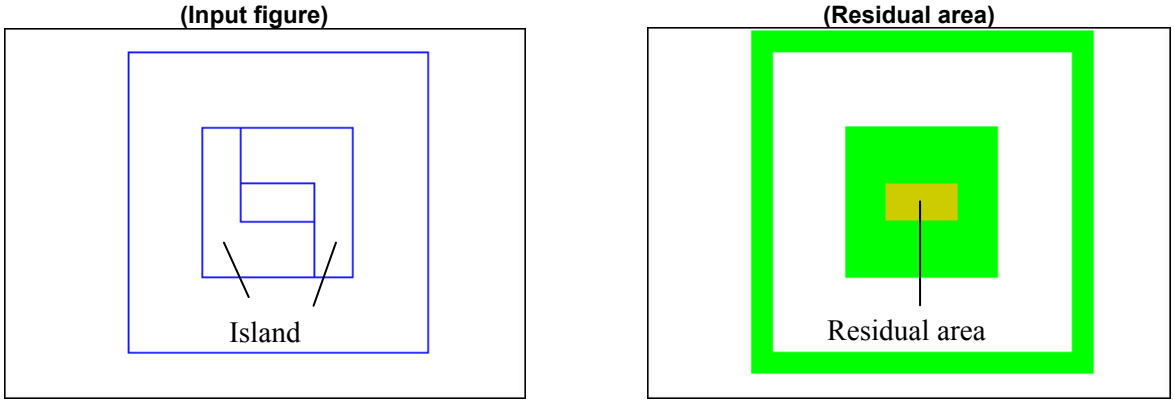
2.5.7.6 Restrictions

The following figure cannot be machined.

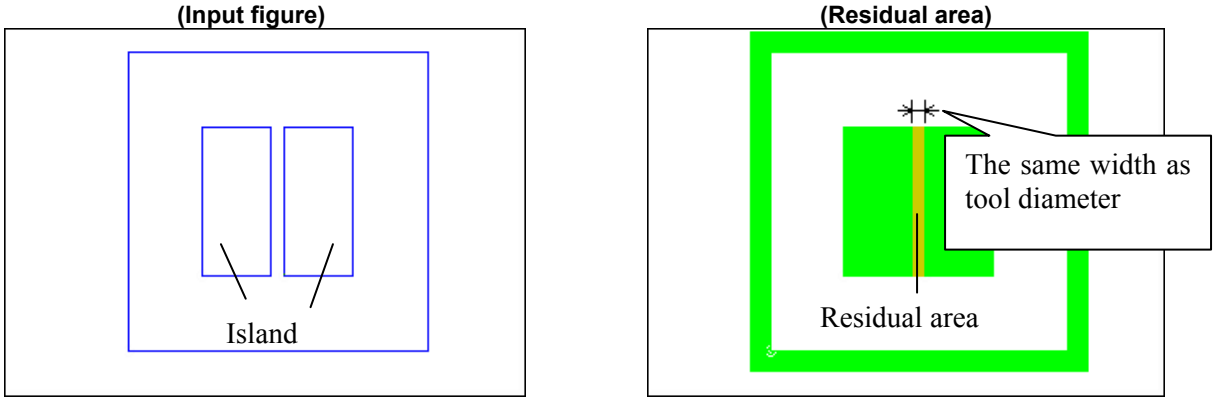
(1) Generated residual area in island figure with concave arc



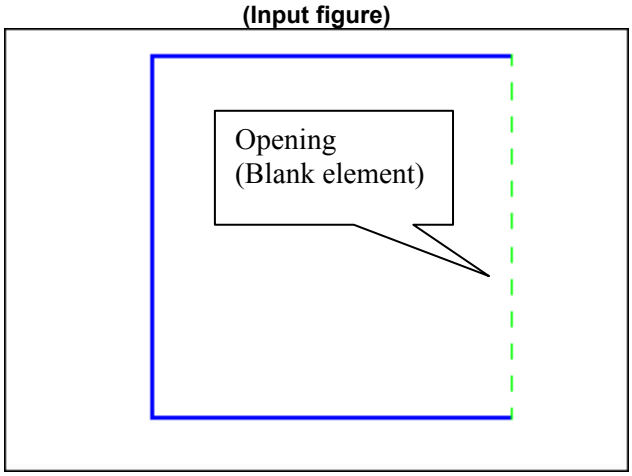
(2) Closed residual area by plural island figure



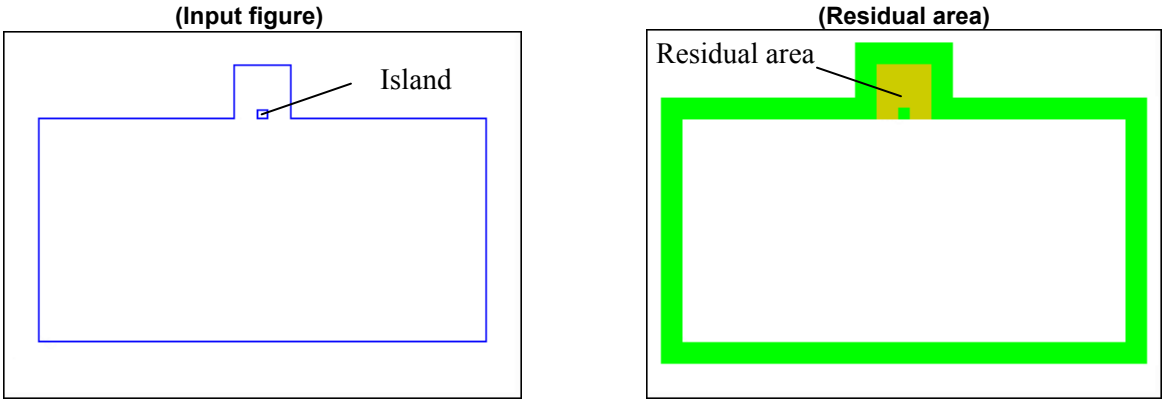
(3) Residual area width equal to diameter of small diameter tool



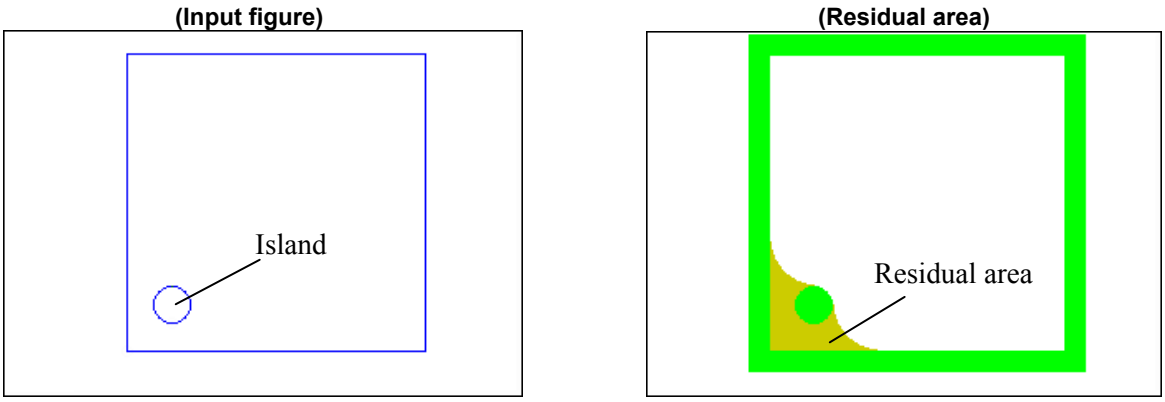
(4) Pocket figure with one or more open elements



(5) Residual area tangent to island figure



(6) Generated residual area in between island figure and a corner of pocket figure



2.6 GROOVING

2.6.1 Machining Command

(1) Input items of ROUGH (G1050)

TOOL COND.		
Data item		Meaning
D	TOOL DIAMETER	End mill diameter

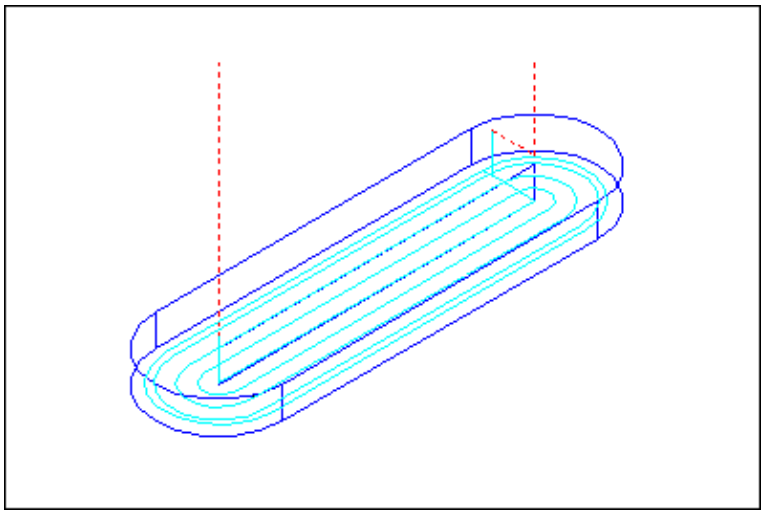
NOTE
1 Tab 'TOOL COND.' is enabled when bit 0 (TLG) of parameter No. 27002 = 1.
2 The operator ordinarily sets the above data on the tab of 'TOOL DATA' in Tool Offset window. Therefore, it is not necessary to display the tab 'TOOL COND.'

CUT COND.		
Data item		Meaning
L	CUT DEPTH OF RADIUS (*)	Depth of cut on the side face (tool radius direction) per cutting operation (radius value, positive value)
J	CUT DEPTH OF AXIS (*)	Depth of cut in the tool axis direction per cutting operation (radius value, positive value)
K*	SIDE FINISH AMOUNT (*)	Finishing allowance on the side face. The blank is regarded as 0. (radius value, positive value)
H*	BOTTOM FINISH AMT. (*)	Finishing allowance at the bottom in side face machining. The blank is regarded as 0. (radius value, positive value)
F	FEED RATE- SING.CUT	Feedrate applicable when only the one-side cutter portion of an end mill is used for cutting. This feedrate is used for cutting in retract operation and on the side face other than initial cutting.
V	FEED RATE- BOTH.CUT	Feedrate applicable when the entire front side of an end mill is used for cutting. This feedrate is used for initial cutting.
E	FEED RATE- AXIS	Feedrate applicable when cutting is performed in the tool axis direction toward the bottom of a side face being machined

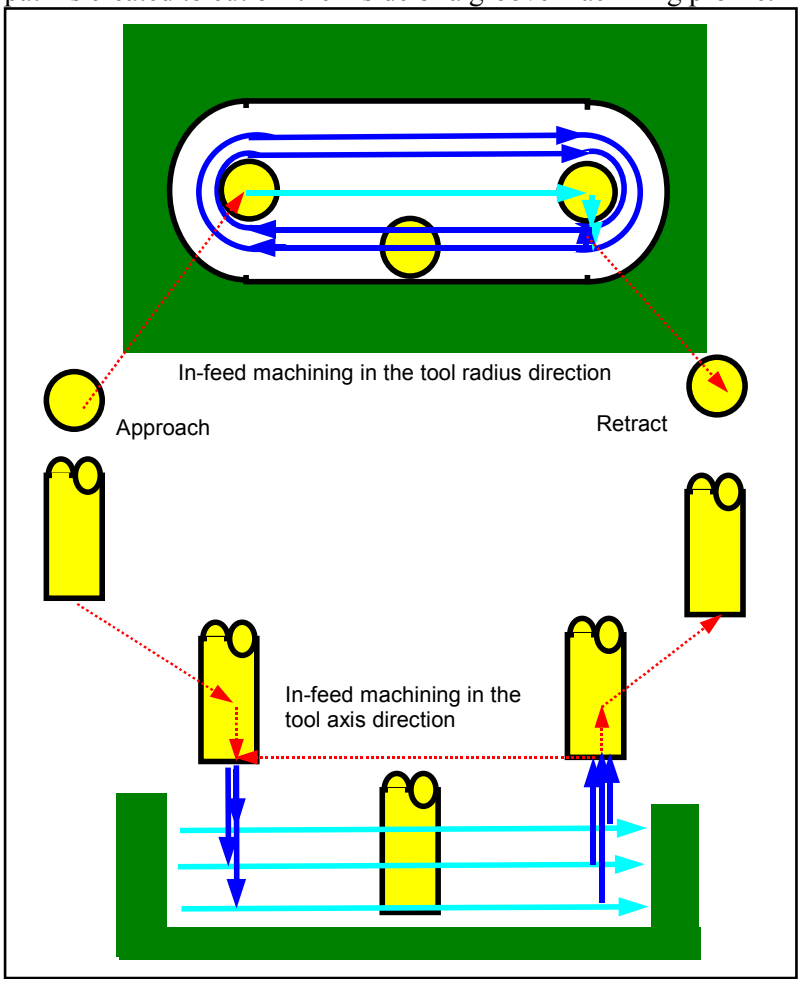
(*) When bit 5 (RCF) of parameter No. 27002 is set to 1, the displayed order is changed by exchanging the input items "CUT DEPTH OF RADIUS" and "CUT DEPTH OF AXIS" and the input items "SIDE FINISH AMOUNT" and "BOTTOM FINISH AMT.".

DETAIL		
Data item		**Meaning**
W	UP CUT/DOWN CUT	[UP CUT] : Performs machining in up-cutting mode, assuming that the tool is rotating clockwise. [DWNCUT] : Performs machining in down-cutting mode, assuming that the tool is rotating clockwise. (COPY)
B	CLEARANCE OF RADIUS	Distance between the wall of a groove and a tool retract position in the tool radius direction (radius value, positive value) Remark1) When one groove cutting operation is completed, the tool performs a retract operation in the tool axis direction from the side face of the groove by this clearance amount. Remark2) By referring to the parameter No.27009 (minimum clamp value), the system sets the data as Initial value.
C	CLEARANCE OF AXIS	Distance between the surface of a blank being machined and a cutting start point (point R) in the tool axis direction (radius value, positive value) Remark) By referring to the parameter No.27009 (minimum clamp value), the system sets the data as Initial value.
Z	APPROCH MOTION	[2 AXES] : When moving from the current position to the machining start point, the tool first moves in the machining plane in two-axis synchronous operation and then moves along the tool axis. (initial value) [3 AXES] : The tool moves from the current position to the machining start point in three-axis synchronous operation. Remark) This input item is not displayed when the option for expanding the number of simultaneously controlled axes is not provided.

(a) Tool path

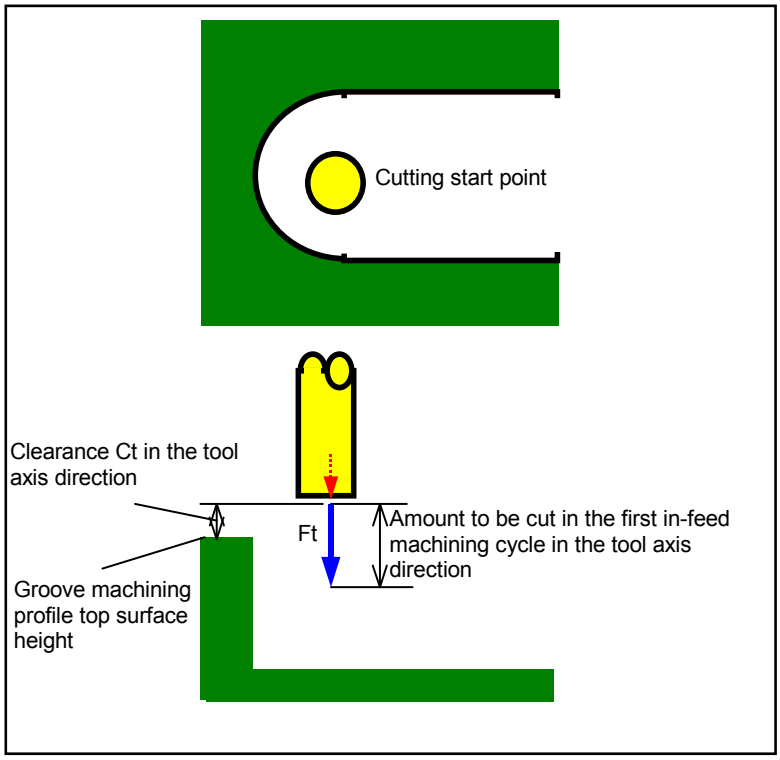


The following tool path is created to cut off the inside of a groove machining profile.



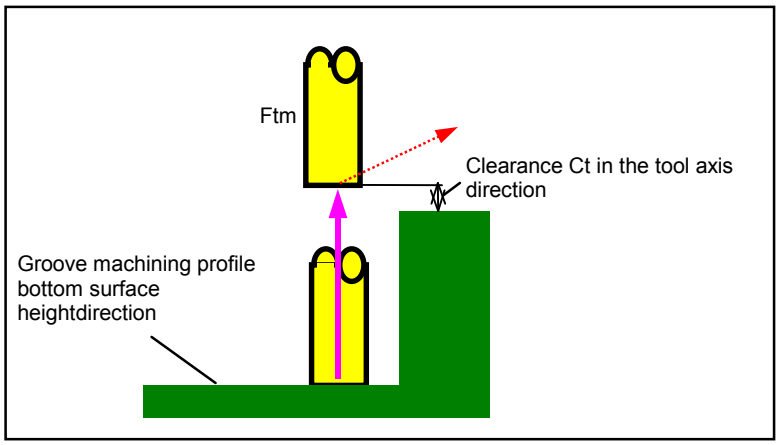
<1> The tool approaches a point above the cutting start point of a groove machining profile.
<2> The tool cuts in the groove machining profile in the tool radius direction.
<3> The tool cuts in the groove machining profile in the tool axis direction.
<4> Step <2> and <3> are repeated until the cutting allowance is removed.
<5> The tool retracts.

(i) Approach



<1> The tool moves to the position "groove machining profile top surface height + clearance (Ct) in the tool axis direction" by rapid traverse.
<2> The tool cuts in by the amount to be cut in the first in-feed machining cycle in the tool axis direction at the feedrate (Ft) specified for cutting in the tool axis direction.

(ii) Retract



<1> The tool retracts from the groove machining profile bottom surface height to the position "groove machining profile top surface height + clearance (Ct) in the tool axis direction" at the feedrate (Ftm) specified for movement in the tool axis direction.

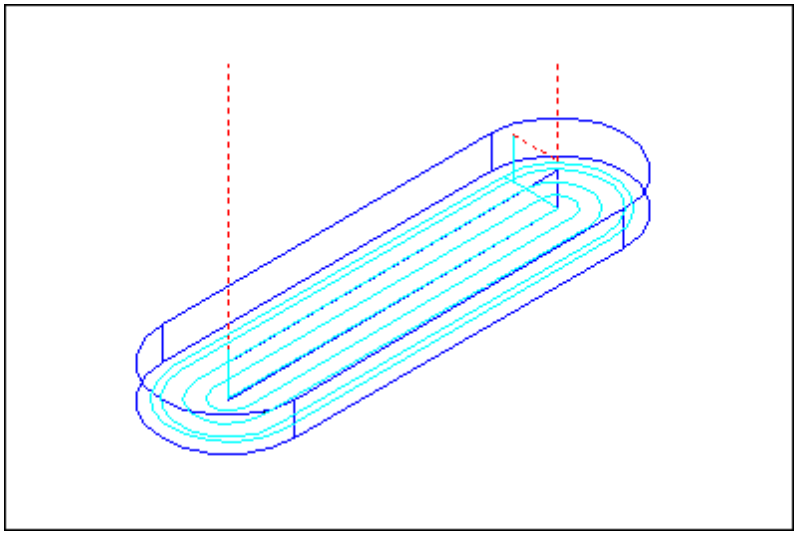
(2) Input items of BOTTOM FINISH (G1051)

TOOL COND.		
Data item		Meaning
D	TOOL DIAMETER	See the descriptions about Roughing: G1050.

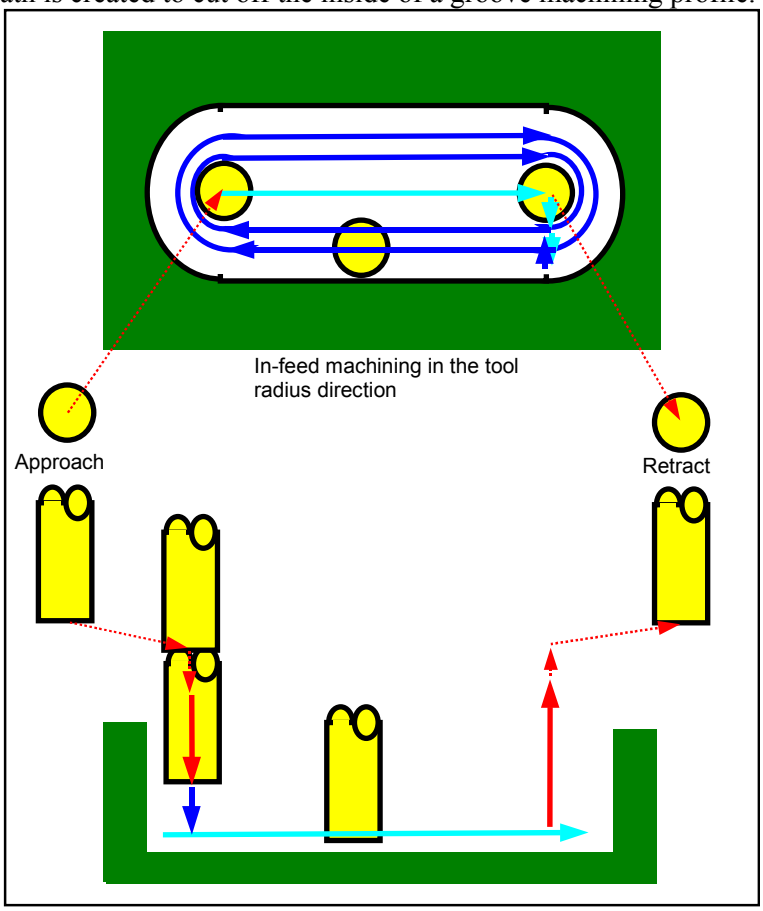
CUT COND.		
Data item		Meaning
T*	BOTTOM THICKNESS	Cutting allowance of the bottom of a groove. The blank is regarded as 0. (radius value, positive value)
K*	SIDE FINISH AMOUNT	See the descriptions about Roughing: G1050.
H*	BOTTOM FINISHI AMT.	-
L	CUT DEPTH OF RADIUS	-
F	FEED RATE- SING.CUT	-
V	FEED RATE- BOTH.CUT	-
E	FEED RATE- AXIS	-

DETAIL		
Data item		Meaning
W	UP CUT/DOWN CUT	See the descriptions about Roughing: G1050.
B	CLEARANCE OF RADIUS	-
C	CLEARANCE OF AXIS	-
Z	APPROCH MOTION	-

(a) Tool path

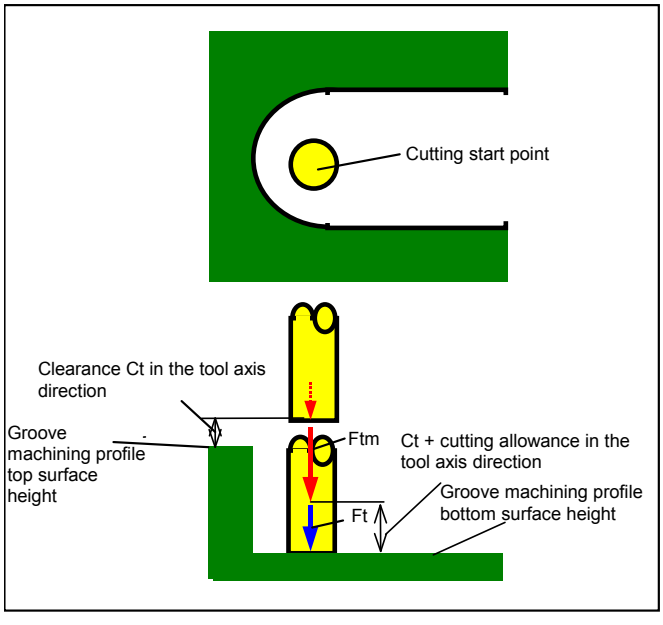


The following tool path is created to cut off the inside of a groove machining profile.



<1> The tool approaches a point above the cutting start point of a groove machining profile.
<2> The tool cuts in the groove machining profile in the tool radius direction.
<3> The tool retracts.

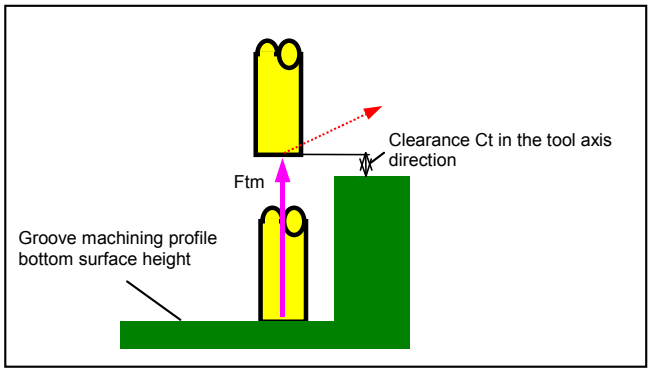
(i) Approach



<1> The tool moves to the position "groove machining profile top height + clearance (Ct) in the tool axis direction" by rapid traverse.

<2> The tool moves to the position "groove machining profile bottom surface height + clearance (Ct) in the tool axis direction" at the feedrate (Ftm) specified for cutting in the tool axis direction.
<3> The tool cuts in to the height of the groove machining profile bottom surface at the feedrate (Ft) specified for cutting in the tool axis direction.

(ii) Retract



<1> The tool retracts from the height of the groove machining profile bottom surface to the position "groove machining profile top surface height + clearance (Ct) in the tool axis direction" at the feedrate (Ftm) specified for movement in the tool axis direction.

(iii) In-feed machining in the tool radius direction

This operation is the same as grooving (roughing).
See descriptions about grooving (roughing) for details.

(3) Input items of SIDE FINISH (G1052)

TOOL COND.		
Data item		**Meaning**
I	INPUT	[INPUT] : Inputs a cutter compensation value directly. [REF.] : Inputs a cutter compensation number to read a cutter compensation value by that number.
D	TOOL DIAMETER	End mill diameter (positive value) Remark) This item is indicated only when [INPUT] is selected for INPUT.
M	CUTTER COMP.NO.	Cutter compensation number of an end mill (positive value) Remark) This item is indicated only when [REF.] is selected for INPUT.

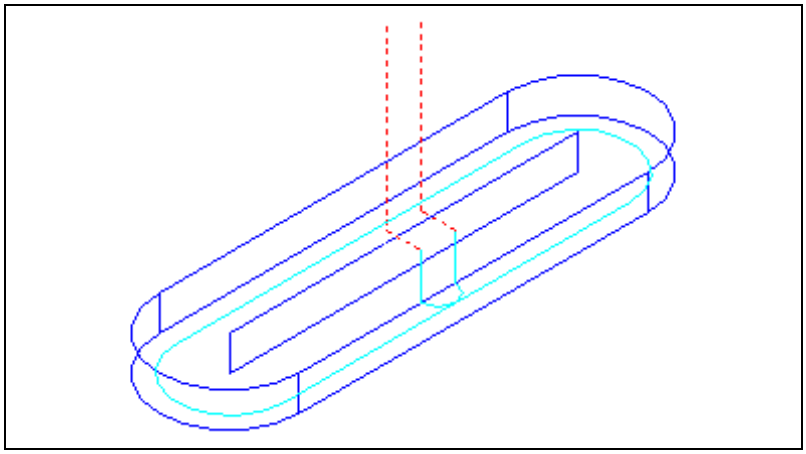
NOTE
Tab 'TOOL COND.' is not enabled when bit 3 (GF3) of parameter No. 27081 = 1.

CUT COND.		
Data item		**Meaning**
S*	SIDE THICKNESS	Cutting allowance of the side face. The blank is regarded as 0. (radius value, positive value)
K*	SIDE FINISH AMOUNT	See the descriptions about Roughing: G1050.
B	NUMBER OF FINISHING	Number of cuts for finishing (positive value) Remark) Depth of each cut = (side surplus thickness)/(number of finishing cuts)
F	FEED RATE-SING.CUT	See the descriptions about Roughing: G1050.
V	FEED RATE- BOTH.CUT	-
E	FEED RATE- AXIS	-

NOTE
When bit 4 (FSP) of parameter No. 27002 is set to 1 to make it possible to set the feedrate for side face finishing for each element, a cutting feedrate that is usually set to "FEED RATE-SINGLE.CUT" specified in the machining type block is set to “SIDE FIN. FEEDRATE” if it is specified.

DETAIL		
Data item		**Meaning**
W	UP CUT/DOWN CUT	See the descriptions about Roughing: G1050.
C	CLEARANCE OF AXIS	-
P	APPROACH TYPE	[ARC] : Approaches a side face along an arc. [TANGEN]: Approaches a side face along the straight line tangent to the first figure in side face cutting. [VERTIC] : Approaches a side face along the straight line normal to the first figure in side face cutting. (COPY)
R	APPROACH RAD./DIST.	Radius when [ARC] is specified. Straight line length when [TANGEN] or [VERTIC] is specified. (radius value, positive value) Remark) By referring to the parameter No.27010 (minimum clamp value), the system sets the data as Initial value.
A*	APPROACH ANGLE	Center angle of the arc when [ARC] is specified. The default is 90 degrees. (positive value) Remark) This data item is indicated only when [ARC] is selected for APPROACH TYPE. (COPY)
Q	ESCAPE TYPE	[ARC] : Retracts from a side face along an arc. [TANGEN]: Retracts from a side face along the straight line tangent to the last figure in side face cutting. [VERTIC] : Retracts from a side face along the straight line normal to the last figure in side face cutting. (COPY)
X	ESCAPE RAD./DIST.	Radius when [ARC] is specified. Straight line length when [TANGEN] or [VERTIC] is specified. (radius value, positive value) Remark) By referring to the parameter No.27010 (minimum clamp value), the system sets the data as Initial value.
Y*	ESCAPE ANGLE	Center angle of the arc when [ARC] is specified. The default is 90 degrees. (positive value) Remark) This data item is indicated only when [ARC] is selected for ESCAPE TYPE. (COPY)
Z	APPROCH MOTION	[2 AXES] : When moving from the current position to the machining start point, the tool first moves in the machining plane in two-axis synchronous operation and then moves along the tool axis. (initial value) [3 AXES] : The tool moves from the current position to the machining start point in three-axis synchronous operation.

(a) Tool path



The side-face contour of a groove machining profile is finished.
The tool path for it is the same as for contouring (side-face finishing).
See descriptions about contouring (side-face finishing) for details.

(4) Input items of CHAMFER (G1053)

TOOL COND.		
Data item		**Meaning**
K	TOOL SMALL DIAMETER	Diameter of the tip of a chamfering tool (positive value)

CUT COND.		
Data item		**Meaning**
S	CHAMFER AMOUNT	Chamfer length (radius value, positive value)
H	EJECTION STROKE	Distance between the tip of a chamfering tool and an actually cut position in the tool axis direction (radius value, positive value)
F	FEED RATE - RADIUS	Feedrate applicable when cutting is performed in the tool radius direction
E	FEED RATE - AXIS	Feedrate applicable when cutting is performed in the tool axis direction

DETAIL		
Data item		**Meaning**
W	UP CUT/DOWN CUT	See the descriptions about Roughing: G1050.
C	CLEARANCE OF AXIS	-
P	APPROACH TYPE	See the descriptions about Side finishing: G1052.
R	APPROACH RAD./DIST.	-
A*	APPROACH ANGLE	-
Q	ESCAPE TYPE	-
X	ESCAPE RAD./DIST.	-
Y*	ESCAPE ANGLE	-
Z	APPROCH MOTION	-

(a) Tool path

The top surface of a wall of a groove is chamfered. The tool path for it is the same as for contouring (chamfering).
See descriptions abut contouring (chamfering) for details.

2.6.2 Fixed Form Figure (XY Plane)

As fixed form figures for grooving, a "square", "circle", "track", and "radial grooves" are available. When any of these pattern figures is specified, a groove with a specified width is cut along the contour.

NOTE
Each of a "square", "circle", and "track" is a closed figure. In actual grooving, sets of a start point and end point are determined, and cutting from the start point to the end point of a set is repeated.

(1) Input items of SQUARE (XY plane) (G1220)

POS./SIZE		
Data item		**Meaning**
T	FIGURE TYPE	[FACE] : Used as a figure for facing [CONVEX] : Used as an outer figure for contouring [CONCAVE]: Used as an inner figure for contouring or as a figure for pocketing [GROOVE] : Used as a figure for grooving Remark) When grooving is selected as machining type, be sure to select [GROOVE].
B	BASE POSITION	Z coordinate of the top surface or the bottom of the side face of a part to be subject to contouring (in the tool axis direction)
L	HEIGHT/DEPTH	When the top surface of a workpiece is selected as the BASE POSITION, specify the distance to the bottom of the side face by using a negative value (radius value). → Depth When the bottom of a side face is selected as the BASE POSITION, specify the distance to the top surface of the workpiece by using a positive value (radius value). → Height
H	CENTER POINT (X)	X coordinate of the center position of a rectangular figure
V	CENTER POINT (Y)	Y coordinate of the center position of a rectangular figure
U	LENGTH FOR X AXIS	Length of the side in the X-axis direction (radius value, positive value)
W	LENGTH FOR Y AXIS	Length of the side in the Y-axis direction (radius value, positive value)
R*	CORNER RADIUS	Radius for corner rounding (positive value)
A*	ANGLE	Inclination angle of a rectangular figure relative to the X-axis (positive or negative value)
D	GROOVE WIDTH	Groove width (radius value, positive value)

(2) Input items of CIRCLE (XY plane) (G1221)

POS./SIZE		
Data item		**Meaning**
T	FIGURE TYPE	See the descriptions about SQUARE: G1220.
B	BASE POSITION	-
L	HEIGHT/DEPTH	-
H	CENTER POINT (X)	X coordinate of the center position of a circular figure
V	CENTER POINT (Y)	Y coordinate of the center position of a circular figure
R	RADIUS	Radius of a circular figure (radius value, positive value)
D	GROOVE WIDTH	Groove width (radius value, positive value)

(3) Input items of TRACK (XY plane) (G1222)

POS./SIZE		
Data item		Meaning
T	FIGURE TYPE	See the descriptions about SQUARE: G1220.
B	BASE POSITION	-
L	HEIGHT/DEPTH	-
H	CENTER POINT (X)	X coordinate of the center position of a left semicircle
V	CENTER POINT (Y)	Y coordinate of the center position of a left semicircle
U	BETWEEN DISTANCE	Distance between the centers of the right and left semicircles (radius value, positive value)
R	RADIUS	Radius of the left and right semicircles (radius value, positive value)
A*	ANGLE	Inclination angle of a track figure relative to the X-axis (positive or negative value)
D	GROOVE WIDTH	Groove width (radius value, positive value)

(4) Input items of RADIAL GROOVE (XY plane) (G1223)

POS./SIZE		
Data item		Meaning
B	BASE POSITION	See the descriptions about SQUARE: G1220.
L	HEIGHT/DEPTH	-
H	CENTER POINT (X)	X coordinate of the center position of a circle where radial grooves are placed
V	CENTER POINT (Y)	Y coordinate of the center position of a circle where radial grooves are placed
U	BETWEEN DISTANCE	Distance between the centers of two semicircles along a radial groove (in a track figure) (radius value, positive value)
D	GROOVE WIDTH	Width of a radial groove (radius value, positive value)
E*	GROOVE ANGLE	Groove inclination angle in an arc where radial grooves are placed, in the radial direction (positive or negative value)
R	RADIUS	Radius of an arc where radial grooves are placed (radius value, positive value)
A*	START ANGLE	Center angle of the first groove position relative to the X-axis (positive or negative value)

REPEAT		
Data item		Meaning
C*	PITCH ANGLE	Center angle formed by two adjacent grooves (positive or negative value)
M*	BROOVE NUMBER	Number of grooves to be cut (positive value)

(5) Input items of POLYGON (XY plane) (G1225)

See Subsection 4.2.2, "Fixed Form Figure Blocks (XY Plane)", in Section 4.2, "FACING".

(6) Input items of LINE (XY plane) (G1224)

REPEAT		
Data item		Data item
B	BASE POSITION	See the descriptions about SQUARE: G1220.
L	HEIGHT/DEPTH	—
P	STD PT OF POSITION	Position of the X and Y coordinates of the reference position 1:Groove center 2:Groove end 3:Circle center at groove end
H	BASE POINT (X)	X coordinate value of the reference position
V	BASE POINT (Y)	Y coordinate value of the reference position
U	GROOVE LENGTH	Length of a groove

REPEAT		
Data item		Data item
D	GROOVE WIDTH	Width of a groove
W	CUTTING DIRECTION	Cutting direction
A	ANGLE	Inclination angle with respect to STD PT OF POSITION

2.6.3 Fixed Form Figure (YZ Plane)

The same fixed-figure block types as for the XY plane explained in the previous subsection are available for the YZ plane. They are provided with the following G codes.
The data to be set for the YZ planes is the same as for the XY plane except that the XY plane is changed to the YZ plane and that the direction in which the tool moves to cut is changed to the X-axis (YZ plane).

Square : G1320 (YZ plane)
Circle : G1321 (YZ plane)
Track : G1322 (YZ plane)
Radial groove : G1323 (YZ plane)
Polygon : G1325 (YZ plane)

2.6.4 Arbitrary Figure (XY Plane)

When an arbitrary figure for grooving is input, data such as a figure type and machining reference position is specified in the start point block. Other data items to be input such as a straight line and arc are exactly the same as for arbitrary figures of other machining types.
So, this subsection describes only the start point block of an arbitrary figure for pocketing.
For other arbitrary figures, see Chapter 5, "DETAILED DESCRIPTIONS ABOUT ENTERING ARBITRARY FIGURES".
When an arbitrary figure is specified for grooving, a groove with a specified width is cut along the arbitrary figure line.

(1) Input items of START POINT (XY plane) (G1200)

ELEMENT		
Data item		Meaning
T	FIGURE TYPE	[FACE] : Used as a figure for facing [CONVEX] : Used as an outer figure for contouring [CONCAVE]: Used as an inner figure for contouring or as a figure for pocketing [GROOVE] : Used as a figure for grooving Remark) When grooving is selected as machining type, be sure to select [GROOVE].
X	START POINT X	X coordinate of the start point of an arbitrary figure
Y	START POINT Y	Y coordinate of the start point of an arbitrary figure
Z	BASE POSITION	Z coordinate of the bottom of a groove or the top surface of a workpiece subject to grooving (in the tool axis direction)
D	HEIGHT/DEPTH	When the top surface of a workpiece is selected as the BASE POSITION, specify the distance to the bottom of the groove by using a negative value (radius value). → Depth When the bottom of a groove is selected as the BASE POSITION, specify the distance to the top surface of the groove by using a positive value (radius value). → Height
W*	GROOVE WIDTH	Groove width (radius value, positive value)

(a) Side face finishing feedrate for each figure element

When bit 4 (FSP) of parameter No. 27002 is set to 1, the feedrate for side face finishing can be set for each element.
When the side face finishing feedrate can be set, the input item for the side face finishing feedrate is displayed on the “ATTRIBUTE” tab for each of the figure elements that are straight line, arc, chamfer, and corner rounding elements.
For details, see Subsection 4.3.4, "Arbitrary Figure (XY Plane)", in Section 4.3, "CONTOURING".

2.6.5 Arbitrary Figure (YZ Plane)

The same arbitrary-figure block types as for the XY plane explained in the previous subsection are available for the YZ plane. Their start points are specified with the following G codes.
The data to be set for the YZ planes is the same as for the XY plane except that the XY plane is changed to the YZ plane and that the direction in which the tool moves to cut is changed to the X-axis (YZ planes).

Start point : G1300 (YZ plane)

2.7 ENGRAVING CYCLE

It is possible to engrave alphanumeric characters and some special character.



Machining example
(Serial number stamps of parts)

2.7.1 Operation

The operations to create an engraving cycle are showed below.

(1) Select "ENGRAVING" on the SPECIAL cycle menu of the milling cycle menu window.
(2) The data input window for Engraving Cycle appears.
(3) There are four tabs in the data input window. Input the data in each tab.

2.7.2 Input Item

Input items of Engraving Cycle are as follows.

(1) [TOOL COND.] Tab

TOOL COND.		
Data item		Meaning
D	TOOL DIAMETER	Tool diameter of Ball end mill (drill)

NOTE
[TOOL COND.] tab is displayed when the parameter No.27002#0 is 1.

(2) [CUT COND.] Tab

CUT COND.		
Data item		**Meaning**
C	CLEARANCE	Distance between the surface of work and cutting start point in direction of tool axis
F	FEED RATE-RADIUS	Feed speed when cutting in direction of tool radius
E	FEED RATE-AXIS	Feed speed when cutting in direction of tool axis
P	PLANE SELECT	Cutting plane : 1= XY plane 2= YZ plane

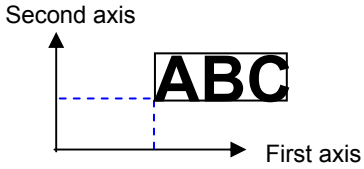
NOTE
1 The plane can be selected according to the parameter No.27004.

(3) [POSITION] Tab
Each input item of the tab [POSITION] is as follows.

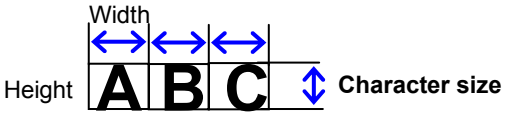
POSITION		
Data item		**Meaning**
H	START PT.(1ST AXIS)	Position of the first character along the first axis (Note1)
V	START PT.(2ND AXIS)	Position of the first character along the second axis (Note1)
B	BASE POSITION	Coordinates value of surface of work
L	CHAR. DEPTH	Depth of the character
U	CHAR. SIZE	Height of the character (Note2)
A	ANGLE	Angle of character (Note3) 1=0°, 2=90°, 3=180°, 4=270°
Z	FACE POSITION	Position for face (Note4) 1= +FACE, 2= -FACE

NOTE

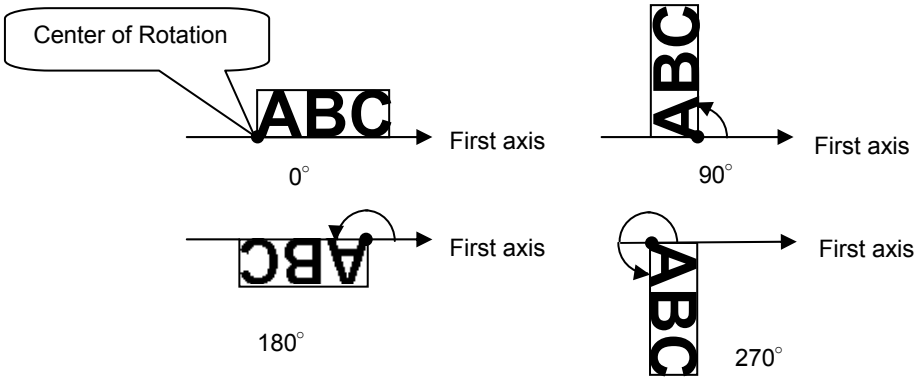
1 The position of beginning character can be specified.



2 The size of the character can be specified. And height and width of the character is the same size.



3 Angle (0°/90°/180°/270°) of the character can be specified. The center of rotation is the start point of the first character.



4 The input item [face position] is displayed, when all the following conditions are meet.
(1) (MC4) of parameter No.27000#4 sets 1.
(2) The XY plane is selected by a plane selection of the cutting condition tab.

(4) [STRING] Tab

STRING		
Data item		Meaning
-	STRING	String to be engraved (80 characters maximum)

(a) The STRING is input directly from the MDI key.
(b) The characters input with the MDI key are immediately displayed into the guidance window.

2.7.3 Enable Characters

Character which can be engraved are as follows.

- Alphabet : A-Z
- Number : 0-9
- Special Character : ? , @#=*+ /. SP: ‘_“ <>

NOTE
When an incorrect character is inserted into the string, then the warning message "The format is not correct" appears. The corresponding character is displayed as red character. All characters, from the beginning of the string until the incorrect characters, are automatically deleted.

2.7.4 Format of G4 Digit Cycle for Engraving Cycle

The format of Engraving Cycle is as follows. The strings specified in data input window are outputted as a comment.

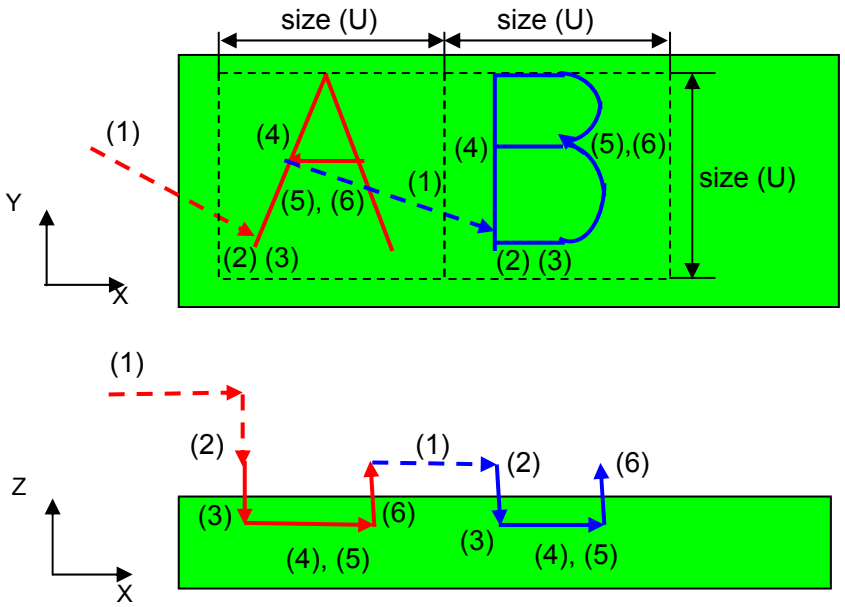
```
G1025 Dd Cc Ff Ee Pp Hh Vv Bb Ll Uu Aa(**********);
```

2.7.5 Tool Path of Engraving Cycle

The tool paths of Engraving Cycle are as follows.

(1) The tool moves to the start point of the first character in rapid traverse in the direction of the tool radius.
(2) Next, the tool moves to the position in rapid traverse (base position (B) + clearance (C)) in the direction of the tool axis.
(3) The tool moves to the position (base position (B) - character depth (L)) in cutting feedrate (E) in the direction of the tool axis.
(4) One character of the string (*****) is engraved in the direction of the tool radius by cutting feedrate (F) along the shape of the character. If the tool cannot engrave the character in one stroke, then the tool escapes at the position (base position (B) + clearance (C)), and moves to another edge of the character, and continues engraving.
(5) The tool moves to the position (base position (B) + clearance (C)) in cutting feedrate (E) in the direction of the tool axis after cutting of one character.
(6) Steps from (1) to (5) are repeated for all the characters.

Example) For the XY plane (It is a stamp as for A and B)



The width of the groove of the character is equal to the tool diameter used to engrave the characters . The shape of the characters is as follows.



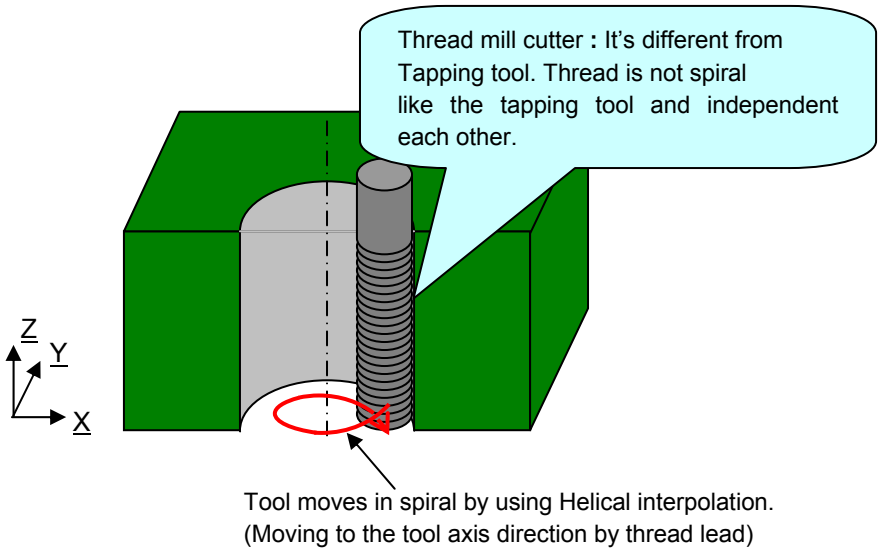
2.7.6 Restrictions

Engraving Cycle has the following restrictions.

(1) The font of the character cannot be changed.
(2) The tools that can be selected with Engraving Cycle are only drilling tool and ball-end-mill tool.
(3) Engraving cycle during DNC operation and Engraving Cycle in a subprogram called from M198 cannot be executed. The following alarm appears.
Series0*i*-C and Series16*i*/18*i*/21*i*-B: “3014 WRONG FIGURE DATA”
Series0*i*-D and Series30*i*/31*i*/32*i*: “3514 WRONG FIGURE DATA”

2.8 THREAD MILLING CYCLE

Thread milling cuts a thread by using the Helical interpolation function and moving the tool, called thread mill cutter, along a spiral.
This function is enabled when the parameter No.27095#5 is set to 1.

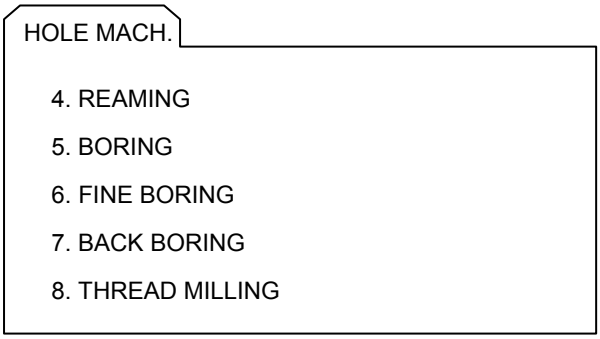


Machining Movements of Thread milling cycle

NOTE
This function requires Helical interpolation Optional Function.

2.8.1 Screens and Operations

In case this function is activated, in hole machining TAB of milling cycle menu, thread milling menu is displayed.



When the menu "THREAD MILLING" is selected, the data input screen of the thread milling cycle is displayed.

2.8.2 Input Items of Thread Milling

Input items of thread milling cycle (G1010) are as follows.

TOOL CONDITION		
Data item		**Meaning**
D	TOOL DIAMETER	Diameter of thread milling tool

FIGURE		
Data item		**Meaning**
U	THREAD DIAMETER	Thread diameter
M	RIGHT HAND /LEFT HAND THREAD	M=1 Right hand thread M=2 Left hand thread
R	THREAD TYPE	R=1 Metric R=2 Unified
T	THREAD LEAD	Tread lead. This item is displayed only in case thread type is "Metric".
S	NUMBER OF THREAD	Number of thread. This item is displayed only in case thread type is "Unified".

CUT COND-1		
Data item		**Meaning**
X	MACHINING TYPE	X=1, one revolution X=2, the number of revolutions is equal to the number of threads X=3 multiple revolutions
Y	NUMBER OF THREAD OF TOOL	Thread Number of tool This item is displayed in case Machining type X is equal to 3. Set data in order to satisfy all the following conditions. Y > Parameter No.27451 overlapped amount Y > Parameter No.27452 start clearance ABS(Cut depth(L)) ≧ (Y x Thread Lead(T))

CUT COND-1		
Data item		Meaning
I	RETURN MODE	I=1 I point-1 When the tool moves between the holes, the tool return to R-point position. Last time, return to I-point. I=2 I point-2 When the tool moves between the holes, the tool returns to I-point. I=3 R point When the tool moves between the holes, the tool returns to R-point.
J	I POINT	I point position This item is displayed only in case Return mode is ”I point-1” or “2”
L	CUT DEPTH	Cut depth of thread milling In case MACHINING TYPE is ‘THRD REV.’ or ‘MULTI REV.’, set data in order to satisfy the following condition. (Number of thread of tool “Y” x Thread lead ”T”) ≦ ABS”L”
C	CLEARANCE	Clearance amount from work surface. In case MACHINING TYPE is ‘THRD REV.’ or ‘MULTI REV.’, set data in order to satisfy the following condition. C ≧ (Thread lead”T”/4＋Parameter ”start clearance” x Thread lead ”T”) See “Note” bellow.
V	FEED RATE OF TOOL AXIS	-
F	FEED RATE OF THREADING	-
Z	APPROACH MOTION	Z=2 Simultaneous 2 axes Z=3 Simultaneous 3 axes

Note: Thread Lead
When unified thread is specified in Thread Type, Thread Lead is calculated from the entered Number of Thread.

CUT COND-2		
Data item		Meaning
W	UP CUT/DOWN CUT	W=1 Up cut W=2 Down cut
K	END CLEARANCE	-

2.8.3 Specifiable Combinations of Figure

Specifiable combinations of figure and thread milling cycle are as follows.

Kind of figure (Hole pattern)	Hole Machining	Thread milling cycle
XY plane points	○	○
XY plane points on the line “Equal intervals”	○	○
XY plane points on the line ”Not equal intervals”	○	○
XY plane points on the grid	○	○
XY plane points on the square	○	○
XY plane points on the circle	○	○
XY plane points on the arc “equal intervals”	○	○
XY plane points on the arc “Not equal intervals”	○	○
YZ plane points	○	○
YZ plane points on the line “Equal intervals”	○	○
YZ plane points on the line “Not Equal intervals”	○	○
YZ plane points on the grid	○	○
YZ plane points on the square	○	○

Kind of figure (Hole pattern)	Hole Machining	Thread milling cycle
YZ plane points on the circle	○	○
YZ plane points on the arc “equal intervals”	○	○
YZ plane points on the arc “Not equal intervals”	○	○
B axis hole in Cylinder ”on the arc”	○	×
B axis hole in Cylinder ”points”	○	×

2.8.4 Tool Kind

In the thread milling cycle, specify a tapping tool as the Tool Kind.

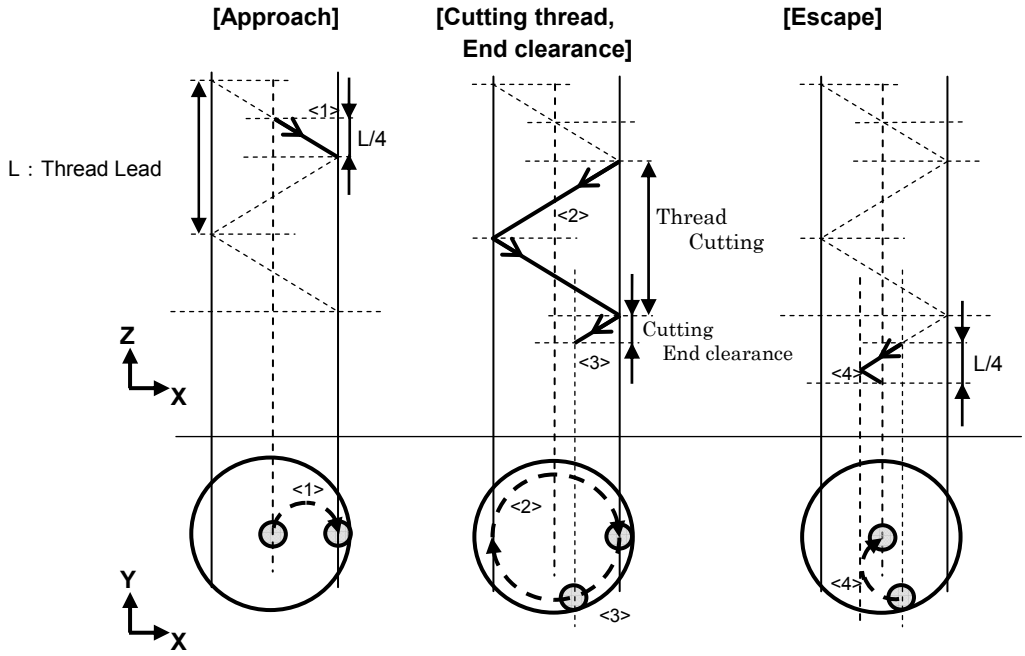
2.8.5 Basic Motions of Tool

The basic motions of thread milling are the following steps from <1> to <4>.

<1> Approach with tangential arc in Helical interpolation
<2> Thread cutting in Helical interpolation
<3> Cutting end clearance in Helical interpolation
<4> Escape in Helical interpolation

About the amount of tool movement with tool direction in approach arc/ escape arc

In approach with tangential arc, as the tool approaches from the center of hole along the path of Helical interpolation for threading, the amount of the tool movement along the tool axis direction is L/4 (L: thread pitch).
In escape with tangential arc, as the tool escapes to the center of hole along the path of Helical interpolation, the amount of the tool movement along the tool axis direction is L/4, too.



2.8.6 Machining Type

In thread milling cycle, there are three machining type as follows.

(1) Machining type 1
The cycle is completed by executing one spiral movement only.

(2) Machining type 2
The cycle is completed by executing a number of spirals equal to the number of programmed threads.

(3) Machining type 3
Tool moves in parallel to the tool axis by the length of tool. Then, the tool approaches to the wall of hole and performs threading.
The above motions are repeated until the tool reaches to the bottom of hole.

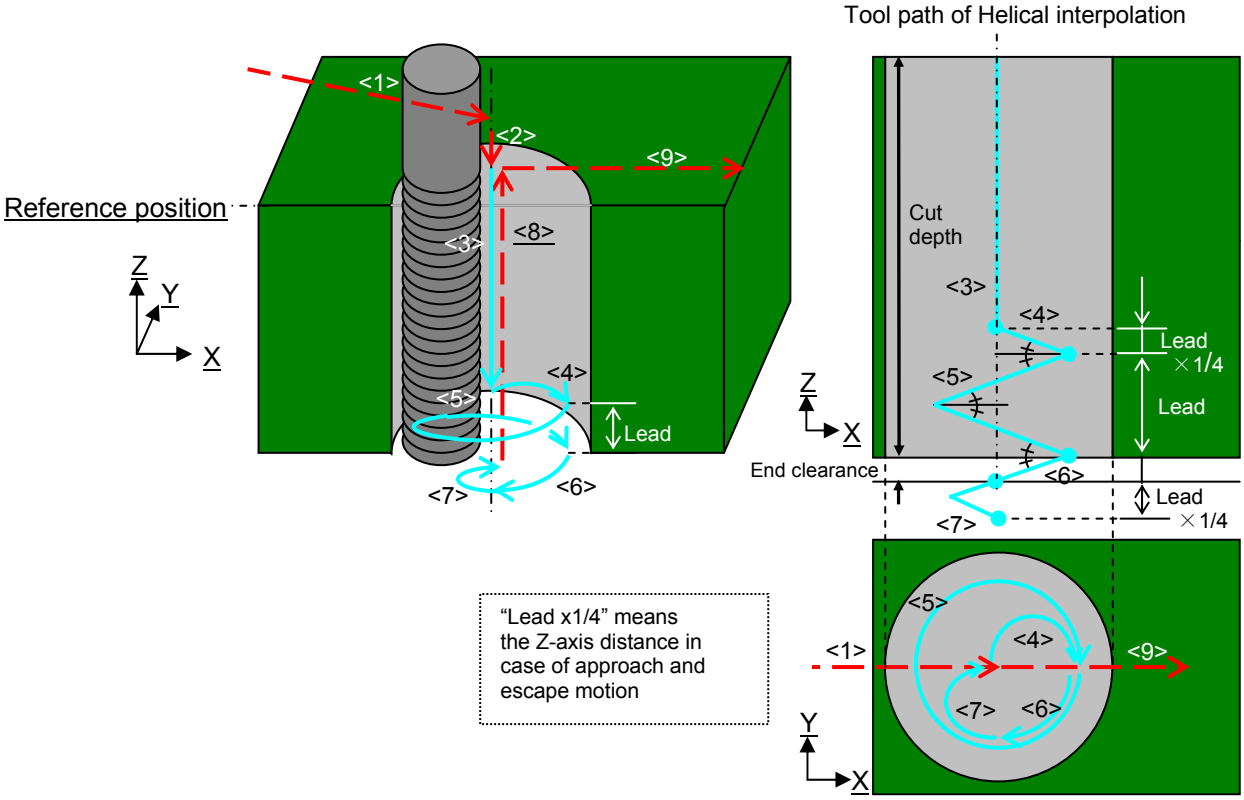
2.8.7 Pattern of Tool Path

Each machining type has 4 patterns of the tool path according to the combination of the motion of the tool along the tool axis direction"+/-" and the direction of the tool along the spiral is "CW/CCW" .

End Face Position	Thread Figure and Cutting Method setting		Tool Path	
	Right Hand Screw or Left Hand Screw	Up Cut or Down Cut	Movement Direction of Tool Axis Direction	Movement Direction of Tool Radius Direction
+ End Face	Right Hand Screw	Up Cut	- Direction	CW
	Left Hand Screw	Down Cut	- Direction	CCW
	Right Hand Screw	Down Cut	+ Direction	CCW
	Left Hand Screw	Up Cut	+ Direction	CW
- End Face	Right Hand Screw	Up Cut	+ Direction	CCW
	Left Hand Screw	Down Cut	+ Direction	CW
	Right Hand Screw	Down Cut	- Direction	CW
	Left Hand Screw	Up Cut	- Direction	CCW

2.8.8 Tool Path (Machining Type 1)

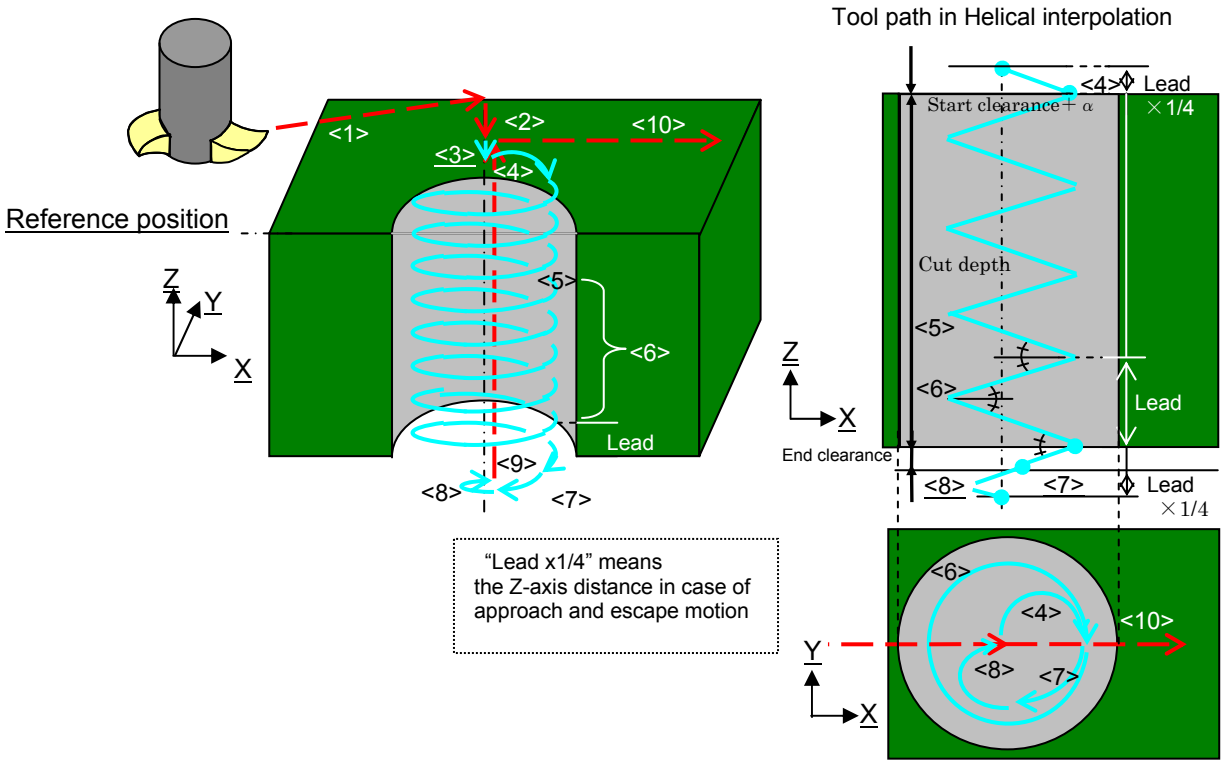
Tool paths in case of machining type 1 generate following movements. The following explanations are in case that the tool moves to -Z-axis direction in cutting feed rotating CW in helical interpolation.



<1> Approach to the I-point of the center of hole
<2> Move to the R-point "=clearance position" in rapid.
<3> Move to the "Reference position –Cut depth + Lead +(Leadx1/4)" in cutting feed rate.
<4> In order to approach to the start point of thread, move CW in Helical interpolation. And then, move to the "-Z" direction by "Lead/4"
<5> To execute the threads, the tool moves CW in Helical interpolation and move to the "-Z" direction by the lead.
<6> Move CW in Helical interpolation by the amount of the end clearance.
<7> In order to return to the center of hole, the tool moves CW in Helical interpolation and moves to the "-Z" direction by "Lead/4".
<8> Move to the I-point or R-point in rapid. In case of movements to the next hole position, the tool moves to the R-point. In other case, it moves to the I-point.
<9> The tool moves to the next hole position in rapid.

2.8.9 Tool Path (Machining Type 2)

Tool paths in case of machining type 2 generate as follows. The following explanations are in case that the tool moves to -Z-axis direction in cutting feed with CW in helical interpolation.



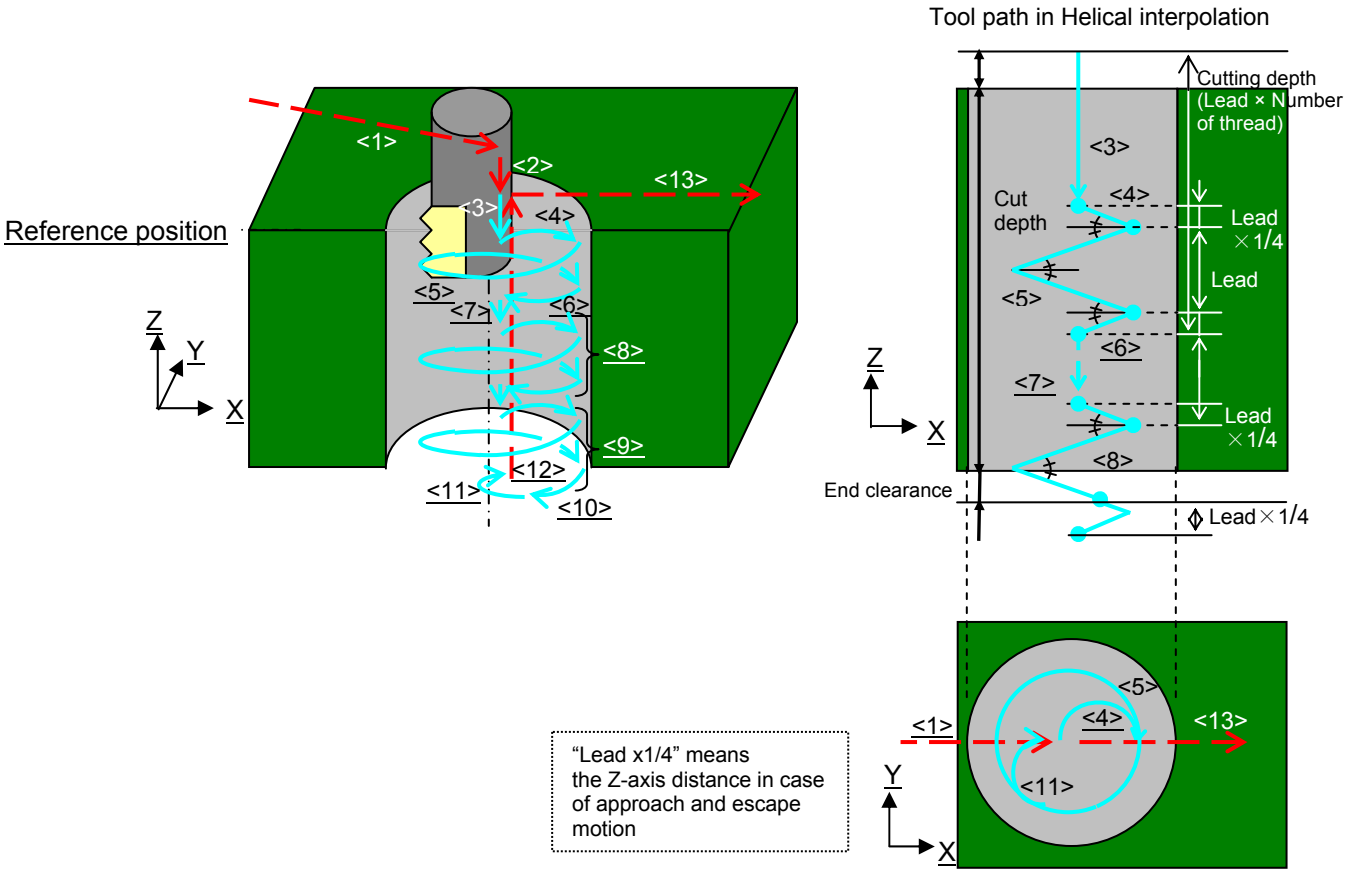
<1> Approach to the I-point of the center of hole
<2> Move to the R-point "=clearance position" in rapid.
<3> Move to the "Reference position +(Start clearance x Lead) + α + (Lead/4)" in cutting feed rate. Start clearance is used in order to cut fine the thread start point on the work face. Start clearance is specified by the parameter No.27452.
<4> In order to approach to the position "Reference position + Start clearance + α " start point of thread, move CW in Helical interpolation. And then, move to the "-Z" direction by "Lead/4"
<5> To execute the threads, the tool moves CW in Helical interpolation. And then, it moves to the "-Z" direction by the lead.
<6> Repeat the motion <5> until tool reaches to the Cut depth.
<7> Move CW in Helical interpolation by the amount of the end clearance.
<8> In order to return to the center of hole, the tool moves CW in Helical interpolation and moves to the "-Z" direction by "Lead/4".
<9> The tool moves to the I-point or R-point in rapid. In case of movement to the next hole position, the tool moves to the R-point. In other cases, the tool moves to the I-point.
<10> The tool moves to the next hole position in rapid

On the meaning of α

By cutting from the height of (work surface + α), the tool movement of the tool axis direction is adjusted in order to position the tool bottom at the position (work surface - CUT DEPTH) when the tool finishes cutting the last lead of the thread.

2.8.10 Tool Path (Machining Type 3)

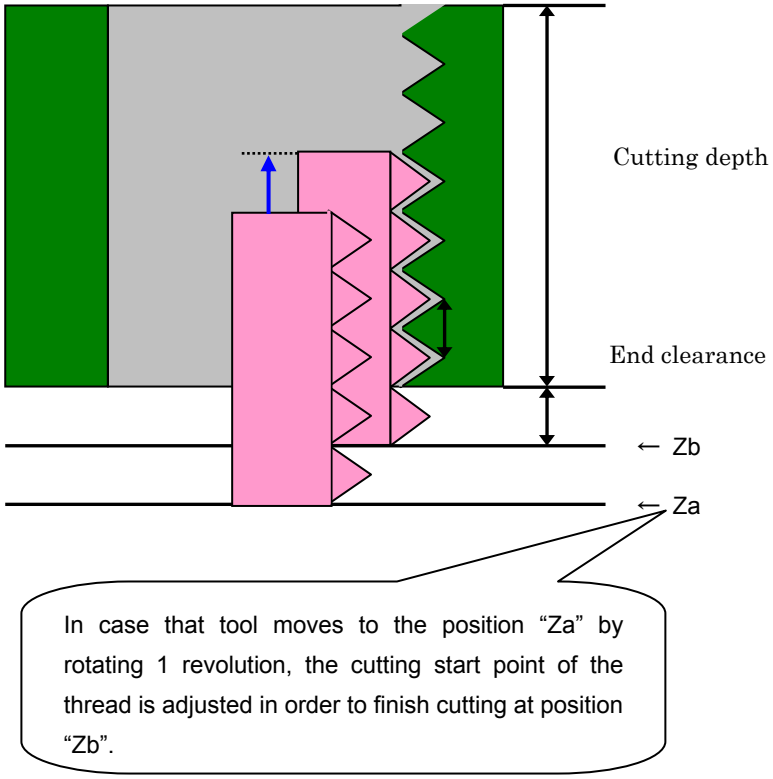
Tool paths in case of machining type 3 generate as follows. The following explanations are in case that a tool moves to -Z-axis direction in cutting feed with rotating CW in helical interpolation.



<1> Approach to the I-point of the center of hole.
<2> Move to the R-point "=clearance position" in rapid.
<3> Move to the "Reference position -(Lead x Number of thread) +Lead+(Start clearance x Lead)+(Lead/4)" in cutting feed rate.
<4> In order to approach to the start point of thread, move CW in Helical interpolation and move to the"-Z" direction by "Lead/4".
<5> To execute the threads, the tool moves CW in Helical interpolation and moves to the"-Z" direction by the lead.
<6> In order to return to the center of hole, the tool moves CW in Helical interpolation and moves to the "-Z" direction by "Lead/4".
<7> From the end point of <6> to the "-Z" direction, the tool moves by the amount "Lead x ((number of thread)-1-(overlapped number of thread))–Lead/4-Lead/4".
The Number of overlapped of threads depends on the parameter No.27451.
<8> Repeat the motion <4> through <7>.
<9> In the motion <5>, Helical interpolation is performed until tool reaches the cut depth, and then stops the repetition.
<10> The tool moves CW in Helical interpolation by the amount of end clearance.
<11> In order to return to the center of hole, the tool moves CW in Helical interpolation and moves to the "-Z" direction by "Lead/4".
<12> The tool moves to the I-point or R-point in rapid. In case of movement to the next hole position, it moves to the R-point. In other case, it moves to the I-point.
<13> The tool moves to the next hole position in rapid.

NOTE

At the motions <9> and <10>, in case that tool moves over “Cutting depth + End clearance” as the figure below, the cutting start point of the thread is adjusted so that the thread is cut correctly.



2.8.11 Machining Simulation Screen

In animated drawing of machining simulation, treads are not displayed on the wall of hole which threads are cut after threading.
In drawing during machining, tool path of thread milling cycle is displayed in spiral.

2.8.12 Parameters

The following parameters are concerning to this function.

	#7	#6	#5	#4	#3	#2	#1	#0
27095			TRM					

TRM Thread milling cycle is
0: Not available.
1: Available.

27451	Over lapped amount for thread milling cycle

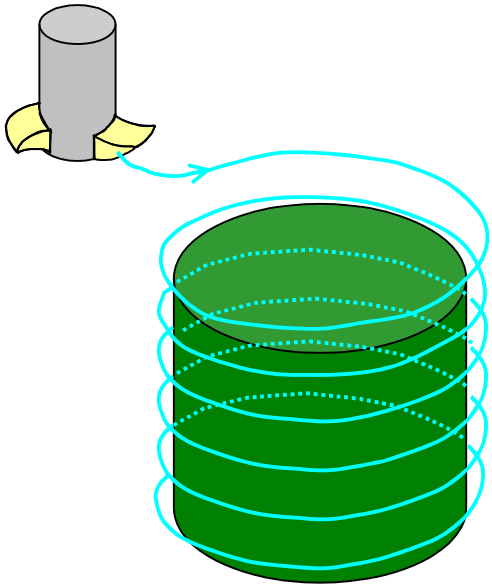
Set over lapped amount for thread milling cycle.

27452	Start clearance for thread milling cycle

Set start clearance for thread milling cycle.

2.9 OUTER THREAD MILLING CYCLE

The thread can be machined by moving tool along surface of work-piece in helical interpolation.
It's possible to machine a thread on the work-piece with large radius.

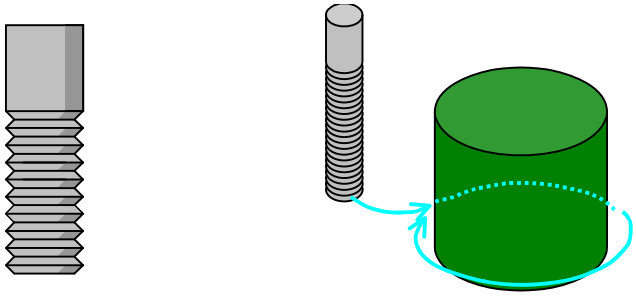


This function is enabled when the parameter No.27096#4 is set to 1.
In order to use this function, helical interpolation function is needed.
Helical interpolation function is optional function in FS30i-A/FS30i-B/FS0i-TD.

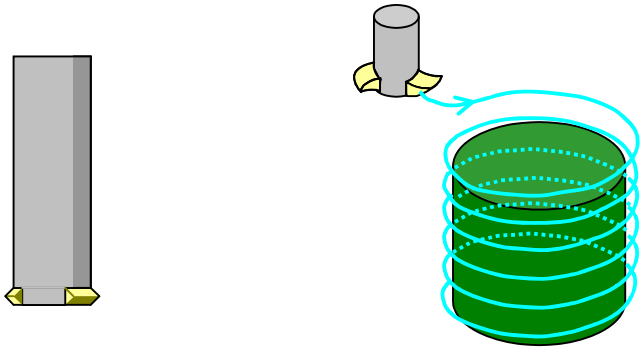
2.9.1 Machining Types

The following three machining types can be selected.

(1) Machining type 1
Tool moves helically 1 revolution along the surface of work-piece. All threads are cut in 1 revolution of tool.



(2) Machining type 2
Tool cuts thread as it moves helically revolutions with number of thread along the surface of work-piece.



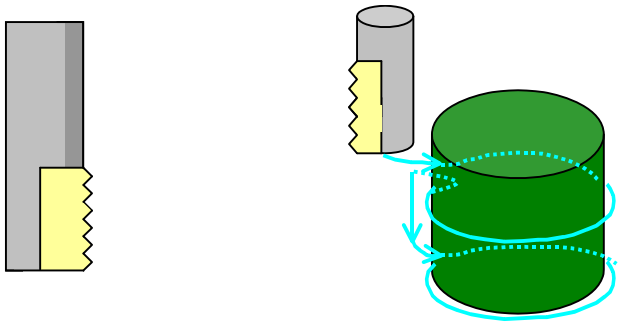
(3) Machining type 3

<1> Tool moves helically 1 revolution along the surface of work-piece. And, tool cuts threads equals to the number of tool edge.

<2> Once tool removes from work-piece, tool moves along the tool axis direction by the distance equal to the length of tool edge.

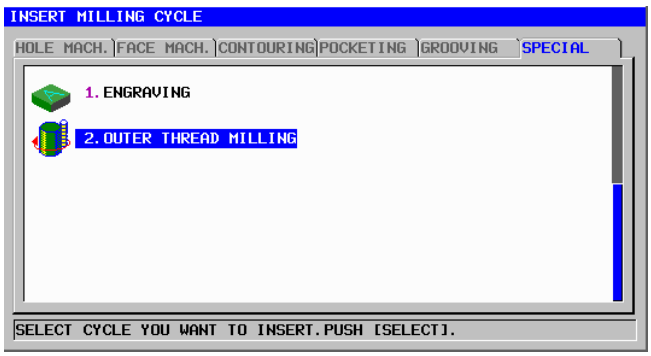
<3> Again, tool approaches to work-piece. Then continuation of threading <2> is performed.

<4> From <1> through <3> is repeated by the thread length.



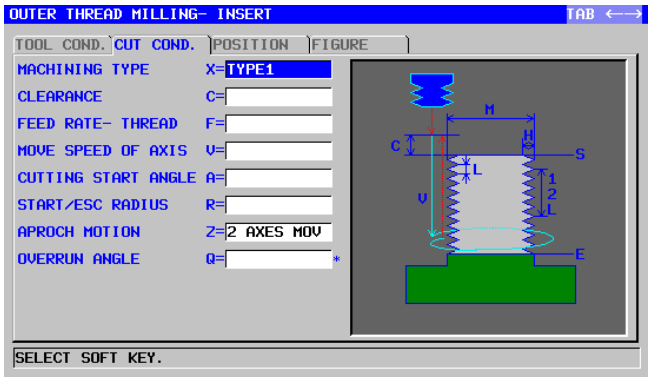
2.9.2 Method of Program Creation

In case that this function is available, “OUTER THREAD MILLING” menu is displayed on the [SPECIAL] tab of milling cycle menu screen.



When the menu “OUTER THREAD MILLING” is selected, the creation screen of the outer thread milling cycle is displayed.

By inputting data according to the guidance window, the program of thread milling cycle is created.



2.9.3 Outer Thread Milling Cycle Command

Outer thread milling cycle is commanded in the following format.

G1026 Dd Xx Cc Ff Vv Aa Rr Zz Qq Ww Yy Uu Pp Ii Jj Ss Ee Bb Nn Tt Mm Hh Ll

TOOL CONDITION		
Input item		**Meaning**
D	TOOL DIAMETER	Diameter of thread milling tool. Specify only in case of No.27002#0=1.

CUTTING CONDITION		
Input item		**Meaning**
X	MACHINING TYPE	Machining type. X=1: Type1 X=2: Type2 X=3: Type3
C	CLEARANCE	Clearance amount. In case of “START PT.(3RD AXIS)” > “END PT.(3 RD AXIS)”, Specify clearance amount from the position “START PT.(3 RD AXIS)”. In case of X=2, Specify clearance amount from the position “START PT.(3RD AXIS) + RUN-UP)”. In case of “START PT.(3RD AXIS)” < “END PT.(3RD AXIS)”, Specify clearance amount from “END PT.(3RD AXIS)”.
F	FEED RATE THREAD	Feed rate of threading
V	MOVE SPEED OF AXIS	Speed of moving tool to the tool-axis direction. In case of specifying 0, Rapid feed rate is set.
A	CUTTING START ANGLE	Angle of starting threading.
R	START/ESC RADIUS	Radius of the approach or escape.
Z	APROACH MOTION	Approach motion of tool, Z=2: simultaneous 2 axes Z=3: simultaneous 3 axes
Q	OVERRUN ANGLE	Angle of OVERRUN.
W	RUN-UP	Amount of Running up. Specify the amount from the position START PT.(3RD AXIS). Specify only in case of X=2 and START PT.(3RD AXIS) > END PT.(3RD AXIS).
Y	THREAD NUM. OF TOOL	Thread number of tool. Specify only in case of X=3.
U	MOVE SPEED OF RAD	Speed of moving to the tool radius direction. Specify only in case of X=3.

POSITION		
Input item		Meaning
P	PLANE SELECTION	Cutting plane. P=1 : XY plane P=2 : YZ plane
I	CENTER PT. (1ST AXIS)	Center point of threading (1'st axis)
J	CENTER PT. (2ND AXIS)	Center point of threading (2'nd axis)
S	START PT. (3RD AXIS)	Start point of threading (3'rd axis) In case of START PT.(3RD AXIS) < END PT.(3RD AXIS), Specify the start point included "incorrect thread length".
E	END PT. (3RD AXIS)	End point of threading (3'rd axis)
B	FACE POS.	End face position. B=1 : +End face B=2 : -End face Specify only in case of P=1(XY plane) & the parameter No.27000#4=1.

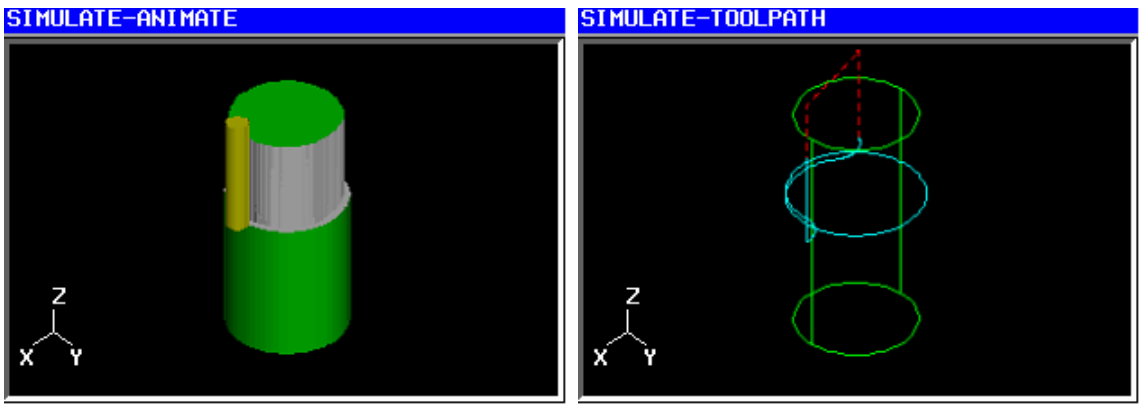
FIGURE		
Input item		Meaning
N	RIGHT-HAND/LEFT-HAND THREAD	Figure of threading N=1 : Right-hand thread N=2 : Left-hand thread
T	THREAD TYPE	Kind of threading T=1 : Metric thread T=2 : Unified thread
M	THREAD DIAMETER	Diameter of threading (Top of thread)
H	THREAD DEPTH	Depth of threading
L	THREAD LEAD	Lead of threading Specify only in case of T=1
	THREAD NUMBER	Number of threading per 1inch. Specify only in case of T=2.

2.9.4 Tool Type

In outer thread milling cycle, specify "Tapping Tool" as the tool type.

2.9.5 Machining Simulation

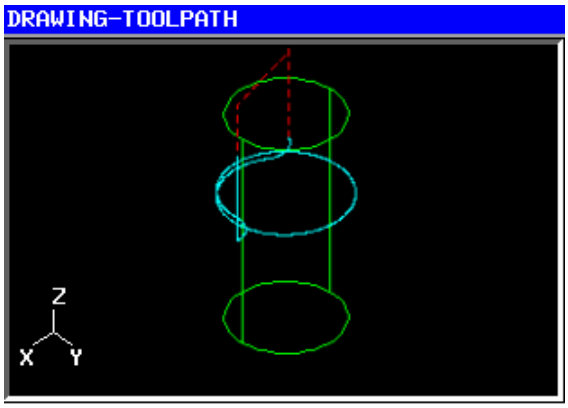
Animated drawing and tool path drawing are displayed as follows.



In animated drawing, machined surface of work-piece is displayed with gray color, and the shape of thread is not displayed.
Then the motion which tool moves around work-piece is not displayed.

2.9.6 Drawing-During-Machining

The drawing-during-machining is displayed as follows.



2.9.7 Parameters

The following parameters are concerning to this function.

<1> Parameter No.27459
Angle of machining start/escape for outer thread milling cycle
<2> Parameter No.27464
Number of overlapped threads in Machining type 3 for outer thread milling cycle

2.9.8 Alarms

Alarms related to this function are as follows.

Alarm No.	Message	Content
0021	ILLEGAL PLANE SELECT	Helical interpolation function is not available.
3507	WRONG CUTTING CONDITION	In the cutting condition, there is the following mistake. - In “CUTTING START ANGLE”, the value lower than 0 or grater than 360 is entered. - Tool interferes work-piece at the Start point for moving to the machining start point in arc. [“Distance from Center point to Start point for moving to the machining start point in arc”-”TOOL DIAMETER/2.0”] < ”THREAD DIAMETER/2.0” - In “RUN-UP”, the value lower than 0 or grater than 99999999.0 is entered. - In “OVERRUN ANGLE”, the value lower than 0 or grater than 360 is entered.
3510	NO MACHINING CYCLE BLOCK	Program is incorrect.
3511	UNCOMBINED G-CODE	Outer thread milling cycle is not available.

Alarm No.	Message	Content
3514	WRONG FIGURE DATA	In the figure data, there is the following mistake. - In “L THREAD/R THREAD”, the value except 1 or 2 is entered. - Address of “L THREAD/R THREAD” is not entered. - In “THREAD TYPE”, the value except 1 or 2 is entered. - Address of “THREAD TYPE” is not entered. - In “THREAD DEPTH”, the value lower than 0 or grater than thread diameter is entered. - In ”START PT.” and “END PT.”, the same value is entered. - Distance between “START PT.” And “END PT.” Is lower than 1 lead. - In case of “MACHINING TYPE”=3, the area smaller than “THREAD NUM. OF TOOL”*”THREAD LEAD” cannot be machined. (Distance of [“START PT. – END PT.] < “THREAD NUM. OF TOOL”*”THREAD LEAD”)
3516	NO NECESSARY ADDRESS	Necessary address is not entered.
3525	WRONG PARAMETER SETTING	In setting parameters, there is following mistake. - In the parameter “Angle of machining start/escape”, the value lower than 0 or grater than 360 is entered. - In the parameter “Number of overlapped threads”, the value lower than 0 is entered.
3529	WRONG SELECTING THE PLANE	In “PLANE SELECTION”, the value except 1 or 2 is entered.
3530	WRONG MACHINING TYPE	In “MACHINING TYPE”, the value except 1 to 3 is entered or address of “MCHINING TYPE” is not entered.
3535	WRONG THREADING LEAD	In “THREAD LEAD”, the value lower than 0 or grater than 99999999.0 is entered.
3544	WRONG TOOL DIAMETER	In “TOOL DIAMETER”, invalid value is entered.
3548	WRONG CLEARANCE	In “CLEARANCE”, the value lower than 0 or grater than 99999999.0 is entered.
3550	WRONG FEED RATE	In feed rate, there is the following mistake. - In “FEED RATE THREAD” or “MOVE SPEED OF RAD”, the value lower than 0 or grater than 99999999.0 is entered. - In “MOVE SPEED OF AXIS”, the value lower than 0 or grater than 99999999.0 is entered.
3552	WRONG APPROACH SETTING	In the data of approach, there is the following mistake. - In “START/ESC RADIUS”, the value lower than 0 or grater than 99999999.0 is entered. - In “APROACH MOTION”, the value except 2 or 3 is entered.
3564	TOOL DATA IS SHORT	The following tool data is short. - Necessary tool data for executing outer thread milling cycle - Tool compensation No.
3565	WRONG TOOL TYPE	Wrong tool is selected.
3586	WRONG COORDINATE POSITION	In the position, there is the following mistake. - In “CENTER PT.(1st AXIS)”, the value grater than 99999999.0 or lower than -99999999.0 is entered. - In “CENTER PT.(2nd AXIS)”, the value grater than 99999999.0 or lower than -99999999.0 is entered. - In “START PT.(3rd AXIS)”, the value grater than 99999999.0 or lower than -99999999.0 is entered. - In “END PT.(3rd AXIS)”, the value grater than 99999999.0 or lower than -99999999.0 is entered.
3590	WRONG THREAD/HOLE DIAMETER	In “THREAD DIAMETER”, the value lower than 0 or grater than 99999999.0 is entered.

Alarm No.	Message	Content
3591	WRONG THREAD NUMBER	In “THREAD NUMBER”, there is the following mistake. - In case of unified thread, the value lower than 0 or grater than 99999999.0 is entered in the “THREAD NUMBER” - In “THREAD NUM. OF TOOL”, the value lower than 1 or grater than 32767 is entered. - In case of “MACHINING TYPE”=3, the value of “THREAD NUMBER” lower than the parameter “Number of overlapped threads” is entered.
3595	WRONG FACE POSITION	In “FACE POS.”, the value except 1 or 2 is entered.

2.10 REAR END FACING BY MILLING

2.10.1 Rear End Facing

By setting bit 4 of parameter No. 27000 to 1, the input item "FACE POSITION" is displayed on the figure menu for milling below. By entering this data, rear end facing is enabled.

(1) Arbitrary-figure (XY plane) - Start point : G1200
(2) Hole position (XY plane) - Random points : G1210
(3) Hole position (XY plane) - Linear points (same interval) : G1211
(4) Hole position (XY plane) - Linear points (different interval) : G1212
(5) Hole position (XY plane) - Grid points : G1213
(6) Hole position (XY plane) - Rectangle points : G1214
(7) Hole position (XY plane) - Circle points : G1215
(8) Hole position (XY plane) - Arc points (same interval): G1216
(9) Hole position (XY plane) - Arc points (different interval) : G1217
(10) Fixed-figure (XY plane) - Rectangle : G1220
(11) Fixed-figure (XY plane) - Circle : G1221
(12) Fixed-figure (XY plane) - Track : G1222
(13) Fixed-figure (XY plane) - Radial groove : G1223

2.11 MIRROR / ROTATION / SCALING / FIGURE COPY COMMAND

New menus for mirror, rotation and scaling are added to COORD CONV tab in START window. REPETITION tab was added to START window, and in this tab, figure copy menu of rotation and linear movement menus are added.

NOTE
1 These functions are available on 16*i*/18*i*/21*i*-MB. On Series 30i for lathe machine and G-code system A, scaling and scaling cancel cannot be used.
2 In order to use these functions, milling cycle machining option is needed on Series 16*i*/18*i*/21*i*-B. On Series 30i, turning or milling cycle machining option is needed.
3 In order to use these functions, corresponding CNC option for each function is needed.

2.11.1 Coordinate Conversion (Mirror Image)

COORDINATE CONVERSION	
G1980	COORDINATE COBVERSION (MIRROR IMAGE)
G1981	COORDINATE CONVERSION CANCEL (MIRRORIMAGE)

NOTE
1 Coordinate conversion cancel (mirror image) must be done after finishing the machining using coordinate conversion (mirror image).
2 In order to use this function, set parameters No.27000#6(MC6) to 0.
3 Programmable mirror image option is needed for this function.

COORDINATE CONVERSION (MIRROR IMAGE) : G1980

Mirror image can be set by entering the symmetry axis data.

COORDINATE CONVERSION (MIRROR IMAGE) - INSERT		
Data item		Meaning
X	SYMMETRY AXIS X	X coordinate of mirror image symmetry axis
Y	SYMMETRY AXIS Y	Y coordinate of mirror image symmetry axis
Z	SYMMETRY AXIS Z	Z coordinate of mirror image symmetry axis

COORDINATE CONVERSION CANCEL (MIRROR IMAGE) : G1981

Cancel mirror image.

COORDINATE CONVERSION CANCEL (MIRROR IMAGE) - INSERT		
Data item		Meaning
X	CANCEL X	DO : Cancel mirror image of X-axis
Y	CANCEL Y	DO : Cancel mirror image of Y-axis
Z	CANCEL Z	DO : Cancel mirror image of Z-axis

2.11.2 Coordinate Conversion (Rotation)

COORDINATE CONVERSION	
G1957	COORDINATE CONVERSION (ROTATION)
G1958	COORDINATE CONVERSION CANCEL (ROTATION)

NOTE
1 Coordinate conversion cancel (rotation) must be done after finishing the machining using coordinate conversion (rotation).
2 In order to use this function, set parameters No.27000#6(MC6) to 0.
3 Coordinate rotation option is needed for this function.

XY PLANE COORDINATE CONVERSION (ROTATION) : G1957

Coordinate rotation on XY plane can be set by entering the center point and angle data.

COORDINATE CONVERSION (ROTATION) - INSERT		
Data item		Meaning
L	PLANE SELECTION	Rotation plane (only XY can be set)
X	CENTER X OF ROTATION	Center point X coordinate of rotation
Y	CENTER Y OF ROTATION	Center point Y coordinate of rotation
R	ANGLE	Rotation angle

ZX PLANE COORDINATE CONVERSION (ROTATION) : G1957

Coordinate rotation on ZX plane can be set by entering the center point and angle data.

COORDINATE CONVERSION (ROTATTION) - INSERT		
Data item		Meaning
L	PLANE SELECTION	Rotation plane (only ZX can be set)
X	CENTER X OF ROTATION	Center point X coordinate of rotation
Z	CENTER Z OF ROTATION	Center point Z coordinate of rotation
R	ANGLE	Rotation angle

YZ PLANE COORDINATE CONVERSION (ROTATION) : G1957

Coordinate rotation on YZ plane can be set by entering the center point and angle data.

COORDINATE CONVERSION (ROTATTION) - INSERT		
Data item		Meaning
L	PLANE SELECTION	Rotation plane (only YZ can be set)
Z	CENTER Z OF ROTATION	Center point Z coordinate of rotation
Y	CENTER Y OF ROTATION	Center point Y coordinate of rotation
R	ANGLE	Rotation angle

COORDINATE CONVERSION CANCEL (ROTATION) : G1958

Cancel coordinate rotation mode.

COORDINATE CONVERSION CANCEL (ROTATION) - INSERT		
Data item		Meaning
C	CANCEL	Cancel coordinate rotation mode

2.11.3 Scaling

COORDINATE CONVERSION	
G1982	SCALING
G1983	SCALING CANCEL

NOTE
1 Scaling cancel must be done after finishing the machining using coordinate conversion (rotation).
2 In order to use this function, set parameters No.27000#6(MC6) and No.27000#2(FGC) to 0.
3 Scaling option is needed for this function.

SCALING : G1982

By selecting ALL AXIS on SCALE TYPE, scaling around the entered center point can be done.

SCALING		
Data item		Meaning
L	SCALE TYPE	Type of scaling and ALL axis is selected
X	CENTER X	Center point X coordinate of scaling
Y	CENTER Y	Center point Y coordinate of scaling
Z	CENTER Z	Center point Z coordinate of scaling
P	SCALIE	Scaling amount for all axis

By selecting EACH AXIS on SCALE TYPE, scaling for each selected axis is done.

SCALING		
Data item		Meaning
L	SCLAE TYPE	Type of scaling and EACH AXIS is selected
X	CENTER X	Center point X coordinate of scaling
Y	CENTER Y	Center point Y coordinate of scaling
Z	CENTER Z	Center point Z coordinate of scaling
I	SCALE X	Scaling amount for X-axis
J	SCALE Y	Scaling amount for Y-axis
K	SCALE Z	Scaling amount for Z-axis

NOTE
When EACH AXIS is selected and minus value for axis scale amount, mirror image becomes available.

SCALING CANCEL : G1983

Cancel scaling.

SCALING		
Data item		Meaning
C	CANCEL	Cancel scaling mode

2.11.4 Figure Copy (Rotation Copy/ Parallel Copy)

REPETITION	
G72.1	FIGURE COPY (ROTATION COPY)
G72.2	FIGURE COPY (PARALLEL COPY)

NOTE
1 In order to use this function, set parameters No.27000#2(FGC) to 0.
2 Rotational copy option is needed for figure copy (rotation copy), and linear copy option is needed for figure copy (parallel copy).
3 Specify a plane selection command (G17, G18 or G19) to select the plane on which the rotation copy is made.

FIGURE COPY (ROTATION COPY) : G72.1

On the plane specified before this command, selected sub-program can be executed with rotating by entered angle around center point. This operation can be repeated for entered repeating number.

In case of T system and parameter No.27002#3=0

FIGURE COPY (ROTATION COPY) - INSERT		
Data item		Meaning
P	SUB PROGRAM NO.	Sub program number to be executed for repetition with rotation
L	NUMBER OF REPETITION	Number of repetition
X	CENTER X OF ROTATION	Center point X coordinate of rotation
Z	CENTER Z OF ROTATION	Center point Z coordinate of rotation
R	ANGLE	Rotation angle

In case of T system and parameter No.27002#3=1

FIGURE COPY (ROTATION COPY) - INSERT		
Data item		Meaning
P	SUB PROGRAM NO.	Sub program number to be executed for repetition with rotation
L	NUMBER OF REPETITION	Number of repetition
X	CENTER X OF ROTATION	Center point X coordinate of rotation
Y	CENTER Y OF ROTATION	Center point Y coordinate of rotation
Z	CENTER Z OF ROTATION	Center point Z coordinate of rotation
R	ANGLE	Rotation angle

FIGURE COPY (PARALLEL COPY) : G72.2

On the plane specified before this command, selected sub-program can be executed with shifting by entered shift amount. This operation can be repeated for entered repeating number.

In case of T system and parameter No.27002#3=0

FIGURE COPY (PARALLEL COPY) - INSERT		
Data item		Meaning
P	SUB PROGRAM NO.	Sub program number to be executed for repetition with rotation
L	NUMBER OF REPETITION	Number of repetition
I	SHIFT VALUE X	Shifting amount of X direction
K	SHIFT VALUE Z	Shifting amount of Z direction

In case of T system and parameter No.27002#3=1

FIGURE COPY (PARALLEL COPY) - INSERT		
Data item		Meaning
P	SUB PROGRAM NO.	Sub program number to be executed for repetition with rotation
L	NUMBER OF REPETITION	Number of repetition
I	SHIFT VALUE X	Shifting amount of X direction
J	SHIFT VALUE Y	Shifting amount of Y direction
K	SHIFT VALUE Z	Shifting amount of Z direction

2.11.5 Parameter

	#7	#6	#5	#4	#3	#2	#1	#0
27000		MC6						

MC6 0 : In [START] menu screens, the tab of [COORD CONV] is displayed.
1 : In [START] menu screens, the tab of [COORD CONV] is not displayed.

	#7	#6	#5	#4	#3	#2	#1	#0
27002					YMN	FGC		

FGC 0 : In [START] menu screens, the tab of [FIG. COPY] is displayed.
1 : In [START] menu screens, the tab of [FIG. COPY] is not displayed.

YMN 0 : In the cycle menu screen or cycle input screen, the items related to the Y-axis are not displayed. (Only for the T series CNC system).
1 : In the cycle menu screen or cycle input screen, the items related to the Y-axis are displayed. (Only for the T series CNC system).

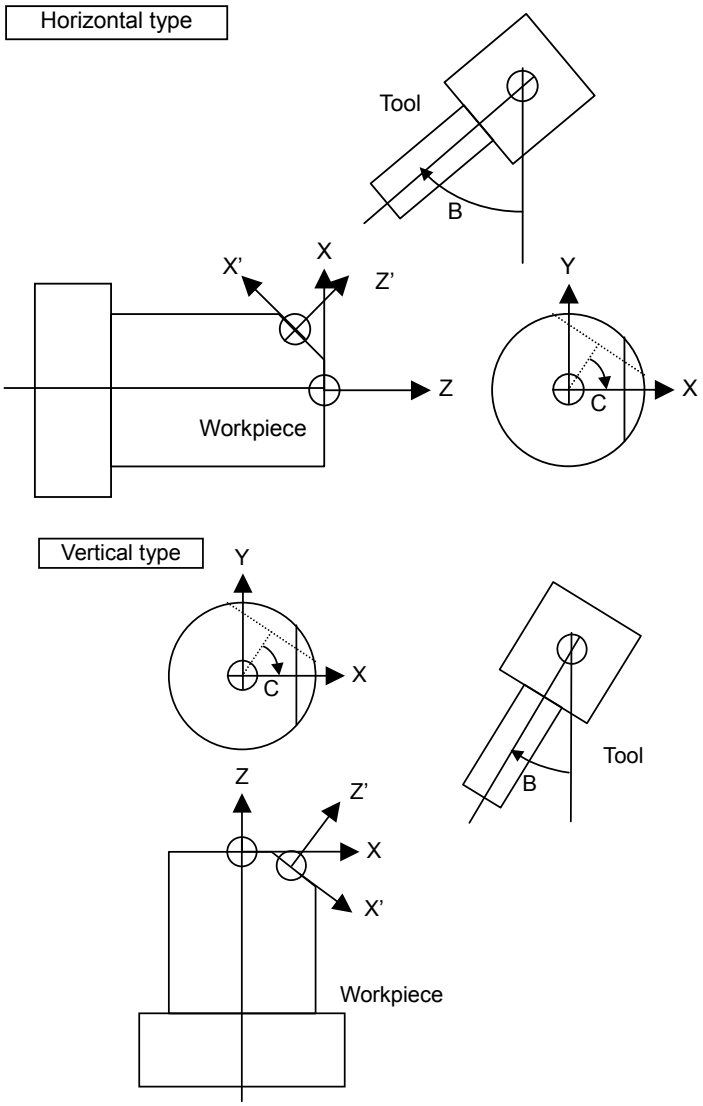
2.12 SLANT FACE MACHINING CYCLE (COORDINATE CONVERSION)

NOTE
To use slant face machining with MANUAL GUIDE *i*, the option for the three-dimensional coordinate conversion function is required.
For details, refer to the relevant manual issued by the machine tool builder.

2.12.1 Supportable Machine Configuration

With MANUAL GUIDE *i*, slant face machining, which is a mixture of table rotation and tool rotation, can be specified.
Those parameters that support a machine configuration used must be set beforehand.

(1) Mixture type
This type of machining machines a slant face by workpiece rotation and tool rotation.



NOTE
It is assumed that the C-axis rotates about the Z-axis, and the rotation center is on the Z-axis.
Moreover, it is assumed that the B-axis rotates about the Y-axis, and the rotation center is on the Y-axis.

2.12.2 Slant Face Machining Command (Coordinate Conversion)

When slant face machining is performed with MANUAL GUIDE *i*, a slant face to be machined must be first specified with the coordinate conversion command, then a machining program for milling to be performed must be input.
Upon completion of slant face machining, coordinate conversion cancellation must be specified.

NOTE
Slant face machining can be performed with the following types of milling (all on the XY plane):
1. Hole machining
2. Facing
3. Contouring
4. Pocketing
5. Grooving

With MANUAL GUIDE *i*, the G code commands listed below are available to enable coordinate conversion for slant face machining.

Coordinate conversion			
	Command block	G1952	Direct origin specification (with the rotation center on the Y-axis)
		G1953	Indirect origin specification (with the rotation center on the Y-axis)
		G1954	Direct origin specification (with the rotation center on the Z-axis)
		G1955	Indirect origin specification (with the rotation center on the Z-axis)
		G1959	Coordinate conversion cancel

NOTE
G code for coordinate conversion can be selected from the "COORDINATE CONVERSION" tab on the milling start command menu (displayed by pressing [START] on the milling menu).

(1) Input items of DIRECT ORIGIN POINT TYPE (ROT.CENTER AXIS-Y) (G1952)

A machining surface rotates about the point (reference point) that serves as the new workpiece origin of the slant face.

COORD CONVERSION		
Data item		Meaning
U	NEW WORK ORIGIN X'	X coordinate of the workpiece origin of a slant face
V	NEW WORK ORIGIN Y'	Y coordinate of the workpiece origin of a slant face
W	NEW WORK ORIGIN Z'	Z coordinate of the workpiece origin of a slant face
R	SLANT ANGLE	Signed angle relative to the Z-axis. The CW direction viewed from the plus direction of the rotation axis is positive. ($-90 \leq R \leq 90$).

(2) Input items of INDIRECT ORIGIN POINT TYPE (ROT.CENTER AXIS-Y) (G1953)

A machining surface rotates about a specified reference point, and the workpiece origin of a slant face, that is, a new machining surface, is to be specified based on the distance from the reference point (positive/negative value).

COORD CONVERSION		
Data item		Meaning
U	REF. POINT X	X coordinate of a reference point (workpiece coordinate system before rotation)
W	REF. POINT Z	Z coordinate of a reference point (workpiece coordinate system before rotation)
I	NEW WORK ORIGIN X'	X coordinate of the workpiece origin of a slant face
J	NEW WORK ORIGIN Y'	Y coordinate of the workpiece origin of a slant face
K	NEW WORK ORIGIN Z'	Z coordinate of the workpiece origin of a slant face

COORD CONVERSION		
Data item		Meaning
R	SLANT ANGLE	Signed angle relative to the Z-axis. The CW direction viewed from the plus direction of the rotation axis is positive. (-90≤R≤90).

(3) Input items of DIRECT ORIGIN POINT TYPE (ROT.CENTER AXIS-Z) (G1954)

The machining plane rotates about the new point (reference point) that is to become the workpiece origin of the XY plane.

COORD CONVERSION		
Data item		Meaning
U	REF. POINT X	X coordinate of a new workpiece origin
V	REF. POINT Y	Y coordinate of a new workpiece origin
W	REF. POINT Z	Z coordinate of a new workpiece origin
R	SLANT ANGLE	Signed angle with respect to the X-axis. The CW direction viewed from the plus rotation axis direction is positive (-90 ≤ R ≤ 90).

(4) Input items of INDIRECT ORIGIN POINT TYPE (ROT.CENTER AXIS-Z) (G1955)

The machining plane rotates about a specified reference point. Define the workpiece origin of the XY plane, which is a new machining plane, by specifying the distance (positive/negative value) from the reference point.

COORD CONVERSION		
Data item		Meaning
U	REF. POINT X	X coordinate of a reference point (workpiece coordinate system before rotation)
W	REF. POINT Y	Y coordinate of a reference point (workpiece coordinate system before rotation)
I	NEW WORK ORIGIN X'	X coordinate of the workpiece origin of the slant
J	NEW WORK ORIGIN Y'	Y coordinate of the workpiece origin of the slant
K	NEW WORK ORIGIN Z'	Z coordinate of the workpiece origin of the slant
R	SLANT ANGLE	Signed angle with respect to the Z-axis. The CW direction viewed from the plus rotation axis direction is positive (-90 ≤ R ≤ 90).

(5) Input items of COORDINATE CONVERSION CANCEL (G1959)

After coordinate conversion is canceled, the tool moves to the entered end point by rapid traverse. If no end point is specified, the tool will not move.

CANCEL		
Data item		Meaning
X*	END POINT X	X coordinate to which the tool is to move by rapid traverse after coordinate conversion is canceled. If this data is omitted, the tool will not move.
Y*	END POINT Y	Y coordinate to which the tool is to move by rapid traverse after coordinate conversion is canceled. If this data is omitted, the tool will not move.
Z*	END POINT Z	Z coordinate to which the tool is to move by rapid traverse after coordinate conversion is canceled. If this data is omitted, the tool will not move.

2.13 TILTED WORKING PLANE MACHINING (FOR Series 30*i*/31*i*)

The following tilted working plane machining cycles are enabled in MANUAL GUIDE *i*.

(1) Euler's angle (G1850)
(2) Roll-Pitch-Yaw angle (G1851)
(3) 3 points specification (G1852)
(4) 2 vectors specification (G1853)
(5) Projection angle (G1854)
(6) Tool Axis direction (G1855)
(7) Tilted working plane canceled (G1859)

NOTE
This function requires the following optional functions.
(1) Tilted Working Plane command with Guidance (R522)
(2) MANUAL GUIDE *i* Tilted Plane Machining function (S788)

NOTE
The following software is necessary to use this function in Series 30*i* /31*i* –A.
Series G003/G103/G123: Ver.10 or later

2.13.1 Operation

Display tilted working plane function menu screen by the following procedures.

<1> Press the [START] soft key for the milling cycle.
<2> Select "TWP" TAB, then tilted working plane function menu appears.
<3> Move the cursor on a tilted working plane command and press the [SELECT] soft key.

2.13.2 Tilted Working Plane Input Screen by Euler's Angle

Input items of this cycle G1850 are as follows.

TWP		
Data item		Meaning
A	MULTI TYPE	A1: Values of specified data are in work coordinate system. A2: Values of specified data are in feature coordinate system.
X	ORIGIN OF FEATURE COORDINATE	Origin X of feature coordinate
Y	ORIGIN OF FEATURE COORDINATE	Origin Y of feature coordinate
Z	ORIGIN OF FEATURE COORDINATE	Origin Z of feature coordinate
I	EULER'S ANGLE	Angle of rotation around Z-axis of original coordinate. The rotated coordinate becomes coordinate 1.
J	EULER'S ANGLE	Angle of rotation around X-axis of coordinate 1.The rotated coordinate becomes coordinate 2.
K	EULER'S ANGLE	Angle of rotation around Z-axis of coordinate 2.

TL.DIR.CTL		
Data item		Meaning
T	TOOL AXIS DIRECTION CONTROL	T1: None T2: Control point hold type T3: TCP hold type
H	TOOL OFFSET NUMBER	Tool length offset number
R	FROM TCP TO CENTER	Distance from tool center point to rotation center

The machining program is created as follows.

```
(Example)
  G1850A1.Q1.X0.Y0. Z0. I0. J0. K90. T1.       : Tilted plane machining
  G1040L10.F500.V500.E500.W1.B0.3C0.3Z2 : Pocket machining
  M98 P1001                                    : Figure block
  G1859                                        : Tilted plane machining cancel
```

2.13.3 Tilted Working Plane Input Screen by Roll-Pitch-Yaw

Input items of this cycle G1851 are as follows.

TWP		
Data item		Meaning
A	MULTI TYPE	A1: Values of specified data are in work coordinate system. A2: Values of specified data are in feature coordinate system.
Q	ORDER OF ROTATION	Order of rotating X-axis, Y-axis, Z-axis Q1: X-axis→Y-axis→Z-axis Q2: X-axis→Z-axis→Y-axis Q3: Y-axis→X-axis→Z-axis Q4: Y-axis→Z-axis→X-axis Q5: Z-axis→X-axis→Y-axis Q6: Z-axis→Y-axis→X-axis
X	ORIGIN OF FEATURE COORDINATE	Origin X of feature coordinate
Y	ORIGIN OF FEATURE COORDINATE	Origin Y of feature coordinate
Z	ORIGIN OF FEATURE COORDINATE	Origin Z of feature coordinate
I	ROTATION ANGLE ABOUT X	Rotation angle about X
J	ROTATION ANGLE ABOUT Y	Rotation angle about Y
K	ROTATION ANGLE ABOUT Z	Rotation angle about Z

TL.DIR.CTL		
Data item		Meaning
T	TOOL AXIS DIRECTION CONTROL	T1: none T2: control point hold type T3: TCP hold type
H	TOOL OFFSET NUMBER	Tool length offset number
R	FROM TCP TO CENTER	Distance from tool center point to rotation center

2.13.4 Tilted Working Plane Input Screen by Three Points

Input items of this cycle G1852 are as follows.

TWP		
Data item		Meaning
A	MULTI TYPE	A1: Values of specified data are in work coordinate system. A2: Values of specified data are in feature coordinate system.
B	1ST POINT X	Coordinate X of the origin of the new Feature Coordinate System
C	1ST POINT Y	Coordinate Y of the origin of the new Feature Coordinate System
D	1ST POINT Z	Coordinate Z of the origin of the new Feature coordinate System
E	2ND POINT X	Coordinate X of the second point on the tilted plane
F	2ND POINT Y	Coordinate Y of the second point on the tilted plane
I	2ND POINT Z	Coordinate Z of the second point on the tilted plane
J	3RD POINT X	Coordinate X of the third point on the tilted plane
K	3RD POINT Y	Coordinate Y of the third point on the tilted plan
L	3RD POINT Z	Coordinate Z of the third point on the tilted plan

SFT. ORIG.		
Data item		Meaning
M	SHIFT OF ORIGIN X	Shift amount X for 1st point
N	SHIFT OF ORIGIN Y	Shift amount Y for 1st point
P	SHIFT OF ORIGIN Z	Shift amount Z for 1st point
Q	ROTATION ANGLE ABOUT Z-AXIS	Rotation angle about Z-axis

TL.DIR.CTL		
Data item		Meaning
T	TOOL AXIS DIRECTION CONTROL	T1: none T2: control point hold type T3: TCP hold type
H	TOOL OFFSET NUMBER	Tool length offset number
R	FROM TCP TO CENTER	Distance from tool center point to rotation center

2.13.5 Tilted Working Plane Input Screen by Two Vectors

Input items of this cycle G1853 are as follows.

TWP		
Data item		Meaning
A	MULTI TYPE	A1: Values of specified data are in work coordinate system. A2: Values of specified data are in feature coordinate system.
X	ORIGIN OF FEATURE COORDINATE X	Origin X of feature coordinate
Y	ORIGIN OF FEATURE COORDINATE Y	Origin Y of feature coordinate
Z	ORIGIN OF FEATURE COORDINATE Z	Origin Z of feature coordinate
I	1ST VECTOR I	The direction of X axis of Feature coordinate system direction on Work coordinate system (1st Vector)
J	1ST VECTOR J	
K	1ST VECTOR K	
B	2ND VECTOR I	The direction of Z axis of Feature coordinate system direction on Work coordinate system (2nd Vector)
C	2ND VECTOR J	
D	2ND VECTOR K	

TL.DIR.CTL		
Data item		Meaning
T	TOOL AXIS DIRECTION CONTROL	T1: none T2: control point hold type T3: TCP hold type
H	TOOL OFFSET NUMBER	Tool length offset number
R	FROM TCP TO CENTER	Distance from tool center point to rotation center

2.13.6 Tilted Working Plane Input Screen by Projection Angles

Input items of this cycle G1854 are as follows.

TWP		
Data item		Meaning
A	MULTI TYPE	A1: Values of specified data are in work coordinate system. A2: Values of specified data are in feature coordinate system.
X	ORIGIN OF FEATURE COORDINATE X	Origin X of feature coordinate
Y	ORIGIN OF FEATURE COORDINATE Y	Origin Y of feature coordinate
Z	ORIGIN OF FEATURE COORDINATE Z	Origin Z of feature coordinate
I	1ST PROJECTION ANGLE	Projection angle between X-axis
J	2ND PROJECTION ANGLE	Projection angle between Y-axis
K	3RD PROJECTION ANGLE	Rotation angle around Z-axis

TL.DIR.CTL		
Data item		Meaning
T	TOOL AXIS DIRECTION CONTROL	T1: none T2: control point hold type T3: TCP hold type
H	TOOL OFFSET NUMBER	Tool length offset number
R	FROM TCP TO CENTER	Distance from tool center point to rotation center

2.13.7 Tilted Working Plane Input Screen by Tool Axis Direction

Input items of this cycle G1855 are as follows.

TWP		
Data item		Meaning
A	ORIGIN COMMAND OF FEATURE COORDINATE	A1: Feature coordinate is specified. A2: No feature coordinate is specified.
X	ORIGIN X OF FEATURE COORDINATE	Origin X of feature coordinate in work coordinate system.
Y	ORIGIN Y OF FEATURE COORDINATE	Origin Y of feature coordinate in work coordinate system.
Z	ORIGIN Z OF FEATURE COORDINATE	Origin Z of feature coordinate in work coordinate system.
R	ROTATION ANGLE AROUND Z-AXIS	Rotation angle around Z-axis in feature coordinate system

2.13.8 Tilted Working Plane Input Screen by Tilted Plane Cancel

Input items of this cycle G1859 are as follows.

TWP		
Data item		Meaning
C	CANCEL	Tilted Plane Cancel (Input Disable)

2.13.9 Restriction of Machining Simulation

The restrictions are in the followings.

(1) The machining simulation only works when the mechanical unit type of machine is the table rotation type and the parameter No.19680 is set to be 12.
(2) The machining simulation works if rotary axes are specified as follows.
A is the rotary master axis;
C is the rotary slave axis;
In all other rotary axis configurations machining simulation does not work.
(3) In case that “Multi type” is “INC”, the machining simulation does not work correctly.
(4) Machining simulation by [SINGLE] soft key does not work correctly.
(5) Set-up Guidance Functions of MANUAL GUIDE *i* do not work correctly during the tilted working plane command mode.

2.14 MILLING BY ROTATING A WORKPIECE ABOUT THE Y-AXIS

The following milling operations are performed by rotating a workpiece about the Y-axis:

- B-axis hole machining on a cylindrical surface (YB)
- B-axis or Y-axis grooving on a cylindrical surface

NOTE
1 The axis number of the B-axis must be assigned to the axis about which the workpiece is rotated.
2 Turning cycles cannot be used.
3 Polar coordinate interpolation cannot be used.
4 Circular interpolation cannot be used.
5 It is possible to use this function in only horizontal machining center.

2.14.1 Machining Type

(1) Input items of hole machining (B-axis holes on a cylindrical surface)

Same as in Subsection 4.1.1, "Machining Command (Tool Rotation: M Series)".

(2) Input items of grooving (BY axis on a cylindrical surface) (G1054)

CUT COND.		
Data item		Meaning
J	CUT DEPTH OF AXIS	Depth of cut in the tool axis direction per cutting operation (radius value, positive value)
F	FEED RATE - RADIUS	Feedrate applicable when cutting is performed in the tool radius direction

CUT COND.		
Data item		Meaning
E	FEED RATE - AXIS	Feedrate applicable when cutting is performed in the tool axis direction toward the bottom of a side face being machined
C	CLEARANCE OF AXIS	Distance between the surface of a blank being machined and cutting start point (point R) in the tool axis distance (radius value, positive value)
Z	APPROCH MOTION	[2 AXES]: When moving from the current position to the machining start point, the tool first moves in the machining plane in two-axis synchronous operation and then moves along the tool axis. (Initial value) [3 AXES]: The tool moves from the current position to the machining start point in three-axis synchronous operation.

2.14.2 Figure

(1) Input items of B-axis holes (arc points) on a cylindrical surface (YB) (G1777)

POS / SIZE		
Data item		Meaning
B	BASE POSITION	Z-axis coordinate value on the workpiece surface
Z	Y AXIS POS.	Y-axis coordinate value of the hole position (common to all holes)
A	STARTING ANGLE	Central angle of the first hole to the B0 axis (positive or negative value)

REPEAT		
Data item		Meaning
C*	PITCH ANGLE	Angle between hole positions
M*	HOLE NUMBER	Number of holes

(2) Input items of B-axis holes (random points) on a cylindrical surface (YB) (G1778)

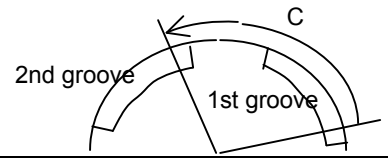
HOLE POS-1		
Data item		Meaning
B	BASE POSITION	Z-axis coordinate value on the workpiece surface
H	Y AXIS POS.1	Y coordinate value of the first hole
V	B AXIS POS.1	B coordinate value of the first hole
A*	Y AXIS POS.2	Y coordinate value of the second hole
C*	B AXIS POS.2	B coordinate value of the second hole
D*	Y AXIS POS.3	Y coordinate value of the third hole
E*	B AXIS POS.3	B coordinate value of the third hole
F*	Y AXIS POS.4	Y coordinate value of the fourth hole
I*	B AXIS POS.4	B coordinate value of the fourth hole

HOLE POS-2		
Data item		Meaning
J*	Y AXIS POS.5	Y coordinate value of the fifth hole
K*	B AXIS POS.5	B coordinate value of the fifth hole
M*	Y AXIS POS.6	Y coordinate value of the sixth hole
P*	B AXIS POS.6	B coordinate value of the sixth hole
Q*	Y AXIS POS.7	Y coordinate value of the seventh hole
R*	B AXIS POS.7	B coordinate value of the seventh hole
S*	Y AXIS POS.8	Y coordinate value of the eighth hole
T*	B AXIS POS.8	B coordinate value of the eighth hole

NOTE
Values need not be entered for all hole positions, but it is necessary to enter both the Y and B coordinate values for one hole position.

(3) Input items of B-axis grooves on a cylindrical surface (G1775)

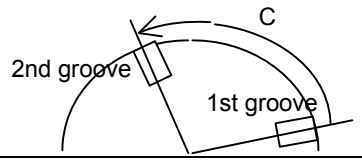
POS / SIZE		
Data item		Meaning
B	BASE POSITION	Z-axis coordinate value of the top surface of the workpiece to be grooved or the groove bottom (in the tool axis direction)
L	HEIGHT/DEPTH	When the top surface of the workpiece is selected for BASE POSITION, the distance to the groove bottom is specified with a negative value (radius value). → Depth When the groove bottom is selected for BASE POSITION, the distance to the upper surface of grooves is specified with a positive value (radius value). → Height
Z	Y AXIS POS.	Y-axis coordinate value of the center position around which grooves are placed
Q	GROOVE ANGLE	Inclination angle of a groove with respect to the radial direction of the arc along which grooves are placed (positive or negative value)
A	STARTING ANGLE	Central angle of the first groove position with respect to the B0 axis (positive or negative value)

REPEAT		
Data item		Meaning
C*	PITCH ANGLE	Angle between grooves 
M*	GROOVE NUMBER	Number of grooves

(4) Input items of Y-axis grooves on a cylindrical surface (G1776)

POS / SIZE		
Data item		Meaning
B	BASE POSITION	Z-axis coordinate value of the top surface of the workpiece to be grooved or the groove bottom (in the tool axis direction)

POS / SIZE		
Data item		Meaning
L	HEIGHT/DEPTH	When the top surface of the workpiece is selected for BASE POSITION, the distance to the groove bottom is specified with a negative value (radius value). → Depth When the groove bottom is selected for BASE POSITION, the distance to the upper surface of grooves is specified with a positive value (radius value). → Height
Z	Y AXIS POS.	Y-axis coordinate value of the center position around which grooves are placed
P	GROOVE LENGTH	Groove length in the Y-axis direction
A	STARTING ANGLE	Central angle of the first groove position with respect to the B0 axis (positive or negative value)

REPEAT		
Data item		Meaning
C*	PITCH ANGLE	Angle between grooves 
M*	GROOVE NUMBER	Number of grooves

2.14.3 Machining Simulation

Milling of a workpiece rotating around the Y-axis can be simulated.

NOTE
1 Turning cannot be simulated.
2 Simulation using polar coordinate interpolation cannot be performed.
3 Simulation using cylindrical interpolation cannot be performed.
4 Because the workpiece rotation axis is determined by parameter setting, it is impossible to change the workpiece rotation axis during simulation.
5 While rotation of the workpiece about the Y-axis is being simulated, the coordinate axes are displayed as X, Y, and -Z.

The following types of blank figures rotating about the Y-axis are available:

(1) Column (rotating about the Y-axis) : G1756
(2) Column with a hole (rotating about the Y-axis) : G1757
(3) Prism (rotating about the Y-axis) : G1758
(4) Prism with a hole (rotating about the Y-axis) : G1759
(5) Arbitrary column figure (rotating about the Y-axis)
Start point (rotating about the Y-axis) : G1780
Line (rotating about the Y-axis) : G1781
Arc (CW) (rotating about the Y-axis) : G1782
Arc (CCW) (rotating about the Y-axis) : G1783
Chamfering (rotating about the Y-axis) : G1784
Corner R (rotating about the Y-axis) : G1785
End point (rotating about the Y-axis) : G1786

3 ENTERING ARBITRARY FIGURES

This chapter explains how to enter arbitrary figure data.

NOTE
1 When entering arbitrary figures, enter all the data for each figure specified on the drawing.
If superfluous data is entered, it will not be used in figure calculation; only the data necessary for figure calculation is used.
2 If all the data necessary for figure calculation is specified on the drawing, entering it enables figure calculation. If data is insufficient, however, it is necessary to calculate and enter the necessary figure data separately.
3 If arbitrary-figure calculation according to MANUAL GUIDE *i* produces more than one candidate figure, the figure that passes the shortest path will be selected automatically.
In calculating a contact between an arc and straight line or between two arcs, a figure that links two figures smoothly is selected automatically.
4 On the program editing screen, after moving the cursor on the sub program call command (M98 P****) which is composed by the arbitrary figure blocks, pressing the <INPUT> key displays the window for editing the arbitrary figures to edit directly.
5 When positioning the cursor on the block of subprogram calling in the program-editing screen, arbitrary figures of the subprogram are drawn on the graphic window.
6 Tool path of machining cycle is calculated strictly form input data of free figure. When input data of free figure has error, an alarm may be occurs, a figure cannot be machined correctly and so on.
Therefore, input the vale with the least error possible.

3.1 ENTERING ARBITRARY FIGURE DATA

In entering arbitrary figures of element “LINE” or “ARC”, the end point can be set as an incremental programming.

When positioning the cursor on “END POINT”, the soft key [ST.P+I] and [ST.P-I] are displayed.

(1) When the operator pushes the soft key [ST.P+I], “END POINT” is set to
“Start Point of the figure (=End Point of last figure) +”.
And input the increment or decrement in succession. As result of pushing [INPUT] key, the plus incremental value is set.

(2) When the operator pushes the soft key [ST.P-I], “END POINT” is set to
“Start Point of the figure (=End Point of last figure) -”.
And input the increment or decrement in succession. As result of pushing [INPUT] key, the minus incremental value is set.

NOTE

1 If the last figure element is “CORNER R” or “CHAMFER”, The point to set as the start point in this function is as follows.

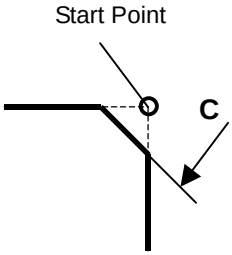


Fig1 Last figure is a chamfer.

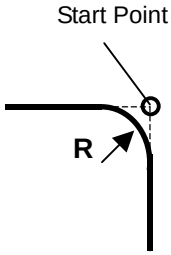


Fig2 Last figure is a corner R.

2 If “END POINT” of the last figure is pending, “END POINT” is not set to the start point by pressing the software key [ST.P+I] and [ST.P-I]. The warning message “START POINT IS PENDING.” is displayed.

3.1.1 Arbitrary Figures for the XY Plane

Arbitrary figures in the XY plane can be used in the following types of milling.

<1> Facing
<2> Contouring (Side cutting)
<3> Pocketing
<4> Grooving
<5> Emboss machining

NOTE

1 See Chapter 4, "MILLING" for detailed descriptions about data to be entered for each type of cycle machining.
2 When entering an arbitrary figure for pocketing, make its end point coincide with its start point. To put another way, combine parts figures with blank figures in such a way that a closed curve is created.
Pocketing is carried out in this closed curve.
3 It is possible to specify "PART" and "BLANK" as "ELEMENT TYPE" for an individual arbitrary figure in pocketing.
If a pocket is open in part, this "BLANK" can be used to enter an arbitrary figure resembling a blank figure for the open portion, thus carrying out optimum pocketing.

(1) Input items of START POINT (XY plane) (G1200)

ELEMENT (INPUT DATA) (Note 1)		
Data item		**Meaning**
T	FIGURE ATTRIBUTE **(Note 2)**	[FACE] : Used as a figure in facing [CONVEX] : Used as an outer-perimeter figure in contouring [CONCAV] : Used as an inner-perimeter figure in contouring and emboss machining or a figure in pocketing [GROOVE] : Used as a figure in grooving
X	START POINT X	X coordinate of the start point of an arbitrary figure
Y	START POINT Y	Y coordinate of the start point of an arbitrary figure
Z	BASE POSITION	Position of the machining surface of an arbitrary figure

ELEMENT (INPUT DATA) (Note 1)		
Data item		**Meaning**
D	HEIGHT/DEPTH	Height or Depth from Base position to cutting surface Remarks) This item is displayed in Contouring, Pocketing, Grooving and Emboss machining.
W	GROOVE WIDTH	Groove width Remarks) This item is displayed in Grooving.
P	FIGURE ATTRIBUTE	[RIGHT] : The right side of an entered figure as cutting [LEFT] : The left side of an entered figure as cutting Remarks) This item is displayed in Open figure of Contouring.

NOTE
1 'INPUT DATA' means the items, which are displayed on the input data window in editing or altering.
Similarly, this applies to G1201 to G1206 described later.
2 Select, with a figure type for a start point, which milling operation is to use an arbitrary figure. Data to be entered for the start point varies depending on what figure type is selected. For details, see descriptions about the respective milling types.

When the cursor is placed on BASE POSITION or HEIGHT/DEPTH, the soft key [GUIDE] appears. When the soft key is pressed, the guidance window for each input item is displayed.

ELEMENT (OUTPUT DATA) (Note 3)		
Data item		**Meaning**
T	FIGURE ATTRIBUTE	[1] : Used as a figure in facing [2] : Used as an outer-perimeter figure in contouring [3] : Used as an inner-perimeter figure in contouring and emboss machining or a figure in pocketing [4] : Used as a figure in grooving (input value)
H	START POINT X	X coordinate of the start point (input value)
V	START POINT Y	Y coordinate of the start point (input value)
B	BASE POSITION	Position of the machining surface (input value)
L	HEIGHT/DEPTH	Height or Depth from Base position to cutting surface (input value) Remarks) This item is displayed in Contouring, Pocketing, Grooving and Emboss machining.
D	GROOVE WIDTH	Groove width (input value) Remarks) This item is displayed in Grooving.
P	FIGURE ATTRIBUTE	[1] : The right side of an entered figure as cutting [2] : The left side of an entered figure as cutting (input value) Remarks) This item is displayed in Open figure of Contouring.

NOTE
3 'OUTPUT DATA' means the items, which are displayed on the program window as creating program. It can be referenced only for program display purposes.
Similarly, this applies to G1201 to G1206 described later.

(2) Input items of LINE (XY plane) (G1201)

ELEMENT (INPUT DATA)		
Data item		**Meaning**
D	LINE DIRECTION	The direction of a straight line is selected from a menu indicated on a soft key.
X*	END POINT X	X coordinate of the end point of a straight line Remarks1) This data may not be displayed depending on the value entered as the direction of the straight line. Remarks2) Incremental programming is possible.
Y*	END POINT Y	Y coordinate of the end point of a straight line Remarks1) This data may not be displayed depending on the value entered as the direction of the straight line. Remarks2) Incremental programming is possible.
A*	ANGLE	Straight-line angle Remarks) This data may not be displayed depending on the value entered as the direction of the straight line.
L	LAST CONNECTION	[TANGNT] : In contact with the immediately preceding figure [NO SET] : Not in contact with the immediately preceding figure (initial value)
M	NEXT CONNECTION	[TANGNT] : In contact with the immediately following figure [NO SET] : Not in contact with the immediately following figure (initial value)

ATTRIBUTE (INPUT DATA)		
Data item		**Meaning**
T	ELEMENT TYPE	[PART] : Cut as parts [BLANK] : Cut as a blank portion Remarks) The system sets the last data as an initial value.

ELEMENT & ATTRIBUTE (OUTPUT DATA)		
Data item		**Meaning**
H	END POINT X	X coordinate of the end point of a straight line (calculation result)
V	END POINT Y	Y coordinate of the end point of a straight line (calculation result)
K	LINE DIRECTION	The direction of a straight line is selected from a menu indicated on a soft key. (input value)
C*	END POINT X	X coordinate of the end point of a straight line (input value)
D*	END POINT Y	Y coordinate of the end point of a straight line (input value)
A*	ANGLE	Straight-line angle (input value)
L	LAST CONNECTION	[1] : In contact with the immediately preceding figure [0] : Not in contact with the immediately preceding figure (input value)
M	NEXT CONNECTION	[1] : In contact with the immediately following figure [0] : Not in contact with the immediately following figure (input value)
T	ELEMENT TYPE	[1] : Cut as parts [2] : Cut as a blank portion (input value)
S	SELECT FIG. INFO.	In the case of plural intersection or contact, the operator sets a candidate. (input value)

(3) Input items of ARC (CW) (XY plane) (G1202)
Input items of ARC (CCW) (XY plane) (G1203)

ELEMENT (INPUT DATA)		
Data item		**Meaning**
X*	END POINT X	X coordinate of an arc end point Remarks) Incremental programming is possible.
Y*	END POINT Y	Y coordinate of an arc end point Remarks) Incremental programming is possible.
R*	RADIUS	Arc radius
CX*	CENTER POINT CX	X coordinate of an arc center
CY*	CENTER POINT CY	Y coordinate of an arc center
L	LAST CONNECTION	[TANGNT] : In contact with the immediately preceding figure [NO SET] : Not in contact with the immediately preceding figure (initial value)
M	NEXT CONNECTION	[TANGNT] : In contact with the immediately preceding figure [NO SET] : Not in contact with the immediately preceding figure (initial value)
U	ROUTE TYPE	[LONG] : An arc with a long route is made [SHORT] : An arc with a short route is made (initial value)

ATTRIBUTE (INPUT DATA)		
Data item		**Meaning**
T	ELEMENT TYPE	[PART] : Cut as parts [BLANK] : Cut as a blank portion Remarks) The system sets the last data as an initial value.

ELEMENT & ATTRIBUTE (OUTPUT DATA)		
Data item		**Meaning**
H	END POINT X	X coordinate of an arc end point (calculation result)
V	END POINT Y	Y coordinate of an arc end point (calculation result)
R	RADIUS	Arc radius (calculation result)
I	CENTER POINT X	X coordinate of an arc center (calculation result)
J	CENTER POINT Y	Y coordinate of an arc center (calculation result)
C*	END POINT X	X coordinate of an arc end point (input value)
D*	END POINT Y	Y coordinate of an arc end point (input value)
E*	RADIUS	Arc radius (input value)
P*	CENTER POINT CX	X coordinate of an arc center (input value)
Q*	CENTER POINT CY	Y coordinate of an arc center (input value)
L	LAST CONNECTION	[1] : In contact with the immediately preceding figure [0] :Not in contact with the immediately preceding figure (input value)
M	NEXT CONNECTION	[1] : In contact with the immediately preceding figure [0] :Not in contact with the immediately preceding figure (input value)
U	ROUTE TYPE	[1] : An arc with a long route is made [0] :An arc with a short route is made (input value)
T	ELEMENT TYPE	[1] : Cut as parts [2] : Cut as a blank portion (input value)
S	SELECT FIG. INFO.	In the case of plural intersection or contact, the operator sets a candidate. (input value)

(4) Input items of CHAMFER (XY plane) (G1204)

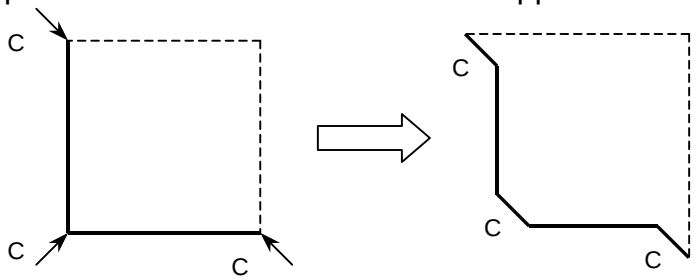
ELEMENT (INPUT DATA)		
Data item		**Meaning**
C	CHAMFER	Amount of chamfering (radius value, positive value)

ATTRIBUTE (INPUT DATA)		
Data item		**Meaning**
T	ELEMENT TYPE	[PART] : Cut as parts (initial value) [BLANK] : Cut as a blank portion Remarks) The system sets the last data as an initial value.

ELEMENT & ATTRIBUTE (OUTPUT DATA)		
Data item		**Meaning**
H	END POINT X	X coordinate of the end point of a straight line (calculation result)
V	END POINT Y	Y coordinate of the end point of a straight line (calculation result)
C	CHAMFER	Amount of chamfering (radius value, positive value) (input value)
T	ELEMENT TYPE	[1] : Cut as parts [2] : Cut as a blank portion (input value)

NOTE

By setting the parameter No.14851#0=1, Corner element between a blank element and a part element can be created in the opposite direction.



Dotted line : blank element
Normal line : part element

(5) Input items of CORNER R (XY plane) (G1205)

ELEMENT (INPUT DATA)		
Data item		**Meaning**
R	CORNER RADIUS	Corner rounding (radius value, positive value)

ATTRIBUTE (INPUT DATA)		
Data item		**Meaning**
T	ELEMENT TYPE	[PART] : Cut as parts [BLANK] : Cut as a blank portion Remarks) The system sets the last data as an initial value.

ELEMENT & ATTRIBUTE (OUTPUT DATA)		
Data item		Meaning
H	END POINT X	X coordinate of an arc end point (calculation result)
V	END PINT Y	Y coordinate of an arc end point (calculation result)
R	RADIUS	Arc radius (calculation result)
I	CENTER POINT X	X coordinate of an arc center (calculation result)
J	CENTER POINT Y	Y coordinate of an arc center (calculation result)
K	ROTATION DIRECTION	[2] : clockwise [3] : counterclockwise
T	ELEMENT TYPE	[1] : Cut as parts [2] : Cut as a blank portion (input value)

NOTE
By setting the bit 0 (CNR) of parameter No.14851 = 1, Corner element between a blank element and a part element can be created in the opposite direction.

(6) Input items of END OF ARBITRARY FIGURES (XY plane) (G1206)

This block is output at the end of a series of arbitrary-figure blocks.

NOTE
A series of arbitrary figures must be enclosed between G1200 (start point) and G1206 (end of arbitrary figures).

3.1.2 Arbitrary Figures for the YZ Plane

Arbitrary figures in the YZ plane can be used in the following types of milling.
<1> Facing
<2> Contouring (Side cutting)
<3> Pocketing
<4> Grooving
<5> Emboss machining

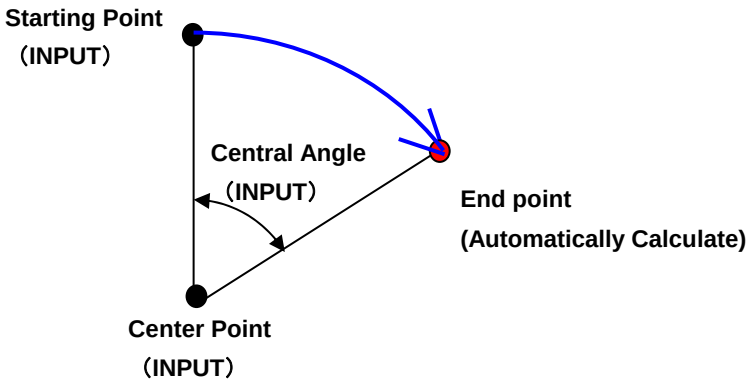
NOTE
The data of the G codes of arbitrary figures for the YZ plane given below is the same as that of the G codes of arbitrary figures for the XY plane except that X coordinate values are replaced by Z coordinate values. For details, see Subsection 5.1.1, "Arbitrary Figures for the XY Plane", in Section 5.1, "ENTERING ARBITRARY FIGURE DATA".

The arbitrary figures of the following G codes are the same as the arbitrary figures for the XY plane:

- Start point : G1300 (YZ plane)
- Line : G1301 (YZ plane)
- Arc (CW) : G1302 (YZ plane)
- Arc (CCW) : G1303 (YZ plane)
- Chamfer : G1304 (YZ plane)
- Corner R : G1305 (YZ plane)
- Corner R : G1305 (YZ plane)
- End of arbitrary figures : G1306 (YZ plane)

3.2 ARC SPECIFICATION BY CENTRAL ANGLE

In the Arbitrary Figure entering screen it is possible to specify the arcs using their central angle. An apposite window is provided to the machine operator to insert the necessary data. The arc end point is automatically calculated by specifying the central angle, the starting point coordinates, and the center point coordinates.



3.2.1 Operations

<1> If the cursor is located in any of the two input items corresponding to the coordinate of the end point of an Arc window, then the [AUXCAL] soft key for Auxiliary Calculation is displayed as follows.

AUXCAL			ST.P+I	ST.P-I	CHCURS			OK	CANCEL

<2> If the [AUXCAL] soft key is pressed, then the Auxiliary Calculation Window is displayed.

<3> The end point are calculated by inputting the arc direction, the start point, the center point, the central angle of the arc, and finally pressing the [CALC] soft-key.
If the [CANCEL] soft key is pressed, then the Auxiliary Calculation Window is closed and no calculation is performed.

3.2.2 Input Items

The input items for Auxiliary Calculation Window are as follows

ARC END PT		
Data item		**Meaning**
R	ARC DIRECTION	A circular arc direction is selected with the software key. [ARC ↻] : CW [ARC ↺] : CCW
X	START POINT X	X axis coordinates value of circular arc starting point.
Y	START POINT Y	Y axis coordinates value of circular arc starting point.
CX	CENTER POINT CX	X axis coordinates value of circular arc center
CY	CENTER POINT CY	Y axis coordinates value of circular arc center
A	ANGLE	Circular arc angle

NOTE
If the END POINT values of the previous element are already defined then START POINT in the Auxiliary Calculation Window is equal to the END POINT of the previous figure element. If the CENTER POINT values in Arc Window are already defined then CENTER POINT in the Auxiliary Calculation Window is equal to the CENTER POINT values in Arc Window.

3.2.3 Enable Arbitrary Figures

The arc can be specified by using the central angle in following machining planes and functions.

(1) XY plane arbitrary figure
(2) YZ plane arbitrary figure
(3) ZX plane arbitrary figure
(4) XY plane contour programming
(5) YZ plane contour programming
(6) ZX plane contour programming
(7) Blank figure specified as arbitrary figure (rotating around the Z-axis)
(8) Blank figure specified as arbitrary figure (rotating around the Y-axis)
(9) Blank figure specified as arbitrary figure (rotating around the X-axis)

3.3 ENLARGEMENT AND REDUCTION ON THE ARBITRARY FIGURE INPUT SCREEN

When bit 1 (GCN) of parameter No. 27310 is set to 0, a drawing figure is enlarged or reduced with respect to the center position of the drawing window on the screens for entering arbitrary figure data, machining simulation, and drawing during machining.
When bit 1 (GCN) of parameter No. 27310 is set to 1, enlargement and reduction are performed with respect to the center of the figure to be drawn, as done conventionally.

3.3.1 Machining Simulation Screen

When bit 1 (GCN) of parameter No. 27310 is set to 0, enlargement and reduction are performed with respect to the center of the drawing window.

<1> Move the part to be enlarged or reduced to the center of the screen by using soft keys such as [←MOVE] and [MOVE→].
<2> When soft key [LARGE] is pressed, enlargement is performed with respect to the center position of the simulation screen.
<3> When soft key [SMALL] is pressed, reduction is performed with respect to the center position of the simulation screen.

3.3.2 Arbitrary Figure Input Screen

When bit 1 (GCN) of parameter No. 27310 is set to 0, enlargement and reduction are performed with respect to the center of the drawing window.

<1> Move the part to be enlarged or reduced to the center of the screen by using soft keys such as [←MOVE] and [MOVE→].
<2> When soft key [LARGE] is pressed, enlargement is performed with respect to the center position of the arbitrary figure drawing window.

<3> When soft key [SMALL] is pressed, reduction is performed with respect to the center position of the arbitrary figure drawing window.

3.3.3 Screen for Drawing during Machining

Also on the screen for drawing during machining, enlargement and reduction are performed in the same way as described in Subsection 5.2.1, "Machining Simulation Screen".

3.4 GROUPING ARBITRARY FIGURE PROGRAMS

3.4.1 Group Range Selection Commands

When a pocketing cycle or emboss machining cycle program is used, a pocket figure and island figures included in the pocket figure are grouped using group range selection commands.
The grouped figures are then assumed to be target figures for the pocketing cycle or emboss machining cycle.

G1990: Group range selection start command
G1991: Group range selection end command

(Sample program)

```
G1990 ; (GROUP START) ←        Group range selection start command
G1200 Hh Vv Bb Ll Dd ; (POCKET) ←Pocket figure
G1201 Hh Vv Kk Cc Dd Aa Ll Mm Ss ;
G1202 Hh Vv Rr Ii Jj Cc Dd Ee Pp Qq Ll Mm Ss ;
G1203 Hh Vv Rr Ii Jj Cc Dd Ee Pp Qq Ll Mm Ss ;
G1206 ;
G1200 Hh Vv Bb Ll Dd ; (ISLAND-1) ←Island figure
G1201 Hh Vv Kk Cc Dd Aa Ll Mm Ss ;
G1202 Hh Vv Rr Ii Jj Cc Dd Ee Pp Qq Ll Mm Ss ;
G1203 Hh Vv Rr Ii Jj Cc Dd Ee Pp Qq Ll Mm Ss ;
G1206 ;
G1200 Hh Vv Bb Ll Dd ; (ISLAND-2) ←Island figure
G1201 Hh Vv Kk Cc Dd Aa Ll Mm Ss ;
G1202 Hh Vv Rr Ii Jj Cc Dd Ee Pp Qq Ll Mm Ss ;
G1203 Hh Vv Rr Ii Jj Cc Dd Ee Pp Qq Ll Mm Ss ;
G1206 ;
G1991 ; (GROUP END) ←          Group range selection end command
```

When a G code for group range selection is specified for other than pocketing and emboss machining, an alarm (P/S3598 “GROUP SET MISMATCH”) is issued.

3.4.2 Automatic Insertion of Group Range Selection Commands

When arbitrary figures used in a pocketing cycle or emboss machining cycle are created, the G codes for group range selection commands (G1990 and G1991) are automatically inserted to the program by pressing soft key [OK] on the [METHOD OF FREE FIGURE CREATION] screen.

3.5 ARBITRARY FIGURE COPY FUNCTIONS

A specified area of figure can be copied (parallel copy, mirror copy, or rotational copy) for addition as a new figure on the arbitrary figure creation screen or the contour program input screen.

(1) Parallel copy
A specified area of arbitrary figure is copied specified times after the selected area.
(2) Rotational copy
A specified area of arbitrary figure is copied specified times around specified coordinates after the selected area.
(3) Mirror copy
A specified area of arbitrary figure is mirror-copied relative to a specified symmetry axis.

3.5.1 Selecting an Arbitrary Figure Copy Function

<1> An arbitrary figure copy function can be selected by pressing the [PARAL], [ROTATE], or [MIRROR] soft key displayed when the [>] soft key is pressed on the arbitrary figure editing screen.
<2> The arbitrary figure editing screen is displayed:
<3> Pressing the rightmost soft key [>] displays the soft keys for arbitrary figure copy operation.

3.5.2 Copy Condition Input Screen

(1) Parallel copy
Pressing the [PARAL] soft key displays a parallel copy screen.

NUMBER OF REPETITIONC :
Enter the number of times a selected figure is to be copied.

(2) Rotational copy
Pressing the [ROTATE] soft key displays a rotational copy screen.

INPUT TYPE :
Select [CENTER] or [RADIUS] to specify the method of condition input.
CENTER POINT CX, CENTER POINT CY :
Enter the X coordinate and Y coordinate of a rotation center around which a rotation is made.
NUMBER OF REPETITIONC :
Enter the number of times a selected figure is to be copied.

When "BY RADIUS" is selected for "INPUT TYPE", the input items for entering a radius and rotation direction are displayed.

RADIUS :
Enter the distance from the start or end point of a selected figure to the rotation center point.

ROTATION DIRECTION :
Select [CW] or [CCW] to specify the direction of rotation copy operation.

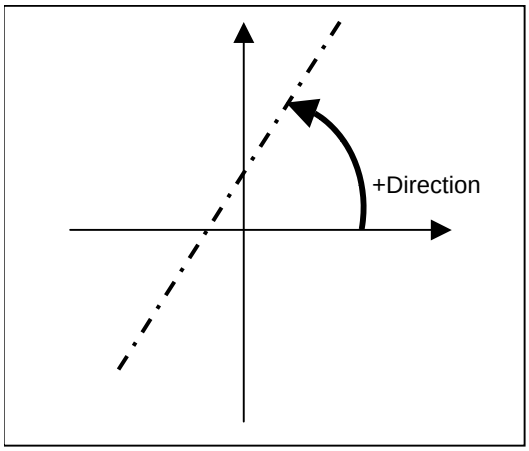
(3) Mirror copy
Pressing the [MIRROR] soft key displays a mirror copy screen.

SPECIFY OF SYMMETRY :
Use the [COORD] or [ANGLE] soft key to select the method for specifying a symmetry axis.
SYMMETRY AXIS X, SYMMETRY AXIS Y :
Enter X and Y coordinates through which the symmetry axis for mirror copying passes. The symmetry axis should pass through the end point of a specified figure and these X and Y coordinates.

When "BY ANGLE" is selected for "SPECIFY OF SYMMETRY", the input item for entering an angle is displayed.

ANGLE :

Enter the angle between a symmetry axis used for mirror copy operation and the horizontal axis. The plus direction of the horizontal axis represents 0°. Enter a positive value for an angle made toward the plus vertical axis direction. Enter a negative value for an angle made toward the minus vertical axis direction.



NOTE

Data entered on the parallel copy input screen, rotation copy input screen, or mirror copy input screen is not output to the program.
Each figure entered by a type of copy operation is automatically expanded as an arbitrary figure element, so that each figure can be modified. However, the state before copy operation cannot be automatically resumed.

3.5.3 Executing Arbitrary Figure Copy Operation

Use the operation procedure described below.

(Example)

Parallel copy

<1> Enter the number of repetition on the copy input screen then press the [OK] soft key.

<2> The screen for specifying an area of figure used as a copy source appears. Move the cursor to the icon of the figure element at the start of the area, then press the [SELECT] soft key.
Pressing the [CANCEL] soft key returns the screen display to the arbitrary figure editing screen.

NOTE

A selected copy source ranges from the figure where the cursor is placed when the [SELECT] soft key is pressed to the last figure.

<3> At this time, the graphic window displays a figure produced by executing copy operations repeatedly. A figure produced by copying is inserted after the selected copy source.
Pressing the [NO] soft key returns the screen display to the previous step.

<4> Pressing [YES] at the end fixes a figure produced by repeated copy operations.

3.5.4 Figure after Arbitrary Figure Copy Execution

When the screen used for entering a copied figure is opened by selecting [ALTER], the set input items differ from those set for the figure before being copied, as described below.
For a straight line:

Only the input items "LINE DIRECTION" and "END POINT" are set.

For an arc:

Only the input items "RADIUS" and "END POINT " are set.

If a copy source figure includes a corner R and chamfer, the corner R and chamfer are an arc and straight line, respectively, in a figure produced by copying.
This is because parallel, rotation, or mirror copy operation is performed according to the end point information of the copy source figure. So, information such as "NEXT CONNECTION" set for a copy source figure is not reflected in a figure produced by copying.

NOTE
1 A warning is output if the result of expansion based on an attempt for copy operation exceeds the maximum allowable number of blocks.
2 The end point of the last selected copy source figure needs to be fixed.
3 The element immediately before a selected copy source figure must not be a corner R or chamfer. Moreover, the starting element and ending element of a selected copy source figure must not be a corner R and chamfer.

3.6 EDITING AN ARBITRARY FIGURE SUBPROGRAM

3.6.1 Editing

Follow the steps below.

<1> On the program edit screen, position the cursor on a block specifying an arbitrary figure subprogram call command (M98P xxxx), and press the <INPUT> key or the [ALTER] soft key.
<2> The arbitrary figure edit screen for the subprogram is displayed.
<3> When editing of an arbitrary figure is completed, the screen display is changed as follows:
- When soft key [OK] is pressed, an existing block in program memory is replaced by the figure block being entered, then the screen display returns to the main program screen. When another arbitrary figure is present after the arbitrary figure being edited (when an island figure is present after an arbitrary figure used as a pocket outer wall figure, for example), the edit screen for the next arbitrary figure is displayed.
- In pocketing, when soft key [ISLAND] is pressed, an existing block in program memory is replaced by the figure block being entered, then a new arbitrary figure screen (start point) is displayed, allowing the operator to enter a new island figure.

<4> On the arbitrary figure edit screen, pressing soft key [CANCEL] displays a confirmation screen asking the operator whether to cancel editing. If soft key [YES] is selected, the edit operation is canceled, and the screen display returns to the main program display screen.

NOTE
1 In the subprogram edit mode (when a screen for editing an arbitrary figure is displayed), any of the operations listed below cancels the subprogram edit mode. At this time, the selected subprogram is maintained.
   - A mode change is made during editing in the foreground.
     (Example: When the mode is changed from the edit to MEM mode)
   - The screen is changed to an NC screen.
   - The power is turned off.
2 In this case, the arbitrary figure edit screen is closed. The program displayed in the program display area is the subprogram.

3.6.2 Warning

During editing of a subprogram, the following warning messages may be displayed:

- "SUBPROGRAM NUMBER IS NOT FOUND."
  The <INPUT> key or soft key [ALTER] is pressed when the cursor is positioned on a block specifying a subprogram call command, but the program number specified with address P is not present.
- "PROGRAM IS PROTECTED."
  The <INPUT> key or soft key [ALTER] is pressed when the cursor is positioned on a block specifying a subprogram call command, but the program number specified with address P is protected and therefore cannot be displayed.
- "NOT SUBPROGRAM OF FREE FIGURE."
  The <INPUT> key or soft key [ALTER] is pressed when the cursor is positioned on a block specifying a subprogram call command, but the program specified with address P does not contain an arbitrary figure block.
- "THIS PROGRAM USED AT F.G."
  In the subprogram edit mode in the background editing, soft key [OK] or [CANCEL] is pressed to display a calling program, but the calling program is already selected in the foreground.

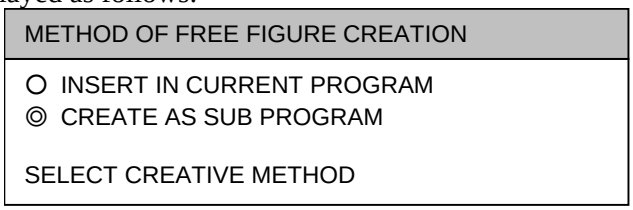
3.7 SAVING SUBPROGRAMS

Arbitrary figures and fixed-form figures can be saved as subprograms.

NOTE
1 This function is available only when the device used is a CNC memory. When the device is a data server or a memory card, this function cannot be used.
2 Use of the undo and redo functions cannot restore an overwritten program to its original state.

3.7.1 Saving a New Subprogram

<1> After creating an arbitrary figure, press soft key [CREATE]. Then, “METHOD OF FREE FIGURE CREATION” is displayed as follows:

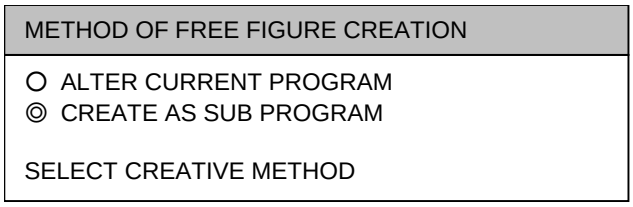


<2> As the method for creating an arbitrary figure, select “CREATE AS SUB PROGRAM”.
<3> Enter a subprogram name (O number), then press soft key [OK].
The block M98P xxxx; is then inserted at the current cursor position automatically, and a subprogram for the entered figure (xxxx) is created automatically.

Remark)
If a subprogram having the same number (name) is already present, the following soft keys are displayed:
[YES] : The existing subprogram is overwritten.
Upon completion of the overwrite operation, all windows for the arbitrary figure are closed.
[NO] : The screen display returns to the “METHOD OF FREE FIGURE CREATION” window.

3.7.2 Saving an Existing Subprogram

<1> Place the cursor on an existing block specifying a subprogram call command (M98P xxxx), then press soft key [ALTER] to modify a figure.
<2> After modifying the arbitrary figure, press soft key [CREATE]. Then, “METHOD OF FREE FIGURE CREATION” is displayed as follows:



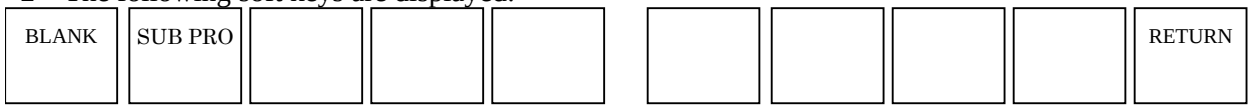
<3> As the method for creating an arbitrary figure, select “CREATE AS SUB PROGRAM”.
<4> Enter a subprogram name, then press soft key [OK]. A subprogram for the entered figure is created automatically.
The call number (xxxx) of the block M98P xxxx ; currently at the cursor position is not changed. As necessary, change the input call number.

3.8 REUSING ARBITRARY FIGURES

When arbitrary figures for cycle machining are created, an arbitrary figure programmed as a subprogram can be read for reuse.

3.8.1 Calling an Arbitrary Figure

<1> On the “START POINT - INSERT” screen for an arbitrary figure, press soft key [REUSE].
<2> The following soft keys are displayed:



[BLANK] : Reuses blank figure data included in the currently edited program.
[SUB PRO] : Calls an arbitrary figure defined as a subprogram.
[RETURN] : Returns to the “START POINT - INSERT” screen.

3.8.2 Selecting a Subprogram

When soft key [SUB PRO] is pressed, the “REUSE” window for existing figures is opened, displaying a list of registered arbitrary figure subprograms. Move the cursor to the subprogram for the arbitrary figure to be reused.

(1) Items in the REUSE window
  (a) CURRENT FOLDER
    A list of subprograms stored in the folder of the program currently edited is displayed.
  (b) CNC_MEM/USER/LIBRARY
    A list of programs stored in a common folder is displayed.

(2) MDI keyboard
  <1> Cursor move keys <←><→>
    Switches between radio buttons.
  <2> Cursor move keys <↑><↓>
    Moves the cursor for selecting a subprogram.

(3) Soft keys
  <1> Soft key [OK]
    Inserts the figure of a selected subprogram, and displays the “START POINT - ALTER” window.
  <2> Soft key [CANCEL]
    Cancels a selected subprogram, and returns to the “START POINT - INSERT” window.

NOTE
1 The following subprograms cannot be selected:
- Subprograms containing no arbitrary figure
- Subprograms containing an incomplete arbitrary figure

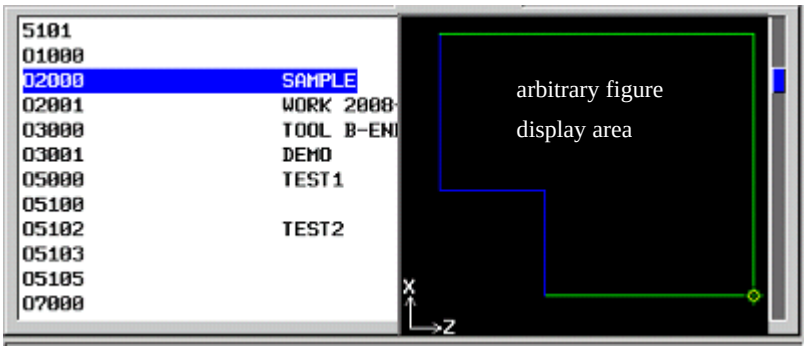
If such a subprogram is selected, the error message “THERE IS NO FIGURE CODE TO REUSE.” is displayed.
2 The subprogram including contour program cannot be recycled.

3.9 DRAWING OF ARBITRARY FIGURE ON SUB-PROGRAM LIST SCREEN

The arbitrary figure, stored in a subprogram, can be drawn on the subprogram list screen.
This function is enabled when the parameter No.14855#2 is set to 1.

3.9.1 Display Sub-Program List Screen

(1) When the sub-program list screen is opened and parameter No.14855#2=1, then the arbitrary figure display area and [VIEW] soft key are displayed.
When [VIEW] soft key is pressed during displaying arbitrary figure area, then the arbitrary figure display area closes and comment display area is displayed all over the window.



(2) As soon as the cursor is positioned over a sub-program, the retrieval of the arbitrary figure contained in the sub-program is started. When there is an arbitrary figure in the sub-program, the message "NOW DRAWING" is displayed at the center of the arbitrary figure display area during the retrieval. The retrieval is interrupted by pressing the cursor keys [↑] [↓], and the retrieval of the arbitrary figure of the next sub-program starts.
(3) After finishing the retrieval, its shape is drawn in the arbitrary figure display area.

NOTE
When the following operations are executed while the message "NOW DRAWING" is displayed, the behavior by the operation is done after drawing arbitrary figure.
(1) Changing of NC screen by function key.
(2) Mode switching
(3) Changing of path
(4) Pressing of each soft-key
(5) Short cut operation (MDI key operation) and cursor key operation
(6) Pressing of edit operation key (ALTER, DELETE, and INSERT)

3.9.2 Available Drawing Figure

The available drawing figure is as follows.
(1) When the sub-program contains an arbitrary figure, a free contour cylinder blank figure or a contour program, then the arbitrary figure is drawn on the arbitrary figure display area.
(2) When there is more than one figure in the sub-program, then only the first figure is drawn. However, all figures belonging to the same group are drawn together. For example, the island is drawn together with its corresponding pocket.
(3) When the sub-program does not contain any figure, then no arbitrary figure is drawn.

4 NC PROGRAM CONVERSION FUNCTION

Pressing the soft key [NC CNV] starts the NC program conversion function.
With the NC program conversion function, a 4-digit G cycle machining command can be dissolved into a single move command and stored in the part program storage area of the NC.

4.1 OVERVIEW

(1) With the NC program conversion function, only a 4-digit G cycle machining command can be dissolved into a single move command. Any other types of commands are output without modification.
(2) The NC program conversion function can be used in the MEM mode only.
(3) Only an executed block becomes an NC program conversion target.
(4) A block containing an M98, M99, or custom macro program is not output to the conversion destination program.
(5) In the case of multiple paths, NC program conversion is performed for each path.
(6) If there is a loop or conditional branch program of a custom macro program, executed blocks only are output. A block containing a conditional branch program of a custom macro is not output to the conversion destination program.
(7) If a loop or conditional branch program of a custom macro program includes a 4-digit G cycle, the 4-digit G cycle is expanded as many times as the number of repeats. A block containing a conditional branch program of a custom macro is not output to the conversion destination program.
(8) Whether to output a 4-digit G cycle machining command before expansion as a comment can be chosen using bit 5 (NCC) of parameter No. 14703.
Bit 5 (NCC) of parameter No. 14703 = 0:
Outputs a 4-digit G cycle machining command before expansion as a comment in NC program conversion.
Bit 5 (NCC) of parameter No. 14703 = 1:
Does not output a 4-digit G cycle machining command before expansion as a comment in NC program conversion.
(9) In the case of a subprogram call, see the examples below. A block containing M98 or M99 is not output to the conversion destination program.

(Example 1)

```
(Before conversion)
O0001
M98 P0002;                →   O0002
M30;                          G0 X100. ;
%                             G0 X200. ;
                              G0 X300. ;
(After conversion)            M99;
O0001                         %
G0 X100. ;
G0 X200. ;
G0 X300. ;
M30;
%
```

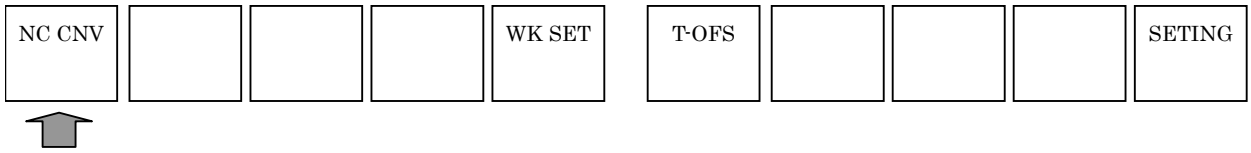
(Example 2)

```
(Before conversion)
O0001
G1128I1.R0.8A95.B80.J3.P3.L3.M0.F0.5X1.Y1.Z10. ;
```

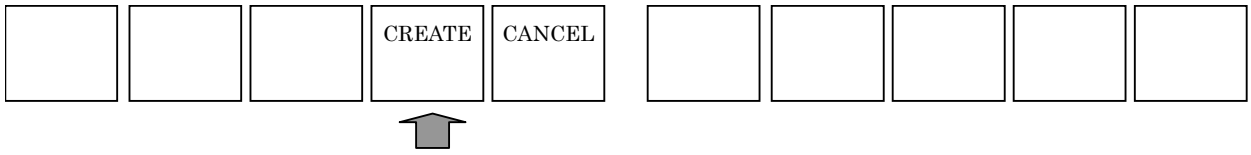
```
M98 P0002;                    →   O0002
M30;                              G1450H0.V75. ;
%                                 G1451H0.V0.K7.D0.L0.M0.T1. ;
                                  G1451H5.V0.K1.C5.L0.M0.T1. ;
                                  G1451H5.V75.K3.D75.L0.M0.T2. ;
                                  G1456;
(After conversion)                M99;
O0001                             %
(NC PROGRAM CONVERSION-START);
G01X963Z1616;
G01X896 Z1654;
 :
(NC PROGRAM CONVERSION-END);
M30;
%
```

4.2 BASIC OPERATION

Selecting MEM mode on the machine operator's panel, and pressing the leftmost soft key [<] or rightmost soft key [>] several times displays the soft-keys including soft key [NC CNV]



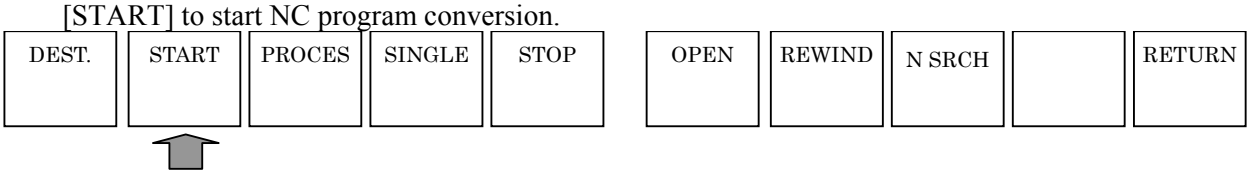
<1> The [CREATE PROGRAM FOR DESTINATION] screen appears. Enter the number of a conversion destination program, then press the [CREATE] soft key.



<2> If the program already exists, a message for checking if the program may be overwritten is displayed. If the program may be overwritten, press soft key [YES]. If you press soft key NO], the screen goes back to the memory program screen, so press soft key [NC CNV] again and enter other program number.

<3> Press the soft key [CREATE] soft key. A new program with the entered number is created. The NC program conversion function screen is displayed, on which the conversion source program window and the resultant program window are displayed at the same time. Pressing soft key [CANCEL] switches the screen display from the NC program conversion function screen to the program screen.

<4> The following soft keys appear on the NC program conversion function screen. Press soft key [START] to start NC program conversion.



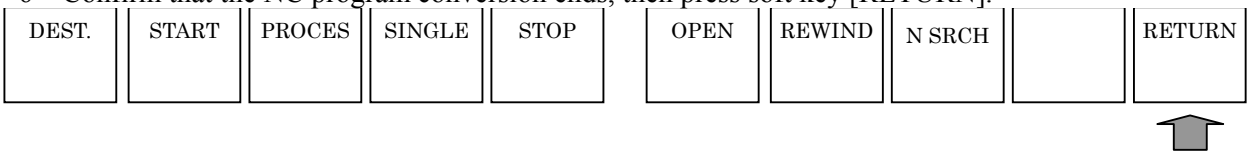
[DEST.] : Creates a new conversion destination program.
[START] : Starts NC program conversion execution.
[PAUSE] : Stops NC program conversion execution temporarily.
[SINGLE] : Executes NC program conversion block by block.
[STOP] : Stops NC program conversion execution.
[OPEN] : Selects a conversion source program from the program list screen.

[REWIND] : Searches for a conversion source program.
[N SRCH] : Makes an N search for a conversion source program.
[CHPATH] : Displayed only when the multi-path option is specified. This soft key switches between paths.
[RETURN] : Ends the NC program conversion function and returns the screen display to the ordinary MEM mode screen.

* When bit 0 (PRC) of parameter No. 27310 = 1, the soft key [PAUSE] is replaced with the soft key [PROCES].

[PROCES] : Executes NC program conversion process by process.

<5> During executing NC program conversion, the converted part program blocks are displayed in the destination window.
<6> Confirm that the NC program conversion ends, then press soft key [RETURN].



NOTE
In Series 0*i*-D/30*i*/31*i*/32*i*, the soft key [N SRCH] is not displayed.

4.3 RESTRICTIONS

(1) The NC program conversion function cannot be used during background editing.
(2) Blocks containing the following words are not output to the conversion destination program:
- M98
- M99
- Custom macro conditional branch program
  <1> GOTO
  <2> IF
  <3> THEN
  <4> WHILE
  <5> END
- Custom macro variable assignment program
  #?=~
- Custom macro external output command program
  <1> POPEN
  <2> PCLOS
  <3> BPRNT[~]
  <4> DPRNT[~]

(3) If an alarm is issued during NC program conversion, the results of conversion at that time are output to the conversion destination program.
(4) During NC program conversion, the following soft keys cannot be used:
[DEST.], [START], [OPEN], [REWIND], [SRCH], [CHPATH], [RETURN]
(5) If an execution mode other than MEM mode is set during NC program conversion, the execution of NC program conversion is terminated forcibly.
(6) During NC program conversion, the screen display cannot be switched to the NC screen.
(7) If only one block is inserted between WHILE programs, blocks as many as the loop count are not output but only one block is output.
(Before conversion)
WHILE [#1 EQ #2]DO1;
G0 X0. ;
END1 ;

(After conversion)
Only one block is output even if three loop operations are performed.
G0 X0. ;

(8) In Series 30*i*/31*i*/32*i*, if the is the block of M198, the alarm will occurred.

(9) The NC program conversion function is designed to expand a 4-digit G cycle machining code. So, NC program conversion is not performed as expected in cases other than the cases indicated below.

(Example 1) When both the machining command and figure command of a cycle are present on the main program

```
O0001
G1128I1.R0.8A95.B80.J3.P3.L3.M0.F0.5X1.Y1.Z10. ;
G1450H0.V75. ;
G1451H0.V0.K7.D0.L0.M0.T1. ;
G1451H5.V0.K1.C5.L0.M0.T1. ;
G1451H5.V75.K3.D75.L0.M0.T2. ;
G1456;
M30;
%
```

(Example 2) When the figure command of a cycle is present on a subprogram

```
O0001
G1128I1.R0.8A95.B80.J3.P3.L3.M0.F0.5X1.Y1.Z10. ;
M98 P0002;                          →   O0002
M30;                                    G1450H0.V75. ;
%                                       G1451H0.V0.K7.D0.L0.M0.T1. ;
                                        G1451H5.V0.K1.C5.L0.M0.T1. ;
                                        G1451H5.V75.K3.D75.L0.M0.T2. ;
                                        G1456;
                                        M99;
                                        %
```

* It is assumed that the subprogram includes only the figure command of a cycle.

(10) Executed program blocks are the target of NC statement conversion.

(11) Feedrate command is converted with decimal point

(12) When custom macro block exits before executing stop command (M00, M01, M02, M30, M99), executing stop command (M00, M01, M02, M30, M99) is not converted.

(13) When M98 block exits before executing stop command (M00, M01, M02, M30, M99), executing stop command (M00, M01, M02, M30, M99) is not converted.

(14) A block including M198 is not converted.

(15) In order to convert tool radius/tool nose radius compensation (G41,G42) , it is necessary to set parameters No.27480#0=1 and No.27483#3=1.

(16) The following commands can not be converted.

<G-code command>

Command	Function	machine control type
G02.4 G03.4	3-dimensional coordinate system conversion CW/CCW	Common to Lathe System / Machining Center System
G05	AI contour control (high-precision contour control compatible command), High-speed cycle machining, High-speed binary program operation	Common to Lathe System / Machining Center System
G05.1	AI contour control / Nano smoothing / Smooth interpolation	Common to Lathe System / Machining Center System

Command	Function	machine control type
G05.4	HRV3, 4 on/off	Common to Lathe System / Machining Center System
G06.2	NURBS interpolation	Common to Lathe System / Machining Center System
G08	AI contour control (advanced preview control compatible command)	Machining Center System
G39	Tool radius/tool nose radius compensation : corner circular interpolation	Common to Lathe System / Machining Center System
G41.2	3-dimensional cutter compensation : left (type 1)	Common to Lathe System / Machining Center System
G41.3	3-dimensional cutter compensation : leading edge offset	Common to Lathe System / Machining Center System
G41.4	3-dimensional cutter compensation : left (type 1) (FS16i-compatible command)	Common to Lathe System / Machining Center System
G41.5	3-dimensional cutter compensation : left (type 1) (FS16i-compatible command)	Common to Lathe System / Machining Center System
G41.6	3-dimensional cutter compensation : left (type 2)	Common to Lathe System / Machining Center System
G42.2	3-dimensional cutter compensation : right (type 1)	Common to Lathe System / Machining Center System
G42.4	3-dimensional cutter compensation : right (type 1) (FS16i-compatible command)	Common to Lathe System / Machining Center System
G42.5	3-dimensional cutter compensation : right (type 1) (FS16i-compatible command)	Common to Lathe System / Machining Center System
G42.6	3-dimensional cutter compensation : right (type 2)	Common to Lathe System / Machining Center System
G43.4	Tool center point control (type 1)	Common to Lathe System / Machining Center System
G43.5	Tool center point control (type 2)	Common to Lathe System / Machining Center System
G43.7	Tool offset	Common to Lathe System / Machining Center System
G44.1	Tool offse conversion	Common to Lathe System / Machining Center System
G66	Macro modal call A	Common to Lathe System / Machining Center System
G66.1	Macro modal call B	Common to Lathe System / Machining Center System
G72.1	Figure copying (rotary copy)	Common to Lathe System / Machining Center System
G72.2	Figure copying (linear copy)	Common to Lathe System / Machining Center System
G73	Peck drilling cycle	Machining Center System
G73.7	Closed loop cutting cycle	Machining Center System
G72	Traverse grinding cycle	Lathe System
G73	Traverse direct sizing/grinding cycle	Lathe System
G74	Oscillation grinding cycle	Lathe System
G75	Oscillation direct sizing/grinding cycle	Lathe System

<Other command>

Command	Function	machine control type
,C ,R	OPTIONAL ANGLE CHAMFERING AND CORNER ROUNDING	Machining Center System
C R	CHAMFERING AND CORNER ROUNDING	Lathe System
,A ,C ,R	DIRECT DRAWING DIMENSIONS PROGRAMMING	Lathe System

4.4 OMITTING MODAL COMMANDS IN NC PROGRAM CONVERSION

During NC program conversion, if a command to output is a modal command, its output can be omitted.

4.4.1 Omissible Modal Commands

During NC program conversion, if a command to output is a modal command, it can be omitted.

The following G code commands can be omitted:
G code command: G0 G1 G2 G3

For the G code commands G0, G1, G2, and G3, the following axis addresses and feedrate command can be omitted:
Axis address : X Y Z A B C E
Feedrate command : F

4.4.2 Specifications for Omitting Modal Commands

When bit 4 (MOT) of parameter No. 14700 is set to 1, the output of a command is suppressed if the command is a modal command.
Only commands that are output during 4-digt G cycle execution are regarded as the target commands for modal command omission.

* When F100 has been specified before the execution of a 4-digit G cycle, for example, the first F100 command output during the 4-digit G cycle is not omitted..
When there becomes nothing to specify with a G code as a result of modal command omission, the G code is not output.
G0 X100;
G1 Y200;
G0 X100;----------<1>
In the above case, G0 X100 in <1> above is omitted as a result of the omission of a modal command as follows:
G0 X100;
G1 Y200;

When a G code other than G0, G1, G2, and G3 is output, the modal value is cleared.
G0 X100;
G80; ----------<2>
G0 X100; ----------<3>
In the above case, the modal value X100 is cleared in <2>, so X100 in <3> is not omitted.

Conversion example 1)

Bit 4 (MOT) of parameter No. 14700 = 0
(Modal commands are output during NC program conversion)

```
 (NC PROGRAM CONVERSION-START)
G53
G40
G90
G17
G0X52.495Y-18.015Z5.
G0X52.495Y-18.015Z-5.
G1X52.495Y18.015F1000.
G0Z5.
G0X47.455Y-28.78Z5.
G0Z-5.
G3X47.496Y-27.5I-19.955J1.28F1000.
G1X47.496Y27.5F1000.
G3X47.455Y28.78I-19.996J0.F1000.
G0Z5.
G0X-18.015Y-52.495Z5.
```

*Commands in boldface denote modal commands.

Conversion example 2)

Bit 4 (MOT) of parameter No. 14700 = 1
(Modal commands are not output during NC program conversion)

```
(NC PROGRAM CONVERSION-START)
G53
G40
G90
G17
G0X52.495Y-18.015Z5.
Z-5.
G1Y18.015F1000.
G0Z5.
X47.455Y-28.78
Z-5.
G3X47.496Y-27.5I-19.955J1.28
G1Y27.5
G3X47.455Y28.78I-19.996J0.
G0Z5.
X-18.015Y-52.495
```

4.5 SUPPRESSING SUBPROGRAM EXPANSION IN NC PROGRAM CONVERSION

When a subprogram call command is executed during NC program conversion, the call command can be output without expanding subprogram contents.

NOTE
To use this function, set bit 0 (CSP) of parameter No. 27480 to 1.
When CSP is set to 0, the block of G41/G42 is not outputted.

Use of this function changes the operation of some commands as well as subprogram call commands when NC program conversion is performed. The specifications of the function for suppressing expansion in NC program conversion are explained below.

Conversion target			NC program conversion with subprogram expansion	NC program conversion with subprogram expansion suppressed
4-digit G cycle	4-digit G cycle in main program		Output after conversion	Output after conversion
	4-digit G cycle in main program (Editing prohibited and display enabled)		Output after conversion	Output after conversion
	4-digit G cycle not executed (Execution is prevented by GOTO statement, etc.)		Not output	4-digit G command is output without conversion
	4-digit G cycle repeated by macro statement		4-digit G cycle is output after converted as many times as number of times the cycle is executed, and macro statement is not output.	4-digit G cycle is output after converted as many times as number of times the cycle is executed, and macro statement is also output.
4-digit G cycle	4-digit G cycle in subprogram (Subprogram is not edit-prohibited)	Machining commands are specified in main program, and figure commands are specified in subprogram.	Output after conversion	Output after conversion
		Both machining and figure commands are specified in subprogram.	Output after conversion	Subprogram call command is output.
	4-digit G cycle in subprogram (Subprogram is edit-prohibited and cannot be displayed)	Machining commands are specified in main program, and figure commands are specified in subprogram.	Not output	Subprogram call command is output.
		Both machining and figure commands are specified in subprogram.		
	Figure commands and NC statements are specified in subprogram.		4-digit G commands are converted and output, and NC statements are also output.	4-digit G commands are converted and output, and NC statements are not output.
Subprogram call (M98, G/M/T code)	Subprogram is not edit-prohibited.	4-digit G is not present in subprogram.	Subprogram contents are output.	Call command is output.
	Subprogram is edit-prohibited. (Display enabled)	Call by M98	Subprogram contents are output.	Call command is output.
		Call by G/M/T code	Subprogram contents are output.	Call command is output.
	Subprogram is edit-prohibited. (Display disabled)	Call by M98	Not output	Call command is output.
		Call by G/M/T code	Call command is output.	Call command is output.
Custom macro statement	Macro statement specified between 4-digit G machining command and figure command		Not output	Not output
	Macro statement in a place other than the above		Not output	Output
Finishing cycle (G70)			Not output correctly	Output correctly

4.6 CONVERSION OF G41/G42 COMMAND BY NC PROGRAM CONVERSION (WITH SUBPROGRAM DECOMPOSITION) (for only Series 30*i*/31*i*/32*i*-B, 0*i*-D)

When NC program conversion is executed with the subprogram decomposition, G41/G42 command can be converted.

4.6.1 Parameter Setting for Use

(1) Parameter to select whether the subprogram is decomposed or not in NC program conversion
Parameter No.27480#0=0 is set.

(2) Parameter to make this function effective
Parameter No.27483#4=1 is set.

(3) Parameter to set the number of blocks to be read in the cutter compensation/tool nose radius compensation mode
Parameter No.19625=0 or 3 ~ 8 is set.

4.6.2 Restriction

NC program conversion doesn't execute correctly in the following cases.
(Unnecessary block is output or necessary block is not output.)

(1) G41/G42 is executed with offset mode cancel by D0/T0.
(2) G41/G42 is executed as duplication.
(3) G41/G42 is executed at the next block of subprogram call command or macro call command.
[Solution]
Insert empty block (EOB) next to subprogram call command or macro call command.

(4) On the following condition
(Condition)
Number of blocks from G41/G42 to macro statement ≤ Value of parameter No.19625

[Solution]
Insert empty block (EOB) for satisfying the following condition.
(Condition)
Number of blocks from G41/G42 to macro statement > Value of parameter No.19625

(Example)
Case of "IF" statement (Parameter No.19625=3)

```
****** ;
G00 G41 X250. Y550. D01;
G01 Y900. F150 ;
IF[#4120 EQ 0] GOTO99 ;
X450. ;
G03 X500. Y1150. R650. F150 ;
****** ;
****** ;
N99 M30;
```

Modify

```
****** ;
G00 G41 X250. Y550. D01;
G01 Y900. F150 ;
;
;
IF[#4120 EQ 0] GOTO99 ;
X450. ;
G03 X500. Y1150. R650. F150 ;
****** ;
****** ;
N99 M30;
```

(5) On the following condition
(Condition)
Number of blocks from G41/G42 to command of [list 1] ≤ Value of parameter No.19625

[Solution]
Insert empty block (EOB) for satisfying the following condition.
(Condition)
Number of blocks from G41/G42 to command of [list 1] > Value of parameter No.19625

(Example)
Case of "M98" command (Parameter No.19625=3)

```
****** ;
G00 G41 X250. Y550. D01;
G01 Y900. F150 ;
M98 P8000 ;
G00 X0. Y0. ;
****** ;
```

Modify

```
****** ;
G00 G41 X250. Y550. D01;
G01 Y900. F150 ;
;
;
M98 P8000 ;
G00 X0. Y0. ;
****** ;
```

[List 1] Command list

Command	Function
M98 P****	M98 Subprogram call
M99	Program end
G65 A* B* ~ P****	Simple macro call
G66 A* B* ~ P****	Macro modal call : call after move command
G66.1 A* B* ~ P****	Macro modal call : each block call
G*** A* B* ~	Macro call using G code
M*** A* B* ~	Macro call using M code
M***	Subprogram call using M code
T***	Subprogram call using T code
S***	Subprogram call using S code
B***	Subprogram call using second auxiliary function code
A*** (B***/ C*** / ~)	Subprogram call using specific code

Command	Function
G02.3 / G03.3	Exponential interpolation CW/CCW
G05	AI contour control (high-precision contour control compatible command)
G07.1 / G107	Cylindrical interpolation
G10	Programmable data input
G10.8	Programmable inner data input
G10.9	Programmable switching of diameter/radius specification
G12.1 / G13.1 /G112 / G113	Polar coordinate interpolation mode/ cancer mode
G20 / G70	Inch input
G21 / G71	Metric input
G22 / G23	Stored stroke check function on / off
G25 / G26	Spindle speed fluctuation detection off / on
G28	Automatic return to reference position
G28.2	In-position check disable reference position return
G29	Movement from reference position
G30	2nd, 3rd, and 4th reference position return
G30.1	Floating reference position return
G30.2	In-position check disable 2nd, 3rd, or 4th reference position Return
G31	Skip function
G31.8	EGB-axis skip
G37	Automatic tool length measurement (M system)
G36 / G37 / G37.1 / G37.2	Automatic tool offset (X axis / Z axis) (T system)
G40	Tool radius/tool nose radius compensation cancel mode
G40.1	Normal direction control cancel mode
G43.4 / G43.5	Tool center point control (type 1/type 2)
G43.4 P1 / G43.5 P1	Tool posture control (type 1 / type 2)
G43.4 L1 / G43.5 L1	High-speed smooth TCP - rotation axes compensation (type 1 / type 2)
G43.4 P3 / G43.5 P3	High-speed smooth TCP - smooth control (type 1 / type 2)
G43.8 / G43.9	Cutting point command (type 1 / type 2)
G92	Setting for workpiece coordinate system (M system)
G50 / G92	Setting for workpiece coordinate system (T system)
G92.1	Work-piece coordinate system preset (M system)
G50.3 / G92.1	Work-piece coordinate system preset (T system)
G50.2	Polygon milling cancel
G50.2 / G250	Polygon turning cancel (T system)
G51.4 / G50.4	Start synchronous control / cancel
G51.5 / G50.5	Start composite control / cancel
G51.6 / G50.6	Start superimposed control / cancel
G52	Local coordinate system setting
G53	Machine coordinate system setting
G53.1	Tool axis direction control
G53.6	Tool center point retention type tool axis direction control
G54.4	Work setting error compensation
G68 / G69	Coordinate system rotation start or 3-dimensional coordinate conversion mode on / off (M system)
G68.1 / G69.1	Coordinate system rotation start or 3-dimensional coordinate conversion mode on / off (T system)
G68.2	Tilted working plane command
G68.3	Tilted working plane specification in tool axis direction
G68.4	Tilted working plane indexing (incremental multi-command)
G70.7	Finishing cycle (M system)

Command	Function
G71.7	Stock removal in turning (M system)
G72.7	Stock removal in facing (M system)
G73.7	Pattern repeating cycle (M system)
G76.7	Multiple-thread cutting cycle (M system)
G77	Plunge direct sizing/grinding cycle (M system)
G70 / G72	Finishing cycle (T system)
G71 / G73	Stock removal in turning (T system)
G72 / G74	Stock removal in facing (T system)
G73 / G75	Pattern repeating cycle (T system)
G75 / G77	Outer diameter/internal diameter drilling cycle (T system)
G76 / G78	Multiple-thread cutting cycle (T system)
G72 / G73	Traverse direct sizing/grinding cycle (T system)
G80.4	Electronic gear box: synchronization cancellation
G80.5	Electronic gear box 2 pair : synchronization cancellation
G81.1	Chopping
G92 / G78 / G21	Thread cutting cycle (T system)
M00	Program stop
M01	Optional stop
M02	End of program
M06	Tool change (Tool management function or Tool life management is available)
M30	End of tape
M**	M code preventing buffering (parameter No.3411 ~ 3432, No.11290 ~ No.11299)
	M code for tool life count restart (parameter No.13221)
	M code for tool life count restart (parameter No.6811)
	The top number of M code used to turn on each axis workpiece coordinate system preset signal (parameter No.11275、No.11276)
	M code for turning on the flexible synchronous control mode (parameter No.5670~No.5677)
	M code for tool life count restart (parameter No.6811)
	M code to start arbitrary speed threading mode (parameter No.11487)
	M code to cancel arbitrary speed threading mode (parameter No.11488)
	M code for high-speed switching of Cs contour control (parameter No.3795)
T**	T code for Tool management function
	T code for Tool life management function

V. HANDLE MACHINING

1 HANDLE MACHINING (For only Series 30*i*/31*i*/32*i*-B)

Handle machining has the following two functions.

(1) Machining by handle
This function enables operators to machine a slanted line or an arc with simultaneous two axes motion by one handle.
(2) Programming by handle
The handle operation can be memorized as a program, and the tool motion can be played back.

1.1 WORKING CONDITION

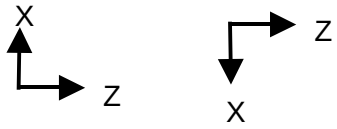
Handle machining can be used only by the following conditions.

(1) Necessary optional function
- MANUAL GUIDE *i* function (S790)
- Handle machining function (S797)
- Manual liner/circular interpolation function (J774)

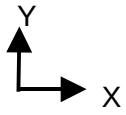
(2) Machine configurations
- Lathe : One path system 2 axes (X, Z)
- Machining center : One path system 3 axes (X, Y, Z)

(3) Available mode
HND mode, JOG mode, THND mode, and TJOG mode

(4) Available coordinate system for handle machining
- Lathe:



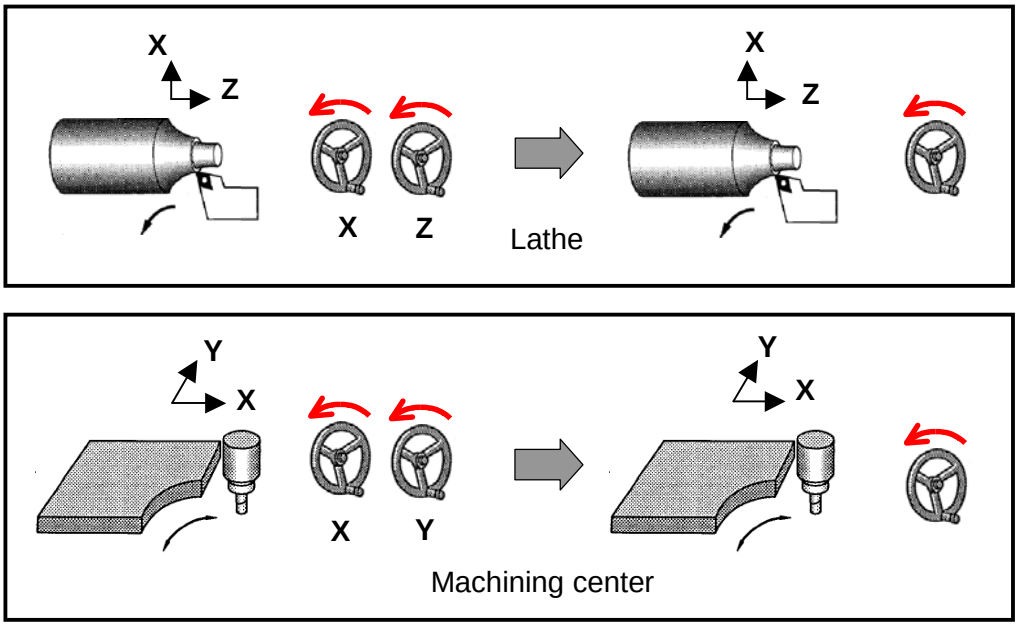
- Machining center:



1.2 MACHINING BY HANDLE

1.2.1 Specification of Function

This function enables operators to machine a slanted line or an arc with simultaneous two axes motion by one handle.



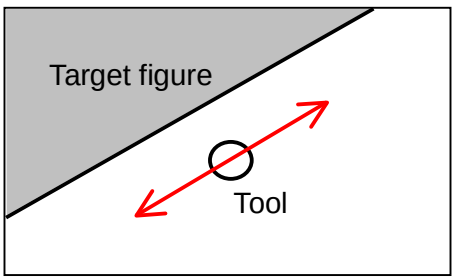
NOTE
If the signals G544 (MHLCn) and G545 (MHUSn) for Approach/Guidance handle are not set, handle machining function is not available.

Handle and tool motion

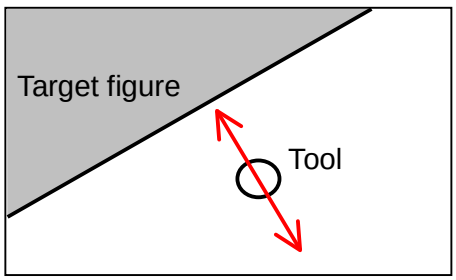
There are the following two type of using handle for handle machining.
- Guidance handle: the handle for moving tool in parallel along the target figure
- Approach handle: the handle for moving tool vertically with the target figure

It is possible to machine the line figure and circle figure by Guidance handle and Approach handle.

(1) Line figure
- The tool moves in parallel along the target figure when Guidance handle is moved.
- The tool moves vertically with the target figure when Approach handle is moved.

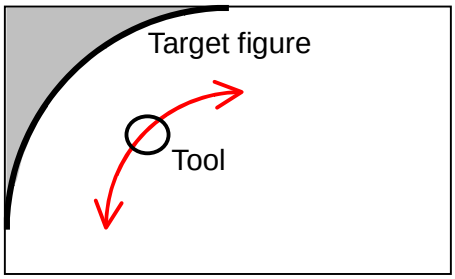


Tool motion by Guidance handle

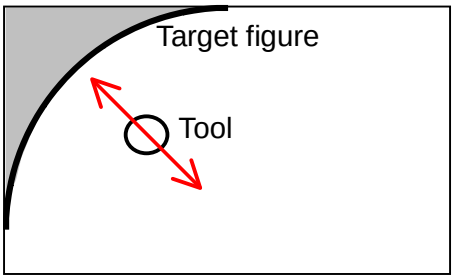


Tool motion by Approach handle

(2) Circle figure
- The tool moves along the circumference of the target figure when Guidance handle is moved.
- The tool moves toward the center of the circle when Approach handle is moved.



Tool motion by Guidance handle

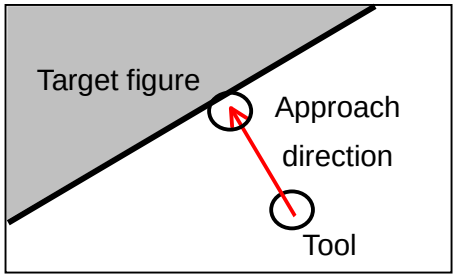


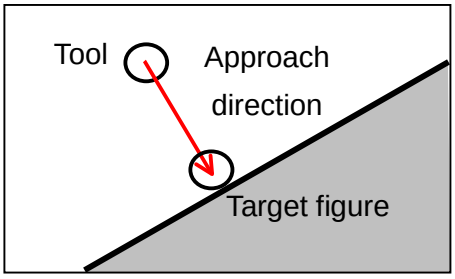
Tool motion by Approach handle

Limit function

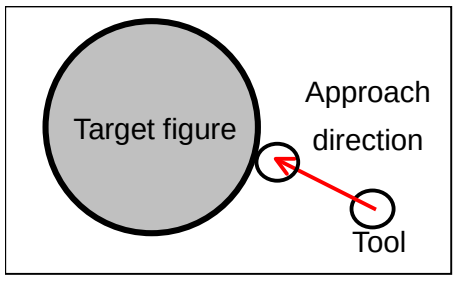
When the limit function is available, the tool is stopped at the position of which a tool touches the target figure. It is possible to prevent over cutting by using the limit function.

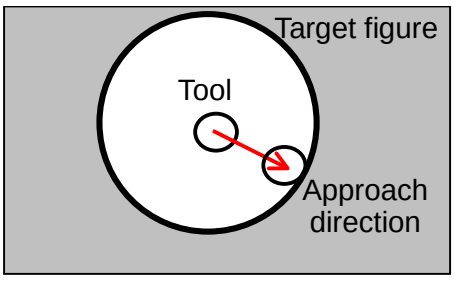
(1) Line figure





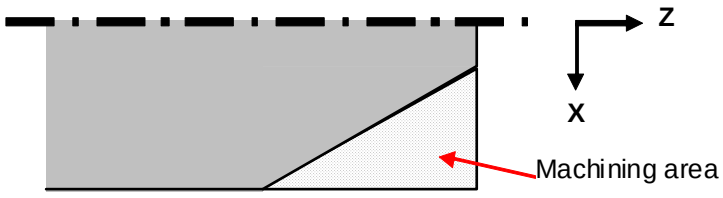
(2) Circle figure



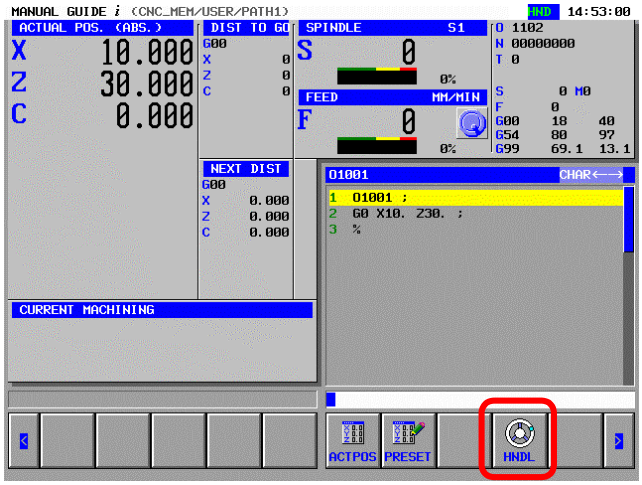


1.2.2 Example of Operation

It is an operation example of machining part of the below figure by handle.

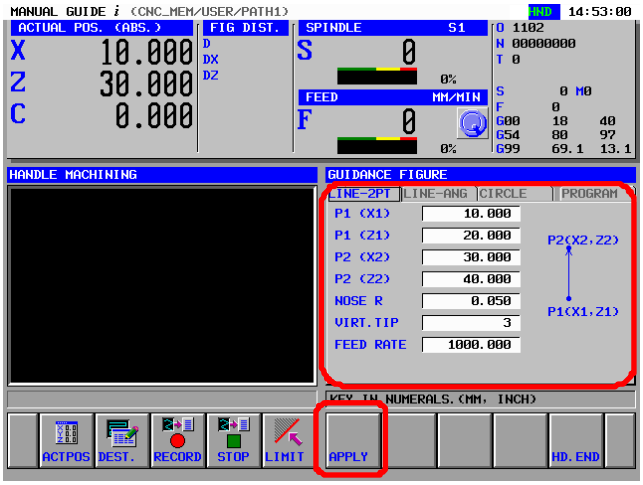


(1) The software key [HNDL] is pressed on the base screen of any one of HND mode, JOG mode, THND mode and TJOG mode.



Base screen

(2) After the two point coordinate system data of the target line figure is input on the figure input screen, the software key [APPLY] is pressed.



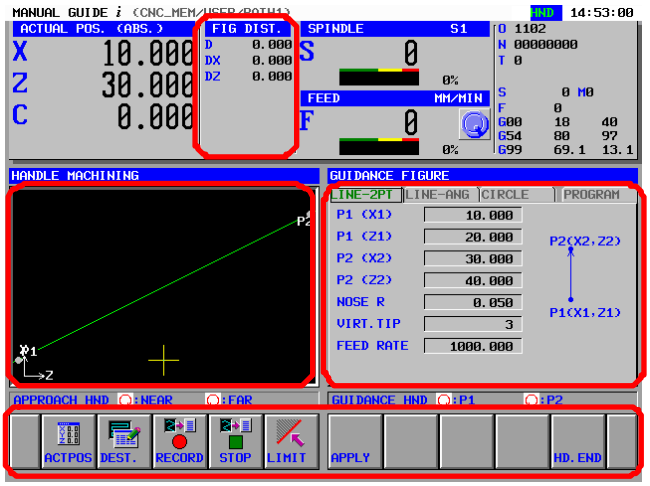
Input data of target figure (line)

Handle machining screen

(3) The following information is displayed on the execution screen of handle machining.

Display of distance between target figure and present tool position

Display of target figure and tool position (+) on graphic window

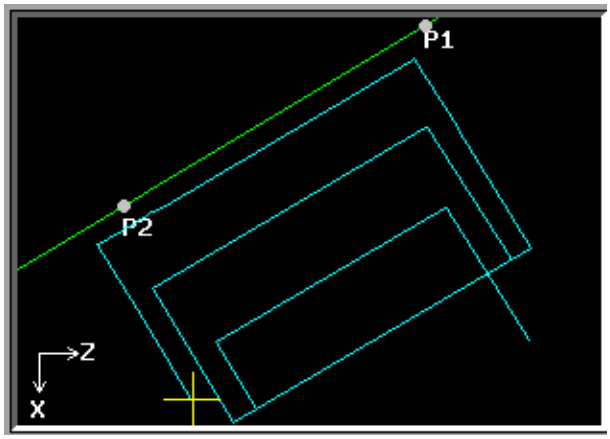


Display of target figure data and state of limit function

Display of explanation for approach/guidance handle

Handle machining screen

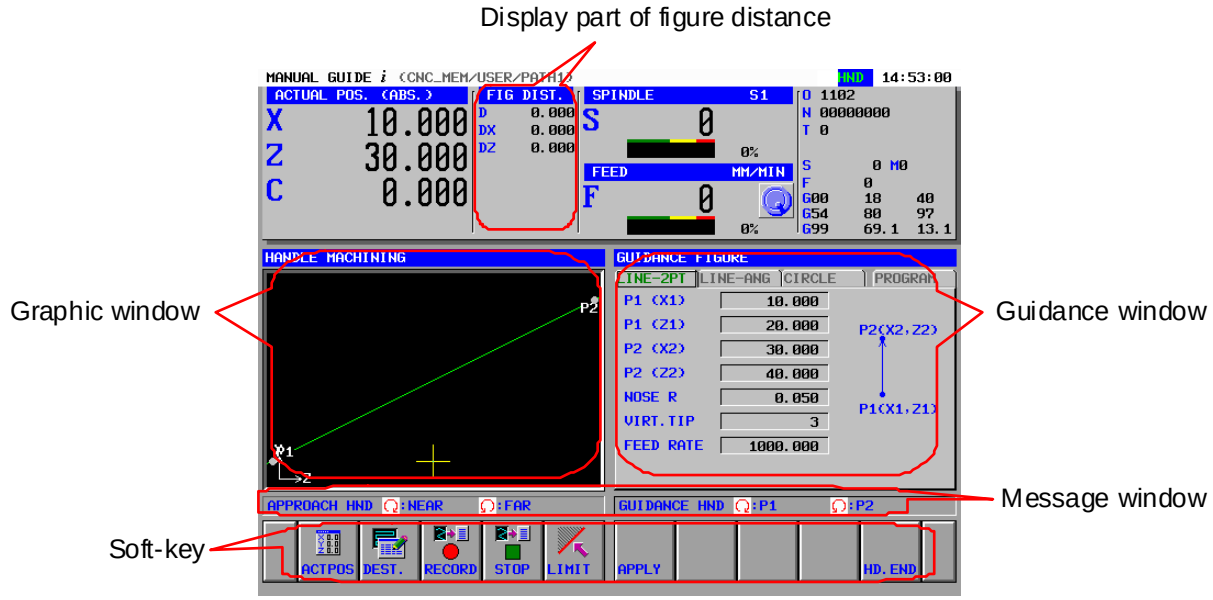
(4) When the approach handle is moved, the tool moves vertically with the target figure (line). When the guidance handle is moved, the tool moves in parallel along the target figure. And the tool path is displayed on the graphic window.



Graphic window

1.2.3 Handle Machining Screen

The construction of handle machining screen is as follows.



Construction of handle machining screen

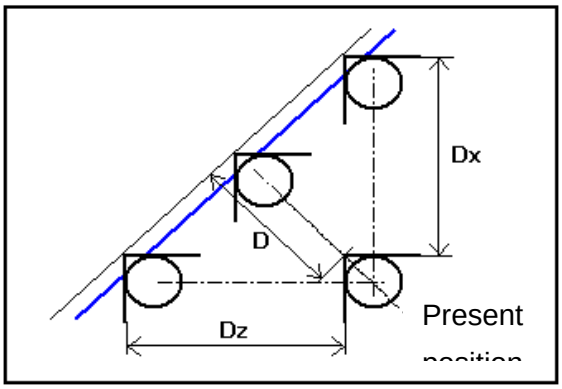
Soft-key

Soft-key name	Outline
ACTPOS	The display of the actual position is switched. Refer to the specification of MANUAL GUIDE i for details.
DEST.	Specification of recorded program (This item is displayed only for No.14858#3=1.)
RECORD	The program making begins. (This is displayed only for No.14858#3=1.)
STOP	The program making is ended. (This is displayed only for No.14858#3=1.)
LIMIT	On/Off of the limit function is switched. (Impropriety when guidance figure is not determined.)
APPLY	The guidance figure is determined by the input value, and the figure is drawn by the value.
HD.END	Handle machining function is ended, and it returns to the base screen of MANUAL GUIDE i.

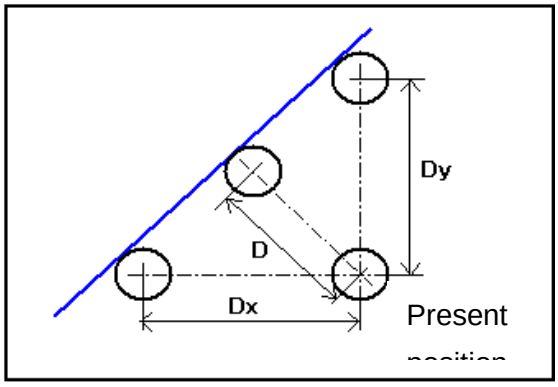
Display part of figure distance

The distance between a target figure and present tool position is displayed in an upper state display window.

(1) Lathe
The distance from the virtual tip of tool nose to the target figure is displayed.
D: Distance on vertical straight line to target figure from present position
Dx: Distance for X axis direction
Dz: Distance for Z axis direction



(2) Machining center
The distance from the center of the tool to the target figure is displayed.
D: Distance on vertical straight line to target figure from present position
Dx: Distance for X axis direction
Dy: Distance for Y axis direction



Guidance window

The shape input screen of the steering wheel processing includes three input screens of yen of straight line-angle of the straight line -2 points. It is possible to switch to each screen by moving the cursor to the tab, and pressing a right and left cursor key.

(1) Tab [LINE-PT]
Coordinates of two points are specified, and a straight line target figure is input.

No.	Input item		Meaning
	Lathe	Machining center	
1	P1 (X1)	P1 (X1)	X coordinates value of the first point
2	P1 (Z1)	P1 (Y1)	Z coordinates value / Y coordinates value of the first point
3	P2 (X2)	P2 (X2)	X coordinates value of the second point
4	P2 (Z2)	P2 (Y2)	Z coordinates value / Y coordinates value of the second point
5	NOSE R	TOOL DIA.	Tool noise radius / Diameter of tool
6	VIRT.TIP	-	Virtual tip of tool nose (0 - 9)
7	OFFSET NO.	OFFSET NO.	Offset number (Note 1)
8	FEED RATE	FEED RATE	Feed rate (Note 2)

(Note 1) Refer to parameter No.14858 #2.
(Note 2) Refer to parameter No.14858 #3. (Used to Programming by handle)

(2) Tab [LINE-ANG]

A straight line target figure is input specifying coordinates of one point and the angle.

No.	Input item		Meaning
	Lathe	Machining center	
1	P (X)	P (X)	X coordinates value of the first point
2	P (Z)	P (Y)	Z coordinates value / Y coordinates value of the first point
3	ANGLE (A)	ANGLE (A)	Angle with horizontal axis (Anti-clockwise is positive)
4	NOSE R	TOOL DIA.	Tool noise radius / Diameter of tool
5	VIRT.TIP	-	Virtual tip of tool nose (0 - 9)
6	OFFSET NO.	OFFSET NO.	Offset number (Note 1)
7	FEED RATE	FEED RATE	Feed rate (Note 2)

(Note 1) Refer to parameter No.14858 #2.
(Note 2) Refer to parameter No.14858 #3. (Used to Programming by handle)

(3) Tab [CIRCLE]

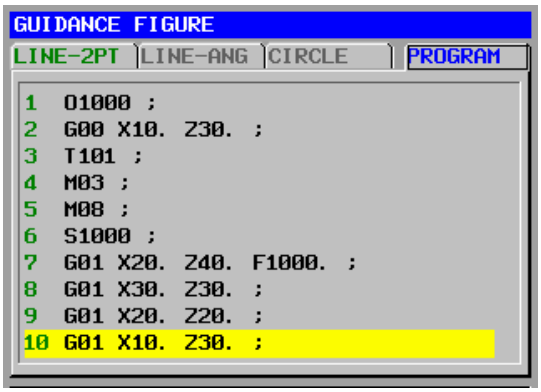
The target figure of circle is input specifying center coordinates and the radius.

No.	Input item		Meaning
	Lathe	Machining center	
1	Center (X)	Center (X)	X coordinates value of circle center
2	Center (Z)	Center (Y)	Z coordinates value / Y coordinates value of circle center
3	RADIUS (R)	RADIUS (R)	Radius of circle
4	NOSE R	TOOL DIA.	Tool noise radius / Diameter of tool
5	VIRT.TIP	-	Virtual tip of tool nose (0 - 9)
6	OFFSET NO.	OFFSET NO.	Offset number (Note 1)
7	FEED RATE	FEED RATE	Feed rate (Note 2)

(Note 1) Refer to parameter No.14858 #2
(Note 2) Refer to parameter No.14858 #3 (Used to Programming by handle)

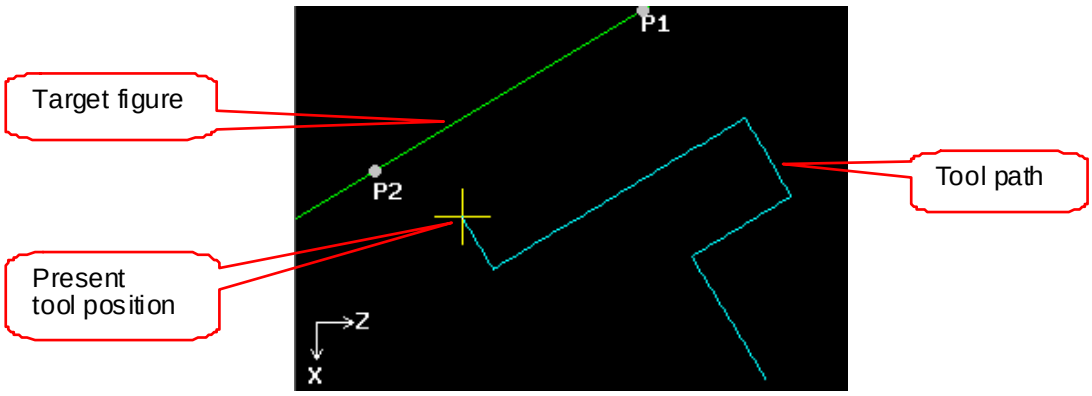
(4) Tab [PROGRAM]

This tab is displayed only for No.14858 #3 = 1, used to Programming by handle.
The recorded program is displayed in this tab.



Graphic window

A target figure and present tool position and tool path are displayed in a left graphic window.



(1) Display of target figure
The target figure is displayed in green.

(2) Display of present tool position
A present tool position is displayed in yellow.
The tool position which is displayed by circle or cross, can be switched by setting the parameter No.14858 #1.

(3) Drawing of tool path
Tool path is displayed in blue.

Lathe: Tool path draws in tracks of the tool where the virtual tip position moves.
Machining center: Tool path draws in tracks of the tool where the tool center moves.

Message window

The guidance, that the movement of the tool and the rotation direction for Approach handle and Guidance handle is notified, is displayed in a message window.
And the position of guidance follows parameter No.14858 #0.

- For parameter No.14858 #0 = 0



- For parameter No.14858 #0 = 1

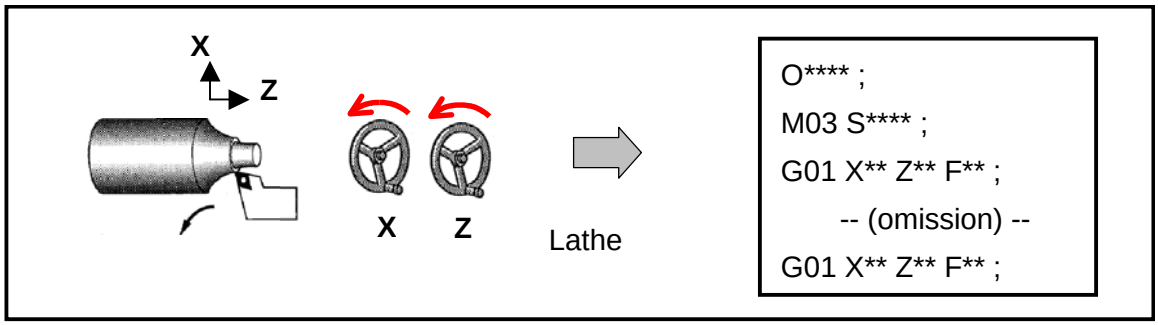


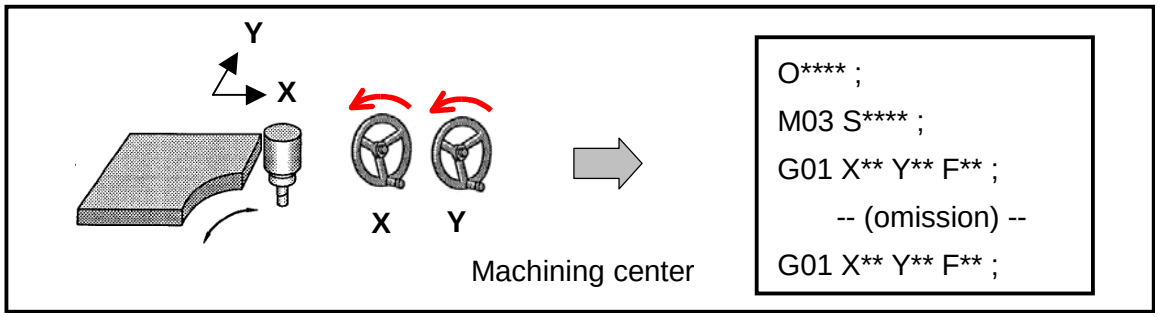
NOTE
1 Which handle is considered to be an approach handle or guidance handle is decided by setting signal G544 (MHLCn) and signal G545 (MHUSn).
2 Guidance message does not display on the guidance window, when the signal of Approach handle or Guidance handle are not set.

1.3 PROGRAMMING BY HANDLE

1.3.1 Specification of Function

This function can convert handle operation to program, and can record the CNC command into Tape storage memory.





NOTE
Programming by handle makes the cutting feed command (G01,G02,G03). When the operator want to change it to the rapid traverse command (G00), please check the converted program by machining simulation and edit the command directly.

Handle operation and converted CNC command

The each handle operation is converted to the following CNC command.

Handle operation		Converted CNC command (Note 1)	Timing of converting command
Set up (Auxiliary function)	S code	S**	When the change amount of the actual spindle speed exceeds a constant value.
	T code	T**	When the signal corresponding to the auxiliary function command with the operation panel button is turned on.
	Spindle rotation	M3	
	Spindle rotation	M4	
	Stopping spindle	M5	
	Coolant ON	M8	
	Coolant OFF	M9	
Machining of workpiece	Axis direction feed handle (1axis)	G01 X** (Y**) (Z**) F**	When the following conditions are meet. (1) When the condition which becomes the pause (Note2) of the block is meet. (2) When the soft-key [STOP] is pushed. (3) When the guidance figure is changed.
	Axis direction feed handle (Simultaneous 2axes)	G01 X** Z** F** G01 X** Y** F**	
	Approach handle	G01 X** Z** F** G01 X** Y** F**	
	Guidance handle	(Line figure) G01 X** Z** F** (Circle figure) G02/03 X** Z** I** K** F** G02/03 X** Z** I** J** F**	

(Note1)

The following information is input to "**" at the converted command respectively.
S: Actual spindle speed (Feed per minute)
T: Tool information (Lathe : Tool number + Offset number, Machining Center : Tool number)
X, Y, and Z: Present position coordinates when the command is made (absolute coordinates)
I, J, and K: Distance from starting point to circular arc center (with sign)
F: Feed speed input by item [FEED RATE]

(Note 2)

It is time when the condition, which becomes the pause of the block meets either of the following requirements.
- When the operated handle is switched.
- When the number of operated handle is changed.
- When the sending direction of handle is reversed.
- When the stopping time is longer than given time after the handle is operated.
- When the soft-key [STOP] is pushed.
- When the guidance figure is changed.

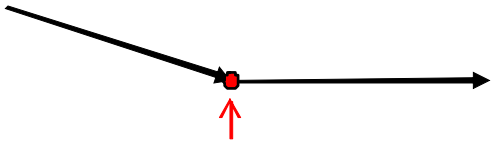
Detail of converted movement command

(1) Coordinates value

The coordinates value, which is converted as an end point, is one that is acquired at the pause of the block.

This coordinates value is a position where the movement stopped temporarily. (The servo delay amount and the acceleration/deceleration delay amount are 0 at the time.)

Or, it is a position where the direction of the movement was reversed. (Only by using axis direction feed handle)



This point of the present position is converted as an end point.

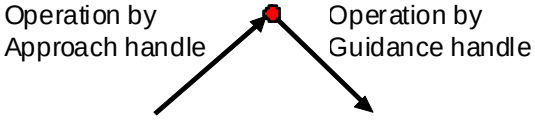
(2) Feed speed

Feed speed is converted as the data, which is input to the input item on the guidance input window. "F?" is instructed when nothing is input to the input item.

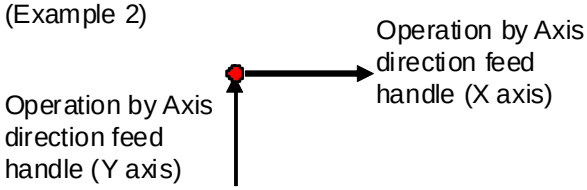
(3) The pause of the block

(a) When the kind of the operated handle is switched, it is assumed the pause of the block.
(The coordinate, when the movement stops temporarily, is assumed to be a terminal position.)

(Example 1)

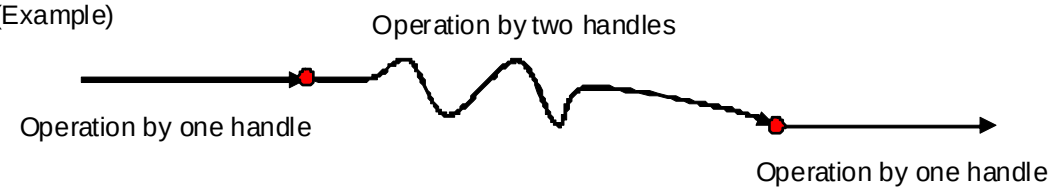


(Example 2)



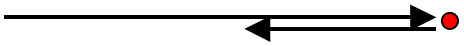
(b) When the number of the operated handle is changed, it is assumed the pause of the block.
(The coordinate, when the movement stops temporarily, is assumed to be a terminal position.)

(Example)

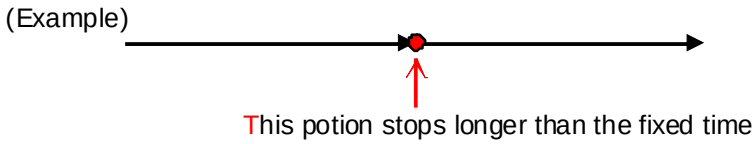


(c) When sending the direction of one axis is reversed with one axis direction feed handle, it is assumed the pause of the block.
The coordinate, when sending is reversed, is assumed to be a terminal position.)

(Example)



(d) hen the movement is stopped longer than the fixed time after the handle is operated, it is assumed the pause of the block.



(e) hen "STOP" of soft-key is pressed, it is assumed the pause of the block.
The coordinate, when the soft-key [STOP] is pressed, is assumed to be a terminal position.)

(f) hen the data of the guidance figure is changed, it is assumed the pause of the block.
The coordinates when the data of guidance figure is changed, is made the terminal position.)

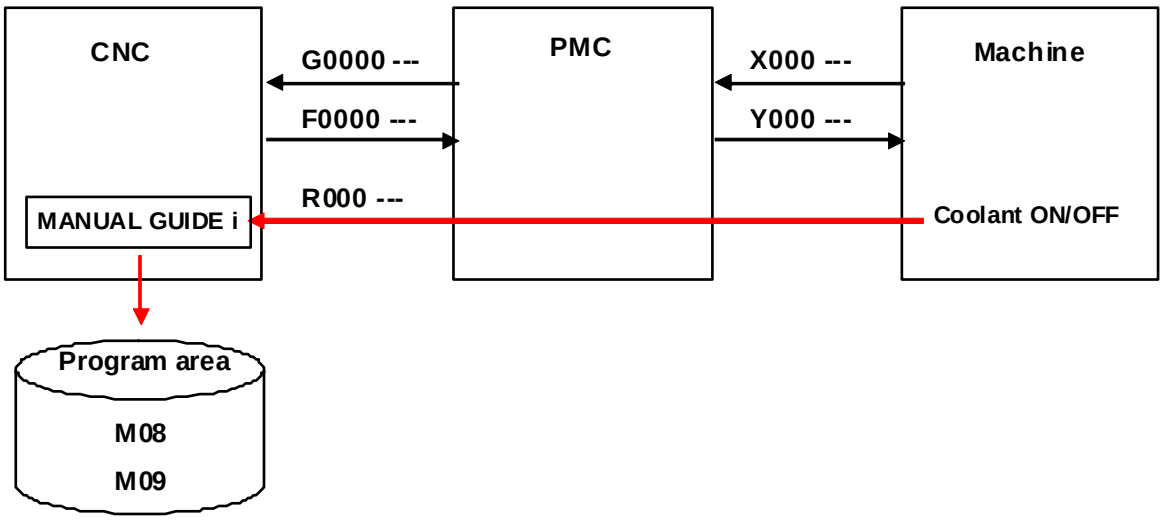
(Note 1)
The converted movement instruction is an absolute programming. So, when the modal information is an incremental programming, please specify an absolute instruction (G90).
(Note 2)
The following movement instruction of circle can not be converted.
- one time circle
- circular arc that exceeds 360 °

(Note 3)
Because the terminal position cannot be fixed according to the following timing, the movement instruction is not converted.
- never the tool does not stop temporarily, and the tool always moves
- when the operator operates two handles at the same time

(Note 4)
When the soft-key [RECORD] is pressed, the movement to start point is converted as the rapid traverse command (G00), which the terminal coordinate is the present position. The other movement is converted as the cutting feed command (G01,G02,G03).

Detail of converted auxiliary function command

The auxiliary function command with the button of operation panel is converted by detecting the DI signal, which is allocated to the switch of operation panel.



(Note)

As to S code command, when accumulated of the change amount for real spindle speed exceeds a constant value (Set it in parameter No.27462) after pressing the soft-key [RECORD], S code command is made by real spindle speed.

Command corresponded to auxiliary function command call signal

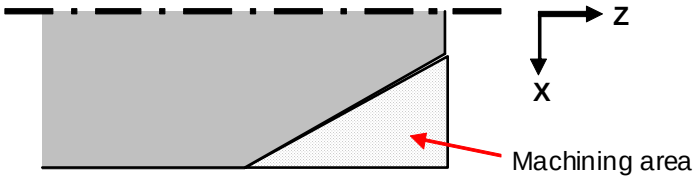
To call the auxiliary function command, the maximum DI signals (R-signal) of 24 are prepared. The number of the R-signal is specified by the parameter No.27460.

The text data of command is made with personal computer beforehand, and the made command is registered on the setting screen of MANUAL GUIDE *i* with a memory card.

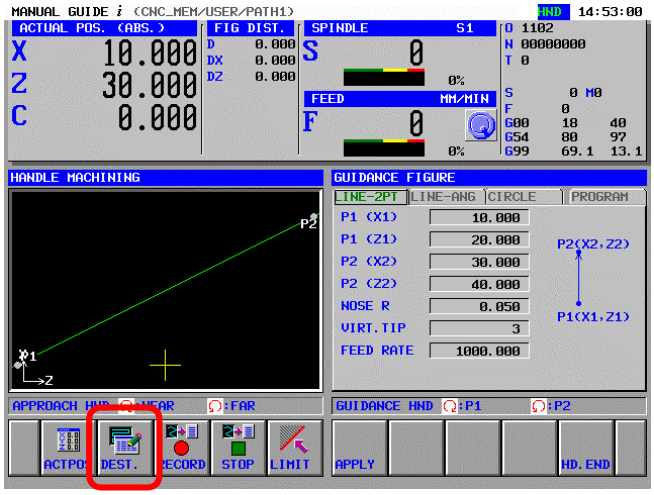
(Refer to "1.3.3 Registration of auxiliary function command" for details.)

1.3.2 Sample of Operation

It is an operation example of programming machining part of the below figure.

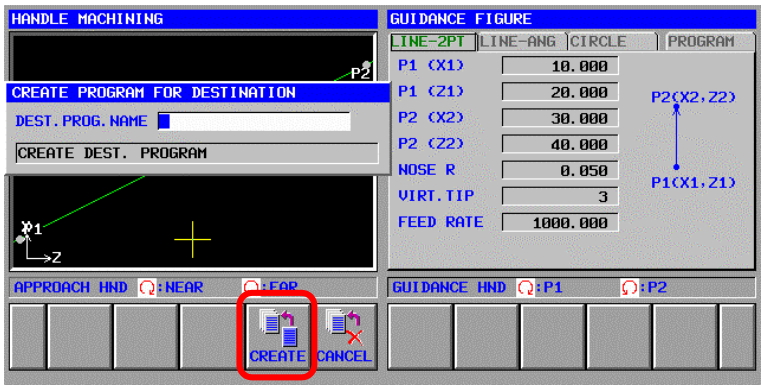


(1) The setting window of "CREATE PROGRAM FOR DESTINATION" is displayed by pressing the soft-key [DEST.] on the handle machining screen.



Handle machining screen

(2) The destination of created program is defined by pressing the soft-key [CREATE] after inputting the program name on the setting window of recording program.

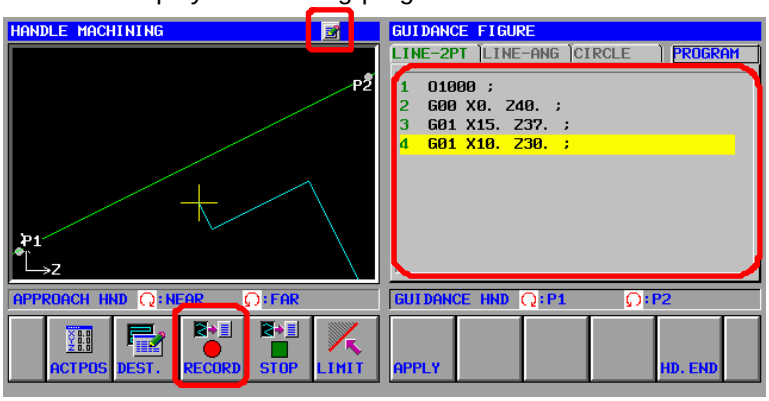


Setting window of recording program

(3) Recording program mode is tuned on by pressing soft key [RECORD] on the execution screen of handle machining.

(4) Execute the handle operation that is written in Chapter "4.2.2 Screen and operation". Handle operation is converted to CNC command and, the command is recorded into the program. And the content of converted program can be displayed by switching the tab [PROGRAM].

Display of recording program mode



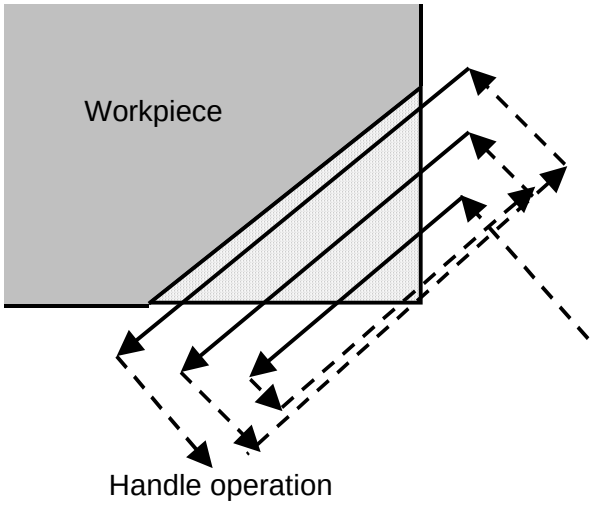
Display of recorded program

Handle machining screen

(5) When the handle operation is finished, turn the recording program mode off by pressing the soft key [STOP].

Example of converted program is as follows.

(Example of converted program)




```
O1111
M03 S1000 ;
G00 X20. Z60. ;
G01 X5. Z40. F0.8 ;
G01 X-20. Z60. ;
G01 X-15. Z70. ;
G01 X10. Z40. ;
G01 X5. Z30. ;
G01 X-25. Z60. ;
G01 X-20. Z75. ;
G01 X15. Z35. ;
G01 X5. Z20. ;
G01 X-30. Z60. ;
```

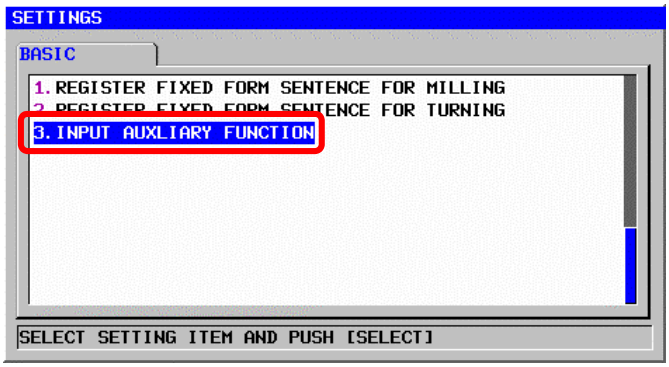
1.3.3 Registration of Auxiliary Function Command

The auxiliary function command with the button of operation panel is made as the fixed text data by using personal computer beforehand, and is registered by reading from the memory card.

Registration of auxiliary function command

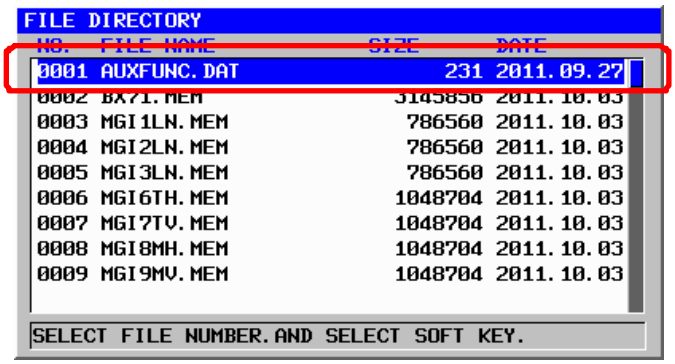
The registration of the auxiliary function command becomes the following operations.

(1) The window of "SETTINGS" is displayed by pushing the soft-key [SETING] on the base screen of MANUAL GUIDE *i*.



Settings Window

(2) The file directory window, that file is memorized into the memory card, is displayed by pushing the soft-key [SELECT] after moving cursor to the item of "3. INPUT AUXILIARY FUNCTION".



File for auxiliary function command

(3) The message of the writing confirmation is displayed by pushing the soft-key [INPUT] after moving the cursor to the file that made as the auxiliary function command beforehand. And the registration is executed by pressing the soft-key [YES].

(4) The message of "INPUT COMPLETE" is displayed on the file directory window after finishing the registration of the auxiliary function.

Making of registration data for auxiliary function command

The format of the auxiliary function command is as follows. Following text format data is made with personal computer.

<Title>=_,<Code>=_

Title: The registration name of the auxiliary function command is set.
Code: The auxiliary function command is set.

(Example)

```
<Title>=TOOL CHANGE,<Code>=T?;M06;
<Title>=COOLANT ON,<Code>=M08;
<Title>=COOLANT OFF,<Code>=M09;
....
....
....
```

Maximum 24 auxiliary function commands can be registered.

(Note)

Please specify "T?" as the T code command. When the T code command is made, "?" is replaced with the tool number (Head tool number : "HD.T") which has adhered to a present turret.
In the lathe, the tool number is added the offset number. (Because T code makes "tool selection number + offset number" assumption, the tool management function cannot be used.)

Addition of ladder program

The maximum 24 call signals are prepared for auxiliary function command, which is allocated to the switch of operation panel.
Please add to the ladder program to turn on the R-signal when the switch of operation panel is pushed.
MANUAL GUIDE *i* observes at the cycle of 8 msec. If the R-signal is turning on, MANUAL GUIDE *i* turns off the R-signal after acquiring the registered auxiliary function command.

Call signals for auxiliary function command	Converting command
R***.0	Command of No.1
R***.1	Command of No.2
- - -	- - -
R***.7	Command of No.8
R(***+1).0	Command of No.9
- - -	- - -
R(***+1).7	Command of No.16
R(***+2).0	Command of No.17
- - -	- - -
R(***+2).7	Command of No.24

1.4 PARAMETERS

(1) To use this function, be sure to set the following parameters.
- No.27457 Work coordinate system during handle machining
- No.27458 View coordinate of graphic window for handle machining
- No.27462 Standard value which is assumed that spindle speed is changed (r.p.m)

(2) The relating parameters are as follows.
- No.7106#3 Internal relay (the R-signal) of PMC that uses it with input data in manual linear/circular interpolation
- No.7106#4 Internal relay (the R-signal) of PMC that uses it with output data in manual linear/circular interpolation
- No.13541 Head address of the R-signal used by the input data in the manual linear/circular interpolation
- No.13542 Head address of the R-signal used by the output data in the manual linear/circular interpolation
- No.14858#0 The position of Guidance message
- No.14858#1 The tool position on the graphic window
- No.14858#2 The tool information
- No.14858#3 Programming by handle
- No.27460 Head address of the R-signal used by the output data for auxiliary function commands
- No.27461 Time which is assumed that handle machining is stop (msec)

1.5 RELATED SIGNALS

The relating signals are as follows.

(1) Feed axis and direction selection signal +Jg, -Jg, +Ja, -Ja < G086 >

[Classification] Input signal
[Function] Select a desired feed axis and direction in jog feed or incremental feed. The sign (+ or -) in the signal name indicates the feed direction. “Jg” and “Ja” indicates Guidance JOG and Approach JOG, respectively.

	#7	#6	#5	#4	#3	#2	#1	#0
G086					-Ja	+Ja	-Jg	+Jg

(2) Manual linear/circular interpolation signal MHLC1 to MHLC5 < G544.0 to G544.4 >

[Classification] Input signal

[Function] Each of these signals selects the manual linear/circular interpolation function for the n-th manual pulse generator.

[Operation] 0: Manual linear/circular interpolation is disabled. The manual handle is used for axis direction feed.

1: Manual linear/circular interpolation is enabled. The manual handle is used for the purpose selection by the usage selection of manual linear/circular interpolation signal (MHUSn).

	#7	#6	#5	#4	#3	#2	#1	#0
G544				MHLC5	MHLC4	MHLC3	MHLC2	MHLC1

(3) Usage selection of manual linear/circular interpolation signal MHUS1 to MHUS5 < G545.0 to G545.4 >

[Classification] Input signal

[Function] Selects the guidance handle or approach handle for the n-th manual pulse generator when the manual linear/circular interpolation signal (MHLCn) is set to 1 for the manual linear/circular interpolation function.

[Operation] 0: The manual handle is used as the approach handle.

1: The manual handle is used as the guidance handle.

	#7	#6	#5	#4	#3	#2	#1	#0
G545				MHUS5	MHUS4	MHUS3	MHUS2	MHUS1

(4) Auxiliary function command signal AUXF1 to AUFX24 < R***.0 to R(***+2).7 >
The number of R*** is specified by setting the parameter No.27460.

[Classification] Input signal

[Function] The auxiliary function command is executed by pushing the button of operation panel.

[Operation] 0: The button of operation panel is not pushed.(The auxiliary function command is not executed.)

1: The button of operation panel is pushed.
(The auxiliary function command is executed.)

When the button of operation panel is pushed, MTB ladder program set to 1 of signal AUXFn.
When the signal AUXFn is set to 1, MANUAL GUIDE *i* sets to 0 of signal AUXFn in cycles of 8ms.

	#7	#6	#5	#4	#3	#2	#1	#0
R***	AUFX8	AUFX7	AUFX6	AUFX5	AUFX4	AUFX3	AUFX2	AUFX1
R(***+1)	AUFX16	AUFX15	AUFX14	AUFX13	AUFX12	AUFX11	AUFX10	AUFX9
R(***+2)	AUFX24	AUFX23	AUFX22	AUFX21	AUFX20	AUFX19	AUFX18	AUFX17

VI. OTHER FUNCTIONS

1 TOOL MANAGEMENT FUNCTION

1.1 OVERVIEW

This function manages various tools using tool numbers specific to the MANUAL GUIDE *i*. The NC standard tool management function manages tools using tool type numbers, but this function manages them using tool numbers.
Multiple tool offset numbers can be assigned to each tool number. Therefore, this function is also available when assigning an offset number for each tool attitude during machining.

NOTE
When using this function, the tool management function option is required.
For details, refer to the manual provided by the machine tool builder.

1.2 ASSOCIATING TOOL NUMBERS WITH OFFSET NUMBERS

On the screen for associating a tool number with a offset number, set the number of the tool that will use a offset number. When registered on this screen, the tool number is registered in the tool management data table, together with the offset number and the tool type.
This screen is effective only if bit 0 (ORT) of parameter No. 14823 is 1.

1.2.1 Selecting the Screen for Associating a Tool Number with a Offset Number

(1) On the initial screen in each mode, press soft key [SETING]. If soft key [SETING] does not appear, press the rightmost soft key [>] to display it.

(2) Selecting [SETTING OF OFFSET AND TOOL NO.] on the [SETTINGS] menu selection screen that appears causes the screen for corresponding between tool offset numbers and tool numbers to appear.

1.2.2 Screen Display Items

1.2.2.1 Display items

OFS NO.:
You can only view offset numbers, and cannot set new ones.
The range of available offset numbers depends on the setting of parameter No. 14824.

TOOL NO:
To register a new tool number in an empty tool number field, enter a new number and press the <INPUT> key.
To invalidate an existing tool number, enter 0.

NOTE

When a tool number is entered, the associated offset number is set in the “offset number” item of the data having that tool number in the tool management data table.
If the tool management data table does not contain data having that tool number, the system searches for tool management data having no tool number, and the offset number associated with the entered tool number is set in that tool management data.
If a tool number is deleted (0 is entered), all of the tool management data having that tool number is deleted.

TYPE:

To select the desired one, press the corresponding soft key.

TURN	ROTATE	OTHERS			CHCURS			NO.SRH	TO MNU

1.3 VIEWING AND SETTING TOOL OFFSET VALUES

In addition to the conventional tool offset setting screen, a screen is available which allows you to view and set tool offset values using tool numbers and offset types.
This screen is effective only if bit 1 (TOF) of parameter No. 14823 is 1.

1.3.1 Selecting the Tool Number-by-tool Number Tool Offset Setting Screen

(1) On the initial screen in each mode, press soft key [T-OFS]. If soft key [T-OFS] does not appear, press the rightmost soft key [>] to display it.
(2) The tool offset setting screen appears.
(3) Moving the cursor to “M:OFS-TOL” tab causes the “Milling offset screen (on a tool number by tool number basis)” to appear.

1.3.2 Offset Screen (on a Tool Number by Tool Number Basis)

TOOL NO.:

Tool numbers are displayed.

TYPE:

Offset types are displayed.
Values ranging from 1 to the “number of offset types” set in parameter No. 14825 are displayed sequentially.
If the setting of parameter No. 14825 is 0, nothing is displayed in the offset type column.

TIPTOOL LENGTH COMPENSATION, CUTTER COMPENSATION:

The offset number corresponding to each combination of tool number and offset type is determined, and the offset values of the offset number are displayed.
The settings are made in the data for the offset number determined with the combination of tool number and offset type.

1.4 VIEWING AND SETTING TOOL MANAGEMENT DATA

This screen is effective only if bit 3 (TMG) of parameter No. 14823 is 1.

1.4.1 Selecting the Tool Management Data Setting Screen

(1) On the initial screen in each mode, press soft key [SETING]. If soft key [SETING] does not appear, press the rightmost soft key [>] to display it.
(2) Selecting [TOOL MANAGEMENT DATA] on the [SETTINGS] menu selection screen that appears causes the tool management data setting screen to appear.

1.4.2 Magazine Data Screens (Magazine 1 to 4)

The tool number corresponding to each pot, type, group number, and offset number are displayed.
You can change tool numbers, types, and group numbers.

1.4.2.1 Display items

POT:
Pot numbers are displayed.

NOTE
On the individual magazine data screens, as many pots as the “number of data items” set in parameters Nos. 13222, 13227, 13232, and 13237 are displayed, beginning with the “start pot number” set in NC parameters Nos. 13223, 13228, 13233, and 13238.

TOOL NO.:
Set a tool number.
TOOL KIND:
The “tool type” corresponding to each tool number, as determined from the tool management data table, is displayed.
This item can be set with soft keys.
GROUP:
The “group number” corresponding to each tool number, as determined from the tool management data table, is displayed.
To set a new one, enter a value.
OFFSET NO.:
The “offset number” corresponding to each tool number, as determined from the tool management data table, is displayed.

1.4.3 Spindle and Standby Position Tool Display Screen

This screen displays the tools at spindle positions and at subpots (standby positions).

The number of displayed spindle positions and the number of displayed standby positions vary depending on the settings of
Parameter No. 13250 (number of effective spindles)
Parameter No. 13251 (number of effective standby positions)

The display items are the same as those of the magazine data screen.

1.5 VIEWING AND SETTING LIFE MANAGEMENT DATA

This screen is effective only if bit 4 (TLF) of parameter No. 14823 is 1.

1.5.1 Selecting the Life Management Data Setting Screen

(1) On the initial screen in each mode, press soft key [SETING]. If soft key [SETING] does not appear, press the rightmost soft key [>] to display it.
(2) Selecting [TOOL LIFE DATA] on the [SETTINGS] menu selection screen causes the life management data setting screen to appear.

1.5.1.1 Display items

ORDER:
In the first column for each tool, the value indicating the priority of the tool is displayed.
By positioning the cursor on this item and entering a new value, you can change the priority of that tool.

GROUP:
The group numbers in the tool management data table are displayed.

COUNT:
The count types (time or number of times of use) in the tool management data table are displayed.
For each group, the life count type (time or number of times of use) can be specified.
Select one of soft keys [TIME] and [COUNT].

TOOL NO.:
The tool numbers with the same group number are displayed.
The numbers are displayed in the order in which the tools will be used.
To register new tool in a group, move the cursor at the bottom (blank portion) of the tool number column for that group and press [REGIST] or press <INPUT> key. On the contrary, if you want to delete an existing tool, move the cursor to the number of that tool and press [DELETE].

LIFE:
The lives corresponding to the tool numbers on the tool management table are displayed.
You can set the life of each tool.
By pressing [GRPALL] after entering a value, you can set the same life for all the tools in the group.
In the first row for each group, the sum of the lives of the tools registered with that group is displayed.

REST LIFE:
The rest of the life of each tool, as determined from the tool management data table, is displayed.
Data can be set by entering values.
In the first row for each group, the sum of the rests of the lives of the tools registered with that group is displayed.

NOTICE LIFE:
The notice lives (the rest of life after which the notice signal turns on) corresponding to the tool numbers in the tool management data table are displayed.
Data can be set by entering values.
By pressing [GRPALL] after entering a value, you can set the same announced life for all the tools in the group.
In the first row for each group, the rest of life after which the life state of that group changes to “announced” is displayed.

STATE:
The life states (invalid, present, absent, in use, and skipped (tool damaged)) corresponding to the tool numbers in the tool management data table are displayed.
To select the desired one, press the corresponding soft key.

In the first row for each group, the life state of that group (not announced or announced) is displayed.

1.5.2 Changing Tool Priority

You can change the priority of the tools in a group.
The procedure for changing the priority is shown below.

(1) Position the cursor on the priority value in the first column for the desired tool and enter a new value.
(2) Press [ORDER], and the priority of the tool changes to the new value. Those tools that have priority values greater than the entered new value before the change are assigned their previous values plus one, respectively.

1.5.3 Updating Life Values Displayed on the Tool Life Data Screen

When the tool life data is changed with operating program, the tool life data is updated.

1.5.4 Group Number List Display

A list of the life states of groups can be displayed. Groups can be sorted in the order of number or life state.

Pressing [GRPLST] when the life management data screen is displayed displays the GROUP NO. LIST screen:

1.5.4.1 Display items

"PREVIOUS NOTICE"
"NOTICED" is displayed only for those groups with the Previous Notice Flag set.
"STATE"
When the life states of all tools belonging to a group are "NO-MNG", the life state of the group is "NO-MNG". In another group, the life state of the group not including a tool with a life state of "ENABLE" or "USING" is "OVER".

1.5.4.2 Display of groups sorted in the order of number or life state

When [S SORT] is pressed on the group number list screen, the group numbers are displayed in the life state/previous notice order.

NOTE
When group numbers are displayed in the order of the life state, the following priority is used.
(1) Groups for which "OVER" is displayed in the "STATE" column
(2) Groups for which "NOTICED" is displayed in the "PREVIOUS NOTICE" column
(3) Groups other than (1) and (2)
All the groups for which "NO-MNG" is displayed in the "STATE" column are displayed finally.

Pressing [N SORT] displays groups sorted by group number.

1.5.4.3 Group selection

Place the cursor over a group number to be selected, then press [SELECT]. The life management data screen of the selected group number appears.

1.5.4.4 Restore group life

Move the cursor to the group of which state is “OVER”, and press [G FILL]. And the life states of the tools, which belong to the group, change to “ENABLE”, and the rest life value becomes same as the life value.
The tool life state which is “NO-MNG” or “SKIP” is not updated when [G FILL] is pressed.

1.5.5 Setting the Life Notice Flag

To display the life state ("NOTICED" or "UN-NOTICE") of a group on the life management data screen, the "Previous Notice Flag" of tool management data needs to be set with the PMC.

The MANUAL GUIDE *i* system displays "NOTICED" as the state of a group when the "Previous Notice Flag" is set to the state described below.

If bit 3 (ETE) of parameter No. 13200 = 0
When the "Previous Notice Flag" of one of the tools that belong to a group is set to "NOTICED"
If bit 3 (ETE) of parameter No. 13200 = 1
When the "Previous Notice Flags" of all tools that belong to a group are set to "NOTICED"

1.6 TOOL LIFE DATA LIST SCREEN

The tool life state of all tools can be displayed on the tool life management data list screen.

1.6.1 Selecting the Life Management Data List Screen

(1) On the initial screen in each mode, press soft key [SETTING]. If soft key [SETTING] does not appear, press the rightmost soft key [>] to display it.
(2) Selecting [TOOL LIFE DATA LIST] on the [SETTINGS] menu selection screen that appears causes the life management data list screen to appear.

This item is displayed when the bit 5 (LIA) of parameter No.14823 is ’1’.

1.6.2 Life Management Data List Screen

On this screen, you can change the priority of the tools which are belong to the same group with cursor pointed tool. The action performed by pressing each soft key is the same as that on the conventional life management data screen.

(1) Selecting TOOL LIFE DATA LIST on the "BASIC" menu selection screen of SETTINGS causes the TOOL LIFE DATA LIST screen on which group numbers are displayed at the top.
(2) Pressing [GRPLST] displays the list of the life states of groups. Selecting a group number on the [GROUP NUMBER LIST] screen causes the life management data list screen of the selected group to appear.

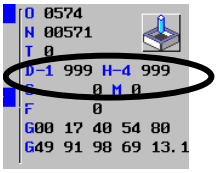
When the tool life data is changed with operating program, the tool life data is updated on the tool life data list screen.

1.7 MODAL DISPLAY OF OFFSET TYPES

When an offset type is used to specify a tool offset number, the offset type is displayed in the modal information display position while the offset type is enabled after the offset type is specified.

1.7.1 Screen Display Items

Screen that appears when a offset type is specified (on the milling machine)



If bit 7 (STS) of parameter No. 14823 is 1 and the offset type is displayed, the offset type is displayed after 'D-' and 'H-' in the status display section.

1.7.2 Displayed Offset Types (Set by the Machine Tool Builder)

In the status display section, offset types are displayed by referencing the following variables:

#90248, D code offset type on the milling machine
#90249, H code offset type on the milling machine

When specifying a tool offset number, the machine tool builder is required to set a offset type in variables #90248 and #90249 in the called macro program, using T, D, and H codes.
If directly specifying a offset number, instead of specifying a offset type, the machine tool builder is required to set the above variables to null.

1.8 DISPLAY TOOL MANAGEMENT DATA OF CNC STANDARD SCREEN

By pressing the soft key displayed on the MANUAL GUIDE *i* screen, it is possible to change the screen to the tool management data table on the NC side.
In order to use this feature, it is necessary to set the TLD(No.14823#6).
Follow the steps below.

(1) At the case of the parameter TLD (No.14823#6) setting '1', soft key [TL-MNG] is displayed on the base screen in the each mode. If soft key [TL-MNG] does not appear, press the rightmost soft key [>] to display it.
(2) Pressing soft key [TL-MNG] on this screen displays the NC tool management data table screen.

NOTE
Either “Magazine management table screen” or “Tool management data table screen” is displayed. The screen previously displayed is appeared.

(3) On this screen, if the function keys for startup MANUAL GUIDE *i* are pressed, the screen returns to the MANUAL GUIDE *i* screen.

1.9 OTHERS

1.9.1 Return to Menu Screen

Pressing soft key [TO MNU] on the tool management data table screen returns to the menu selection screen.
When the bit 2 (RET) of parameter No.14850 is '1', [CLOSE] is displayed instead of [TO MNU]. Pressing [CLOSE] returns to the base screen as before.

1.9.2 Inhibition of Editing Tool Management Data at CNC Standard Screen

On the tool management data screen of NC side, it is possible to inhabit to edit the tool management data.
In the case of the bit 7 (TME) of parameter No.14851 on, when [EDIT] is pressed on the tool management data screen, the following waning is displayed. And the tool management data cannot be altered at NC screen.

"WRITE PROTECTED"

1.9.3 Using Tool Management Data

The following customization data on the tool management data table are used by the tool management functions for MANUAL GUIDE *i*.
Therefore MTB can not use these customization items when the tool management functions for MANUAL GUIDE *i* is used.

Item	Bit	Content
Customization item 0	7	Previous Notice Flag
	6	
	5	
	4	
	3	
	2	
	1	
	0	
Customization item 1		TOOL NO.
Customization item 2		OFFSET NO.
Customization item 3		TOOL KIND
Customization item 4		TOOL USING ORDER

NOTE
In order to enable "Tool Using Order", it is necessary to set the bit 6 (SWC) of parameter No.13203 to 1 and parameter No.13260 to 4. When these parameters are 0, the shortest life tool is searched not according to this order.

1.10 WARNING MESSAGES

The following warnings are displayed about the tool management function.

- WRONG VALUE OF PARAMETER NO. 14824 :
  When parameter No.l4824 is set to a value other than 1 to 999.
- INVALID INPUT :
  When the input value of each input data item falls outside the valid range.
- TOOL NUMBER ALREADY EXISTS :
  Displayed if the same tool number as that entered has already been set.
- TOOL MANAGEMENT DATA ACCESS ERROR :
  When tool management data such as a tool number, offset number, and group number cannot be read or written.
- MAGAZINE MANAGEMENT DATA ACCESS ERROR :
  When a pot number in the magazine management data table or data number relative to the spindle/waiting position cannot be read or written normally.
- GROUP LIFE STATE IS NOT 'OVER'. :
  When [G FILL] is pressed on the tool life data screen, this warning is displayed if the group state of the current cursor position is not “OVER”.

2 Series 16*i*-MB 2-PATH FUNCTION

2.1 OVERVIEW

This function allows the 2-path machining center CNC to use the MANUAL GUIDE *i*. However, the MANUAL GUIDE *i* can be used only when path 1 is selected.

NOTE
1 The target CNC of this function is the FANUC Series 16*i*-MB 2CPU2-path system.
2 When using this function, cancel the multi-path lathe option.

2.2 DISPLAY SCREEN

The MANUAL GUIDE *i* screen appears only when the path 1 side is selected.
The display items are the same as in the case where it is used with the FANUC Series 16*i*-MB 1-path system.

Display screen on the path 2 side

(1) If switching to the path 2 side is performed when the MANUAL GUIDE *i* screen appears, the NC screen appears.
(2) If the MANUAL GUIDE *i* startup function key is pressed when path 2 is selected, the MANUAL GUIDE *i* screen does not appear.

3 PROGRAMMABLE DIAMETER/RADIUS SWITCHING FUNCTION

3.1 OVERVIEW

The Series 30*i*/31*i*/32*i* diameter/radius programmable changing function can be used to dynamically switch between the diameter and the radius.

NOTE
1 This function requires the NC diameter/radius dynamic switching option.
2 The function for dynamically switching between the diameter and the radius is not supported.

3.2 BASE SCREEN DISPLAY

When the specification method is changed by diameter/radius programmable switching, switching between diameter values and radius values is performed in the following display.

(1) Absolute coordinates on the base screen
(2) Relative coordinates on the base screen
(3) Machine coordinates on the base screen
(4) Remaining motion amount on the base screen
(5) Next block motion amount on the base screen

Even when switching between diameter values and radius values is performed, the diameter values/radius values according to bit 3 (DIAx) of parameter No. 1006 are used in the following display.

(a) Manual handle interrupt amount on base screen
(b) Offset display (including the offset measurement screen and C input screen)
(c) Workpiece coordinate system setting screen (including the measurement screen)
(d) Relative coordinate system preset screen

NOTE
1 To make the display for (1), (2), (4), and (5) support the diameter/radius switching state, set bit 2 (DPA) of parameter No. 3194 to 0.
2 To make the display for (3) support the diameter/radius switching state, set bit 3 (DPM) of parameter No. 3194 to 1.

3.3 MACHINING CYCLE

1. Machining cycle input data

Use of diameter/radius is always the same on the machining cycle data input screen regardless of the state of diameter/radius programmable switching.

(1) Turning cycle

Diameter values are used to input X axis values and radius values are used to input data of the other axes.

(2) Milling cycle

Radius values are used to input data of all axes.

2. Execution of a machining cycle

According to the diameter/radius programmable switching state, a movement command is output using diameter values during diameter specification or using radius values during radius specification.

3.4 NC STATEMENT CONVERSION

When the G4 digit cycle is subjected to NC statement conversion, switching between diameter values and radius values is performed for a specification value to be output according to the diameter/radius programmable switching state.

4 ALLOCATION SCREEN ON SMALL MDI UNIT (FOR Series 16*i*/18*i*/21*i*)

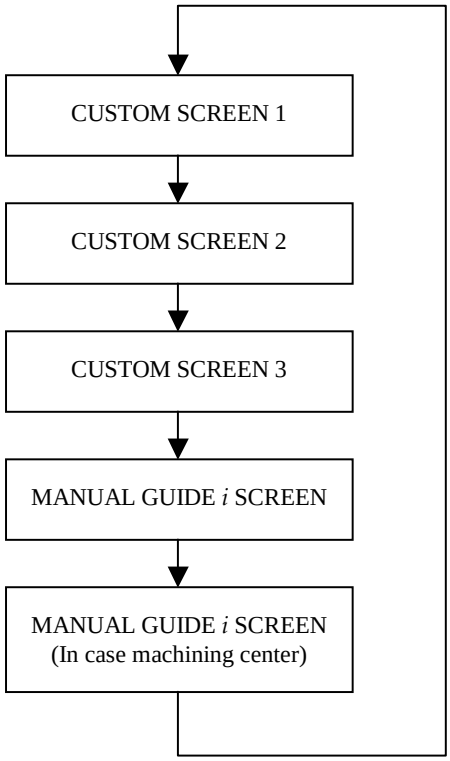
4.1 OVERVIEW

When the system meets the following all conditions, the screens of MANUAL GUIDE *i* and conversational macro can be allocated together to <CUSTOM/GRAPH> key.

(1) Small MDI unit is available.
(2) Conversational macro is available.
(3) MANUAL GUIDE *i* is available.
(4) The screen of MANUAL GUIDE *i* is allocated to GRAPH key.

4.2 SCREEN CHANGING

When the <CUSTOM/GRAPH> key is pressed, the screen is changed as the following order.



4.3 NOTES

Please take notice of the followings.

1) MANUAL GUIDE *i* Software BX00/BX01 (Series 16*i*/18*i*/21*i*-B) after version 21 is available.
2) During displaying a window of MANUAL GUIDE *i*, the screen will not be changed when the <CUSTOM/GRAPH> key is pressed.
3) In case of machining center, it is necessary that <CUSTOM/GRAPH> key is pressed two times in order to display the screen of conversational macro after the screen is changed from conversational macro to MANUAL GUIDE *i*.

5 USAGE IN 8.4" LCD (For only Series 30*i*/31*i*/32*i*-B, Series 0*i*-D)

MANUAL GUIDE *i* can be used in 8.4" LCD (7 soft-keys).

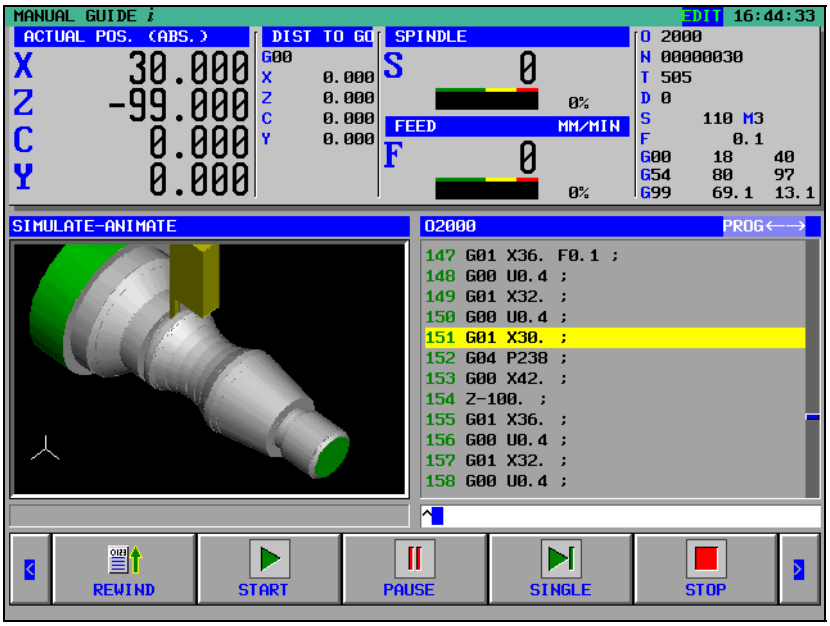
5.1 WORKING CONDITION

This function can be used only by the following conditions.

(1) Available display unit
8.4" LCD (7 soft-keys) without touch panel
(2) Necessary parameter setting
Please set the following parameters.
- No.14791#3 (SKY) = 1
- No.27481#4 (CSP) = 0
- No.3100#2 (SKY) = 1 (Series 0*i*-D or small MDI unit in Series 30*i*-B)

5.2 SCREEN DISPLAY

7 soft-keys (5 operation keys + 2 switch keys for soft-key page) are displayed.
The content displayed on screens other than the soft-key is the same as 10.4" LCD.



5.3 RESTRICTION

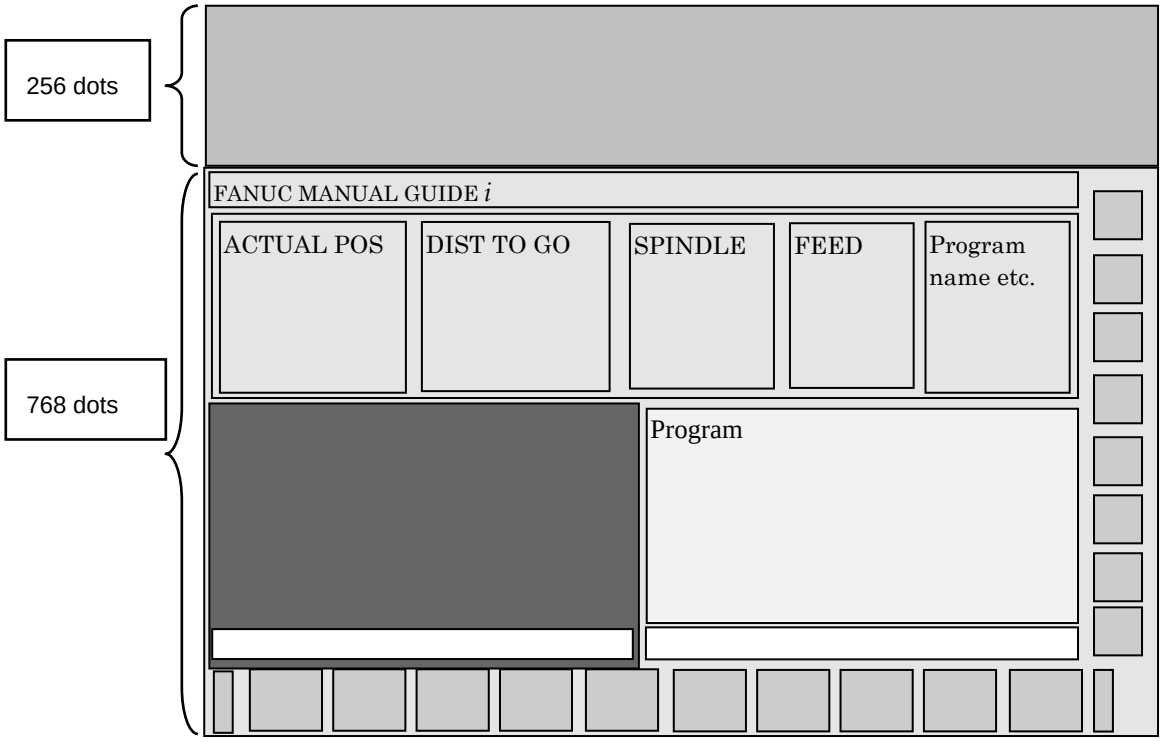
The following functions cannot be used.

(1) Customizing soft-key (for enhancing customizing and user customizing)
(2) Setting screen for vertical soft-key
Because there is no vertical soft-key in 8.4” LCD, the menu of vertical soft-key is not displayed on the setting screen.
(3) Help screen
Even if the HELP key is pressed, the help screen is not displayed.
(4) Navigation menu of FANUC standard
In the navigating function, the navigation menu of FANUC standard is not displayed.

6 19” DISPLAY UNIT (FOR Series 30*i*/31*i*/32*i* ONLY)

MANUAL GUIDE *i* is available on 19” display unit.
In case of using 19” display unit, MANUAL GUIDE *i* is able to be displayed as 19 inch display mode, 10.4 inch display mode or 15 inch display mode.
In case of 10.4 inch display mode or 15 inch display mode, CNC screen display function that has smaller window than display unit is displayed and MANUAL GUIDE *i* is displayed in it. The contents of screen are same as in the past.

In 19” display mode, the following screen is displayed.



The screen is divided to upper section sized 256 dots and lower section sized 768 dots. In upper section, user application made by machine tool builder is displayed and MANUAL GUIDE *i* is displayed in lower section. Refer to the manual provided by the machine tool builder about the detail of user application.

NOTE
1 Only PANEL i supports 19” display unit.
2 The option “19 inch display” is necessary to display the screen of 19 inch mode.
3 It is necessary to set parameter 13114#0=0 and #1=1 to display the screen of 19 inch mode.
4 The following software is necessary to display the screen of 19 inch mode.
MANUAL GUIDE *i* : System BY70 version 32 or later
CNC screen display function : version 3.3 or later

6.1 19 INCH MODE DISPLAY OF MANUAL GUIDE *i*

The screen of 19” display mode was improved from the screen of 15” mode about following points.

(1) Enlarging the width of the “MANUAL GUIDE *i*” title part
(2) Enlarging characters of distance to go and next distance (*)
(3) Enlarging characters of program name and sequence number in upper and right part of the screen
(4) Enlarging the width of graphic window
(5) Enlarging the width of program window
(6) Enlarging the width of current machining (*)
(7) Enlarging the width of key-in buffer
(8) Enlarging the width of the message area
(9) Enlarging the width of the cell in the process list

(*) Information of MDI / MEM / HND / JOG / TJOG / THND / INC / REF / RMT mode

NOTE
1 When the CNC displays 7 axes or more, the characters of distance to go and next distance are not enlarged.
2 The characters of next distance are not enlarged during machining drawing.
3 The characters of distance to go, next distance, program name and sequence number are not enlarged during concurrent display and edit of programs of all paths.
4 When the characters of program name are 13 or more, they not enlarged.
5 The size of pop up windows is same as past.

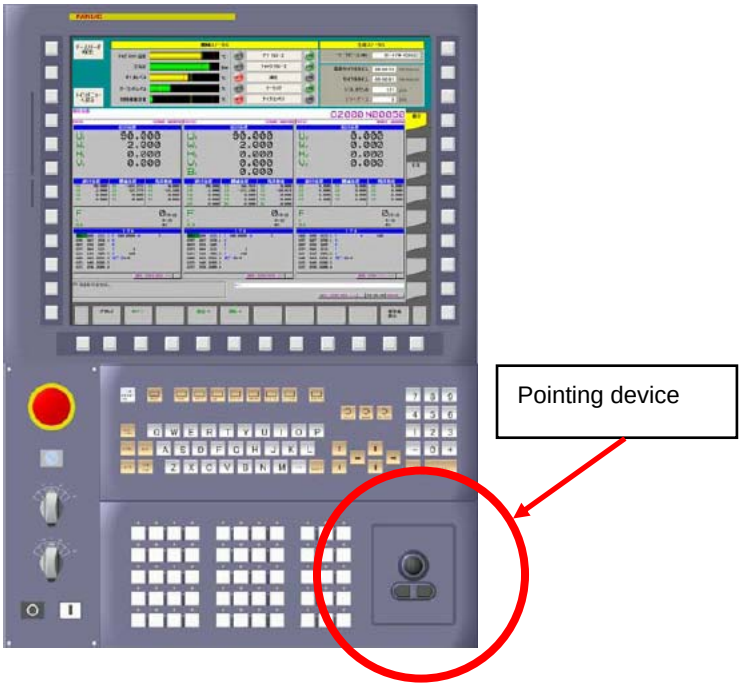
6.2 APPLICATION SELECTION TO WORK MDI KEYS AND SOFT-KEYS

When MDI key or soft-key is pushed, only being selected application can react the input.
Thus, MANUAL GUIDE *i* must be selected when the operator want to operate MANUAL GUIDE *i*. And user application must be selected when the operator want to operate the user application.
It is possible to select either by doing one of following three ways.

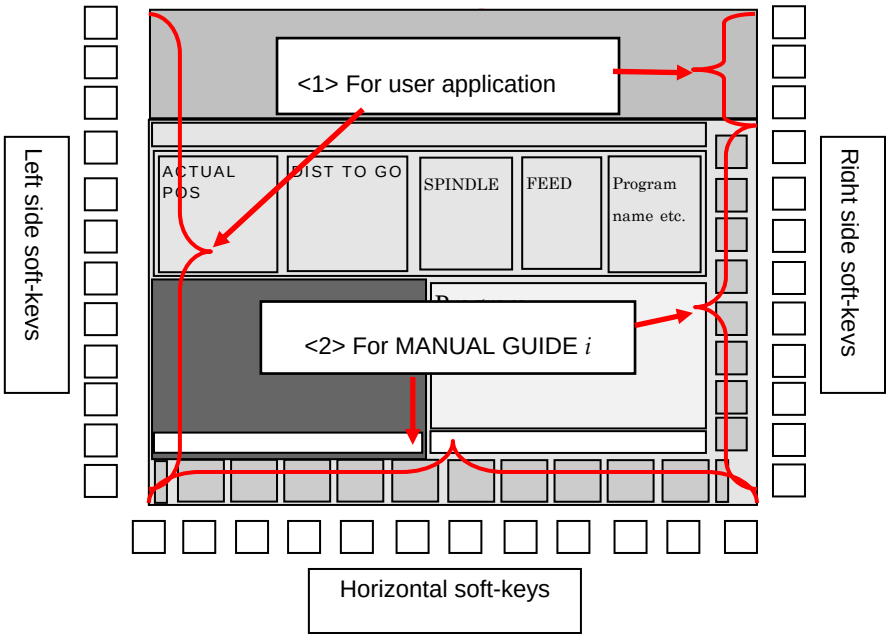
(1) Selecting by pointing device (Selecting by left button click)
(2) Selecting by “KEY SERVER” that is optional software offered by FANUC (Selecting by soft-keys)
(3) Selecting by user application

The details of each way are followings.

(1) Application selection by pointing device
After a pointer (arrow) used in personal computer is moved on the window of either application by pointing device, if left button is clicked, the application is selected.



(2) Application selection by key server
"KEY SERVER" is a optional tool software that controls selecting application and it can be used in only 19" display unit. "KEY SERVER" makes user application selected if the soft-keys of <1> are pushed. And it makes MANUAL GUIDE *i* selected if the soft-keys of <2> are pushed.



(3) Application selection by user application
User application can switch selection freely.

NOTE
1 It is not indicated which MANUAL GUIDE *i* is being selected or not.
2 MANUAL GUIDE *i* can't be controlled while user application is selected.

6.3 PARAMETER

	#7	#6	#5	#4	#3	#2	#1	#0
13114							P19	P15

P15 When the screen is displayed using the CNC screen display function, the 15 inch mode is
0: Not used.
1: Used.

P19 When the screen is displayed using the CNC screen display function, the 19 inch mode is
0: Not used.
1: Used.

P15	P19	Screen mode
0	0	10.4”
1	0	15”
0	1	19”

7 USE WITH PERIPHERAL AXIS CONTROL (for only Series 30*i*/31*i*/32*i*-B)

7.1 OUTLINE

MANUAL GUIDE *i* can be used together with peripheral axis control.

7.2 CONDITION

It is possible to use this function on the following conditions.

(1) Necessary option
Peripheral axis control: R725

(2) Necessary parameter setting
Please set parameter CSP(No.27480#0) to 1.

(Method of NC program conversion)

CSP	Subprogram calling	Commands in subprogram	Custom macro	Finishing cycle (G70)	Commands of peripheral axis control
0	×	○	×	×(*)	×
1	○	×	○	○	○

○ : Output
× : No output
(*) Figure commands to be executed G70 are output.

(3) Necessary setting of peripheral axis
Setting of peripheral axis meets the following conditions.
- Peripheral axis is the linear axis.
- Peripheral axis is any of A-axis, B-axis, C-axis, E-axis, U-axis, V-axis or W-axis.

7.3 MACHINING SIMULATION

Commands of peripheral axis control are ignored.

7.4 RESTRICTION

(1) Set-up guidance function cannot be used.
(2) Lathe/Machining center G code system switching function cannot be used.
(3) Machining cycles cannot be used in peripheral axis control program.

(Example)

```
O1234 ;
  :
G101 ;      ← Start of registration of peripheral axis control
×G1001 W1. I1. J2. L-10. K0. C2. F10. Z2. ;  } Can not command
×G1210 B0. H100. V100. ;                     }
G100 ;      ← End of registration of peripheral axis control
  :
M100 ;
  :
```

8 WARNING MESSAGE DISPLAY WHEN RESET KEY IS PUSHED (for only Series 30*i*/31*i*/32*i*-B, 31*i*-B5)

When address G, F, H, D, T, S, M, and B(Secondary Auxiliary Function) are changed by pushing the key <RESET> during automatic operation, the warning of "MODAL DATA IS CHANGED BY BLOCK STOP" can be displayed on the message display part of MANUAL GUIDE *i* screen.
Warning "MODAL DATA IS CHANGED BY BLOCK STOP" is cleared by the following operations.

(1) Key operation
(2) Reset

8.1 NECESSARY SOFTWARE

CNC software (Series 30*i*/31*i*/32*i*-B, 31*i*-B5)
Series：G301/G311/G401/G411/G421/G431/G501/G511、edition: 37 or later

8.2 NECESSARY PARAMETER

Please set parameter WMD(No.10334#1) to 1.

8.3 RESTRICTIONS

(1) If the CNC is reset by reset and rewind signal RRW<Gn008.6>, this warning is not displayed.
(2) Even if this warning is cleared on MANUAL GUIDE *i* screen, this warning is not cleared on CNC screen.
(3) Even if CNC is reset in waiting of the balance cutting, this warning is not displayed because waiting of the balance cutting is not the middle of a block.
(4) The display and cancellation of warning doesn't depend on the path.
   (i) Even if the operator selects the path of which modal data has not changed, this warning is displayed in the case that modal data is changed by resetting of MDI key in other path.
   (ii) Even if the operator selects the path of which this warning is not issued, this warning is cleared by key operation and resetting.

9 DISABLING MANUAL GUIDE *i*

This function can disable all functions of MANUAL GUIDE *i* .
Please set parameter No.14836#7 to 1.

* This function perform the same operation as in the case where the option of MANUAL GUIDE *i* is disabled.

10 EXTENDED AXIS NAME FUNCTION

The extended axis name can be used on MANUAL GUIDE *i* screen.

10.1 CONDITIONS TO USE THIS FUNCTION

This function requires the following all conditions.

(1) The option “MANUAL GUIDE *i* extended axis name function” is valid.
(2) Extended axis name is applied for linear axis other than basic three axis or rotary axis of workpiece rotation.
(3) Extended axis name is composed 2 characters. (Ex. B2, CS)
(4) Set-up guidance function is invalid.

10.2 AXIS NAME DISPLAY

Extended axis name can be displayed in the following screens.

(1) Base screen
(2) Relative coordinates presetting screen
(3) Workpiece coordinate data screen
(4) Machining simulation screen

NOTE
1 In “CALCULATE OFFSET(MEASURE)” screen of workpiece coordinate data screen, extended axis name is not displayed.
2 In “CALCULATE OFFSET(+INPUT)” screen of workpiece coordinate data screen, extended axis name is not displayed.

10.3 PROGRAM DISPLAY AND EDITING FUNCTION

A program included the extended axis name can be displayed and edited.
And when the command of “axis name” + ”=number” (ex. C2=100.) is edited, the command is regarded as a word.

10.4 MACHINING CYCLE

When the name of workpiece rotation axis is extended, the following MANUAL GUIDE *i* machining cycles can be used.

(Available machining cycles)

Type of Machining cycle	Available figure	
Hole machining	C axis hole on face (arc points)	G1572
	C axis hole on face (random points)	G1573
	C axis hole on cylinder (arc points)	G1672
	C axis hole on cylinder (random points)	G1673
Facing	Polar coordinate plane Square fixed form figure	G1520
	Polar coordinate plane Circle fixed form figure	G1521
	Polar coordinate plane Track fixed form figure	G1522
	Polar coordinate plane Polygon fixed form figure	G1525
	Polar coordinate plane Arbitrary figure	G1500 - G1506
	Cylindrical Surface Arbitrary figure	G1600 - G1606
Contouring	Polar coordinate plane Square fixed form figure	G1520
	Polar coordinate plane Circle fixed form figure	G1521
	Polar coordinate plane Track fixed form figure	G1522
	Polar coordinate plane Polygon fixed form figure	G1525
	Polar coordinate plane Arbitrary figure	G1500 - G1506
	Cylindrical Surface Arbitrary figure	G1600 - G1606
Emboss machining	Polar coordinate plane Arbitrary figure	G1500 - G1506
	Cylindrical Surface Arbitrary figure	G1600 - G1606
Pocketing	Polar coordinate plane Square fixed form figure	G1520
	Polar coordinate plane Circle fixed form figure	G1521
	Polar coordinate plane Track fixed form figure	G1522
	Polar coordinate plane Polygon fixed form figure	G1525
	Polar coordinate plane Arbitrary figure	G1500 - G1506
	Cylindrical Surface Arbitrary figure	G1600 - G1606
Grooving	Polar coordinate plane Square fixed form figure	G1520
	Polar coordinate plane Circle fixed form figure	G1521
	Polar coordinate plane Track fixed form figure	G1522
	Radial groove on the polar coordinate plane	G1523
	Polar coordinate plane Polygon fixed form figure	G1525
	Polar coordinate plane Arbitrary figure	G1500 - G1506
	Cylindrical Surface Arbitrary figure	G1600 - G1606
	C-axis groove on the polar coordinate plane	G1570
	X-axis groove on the polar coordinate plane	G1571
	C-axis groove on the cylindrical surface	G1670
	X-axis groove on the cylindrical surface	G1671
Engraving (*)	Nothing	

(*) When XC plane or ZC plane is selected

10.5 SELECTING ROTATION AXIS IN FIGURE DATA INPUT SCREEN OF MACHINING CYCLE

Workpiece rotation axis that works while executing machining cycle can be selected in “ROT.AXIS” tab of figure data input screen by pressing soft-key [C] or [A].

(1) In case soft-key [C] is pressed
The axis whose axis number is lower 2 digits of parameter No.27453 works.
(2) In case soft-key [A] is pressed
The axis whose axis number is upper 2 digits of parameter No.27453 works.

NOTE
1 Set the bit 0 of parameter No.27001 to 1 to display “ROT.AXIS” tab.
2 Soft-key [C] or [A] might be changed to actual name of workpiece rotation axis.
(Ex. [C]→[C1], [A]→[C2])

10.6 METHOD OF CHANGE TO EXTENDED AXIS NAME IN FIGURE DATA INPUT SCREEN OF MACHINING CYCLE

When the bit 0 and bit 1 of parameter No.27001 are set to 1, rotation axis name is displayed in the “ROT.AXIS” tab of figure data input screen of machining cycle. It is possible to change the name to extended axis name by the following method.

(Method)
(1) Change of soft-key name
Change the messages of number 322 and 323 in customization definition file dlgmsg*u.txt.
(Ex.)
322,C,Y1 → 322,C1,Y1
323,A,Y2 → 323,C2,Y2

(2) Change of rotation axis name selected in the tab “ROT.AXIS” of the window “START POINT - INSERT”
Change the messages of number 55 and 56 in customization definition file keymsg*u.txt.
(Ex.)
55, C → 55, C1
56, A → 56, C2

10.7 PARAMETER SETTING FOR EXTENDING AXIS NAME OF WORKPIECE ROTATION

Set the following parameters when the axis name of workpiece rotation is extended.

Parameter 27453 = Axis number of rotation axis
Parameter 27001#0, #1 = 1
Parameter 27001#4 = 0

11 DYNAMIC AXIS NAME CHANGING (For only Series 30*i*/31*i*/32*i*-B)

The display of axis name can be dynamically changed by changing the axis name (parameter No.1020 and No.1022) with the programmable parameter input (G10L52 command). When this function is used, please set the parameter No.14791#1 to 1.
The following functions are available.

(1) The axis name can be dynamically changed in the following screen and the window of MANUAL GUIDE *i*.
  (a) Base screen
  (b) Machining simulation screen (animation drawing / tool path drawing)
  (c) Drawing during machining screen
  (d) NC program conversion screen
  (e) Workpiece coordinate data window
  (f) Relative coordinates presetting window
(2) The machining cycle of MANUAL GUIDE *i* can be executed in the axis composition changed by the programmable parameter input (G10L52 command).

11.1 WORKING CONDITION

This function can be used only by the following conditions.

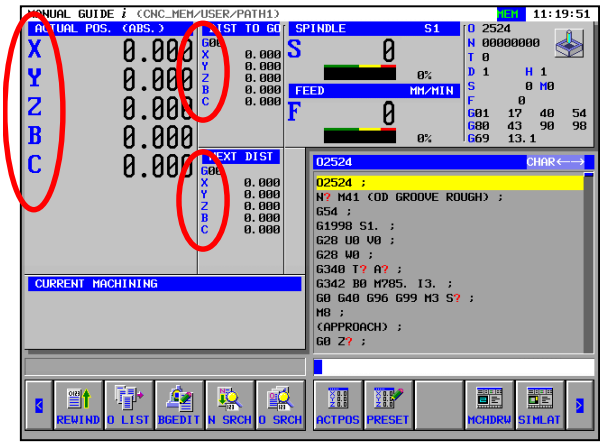
(1) Necessary optional function
Lathe/Machining center G code system switching function (R597)

NOTE
1 The axis name that can be dynamically changed is only an alphabetic character of one character such as "X", "Y", and "Z".
2 The enhancing axis name cannot be dynamically changed.

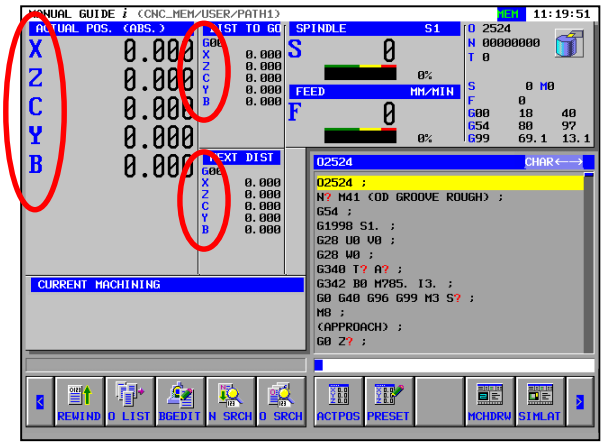
11.2 BASE SCREEN

The display of the following axis names of the base screen can be dynamically changed according to the change in the parameter No.1020 and No.1022.

Axis names
Axis name of actual position (Absolute coordinate, Relative coordinate, Machine coordinate, Manual handle interruption(input unit), Manual handle interruption (output unit))
Axis name of distance to go
Axis name of next distance
Axis name of feed (Load meter display of each axis for 15” LCD and 19” LCD)



Base screen (Milling mode)



Base screen (Turning mode)

11.3 MACHINING SIMULATION SCREEN

When this function is used, please set the parameter No.27482 # 2 and # 1 to 1.

Machining simulation is done only during Milling mode (#3940=1) of Lathe/Machining center G code system switching function, and Machining simulation is not done for Turning mode (#3940=0).

11.4 DRAWING DURING MACHINING SCREEN

When this function is used, please set No.27482 # 1 to 1.

Drawing during machining is done only during Milling mode (#3940=1) of Lathe/Machining center G code system switching function, and Drawing during machining is not done for Turning mode (#3940=0).

11.5 NC PROGRAM CONVERSION SCREEN

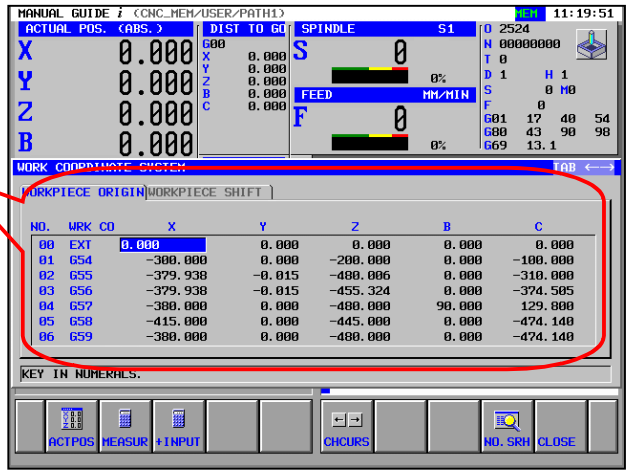
When this function is used, please set the parameter No.27482 # 2 to 1.

11.6 WORKPIECE COORDINATE DATA WINDOW

The display of the following axis names of the workpiece coordinate data window can be dynamically changed according to the change in the parameter No.1020 and No.1022.
It follows the axis name when the window opens.

Axis names
Axis name of workpiece coordinate data window
Axis name of calculate offset (measure) window
Axis name of calculate offset (+input) window

Workpiece coordinate data window



Base screen (Milling mode)

When it meets all following conditions (1) and (2), the data cannot be input during operation (automatic operation start-up state, feed hold state, automatic operation stop state, tool retract & recover operation, repositioning or manual numerical command operation).

< Condition >

(1) Setting that assumes dynamic change of axis name to be effective (parameter No.14791#1=1)

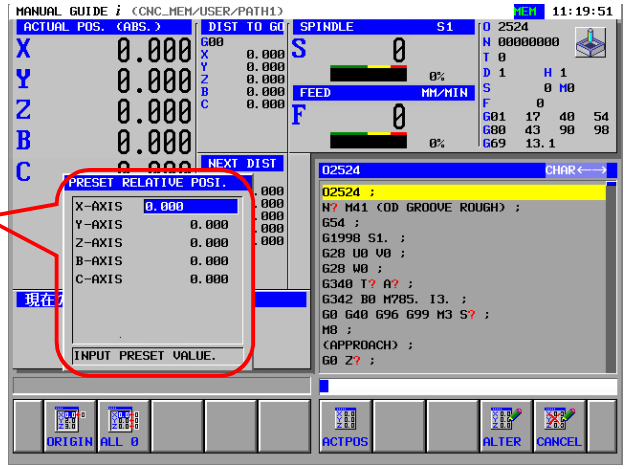
(2) Setting that prohibits data entry on the workpiece origin offset setting screen during operation (parameter No.14704#1=1)

Moreover, it is not possible to input the data after operation end until the window is displayed again.

11.7 RELATIVE COORDINATES PRESETTING WINDOW

The display of the axis names of the relative coordinates presetting window can be dynamically changed according to the change in the parameter No.1020 and No.1022.
It follows the axis name when the window opens.

Relative coordinates presetting window



Base screen (Milling mode)

When it meets the following conditions, presetting cannot be done during operation (automatic operation start-up state, feed hold state, automatic operation stop state, tool retract & recover operation, repositioning or manual numerical command operation).

< Condition >

(1) Setting that assumes dynamic change of axis name to be effective (parameter No.14791#1=1)

Moreover, presetting cannot be done after operation end until the window is displayed again.

12 HIGH SPEED PROGRAM MANAGEMENT (For only Series30*i*/31*i*/32*i*-A, B)

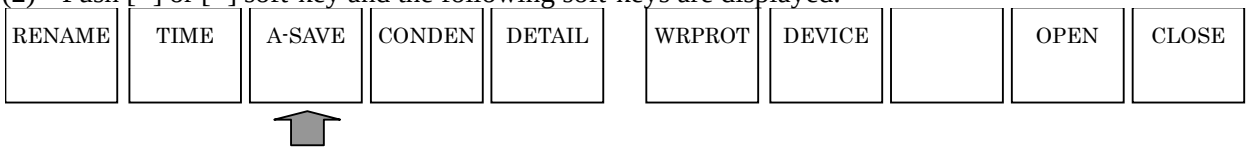
In case high speed program management is available, programs can be saved on MANUAL GUIDE *i* screen.

NOTE
1 Refer to A-92025 ”High speed program management” for details of high speed program management.
2 Set bit 7 of parameter No.11354 to 1 to use this function.
3 If this function is available, registering program speed or deleting program speed is faster. But the alterations of programs are canceled if programs are not saved.

12.1 OPERATION OF SAVING PROGRAMS

Programs can be saved by the following operation on EDIT mode.

(1) Push [O LIST] soft-key and the program list is displayed.
(2) Push [<] or [>] soft-key and the following soft-keys are displayed.



(3) Push [A-SAVE] soft-key and message “SAVE PROGRAMS?” is displayed.
(4) Push [YES] soft-key and programs are saved.

12.2 STATUS DISPLAY

In the following case, the status is displayed on the left lower of MANUAL GUIDE *i* screen.

1. The case while saving programs
   The status “ALLSAVE” is displayed.

2. The case that programs haven’t saved yet after altering programs
   The status “NOTSAVE” is displayed.

NOTE
Set bit 5 of parameter No.11370 to 1 to display “NOTSAVE”.

APPENDIX

A EXAMPLES OF CREATING PROGRAMS FOR THE MACHINING CENTER

> ⚠ **WARNING**
> All data described in this Part such as parameter, offset data and part program cannot be used for actual machining. Actual data varies from one machine model to another. Refer to the applicable manual supplied by the respective machine tool builders for details.
> If the set data does not match the characteristic of a specific machine, the tool may bump against the workpiece, and the machine may be forced to perform unnatural machining, possibly causing damage to the tool and/or machine, and even injuries.

The part program made by using MANUAL GUIDE *i* has a form of ISO-code program with G-code and so on.
You must enter the program used for such as tool changing, tool offset, spindle rotation, approaching and releasing in the form of ISO-code program.
In addition to those actions, you can enter the part program for complicated machining motions, which are usually difficult to make by ISO-code form, as a “Machining cycle” by using menu programming method. This machining cycle is made in form of a block including G-4digits and necessary data items.

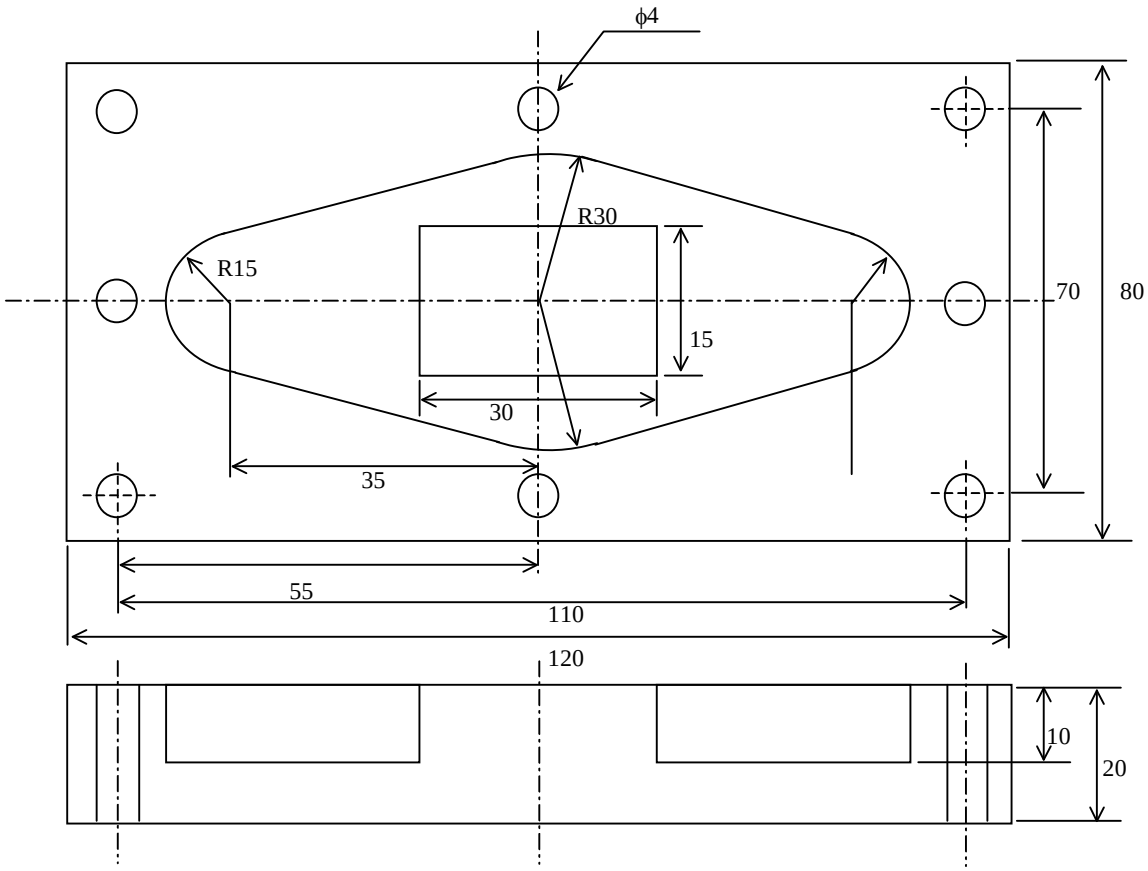
In the following explanations, the contents of a square frame mean the actual operations, and each operation are described as follows.

[NEWPRG]	: Push a soft key
12345	: Enter numeric data
INPUT	: Push an <INPUT> key
↓ ↑ → ←	: Push a cursor move key
⇓ ⇑	: Push a page change key
(CREATE NEW PROG)	: Name of Window or Data item
<START>	: Name of Tab
<<1.CYLINDER>>	: Menu item

Example) Outer wall contouring, Pocketing, Drilling

Workpiece : 90×130×30

1st Process : Outer wall contouring by Flat end mill (T01)
2nd Process : Pocket roughing by Flat end mill (T01)
3rd Process : Pocket finishing by Flat end mill (T02) (side, bottom)
4th Process : Drilling by Drill tool (T03)



A.1 SETTING OF TOOL LENGTH OFFSET DATA

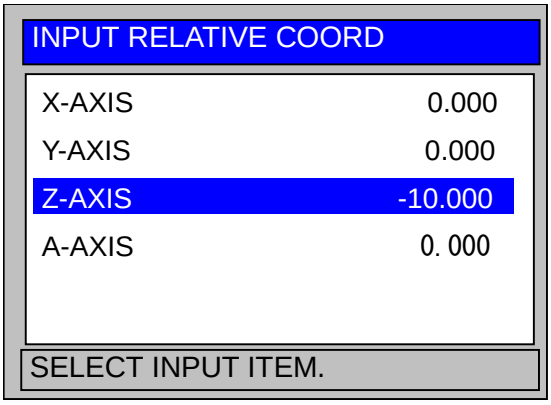
⚠ WARNING

1 Operation of tool offset setting varies from one machine model to another. So operations described in this section may differ from those ion actual machine. As to the actual operation of tool offset setting on the actual machine, refer to the applicable manual supplied by the respective machine tool builders for details. If the set data does not match the characteristic of a specific machine, the tool may bump against the workpiece, and the machine may be forced to perform unnatural machining, possibly causing damage to the tool and/or machine, and even injuries.

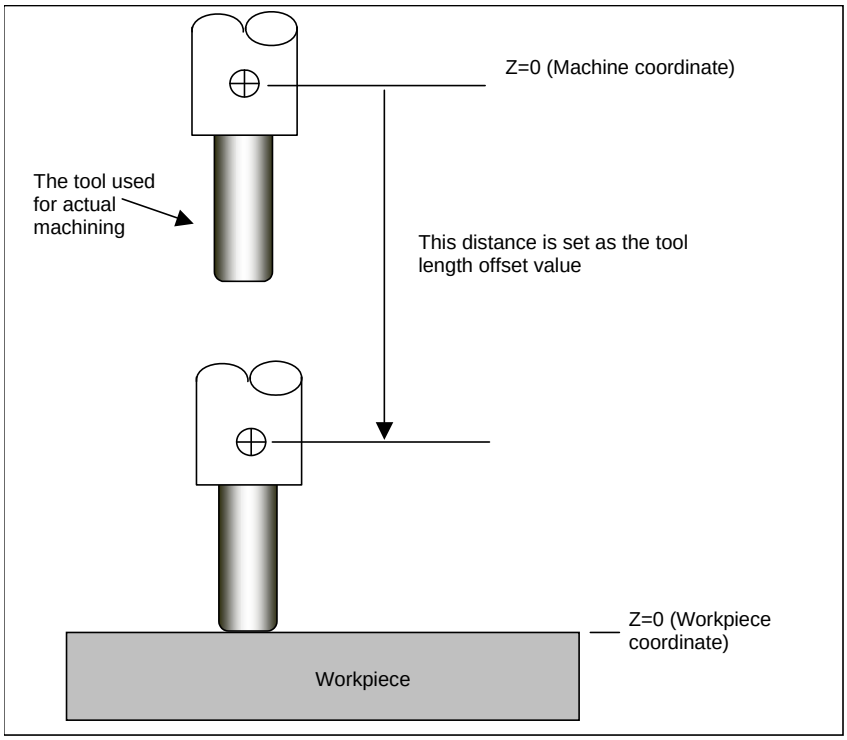
2 As to the operations on a machine described in this chapter, refer to the applicable manual supplied by the respective machine tool builders for details. If the operation does not match the characteristic of a specific machine, the tool may bump against the workpiece, and the machine may be forced to perform unnatural machining, possibly causing damage to the tool and/or machine, and even injuries.

Assume that the Z=0 position of the workpiece coordinate is the surface of the workpiece, while the Z=0 position of the machine coordinate is the machine origin.
By setting the distance between these 2 points to the tool length offset data, workpiece coordinate of Z-axis can be defined.
Tool length is different between each tool for the actual machining, so set the offset data respectively.
The tool length offset value is the relative coordinate value of Z-axis at the point where the tool tip is touched to the surface of the workpiece.

<1> Select the tool used for actual machining by switch on the machine operator's panel.
<2> Execute the machine reference position return of Z-axis.
<3> Press soft key [ACTPOS] and make the relative coordinate data display.
<4> Press soft key [PRESET], then press soft key [ALL 0], and press soft key [ALTER], then the relative coordinate value of all axes are reset to 0. But, the Z-axis coordinate only is used for measuring.
<5> Press soft key [T-OFS] and make the tool offset widow open.
<6> Make the tool tip touch at the surface of the workpiece by JOG or Manual handwheel. Then, the distance from the machine origin is displayed as the Z-axis relative coordinate value.
<7> Select the tool length compensation column of the tool used for measuring by moving the cursor by cursor key.
<8> Pressing soft key [INP.C.] displays the window of INPUT RELATIVE COORD. window, then move the cursor to the Z-axis.



<9> Pressing soft key [INPUT] makes the Z-axis relative coordinate value enter as the tool offset length data.

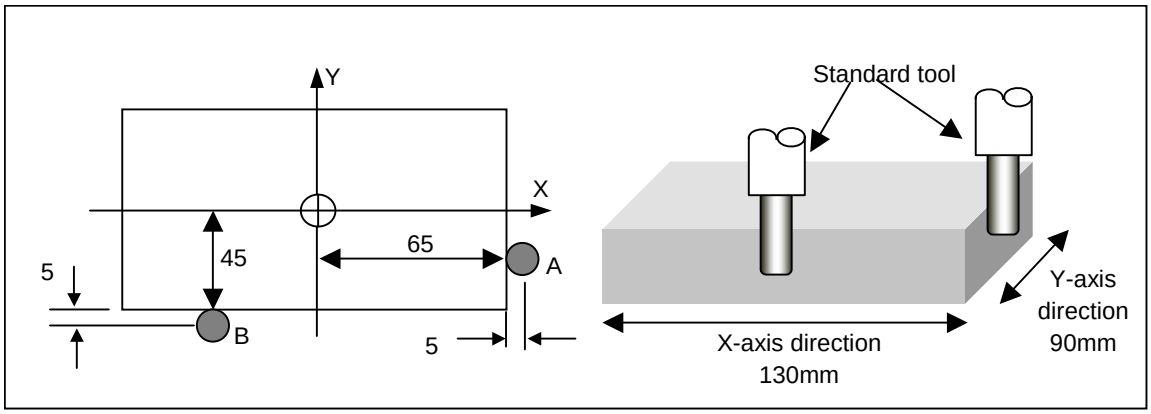


NOTE
There is a case such as using a sensor called base master instead of touching the tool to the workpiece surface, so into details and actual operations, refer to the manual made by machine tool builder.

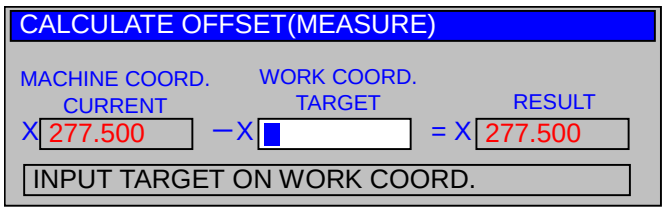
A.2 SETTING OF WORKPIECE ORIGIN OFFSET VALUE

After setting the geometry offset data for necessary tools, set the workpiece origin on the actual workpiece used for machining.
In order to carry out the actual machining by using the part program made on the workpiece coordinate, set the distance between machine coordinate and workpiece coordinate to the workpiece coordinate system as the workpiece origin offset value.
In this section, operations, of which setting the center of the workpiece (90×130) as the workpiece origin of X/Y/Z-axis by using standard too with the radius 5mm, are described.
And, G54 is used as the workpiece coordinate system.

<1> Select the standard tool by using the switch on the machine operator's panel. There is a case of using special tool such as the centering lever as the standard tool for easy operations and high precision accuracy, so refer to the manual made by machine tool builder for actual operations.
<2> After then, for safety, take measure to keep fully safety such as closing the machine door.
<3> Select the manual handwheel mode on the machine operator's panel, and make the standard tool touch to the right side of the workpiece (A-position in the drawing) by referencing the following drawing. At this time, make a spindle rotate by fully safety speed if necessary, then you can avoid over-cutting by stopping the tool moving at the position the tool begins cutting. As to actual operations, refer to the manual made by the machine tool builder.

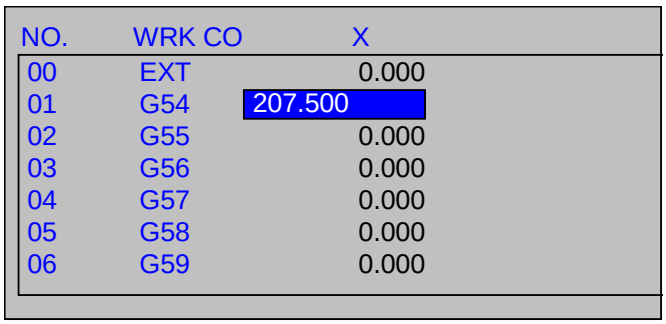


<4> Pressing soft key [WK SET] displays the window of the workpiece coordinate system.
<5> Select the G54 X-axis data by moving the cursor. When other tab is displayed in the window, the cursor motion type might be the tab transition type, so press soft key [CHCURS] for changing to item transition type of cursor.
<6> Pressing soft key [MEASUR] displays the window of offset calculating.



<7> When the tool touches to the right side of the workpiece, the X-axis position should be X=70.0mm, 65mm of the right side position + 5mm of the tool radius, so enter 70.0 to the target workpiece coordinate value. Then, the calculation result value is displayed at the RESULT column, so confirm the result is correct.

<8> Pressing soft key [INPUT] sets the above calculated value to the workpiece origin offset data.



NO.	WRK CO	X
00	EXT	0.000
01	G54	207.500
02	G55	0.000
03	G56	0.000
04	G57	0.000
05	G58	0.000
06	G59	0.000

<9> Release the tool from the workpiece once, and set the Y-axis workpiece origin offset data by similar operations of <3> to <8>. In this case, touch the point B in the drawing and the entering target value is Y=-50.0mm.

A.3 PREPARING OF THE FIXED FORM SENTENCE MENU

As to the fixed form sentence, machine tool builder usually sets the suitable menu for specified respective machine. But, you can enter his own menu on the MANUAL GUIDE *i* screen by yourself. The procedure is as follows.

Enter the fixed form sentence menu which will be called by the soft key [FIXFRM] in the soft key group for milling.

Enter programs for START and END.

[SETTING]
(SETTING)
<BASIC>
↓
<<1. REGISTER FIXED FORM SENTENCE FOR MILLING>>
[SELECT]
(REGISTER FIXED FORM SENTENCE FOR MILLING)
<FORM 1>
[NEW]
(CREATE NEW FIXED FORM SENTENCE)
<REGISTERED NAME : >
MILLING START **INPUT**
<REGISTERED SENTENCE : >
G28 G91 Z0. ; G28 X0. Y0. ; T? M06 ; D? ; M03 S? ; G00 G90 G54 X0. Y0.; G43 Z100. H? ;
[ADD]
→
<FORM5>
[NEW]
(CREATE NEW FIXED FORM SENTENCE)
<REGISTERED NAME : >
PROGRAM END **INPUT**
<REGISTERED SENTENCE : >
M05. ; G00 G90 Z100. ; G28 G91 Z0, ; M06 T0 ;M30 ;
[ADD]
[TO MNU]

A.4 SETTING OF THE TOOL DATA

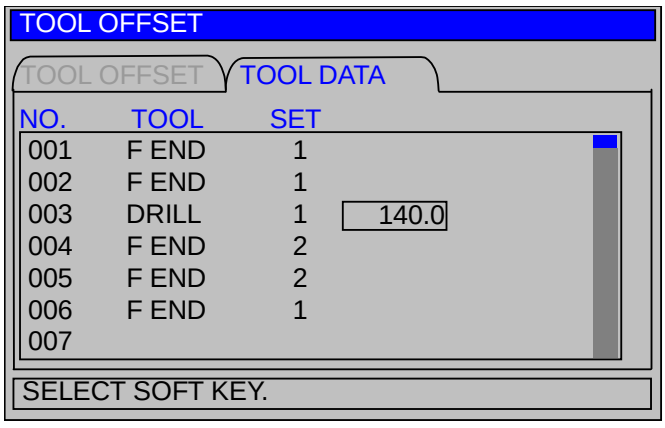
Set the necessary tool data. These tool data are used for displaying tool form of animation and calculation of cutting angle in the machining cycle.
The tool length offset data were already set in Appendix A.1, “SETTING OF TOOL LENGTH OFFSET DATA”.
T01 : Flat end mill for roughing
T02 : Flat end mill for finishing
T03 : Drill
Display “TOOL OFFSET” window by [T-OFS]

TOOL OFFSET				
NO.	TOOL LENGTH. COMP.		CUTTER COMPENSATOIN	
	GEOMETRY	WEAR	GEOMETRY	WEAR
001	Measured value	0.000	4.000	0.000
002	Measured value	0.000	3.000	0.000
003	Measured value	0.000	2.000	0.000

NOTE
1 The above offset data are just an example, and they cannot be used for actual machining.
2 The above offset number 003 is used for drilling tool, and if you use it for animation, you must set the radius amount of the drill to radius offset data.
3 There is a case that the above offset screen might not be displayed depending on the attached optional function. Into details, refer to the manual made by machine tool builder.

Display “TOOL DATA” tab by cursor key <→>

TOOL DATA				
NO.	TOOL	SET		
001	F END	1		
002	F END	1		
003	DRILL	1	140.0	



A.5 CREATING OF PART PROGRAM

In the MANUAL GUIDE *i* , background editing can be used, but in this section, part program creating operations are described by using foreground editing on the EDIT mode.

A.5.1 Creating New Part Program

Create a new part program of O1234.

(1) In case of creating a new part program directly

Select EDIT mode by using a mode-selecting switch on a machine-operating panel
[NEWPRG]
(CREATE NEW PROGRAM)
1234 **[CREATE]** (NEW PROGRAM NO.)

(2) In case of creating a new part program on the program list window

Select EDIT mode by using a mode-selecting switch on a machine-operating panel
[O-LIST]
(OPEN PROGRAM)
[NEW]
(CREATE NEW PROGRAM)
1234 **[CREATE]** (NEW PROGRAM NO.)
Select the newly entered program by ↓
[OPEN]

A.5.2 Operations of “START” Menu

By pressing soft key [START] in the soft key menu for milling, the window “INSERT STARTING COMMAND FOR MILLING” with the following tabs is displayed.

<START> : Fixed form sentence menu used for the top of part program or each machining process.
<COORD CONV> : Coordinate conversion menu, which is necessary for the machine on the inclined surface, this is not used in this section.
<BLANK> : Blank form menu, which is necessary for animation.

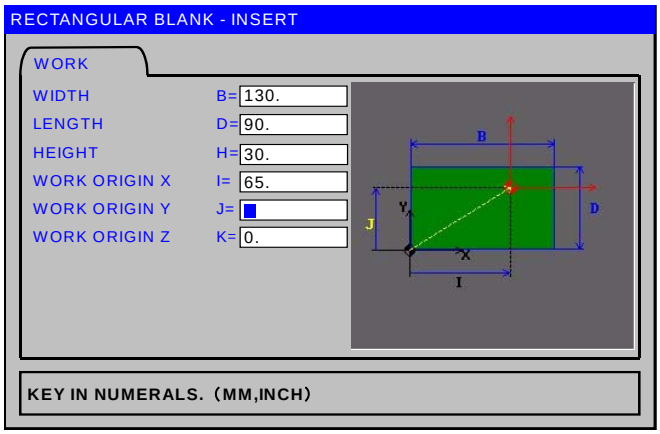
A.5.2.1 Entering blank form data

By selecting a tab <BLANK> by cursor key, blank form menu is displayed.

(INSERT STARTING COMMAND FOR MILLING)
<BLANK>
↓
<<1. RECTANGULAR BLANK FIGURE>>
[SELECT]

Enter blank form data as a round bar.

(RECTANGULAR BLANK FIGURE - INSERT)
130 **INPUT** (WIDTH)
90 **INPUT** (LENGTH)
30 **INPUT** (HEIGHT)
65 **INPUT** (WORK ORIGIN X)
45 **INPUT** (WORK ORIGIN Y)
0 **INPUT** (WORK PROGIN Z)
[INSERT]



A.5.3 Entering Tool Changing and Spindle Rotating Blocks for Roughing Flat End Mill

A.5.3.1 Entering in ISO-code form directly

It is difficult to define the action of tool changing, spindle rotation, approaching and releasing generally because there are many difference depending on machine configuration. So, using ISO-code form program can realize more flexible and safety part program.

G28 G91 Z0. ; **INSERT**	(Z-axis reference position return)
G28 G91 X0. Y0. ; **INSERT**	(X/Y-axis reference position return)
T01 M06 ; **INSERT**	(Tool change)
D1 ; **INSERT**	(Selecting tool data)
M03 S1000 ; **INSERT**	(Spindle rotation)
G00 G90 G54 X0. Y0. ; **INSERT**	(X/Y-axis approaching)
G43 Z100. H1 ; **INSERT**	(Z-axis approaching)

A.5.3.2 Entering by fixed form sentence menu

ISO-code form part program can be entered from the fixed form sentence menu. But, in such a case, proper fixed form sentence must be prepared in advance.

(Soft-key group for milling cycle menu)
[START]
(INSERT STARTING COMMAND FOR MILLING)
<START>
↓
<<3.MILLING START>>
[INSERT]

NOTE

There may be a case that undefined value is entered by “?” in the fixed form sentence menu, so in this case, you need to replace the “?” by proper value can be used in actual machining.
Place the cursor to the address with “?”, enter numeric data, then press <ALTER> key.
After altering the data, return the cursor to the EOB at the end of the program.

A.5.4 Entering Outer Wall Contouring Process

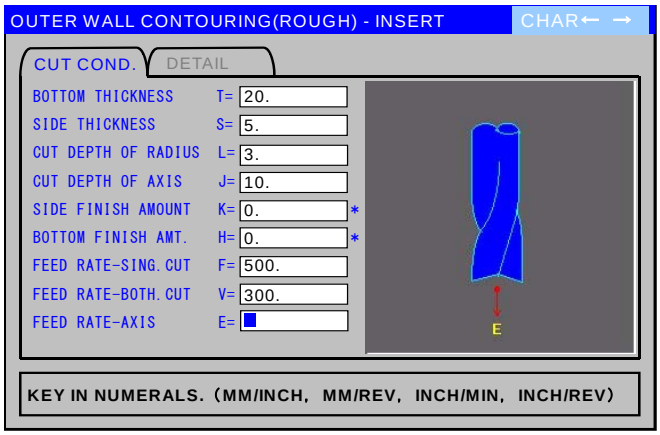
A.5.4.1 Entering outer wall contouring (rough) cycle block

Enter the 1st process : outer wall contouring process by the roughing flat endmill (T01).
Enter machining type, cutting condition and so on.

(Soft-key menu for milling cycle menu)
[CYCLE]
(INSERT MILLING CYCLE)
→
<CONTOURING>
↓
<<1.OUTER WALL CONTOURING(ROUGH)>>
[SELECT]
(OUTER WALL CONTOURING(ROUGH) - INSERT)
<CUT COND.>

20 **INPUT**	(BOTTOM THICKNESS)
5 **INPUT**	(SIDE THICKNESS)
3 **INPUT**	(CUT DEPTH OF RADIUS)
10 **INPUT**	(CUT DEPTH OF AXIS)
0 **INPUT**	(SIDE FINISH AMOUNT)
0 **INPUT**	(BOTTOM FINISH AMOUNT)
500 **INPUT**	(FEED RATE - SING.CUT)
300 **INPUT**	(FEED RATE - BOTH CUT)
200 **INPUT**	(FEED RATE - AXIS)

[INSERT]



NOTE
1 In the machining cycle data menu window, all data excepting cutting condition data are set automatically. However, the data entered at previously entered cycle of same kind are copied, so you must enter the data if you have not entered the same kind of cycle.
2 In the machining cycle data menu window, 2 tabs, <CUT COND.> and <DETAIL>, are displayed.
In the <CUT COND.>, all data must be set. In the <DETAIL>, all data are automatically set, so check those data and modify them if necessary only

A.5.4.2 Entering figure for outer wall contouring (rough)

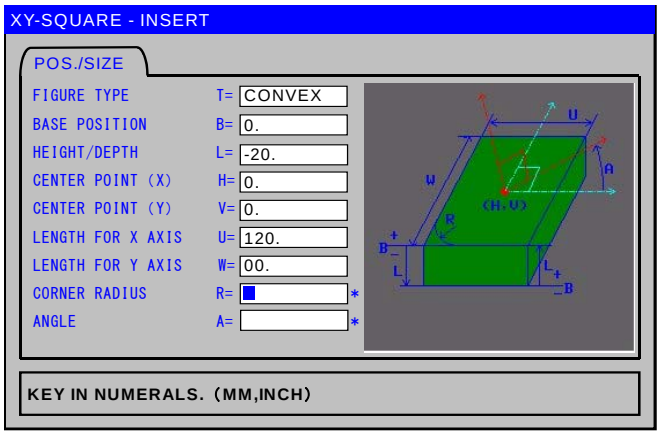
By inserting the machining cycle block, the window of contouring figure menu is displayed, so select the XY-SQUARE CONVEX

(INSERT MILLING FIGURE)
<CONT. FIG.>

<<1.XY-SQUARE CONVEX>>
[SELECT]
(XY-SQUARE - INSERT)
<POS./SIZE>

[CONVEX]	(FIGURE TYPE)
0 **INPUT**	(BASE POSITION)
-20 **INPUT**	(HEIGHT/DEPTH)
0 **INPUT**	(CENTER POINT (X))
0 **INPUT**	(CENTER POINT (Y))
120 **INPUT**	(LENGTH FOR X AXIS)
80 **INPUT**	(LENGTH FOR Y AXIS)

[INSERT]



A.5.5 Entering Pocket Roughing Process

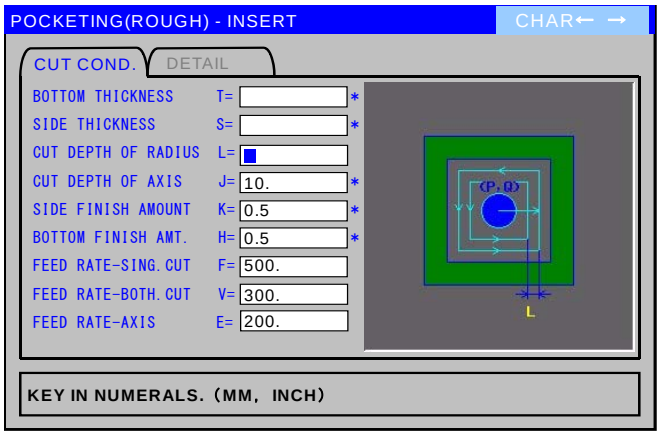
A.5.5.1 Entering pocket roughing cycle block

Enter the 2nd process : pocket roughing by a roughing flat endmill (T01).
Since same tool with the 1st process is used, tool changing blocks are not necessary.
Enter machining type, cutting condition and so on.

(Soft-key group for Milling cycle menu)
[CYCLE]
(INSERT MILLING CYCLE)
→
<POCKETING>
↓
<<1.POCKETING(ROUGH)>>
[SELECT]
(POCKETING(ROUGH) - INSERT)
<CUT COND.>

INPUT	(BOTTOM THICKNESS)
INPUT	(SIDE THICKNESS)
3 **INPUT**	(CUT DEPTH OF RADIUS)
10 **INPUT**	(CUT DEPTH OF AXIS)
0.5 **INPUT**	(SIDE FINISH AMOUNT)
0.5 **INPUT**	(BOTTOM FINISH AMOUNT)
500 **INPUT**	(FEED RATE - SING.CUT)
300 **INPUT**	(FED RATE - BOTH CUT)
200 **INPUT**	(FEED RATE - AXIS)

[INSERT]



NOTE
1 In the machining cycle data menu window, all data excepting cutting condition data are set automatically. However, the data entered at previously entered cycle of same kind are copied, so you must enter the data if you have not entered the same kind of cycle.
2 In the machining cycle data menu window, 2 tabs, <CUT COND.> and <DETAIL>, are displayed.
In the <CUT COND.>, all data must be set. In the <DETAIL>, all data are automatically set, so check those data and modify them if necessary only

A.5.5.2 Entering figure for pocket roughing

By inserting the machining cycle block, the window of pocketing figure menu, so select the XY-FREE CONCAVE FIGURE.

(INSERT MILLING FIGURE)	
<POCKET FIG>	
↓	
<<4.XY-FREE CONCAVE FIGURE>>	
[SELECT]	
(XY PLANE FREE FIGURE - INSERT)	
(START POINT - INSERT)	
INPUT	(FIGURE TYPE)
-50 **INPUT**	(START POINT X)
0 **INPUT**	(START POINT Y)
0 **INPUT**	(BASE POSITION)
-10 **INPUT**	(HEIGHT/DEPTH)
[OK]	
[ARC↻]	
(ARC (CW) - INSERT)	
INPUT	(END POINT X)
INPUT	(END POINT Y)
INPUT	(RADIUS)
-35 **INPUT**	(CENTER POINT CX)
0 **INPUT**	(CENTER POINT CY)
INPUT	(LAST CONNECTION)
[TANGNT]	(NEXT CONNECTION)
[OK]	
[LINE]	
(LINE - INSERT)	
[R-UP]	(LINE DIRECTION)
INPUT	(END POINT X)
INPUT	(END POINT Y)
INPUT	(ANGLE)
[TANGNT]	(NEXT CONNECTION)
[OK]	
[ARC↻]	
(ARC (CW) - INSERT)	
INPUT	(END POINT X)
INPUT	(END POINT Y)
30 **INPUT**	(RADIUS)
0 **INPUT**	(CENTER POINT CX)
0 **INPUT**	(CENTER POINT CY)
INPUT	(LAST CONNECTION)
[TANGNT]	(NEXT CONNECTION)
[OK]	
[LINE]	
(LINE - INSERT)	
[R-DOWN]	(KINE DIRECTION)
INPUT	(END POINT X)
INPUT	(END POINT Y)
INPUT	(ANGLE)
[TANGNT]	(NEXT CONNECTION)
[OK]	

[ARC]	
(ARC (CW) - INSERT)	
INPUT	(END POINT X)
INPUT	(END POINT Y)
15 **INPUT**	(RADIUS)
35 **INPUT**	(CENTER POINT CX)
0 **INPUT**	(CENTER POINT CY)
INPUT	(LAST CONNECTION)
[TANGNT]	(NEXT CONNECTION)
[OK]	
[LINE]	
(LINE - INSERT)	
[L-DOWN]	(LINE DIRECTION)
INPUT	(END POINT X)
INPUT	(END POINT Y)
INPUT	(ANGLE)
[TANGNT]	(NEXT CONNECTION)
[OK]	
[ARC]	
(ARC (CW) - INSERT)	
INPUT	(END POINT X)
INPUT	(END POINT Y)
30 **INPUT**	(RADIUS)
0 **INPUT**	(CENTER POINT CX)
0 **INPUT**	(CENTER POINT CY)
INPUT	(LAST CONNECTION)
[TANGNT]	(NEXT CONNECTION)
[OK]	
[LINBE]	
(LINE - INSERT)	
[L-UP]	(LINE DIRECTION)
INPUT	(END POINT X)
INPUT	(END POINT Y)
INPUT	(ANGLE)
[TANGNT]	(NEXT CONNECTION)
[OK]	
[ARC]	
(ARC (CW) - INSERT)	
-50 **INPUT**	(END POINT X)
0 **INPUT**	(END POINT Y)
15 **INPUT**	(RADIUS)
-35 **INPUT**	(CENTER POINT CX)
0 **INPUT**	(CENTER POINT CY)
INPUT	(NEXT CONNECTION)
[OK]	

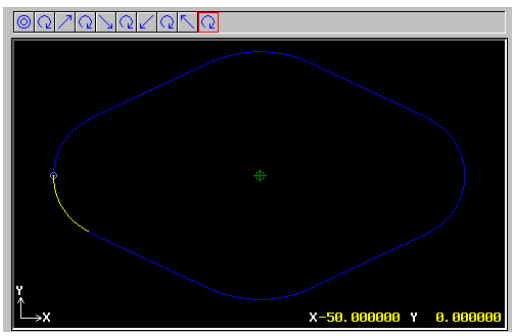
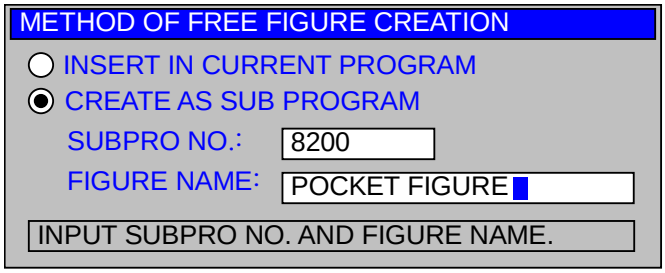


Figure blocks can be registered into the current part program directly, and also can be registered as another sub program.
Registered figure blocks can be used also for finishing, so in this example, register them as a sub program.

(XY PLANE FREE FIGURE - INSERT)	
[CREATE]	
(METHOD OF FREE FIGURE CREATION)	
↓	
<CREATE AS SUB PROGRAM>	
8200 **INPUT**	(SUBPRO NO.)
POCKET FIGURE	(FIGURE NAME)
[ISLAND]	



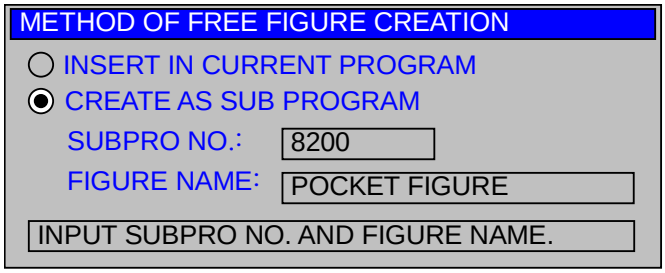
There is an island in a pocket, so enter island figure continuously.

(START POINT - INSERT)	
INPUT	(FIGURE TYPE)
-15 **INPUT**	(START POINT X)
-7.5 **INPUT**	(START POINT Y)
0 **INPUT**	(BASE POSITION)
-10 **INPUT**	(HEIGHT / DEPTH)
[OK]	
[LINE]	
(LINE - INSERT)	
[UP]	(LINE DIRETION)
7.5 **INPUT**	(END POINT Y)
[OK]	
[LINE]	
(LINE - INSERT)	
[RIGHT]	(LINE DIRECTION)
15 **INPUT**	(END POINT X)
[OK]	
[LINE]	
(LINE - INSERT)	
[DOWN]	(LINE DIRECTION)
-7.5 **INPUT**	(END POINT Y)
[OK]	
[LINE]	
(LINE - INSERT)	
[LEFT]	(LINE DIRECTION)
-15 **INPUT**	(END POINT X)
[OK]	



Entered figure blocks were already defined to make as a subprogram, so check the contents of them and create it as a subprogram by pressing soft key [OK]

(XY PLANE FREE FIGURE - INSERT)
[CREATE]
(METHOD OF FREE FIGURE CREATION)
[OK]



NOTE
Registered sub program can be displayed in a figure menu tab, “SUBPROGRAM”.
In this case, set the parameters No14720 to 14723 in advance.
For this example, set those parameters as follow.
No.14722=8500 (Minimum program number of sub programs displayed in the milling machining sub program menu)
No.14723=8999 (Maximum program number of sub programs displayed in the milling machining sub program menu)

A.5.6 Entering Tool Changing and Spindle Rotating Blocks for Finishing Flat End Mill

A.5.6.1 Entering in ISO-code form directly

For pocket finishing, enter commands for operations including changing tools, specifying the spindle, and approaching the machining start point.

G28 G91 Z0. ; **INSERT**	(Z-axis reference position return)
G28 G91 X0. Y0. ; **INSERT**	(X/Y-axis reference position return)
T02 M06 ; **INSERT**	(Tool change)
D2 ; **INSERT**	(Selecting tool data)
M03 S1500 ; **INSERT**	(Spindle rotation)
G00 G90 G54 X0. Y0. ; **INSERT**	(X/Y-axis approaching)
G43 Z100. H2 ; **INSERT**	(Z-axis approaching)

A.5.6.2 Entering by fixed form sentence menu

ISO-code form part program can be entered from the fixed form sentence menu. The already prepared fixed form sentence as 3.MILLING START was made with entering undefined data as “?”, so it can be used also for finishing.

(Soft-key group for milling cycle menu)
[START]
(INSERT STARTING COMMAND FOR MILLING)
<START>
↓
<<3.MILLING START>>
[INSERT]

NOTE
There may be a case that undefined value is entered by “?” in the fixed form sentence menu, so in this case, you need to replace the “?” by proper value can be used in actual machining.
Place the cursor to the address with “?”, enter numeric data, then press <ALTER> key.
After altering the data, return the cursor to the EOB at the end of the program.

A.5.7 Entering Pocket Bottom and Side Finishing Process

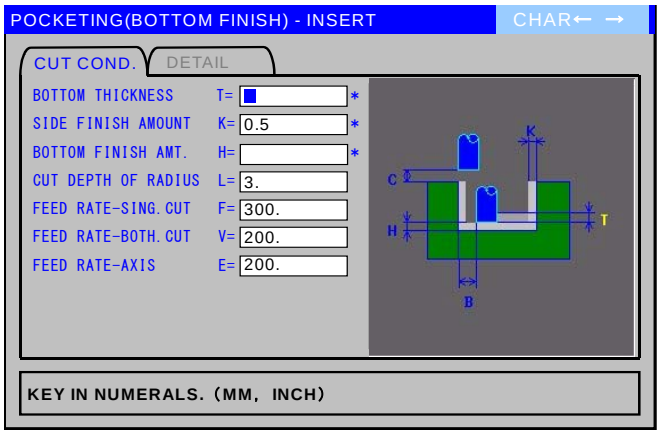
A.5.7.1 Entering pocket bottom finishing cycle block

Enter 3rd process : pocket bottom finishing process by the roughing flat end mill (T01).
Enter machining type, cutting condition and so on.

(Soft-key menu for milling cycle menu)
[CYCLE]
(INSERT MILLING CYCLE)
→
<POCKETING>
↓
<<2.POCKETING (BOTTOM FINISH)>>
[SELECT]
(POCKETING (BOTTOM FINISH) - INSERT)
<CUT COND.>

INPUT	(BOTTOM THICKNESS)
0.5 **INPUT**	(SIDE FINISH AMOUNT)
INPUT	(BOTTOM FINISH AMT.)
3 **INPUT**	(CUT DEPTH OF RADIUS)
300 **INPUT**	(FEED RATE - SING.CUT)
200 **INPUT**	(FEED RATE - BOTH.CUT)
200 **INPUT**	(FEED RATE - AXIS)

[INSERT]



NOTE

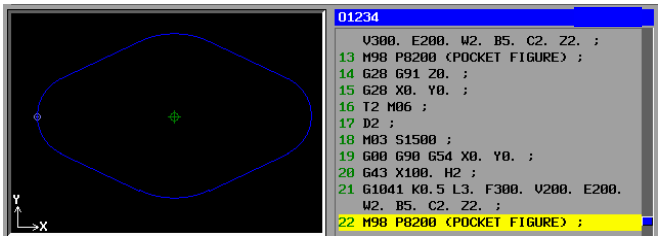
1 In the machining cycle data menu window, all data excepting cutting condition data are set automatically. However, the data entered at previously entered cycle of same kind are copied, so you must enter the data if you have not entered the same kind of cycle.

2 When you select the machining cycle type form machining cycle menu, instead of selecting by the cursor, you can use direct inputting of item number and <INPUT> key. (In this case, enter 2 INPUT)

A.5.7.2 Entering figure for pocket bottom finishing

By inserting the cycle block, the window of pocketing figure is displayed, so enter the figure for finishing. But, the former registered figure blocks for roughing can be used, so select from the subprogram menu.

(INSERT MILLING FIGURE)
→
<SUBPROGRAM>
↓
<<08200: POCKET FIGURE>>
[SELECT]



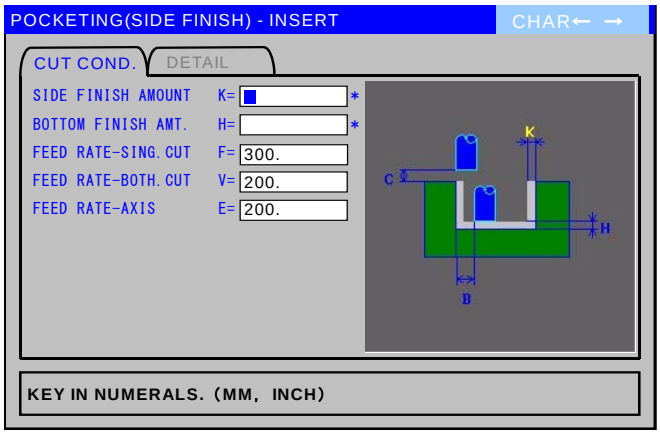
A.5.7.3 Entering pocket side finishing cycle block

Enter 3rd process : pocket side and bottom finishing process by the roughing flat end mill (T01).
Enter machining type, cutting condition and so on.

(Soft-key menu for milling cycle menu)
[CYCLE]
(INSERT MILLING CYCLE)
→
<POCKETING>
↓
<<3.POCKETING (SIDE FINISH)>>
[SELECT]
(POCKETING (SIDE FINISH) - INSERT)
<CUT COND.>

INPUT	(SIDE FINISH AMOUNT)
INPUT	(BOTTOM FINISH AMT.)
300 **INPUT**	(FEED RATE - SING.CUT)
200 **INPUT**	(FEED RATE - BOTH.CUT)
200 **INPUT**	(FEED RATE - AXIS)

[INSERT]

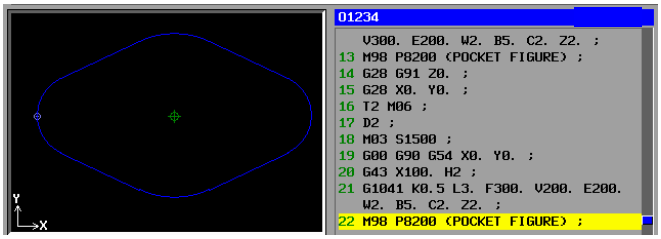


NOTE
1 In the machining cycle data menu window, all data excepting cutting condition data are set automatically. However, the data entered at previously entered cycle of same kind are copied, so you must enter the data if you have not entered the same kind of cycle.
2 When you select the machining cycle type form machining cycle menu, instead of selecting by the cursor, you can use direct inputting of item number and <INPUT> key. (In this case, enter 3 INPUT)

A.5.7.4 Entering figure for pocket side finishing

By inserting the cycle block, the window of pocketing figure is displayed, so enter the figure for finishing. But, the former registered figure blocks for roughing can be used, so select from the subprogram menu.

(INSERT MILLING FIGURE)
→
<SUBPROGRAM>
↓
<<08200: POCKET FIGURE>>
[SELECT]



A.5.8 Entering Tool Changing and Spindle Rotating Blocks for Drilling

A.5.8.1 Entering in ISO-code form directly

For drilling, enter commands for operations including changing tools, specifying the spindle, and approaching the machining start point.

G28 G91 Z0. ; **INSERT**	(Z-axis reference position return)
G28 G91 X0. Y0. ; **INSERT**	(X/Y-axis reference position return)
T03 M06 ; **INSERT**	(Tool change)
D3 ; **INSERT**	(Selecting tool data)
M03 S800 ; **INSERT**	(Spindle rotation)
G00 G90 G54 X0. Y0. ; **INSERT**	(X/Y-axis approaching)
G43 Z100. H3 ; **INSERT**	(Z-axis approaching)

A.5.8.2 Entering by fixed form sentence menu

ISO-code form part program can be entered from the fixed form sentence menu. The already prepared fixed form sentence as 3.MILLING START was made with entering undefined data as “?”, so it can be used also for finishing.

(Soft-key group for milling cycle menu)
[START]
(INSERT STARTING COMMAND FOR MILLING)
<START>
↓
<<3.MILLING START>>
[INSERT]

NOTE
There may be a case that undefined value is entered by “?” in the fixed form sentence menu, so in this case, you need to replace the “?” by proper value can be used in actual machining.
Place the cursor to the address with “?”, enter numeric data, then press <ALTER> key.
After altering the data, return the cursor to the EOB at the end of the program.

A.5.9 Entering Drilling Process

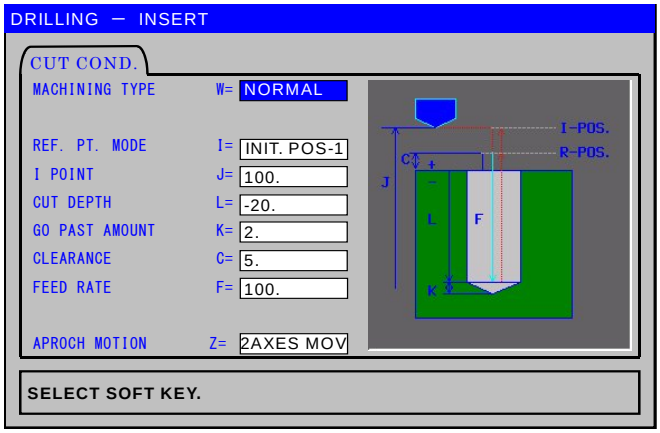
A.5.9.1 Entering drilling cycle block

Enter 4th process : Drilling by drill (T0303).
Enter machining type, cutting condition and so on.

(Soft-key menu for milling cycle menu)
[CYCLE]
(INSERT MILLING CYCLE)
<HOLE MACH.>
↓
<<2.DRILLING>>
[SELECT]
(DRILLING - INSERT)
<CUT COND.>

INPUT	(MACHINING TYPE)
INPUT	(REF.PT.MODE)
100 **INPUT**	(I POINT)
-20 **INPUT**	(CUT DEPTH)
2 **INPUT**	(GO PAST AMOUNT)
5 **INPUT**	(CLEARANCE)
100 **INPUT**	(FEED RATE)

[INSERT]



NOTE
1 In the machining cycle data menu window, all data excepting cutting condition data are set automatically. However, the data entered at previously entered cycle of same kind are copied, so you must enter the data if you have not entered the same kind of cycle.
2 When you select the machining cycle type form machining cycle menu, instead of selecting by the cursor, you can use direct inputting of item number and <INPUT> key. (In this case, enter 2 INPUT)

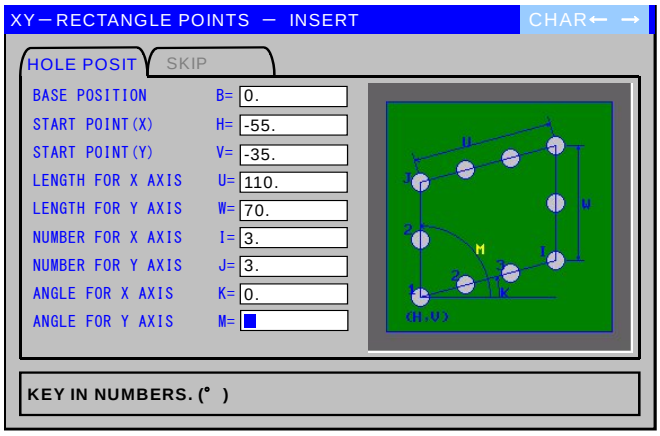
A.5.9.2 Entering hole position block

By inserting the drilling cycle block, the window of hole position menu is displayed, so select the "YY-RECTANGLE POINTS" item.

(INSERT MILLING FIGURE)
<HOLE POSI.>
↓
↓
<<5.XY-RECTANGLE POINTS>>
[SELECT]
(XY-RECTANGLE POINTS - INSERT)
<HOLE POSIT>

0 **INPUT**	(BASE POSITION)
-55 **INPUT**	(START POINT (X))
-35 **INPUT**	(START POINT (Y))
110 **INPUT**	(LENGTH FOR X AXIS)
70 **INPUT**	(LENGTH FOR Y AXIS)
3 **INPUT**	(NUMBER FOR X AXIS)
3 **INPUT**	(NUMBER FOR Y AXIS)
0 **INPUT**	(ANGLE FOR A AXIS)
90 **INPUT**	(ANGLE FOR Y AXIS)

[INSERT]



A.5.10 Operations in the “END” Menu

All necessary machining program have been entered, so enter end procedure.

A.5.10.1 Entering in ISO-code form directly

Enter blocks for spindle stop, releasing and end M-code in ISO-code form with G-code and son on.

M05. ; **INSERT**	(Spindle stop)
G00 G90 Z100.. ; **INSERT**	(Releasing motion)
G28 G91 Z0. ; **INSERT**	(Reference position return)
M06 T0 ; **INSERT**	(Tool changing)
M30 ; **INSERT**	(End M-code)

A.5.10.2 Entering by fixed form sentence menu

ISO-code form part program can be entered from the fixed form sentence menu. But, in such a case, proper fixed form sentence must be prepared in advance.

(Soft-key group for Milling cycle menu)
[END]
<END>
↓
<<1.END OF PROGRAM>>
[INSERT]

NOTE
There may be a case that undefined value is entered by “?” in the fixed form sentence menu, so in this case, you need to replace the “?” by proper value can be used in actual machining.

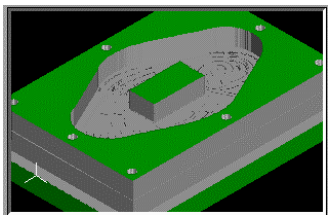
A.6 CHECKING OF THE PART PROGRAM

You can check the entered part program by animation.

A.6.1 Checking by Animation

Select MEM mode by using a mode-selecting switch on a machine-operating panel
[SIMLAT]
(SIMULATE - ANIMATE)
[REWIND]
[START]

NOTE
After checking by animation, in order to do other operation, you must close the animation window by pressing soft key [GRPOFF] always.



B ALARMS

If the input program or one or more parameter settings are not correct, the following P/S alarms are raised.
When an alarm other than the following P/S alarms is raised, refer to the relevant NC operator's manual.

NOTE
In Series 0*i* -D/30*i*/31*i*/32*i*, the alarm is not P/S, but MC.

Alarm		Description	
Series 0*i*-C/ 16*i*/18*i*/21*i*	**Series 0*i*-D/ 30*i*/31*i* /32*i***		
3003	3503	Cause	There is no area that can be machined. The tool is too large for the specified machining area.
		Action	Modify the machining program to use a smaller tool.
3004	3504	Cause	The number of cuts has exceeded the limit.
		Action	Modify the machining program to reduce the number of cuts by, for example, dividing the machining profile.
3005	3505	Cause	The machining start point is invalid.
		Action	Either change the machining start point to a normal one or modify the machining program to automatically decide the point.
3006	3506	Cause	The machining area is invalid.
		Action	Modify the machining program to specify a correct machining area, such as a part figure and a blank figure in a machining profile.
3007	3507	Cause	The cutting conditions are invalid.
		Action	Modify the machining program to specify normal cutting conditions such as the feedrate.
3008	3508	Cause	Finishing is not possible under the specified conditions.
		Action	Review the finishing program.
3009	3509	Cause	An interference occurred. For example, the tool path may interfere with the opposite machining area.
		Action	For example, the cutter radius may be too large for the machining area. Modify the machining program.
3010	3510	Cause	No machining cycle is found. No machining type block is found; only a figure block is specified.
		Action	Modify the machining program by, for example, adding the necessary machining type block.
3011	3511	Cause	No machining cycle is found. An unavailable machining cycle is specified.
		Action	Necessary options may not be added. Contact the machine tool builder.
3012	3512	Cause	G codes that cannot be combined together. A figure block not available to the machining type block is specified.
		Action	Modify the machining program to specify a correct combination of machining type block and figure block.
3013	3513	Cause	The arbitrary figure is invalid.
		Action	Modify the machining program to specify a correct arbitrary figure.
3014	3514	Cause	The figure data is invalid.
		Action	Modify the machining program to specify correct figure data.
3015	3515	Cause	The tool offset cannot be read correctly.
		Action	Necessary options, such as the number of offset sets, may not be set. Modify the machining program by, for example, changing the offset number to an available one.

Alarm		Description	
Series 0*i*-C/ 16*i*/18*i*/21*i*	**Series 0*i*-D/ 30*i*/31*i* /32*i***		
3016	3516	Cause	With a cycle machining command or other 4-digit G commands, necessary arguments are not entered.
		Action	Modify the machining program by, for example, adding necessary arguments.
3025	3525	Cause	The parameter setting is invalid.
		Action	Check the parameter setting.
3029	3529	Cause	Spiral plunge of pocket machining cannot be used on ZC, XC or XA plane.
		Action	Use plunge in line on ZC, XC or XA plane.
3030	3530	Cause	The machining type specification is invalid.
		Action	Modify the machining program to specify an appropriate machining type.
3031	3531	Cause	The return mode is invalid.
		Action	Modify the machining program to specify an appropriate return mode.
3032	3532	Cause	The I point coordinate specification is invalid.
		Action	Modify the machining program to specify an appropriate I point coordinates.
3033	3533	Cause	The dwell time specification is invalid.
		Action	For example, a negative value may be entered as the dwell time. Modify the machining program to specify an appropriate dwell time.
3034	3534	Cause	The overrun specification is invalid.
		Action	For example, a negative value may be entered as the overrun. Modify the machining program to specify an appropriate overrun.
3035	3535	Cause	The thread lead specification is invalid.
		Action	For example, a negative value may be entered as the thread lead. Modify the machining program to specify an appropriate thread lead.
3036	3536	Cause	The shift specification is invalid.
		Action	For example, a negative value may be entered as the shift. Modify the machining program to specify an appropriate shift.
3037	3537	Cause	The cutting method specification is invalid.
		Action	A value not specifiable as threading or other cutting methods is entered. Modify the machining program to specify an appropriate cutting method.
3038	3538	Cause	The cutting direction specification is invalid.
		Action	A value not specifiable as turning or other cutting directions is entered. Modify the machining program to specify an appropriate cutting direction.
3039	3539	Cause	The cut direction specification is invalid.
		Action	A value not specifiable as turning or other cut directions is entered. Modify the machining program to specify an appropriate cut direction.
3040	3540	Cause	The movement method specification is invalid.
		Action	A value not specifiable as facing or other movement methods is entered. Modify the machining program to specify an appropriate movement method.
3041	3541	Cause	The chamfer amount specification is invalid.
		Action	A value not specifiable as chamfering or other chamfer amounts is entered, such as a negative value. Modify the machining program to specify an appropriate chamfer amount.
3042	3542	Cause	The ejection amount specification is invalid.
		Action	A value not specifiable as chamfering or other face ejection amounts is entered, such as a negative value. Modify the machining program to specify an appropriate ejection amount.
3043	3543	Cause	The surplus thickness specification is invalid.
		Action	A value not specifiable as pocketing or other surplus thicknesses is entered, such as a negative value. Modify the machining program to specify an appropriate surplus thickness.

Alarm		Description	
Series 0*i*-C/ 16*i*/18*i*/21*i*	**Series 0*i*-D/ 30*i*/31*i* /32*i***		
3044	3544	Cause	The cutter radius is invalid.
		Action	A value not specifiable as a cutter radius is entered, such as a negative value. Modify the machining program to specify an appropriate cutter radius.
3045	3545	Cause	The depth of cut is invalid.
		Action	A value not specifiable as a depth of cut is entered, such as a negative value. Modify the machining program to specify an appropriate depth of cut.
3046	3546	Cause	The cutting angle is invalid.
		Action	A value not specifiable as a cutting angle is entered. Modify the machining program to specify an appropriate cutting angle.
3047	3547	Cause	The cut angle is invalid.
		Action	A value not specifiable as pocketing or other cut angles is entered. Modify the machining program to specify an appropriate cut angle.
3048	3548	Cause	The clearance is invalid.
		Action	A value not specifiable as a clearance is entered, such as a negative value. Modify the machining program to specify an appropriate clearance.
3049	3549	Cause	The finishing allowance is invalid.
		Action	A value not specifiable as a finishing allowance is entered, such as a negative value. Modify the machining program to specify an appropriate finishing allowance.
3050	3550	Cause	The feedrate is invalid.
		Action	A value not specifiable as a feedrate is entered, such as zero. Modify the machining program to specify an appropriate feedrate.
3051	3551	Cause	The number of finishing operations is invalid.
		Action	A value not specifiable as the number of threading or other finishing operations is entered, such as 0. Modify the machining program to specify an appropriate number of finishing operations.
3052	3552	Cause	The approach specification is invalid.
		Action	A value not specifiable as contouring or other approaches is entered. Modify the machining program to specify an appropriate approach.
3053	3553	Cause	The escape specification is invalid.
		Action	A value not specifiable as contouring or other escapes is entered. Modify the machining program to specify an appropriate escape.
3054	3554	Cause	The compensation number is invalid.
		Action	A value not specifiable as contour finishing or other compensation numbers is entered. Modify the machining program to specify an appropriate compensation number.
3055	3555	Cause	The radius of the tool nose is invalid.
		Action	A value not specifiable as turning or other tool nose radii is entered. Modify the machining program to specify an appropriate radius of the tool nose.
3056	3556	Cause	The cutting edge angle/tool angle is invalid.
		Action	A value not specifiable as turning or other cutting edge angles or tool angles is entered. Modify the machining program to specify an appropriate value.
3057	3557	Cause	The virtual tool tip position is invalid.
		Action	A value not specifiable as turning or other virtual tool tip positions is entered. Modify the machining program to specify an appropriate virtual tool tip position.
3058	3558	Cause	The depth-of-cut change ratio is invalid.
		Action	A value not specifiable as turning or other depth-of-cut change ratios is entered. Modify the machining program to specify an appropriate value.

Alarm		Description	
Series 0*i*-C/ 16*i*/18*i*/21*i*	**Series 0*i*-D/ 30*i*/31*i* /32*i***		
3059	3559	Cause	The return clearance is invalid.
		Action	A value not specifiable as turning or other return clearances is entered. Modify the machining program to specify an appropriate value.
3060	3560	Cause	The tool width is invalid.
		Action	A value not specifiable as turning and grooving or other tool widths is entered. Modify the machining program to specify an appropriate value.
3061	3561	Cause	The pecking is invalid.
		Action	A value not specifiable as turning and grooving or other peckings is entered. Modify the machining program to specify an appropriate value.
3062	3562	Cause	The first feed override is invalid.
		Action	A value not specifiable as turning or other first feed overrides is entered. Modify the machining program to specify an appropriate value.
3063	3563	Cause	The spindle speed is invalid.
		Action	A value not specifiable as the spindle speed, for example 0, is entered. Modify the machining program to specify an appropriate spindle speed.
3064	3564	Cause	Tool data aren't enough to execute cycle.
		Action	Set tool data in offset window or tool condition in machining blocks.
3065	3565	Cause	Tool type doesn't fit the cycle type.
		Action	Modify tool data in offset window or the cycle type in machining program.
3066	3566	Cause	Program coordinate system changing function is invalid.
		Action	Necessary options or parameters concerning with program coordinate system changing function may not be set. Confirm these setting.
3070	3570	Cause	Modal can't be sheltered and restored.
		Action	Command G00, G01, G02 or G03 before the cycle is executed.
3071	3571	Cause	Modal can't be sheltered and restored.
		Action	Command G40 before the cycle is executed.
3072	3572	Cause	Modal can't be sheltered and restored.
		Action	Command G80 before the cycle is executed.
3073	3573	Cause	In reverse tapping cycle, SPINDLE Q is not specified in case of using plural spindles.
		Action	Specify the entry data item SPINDLE Q.
3074	3574	Cause	In reverse tapping cycle, parameters for M code to be specified before reverse tapping are not set.
		Action	Set the parameters No.27197, 27199, 27201 and 27209 for M code to be specified before reverse tapping.
3075	3575	Cause	The figure type is invalid.
		Action	A type not specifiable as the figure type of a figure block is selected. Modify the machining program to specify an appropriate type.
3076	3576	Cause	The reference position is invalid.
		Action	A value not specifiable as the reference position of a figure block is entered. Modify the machining program to specify an appropriate value.
3077	3577	Cause	The height/depth is invalid.
		Action	A value not specifiable as the height/depth of a figure block is entered. Modify the machining program to specify an appropriate value.
3078	3578	Cause	The figure length is invalid.
		Action	A value not specifiable as the length of a figure block is entered. Modify the machining program to specify an appropriate value.
3079	3579	Cause	The corner rounding specification is invalid.
		Action	A value not specifiable as the corner rounding of a figure block is entered. Modify the machining program to specify an appropriate value.

Alarm		Description	
Series 0*i*-C/ 16*i*/18*i*/21*i*	**Series 0*i*-D/ 30*i*/31*i* /32*i***		
3080	3580	Cause	The figure angle specification is invalid.
		Action	A value not specifiable as the angle of a figure block is entered. Modify the machining program to specify an appropriate value.
3081	3581	Cause	The groove width specification is invalid.
		Action	A value not specifiable as the groove width of a figure block is entered. Modify the machining program to specify an appropriate value.
3082	3582	Cause	The figure radius specification is invalid.
		Action	A value not specifiable as the arc radius of a figure block is entered. Modify the machining program to specify an appropriate value.
3083	3583	Cause	The center-to-center distance specification is invalid.
		Action	A value not specifiable as the center-to-center distance of track or other figures is entered. Modify the machining program to specify an appropriate value.
3084	3584	Cause	The pitch specification is invalid.
		Action	A value not specifiable as the pitch angle of a figure block is entered. Modify the machining program to specify an appropriate value.
3085	3585	Cause	The number of holes/grooves specification is invalid.
		Action	A value not specifiable as the number of holes or grooves in a figure block is entered. Modify the machining program to specify an appropriate value.
3086	3586	Cause	The coordinate specification is invalid.
		Action	A value not specifiable as a coordinate of a figure block is entered. Modify the machining program to specify an appropriate value.
3087	3587	Cause	The groove depth specification is invalid.
		Action	A value not specifiable as the groove depth of a figure block is entered. Modify the machining program to specify an appropriate value.
3088	3588	Cause	In selecting a range of arbitrary figures, the selected range is wrong.
		Action	Select a correct range. Concerning the details, refer “Settings that disable machining” in chapter “Machining of a Divided Area” of Selecting a Range of Arbitrary Figures.
3089	3589	Cause	The groove bottom specification is invalid.
		Action	A value not specifiable as the groove bottom of a figure block for grooving is entered. Modify the machining program to specify an appropriate value.
3090	3590	Cause	The thread/hole diameter specification is invalid.
		Action	A value not specifiable as a thread or hole diameter in a figure block is entered. Modify the machining program to specify an appropriate value.
3091	3591	Cause	The number-of-threads specification is invalid.
		Action	A value not specifiable as the number of threads in a figure block to be threaded is entered. Modify the machining program to specify an appropriate value.
3092	3592	Cause	An arbitrary figure is not closed.
		Action	An arbitrary figure entered for facing, pocketing, or turning is not closed. Modify the machining program to specify a closed figure that uses the same point both as the start and end points.
3093	3593	Cause	All figure elements of an arbitrary figure are specified as parts.
		Action	All figure elements of an arbitrary figure entered for turning are specified as "parts." Modify the machining program to specify the figure elements corresponding to actual blanks as "blanks."
3094	3594	Cause	The end face specification is invalid.
		Action	A value not specifiable as the end face of the cycle machining is entered. Modify the machining program to specify an appropriate value.

Alarm		Description	
Series 0*i*-C/ 16*i*/18*i*/21*i*	**Series 0*i*-D/ 30*i*/31*i* /32*i***		
3095	3595	Cause	The arbitrary figure group specification is not correct.
		Action	In the group specification used in, for example, pocketing with islands, either the start or end block is not entered. Enter an appropriate block.
3098	3598	Cause	The arbitrary figure group specification is not correct.
		Action	In the group specification used in, for example, pocketing with islands, either the start or end block is not entered. Enter an appropriate block.

NOTE

Alarm numbers 3503 through 3598 for the Series 30*i*/31*i*/32*i* are changed to 0503 through 0598 when bit 1 (MCA) of parameter No. 6008 is 1.

C MACHINING CYCLE PARAMETERS

> ⚠ **WARNING**
> Be sure to use the parameters set by the machine tool builder.
> If you change the setting of a parameter, the machining program may not work correctly.
> If the machining program does not work correctly, the tool may bump against the workpiece, and the machine may be forced to perform unnatural machining, possibly causing damage to the tool and/or machine, and even injuries.

Appendix C, "MACHINING CYCLE PARAMETERS", consists of:

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C.1 PARAMETERS FOR MILLING CYCLE

C.1.1 Parameters Common to Milling Cycles

These parameters are for the settings common to Milling cycles.

	#7	#6	#5	#4	#3	#2	#1	#0
27000	MC7	MC6	MC5	MC4	MC3	MC2	MC1	MC0

MC0 0 : In ZC or XA plane cycle output, G02/G03 are inverted.
1 : In ZC or XA plane cycle output, G02/G03 are not inverted.

MC1 0 : The hole machining menu has hole machining items for M.
1 : The hole machining menu has hole machining items for T.

MC2 0 : In XC plane cycle output, G12.1/G13.1 are not output.
1 : In XC plane cycle output, G12.1/G13.1 are output.
Remark) Refer to the paragraph of XC plane figure of Facing, Contouring, Pocketing or Grooving.

MC3 0 : In ZC or XA plane cycle output, G07.1 is not output.
1 : In ZC or XA plane cycle output, G07.1 is output.
Remark) Refer to the paragraph of ZC plane figure of Facing, Contouring, Pocketing or Grooving.

MC4 0 : In cycle input screens, the tab of [FACE POSIT] for end face position is not displayed.
1 : In cycle input screens, the tab of [FACE POSIT] for end face position is displayed.
Remark) Refer to Section 4.7, "REAR END FACING BY MILLING" in the Part IV.

MC5 0 : In the soft-key of [START],[CYCLE],[END] and [FIGURE], the icons for Milling are displayed.
1 : In the soft-key of [START],[CYCLE],[END] and [FIGURE], the icons for Turning are displayed.

MC6 0 : In [START] menu screens, the tab of [COORD CONV] is displayed.
1 : In [START] menu screens, the tab of [COORD CONV] is not displayed.

Remark) Refer to Section 2.11, “COORDINATE SYSTEM CONVERSION CYCLE” in the Part IV.

MC7 0 : The tab of [EMBOSSING] is not displayed on the cycle machining menu.
1 : The tab of [EMBOSSING] is displayed on the cycle machining menu.

	#7	#6	#5	#4	#3	#2	#1	#0
27001				P4	P3	P2	P1	P0

P0 0 : The tab of [ROT. AXIS] for rotation axis names are not displayed.
1 : The tab of [ROT. AXIS] for rotation axis names are displayed.

P1 0 : Rotation axis name selection soft keys are disabled.
1 : Rotation axis name selection soft keys [C] and [A] are used.
(It is necessary to set P0 to 1.)

P2 0 : Rotation axis name selection soft keys are disabled.
1 : Rotation axis name selection soft keys [C] and [B] are used.
(It is necessary to set P0 to 1.)

P3 0 : Rotation axis name selection soft keys are disabled.
1 : Rotation axis name selection soft keys [C] and [E] are used.
(It is necessary to set P0 to 1.)

Remark) Refer to Section 2.11, “COORDINATE SYSTEM CONVERSION CYCLE” in the Part IV.

P4 0 : When A, B, or E is specified as a rotation axis name in a figure command, A, B, or E is output on a tool path.
1 : When A, B, or E is specified as a rotation axis name in a figure command, C is output on a tool path.

	#7	#6	#5	#4	#3	#2	#1	#0
27002	ESC	MDL	RCF	FSP	YMN	FGC	TYP	TLG

TLG 0 : In Milling cycle menu, the tab of [TOOL COND.] is not displayed.
1 : In Milling cycle menu, the tab of [TOOL COND.] is displayed.

TYP 0 : When the data input window for a milling figure block is opened for modification, the item "FIGURE TYPE" displays all figure types.
1 : When the data input window for a milling figure block is opened for modification, the item "FIGURE TYPE" displays only the figure type that is applicable to the immediately preceding machining process.

FGC 0 : In [START] menu screens, the tab of [FIG. COPY] is displayed.
1 : In [START] menu screens, the tab of [FIG. COPY] is not displayed.

YMN 0 : In the cycle menu screen or cycle input screen, the items related to the Y-axis are not displayed.
1 : In the cycle menu screen or cycle input screen, the items related to the Y-axis are displayed. (Only for the T series CNC system)

FSP 0 : In the arbitrary figure creation screen, the input item for the finishing feedrate is not displayed.
1 : In the arbitrary figure creation screen, the input item for the finishing feedrate is displayed.

RCF 0 : In the pocketing, embossing, contouring, or grooving cycle input screen, the depth of a tool cut and the finishing allowance are shown in conventional order.
1 : In the pocketing, embossing, contouring, or grooving cycle input screen, the depth of a tool cut and the finishing allowance are shown in reverse order.

MDL 0 : In Milling cycles, after completing the whole cycle motions, the modal will be returned to the state of beginning cycle.
1 : In Milling cycles, after completing the whole cycle motions, the modal will not be returned to the state of beginning cycle.

ESC 0 : In Milling cycles, after completing the whole cycle motions, the tool will go back to the point where machining cycle started.
1 : In Milling cycles, after completing the whole cycle motions, the tool will not go back to the point where machining cycle started.

	#7	#6	#5	#4	#3	#2	#1	#0
27003						ML2	ML1	ML0

By setting this parameter, the optimum cycle menus can be displayed on the screen. Please set 1 bit only according to the machine configuration.

ML0 0 : The machining center machine does not include the X-axis, Y-axis, and Z-axis
1 : The machining center machine includes the X-axis, Y-axis, and Z-axis
ML1 0 : The machining center machine does not include the X-axis, Y-axis, Z-axis, C-axis, and B-axis (axis around which the tool swivels).
1 : The machining center machine includes the X-axis, Y-axis, Z-axis, C-axis, and B-axis (axis around which the tool swivels).
ML2 0 : The machining center machine does not include the X-axis, Y-axis, Z-axis, A-axis, and B-axis (axis around which the tool swivels).
1 : The machining center machine includes the X-axis, Y-axis, Z-axis, A-axis, and B-axis (axis around which the tool swivels).
ML3 0 : The machining center machine includes the X-axis, Y-axis, Z-axis, and B-axis.
1 : The machining center machine includes the X-axis, Y-axis, Z-axis, and B-axis.

NOTE
1 When the parameter No.27003 is set, please be sure to push [F] key on NOW LOADING screen after Power ON. The necessary parameters are set automatically. (When the necessary parameters are set, the message of “NOW SETTING PARAMETERS” is displayed on the left side of NOW LOADING screen.)
2 When bit2(ML2) of the parameter No.27003 is set to 1, please be sure to set bit0 of the parameter No.27003 to 1, bit0 of the parameter No.27001 to 0 and bit3 of the parameter No.27001 to 0.
3 In case of machines other than vertical machining center, please set bit2 of the parameter No.27003 to 0.

	#7	#6	#5	#4	#3	#2	#1	#0
27004						MM2	MM1	MM0

If the machine configuration does not meet the parameter No.27003 or No.27103, this parameter is used to display the optimum menu on the screen.

MM0 0 : Hole machining, facing, contouring, pocketing, grooving, and XY plane figures are not displayed.
1 : Hole machining, facing, contouring, pocketing, grooving, and XY plane figures are displayed.
MM1 0 : Hole machining, facing, contouring, pocketing, grooving, C-axis grooving, polar coordinate (XC) surface figures, and cylindrical (ZC) surface figures are not displayed.
1 : Hole machining, facing, contouring, pocketing, grooving, C-axis grooving, polar coordinate (XC) surface figures, and cylindrical (ZC) surface figures are displayed.
MM2 0 : Cylindrical (XA) surface figures are not displayed.
1 : Cylindrical (XA) surface figures are displayed. (Only when MM0 = 1)

MM3 0 : Hole machining, facing, contouring, pocketing, grooving, B-axis grooving, polar coordinate (XC) surface figures, and cylindrical (YB) surface figures are not displayed.
1 : Hole machining, facing, contouring, pocketing, grooving, B-axis grooving, polar coordinate (XC) surface figures, and cylindrical (YB) surface figures are displayed.

NOTE
1 When these parameter are all 0, the all milling cycle menu except the menu of XA plame are displayed on the screen
2 When the parameter No.27003 or No.27103 is set, this parameter are initialize at the power ON and pushing “F” key.

27005	CLMPM

CLMPM =0 : M code for Main spindle C-axis clamping is not output.
≠0 : M code for Main spindle C-axis clamping.
(Path-specific parameter)

27006	UCLMPM

UCLMPM =0 : M code for Main spindle C-axis unclamping is not output.
≠0 : M code for Main spindle C-axis unclamping.
(Path-specific parameter)

27007	CFCODM

CFCODM : Feedrate to replace all rapid traverse feedrate during C-axis machining for feed per minute.
If 0 is set, the feedrate is assumed 2000 (mm/min) or 78.7 (inch/min) .
[Unit of data] For metric input (INI (No.0000#2)=0) : 1(mm/min)
For inch input (INI (No.0000#2)=1) : 0.01(inch/min)

NOTE
In XC plane, Polar coordinate interpolation is used. That is, it is impossible to use G0 command. Therefore, parameter No.27007 is used instead of rapid traverse.

27008	CFCODR

CFCODR : Feedrate to replace all rapid traverse feedrate during C-axis machining for feed per revolution.
If 0 is set, the feedrate is assumed 2 (mm/min) or 0.0787 (inch/min).
[Unit of data] For metric input (INI (No.0000#2)=0) : 0.0001(mm/rev)
For inch input (INI (No.0000#2)=1) : 0.000001(inch/rev)

NOTE
In XC plane, Polar coordinate interpolation is used. That is, it is impossible to use G0 command. Therefore, No.27008 is used instead of rapid traverse.

27009	CLERCLMP

CLERCLMP : Minimum Clump value of the clearance for Milling cycles.
[Unit of data] For metric input (INI (No.0000#2)=0) : 0.001(mm)
For inch input (INI (No.0000#2)=1) : 0.0001(inch)

27010	APESCLMP

APESCLMP : Minimum Clump value of Radius or Distance of Approach or Escape for Milling cycles.
[Unit of data] For metric input (INI (No.0000#2)=0) : 0.001(mm)
For inch input (INI (No.0000#2)=1) : 0.0001(inch)

27011	CLMPMS

CLMPMS =0 : M code for Sub spindle C-axis clamping is not output.
≠0 : M code for Sub spindle C-axis clamping.
(Path-specific parameter)
Remark) Refer to “III-1.10 C-axis Clamping M code Output”.

27012	UCLMPMS

UCLMPMS =0 : M code for Sub spindle C-axis unclamping is not output.
≠0 : M code for Sub spindle C-axis unclamping.
(Path-specific parameter)
Remark) refer to “1.10 C-axis Clamping M code Output”.

27013	MSTOP

MSTOP =0 : M05 is output as an M code for stopping the spindle.
≠0 : The number of an M code for stopping the spindle is set.
(Path-specific parameter)

27014	MORIE

MORIE =0 : M19 is output as an M code for spindle orientation.
≠0 : The number of an M code for spindle orientation is set.
(Path-specific parameter)

27018	SRCHBLNK

SRCHBLNK: Block number when searching a blank form block.
When a standard value is (=0), block number to search is 100.

	#7	#6	#5	#4	#3	#2	#1	#0
27095		P6		P4		P2	P1	P0

P0 0 : The surplus thickness for the side and bottom is entered in the cycle input screen.
1 : The surplus thickness for the side and bottom is entered in the figure input screen.
P1 0 : Blank figure data is not reused as machining profile data.
1 : Blank figure data is reused as machining profile data.
P2 0 : A polygon figure pattern is not displayed in a fixed form figure.
1 : A polygon figure pattern is displayed in a fixed form figure.

P4 0 : In the milling cycle figure input screen, a soft key is not used to select a clamping method. (Conventional specifications)
1 : In the milling cycle figure input screen, a soft key is used to select a clamping method. (New specifications)

P6 0 : At the end of drilling, cutting is performed at a time by the depth of a cut. (Conventional specifications)
1 : At the end of drilling, pecking is performed by the depth of a cut. (New specifications)

	#7	#6	#5	#4	#3	#2	#1	#0
27096			#5	#4			WRA	

WRA 0 : In cycle retract motions of milling machining operations, workpiece rotation axis is returned to the start point of machining cycle.
1 : In cycle retract motions of milling machining operations, workpiece rotation axis is not returned to the start point of machining cycle.

NOTE
Workpiece rotation axis means that the parameter No.1006#0 of axis is set to 1.
This parameter is available when the parameter No.27002#7 is set to 0.

#4 0 : Outer thread milling cycle is not available.
1 : Outer thread milling cycle is available.

#5 0 : Pocket residual machining cycle is available.
1 : Pocket residual machining cycle is not available.

	#7	#6	#5	#4	#3	#2	#1	#0
27140				TML				

TML 0 : Thread milling is disabled.
1 : Thread milling is enabled.

27453	ROTEXAXIS

ROTEXAXIS : Axis numbers of rotation axes that are used during machining cycle after the soft-key [A] or [C] is selected in figure data input screen when the rotation axis names are extended.
The each axis number has 2 digits that are specified by the followings.
"**~~¥¥~~" (** : Axis number that is used when [A] is selected, ~~¥¥~~ : Axis number that is used when [C] is selected)

(The example of setting)
The setting to meet following conditions : 603
(1) When soft-key [C] is selected, the 3rd axis should be used.
(2) When soft-key [A] is selected, the 6th axis should be used.

27459	Angle of machining start/escape for outer thread milling cycle

If 0 is set, 90 degrees are set as default value.

27464	Number of overlapped threads in Machining type 3 for outer thread milling cycle

C.1.2 Parameters for Facing Cycles

These parameters are for the settings of facing cycles.

	#7	#6	#5	#4	#3	#2	#1	#0
27030							FC1	FC0

FC0 0 : The input data item of [PATH MOVE METHOD] and [PATH MOVE FEED RATE] are displayed on Facing cycle menu.
1 : The input data item of [PATH MOVE METHOD] and [PATH MOVE FEED RATE] are not displayed on Facing cycle menu.

FC1 0 : all menu is displayed in “CUT SHIFT DIRECTION”.
1 : only the perpendicular direction to “CUTTING DIRECTION” is displayed in “CUT SHIFT DIRECTION”.

C.1.3 Parameters for Contouring Cycles

These parameters are for the settings of contouring cycles.

	#7	#6	#5	#4	#3	#2	#1	#0
27040		CN6		CN4	CN3	CN2	CN1	CN0

CN0 0 : During in-feed in Roughing, the tool moves by retracting to the height of the top workpiece surface plus the clearance.
1 : During in-feed in Roughing, the tool moves by retracting to the height of the machining surface plus the clearance.

CN1 0 : In Roughing, the tool moves at a hollow by retracting to the height of the top workpiece surface plus the clearance.
1 : In Roughing, the tool moves at a hollow by retracting to the height of the machining surface plus the clearance.

CN2 0 : In Roughing, the tool moves at an opening by retracting to the height of the top workpiece surface plus the clearance..
1 : In Roughing, the tool moves at an opening by retracting to the height of the machining surface plus the clearance.

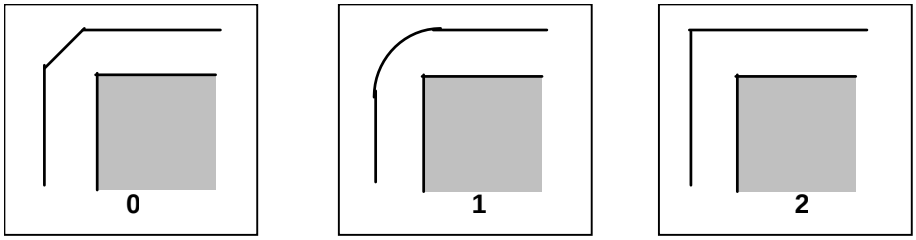
CN3 0 : In side finishing, NC does not perform cutter compensation.
1 : In side finishing, NC performs cutter compensation.
Remark) Tool path is not calculated with Cutter compensation in inside, but G41 or G42 is outputted directly.

CN4 0 : The system performs an interference check.
1 : The system does not perform an interference check.

CN6 0 : In Roughing, when the cutting start point is equal to the end point, the tool cuts directly without escape to the radius direction.
1 : In Roughing, when the cutting start point is equal to the end point, the tool cuts with escape to the radius direction.

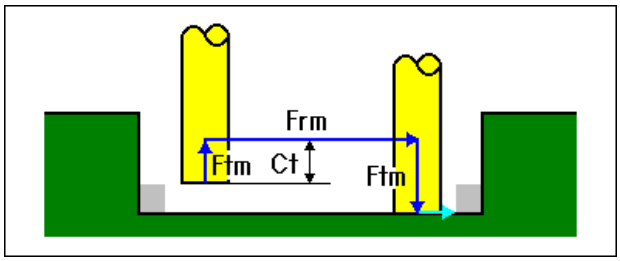
27045	COFSW

COFSW : Offset method for side finishing and chamfering in contouring.
=0 : Corner cut interpolation.
=1 : Circular interpolation.
=2 : Extended straight line.



27046	CMVFR

CMVFR : Feedrate during movement in the cutter radius direction in contouring.
For feed per minute.
If 0 is set, the feedrate is assumed Rapid traverse feedrate.
[Unit of data] For metric input (INI (No.0000#2)=0) : 1(mm/min)
For inch input (INI (No.0000#2)=1) : 0.01(inch/min)
Remark) Feedrate during movement in the cutter radius direction is 'Frm' as the following illustration.

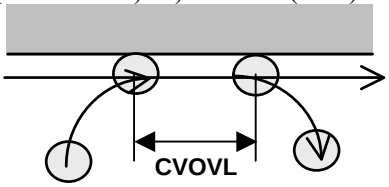


27047	CMVFT

CMVFT : Feedrate during movement in the tool axis direction in contouring.
For feed per minute.
If 0 is set, the feedrate is assumed Rapid traverse feedrate.
[Unit of data] For metric input (INI (No.0000#2)=0) : 1(mm/min)
For inch input (INI (No.0000#2)=1) : 0.01(inch/min)
Remark) Feedrate during movement in the tool axis direction is 'Ftm' as the above illustration.

27048	CVOVL

CVOVL : Amount of overlapping for an approach/escape during contouring.
[Unit of data] For metric input (INI (No.0000#2)=0) : 0.001(mm)
For inch input (INI (No.0000#2)=1) : 0.0001(inch)



27049	CMVFR

CMVFR : Feedrate during movement in the cutter radius direction in contouring for feed per revolution.
If 0 is set, the feedrate is assumed Rapid traverse feedrate.

[Unit of data] For metric input (INI (No.0000#2)=0) : 0.0001(mm/rev)
For inch input (INI (No.0000#2)=1) : 0.000001(inch/rev)
Remark) Refer to parameter No.27046.

27050	CMVFT

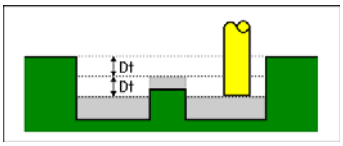
CMVFT : Feedrate during movement in the tool axis direction in contouring for feed per revolution.
If 0 is set, the feedrate is assumed Rapid traverse feedrate.
[Unit of data] For metric input (INI (No.0000#2)=0) : 0.0001(mm/rev)
For inch input (INI (No.0000#2)=1) : 0.000001(inch/rev)
Remark) Refer to parameter No.27047.

C.1.4 Parameters for Pocketing Cycles

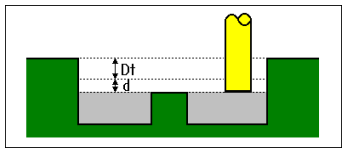
These parameters are for the settings of pocketing cycles.

	#7	#6	#5	#4	#3	#2	#1	#0
27060	PR7	PR6	PR5	PR4	PR3	PR2	PR1	PR0

PR0 0 : Machining starts on the inside during roughing and bottom finishing.
1 : Machining starts on the outside during roughing and bottom finishing.
PR1 0 : The top of an island is not machined during roughing and bottom finishing.
1 : Machining is performed by controlling the depth of cut during roughing and bottom finishing.

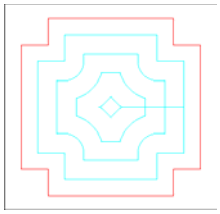


PR1 = 0

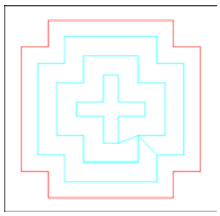


PR1 = 1

PR2 0 : Method of interpolation during roughing and bottom finishing (interpolation of elements with arcs).
1 : Method of interpolation during roughing and bottom finishing (interpolation of elements by extending them).

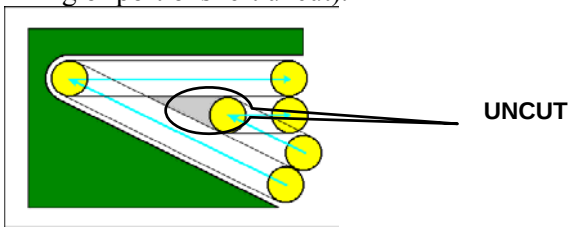


PR2 = 0



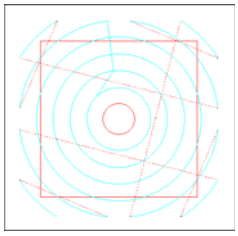
PR2 = 1

PR3 0 : Method of machining portions left uncut during roughing and bottom finishing (no machining of portions left uncut).
1 : Method of machining portions left uncut during roughing and bottom finishing (machining of portions left uncut).

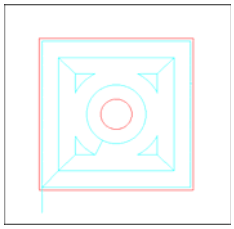


PR4 0 : The tool moves by retracting to the height of the top workpiece surface plus the clearance at an opening during roughing and bottom finishing.
1 : The tool moves by retracting to the height of the machining surface plus the clearance at an opening during roughing and bottom finishing.

PR5 0 : Paths for all material elements are created with an island reference during roughing and bottom finishing.
1 : Paths for all material elements are created with a pocket reference during roughing and bottom finishing.



PR5 = 0



PR5 = 1

PR6 0 : The tool moves by retracting to the height of the top workpiece surface plus the clearance during movement in the tool axis direction during roughing and bottom finishing.
1 : The tool moves by retracting to the height of the machining surface plus the clearance during movement in the tool axis direction during roughing and bottom finishing.

PR7 0 : The input data item of [START PT.(1st AXIS)] and [START PT.(2nd AXIS)] are not displayed on the Pocketing cycle menu..
1 : The input data item of [START PT.(1st AXIS)] and [START PT.(2nd AXIS)] are displayed on the Pocketing cycle menu..

	#7	#6	#5	#4	#3	#2	#1	#0
27061	CDR	#6	PF5	PF4	PF3	PF2	PF1	PF0

PF0 0 : During in-feed in side finishing and chamfering, the tool moves by retracting to the height of the top workpiece surface plus the clearance.
1 : During in-feed in side finishing and chamfering, the tool moves by retracting to the height of the machining surface plus the clearance.

PF1 0 : In side finishing and chamfering, the tool moves at a hollow by retracting to the height of the top workpiece surface plus the clearance.
1 : In side finishing and chamfering, the tool moves at a hollow by retracting to the height of the machining surface plus the clearance.

PF2 0 : In side finishing and chamfering, the tool moves at an opening by retracting to the height of the top workpiece surface plus the clearance.
1 : In side finishing and chamfering, the tool moves at an opening by retracting to the height of the machining surface plus the clearance.

PF3 0 : In side finishing, NC does not perform cutter compensation.
1 : In side finishing, NC performs cutter compensation.
Remark) Tool path is not calculated with Cutter compensation in inside, but G41 or G42 is outputted directly.

PF4 0 : The system performs an interference check.
1 : The system does not perform an interference check.

PF5 0 : A cutting angle is displayed.
1 : A cutting angle is not displayed.

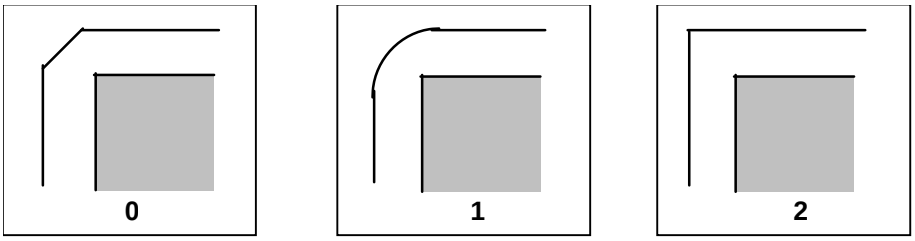
#6 0 : Island figures inside residual area are invalidation. (Island figures are machined)
1 : Island figures inside residual area are left. Maybe time of tool path creation will be long.

CDR 0 : When Pocketing (rough) and emboss machining (rough) is executed, if the cutting condition (SIDE THICKNESS - SIDE FINISH AMOUNT) ≤ CUT DEPTH OF RADIUS is given, an alarm is generated.

1 : When Pocketing (rough) and emboss machining (rough) is executed, if the cutting condition (SIDE THICKNESS - SIDE FINISH AMOUNT) ≤ CUT DEPTH OF RADIUS is given, an alarm isn't generated.

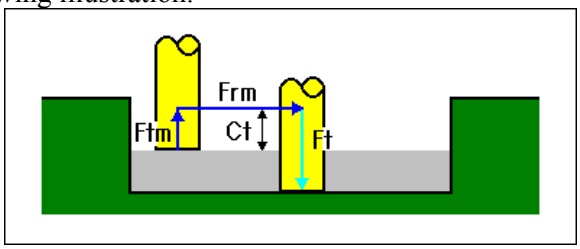
27065	POFSW

POFSW : Offset method for side finishing and chamfering in pocketing.
= 0 : Corner cut interpolation.
= 1 : Circular interpolation.
= 2 : Extended straight line.



27066	PKTFR

PKTFR : Feedrate during movement in the cutter radius direction in in-feed for feed per minute.
If 0 is set, the feedrate is assumed Rapid traverse feedrate.
[Unit of data] For metric input (INI (No.0000#2)=0) : 1(mm/min)
For inch input (INI (No.0000#2)=1) : 0.01(inch/min)
Remark) Feedrate during movement in the cutter radius direction is 'Frm' as the following illustration.

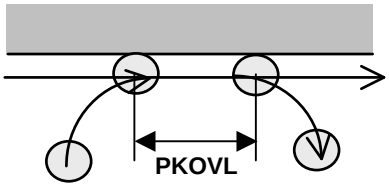


27067	PKTFT

PKTFT : Feedrate during movement in the tool axis direction in in-feed for feed per minute.
If 0 is set, the feedrate is assumed Rapid traverse feedrate.
[Unit of data] For metric input (INI (No.0000#2)=0) : 1(mm/min)
For inch input (INI (No.0000#2)=1) : 0.01(inch/min)
Remark) Feedrate during movement in the tool axis direction is 'Ftm' as the above illustration.

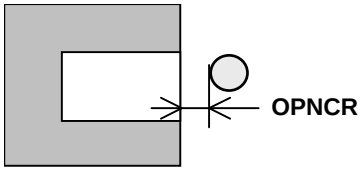
27068	PKOVL

PKOVL : Amount of overlapping for an approach/escape during side finishing and chamfering.
[Unit of data] For metric input (INI (No.0000#2)=0) : 0.001(mm)
For inch input (INI (No.0000#2)=1) : 0.0001(inch)



27069	OPNCR

OPNCR : Clearance for an open portion in pocketing
(When OPNCR = 0)
For metric input (INI (No.0000#2)=0), the clearance for the open portion is assumed to be 3 mm.
For inch input (INI (No.0000#2)=1), the clearance for the open portion is assumed to be 0.3 inch.
[Unit of data] For metric input (INI (No.0000#2)=0) : 0.001(mm)
For inch input (INI (No.0000#2)=1) : 0.0001(inch)



27070	PKTFR

PKTFR : Feedrate during movement in the cutter radius direction in in-feed for feed per revolution.
If 0 is set, the feedrate is assumed Rapid traverse feedrate.
[Unit of data] For metric input (INI (No.0000#2)=0) : 0.0001(mm/rev)
For inch input (INI (No.0000#2)=1) : 0.000001(inch/rev)
Remark) Refer to parameter No.27066.

27071	PKTFT

PKTFT : Feedrate during movement in the tool axis direction in in-feed for feed per revolution.
If 0 is set, the feedrate is assumed Rapid traverse feedrate.
[Unit of data] For metric input (INI (No.0000#2)=0) : 0.0001(mm/rev)
For inch input (INI (No.0000#2)=1) : 0.000001(inch/rev)
Remark) Refer to parameter No.27067.

C.1.5 Parameters for Grooving Cycles

These parameters are for the settings of grooving cycles.

	#7	#6	#5	#4	#3	#2	#1	#0
27080						GR2	GR1	GR0

GR0 0 : During roughing and bottom finishing, in-feed in the cutter radius direction is performed with a uniform depth of cut.
(A uniform depth is the calculated depth automatically.)
1 : During roughing and bottom finishing, in-feed in the cutter radius direction is performed with [CUT DEPTH OF RADIUS].
GR1 0 : During roughing and bottom finishing, in-feed in the tool axis direction is performed with a uniform depth of cut.
(A uniform depth is the calculated depth automatically.)

1 : During roughing and bottom finishing, in-feed in the tool axis direction is performed with [CUT DEPTH OF AXIS].

GR2 0 : During roughing and bottom finishing, tool retracts to the top workpiece surface plus the clearance.

1 : During roughing and bottom finishing, the tool retracts to the position of the machining surface plus the clearance.

	#7	#6	#5	#4	#3	#2	#1	#0
27081				GF4	GF3	GF2	GF1	GF0

GF0 0 : During in-feed in side finishing and chamfering, the tool moves by retracting to the height of the top workpiece surface plus the clearance.

1 : During in-feed in side finishing and chamfering, the tool moves by retracting to the height of the machining surface plus the clearance.

GF1 0 : In side finishing and chamfering, the tool moves at a hollow by retracting to the height of the top workpiece surface plus the clearance.

1 : In side finishing and chamfering, the tool moves at a hollow by retracting to the height of the machining surface plus the clearance.

GF2 0 : In side finishing and chamfering, the tool moves at an opening by retracting to the height of the top workpiece surface plus the clearance.

1 : In side finishing and chamfering, the tool moves at an opening by retracting to the height of the machining surface plus the clearance.

GF3 0 : In side finishing, NC does not perform cutter compensation.

1 : In side finishing, NC performs cutter compensation.

Remark) The NC outputs G41/G42 rather than calculating the tool path through internal cutter compensation. When the finishing count is specified, cutter compensation is performed with G41/G42 only during the last in-feed. Otherwise, cutter compensation by the MANUAL GUIDE *i* is used.

GF4 0 : The system performs an interference check.

1 : The system does not perform an interference check.

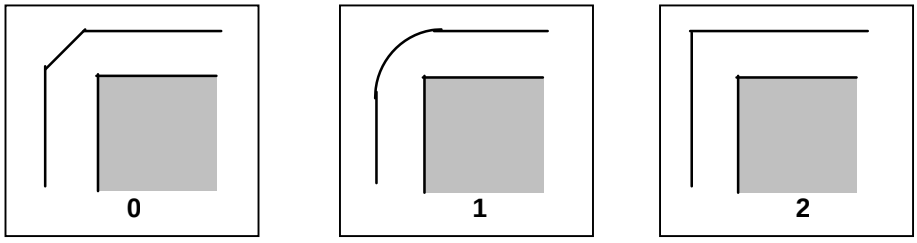
27085	GOFSW

GOFSW : Offset method for groove finishing paths.

= 0 : Corner cut interpolation.

= 1 : Circular interpolation.

= 2 : Extended straight line.



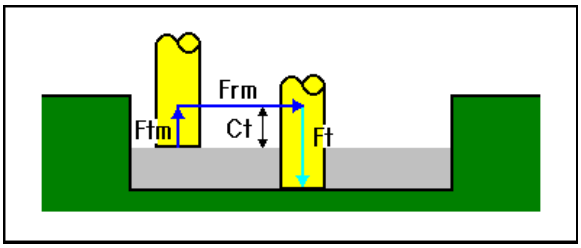
27086	GMVFR

GMVFR : Feedrate during movement in the cutter radius direction in grooving for feed per minute. If 0 is set, the feedrate is assumed Rapid traverse feedrate.

[Unit of data] For metric input (INI (No.0000#2)=0) : 1(mm/min)

For inch input (INI (No.0000#2)=1) : 0.01(inch/min)

Remark) Feedrate during movement in the cutter radius direction is ‘Frm’ as the following illustration.

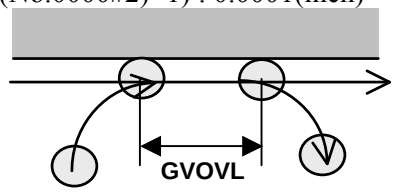


27087	GMVFT

GMVFT : Feedrate during movement in the tool axis direction in contouring for feed per minute.
If 0 is set, the feedrate is assumed Rapid traverse feedrate.
[Unit of data] For metric input (INI (No.0000#2)=0) : 1(mm/min)
For inch input (INI (No.0000#2)=1) : 0.01(inch/min)
Remark) Feedrate during movement in the tool axis direction is 'Ftm' as the above illustration.

27088	GVOVL

GVOVL : Amount of overlapping for an approach/escape during side finishing and chamfering.
[Unit of data] For metric input (INI (No.0000#2)=0) : 0.001(mm)
For inch input (INI (No.0000#2)=1) : 0.0001(inch)



27089	GMVFR

GMVFR : Feedrate during movement in the cutter radius direction in grooving for feed per revolution.
If 0 is set, the feedrate is assumed Rapid traverse feedrate.
[Unit of data] For metric input (INI (No.0000#2)=0) : 0.0001(mm/rev)
For inch input (INI (No.0000#2)=1) : 0.000001(inch/rev)
Remark) Refer to parameter No.27086.

27090	GMVFT

GMVFT : Feedrate during movement in the tool axis direction in contouring for feed per revolution.
If 0 is set, the feedrate is assumed Rapid traverse feedrate.
[Unit of data] For metric input (INI (No.0000#2)=0) : 0.0001(mm/rev)
For inch input (INI (No.0000#2)=1) : 0.000001(inch/rev)
Remark) Refer to parameter No.27087.

C.1.6 Parameters for Hole Machining Cycles

These parameters are for the settings of hole machining cycles.

	#7	#6	#5	#4	#3	#2	#1	#0
27020	HL7					HL2	HL1	HL0

HL0 0 : In the drilling input screen, the input item "GO PATH AMOUNT SET" is not displayed.
1 : In the drilling input screen, the input item "GO PATH AMOUNT SET" is displayed.

HL1 0 : An alarm is issued when the tool database function is used and a hole machining cycle is executed with the chamfering tool and end mill tool.
1 : An alarm is not issued when the tool database function is used and a hole machining cycle is executed with the chamfering tool and end mill tool.

HL2 0 : The type of a tool is not checked when the tool database function is used and a hole machining cycle is executed.
1 : The type of a tool is checked when the tool database function is used and a hole machining cycle is executed.

HL7 0 : Before rigid tapping is specified, "M_ S_ ;" or "S_ ;" is specified.
1 : Before rigid tapping is specified, "G65 P_ M_ S_ ;" is specified (when bit 0 (G84) of parameter No. 5200 is 0) or "G65 P_ S_ ;" is specified (when bit 0 (G84) of parameter No. 5200 is 1).
(The argument of address P is a value set in parameter No.27021.)

27013	MSTOP

FBSTOP M code for stopping the spindle in fine boring
=0 : M05 is output as an M code for stopping the spindle.
≠0 : The number of an M code for stopping the spindle is set.

This parameter is valid for the following cycle:
- Machining center fine boring (G1005), for which "END FEED RATE" is set

27014	MORIE

FBORIENT M code for spindle orientation in fine boring
=0 :M19 is output as an M code for spindle orientation.
≠0 : The number of an M code for spindle orientation is set.

This parameter is valid for the following cycle:
- Machining center fine boring (G1005), for which "END FEED RATE" is set

27021	G65PROGNO

CLERCLMP The number of a P code program to be called by G65 (refer to bit 7 (HL7) of parameter No. 27020)

	#7	#6	#5	#4	#3	#2	#1	#0
27195				PCR			TRV	WRV

WRV 0 : Disabled counter tapping by work rotation.
1 : Enabled counter tapping by work rotation.

TRV 0 : Disabled counter tapping by rotating tool.
1 : Enabled counter tapping by rotating tool.

PCR On drill cycle, pecking clearance of peck drill and pecking return amount of high-speed peck drill is specified
0 : In parameter.
1 : In input window.

27451	THRDMOVL

THRDMOVL Over lap amount for thread milling cycle
[Unit of data] 0 to 99999999

27452	THRDMSCL

THRDMSCL Start clearance for thread milling cycle
[Unit of data] 0 to 99999999

APPENDIX
(FOR MACHINE TOOL BUILDER)

A MANUAL GUIDE *i* SETUP PROCEDURE

Appendix A, "MANUAL GUIDE *i* SETUP PROCEDURE", consists of:

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A.1 GENERAL

In this chapter, the fundamental procedure of starting up MANUAL GUIDE *i* is described. If it is already installed and running correctly, you need not the following operations.

A.2 NECESSARY HARDWARE

The configuration of hardware for running MANUAL GUIDE *i* is determined according to the combination of other CNC functions ordered. In this manual, it is assumed that the necessary hardware is completed.

A.3 NECESSARY SOFTWARE

Software described below is necessary for MANUAL GUIDE *i*.

A.3.1 Machining Center (Series 0*i*)

In case of using MANUAL GUIDE *i* (S790), following software is necessary. Select a machine configuration file depending on the machine configuration. Select one or more language files as necessary.

(1) In case of Series 0*i*-C

File name	Note
BX24*_S1.MEM	Control software
BX24*_MH.MEM	Machine configuration file (for horizontal milling)
BX24*_MV.MEM	Machine configuration file (for vertical milling)
BX24*_L1.MEM	Language file 1
BX24*_L2.MEM	Language file 2 (Note)
BX24*_L3.MEM	Language file 3
BJ25*.MEM	Macro software (for milling)

NOTE
Language file 1: English, Japanese, German, French, Italian, Spanish, Dutch
Language file 2: English, Swedish, Portuguese, Polish, Czech, Russian, Hungarian
Language file 3: English, Chinese (Simplified characters), Chinese (Traditional characters), Korean, Turkish

(2) In case of Series 0*i*-D

File name	Note
BX32*_S1.MEM	Control software
BX32*_MH.MEM	Machine configuration file (for horizontal milling)
BX32*_MV.MEM	Machine configuration file (for vertical milling)
BX32*_L1.MEM	Language file 1
BX32*_L2.MEM	Language file 2 (Note)
BX32*_L3.MEM	Language file 3
BJ32*.MEM	Macro software (for milling)

NOTE
Language file 1: English, Japanese, German, French, Italian, Spanish, Dutch
Language file 2: English, Swedish, Portuguese, Polish, Czech, Russian, Hungarian
Language file 3: English, Chinese (Simplified characters), Chinese (Traditional characters), Korean, Turkish, Bulgarian, Romanian

A.3.2 Machining Center (Series 16*i*/18*i*/21*i*)

(1) In case of using MANUAL GUIDE *i* (S780), following software is necessary. Select a machine configuration software file depending on the machine configuration. Select one or more language software files as necessary.

File name	Note
BX00*_S1.MEM or BXA0*_S1.MEM(Note1)	Control software
BX00*_MH.MEM or BXA0*_MH.MEM(Note1)	Machine configuration file (for horizontal milling)
BX00*_MV.MEM or BXA0*_MV.MEM(Note1)	Machine configuration file (for vertical milling)
BX00*_L1.MEM or BXA0*_L1.MEM(Note1)	Language file 1
BX00*_L2.MEM or BXA0*_L2.MEM(Note1)	Language file 2 (Note2)
BX00*_L3.MEM or BXA0*_L3.MEM(Note1)	Language file 3
BJ00*.MEM or BJA0*.MEM(Note1)	Macro software (for milling)

NOTE
1 When BX00*_S1.MEM is used for Control software, BX00*_*.MEM is used for Machine configuration file and Language file. When BXA0*_S1.MEM is used, BXA0*_*.MEM is used.
2 For option languages, it is necessary to specify NC language options in addition to the following language files.
Language file 1: English, Japanese, German, French, Italian, Spanish, Dutch
Language file 2: English, Swedish, Portuguese, Polish, Czech, Russian, Hungarian
Language file 3: English, Chinese (Simplified characters), Chinese (Traditional characters), Korean, Turkish

(2) When using the MANUAL GUIDE *i* (S777) for the multi C executor, the following software is necessary in place of the control software shown in (1).

File name	Note
BX01*_S1.MEM or BXA1*_S1.MEM(Note)	Control software

NOTE
When BX01*_S1.MEM is used for Control software, BX00*_*.MEM is used for Machine configuration file and Language file. When BXA1*_S1.MEM is used, BXA0*_*.MEM is used.

A.3.3 Machining Center (Series 30*i*/31*i*/32*i*)

In case of using MANUAL GUIDE *i* (S790), following software is necessary. Select a machine configuration file depending on the machine configuration. Select one or more language files as necessary.

(1) In case of Series 30*i*/31*i*/32*i*-A

File name	Note
BY70*.MEM	Control software
BY82*.MEM	Machine configuration file (for horizontal milling)
BY83*.MEM	Machine configuration file (for vertical milling)
BY95*.MEM	Language file 1
BY96*.MEM	Language file 2 (Note)
BY97*.MEM	Language file 3
BJ11*.MEM	Macro software (for milling)

Graphic software is required for the 60VA series.

NOTE
For option languages, it is necessary to specify NC language options in addition to the following language files.
Language file 1: English, Japanese, German, French, Italian, Spanish, Dutch
Language file 2: English, Swedish, Portuguese, Polish, Czech, Russian, Hungarian
Language file 3: English, Chinese (Simplified characters), Chinese (Traditional characters), Korean, Turkish

(2) In case of Series 30*i*/31*i*/32*i*-B

File name	Note
BX71*BS.MEM	Control software
BX71*MH.MEM	Machine configuration file (for horizontal milling)
BX71*MV.MEM	Machine configuration file (for vertical milling)
BX71*L1.MEM	Language file 1
BX71*L2.MEM	Language file 2 (Note)
BX71*L3.MEM	Language file 3
BX71*MM.MEM	Macro software (for milling)

NOTE
For option languages, it is necessary to specify NC language options in addition to the following language files.
Language file 1: English, Japanese, German, French, Italian, Spanish, Dutch
Language file 2: English, Swedish, Portuguese, Polish, Czech, Russian, Hungarian
Language file 3: English, Chinese (Simplified characters), Chinese (Traditional characters), Korean, Turkish, Bulgarian, Romanian

A.3.4 Note

Some functions of MANUAL GUIDE *i* may not work if CNC and Graphic (*) software are old, because MANUAL GUIDE *i* uses the CNC and Graphic functions. So it is recommended to install the latest version of those software.

* In case of Personal computer function with WindowsCE, CNC screen display function corresponds to graphic software.

A.4 SETTING THE MANDATORY PARAMETERS

NOTE
The following parameters are set by the machine tool builder. Do not change the parameters. Otherwise, the machine may operates incorrectly.

To use MANUAL GUIDE *i*, be sure to set the following parameters:

(1) Parameter No.8701#4 = 1
Read of "vacant" P code macro variables is enabled.
(In Series 0*i*-D/30*i*/31*i*/32*i*, this parameter is not necessary.)
(2) Parameter No.3201#6 (NPE) = 1
Program registration does not end with M end codes (M02, M30, and M99).
(3) Parameter No.8650#0 (RSK) = 1
When the <RESET> key is pressed, C-EXE passes a key code to an application program.
(4) Parameter No.8701#6 (CTV) = 1
When the TV check is set to ON, edit operations are not regarded as errors.
(In Series 0*i*-D/30*i*/31*i*/32*i*, this parameter is not necessary.)
(5) Parameter No.3112#0 (SGD) = 0
This parameter enables or disables servo waveform display. When this parameter is set to 1, other graphic functions become unavailable.
(In Series 0*i*-D/30*i*/31*i*/32*i*, this parameter is not necessary.)
(6) Parameter No.9000#0 (SQN) = 0
This parameter is for debugging machining macro program.
And if this parameter is set to 1, MANUAL GUIDE *i* does not work correctly.
(7) Parameter No.3206#6 (DWB) = 1
The function of C language executor refers to this parameter. Bit 6 (DAK) of parameter No.3106, position display change for 3D conversion mode, is effective in MANUAL GUIDE *i* screen
* The system checks this parameter in the case that three-dimensional coordinate conversion is available.)
(In Series 0*i*-D/30*i*/31*i*/32*i*, this parameter is not necessary.)
(8) Parameter No.8650#1 (CNA) = 1
During display of the screen of C language executor, Alarm screen is not changed when the alarm occurs.
(9) Parameter No.9000#7 (L2R) = 0
If this bit is set, MANUAL GUIDE *i* does not update the screen display at the time of operation, and cycle execution is not performed normally.
(In Series 0*i*-D/30*i*/31*i*/32*i*, this parameter is not necessary.)
(10) Parameter No.8650#2 (EKY) = 1
When the standard MDI key for Series 0*i*-D/30*i*/31*i*/32*i*, please sure to set to ON.
(In Series 0*i*-C/16*i*/18*i*/21*i*, this parameter is not necessary.)

(11) Parameter No.14853#7 (PDP) = 1
The program window screen in machining based on the new specifications is used. (Series 16*i*/18*i*/21*i*: Scroll bar is displayed)
(In Series 0*i*-C/16*i*/18*i*/21*i*, this parameter is not necessary.)

(12) Parameter No.9071 ≠ 0
This parameter is set to P-CODE Macro number of MANUAL GUIDE *i* as follows.
Machining Center := 90
Lathe := 91
(In Series 0*i*-C/16*i*/18*i*/21*i*, this parameter is not necessary.)

(13) Parameter No.9072 ≠ 0
This parameter is set to the block number for dealing with Macro sentence continuously in Execute Macro program. MANUAL GUIDE *i* recommend it to 20 for Series 30*i*/31*i*/32*i* or 40 for Series 0*i*-D.
(In Series 0*i*-C/16*i*/18*i*/21*i*, this parameter is not necessary. The setting of 20 may not be appropriate for the multi-path system, so it needs to be adjusted.)

(14) Parameter No.0001#1 (FCV) = 0
Program format is Series 16 standard format.

The following parameters are necessary for machining simulation.

(15) Parameter No.14706≠0
Directions of the three basic axes of spindle 1
Conditions)
- One-path M series

NOTE
Make sure that the number of CNC control axes is correctly set in parameter No.1010. For details on parameter No.1010, refer to the parameter manual of the CNC.

Please the following parameters are set for using small MDI unit.

(16) Parameter No.3100#2 (SKY) = 1
Small MDI unit is used.

Please set the following parameter when AI contour control or Advanced preview control(APC) is used in FS0*i*-D.

(17) Parameter No.13104#0 (AS3) = 1
The return value of FOCAS2 function, “cnc_staticinfo()” and “cnc_statinfo2()”, in edit is compatible with FS16i/FS0i-C/FS30i.

(18) Set the following parameters when FS31iA CPU card A6 or FS32*i*-A CPU card C6 is used.
Turn on CNC after modification of parameter.

P9072=20 Execution block number of the execution macro program.
P14791#0=1 Enable the special function of machining cycle.
P14793=800 Maximum execution time of the special function of machining cycle.
(In case machining cycle is executed when AI contour control I and AI contour control II is off)
P14499=5 Execution block number of the execution macro program when machining cycle is executed.
(In case machining cycle is executed when AI contour control I and AI contour control II is on)

P14499=2 Execution block number of the execution macro program when machining cycle is executed.

Adjust parameter P14499 if performance is not satisfied.
(Criterion of adjustment)

(a) Machining cycle don't stop between block and block when machining cycle is executed.
(b) Time of changing from CNC screen to MANUAL GUIDE i screen is not long when machining cycle is executed.

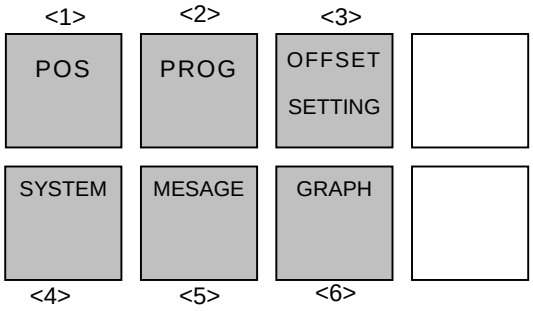
A.5 SETTING THE BASIC PARAMETERS

NOTE
The following parameters are set by the machine tool builder. Do not change the parameters. Otherwise, the machine may operates incorrectly.

A.5.1 Parameters related to the Allocation of Startup Function Keys

	#7	#6	#5	#4	#3	#2	#1	#0
14794			GRP	MES	SYS	OFS	PRG	POS

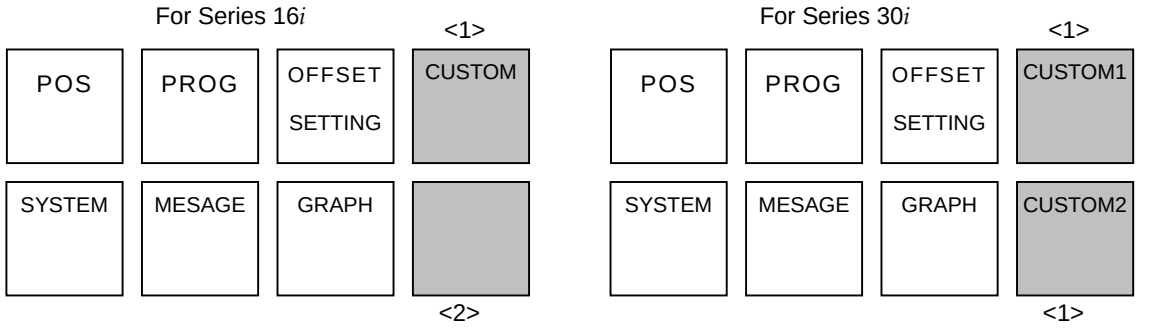
(Path-specific parameter)



POS 0 : The MANUAL GUIDE *i* does not start when function key <1> is pressed.
1 : The MANUAL GUIDE *i* starts when function key <1> is pressed.
PRG 0 : The MANUAL GUIDE *i* does not start when function key <2> is pressed.
1 : The MANUAL GUIDE *i* starts when function key <2> is pressed.
OFS 0 : The MANUAL GUIDE *i* does not start when function key <3> is pressed.
1 : The MANUAL GUIDE *i* starts when function key <3> is pressed.
SYS 0 : The MANUAL GUIDE *i* does not start when function key <4> is pressed.
1 : The MANUAL GUIDE *i* starts when function key <4> is pressed.
MES 0 : The MANUAL GUIDE *i* does not start when function key <5> is pressed.
1 : The MANUAL GUIDE *i* starts when function key <5> is pressed.
GRP 0 : The MANUAL GUIDE *i* does not start when function key <6> is pressed.
1 : The MANUAL GUIDE *i* starts when function key <6> is pressed.

NOTE
For the Series 30*i*/31*i*/32*i* multi-path systems, POS, PRG, OFS, SYS, MES, and GRP cannot be set to different values for each path. Follow the settings of path 1.

	#7	#6	#5	#4	#3	#2	#1	#0
14795					FPT	CS3	CS2	CS1



(Path-specific parameter)

CS1 0 : The MANUAL GUIDE *i* does not start on CRT_CST_AUX screen (Series 0i-C, Series 16i/18i/21i) or CEXE screen (Series 0i-D, Series 30*i*/31*i*/32*i*) when function key <1> is pressed.
1 : The MANUAL GUIDE *i* starts on CRT_CST_AUX screen (Series 0*i*-C, Series 16*i*/18*i*/21*i*) or CEXE screen (Series 0*i*-D, Series 30*i*/31*i*/32*i*) when function key <1> is pressed.

NOTE
1 The CRT_CST_AUX screen and CEXE screen appear when function key <1> is pressed once.
2 When the conversational macro screen is not present in the Series 0*i*-C, Series 16*i*/18*i*/21*i*, set bit 5 (CMEC1) of parameter No.8652 to 1.
3 For the Series 0*i*-D, Series 30*i*/31*i*/32*i* multi-path system, different settings cannot be used for each path. Follow the settings of path 1.

CS2 0 : The MANUAL GUIDE *i* does not start on CRT_CST_MCR screen (Series 0*i*-C, Series 16*i*/18*i*/21*i*) or CEXE2 screen (Series 0*i*-D, Series 30*i*/31*i*/32*i*) when function key <1> is pressed.
1 : The MANUAL GUIDE *i* starts on CRT_CST_MCR screen (Series 0*i*-C, Series 16*i*/18*i*/21*i*) or CEXE2 screen (Series 0*i*-D, Series 30*i*/31*i*/32*i*) when function key <1> is pressed.

NOTE
1 The CRT_CST_MCR screen and CEXE2 screen appear when function key <1> is pressed twice.
2 When the conversational macro screen is not present in the Series 0*i*-C, Series 16*i*/18*i*/21*i*, set bit 6 (CMEC2) of parameter No.8652 to 1.
3 For the Series 0*i*-D, Series 30*i*/31*i*/32*i* multi-path system, different settings cannot be used for each path. Follow the settings of path 1.

CS3 0 : The MANUAL GUIDE *i* does not start on CRT_CST_MENU screen (Series 0i-C, Series 16i/18i/21i) or CEXE3 screen (Series 0i-D, Series 30*i*/31*i*/32*i*) when function key <1> is pressed.
1 : The MANUAL GUIDE *i* starts on CRT_CST_MENU screen (Series 0*i*-C, Series 16*i*/18*i*/21*i*) or CEXE3 screen (Series 0*i*-D, Series 30*i*/31*i*/32*i*) when function key <1> is pressed.

NOTE
1 The CRT_CST_MENU screen and CEXE3 screen appear when function key <1> is pressed three times.
2 When the conversational macro screen is not present in the Series 0*i*-C, Series 16*i*/18*i*/21*i*, set bit 7 (CMEC3) of parameter No.8652 to 1.
3 For the Series 0*i*-D, 30*i*/31*i*/32*i* multi-path system, different settings cannot be used for each path. Follow the settings of path 1.

FPT 0 : The MANUAL GUIDE *i* does not start FAPT screen (Series 0i-C, Series 16i/18i/21i) or CEXE screen (Series 0i-D, Series 30*i*/31*i*/32*i*) when function key <2> is pressed.
1 : The MANUAL GUIDE *i* starts on FAPT screen (Series 0*i*-C, Series 16*i*/18*i*/21*i*) or CEXE screen (Series 0*i*-D, Series 30*i*/31*i*/32*i*) when function key <2> is pressed.

NOTE
1 In the Series 0*i*-C, Series 16*i*/18*i*/21*i*, set bit 4 (CMECF) of parameter No.8652 to 1.
2 For the Series 0*i*-D, 30*i*/31*i*/32*i* multi-path system, different settings cannot be used for each path. Follow the settings of path 1.

A.5.2 Parameters Related to the Display Language

This parameter sets the display language on the MANUAL GUIDE *i* screen.

14712	MSGLANG

(FANUC standard settings = 0)

MSGLANG 0 : The setting of language which NC displays is assumed. When the MANUAL GUIDE *i* does not support the language, English is assumed to be the display language.
1 : English
2 : Japanese
3 : German
4 : French
5 : Italian
6 : Spanish
7 : Swedish
8 : Czech
9 : Portuguese
10 : Polish
11 : Chinese (Simplified characters)
12 : Chinese (Traditional characters)
13 : Korean
14 : Russian
15 : Turkish
16 : Dutch (Dutch is not used in FS16*i*/18*i*/21*i*-TB, FS0*i*-TC.)
17 : Hungarian
18 : Bulgarian (Bulgarian can be used in only FS0*i*-D, FS30*i*/31*i*/32*i*-B.)
19 : Romanian (Romanian can be used in only FS0*i*-D, FS30*i*/31*i*/32*i*-B.)
except above : English

NOTE
1 Language file 1 is necessary to display Japanese, German, French, Italian, Spanish, or Dutch.
Language file 2 is necessary to display Swedish, Czech, Portuguese, Polish, Russian or Hungarian.
Language file 3 is necessary to display Chinese (Simplified characters), Chinese (Traditional characters), Korean, Turkish, Bulgarian, or Romanian.
2 When the display language for the MANUAL GUIDE *i* screen is simplified Chinese, Korean, Bulgarian, or Romanian, alarm messages are displayed in English during execution of the MANUAL GUIDE *i*.
When the display language for the MANUAL GUIDE *i* screen is simplified Chinese and the display language for screens on the NC side is not simplified Chinese, the display language for the MANUAL GUIDE *i* screen is English.

A.5.3 Parameters Related to Current Position Display

14799	DS1AXS

DS1AXS =0 : The first controlled axis is displayed in display area 1.
≠0 : Number of the controlled axis to be displayed in display area 1.

14800	DS2AXS

DS2AXS =0 : The second controlled axis is displayed in display area 2.
≠0 : Number of the controlled axis to be displayed in display area 2.

14801	DS3AXS

DS3AXS =0 : The third controlled axis is displayed in display area 3.
≠0 : Number of the controlled axis to be displayed in display area 3.

14802	DS4AXS

DS4AXS =0 : The fourth controlled axis is displayed in display area 4.
≠0 : Number of the controlled axis to be displayed in display area 4.

14803	DS5AXS

DS5AXS =0 : The fifth controlled axis is displayed in display area 5.
≠0 : Number of the controlled axis to be displayed in display area 5.

14804	DS6AXS

DS6AXS =0 : The sixth controlled axis is displayed in display area 6.
≠0 : Number of the controlled axis to be displayed in display area 6.

14805	DS7AXS

DS7AXS =0 : The seventh controlled axis is displayed in display area 7.
≠0 : Number of the controlled axis to be displayed in display area 7.

14806	DS8AXS

DS8AXS =0 : The eighth controlled axis is displayed in display area 8.
≠0 : Number of the controlled axis to be displayed in display area 8.

A.5.4 Parameters Related to Load Meter

These parameters are used to compensate a CNC controlled axis to which load is applied constantly, such as a vertical axis, for that load, using the load meter.

14815	ELOFS1

(Path-specific parameter)

ELOFS1 : Value of the load current of the first CNC controlled axis in the steady status, as converted into a digital value.
[Valid data range] -6554 to +6554

14816	ELOFS2

(Path-specific parameter)

ELOFS2 : Value of the load current of the second CNC controlled axis in the steady status, as converted into a digital value.
[Valid data range] -6554 to +6554

14817	ELOFS3

(Path-specific parameter)

ELOFS3 : Value of the load current of the third CNC controlled axis in the steady status, as converted into a digital value.
[Valid data range] -6554 to +6554

14818	ELOFS4

(Path-specific parameter)

ELOFS4 : Value of the load current of the fourth CNC controlled axis in the steady status, as converted into a digital value.
[Valid data range] -6554 to +6554

14819	ELOFS5

(Path-specific parameter)

ELOFS5 : Value of the load current of the fifth CNC controlled axis in the steady status, as converted into a digital value.
[Valid data range] -6554 to +6554

14820	ELOFS6

(Path-specific parameter)

ELOFS6 : Value of the load current of the sixth CNC controlled axis in the steady status, as converted into a digital value.
[Valid data range] -6554 to +6554

14821	ELOFS7

(Path-specific parameter)

ELOFS7 : Value of the load current of the seventh CNC controlled axis in the steady status, as converted into a digital value.
[Valid data range] -6554 to +6554

14822	ELOFS8

(Path-specific parameter)

ELOFS8 : Value of the load current of the eighth CNC controlled axis in the steady status, as converted into a digital value.

[Valid data range] -6554 to +6554

A.5.5 Parameters Related to Subprogram Selection Screens

These parameters set the registration start/end numbers of subprogram selection screens.

14722	MFIGSNO

(FANUC standard settings = 0)

MFIGSNO : Registration start number of the milling subprogram selection screen.

14723	MFIGENO

(FANUC standard settings = 0)

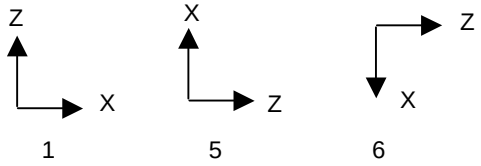
MFIGENO : Registration end number of the milling subprogram selection screen.

A.5.6 Parameters Related to Arbitrary Figures

These parameters are for the settings for arbitrary figures.

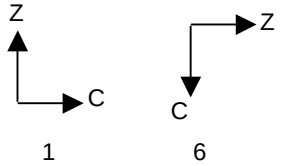
14840	DSPCRDZX

DSPCRDZX : Drawing coordinates when an arbitrary ZX figure is programmed.
= 0 : Same effect as that of setting 5.
= 1 : Plan view, horizontal axis = +X, vertical axis = +Z
= 5 : Plan view, horizontal axis = +Z, vertical axis = +X
= 6 : Plan view, horizontal axis = +Z, vertical axis = -X



14841	DSPCRDZC

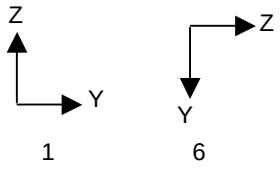
DSPCRDZC : Drawing coordinates when an arbitrary ZC figure is programmed.
= 0 : Same effect as that of setting 6.
= 1 : Plan view, horizontal axis = +C, vertical axis = +Z



= 6 : Plan view, horizontal axis = +Z, vertical axis = -C

14842	DSPCRDYZ

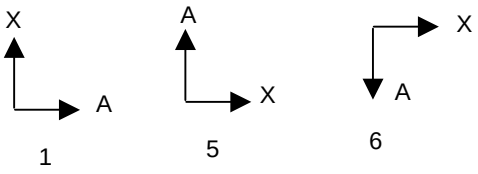
DSPCRDYZ : Drawing coordinates when an arbitrary YZ figure is programmed.
= 0 : Same effect as that of setting 6.
= 1 : Plan view, horizontal axis = + Y, vertical axis = +Z



= 6 : Plan view, horizontal axis = +Z, vertical axis = -Y

14862	DSPCRDXA

DSPCRDXA : Drawing coordinates when an arbitrary XA figure is programmed.
= 0 : Same effect as that of setting 6.
= 1 : Plan view, horizontal axis = +A, vertical axis = +X
= 5 : Plan view, horizontal axis = +X, vertical axis = +A
= 6 : Plan view, horizontal axis = +X, vertical axis = -A



	#7	#6	#5	#4	#3	#2	#1	#0
27104		#6						

#6 : Digit number of output value of arbitrary figure program.
=0: 8 digit.(Normal)
=1: 9 digit.(Extended)

A.5.7 Parameters Related to Operations in General

These parameters set operations in general.

	#7	#6	#5	#4	#3	#2	#1	#0
14700	MGI		LDP	MOT	CS2	CS1		PWD

PWD 0 : When the power is turned on, the system is not switched to the MANUAL GUIDE *i* screen.
1 : When the power is turned on, the system is switched to the MANUAL GUIDE *i* screen.

NOTE
When using a C language executor application created by the machine tool builder in the Series 0*i*-D/30*i*/31*i*/32*i*, PWD assumes the following values:
PWD 0: Settings are disabled.
1: In the initialization (first processing executed after power-on) of a C language executor application created by the machine tool builder, the screen displayed immediately after power-on is changed by the crt_cncscrn() function.

CS1= 0, CS2 = 0 : When the power is turned on, the custom screen is not displayed.
CS1= 1, CS2 = 0 : When the power is turned on, Custom Screen 1 (AUX) is displayed.
CS1= 0, CS2 = 1 : When the power is turned on, Custom Screen 3 (MCR) is displayed.
CS1= 1, CS2 = 1 : When the power is turned on, Custom Screen 2 (MENU) is displayed.

MOT 0 : The modal command is output in NC statement conversion.
1 : The modal command is not output in NC statement conversion.
LDP 0 : Line numbers are displayed during power-on.
1 : Line numbers are not displayed during power-on.
MGI 0 : MANUAL GUIDE *i* is enabled.
1 : MANUAL GUIDE *i* is disabled.

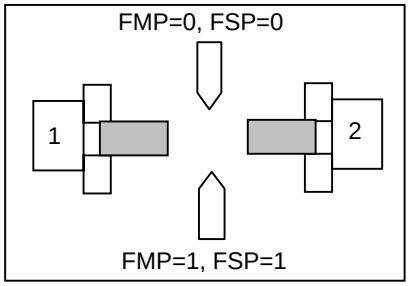
	#7	#6	#5	#4	#3	#2	#1	#0
14701	ALE	PCK	CLP2	CLP1	FSP	FMP	FSE	

FSE 0 : Tool post 1 cannot be used with spindle 2.
1 : Tool post 1 can be used with spindle 2.

NOTE
FSE is available in the case of bit 1 (SUB) of parameter No.14702 = 1 only.

FMP 0 : Tool post 1 is placed above spindle 1.
1 : Tool post 1 is placed under spindle 1.

FSP 0 : Tool post 1 is placed above spindle 2.
1 : Tool post 1 is placed under spindle 2.



NOTE
FSP is available in the case of bit 1 (SUB) of parameter No.14702 = 1 and bit 1 (FSE) of parameter No.14701 = 1 only.

CLP1= 0, CLP2 = 0 The size of the clipboard is set to 1024 bytes.
CLP1= 1, CLP2 = 0 The size of the clipboard is set to 2048 bytes.
CLP1= 0, CLP2 = 1 The size of the clipboard is set to 4096 bytes.
CLP1= 1, CLP 2 = 1 The size of the clipboard is set to 8192 bytes.

PCK 0 : Process List Editing Function : The soft-key [ADD /] and [DEL /] for program check function for each spindle are not displayed.
1 : Process List Editing Function : The soft-key [ADD /] and [DEL /] for program check function for each spindle are displayed.
ALE 0 : When the alarm (PS000 "POWER MUST BE OFF") occurs, the screen display of the MANUAL GUIDE *i* is disabled.
1 : When the alarm (PS000 "POWER MUST BE OFF") occurs, the screen display of the MANUAL GUIDE *i* is not disabled.

	#7	#6	#5	#4	#3	#2	#1	#0
14702	SFA	SFB	SFC	SFD			SUB	MT1

MT1 0 : Vertical.
1 : Horizontal (chuck located on the left side).

SUB 0 : No sub-spindle is provided.
1 : A sub-spindle is provided.

SFD 0 : Normal rotation is assumed if SFRD <G266#5>=0 and SRVD <G266#4>=1
Reverse rotation is assumed if SFRD <G266#5>=1 and SRVD <G266#4>=0
1 : Normal rotation is assumed if SFRD <G266#5>=1 and SRVD <G266#4>=0
Reverse rotation is assumed if SFRD <G266#5>=0 and SRVD <G266#4>=1

SFC 0 : Normal rotation is assumed if SFRC <G204#5>=0 and SRVC <G204#4>=1
Reverse rotation is assumed if SFRC <G204#5>=1 and SRVC <G204#4>=0
1 : Normal rotation is assumed if SFRC <G204#5>=1 and SRVC <G204#4>=0
Reverse rotation is assumed if SFRC <G204#5>=0 and SRVC <G204#4>=1

SFB 0 : Normal rotation is assumed if SFRB <G074#5>=0 and SRVB <G074#4>=1
Reverse rotation is assumed if SFRB <G074#5>=1 and SRVB <G074#4>=0
1 : Normal rotation is assumed if SFRB <G074#5>=1 and SRVB <G074#4>=0
Reverse rotation is assumed if SFRB <G074#5>=0 and SRVB <G074#4>=1

SFA 0 : Normal rotation is assumed if SFRA <G070#5>=0 and SRVA <G070#4>=1
Reverse rotation is assumed if SFRA <G070#5>=1 and SRVA <G070#4>=0
1 : Normal rotation is assumed if SFRA <G070#5>=1 and SRVA <G070#4>=0
Reverse rotation is assumed if SFRA <G070#5>=0 and SRVA <G070#4>=1

	#7	#6	#5	#4	#3	#2	#1	#0
14703	ALP	G62	NCC	TAB	LST	GDM	SFN	FDS

FDS 0 : During feed per revolution, actual feedrate is displayed as that of feed per minute on the base screen.
1 : During feed per revolution, actual feedrate is displayed as that of feed per revolution on the base screen. (Refer to Chapter 1, “BASIC SCREEN” in the Part II.)

SFN 0 : In lower part of the soft keys, shortcut numbers are not displayed.
1 : In lower part of the soft keys, shortcut numbers are displayed.

GDM 0 : When the CNC is a CNC for complex machining, just one guidance message is displayed according to the T or M mode.
1 : When the CNC is a CNC for complex machining, guidance messages for the T and M modes are both displayed regardless of the mode set. (Refer to Chapter 1, “BASIC SCREEN” in the Part II.)

LST 0 : The soft-key [EDTCEL] is not displayed.
1 : The soft-key [EDTCEL] is displayed.

TAB 0 : On the multiple-tab screen, the cursor does not automatically move to the next tab when <INPUT> key is pressed in the lowermost edit box.
1 : On the multiple-tab screen, the cursor automatically moves to the next tab when <INPUT> key is pressed in the lowermost edit box.

NCC 0 : In NC statement conversion, 4-digit G code cycle machining commands before expansion are output as comments. (only for the Series 30*i*/31*i*/32*i*)
1 : In NC statement conversion, 4-digit G code cycle machining commands before expansion are not output as comments. (only for the Series 30*i*/31*i*/32*i*)

G62 0 : The capability of allowing the G62.4 signal to make the screen display visible or invisible is disabled.
1 : The capability of allowing the G62.4 signal to make the screen display visible or invisible is enabled. (only for the Series 0*i*/16*i*/18*i*/21*i*)

ALP 0 : The function for reusing existing figures is enabled on the arbitrary figure screen.
1 : The function for reusing existing figures is disabled on the arbitrary figure screen.

	#7	#6	#5	#4	#3	#2	#1	#0
14704			DXY	IJK	LDM	SLM	PWO	PTO

PTO 0 : During operation, data entry is prohibited on the tool offset setting screen.
1 : During operation, data entry is not prohibited on the tool offset setting screen.

PWO 0 : During operation, data entry is prohibited on the workpiece origin offset setting screen.
1 : During operation, data entry is not prohibited on the workpiece origin offset setting screen.

SLM 0 : The spindle load meter is displayed.
1 : The spindle load meter is not displayed.

LDM 0 : The servo load meter is displayed.
1 : The servo load meter is not displayed.

IJK 0 : The arc command in a contour program is output in IJK format.
1 : The arc command in a contour program is output in R format.
(Refer to Chapter 1, “CONTOUR PROGRAMMING” in the Part III.)

DXY 0 : The X coordinate in the XY plane contour program is output as a radius value.
1 : The X coordinate in the XY plane contour program is output as a diameter value.

	#7	#6	#5	#4	#3	#2	#1	#0
14705	TLJ	OSH	TOL	TOF			CEB	SHT

SHT 0 : The shortcut of number + input and the shortcut of shift + number on the base screen are enabled.
1 : The shortcut of number + input and the shortcut of shift + number on the base screen are disabled.

CEB 0 : When a specified range is copied to the clipboard with soft key [COPY] during while a program is edited, the cursor returns to the start block of the specified range after the copying.
1 : When a specified range is copied to the clipboard with soft key [COPY] while a program is edited, the cursor remains in the end block of the specified range after the copying.

TOF 0 : Soft key [T-OFS] is displayed.
1 : Soft key [T-OFS] is not displayed.

TOL 0 : Soft key [TOOL] is not displayed.
1 : Soft key [TOOL] is displayed.

OSH 0 : Enabled soft key operation by numeric keys on program list window.
1 : Disabled soft key operation by numeric keys on program list window.

TJL 0 : The function for jumping to the tool database and the tool icon display function are disabled.
1 : The function for jumping to the tool database and the tool icon display function are enabled.

	#7	#6	#5	#4	#3	#2	#1	#0
14836	NMG			STK		LOS	MSD	NCO

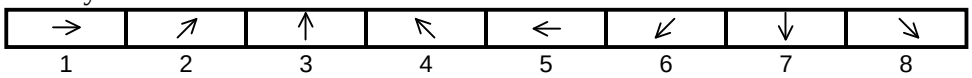
NCO 0 : The NC statement conversion function is enabled.
1 : The NC statement conversion function is disabled.

MSD 0 : The function for displaying the number of revolutions of the milling axis by the servo motor is disabled.
1 : The function for displaying the number of revolutions of the milling axis by the servo motor is enabled.

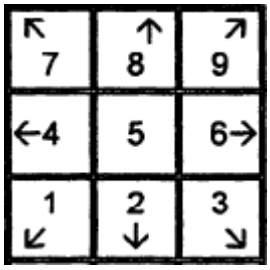
LOS 0 : The new display (support for output switching control) of load meter of the spindle is enabled.

1 : The new display (support for output switching control) of load meter of the spindle is disabled.

STK 0 : In arbitrary figure input / free contour cylinder blank figure input / contour program input, input method of direction by numeric keys of MDI unit is along the layout of soft keys.



1 : In arbitrary figure input / free contour cylinder blank figure input / contour program input, input method of direction by numeric keys of MDI unit is along the layout of MDI unit.



After setting this parameter, please power on again

NMG 0 : MANUAL GUIDE i is enabled

1 : MANUAL GUIDE i is disabled. (This parameter perform the same operation as in the case where the option of MANUAL GUIDE i is disabled)

	#7	#6	#5	#4	#3	#2	#1	#0
14850	RDA	SWH	CHK	MCD	PRT	RET	CUR	TOL

TOL 0 : The [TOOL DATA] tab is displayed in the tool offset window.

1 : The [TOOL DATA] tab is not displayed in the tool offset window.

CUR 0 : Soft key [CHCURS] is displayed ([TAB] is hidden) in the tool offset window and workpiece coordinate system setting window.

1 : Soft key [TAB] is displayed ([CHCURS] is hidden) in the tool offset window and workpiece coordinate system setting window.

RET 0 : The menu selection screen appears when soft key [CLOSE] is pressed.

1 : The base screen appears when soft key [CLOSE] is pressed.

PRT 0 : The fixed form statement is not protected from being edited by the memory protection key.

1 : The fixed form statement is protected from being edited by the memory protection key.

MCD 0 : Soft key [M CODE] is displayed.

1 : Soft key [M CODE] is not displayed.

CHK 0 : Invalid characters are not checked when the program is opened.

1 : Invalid characters are checked when the program is opened. (When bit 5 is set to 1, it takes longer time to open the program.)

SWH 0 : When the MANUAL GUIDE *i* is used with function key <PROG>, the following switching is performed in mode switching: MGi screen → NC screen → MGi screen.

1 : When the MANUAL GUIDE *i* is used with function key <PROG>, the following switching is not performed in mode switching: MGi screen → NC screen → MGi screen.

RDA 0 : During switching from the NC screen to the MANUAL GUIDE *i* screen, the program is not reloaded if the size and time stamp (minutes) are not changed.

1 : During switching from the NC screen to the MANUAL GUIDE *i* screen, the program is always reloaded.

	#7	#6	#5	#4	#3	#2	#1	#0
14851	TME	GCC	PWK	W12	FS2		SBP	CNR

CNR 0 : When an arbitrary figure is created, the corner element sandwiched between the material element and the part element is not the reverse connection corner element.
1 : When an arbitrary figure is created, the corner element sandwiched between the material element and the part element is the reverse connection corner element.

SBP 0 : When soft key [INSERT] is pressed on the fixed figure input screen, the selection screen for output to a subprogram is not displayed.
1 : When soft key [INSERT] is pressed on the fixed figure input screen, the selection screen for output to a subprogram is displayed.

FS2 0 : In an arbitrary figure other than the ZX plane turning figure, when there are two selectable arcs, the selection screen prompts the operator to select one of them.
1 : In an arbitrary figure other than the ZX plane turning figure, when there are two selectable arcs, the arc with the shortest route is automatically selected.

W12 0 : In an editing operation on the process editing screen, W1 or W2 is not output to the argument of the process start command (G1992).
1 : In an editing operation on the process editing screen, W1 or W2 is output to the argument of the process start command (G1992).

PWK 0 : The "PROGRAM COORDINATE" input item is not displayed on the arbitrary figure editing screen.
1 : The "PROGRAM COORDINATE" input item is displayed on the arbitrary figure editing screen.

GCC 0 : When the offset memory switching function is enabled on the tool offset screen or work shift screen, switching between coordinate systems in which data is displayed is performed by soft keys.
1 : When the offset memory switching function is enabled on the tool offset screen or work shift screen, switching between coordinate systems in which data is displayed is performed by coordinate system selection signal GCC.

TME 0 : Editing of data is not prohibited on the NC tool management data screen.
1 : Editing of data is prohibited on the NC tool management data screen.

	#7	#6	#5	#4	#3	#2	#1	#0
14852		TDA	OPH	CFF	EQU	C4E	NTC	DFM

DFM 0 : When the cursor is moved to an arbitrary figure block, figure drawing is performed.
1 : When the cursor is moved to an arbitrary figure block, figure drawing is not performed.

NTC 0 : In drawing during machining, tool compensation is considered (only for the Series 0i-C/16*i*/18*i*/21*i*).
1 : In drawing during machining, tool compensation is not considered (only for the Series 0*i*-C/16*i*/18*i*/21*i*).

G4E 0 : When the cursor is located on an argument in the G4-digit block, a warning is not displayed when an attempt is made to insert the an NC statement.
1 : When the cursor is located on an argument in the G4-digit block, a warning is displayed when an attempt is made to insert the an NC statement.

EQU 0 : When the difference between the values before and after rounding exceeds a tolerance of 0.00001 on the arbitrary figure creation screen, a warning message is not displayed.
1 : When the difference between the values before and after rounding exceeds a tolerance of 0.00001 on the arbitrary figure creation screen, a warning message is displayed.

CFF 0 : The number of registered characters of a fixed form statement is 128. (The maximum number of fixed form statements for each tab is 10.).

1 : The number of registered characters of a fixed form statement is 256. (The maximum number of fixed form statements for each tab is 5.).

OPH 0 : The operation history function is enabled.
1 : The operation history function is disabled.

TDA 0 : The tool data access function is not used.
1 : The tool data access function is used.

	#7	#6	#5	#4	#3	#2	#1	#0
14853	PDP	ONV		MDO	NTC	OSH	PRG	PLT

PLT 0 : The program list screen in the new specification is used. (only for the Series 30*i*/31*i*/32*i*)
1 : The program list screen in the old specification is used. (only for the Series 30*i*/31*i*/32*i*)

PRG 0 : When function key <PROG> is pressed, the NC program screen and program list screen are displayed alternately. (only for the Series 0*i*-D/30*i*/31*i*/32*i*)
1 : When function key <PROG> is pressed, the NC program screen is always displayed. (only for the Series 0*i*-D/30*i*/31*i*/32*i*)

OSH 0 : If [O SRCH] is executed when the key-in buffer is empty, program numbers are searched. (only for the Series 0*i*-D/30*i*/31*i*/32*i*)
1 : If [O SRCH] is executed when the key-in buffer is empty, program numbers are not searched. (only for the Series 0*i*-D/30*i*/31*i*/32*i*)

NTC 0 : Tool compensation is considered in drawing during machining. (only for the Series 0*i*-D/30*i*/31*i*/32*i*)
1 : Tool compensation is not considered in drawing during machining. (only for the Series 0*i*-D/30*i*/31*i*/32*i*)

MDO 0 : Program input/output between the memory card and the data server is disabled. (only for the Series 30*i*/31*i*/32*i*)
1 : Program input/output between the memory card and the data server is enabled. (only for the Series 30*i*/31*i*/32*i*)

ONV 0 : Operation navigating function is not available.
1 : Operation navigating function is available.

PDP 0 : A running program is displayed in the old specification (Series 30*i*/31*i*/32*i*: without the scrollbar). (only for the Series 30*i*/31*i*/32*i*)
1 : A running program is displayed in the new specification (Series 30*i*/31*i*/32*i*: with the scrollbar). (only for the Series 30*i*/31*i*/32*i*)

	#7	#6	#5	#4	#3	#2	#1	#0
14854	MLT	BGO	FDP	TDM	MTN	SPD		USB

SPD 0 : The actual number of revolutions of the spindle is displayed in spindle information on the base screen.
1 : The specified value (modal value) is displayed in spindle information on the base screen.

MTN 0 : The "MAINTE" tab is not displayed in the "SETTINGS" window.
1 : The "MAINTE" tab is displayed in the "SETTINGS" window.

TDM 0 : The tool data specific to the MANUAL GUIDE *i* is used.
1 : The tool data on the NC side is used.

FDP 0 : "JOGF" and "DRY RUN F" are not displayed.
1 : "JOGF" and "DRY RUN F" are displayed.
(This parameter is followed regardless of the setting of bit 7 (JSP) of NC parameter No. 3108 with the same function.)

BGO 0 : Program input/output is disabled in BG editing (conventional specification).
1 : Program input/output is enabled in BG editing.

MLT 0 : The MANUAL GUIDE *i* performs screen display control immediately after power-on in a multi-application.
1 : The MANUAL GUIDE *i* does not perform screen display control immediately after power-on in a multi-application. (Screen display control is performed in the application of the machine tool builder)

(Series 0i-D only)

USB 0 : The usb memory input/output function is not available.
1 : The usb memory input/output function is available.
* This parameter can be set to 1, when all the following requirements is satisfied.
- The main board which supports the USB interface is used.
- The NC basic software which supports the USB interface is used
- Manual guide *i* software which supports the USB memory input/output function is used.

	#7	#6	#5	#4	#3	#2	#1	#0
14855	198			PRC		FFD	TLM	YOF

(Path-specific parameter)

YOF 0 : The Y axis offset value is displayed in the tool offset window.
1 : The Y axis offset value is not displayed in the tool offset window.

NOTE
Bit 0 (YOF) of parameter No. 14855 requires the display option of the Y axis tool offset.

TLM 0 : The tool management screen and magazine management screen are not displayed on the MANUAL GUIDE *i* screen.
1 : The tool management screen and magazine management screen are displayed on the MANUAL GUIDE *i* screen.

FFD 0 : The shape drawing is not displayed on subprogram list screen (specification in the past).
1 : The shape drawing is displayed on subprogram list screen.

PRC 0 : When the program coordinate system switching function is enabled, the spindle 1 side is program coordinate system 1 and the spindle 2 side is program coordinate system 2.
1 : When the program coordinate system switching function is enabled, the spindle 1 side is program coordinate system 2 and the spindle 2 side is program coordinate system 1.

198 0 : The position display ignorance function by PMC signal <G198> is disabled.
1 : The position display ignorance function by PMC signal <G198> is enabled.

	#7	#6	#5	#4	#3	#2	#1	#0
14856					MIL		ANM	GID

(Path-specific parameter)

GID 0 : The setup function is enabled.
1 : The setup function is disabled.

ANM 0 : The animation function is enabled.
1 : The animation function is disabled.

MIL 0 : The milling cycle function is enabled.
1 : The milling cycle function is disabled.

NOTE
Even though you change the settings of bits 0 (GID) and 1 (ANM) and 2(TRN) and 3(MIL) of parameter No. 14856, the alarm PW0000, "POWER MUST BE OFF" is not issued.
You must however turn on the power again before the new settings can take effect.

14870	NAVIGROUP

(FANUC standard settings = 0)

NAVIGROUP : Group ID number of navigation which starts when [NAVI] soft-key is pushed.
When this parameter is set to 0 or the group ID is not specified in definition file, "SELECT NAVIGATION" window is displayed.

[Valid data range] 0 to 127.

NOTE
When operation navigating should start by first method of start, set parameter No.14870 to 0.
When operation navigating should start by second method of start, set parameter No.14870 to a number more than 0.

	#7	#6	#5	#4	#3	#2	#1	#0
27480				OFI				CSP

CSP This bit selects method of NC program conversion when the following commands are executed.

(Method of NC program conversion)

CSP	Subprogram calling	Commands in subprogram	Custom macro	Finishing cycle (G70)	Commands of peripheral axis control
0	×	○	×	×(*)	×
1	○	×	○	○	○

○: Output
×: No output
(*) Figure commands to be executed G70 are outputted.

OFI 0 : Offset data can be inputted / outputted.
1 : Offset data can't be inputted / outputted.

	#7	#6	#5	#4	#3	#2	#1	#0
27481				CSP		NMA		

NMA 0 : The alarm occurs when M198 command is executed in NC statement conversion.
1 : The alarm does not occur when M198 command is executed in NC statement conversion.

CSP 0 : 3-dimensional interference check status display position and other status display positions are made the past position. (existing specification)
1 : 3-dimensional interference check status display position and other status display positions are made the lower right of the screen.

	#7	#6	#5	#4	#3	#2	#1	#0
27483	#7			G41			#1	#0

#0 0 : Automatic operation after changing to MANUAL GUIDE *i* screen from user application is disabled.
1 : Automatic operation after changing to MANUAL GUIDE *i* screen from user application is enabled.

#1 0 : Change to user application screen after closing MAUAL GUIDE *i* window is disabled.
1 : Change to user application screen after closing MAUAL GUIDE *i* window is enabled.

NOTE
When bit 1 of parameter No.27483 to 1, setting the bit 7 of parameter No.27483 to "1" is necessary.

G41 0 : In NC program conversion, G41/G42 command can not be converted.
1 : In NC program conversion, G41/G42 command can be converted.

NOTE
It is effective only at parameter No.27480#0=0.

#7 0 : Change to menu window when blank figure input window is closed by softkey [CANCEL]
1 : Change to base window when blank figure input window is closed by softkey [CANCEL]

A.5.8 Parameters Related to Graphic Display

These parameters set graphic display.

14713	GRPSCALE

(FANUC standard settings = 0)

GRPSCALE : Scaling unit for graphic enlargement and reduction.
[Scaling unit] 64/GRPSCALE
[Valid data range] 0 to 255. (If 0 is set, 64 is assumed.)

14714	GRPMOVEH

(FANUC standard settings = 0)

GRPMOVEH : Graphic horizontal movement unit (dots)
[Valid data range] 0 to 255. (If 0 is set, 64 dots is assumed.)

14715	GRPMOVEV

(FANUC standard settings = 0)

GRPMOVEV : Graphic vertical movement unit (dots)
[Valid data range] 0 to 255. (If 0 is set, 35 dots is assumed.)

14716	GRPROTA

(FANUC standard settings = 0)

GRPROTA : Graphic rotation direction movement unit (angle)
[Valid data range] 0 to 255. (If 0 is set, 10 degrees is assumed.)

A.5.9 Parameters Related to Tool Drawing

The following tool drawing data needs to be defined to perform machining simulation.

(1) Tool drawing data for milling systems

Tool type	Drill	Chamfer	Flat end mill	Ball end mill
Data 1	Tool length	Tool length	Tool length	Tool length
Data 2		Cutter length		
Data 3		Shank length		
Data 4		Shank diameter		

Tool type	Tap	Reamer	Boring	Face mill
Data 1	Tool length	Tool length	Tool length	Tool length

The tool drawing data is defined by parameters No. 27350 to 27383 below. When these parameters are set to no values, simulation is performed by automatically inserting a constant value for each item.

27372	DRILL TIP LENGTH

(2-word type, FANUC standard settings=0)

DRILL TIP LENGTH : Holder width of straight tool for animated drawing

[Unit of data] mm input (INI (No.0000#2)=0) : 0.001(mm)
inch input (INI (No.0000#2)=1) : 0.0001(inch)
Remarks)
If 0 is set in case of metric input (INI (No.0000#2)=0), default data will be 50mm. If 0 is set in case of inch input(INI (No.0000#2)=1), default data will be 1.9865inch.

27373	F E MIL TIP LENGTH

(2-word type, FANUC standard settings=0)

F E MIL TIP LENGTH : Tip length of flat end mill tool for animated drawing

[Unit of data] mm input (INI (No.0000#2)=0) : 0.001(mm)
inch input (INI (No.0000#2)=1) : 0.0001(inch)
Remarks)
If 0 is set in case of metric input (INI (No.0000#2)=0), default data will be 50mm. If 0 is set in case of inch input(INI (No.0000#2)=1), default data will be 1.9865inch.

27374	TAP TIP LENGTH

(2-word type, FANUC standard settings=0)

TAP TIP LENGTH : Tip length of tapping tool for animated drawing

[Unit of data] mm input (INI (No.0000#2)=0) : 0.001(mm)
inch input (INI (No.0000#2)=1) : 0.0001(inch)
Remarks)
If 0 is set in case of metric input (INI (No.0000#2)=0), default data will be 50mm. If 0 is set in case of inch input(INI (No.0000#2)=1), default data will be 1.9865inch.

27375	CHANFER CUTTER ANG

(2-word type, FANUC standard settings=0)

CHANFER CUTTER ANG Tool angle of a chamfering tool for animated drawing

[Unit of data] 0.001°

27376	C SINK TIP LENGTH

(2-word type, FANUC standard settings=0)

C SINK TIP LENGTH : Tip length of counter sink tool for animated drawing

[Unit of data] mm input (INI (No.0000#2)=0) : 0.001(mm)
inch input (INI (No.0000#2)=1) : 0.0001(inch)
Remarks)
If 0 is set in case of metric input (INI (No.0000#2)=0), default data will be 26mm. If 0 is set in case of inch input(INI (No.0000#2)=1), default data will be 1.0236inch.

27377	C SINK CUT LENGTH

(2-word type, FANUC standard settings=0)

C SINK CUT LENGTH : Cutter length of counter sink tool for animated drawing

[Unit of data] mm input (INI (No.0000#2)=0) : 0.001(mm)
inch input (INI (No.0000#2)=1) : 0.0001(inch)
Remarks)
If 0 is set in case of metric input (INI (No.0000#2)=0), default data will be 50mm. If 0 is set in case of inch input(INI (No.0000#2)=1), default data will be 1.9685inch.

27378	C SINK SHANK LENGTH

(2-word type, FANUC standard settings=0)

C SINK SHANK LENGTH : Shank length of counter sink tool for animated drawing

[Unit of data] mm input (INI (No.0000#2)=0) : 0.001(mm)
inch input (INI (No.0000#2)=1) : 0.0001(inch)
Remarks)
If 0 is set in case of metric input (INI (No.0000#2)=0), default data will be 130mm. If 0 is set in case of inch input(INI (No.0000#2)=1), default data will be 5.1181inch.

27379	C SINK SHANK DIA

(2-word type, FANUC standard settings=0)

C SINK SHANK DIA : Shank diameter of counter sink tool for animated drawing

[Unit of data] mm input (INI (No.0000#2)=0) : 0.001(mm)
inch input (INI (No.0000#2)=1) : 0.0001(inch)
Remarks)
If 0 is set in case of metric input (INI (No.0000#2)=0), default data will be 32mm. If 0 is set in case of inch input(INI (No.0000#2)=1), default data will be 1.2598inch.

27380	B E MIL TIP LENGTH

(2-word type, FANUC standard settings=0)

B E MIL TIP LENGTH : Shank diameter of counter sink tool for animated drawing

[Unit of data] mm input (INI (No.0000#2)=0) : 0.001(mm)
inch input (INI (No.0000#2)=1) : 0.0001(inch)
Remarks)
If 0 is set in case of metric input (INI (No.0000#2)=0), default data will be 50mm. If 0 is set in case of inch input (INI (No.0000#2)=1), default data will be 1.9685inch.

27381	REAM TIP LENGTH

(2-word type, FANUC standard settings=0)

REAM TIP LENGTH : Tip length of reamer tool for animated drawing

[Unit of data] mm input (INI (No.0000#2)=0) : 0.001(mm)
inch input (INI (No.0000#2)=1) : 0.0001(inch)
Remarks)
If 0 is set in case of metric input (INI (No.0000#2)=0), default data will be 50mm. If 0 is set in case of inch input(INI (No.0000#2)=1), default data will be 1.9685inch.

27382	BORING TIP LENGTH

(2-word type, FANUC standard settings=0)

BORING TIP LENGTH : Tip length of boring tool for animated drawing

[Unit of data] mm input (INI (No.0000#2)=0) : 0.001(mm)
inch input (INI (No.0000#2)=1) : 0.0001(inch)
Remarks)
If 0 is set in case of metric input (INI (No.0000#2)=0), default data will be 50mm. If 0 is set in case of inch input(INI (No.0000#2)=1), default data will be 1.9685inch.

27383	F MIL TIP LENGTH

(2-word type, FANUC standard settings=0)

F MIL TIP LENGTH : Tip length of face mill tool for animated drawing

[Unit of data] mm input (INI (No.0000#2)=0) : 0.001(mm)
inch input (INI (No.0000#2)=1) : 0.0001(inch)
Remarks)
If 0 is set in case of metric input (INI (No.0000#2)=0), default data will be 63mm. If 0 is set in case of inch input(INI (No.0000#2)=1), default data will be 2.4803inch.

A.5.10 Parameters Related to Machining Simulation (Tool Path Drawing)

These parameters set the colors used during tool path drawing.

- Specify color setting data with a 6-digit number in the format of “xxyyzz”. (xx:Value for red, yy:Value for green, zz:Value for blue)
- The valid data range of each color value is 0 to 63. A value greater than 63 is interpreted as 63.
- For a number with less than six digits, all the unspecified digits are interpreted as high-order ones and are all assumed 0.
- If these parameters are set to 0, their standard colors are assumed.

14772	PARTCOLOR

PARTCOLOR Color of a part figure during tool path drawing.

14773	BLANKCOLOR

BLANKCOLOR Color of blank figure during tool path drawing.

14774	PATHCOLOR

PATHCOLOR Color of a tool path during tool path drawing.

14775	AXISCOLOR

AXISCOLOR Color of a coordinate axis during tool path drawing.

14776	BACKCOLOR

BACKCOLOR Background color during tool path drawing.

14785	PATHCOL1

PATHCOL1 : Color of the path of a rapid traverse tool during tool path drawing.

14786	PATHCOL2

PATHCOL2 : Color of the path of a cutting feed tool during tool path drawing.

14787	PATHCOL3

PATHCOL3 : Color of the path of a threading tool during tool path drawing.

A.5.11 Parameters Related to Machining Simulation (Animation Drawing)

A.5.11.1 Parameters related to AXIS of animation drawing

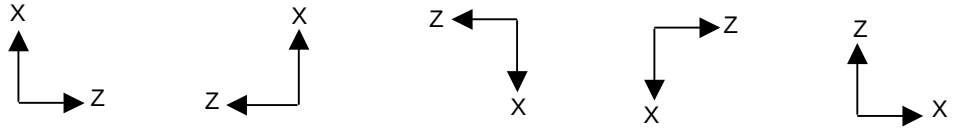
These parameters set the axial configuration of the machine.

14706	DRCTS1

(FANUC standard settings = 20 or 16)

DRCTS1 Number of Workpiece coordinate for main spindle

16 : Right-handed coordinate system, right = +Z, up = +X
17 : Right-handed coordinate system, right = -Z, up = +X
18 : Right-handed coordinate system, right = -Z, up = -X
19 : Right-handed coordinate system, right = +Z, up = -X
20 : Right-handed coordinate system, right = +X, up = +Z



16 17 18 19 20

FANUC standard settings = 20 :Vertical machining center/vertical lathe
FANUC standard settings = 16 :Horizontal machining center/horizontal lathe

14707	DRCTS2

(FANUC standard settings = 0 : Not use)

DRCTS 2 Number of Workpiece coordinate for sub spindle if existing

16 : Right-handed coordinate system, right = +Z, up = +X
17 : Right-handed coordinate system, right = -Z, up = +X
18 : Right-handed coordinate system, right = -Z, up = -X
19 : Right-handed coordinate system, right = +Z, up = -X
20 : Right-handed coordinate system, right = +X, up = +Z

These parameters set machining simulation.

14717	SMLCNO

(FANUC standard settings = 0)

SMLCNO : Rotate (Cs) axis number
[Valid data range] From 0 to the number of controlled axes

NOTE
1 In case of one Cs axis of main spindle, please be sure to set parameter No.14717 only. If no Cs axis or Animation option, it is unnecessary to set.
2 In case of two Cs axis with main and sub spindle, please be sure to set parameters No.27301, No.27302 and No.27312 #0 (SPA). And parameter No.14717 is used as working number for Cs axis, as follows.
3 When parameter No.27312#0 (SPA) = 1, parameter No. 14717 is rewritten to the value of parameter No.27301 or No.27302 by a spindle selection command (G1998). Therefore, in the case of parameter No.27312 #0=1, please be sure to specify Spindle selection command (G1998) before Milling cycles.
4 As to Spindle selection command (G1998), please refer to Section 3.5, “SETTING OF DATA FOR MACHINING SIMULATION” in the Part II.
5 Please be sure to set the intra-path relative axis number.

14718	SMLRTNO

(FANUC standard settings = 0)

SMLRTNO : Rotate (Cs) axis number, which tilts tool head or table
[Valid data range] From 0 to the number of controlled axes

NOTE
Parameter No.14718 is used in Simulation or Turning cycles. Therefore, if the machine has a rotate axis which tilts tool head or table, please sure to set this parameter.
If no such axis or animation option, it is not necessary to set.

	#7	#6	#5	#4	#3	#2	#1	#0
27001				TPC				

TPC When "A", "B", or "E" is set in the "ROTATION AXIS NAME" item on the "ROT. AXIS" tab on the cycle figure data input screen:
0 : Address "A", "B", or "E" is output as the tool path.
1 : Address "C" is output as the tool path.

NOTE
The MANUAL GUIDE *i* identifies the main spindle or subspindle according to the axis name entered in the "ROTATION AXIS NAME" item on the "ROT. AXIS" tab displayed on the cycle figure data input screen: More specifically, it identifies the main spindle when the rotation axis name is set to "C" or the subspindle when the rotation axis name is set to "A", "B", or "E". When the "ROTATION AXIS NAME" item is set to the name on the subspindle side, those on the subspindle side set by parameter No. 27011 and 27012 are output as the C axis clamp/unclamp M codes and address "C" is output as the tool path.

27301	ROTATE AXIS NO. (SP1)

(Byte type, FANUC standard settings = 0)

Rotation axis (Cs axis) number of Main spindle

[Related parameters] No.14717, No.27312#0 (SPA), #1 (INA), #2 (INS), No.27302

NOTE
1 Please be sure to specify Spindle selection command (G1998 S1) before each Milling cycles.
2 In case of one Cs axis of main spindle, please be sure to set parameter No.14717 only. If no Cs axis or Animation option, it is unnecessary to set.
3 In case of two Cs axis with main and sub spindle, please be sure to set parameters No.27301, No.27302 and No.27312 #0 (SPA). And parameter No.14717 is used as working number for Cs axis, as follows.
4 When parameter No.27312#0 (SPA) = 1, parameter No. 14717 is rewritten to the value of parameter No.27301 or No.27302 by a spindle selection command (G1998). Therefore, in the case of parameter No.27312 #0=1, please be sure to specify Spindle selection command (G1998) before Milling cycles.
5 As to Spindle selection command (G1998), please refer to Section 3.5, “SETTING OF DATA FOR MACHINING SIMULATION” in the Part II.
6 Please be sure to set the intra-path relative axis number.

27302	ROTATE AXIS NO. (SP2)

(Byte type, FANUC standard settings = 0)

Rotation axis (Cs axis) number of Sub spindle

[Related parameters] No.14717, No.27312#0 (SPA), #1 (INA), #2 (INS), No.27301

NOTE
1 Please be sure to specify Spindle selection command (G1998 S2) before each Milling cycles.
2 Please be sure to set the intra-path relative axis number.

27303	MTYPE

(Byte type, FANUC standard settings = 0)

MTYPE : Type of machine mechanism

Type	Controlled rotary axis	Parameter setting for the axis
0	Without a rotary axis Or With a tool rotary axis	Parameter No.14178 is a tool rotary axis.
1	With a workpiece table rotary axis	Parameter No.14178 is a workpiece table rotary axis.

[Related parameters] No.27301, No.27305, No.27307

NOTE
1 No. of workpiece rotated axis (usually C-axis) is set to the parameter No.14717. In the case of machine mechanism type=1, this parameter must not be set to workpiece table rotary axis No.
2 Drawing of machining simulation is not available for a machine which takes a tool rotary axis with a workpiece table rotary axis.
3 Drawing of machining simulation is not available for a machine which has the rotary axis slant for the basic three axis.
4 Machine mechanism type=2(With a workpiece table rotary axis) is not available with multi-path system or with a subspindle.
5 The axis to set to the parameter No.14718 is only a rotary axis about Y axis.

27332	QUILAXIS

(2-word type, FANUC standard settings=0)

MINMOV Axis number of the quill axis of the boring facing-and-drilling machine (for machining simulation) (only for the Series 30*i*/31*i*/32*i*)

[Unit of data] Integer

27333	CENTAXIS

(2-word type, FANUC standard settings=0)

MINMOV Axis number of the boring axis of the boring facing-and-drilling machine (for machining simulation) (only for the Series 30*i*/31*i*/32*i*)

[Unit of data] Integer (1 to 3)

NOTE
1 Set any of the three basic axes in parameter No. 27333.
2 In parameter No. 27332, you can set an axis parallel to the axis set in parameter No. 27333.
3 The axes set in parameter Nos. 27332 and 27333 must use the same increment system (IS-A/B/C/D/E and diameter/radius).
4 If the setting of parameter No. 27332 or No. 27333 is not greater than 0 or exceeds the maximum number of axes, quill axis drawing is not performed.
5 Changes to parameter Nos. 27332 and 27333 take effect only after the power is turned on the next time. When you change any of their settings, you should turn the power off and then back on.

27334	RECTAXIS

(2-word type, FANUC standard settings=0)

RECTAXIS Center axis for rotation of the workpiece in rectangular solid material (for machining simulation)
0 : Z axis
1 : Y axis

A.5.11.2 Parameters related to animation drawing in general

27300	SCALE OF THE BLANK

(Byte type, FANUC standard settings = 0)

Scale magnification for automatic scaling in the machining simulation function
[Unit of data] 10 %
[Valid data range] 0 to 20 (When 0 is set, 100% is assumed.)

27305	TBLDISTX

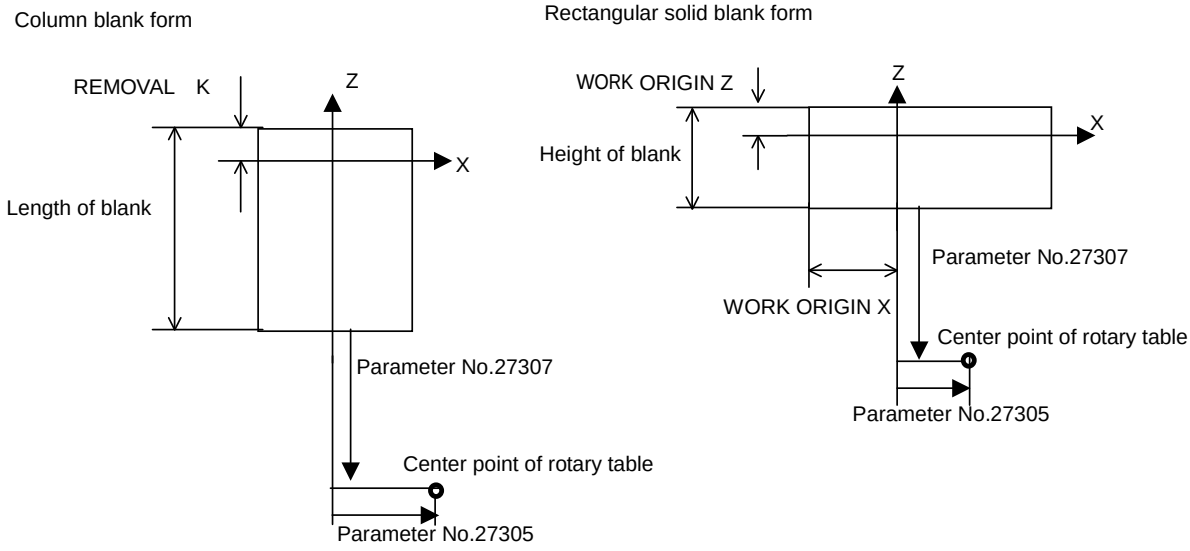
(2-word type, FANUC standard settings = 0)

TBLDISTX : In the case that type of machine mechanism is type 1(With a workpiece table rotary axis), distance (X-axis) from the rotary center point to the rotary standard point of drawing blank figure. The direction of + X axis is "+" for this data.
[Related parameters] No.27301, No.27303, No.27307

27307	TBLDISTZ

(2-word type, FANUC standard settings = 0)

TBLDISTZ : In the case that type of machine mechanism is type 1(With a workpiece table rotary axis), distance (Z-axis) from the rotary center point to the rotary standard point of drawing blank figure. The direction of - Z-axis is "+" for this data.
[Related parameters] No.27301, No.27303, No.27307



27310	#7 ERA	#6 TLD	#5 ATL	#4 WOK	#3 G10	#2 AXN	#1 GCN	#0 PRC

(FANUC standard settings = 00000000)

PRC 0 : The soft-key of [PAUSE] is displayed on the animation soft-key row.
1 : The soft-key of [PROCES] is displayed instead of the soft-key [PAUSE].

GCN 0 : The reference position for enlargement and reduction is the center of the graphic screen (new specification).
1 : The reference position for enlargement and reduction is the center of a figure to be drawn (conventional specification).
(This setting is valid for drawing during machining, machining simulation, and arbitrary figure drawing.)

AXN 0 : The coordinate axis name of animation drawing is not displayed (conventional specification).
1 : The coordinate axis name of animation drawing is displayed (new specification).

G10 0 : When the G10L50, L51, or L52 command is executed during simulation, the simulation is stopped (only for the Series 30*i*/31*i*/32*i*).
1 : When the G10L50, L51, or L52 command is executed during simulation, the simulation is not stopped and the command is ignored (only for the Series 30*i*/31*i*/32*i*).

WOK 0 : A blank figure is displayed on the tool path drawing screen or the machining drawing screen when the drawing screen is opened.
1 : A blank figure is displayed on the tool path drawing screen or the machining drawing screen when a G code for blank figure definition is executed.

ATL 0 : The synchronous drawing tab in the machining start window is not displayed.
1 : The synchronous drawing tab in the machining start window is displayed.

TLD 0 : The tool definition tab in the machining start command window is not displayed.
1 : The tool definition tab in the machining start command window is displayed.

ERA 0 : In machining simulation (tool path), the tool path is deleted after enlargement, reduction, movement, or rotation. (conventional specification)
1 : In machining simulation (tool path), the tool path is not deleted after enlargement, reduction, movement, or rotation. (new specification) (only for the Series 30*i*/31*i*/32*i*)

	#7	#6	#5	#4	#3	#2	#1	#0
27311	BDT	BD2	SPD		TOC	COL	ACD	ITF

(FANUC standard settings = 00000000)

ITF 0 : Animated simulation continues even when the interference check function issues an interference alarm.
1 : Animated simulation is stopped temporarily even when the interference check function issues an interference alarm.

NOTE
Parameters No. 27310 and No. 27311 can not be used in some editions.

ACD 0 : Even If the workpiece coordinate is changed in performing machining simulation, the drawing is performed on the same workpiece coordinate as one on the top of simulation.(The workpiece coordinate by drawing definition of blank form block is same as the continuous-state workpiece coordinate on the top of simulation.)
1 : If the workpiece coordinate is changed in performing machining simulation, the drawing is performed on it.

NOTE
1 This parameter is available only in drawing of machining simulation. Drawing during machining is always performed on the same workpiece coordinate as the top of machining.
2 In multi-paths system, this parameter is a common parameter at all paths.
3 In the case of bit 1 (ACD) of parameter No.27311 = 1, spindle at which animated simulation is performed is the last commanded by G1998 command at one of paths. In the case of bit 1 (ACD) of parameter No.27311 = 0, spindle at which animated simulation is performed is the last commanded by G1992 or G1998 command at one of paths.

COL 0 : The color of a tool path is not changed even when the tool is changed.
1 : The color of a tool path is changed each time the tool is changed.
TOC 0 : Tool drawing is not cleared when the T0 command is issued.
1 : Tool drawing is cleared when the T0 command is issued.
SPD 0 : Anime simulation with the following tools is executed by the speed priority
1 : Anime simulation with the following tools is executed by the cutting off priority.

- Drill
- Chamfering
- Flat end mill
- Ball end mill
- Tap
- Reamer
- Boring
- Face mill

NOTE
When setting as SPD=1, anime simulation with the above-mentioned tool is done by the cutting off priority. In this case, it executes so as not to generate the residual work, and the interference alarm is not generated.

BDT, BD2 Optional block signal BDTn is enabled or disabled as shown below in machining simulation (only for the Series 30*i*/31*i*/32*i*).
The optional block skip addition function is an optional function.

	BDT=0	BDT=1
BD2=0	All disabled	All enabled
BD2=1	BDT7/BDT8/BDT9 are enabled.	All enabled

NOTE
1 This parameter is enabled only in machining simulation.
2 If BDTn is enabled in machining simulation, the same signal is used in actual operation and machining simulation. Accordingly, when using BDTn in machining simulation, make sure that an optional skip is not used in actual operation and exercise extreme care when using it.

	#7	#6	#5	#4	#3	#2	#1	#0
27312	WCA			BRT		INS	INA	SPA

(FANUC standard settings = 00000000)

SPA 0 : The rotation axis number for simulation based on spindle 1 or spindle 2 with a subspindle attached is not switched by a spindle selection command.
1 : The rotation axis number for simulation based on spindle 1 or spindle 2 with a subspindle attached is switched by a spindle selection command.
<Related parameters : No.14717, No.27312#1, #2, No.27301, No.27302>

NOTE
When bit 0 (SPA) of parameter No.27312 = 1, No. 14717 is rewritten to the value of No.27301 (in the case of G1998 S1) or No.27302 (in the case of G1998 S2) by a spindle selection command (G1998). So, please be sure to specify a spindle selection command when bit 0 (SPA) of parameter No.27312 = 1.

INA 0 : Bit 2 (INS) of parameter No.27312 is not available.
1 : Bit 2 (INS) of parameter No.27312 is available.

NOTE
It is necessary to set bit 0 (SPA) of parameter No.27312 to 1.

INS 0 : After animated simulation, No.14717 (the rotation axis number for simulation) is rewritten to the value of No.27301 (axis number of Main spindle).
1 : After animated simulation, No.14717 (the rotation axis number for simulation) is rewritten to the value of No.27302 (axis number of Sub spindle).

NOTE
It is necessary to set bits 0 (SPA) and 1 (INA) of parameter No.27312 to 1.

BRT 0: Clockwise is treated as positive rotation direction of the axis specified in the parameter No.14718.
1: Counterclockwise is treated as positive rotation direction of the axis specified in the parameter No.14718.
WCA 0: Cannot be changed (disabled) the beginning angle for the blank cutout display.
1: Can be changed (enabled) the beginning angle for the blank cutout display.

	#7	#6	#5	#4	#3	#2	#1	#0
27314	#7		#5	#4	#3		SDY	SDZ

(FANUC standard settings=00000000)

SDZ 0 : The material figure about the Z axis is displayed on the material definition screen.
1 : The material figure about the Z axis is not displayed on the material definition screen.
SDY 0 : The material figure about the Y axis is not displayed on the material definition screen.
1 : The material figure about the Y axis is displayed on the material definition screen.
#3 0 : 2 windows machining simulation is not available.
1 : 2 windows machining simulation is available.
After setting this parameter, please power on again to make this function available.
#4 0 : Main spindle is displayed on the left side.
1 : Sub spindle is displayed on the left side.
#5 0 : Main spindle is displayed on the right side.
1 : Sub spindle is displayed on the right side.
Available combination of the parameter No.27314#4 and #5 is as follows.

No.27314#4	No.27314#5	Setting
0	1	OK
1	0	OK
1	1	NG
0	0	NG

One of the following hardware is necessary to use parameters #3,#4,#5.
(1) CNC with 15”color LCD
(2) CNC with Personal computer function with Windows XP
(3) CNC is connected to Personal computer with CNC display screen function.

#7 0 : Large window display function of animated drawing is not available.
1 : Large window display function of animated drawing is available.

One of the following hardware is necessary to use parameters #7.
(1) CNC with 15”color LCD or 10” color LCD B
(2) CNC with Personal computer function with Windows XP
(3) CNC is connected to Personal computer with CNC display screen function.

27320 **BLOCKNO**

(2-word type, FANUC standard settings=0)

BLOCKNO Number of blocks in which macro statements in the execution macro program are continuously processed during execution of a G4-digit command of the MANUAL GUIDE *i* in machining simulation. (only for the Series 0*i*-D, 30*i*/31*i*/32*i*)

[Unit of data] Integer (0 to 20)
(A value of 0 or 20 or greater is assumed to be 20.)

27323 **MINMOV**

(2-word type, FANUC standard settings=0)

MINMOV : Least command increment for the tool in machining simulation

[Unit of data] 1 [dot]

	#7	#6	#5	#4	#3	#2	#1	#0
14859								**CR1**

(Path-specific parameter)

CR1 0 : The forward rotation direction of the C axis (workpiece rotation axis) is defined as the direction from the +X axis to the +Y axis about the Z axis in machining simulation on spindle 1.
1 : The forward rotation direction of the C axis (workpiece rotation axis) is defined as the direction from the +X axis to the -Y axis about the Z axis in machining simulation on spindle 1.

27335 **Number of speed steps for the machining simulation speed level**

(Byte type, FANUC standard settings=0)

[Path attribute] Common to all paths

[Valid data range] 0-9

The number of speed steps for the machining simulation speed level can be specified between 2 and 9.

Example) If you want to have 3 levels of simulation speed , then set parameter No.27335 to 3.

NOTE
If 0, 1, or a negative value is set in parameter No.27335, then the function is disabled. If a value bigger than nine is set in parameter No.27335, then it is equivalent to set parameter No.27335 to 9.

27336 **Waiting time between blocks at lowest speed of simulation**

(Byte type, FANUC standard settings=0)

[Path attribute] Common to all paths

[Unit of data] 8msec

[Valid data range] 0-127
Waiting time between each block at the lowest speed of simulation. A value from 1 (8msec) to 127 (1016msec) can be specified as the waiting time between blocks at the lowest speed

NOTE
When 0 or a negative value is set in parameter No.27336 then it is equivalent to set parameter No.27336 to 64(512msc).

If the machining simulation is executed at the highest speed, then the waiting time between blocks becomes "0". It is the same execution speed as if the function is disabled.
If the machining simulation is executed at the lowest speed, then the waiting time becomes blocks specified by the parameter No.27336.
If the machining simulation is executed in an interval between the minimum and the maximum speed, then the waiting time is adjusted according to the selected speed step.

NOTE
If the setting value of the parameter No.27335 is bigger than the value of the parameter No.27336, then the number of speed steps for the machining simulation speed level is equivalent to be the same value with the setting in No.27336.

A.5.11.3 Parameters related to machining time display

27390	M-CODE AVERAGE TIME

(2-word type, FANUC standard settings=0)

M-CODE AVERAGE TIME Execution time of averaged M code
This parameter is used for calculating machining time.
[Unit of data] 10(msec)
[Valid data range] 0-99999999

27391	S-CODE AVERAGE TIME

(2-word type, FANUC standard settings=0)

S-CODE AVERAGE TIME Execution time of averaged S code
This parameter is used for calculating machining time.
[Unit of data] 10(msec)
[Valid data range] 0-99999999

27392	T-CODE AVERAGE TIME

(2-word type, FANUC standard settings=0)

T-CODE AVERAGE TIME Execution time of averaged T code
This parameter is used for calculating machining time.
[Unit of data] 10(msec)
[Valid data range] 0-99999999

A.5.11.4 Parameters related to color of animation drawing

These parameters set machining simulation animation colors.

- Specify color setting data with a 6-digit number in the format of “xxyyzz”. (xx:Value for red, yy:Value for green, zz:Value for blue)
- The valid data range of each color value is 0 to 63. A value greater than 63 is interpreted as 63.
- For a number with less than six digits, all the unspecified digits are interpreted as high-order ones and are all assumed 0.

14777	ANMCOL1

ANMCOL1 : Color of part figures during animation.

14778	ANMCOL2

ANMCOL2 : Color of material cut-off portions during animation.

14779	ANMCOL3

ANMCOL3 : Color of the tool during animation.

14780	ANMCOL4

ANMCOL4 : Color of the coordinate axes during animation.

14781	ANMCOL5

ANMCOL5 : Color of the background during animation.

If these parameters are set to 0, their standard colors are assumed.

A.5.11.5 Settings for Programmable data input

Machining simulation in the Series 0*i*-D, Series 30*i*/31*i*/32*i*is performed in the background. Therefore, data input by the following G10 commands is prohibited during machining simulation and, if the data input is specified, the stop state is entered.

- Pitch error compensation data input(G10L50)
- 3-dimensional error compensation data input(G10L51)
- Programmable parameter input(G10L52)

If this parameter is set, the following G10 commands specified during machining simulation can be ignored.

- Pitch error compensation data input(G10L50)
- 3-dimensional error compensation data input(G10L51)
- Programmable parameter input(G10L52)

Accordingly, if the custom macro (#3010) determines whether animation drawing is in progress, the G10 command specified during machining simulation can be ignored without the processing for skipping the G10 command being specified in the program.

	#7	#6	#5	#4	#3	#2	#1	#0
27310					NWP			

NWP 0 : When the pitch error compensation data input (G10L50), three-dimensional error compensation data input (G10L51), or programmable parameter input (G10L52) is specified during machining simulation, machining simulation is stopped.
1 : Even when the pitch error compensation data input (G10L50), three-dimensional error compensation data input (G10L51), or programmable parameter input (G10L52) is specified during machining simulation, machining simulation is not stopped and the command is ignored.

A.5.11.6 Parameter related to the T0 command during animation drawing

When offset cancellation is executed concurrently with the T0 command during execution of the operation of machining simulation (animation), the tool moves to a position not including the offset value.
Since the tool does not originally move during execution of offset cancellation, delete the tool at the timing of T0 command to prevent tool drawing from being performed at an invalid position.

	#7	#6	#5	#4	#3	#2	#1	#0
27311					TOC			

TOC 0 : Tool drawing is not deleted at offset number 0 (T0).
1 : Tool drawing is deleted at offset number 0 (T0).

The specifications of machining simulation are described below.

(1) For machining simulation (animation)
<a> Tool drawing is deleted.
When offset number 0 (T0) is detected, tool drawing is deleted.
<b> Tool drawing is restarted.
When the next offset number execution, the tool is displayed from the program specification point in the workpiece coordinate system.

(2) For machining simulation (tool path)
<a> Tool path is stopped.
When offset number 0 (T0) is detected, the tool path is not displayed.
<b> Tool path is restarted.
When the next move command execution, The tool path is displayed from the program specification point in the workpiece coordinate system.

A.5.12 Parameters Related to Tool Management Functions

These parameters are for the settings for tool management functions.

	#7	#6	#5	#4	#3	#2	#1	#0
14823	STS	TLD	LIA	LIF	TMG	MSR	TOF	ORT

ORT 0 : The screen for associating a tool number with a compensation number is not displayed.
1 : The screen for associating a tool number with a compensation number is displayed.
TOF 0 : The per-tool-number tool offset setting screen is not displayed.
1 : The per-tool-number tool offset setting screen is displayed.

MSR 0 : The compensation type, tool number, and group number input fields are not displayed on the measurement screen.
1 : The compensation type, tool number, and group number input fields are displayed on the measurement screen.

TMG 0 : The tool management data setting screen is not displayed.
1 : The tool management data setting screen is displayed.

LIF 0 : The tool life management data setting screen is not displayed.
1 : The tool life management data setting screen is displayed.

LIA 0 : The screen of Tool Life Data List is not displayed.
1 : The screen of Tool Life Data List is displayed.

TLD 0 : The soft-key for changing screen from MANUAL GUIDE *i* to Tool Management Data Table is not displayed.
1 : The soft-key for changing screen from MANUAL GUIDE *i* to Tool Management Data Table is displayed.

STS 0 : When a compensation type is specified, the compensation type is not displayed in the status display area.
1 : When a compensation type is specified, the compensation type is displayed in the status display area.

14824 **OFSRELTL**

OFSRELTL : Start compensation number to be associated with a tool number.
[Valid data range] 0 to 999
A warning is issued when this parameter is set to 0 or when an attempt is made to establish the correspondence between compensation numbers and tool numbers.

14825 **OFSTYPNO**

OFSTYPO : Number of compensation types (0 to 9)
[Valid data range] 0 to 9

14845 **TABORDER1**

TABORDER1 Number representing the order in which the tabs on the tool management screen are displayed
The order in which the tool management data, magazine information, tool offset, and tool information tabs are displayed can be changed.
Set a number from 1 to 4 that represents the order in which the tabs are displayed. This parameter needs to be set together with parameter No. 14846.

Place	Corresponding tab
Ones place	Tool management data tab
Tens place	Magazine information tab
Hundreds place	Tool offset tab
Thousands place	Tool information tab

The default value is 4321.

14846 **TABORDER2**

TABORDER2 Number representing the order in which the tabs on the tool management screen are displayed
The order in which the tool figure data, tool attachment history, and tool detachment history tabs are displayed can be changed.

Set a number from 5 to 7 that represents the order in which the tabs are displayed. This parameter needs to be set together with parameter No. 14845.

Place	Corresponding tab
Ones place	Tool figure data tab
Tens place	Tool attachment history tab
Hundreds place	Tool detachment history tab

The default value is 765.

	#7	#6	#5	#4	#3	#2	#1	#0
14855							TLM	

TLM 0 : The tool management data and magazine management data screens are not displayed on the MANUAL GUIDE *i* screen.
1 : The tool management data and magazine management data screens are displayed on the MANUAL GUIDE *i* screen.

A.5.13 Parameters Related to the Color Palette

A.5.13.1 Parameters related to the color palette for screen display

These parameters set the colors used to display screen components.
Color setting data for a screen display color number* (1 to 16)

- Specify color setting data with a 6-digit number in the format of “xxyyzz”. (xx:Value for red, yy:Value for green, zz:Value for blue)
- The valid data range of each color value is 0 to 63. A value greater than 63 is interpreted as 63.
- For a number with less than six digits, all the unspecified digits are interpreted as high-order ones and are all assumed 0.

14480	DSPCOL0

DSPCOL0 : Color of the special character.
If the value is 0, the color of the special character is red (630000).

14724	DSPCOL1

DSPCOL1 : Color of the cursor in the program display section on the base screen.

14725	DSPCOL2

DSPCOL2 : Used to display alarms in the system title display section.

14726	DSPCOL3

DSPCOL3 : Used to display the mode and line number in the system title display section.

14727	DSPCOL4

DSPCOL4 : Used to display the background of the alarm indication and the load meter.

14728	DSPCOL5

DSPCOL5 : Used to display the background of the title in the status display section, soft key characters, item name characters, and the scroll box of the scroll bar.

14729	DSPCOL6

DSPCOL6 : Used to display the serial numbers of selection screens.

14730	DSPCOL7

DSPCOL7 : Used to display the background of the system title display section.

14731	DSPCOL8

DSPCOL8 : Used to display characters in the title section.

14732	DSPCOL9

DSPCOL9 : Used to display the mode on the base screen and the material elements of arbitrary figures.

14733	DSPCOL10

DSPCOL10 : Used to display frames.

14734	DSPCOL11

DSPCOL11 : Used to display cells that cannot be edited by the process list edit function.

14735	DSPCOL12

DSPCOL12 : Used to display the background of the status display section.

14736	DSPCOL13

DSPCOL13 : Used to display the bar of the load meter.

14737	DSPCOL14

DSPCOL14 : Used to display the background of the cursor mode.

14738	DSPCOL15

DSPCOL15 : Used to display window shadows.

14739	DSPCOL16

DSPCOL16 : Used to display the background of the base screen.

If these parameters are set to 0, the following values are used as their respective initial values.

Parameter No.14724 = 636300 Yellow
Parameter No.14725 = 630000 Red

Parameter No.14726 = 003200	Green
Parameter No.14727 = 636300	Yellow
Parameter No.14728 = 000063	Blue
Parameter No.14729 = 420042	Purple (pinkish)
Parameter No.14730 = 323260	Light ultramarine
Parameter No.14731 = 636363	White
Parameter No.14732 = 163616	Bright green
Parameter No.14733 = 000000	Black
Parameter No.14734 = 121212	Very dark gray
Parameter No.14735 = 484848	Bright gray
Parameter No.14736 = 006363	Bright, light blue
Parameter No.14737 = 202060	Light ultramarine
Parameter No.14738 = 242424	Dark gray
Parameter No.14739 = 404040	Rather bright gray

A.5.13.2 Parameters related to the color palette for icon display

These parameters set the color palette colors used to display icons.
Color setting data for an ICOCOL* screen display color number* (1 to 16)

- Specify color setting data with a 6-digit number in the format of “xxyyzz”.
  (xx:Value for red, yy:Value for green, zz:Value for blue)
- The valid data range of each color value is 0 to 63. A value greater than 63 is interpreted as 63.
- For a number with less than six digits, all the unspecified digits are interpreted as high-order ones and are all assumed 0.

14740	**ICOCOL1**
14741	**ICOCOL2**
14742	**ICOCOL3**
14743	**ICOCOL4**
14744	**ICOCOL5**
14745	**ICOCOL6**
14746	**ICOCOL7**
14747	**ICOCOL8**
14748	**ICOCOL9**
14749	**ICOCOL10**
14750	**ICOCOL11**
14751	**ICOCOL12**
14752	**ICOCOL13**
14753	**ICOCOL14**
14754	**ICOCOL15**

14755	ICOCOL16

If these parameters are set to 0, the following values are used as their respective initial values.

Parameter No.14740 = 630000	Red
Parameter No.14741 = 003200	Green
Parameter No.14742 = 636300	Yellow
Parameter No.14743 = 000063	Blue
Parameter No.14744 = 323260	Light purple
Parameter No.14745 = 480040	Dark pink
Parameter No.14746 = 636363	White
Parameter No.14747 = 164016	Bright green
Parameter No.14748 = 000000	Black
Parameter No.14749 = 006060	Bright, light blue
Parameter No.14750 = 484848	Bright gray
Parameter No.14751 = 121224	Navy blue
Parameter No.14752 = 320000	Dark red
Parameter No.14753 = 161616	Dark gray
Parameter No.14754 = 404040	Rather bright gray
Parameter No.14755 = 000000	Black

A.5.13.3 Parameters related to the color palette for guidance display

These parameters set the colors used to display guides.
Color setting data for a GIDCOL* screen display color* (1 to 16)

- Specify color setting data with a 6-digit number in the format of “xxyyzz”.
  (xx:Value for red, yy:Value for green, zz:Value for blue)
- The valid data range of each color value is 0 to 63. A value greater than 63 is interpreted as 63.
- For a number with less than six digits, all the unspecified digits are interpreted as high-order ones and are all assumed 0.

14756	GIDCOL1

GIDCOL 1 : Material color.

14757	GIDCOL2

GIDCOL2 : Material frame color.

14758	GIDCOL3

GIDCOL3 : Tool color.

14759	GIDCOL4

GIDCOL4 : Tool frame color.

14760	GIDCOL5

GIDCOL5 : Tool path (cutting feed).

14761	GIDCOL6

GIDCOL6 : Tool path (rapid traverse).

14762	GIDCOL7

GIDCOL7 : Dimensional line.

14763	GIDCOL8

GIDCOL8 : Character color (unselected character).

14764	GIDCOL9

GIDCOL9 : Character color (selected character).

14765	GIDCOL10

GIDCOL10 : Portion to be cut.

14766	GIDCOL11

GIDCOL11 : Finishing allowance.

14767	GIDCOL12

GIDCOL12 : Reserved.

14768	GIDCOL13

GIDCOL13 : Reserved.

14769	GIDCOL14

GIDCOL14 : Explanation of measurement (static sentence)

14770	GIDCOL15

GIDCOL15 : Explanation of measurement (dynamic sentence)

14771	GIDCOL16

GIDCOL16 : Background color.

If these parameters are set to 0, the following values are used as their respective initial values.

Parameter No.14756 = 003200	Green
Parameter No.14757 = 000063	Blue
Parameter No.14758 = 000063	Blue
Parameter No.14759 = 006060	Light blue
Parameter No.14760 = 006060	Light blue
Parameter No.14761 = 600000	Red
Parameter No.14762 = 000063	Blue
Parameter No.14763 = 000063	Blue
Parameter No.14764 = 636300	Yellow
Parameter No.14756 = 484848	Bright gray
Parameter No.14766 = 404040	Rather bright gray
Parameter No.14767 = 000000	Black (reserved)

Parameter No.14768 = 000000 Black (reserved)
Parameter No.14769 = 636363 White
Parameter No.14770 = 006060 Light blue
Parameter No.14771 = 242424 Dark gray

A.5.14 Parameters Related to Handle Machining

These parameters are for the settings for handle machining.

	#7	#6	#5	#4	#3	#2	#1	#0
14858					PBH	TST	TLD	HDP

(FANUC standard settings = 00000000)

HDP The position of Guidance message is as follows.
0 : Approach handle is displayed at the left side, and Guidance handle is displayed at the right side.
1 : Approach handle is displayed at the right side, and Guidance handle is displayed at the left side.

TLD 0 : The tool position on the graphic window is displayed as a cross.
1 : The tool position on the graphic window is displayed as a circle. (Note)

NOTE
The radius is a tool nose radius (Lathe) or a tool radius (Machining center).

TST 0 : The tool information is specified by [NOSE R] and [VIRT.TIP] (Lathe), or [TOOL DIA.] (Machining center).
1 : The tool information is specified by [OFFSET NO.].

PBH 0 : Programming by handle is available.
1 : Programming by handle is not available.

27457 **Work coordinate system during handle machining**

Work coordinate system during handle machining is specified as follows.
6 : Right-handed coordinate system, right = +X, up = +Y
16 : Right-handed coordinate system, right = +Z, up = +X
19 : Right-handed coordinate system, right = +Z, up = -X

27458 **View coordinate of graphic window for handle machining**

View coordinate system of graphic window for handle machining is specified as follows.
6 : Right-handed coordinate system, right = +X, up = +Y
16 : Right-handed coordinate system, right = +Z, up = +X
19 : Right-handed coordinate system, right = +Z, up = -X

27460 **Head address of the R-signal used by the output data for auxiliary function commands**

The head address of the internal relay (R-signal) of the PMC used by the output data for auxiliary function commands is set. In output data, the data area in 2 bytes from the address, which is set to this parameter is needed. This parameter is valid when the value is set to 0.

27461	Time which is assumed that handle machining is stop (msec)

When the movement is stopped longer than the specified time after the handle is operated, it is assumed the pause of the block. This parameter is valid when the value is set to 0.

27462	Standard value which is assumed that spindle speed is changed (r.p.m)

When the variation of spindle speed goes over to the specified value, the real spindle speed is outputted automatically. This parameter is assumed to 1 when the value is set to 0.

A.5.15 Other Parameters

	#7	#6	#5	#4	#3	#2	#1	#0
3100						SKY		

(FANUC standard settings = 00000000)

SKY 0 : MANUAL GUIDE *i* is used excluding a small MDI unit.
1 : MANUAL GUIDE *i* is used with a small MDI unit.

NOTE
Because 8.4” LCD of Series 0i-D is a small MDI unit, please set No.3100#2 to 1.
Only when it uses a small MDI unit in Series 30i-B, please set No.3100#2 to 1.

	#7	#6	#5	#4	#3	#2	#1	#0
11354	HPM							

(FANUC standard settings = 00000000)

HPM 0 : High speed program management is disabled.
1 : High speed program management is enabled.

	#7	#6	#5	#4	#3	#2	#1	#0
11370			DHS					

(FANUC standard settings = 00000000)

DHS 0 : The status “NOTSAVE” is not displayed.
1 : The status “NOTSAVE” is displayed.

14489	STARTSIG

STARTSIG R signal used for the application activating signal is set.
The first digit is bit, and the second digit or more are the numbers of R signal.

Setting example) Parameter No.14489=1237 is bit 7 of signal R123.

	#7	#6	#5	#4	#3	#2	#1	#0
14791				IFF	SKY		ANC	

(FANUC standard settings = 00000000)

ANC 0 : Dynamic axis name changing is not available.
1 : Dynamic axis name changing is available.
SKY 0 : 12 numbers of horizontal soft-key are displayed on the screen of MANUAL GUIDE *i*.

1 : 7 numbers of horizontal soft-key are displayed on the screen of MANUAL GUIDE *i*.

IFF 0 : Registering number of fixed form sentences is specified by bit 4 of parameter No.14852.

1 : Registering number of fixed form sentences is as follows. The maximum number of tabs is 7, the maximum number of fixed form sentences per tab is 10, and the maximum number of characters per fixed form sentence is 256.

14843 **SUBPROJUDG**

SUBPROJUDG Number of blocks which is used to judge if a subprogram calling "M98 P****" is arbitrary figure data when a cursor is on the block of the subprogram calling in the program-editing screen.

= 0 All of the subprogram

= -1 None of the subprogram (The arbitrary figures of the subprogram are not drawn.)

14844 **HISTORYTIME**

HISTORYTIME A cycle time when the date and the time are recorded in an operating history.
The date and the time are recorded in an operating history every the set time. If the setting value is 0, it is treated as 10 minutes. However, the date and the time are not recorded if there is no operating data in the time.

[Valid data range] 0 to 1439

[Unit of data] 1 [min]

14860 **SPCLCHAR**

SPCLCHAR Special character for search function in the program display window on the base screen.
Please set the decimal number of ASCII.
If the value is 0, the special character is assumed to be "?".
Special characters are used with the following functions:

(1) In program display, the color of special characters is changed for display.

(2) If a fixed form sentence to be inserted includes special characters, a warning message is indicated.

14861 **UNDOBUF**

UNDOBUF : Specify the total size of the buffer for use with the redo and undo functions.
If 0 is input, 5 KB of buffer space is allocated. If a negative value is input, the redo and undo functions cannot be used.

[Valid data range] -127 to 127

[Unit of data] 1 [KB]

14863 **PAGETIME**

PAGETIME Hold time for cursor movement keys <↑> and <↓>
When cursor movement key <↑> or <↓> is pressed during program editing, the cursor moves one line down. When cursor movement key <↑> or <↓> is pressed and held for a certain period of time, however, the cursor moves on a page-by-page basis.
The time period for which <↑> or <↓> is held before the cursor is moved on a page-by-page basis is set by this parameter.

[Valid data range] -128 to 127

[Unit of data] Second

When this parameter is set to 0 or -128 to -2, the hold time is set to 1 second. When this parameter is set to -1, pressing and holding <↑> or <↓> does not enter the page feed mode.

14866	ANMPNTBF

ANMPNTBF Minimum time (in units of 32 ms only, only for the Series 30*i*/31*i*/32*i*) within which coordinates for drawing can be obtained in drawing during machining. When the time period is less than this value, coordinates for drawing are not obtained.

[Valid data range] 1 to 20: Once every 32 × ANMPNTBF ms at the maximum
20 or greater: Once every 640 ms at the maximum

NOTE
As internal processing during operation is heavy, this minimum time becomes much smaller than the maximum time.

14868	MILLSDSP

MILLSDSP This parameter specifies the subscript that follows S when the milling axis rpm display function by the servo motor is enabled.

	#7	#6	#5	#4	#3	#2	#1	#0
27313				ANM	HLP	IDC	DCM	DUC

(FANUC standard settings = 00000000)

DUC 0 : Decomposed cycle display function is available
1 : Decomposed cycle display function is not available.

DCM 0 : On the window of decomposed cycle display, the command with the same value as the previous command is not displayed by omission.
1 : On the window of decomposed cycle display, all of the command is displayed.

(Example of display in case of parameter No.27313#1=0)
```
G1 X10. Z-20. F30. ;
X40. Z-50. ;
G0 Z60. ;
```

(Example of display in case of parameter No.27313#1=1)
```
G1 X10. Z-20. F30. ;
G1 X40. Z-50. F30. ;
G0 X40. Z60. ;
```

IDC 0 : Input data check by simulation is available.
1 : Input data check by simulation is not available.

HLP 0 : Help window function according to each screen is not available.
1 : Help window function according to each screen is available.

(*) If the display unit is not PANEL *i*, please set this parameter to 0. If the parameter is set to 1, a problem such as the display speed slows might occur.

ANM 0 : Cooperation with animated software that is operated with PANEL i is not available.
1 : Cooperation with animated software that is operated with PANEL i is available.

27453	Rotation axis numbers whose name is extended

Axis numbers of rotation axes that are used for machining cycles in case the soft-key [A] or [C] is selected in figure data input screen when the rotation axis names are extended.
The each axis number has 2 digits that are specified by the followings.
"**¥¥" (** : Axis number that is used when [A] is selected, ¥¥ : Axis number that is used when [C] is selected)

(The example of setting)
Set No.27453 = 603 under the following all conditions.
(1) When soft-key [C] is selected, the 3rd axis should be used.
(2) When soft-key [A] is selected, the 6th axis should be used.

	#7	#6	#5	#4	#3	#2	#1	#0
27482				CMW		PPI	PTE	

(FANUC standard settings = 00000000)

PTE 0 : In Lathe/Machining center G code system switching function, tool path is drawn during Turning mode.
1 : In Lathe/Machining center G code system switching function, tool path is not drawn during Turning mode.

PPI 0 : When the programmable parameter input (G10L52 command) is executed during Machining simulation or NC program conversion,The parameter is not rewrote.
1 : When the programmable parameter input (G10L52 command) is executed during Machining simulation or NC program conversion,The parameter is rewrote.

CMW 0 : Calculation methods of workpiece origin offset value specifications of MANUAL GUIDE *i* (conventional specifications).
1 : Calculation methods of workpiece origin offset value the same specifications as the NC screen.

	#7	#6	#5	#4	#3	#2	#1	#0
27500								FSV

(FANUC standard settings = 00000000)

FSV In the case of using "Controlling conversational macro function screens" (#8510) in a Macro executor which is made by MTB, if #8510 is set in displaying a window screen of MANUAL GUIDE *i*
0 : #8510 is stored and when the window display is closed, #8510 setting is restored.
1 : #8510 is disregarded.
(Parameter common to paths: Enabled only for the Series 16*i*/18*i*/21*i*)

A.6 PARAMETER SETTINGS DEPENDING ON THE MACHINE TYPE

A.6.1 Machining Center

(1) Set following parameters for MANUAL GUIDE *i*.

No.	Value	Note
3106#2 (SBK)	In Series 0*i*-C/ 16*i*/18*i*/21*i*, set to 1 In Series 0*i*-D/ 30*i*/ 31*i*/32*i*, set to 0	<Series 0*i*-C/16*i*/18*i*/21*i*> In SINGLE BLOCK OPERATION, it enables to display distance data of the block to be executed next. It simultaneously enables to buffer 1 block. So if you stop the execution by SINGLE BLOCK signal, then edit the next block and re-start, the block before editing is executed. <Series 0*i*-D/30*i*/31*i*/32*i*> The distance data of the block to be executed next for SINGLE BLOCK OPERATION is displayed as the distance data of the current block regardless of the presence of buffering block to be executed next. However, only when the block to be executed next is buffered and it is immediately after switching from continuous operation to SINGLE BLOCK OPERATION, the distance data of the block to be next block is displayed.
3206#6	1	Bit 6 (DAK) of parameter No.3106, position display change for 3D conversion mode, is effective in MANUAL GUIDE *i* screen (It is unnecessary in Series 0*i* -D/30*i*/31*i*/32*i*.)
3112#0 (SGD)	0	Servo wave display is not available (It is unnecessary in Series 0*i* -D/30*i*/31*i*/32*i*.)
3201#6 (NPE)	1	M02,M30,M99 are not regarded as the end of resister
8661	40	SRAM size for application's backup variables (40KB) (It is unnecessary in Series 0*i* -D/30*i*/31*i*/32*i*.)
8662	23	SRAM size for application's backup (23KB) (It is unnecessary in Series 0*i* -D/30*i*/31*i*/32*i*.)
8664	40	SRAM size used by the MANUAL GUIDE *i* as the variable area (40KB) This parameter is set when multi C executor and the MANUAL GUIDE *i*. (NOTE: Set parameter No.8661 to 0. It is unnecessary in Series 0*i* -D/30*i*/31*i*/32*i*.)
8665	23	SRAM size used by the MANUAL GUIDE *i* as the disk area (23KB) This parameter is set when multi C executor and the MANUAL GUIDE *i*. (NOTE: Set parameter No.8662 to 0. It is unnecessary in Series 0*i* -D/30*i*/31*i*/32*i*.)
8650#1 (CNA)	1	not switched to alarm screen automatically during displaying C executor screen
8650#0 (RSK)	1	C executor transfer <RESET> key code to application
8701#6 (CTV)	1	TV check is skipped during edit operation (It is unnecessary in Series 0*i* -D/30*i*/31*i*/32*i*.)
8701#4	1	Vacant value of P code variables can be read. (It is unnecessary in Series 0*i* -D/30*i*/31*i*/32*i*.)
8781	0	DRAM size for user program of C executor must be zero (It is unnecessary in Series 0*i* -D/30*i*/31*i*/32*i*.)
9000#0 (SQN)	0	Debug parameter for macro program must be zero
14700#7 (MGI)=0	0	Enable to run MANUAL GUIDE *i*
14794-14795	(Note)	Assign function key for start MANUAL GUIDE *i*
9071	90	P code number for MANUAL GUIDE *i* (It is necessary in Series 0*i* -D/30*i*/31*i*/32*i*.)

No.	Value	Note
9072	20 or 40	Block number for dealing with Macro sentence continuously in Execute Macro program for MANUAL GUIDE *i*. (It is necessary in Series 0*i* -D/30*i*/31*i*/32*i*.) Recommended value of Series 30*i*/31*i*/32*i* = 20 Recommended value of Series 0*i*-D = 40

NOTE

Parameter No.14794

#0=1: The MANUAL GUIDE *i* is started by the <POS> key.
#1=1: The MANUAL GUIDE *i* is started by the <PRG> key.
#2=1: The MANUAL GUIDE *i* is started by the <OFS> key.
#3=1: The MANUAL GUIDE *i* is started by the <SYS> key.
#4=1: The MANUAL GUIDE *i* is started by the <MES> key.
#5=1: The MANUAL GUIDE *i* is started by the <GRP> key.

Parameter No.14795

#0=1: The MANUAL GUIDE *i* is started by the <CUSTOM> key (AUX screen)
(When the macro screen is not present, set bit 5 (CM1) of parameter No. 8652 to 1).
#1=1: The MANUAL GUIDE *i* is started by the <CUSTOM> key (MCR screen)
(When the macro screen is not present, set bit 6 (CM2) of parameter No. 8652 to 1).
#2=1: The MANUAL GUIDE *i* is started by the <CUSTOM> key (MENU screen)
(When the macro screen is not present, set bit 7 (CM3) of parameter No. 8652 to 1).
#3=1: The MANUAL GUIDE *i* is started by the blank key.
(Set bit 4 (CMF) of parameter No.8652 to 1)

(2) Set following parameters for displaying load meter. Following parameters depend on machining configuration. Refer to Parameter Manual (B-63530EN) for detail.

No.	Value	Note
1023	≠0	Servo axis number of control axis
2086	≠0	regular current parameter
2165	≠0	maximum value of amplifier current
3151-3158	≠0	Axis number to display load meter (Note)
4127	≠0	Displayed value in maximum load

NOTE

In Series 0*i*-D/30*i*/31*i*/32*i*, there are not the parameters from No.3154 to No.3158.

(3) Set following parameters for optimizing cycle menu and icons of Turning Cycle and Milling Cycle functions according to machine configuration.

No.	Value	Note
14702#0 (MT1)	→	0:Vertical, 1:Horizontal for icons
27003 #1 (MC1) to #0 (MC0)	→	Xxxxxx01: 3 axes (XYZ) Machining Center xxxxxx10: 5 axes (XYZCB) Machining Center
27103#7 (LT7), #3 (LT3) to #0 (LT0)	0xxx0000	0xxx0000: Machining Center

NOTE
1 In case parameters No.27003= xxxxxx00 & No.27103=0xxx0000, all of cycle menu is displayed
2 When the parameters No.27003 or No.27103 is set, please sure to push [F] key on NOW LOADING screen after Power ON. The necessary parameters are set automatically. (When the necessary parameters are set, the message of “NOW SETTING PARAMETERS” is displayed on the left below of NOW LOADING screen.)

(4) Set following parameters for output of polar coordinate interpolation command (G12.1) and cylindrical interpolation command (G7.1) in cycle motion. In case of no Cs axis, it is unnecessary to set.

No.	Value	Note
27000#2 (MC2)	→	0: G12.1/G13.1 are not output in cycle motion on XC plane 1: G12.1/G13.1 are automatically output in cycle motion on XC plane
27000#3 (MC3)	→	0: G7.1Cxx/G7.1C0 are not output in cycle motion on ZC plane 1: G7.1Cxx/G7.1C0 are automatically output in cycle motion on ZC plane

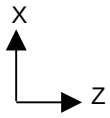
(5) Set following parameters for adapting tool motion to workpiece coordinate in MANUAL GUIDE *i* Animation function.

No.	Value	Note
14706	(Note)	Workpiece coordinate *

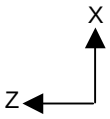
NOTE

Setting value for parameters No.14706

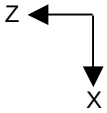
16: Right hand coordinate, right =+Z and upper =+X



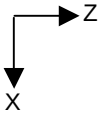
17: Right hand coordinate, right =-Z and upper =+X



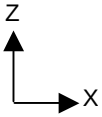
18: Right hand coordinate, right =-Z and upper =-X



19: Right hand coordinate, right =+Z and upper =-X



20: Right hand coordinate, right =+X and upper =+Z



(6) Set following parameters for adapting rotate (Cs) axis motion to workpiece coordinate. If no Cs axis, it is unnecessary to set.

No.	Value	Note
14717	→	Cs axis number

(7) Set following parameters for adapting rotate axis motion, which tilts tool head or table, to workpiece coordinate. If no such axis or Animation option, it is unnecessary to set.

No.	Value	Note
14718	→	Rotate axis number, which tilts tool head or table

(8) To support the function for switching spindle motor output characteristics (winding), set the following parameter to 0.

No.	Value	Note
14836#2 (LOS) =0	0	The load meter new display (support for output switching control) of the spindle is enabled.

NOTE

This function is supported only for the Series 30*i*/31*i*/32*i*.

A.7 M CODE OUTPUT IN CYCLE EXECUTING

Following M code is output in milling cycle

- M code for rigid tapping
- M code for clamping and unclamping rotate axis as “C”

A.7.1 M code for Rigid Tapping

(1) M code is output in following cycles in case that rigid tap is specified and parameter No.5200#0 is zero.

G1002 (Tapping Cycle for Machining Center by live tool)

(2) M code is output in following timing

G90
G00 X(x1) Y(y1) Z(I point)
G99
M29 S(spindle) (Note)
G84 Z(depth) R(R point) P(dwell) F(feed)
X(x2) Y(y2)
X(x3) Y(y3)
G80
G00 Z(I point)

NOTE
M code value depends on parameters No.5210 and No.5212.

A.8 SETTING FOR START-UP WHEN CNC PARAMETER IS UNSET

Please set up by the following methods when the CNC parameter is unset.

<1> The numerical control unit is started while pressing “M”+”0” key.
<2> After it starts, the following parameter is set.
No.8661 = 40
(This parameter sets the size of the static variable area that can share among each task for using in MANUAL GUIDE *i*.(40KB))
No.8662 = 23
(This parameter sets the size of the SRAM disk for using in MANUAL GUIDE *i*.(23KB))
<3> The numerical control unit is rebooted.

B LADDER PROGRAM CREATION

Appendix B, “LADDER PROGRAM CREATION”, consists of:

B.1 TOOL LIFE MANAGEMENT DATA.....450

B.1 TOOL LIFE MANAGEMENT DATA

Modify the ladder program so that the tool management data "Previous Notice Flag" of the tool being used is set to 1 ("NOTICED") when the NC outputs a tool life arrival notice signal.
For the "Previous Notice Flag" of tool management data, bit 7 of customization item 0 is used.

Item	Bit	Meaning	Description of data
Customization item 0	7	Previous Notice Flag	0:UN-NOTICE 1:NOTICED
	6		
	5		
	4		
	3		
	2		
	1		
	0		

C CUSTOMIZATION

Appendix C describes the method of customizing the MANUAL GUIDE *i* to suit the specifications of the machine tool builder.

Appendix C, “CUSTOMIZATION”, consists of:

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C.1 DISPLAYING VERTICAL SOFT KEYS FOR THE 15-INCH LCD

The horizontal soft keys that are frequently used can be displayed on each screen as vertical soft keys. This function can be used only for the CNC with a 15-inch LCD display of the Series 30*i*/31*i*/32*i*.

C.1.1 Displaying the Setting Screen

Press soft key [SETING] to open the screen for setting various types of data and select "SETTING OF VERTICAL SOFT KEY" on the basic setting tab. Then, the vertical soft key setting screen appears.

C.1.2 Operations

On this screen, a vertical soft key can be added or changed. The result is saved and recorded even after the power is turned off and back on again.

- Screens on which vertical soft keys can be used
  (a) Base screen in the EDIT mode
  (b) Base screen in the MEM mode
  (c) Base screen in the MDI mode
  (d) Machining simulation screen (animation)
  (e) Machining simulation screen (path drawing)
  (f) Drawing during machining screen
  (g) Arbitrary figure input screen

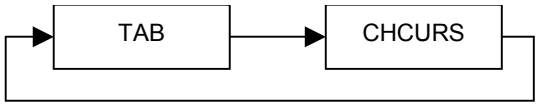
(1) Selecting the setting screen
Select the screen on which a vertical soft key is added or changed by switching between tabs.

(2) Selecting the position in which a soft key is added or changed
Select the position of a soft key to be added or changed from the soft key page items.

(3) Restoring the standard settings
Restore the standard settings of vertical soft keys on all screens as described below.

Press soft key [STAND.] to display the confirmation messages and soft keys in the figure below. When soft key [YES] is pressed, the standard settings of vertical soft keys are restored.
When soft key [NO] is pressed, the changes are canceled.
The standard settings are loaded from the definition file in the system.

(4) Changing the cursor state
Press soft key [CHCURS] to switch between cursor states.
(a) The cursor state is switched as shown below.



(b) The following text is displayed on the screen title.
TAB:



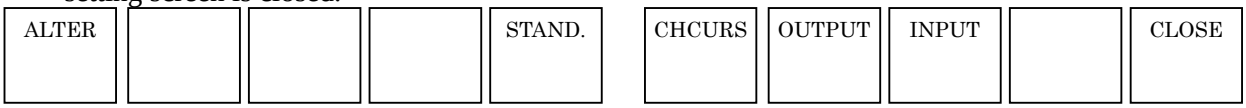
CHCURS:



(5) Finishing settings
If soft key [TO MNU] is pressed when bit 2 (RET) of parameter No. 14850 is 0, the setting screen is closed and the various data setting screen appears.



When bit 2 (RET) of parameter No. 14850 is 1, soft key [CLOSE] appears in place of soft key [TO MNU]. When soft key [CLOSE] is pressed, the various data setting screen does not appear after the setting screen is closed.



C.1.3 Changing Soft Keys

When soft key [ALTER] is pressed on the vertical soft key setting screen, the vertical soft key selection screen appears. The items of the vertical soft keys are displayed in the list on the screen of the selected tab. When an item is selected from the list and soft key [SELECT] is pressed, the soft key position on the selected screen. is replaced.
When (NONE) is selected from the list and soft key [SELECT] is pressed, the soft key position on the selected screen is replaced with (NONE)

C.1.4 Memory Card Input/Output

The settings of vertical soft keys can be output to or input from the outside. In the MDI, MEM, or RMT mode, however, the settings cannot be output or input.

(1) Output

When soft key [OUTPUT] is pressed, the name of an output file is entered, and soft key [OUTPUT] is pressed, the settings of vertical soft keys are output. While the settings are output, a message indicating the progress appears and key-in is disabled.
The output file name is VSKEY.DAT by default.

(2) File name

The file name to be input during external output to a memory card must observe the following rules.
When an extension is present, a file name is 12 characters long or less (including a period and extension). When an extension is not present, a file name is eight characters long or less.
An extension is three characters or less.
If a file name falls outside the range, an alarm message stating "Invalid file name" appears and no output can be performed.
The characters used as a name is the same as those of a program file.

(3) Input

When soft key [INPUT] is pressed, the name of an input file is selected from the list, and soft key [INPUT] is pressed, the settings of vertical soft keys are output. While the settings are input, a message indicating the progress appears and key-in is disabled.

C.1.5 File Format

The format of a definition file and memory card input/output file is given below.

(1) Definition file name

The definition file name used on the vertical soft key setting screen is shown below.

Definition file name : VSKMKUPF.TXT

(2) File format

The file format is given below.

<EDIT> ← Setting screen
P1=SET(1,ID100),SET(2,ID200) ← Soft key setting

- "Setting Screen" is a screen on which vertical soft keys are set.
- "Soft key Setting" includes the number of pages to be set and soft key information.
- When duplicate soft key settings are defined, the latest setting takes effects.
- A soft key in an undefined position is set to the vacant state.

(3) Setting screens

Setting screens are defined as follows.

<EDIT> Setting screens ← (Character string)

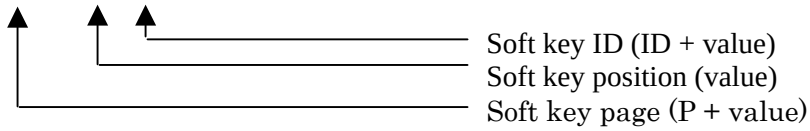
- Setting screens are described with a character string specified in < >.
- The setting screens are represented by the following character strings.
  EDIT : EDIT mode screen

MEM : MEM mode screen
MDI: MDI mode screen
ANIME : Machining simulation screen (animation)
TOOLPATH : Machining simulation screen (path drawing)
DRAWING : Drawing during machining screen
FREEFIG: Arbitrary figure input screen
PRCLAT : Process list screen
MULTIEDIT: definition format of multi part program editing function screen

(4) Soft key setting

The arrangement of a vertical soft key is defined as follow.

P1=SET(1,ID100),SET(2,ID200)



- A soft key page consists of character P followed by a page number. The page number ranges from 1 to 2.
- A soft key ID consists of a character ID followed by a soft key ID number. When a soft key ID of ID0 is specified, a vacant soft key is displayed.
- The soft key positions are 1 to 8, from the top to the bottom.

C.2 TOOL DATABASE ACCESS FUNCTION

The tool database access function reads or writes the tool database registered in the MANUAL GUIDE *i* from a custom macro. This function can be used to access tool data from a program. In addition to read/write operation, tool data can be initialized or tool data with a certain offset number can be copied to tool data with another offset number.

C.2.1 Basic Specifications

The following data can be read and written.

(1) Tool type
(2) Mounting
(3) Tool data 1 (e.g. Cutting edge angle)
(4) Tool data 2 (e.g. Tool angle)

NOTE
1 The tool name cannot be read or written.
2 This function requires the custom macro B option.
3 The read/write operation can be performed only from a custom macro and execution macro.
4 Bit 6 (TDA) of parameter No. 14852 must be set to 1 for this function to take effect.

The following functions can be used in addition to read/write operation.

(a) Copying tool data
All tool data (including the tool name) for one tool can be copied by specifying the offset number of the copy source and copy destination.
(b) Initializing tool data
One or all of tool data items can be initialized.

C.2.2 System Variables

Data access between a custom macro and the MANUAL GUIDE *i* is performed through system variables #5750 to #5756. When accessing tool data, input appropriate values to these system variables.
When the MANUAL GUIDE *i* detects an input value, it accesses tool data.
The system variables are described below.

#5750 : This parameter determines the operation of the tool database access function.

The MANUAL GUIDE *i* reads or writes a system variable when detecting the value of the variable. When an invalid value is input, a value of 3 is returned to the result variable. These valuables are automatically initialized when tool data is accessed.

This variable assumes the following values.

0 : No operation is performed.
1 : Read operation is performed.
2 : Write operation is performed.
3 : The tool data of copy source is copied to the copy destination.
4 : The tool data with the specified offset number is initialized.
5 : All tool data is initialized.

#5751 : Result value

This variable assumes the following values.

0 : Idling
1 : Normal end
2 : The number of registered tools exceeds the limit.
3 : The specified operation is incorrect.
4 : The specified offset number is incorrect.
5 : The specified tool type is incorrect. (only in writing)
6 : The mounting specification is incorrect. (only in writing)
7 : Data 1 is incorrectly specified. (only in writing)
8 : Data 2 is incorrectly specified. (only in writing)

#5752 : Specification of an offset number to be read or written or specification of the offset number of the copy destination.

When an incorrect value is input, a value of 4 is returned to the result variable during write or read operation.

#5753 : Specification of the tool type to be written or the tool type obtained by read operation. It specifies the offset number of the copy source.

When an incorrect value is input, a value of 5 is returned to the result variable during write operation. This variable assumes the following values.

10 : General tool
11 : Threading tool
12 : Grooving tool
13 : Round-nose tool
14 : Straight tool
20 : Drilling tool
21 : Chamfering tool
22 : Flat end mill tool
23 : Ball end mill tool
24 : Tapping tool
25 : Reamer tool
26 : Boring tool B
27 : Face mill tool

#5754 : Specification of the mount to be written or the mount obtained as a result of read operation. When an incorrect value is input, a value of 6 is returned during write operation.

#5755 : Specification of data 1 to be written or data 1 obtained as a result of read operation. When an incorrect value is input, a value of 7 is returned to the result variable during write operation.

#5756 : Specification of data 2 to be written or data 2 obtained as a result of read operation. When an incorrect value is input, a value of 8 is returned to the result variable during write operation.

C.2.3 Read

To read tool data, set #5752 to the offset number of tool data to be read and #5750 to 1. The tool data is output to #5753, #5754, #5755, and #5756.

C.2.4 Write

To write tool data, set #5752 to the offset number of tool data to be written and #5753, #5754, #5755, and #5756 to the tool data. Set #5750 to 2. The tool data is stored in the data storage area of the MANUAL GUIDE *i*.

C.2.5 Copy

To make a copy, set #5752 to the offset number of the copy destination, #5753 to the offset number of the copy source, and #5750 to 3. The tool data of the copy source is copied to the copy destination.

C.2.6 Initialization

To initialize tool data, set #5752 to the offset number of the tool data to be initialized and #5750 to 4. The tool data with this offset number is deleted from the data storage area of the MANUAL GUIDE *i*.

C.2.7 Initializing All Tool Data

To initialize all tool data, set #5750 to 5. All tool data is initialized.
For multi-path systems, however, only the tool data of the path for which a custom macro is executed is initialized.
For combined machines, only the tool data of the mode (turning side or milling side) for which a custom macro is executed is initialized.

C.3 CREATING A P-CODE USER MODULE

This section describes the method of creating a P-code macro user module when the MANUAL GUIDE *i* is used.

C.3.1 Creation Procedure

The files used to create a user module are shown below.

(1) Link files
Link files used to create a user module
- BJ00_U1.LNK For FANUC Series 16*i*/18*i*/21*i*-MB
- BJ25_U1.LNK For FANUC Series 0*i*-MC

(2) Header files
Header files used to create a user module
- F16IMP1M.MEX For FANUC Series 16i/18i/21i-MB
- F0IMP1M.MEX For FANUC Series 0i-MC

(3) Sample macro source file SAMPLE.SRC
This is a sample macro source file, which is not needed when you actually create a user module.

The model-specific procedure for creating a user module is as follows.

C.3.1.1 For the FANUC Series 16*i*/18*i*/21*i*-MB

(1) Change the following items of BJ00_U1.LNK.
(a) Compile parameter
P9000 => Set the bits according to the size of a file to be created.
If the compile parameter needs to be set for reasons of the user macro, set it.
(b) File name of a macro program to be incorporated
Replace "FILE=sample" with the file name of a macro program to be incorporated by the user.
(c) Series and edition
SERN =USR1
VERN =00ZZ
If the above lines are changed, the series and edition displayed on the SYSTEM configuration screen can be changed.
(2) Compile the macro program to be incorporated using MCOMP0.
(3) Link the compile object in BJ00_U1.LNK using MLINK.
(4) Create a user module (MEM file) using MMCARD.
(5) Install the user module in FROM using BOOTSYSTEM.

C.3.1.2 For the FANUC Series 0*i*-MC

(1) Change the following items of BJ25_U1.LNK.
(a) Compile parameter
P9000 => Set the bits according to the size of a file to be created.
If the compile parameter needs to be set for reasons of the user macro, set it.
(b) File name of a macro program to be incorporated
Replace “FILE=sample” with the file name of a macro program to be incorporated by the user.
(c) Series and edition
SERN =USR1
VERN =00ZZ
If the above lines are changed, the series and edition displayed on the SYSTEM configuration screen can be changed.
(2) Compile the macro program to be incorporated using MCOMP.
(3) Link the compile object in BJ25_U1.LNK using MLINK.
(4) Create a user module (MEM file) using MMCARD.
(5) Install the user module in FROM using BOOTSYSTEM.

C.3.1.3 For the FANUC Series 0*i*-D/30*i*/31*i*/32*i*

It is the same as the method of making a standard macro compiler, and refer to the programming manual of Macro Compiler/Executer for Series 0*i*-D or Series 30*i*/31*i*/32*i*.

C.3.2 Notes on Creation

C.3.2.1 For the FANUC Series 0*i*-MC,16*i*/18*i*/21*i*-MB

(1) Set the "Custom software size" option (J738) as necessary.
(2) Since the range from O1000 to O3999 is used by cycle machining etc., it cannot be used by a user macro.
(3) Common variables can be used by a user macro.

C.3.2.2 For the FANUC Series 0*i*-D/30*i*/31*i*/32*i*

(1) Since the range from O1000 to O3999 is used by cycle machining etc., it cannot be used by a user macro.

C.4 SOFT KEY IDS TO BE OPEN TO USERS

The following soft keys become available.

<Soft key ID:30000>

(1) For new creation

GENERA	METRIC	UNIFY	PIPE T	PIPE F				INSERT	CANCEL

(2) For modification

GENERA	METRIC	UNIFY	PIPE T	PIPE F				ALTER	CANCEL

- Address output as an argument of the G code: [R]
- Candidates for input:
  English
  [GENERAL, R1]
  [METRIC, R2]
  [UNIFY, R3]
  [PIPE T, R4]
  [PIPE F, R5]
- KeyMsg ID:
  [GENERA]:No.10000
  [METRIC]:No.10001
  [UNIFY]:No.10002
  [PIPE T]:No.10003
  [PIPE F]:No.10004
  [INSERT]:No.10005
  [CANCEL]:No.10006
  [ALTER]:No.10021

<Soft key ID:30001>

(1) For new creation

MALE	FEMALE							INSERT	CANCEL

(2) For modification

MALE	FEMALE							ALTER	CANCEL

- Address output as an argument of the G code: [W]
- Candidates for input:
  English
  [MALE SCREW, W1]
  [FEMALE SCR, W2]
- KeyMsg ID:
  [MALE]:No.10007
  [FEMALE]:No.10008
  [INSERT]:No.10005
  [CANCEL]:No.10006
  [ALTER]:No.10021

<Soft key ID:30002>

(1) For new creation

FORWRD	REVERS							INSERT	CANCEL

(2) For modification

FORWRD	REVERS							ALTER	CANCEL

- Address output as an argument of the G code: [M]
- Candidates for input:
  English
  [FORWRD, M1]
  [REVERSE, M2]
- KeyMsg ID:
  [FORWRD]:No.10009
  [REVERS]:No.10010
  [INSERT]:No.10005
  [CANCEL]:No.10006
  [ALTER]:No.10021

<Soft key ID:30002>

(1) For new creation

STOP	ORIENT							INSERT	CANCEL

(2) For modification

STOP	ORIENT							ALTER	CANCEL

* This soft key ID has no icons.

- Address output as an argument of the G code: [V]
- Candidates for input:
  English
  [STOP, V1]
  [ORIENT, V2]
- KeyMsg ID:
  [STOP]:No.10011
  [ORIENT]:No.10012
  [INSERT]:No.10005
  [CANCEL]:No.10006
  [ALTER]:No.10021

<Soft key ID:30004>

(1) For new creation

CSS	RPM							INSERT	CANCEL

(2) For modification

CSS	RPM							ALTER	CANCEL

* This soft key ID has no icons.

- Address output as an argument of the G code: [D]
- Candidates for input:
  English
  [CSS, D1]
  [RPM, D2]
- KeyMsg ID:
  [CSS]:No.10013
  [RPM]:No.10014
  [INSERT]:No.10004
  [CANCEL]:No.10005
  [ALTER]:No.10021

<Soft key ID:30005>

(1) For new creation

ON	SECOND	OFF						INSERT	CANCEL

(2) For modification

ON	SECOND	OFF						ALTER	CANCEL

* This soft key ID has no icons.

- Address output as an argument of the G code: [C]
- Candidates for input:
  English
  [ON, C1]
  [SECOND, C2]
  [OFF, C3]
- KeyMsg ID:

[ON]:No.10015
[SECOND]:No.10017
[OFF]:No.10016
[INSERT]:No.10005
[CANCEL]:No.10006
[ALTER]:No.10021

<Soft key ID:30006>

(1) For new creation

ON	O-STOP	OFF						INSERT	CANCEL

(2) For modification

ON	O-STOP	OFF						ALTER	CANCEL

* This soft key ID has no icons.

- Address output as an argument of the G code: [B]
- Candidates for input:
  English
  [ON, B1]
  [O-STOP, B2]
  [OFF, B3]

- KeyMsg ID:
  [ON]:No.10018 (= No.538 [ON])
  [O-STOP]:No.10020
  [OFF]:No.10019
  [INSERT]:No.10005
  [CANCEL]:No.10006
  [ALTER]:No.10021

<Soft key ID:30007>

(1) For new creation

ON	OFF							INSERT	CANCEL

(2) For modification

ON	OFF							ALTER	CANCEL

* This soft key ID has no icons.

- Address output as an argument of the G code: [H]
- Candidates for input:
  English
  [ON, H1]
  [OFF, H2]
- KeyMsg ID:
  [ON]:No.10018
  [OFF]:No.10019
  [INSERT]:No.10005
  [CANCEL]:No.10006
  [ALTER]:No.10021

C.5 MESSAGE FILES

It was necessary to create a customize file for each language to display user-specific messages, but messages in another language became available. For example, customized English messages can be used in place of Swedish messages.

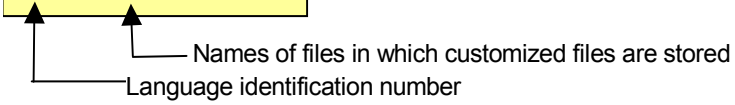
(1) Language-specific data management files

The language-specific data management file manages the locations of customize files used for individual languages.

(Example)

The language-specific data management file, localu.txt, for customize files includes the following.

```
0,CEX3DATU
1,CEX3DATU
2,CEX3DATU
3,CEX3DATU
4,CEX3DATU
5,CEX3DATU
6,CEX3DATU
7,CEX3DATU
8,CEX3DATU
9,CEX3DATU
B,CEX3DATU
C,CEX3DATU
D,CEX3DATU
E,CEX3DATU
F,CEX3DATU
G,CEX3DATU
```



(2) Addition of new format

A new format has been added to the formats of localu.txt.
To display a message in a language for which no customize file is crated using a specified language, make settings as shown below.

(a) Format
Language identification number, Language identification number

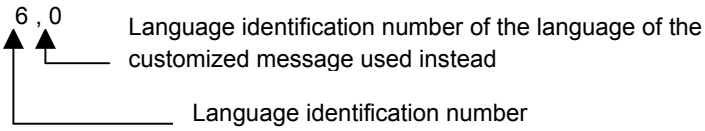
(b) Language identification number

Language identification number	Language
0	English
1	Japanese
2	German
3	French
4	Italian
5	Spanish
6	Swedish
7	Czech
8	Portuguese
9	Polish
A	Russian

Language identification number	Language
B	Chinese (Traditional characters)
C	Chinese (Simplified characters)
D	Korean
E	Turkish
F	Dutch
G	Hungarian
H	Bulgarian
I	Romanian

(Example)

A customized English message is displayed in place of a Swedish message.




```
0,CEX3DATU
1,CEX3DATU
2,CEX3DATU
3,CEX3DATU
4,CEX3DATU
5,CEX3DATU
6,0
7,CEX3DATU
8,CEX3DATU
9,CEX3DATU
```

Set 0 (language number of English).

C.6 SOFT-KEY CUSTOMIZATION FUNCTION

It is possible to replace the soft key position by using user customize data file.

C.6.1 Structure of Customize Data File

The structure of the customize file for replace the soft keys is in the following.

File Name : SKCSTMU.TXT

C.6.2 Format

The structure of the customize data file is as follows.

<EDIT>

(Example)

P1=SET(1,ID100), SET(2,ID200)

Set sentence.
Soft key page number

Soft key page number

Specify the page number of replaced soft keys in the ‘Soft key Page Number’.

See sentence

The format of set sentence as follows.

SET(Soft key Position, Soft key ID)

Soft key ID (ID + Number)
Soft key Position (1 to 10)

If there is no Set sentence, the default soft key is displayed

1) Soft Key Position

The soft key specified in ‘Soft Key Position’ is replaced.
If ‘Soft key Position’ is not specified, the standard soft key is displayed in this position.
The soft key position is counted from left side.



1 2 3 4 5 6 7 8 9 10

2) Soft key ID

Soft key ID can be specified by ‘ID+number’.
If ‘0’ is specified as the number, the soft key is not displayed in specified position.
The numbers of Soft key ID are described in “Soft key ID List”.

（Examples）

P1=SET(1,ID100)
P2=SET(1,ID200), SET(2,ID0)

Soft key is not displayed.
Replace to the ID200 soft key

C.6.3 Note

- The soft keys which are displayed in the base screen in EIDT mode are only possible to customize.
- Don’t specify the soft key which can not be used because of the optional function. If these soft keys are specified, the system doesn’t work correctly.

C.6.4 Soft Key ID List

The following soft-key ID can be used.

ID	KEY	ID	KEY
0	[ ] （Not Display）	126	[FIXFRM]（M）
444	[NEWPRG]	249	[ALTER]
498	[O LIST]	232	[M CODE]
27	[SEARCH BACKWARD]	45	[REWIND]
28	[SEARCH FORWARD]	74	[CHSIZE]
26	[O SRCH]	73	[LINENO]
23	[COPY]	888	[EDWORK]
22	[CUT]	1117	[MLT WIN]
24	[DELETE]	811	[CH PATH]

ID	KEY	ID	KEY
121	[KEYPST]	16	[G-CONT]
69	[PASTE]	1166	[UNDO]
260	[START]（T）	1167	[REDO]
233	[CYCLE]（T）	845	[WK SET]
261	[END]（T）	846	[T-OFS]
262	[MESCYC]（T）	1116	[TL MNG]
123	[FIGURE]（T）	9	[SETING]
105	[FIXFRM]（T）	48	[BG END]
122	[START]（M）		
17	[CYCLE]（M）		
124	[END]（M）		
420	[MESCYC]（M）		
234	[FIGURE]（M）		

C.7 MANUAL GUIDE *i* WINDOW CALL FUNCTION (for only Series 30*i*/31*i*/32*i*-B)

C.7.1 Outline

User application made by C language executer can call the following MANUAL GUIDE *i* windows.

(1) Blank figure selection window
(2) Blank figure data input window (Fixed form and arbitrary free figure)
(3) Free figure input window (Free figure creation window and range selection window)
(4) Machining simulation window
(5) NC program conversion window

C.7.1.1 Available display

This function is available on the following display.

(1) 10.4” size display
(2) 15” size display (Including the mode of enlarging 10.4” display)
(3) 19” size display (Including the mode of enlarging 15” display)

C.7.1.2 Necessary option

All the following options are necessary.

(1) MANUAL GUIDE *i* window call function : S779
(2) Macro executor + C language executor : J734

C.7.1.3 Necessary parameter setting

Set one of the following parameters when MANUAL GUIDE *i* window call function is used, and prohibit to use standard screen of MANUAL GUDIE *i*.

(1) Parameter No.14794#0 - #7 = 0 and parameter No.14795#0 - #7 = 0
(2) Parameter No.14703#6 = 1 and signal G62.4 = 0

C.7.2 Method of Calling

User application calls MANUAL GUIDE *i* window by the following function.

[Syntax]

```
#include <oscall.h>
unsigned short ret = os_chng_appstr( long info, char *str_info );
```

[Arguments]

info Information to be transmitted.
str_info The start address of string information to be transmitted.
(Note) The maximum number of string characters is 256 including a NULL character at the end.

[Return]

ret = 0 Success.
= EC_TSKID(2) MANUAL GUIDE *i* doesn’t exist.

[Description]

When this function is called, screen control is switched to MANUAL GUIDE *i* .
And user application can transmit information to MANUAL GUIDE *i* .
Set values of info and str_info are different from the kind of windows to be called.
(Refer after next chapter about details.)

[Example of code]

```
unsigned short    ret;
long              info;
char              str_info[256];

info = -1000100;
memset(str_info, '¥0', sizeof(str_info));
strcpy(str_info, “B=25836,S=O1001”);

ret = os_chng_appstr( info, &str_info );
```

C.7.2.1 Blank figure selection window

Specify the following info and str_info in case of calling blank figure selection window.

Specifying value of argument info

Specify one of the following values in a table to argument info.

(1) Method of output when free contour cylinder blank figure window is selected

info value	Method of output
-1000100	Select method of output on free figure data input window of MANUAL GUIDE *i* .
-1000200	(In case of EDIT mode) G code is inserted to the current program. (In case of MEM mode) G code is inserted to the specified main program.
-1000300	(In case of EDIT mode) M98 code is inserted to the current program and G code is inserted to the specified subprogram that is newly made in the same folder of the current program. (In case of MEM mode) M98 code is inserted to the specified main program and G code is inserted to the specified subprogram that is newly made in the same folder of the main program.
-1000400	(In case of EDIT mode) M98 code is inserted to the current program and G code is inserted to the specified subprogram that is newly made in the common program folder. (In case of MEM mode) M98 code is inserted to the specified main program and G code is inserted to the specified subprogram that is newly made in the common program folder.

Specifying value of argument info_str

(1) In case of EDIT mode

Link the string characters <1> and <2> with a comma (,) between them.
And specify start address of the linked string characters to info_str.

<1> B=?????

These are string characters to specify a block number where G code or M98 is inserted.
????? are string characters that specify a previous block number to insert G code or M98.
And it is impossible to insert G code or M98 to top of program.

<2> S=!!!!!
These are string characters to specify a subprogram name if G code is inserted to a subprogram. !!!!! are string characters of the subprogram name.

(Example of specifying)
str_info = “B=25836,S=O1001”

(2) In case of MEM mode
Link the string characters <1>, <2> and <3> with a comma (,) between them.
And specify start address of the linked string characters to info_str.
<1> B=?????
Same as EDIT mode.
<2> S=!!!!!
Same as EDIT mode.
<3> P=XXXXX
These are string characters to specify a program name to insert G code or M98 code. XXXXX is main program name of full path format.

(Example of specifying)
str_info = “B=25836,S=O1001,P=//CNC_MEM/USER/PATH1/O1000”

C.7.2.2 Blank figure data input window (Fixed form and arbitrary free figure)

Specify the following info and str_info in case of calling blank figure data input window.

Specifying value of argument info

Specify one of the following values in a table to argument info.

(1) Kind of figure

info value	Kind of figure
-2000*00	Kind of figure isn’t selected (For displaying a window to alter blank figure)
-2000*01	Rectangular
-2000*02	Cylinder (around Z)
-2000*03	Hollow Cylinder (around Z)
-2000*04	Prism (around Z)
-2000*05	Hollow prism (around Z)
-2000*12	Cylinder (around X)
-2000*13	Hollow cylinder (around X)
-2000*14	Prism (around X)
-2000*15	Hollow prism (around X)
-2000*22	Cylinder (around Y)
-2000*23	Hollow cylinder (around Y)
-2000*24	Prism (around Y)
-2000*25	Hollow prism (around Y)
-2000*50	Free contour cylinder (around Z)
-2000*51	Free contour cylinder (around X)
-2000*52	Free contour cylinder (around Y)

(“*” in upper table indicates a arbitrary number of 0 - 9)

(2) Method of output when free contour cylinder blank figure window is selected

info value	Method of output
-20001**	Select method of output on free figure data input window of MANUAL GUIDE *i* .
-20002**	(In case of EDIT mode) G code is inserted to the current program. (In case of MEM mode) G code is inserted to the specified main program.
-20003**	(In case of EDIT mode) M98 code is inserted to the current program and G code is inserted to the specified subprogram that is newly made in the same folder of the current program. (In case of MEM mode) M98 code is inserted to the specified main program and G code is inserted to the specified subprogram that is newly made in the same folder of the main program.
-20004**	(In case of EDIT mode) M98 code is inserted to the current program and G code is inserted to the specified subprogram that is made in the common program folder. (In case of MEM mode) M98 code is inserted to the specified main program and G code is inserted to the specified subprogram that is newly made in the common program folder.

("*" in upper table indicates a arbitrary number of 0 - 9)

Specifying value of argument info_str

(In case of displaying a window to newly create)

Same as blank figure selection window.

(In case of displaying a window to alter)

(1) In case of EDIT mode

Link the string characters <1> and <2> with a comma (,) between them.
And specify start address of the linked string characters to info_str.

<1> B=?????

These are string characters to specify a block number of G code to alter.
????? are string characters of a block number G code or M98 exists.

<2> S=!!!!!

Same as blank figure selection window.

(Example of specifying)

str_info = "B=25836,S=O1001"

(2) In case of MEM mode

Link the string characters <1>, <2> and <3> with a comma (,) between them.
And specify start address of the linked string characters to info_str.

<1> B=?????

Same as EDIT mode.

<2> S=!!!!!

Same as EDIT mode.

<3> P=XXXXX

Same as blank figure selection window.

(Example of specifying)

str_info = "B=25836,S=O1001,P=//CNC_MEM/USER/PATH1/O1000"

C.7.2.3 Free figure input window (Free figure creation window and range selection window)

Specify the following info and str_info in case of calling free figure input window.

Specifying value of argument info

Specify one of the following values in a table to argument info.

(1) Kind of figure

This is specified by first and second figure of info.

In first figure, specify figure type.

info value	Kind of figure
-3000**0	Figure type isn't selected (For displaying a window to alter free figure)
-3000**1	Figure for turning
-3000**2	Face
-3000**3	Convex
-3000**4	Concave
-3000**5	Groove
-3000**6	Open
-3000**7	Figure for pocket machining
-3000**8	Figure for emboss machining

("*" in upper table indicates a arbitrary number of 0 - 9)

In second figure, specify kind of plane

info value	Kind of plane	Figure for turning cycle/ milling cycle
-3000*00	Kind of plane isn't selected (For displaying a window to alter free figure)	
-3000210	Kind of plane isn't selected (For displaying a window to alter range selection)	Figure for turning cycle
-3000*21	ZX plane	
-3000*3*	XY plane	Figure for milling cycle
-3000*4*	YZ plane	
-3000*5*	XC plane figure (Using interpolation)	
-3000*6*	ZC plane figure (Using interpolation)	
-3000*7*	XA plane figure (Using interpolation)	

("*" in upper table indicates a arbitrary number of 0 - 9)

(2) Method of output when free figure input window is selected

info value	Method of output
-30001**	Select method of output on free figure data input window of MANUAL GUIDE *i* .
-30002**	(In case of EDIT mode) G code is inserted to the current program. (In case of MEM mode) G code is inserted to the specified main program.
-30003**	(In case of EDIT mode) M98 code is inserted to the current program and G code is inserted the specified subprogram that is newly made in the same folder of the current program. (In case of MEM mode) M98 code is inserted to the specified main program and G code is inserted to the specified subprogram that is newly made in the same folder of the main program.

info value	Method of output
-30004**	(In case of EDIT mode) M98 code is inserted to the current program and G code is inserted to the specified subprogram that is newly made in the common program folder. (In case of MEM mode) M98 code is inserted to the specified main program and G code is inserted to the specified subprogram that is newly made in the common program folder.

("*" in upper table indicates a arbitrary number of 0 - 9)

Specifying value of argument info_str

(In case of displaying a window to newly create)

(1) In case of EDIT mode

Link the string characters <1> and <2> with a comma (,) between them.
And specify start address of the linked string characters to info_str.

<1> B=?????

These are string characters to specify a block number where G code or M98 is inserted.
????? are string characters that specify a previous block number to insert G code or M98.
And it is impossible to insert G code or M98 to top of program.

<2> S=!!!!!

These are string characters to specify a subprogram name if G code is inserted a subprogram.
!!!!! are string characters of the subprogram.

(Example of specifying)

str_info = "B=25836,S=O1001"

(2) In case of MEM mode

Link the string characters <1>, <2> and <3> with a comma (,) between them.
And specify start address of the linked string characters to info_str.

<1> B=?????

Same as EDIT mode.

<2> S=!!!!!

Same as EDIT mode.

<3> P=XXXXX

These are string characters to specify a program name to insert G code or M98 code.
XXXXX is main program name of full path format.

(Example of specifying)

str_info = "B=25836,S=O1001,P=//CNC_MEM/USER/PATH1/O1000"

(In case of displaying a window to alter)

(1) In case of EDIT mode

Link the string characters <1> and <2> with a comma (,) between them.
And specify start address of the linked string characters to info_str.

<1> B=?????

These are string characters to specify a block number of G code to alter.
????? are string characters of a block number of G code or M98.

<2> S=!!!!!

Same as blank figure selection window.

(Example of specifying)

str_info = "B=25836,S=O1001"

(2) In case of MEM mode
Link the string characters <1>, <2> and <3> with a comma (,) between them.
And specify start address of the linked string characters to info_str.
<1> B=?????
Same as EDIT mode.
<2> S=!!!!!
Same as EDIT mode.
<3> P=XXXXX
Same as blank figure selection window.
(Example of specifying)
str_info = “B=25836,S=O1001,P=//CNC_MEM/USER/PATH1/O1000”

C.7.2.4 Machining simulation window

Specify the following info and str_info in case of calling machining simulation window.

Specifying value of argument info

Specify info = -4000000.

Specifying value of argument info_str

Link the string characters <1>, <2> and <3> with a comma (,) between them.
And specify start address of the linked string characters to info_str.
<1> P1=?????
These are string characters to specify a default program to simulate (for first path).
????? is a program name of full path format.
If the address isn’t specified, current program is a default program to simulate.
<2> P2=!!!!!
These are string characters to specify a default program to simulate (for second path).
The method of specifying !!!!! is same as first path.
<3> P3=XXXXX
These are string characters to specify a default program to simulate (for third path).
The method of specifying !!!!! is same as first path.

(Example of specifying)
str_info = “P1=//CNC_MEM/USER/PATH1/O1000,P2=//CNC_MEM/USER/PATH2/O1000”

(Note)
User application can specify default programs for simulating by switching current programs.

C.7.2.5 NC program conversion window

Specify the following info and str_info in case of calling machining simulation window.

Specifying value of argument info

Specify info = -5000000.

Specifying value of argument info_str

Make string characters <1>, and specify a start address of them.
<1> P=?????
These are string characters to specify a default destination.
????? is a program name of full path format. If the address isn’t specified, specify it on MANUAL GUIDE *i* sreen.
(Example of specifying)
str_info = “P=//CNC_MEM/USER/PATH1/O2000”

C.7.2.6 Number of string characters to be transmitted is 256 or more

If the number of string characters to be transmitted is 256 or more, because all of characters can't be transmitted by only one time, transmit the information by switching screen control from user application to MANUAL GUIDE *i* (3 times is maximum).
Specify the following info and str_info in this case.

Specifying value of argument info

(1) Case of left string characters to be transmitted to MANUAL GUIDE *i* is 256 or more
Specify info=-99000000.

(2) Case of left string characters to be transmitted to MANUAL GUIDE *i* is less than 256
Specify the above-mentioned number corresponding to called window.

Specifying value of argument info_str

(1) Case of left string characters to be transmitted to MANUAL GUIDE *i* is 256 or more
Divide string characters to be transmitted to MANUAL GUIDE *i* to 255 string characters and left string characters, and specify the start address of first 255 string characters to argument info_str.

(2) Case of left string characters to be transmitted to MANUAL GUIDE *i* is less than 256.
Specify start address of the left string characters to argument info_str.

Specifying value of argument info after specifying a part of string characters

If user application specifies info=-99000000, MANUAL GUIDE *i* returns screen control to user application after getting the string characters. After this, specify info=-99000000 if 256 or more string characters should be transmitted again. Specify the above-mentioned number corresponding to called window if less than 256 string characters should be transmitted.

C.7.3 Method to get return information from MANUAL GUIDE *i*

When called window is closed by an operation or the method of calling is wrong, screen control is returned to user application. When screen control is returned to user application, get return information (ret_info) from MANUAL GUIDE *i* by the following function. User application redraws the screen or a message according to ret_info.

[Syntax]

```
#include <oscall.h>
long ret_info = os_get_chngappstr( char *str_info );
```

[Arguments]

str_info Don't use.

[Return]

ret_info Return information from MANUAL GUIDE *i* .
But when NC screen is displayed before calling this function,0 is returned.

[Description]

User application gets information from MANUAL GUIDE *i* .

[Example of code]

```
long    ret_info;
char    str_info[256];
ret_info = os_get_chngappstr( &str_info );
```

C.7.3.1 Blank figure selection window

(1) "ret_info" value when MANUAL GUIDE *i* window is called normally

ret_info	Meaning
-10000	G4 digit code is inserted normally.
-10001	[CANCEL] soft-key is pushed.
-10002	The mode is switched.
-10003	The mode is switched during output of program. The output is interrupted.
-10004	Path selection signal is switched.
-10005	Path selection signal is switched during output of program. The output is interrupted.
-10006	[POSITION] function key is pushed.
-10007	[PROGRAM] function key is pushed.
-10008	[OFFSET/SETTING] function key is pushed.
-10009	[SYSTEM] function key is pushed.
-10010	[MESSAGE] function key is pushed.
-10011	[GRAPH] function key is pushed.
-10012	[CUSTOM1] function key is pushed.
-10013	[COSTOM2] function key is pushed.
-10014	Unexpected error occurred on MANUAL GUIDE *i*.

(2) "ret_info" value when MANUAL GUIDE *i* window isn't called normally

ret_info	Meaning
-10101	An alarm is issued.
-10102	The emergency stop state is present.
-10103	The mode isn't correct. (Other than EDIT or MEM)
-10104	The seventh digit of "info" aren't within the range. (Other than 1-5 or 99)
-10106	Specified program name to output G code is not correct.
-10107	Main program name isn't specified. (On only MEM mode)
-10108	MEM mode : Specified program doesn't exist. EDIT mode : No program is selected.
-10109	The specified main program is being operated. (On only MEM mode)
-10110	The specified main program is protected. (On only MEM mode)
-10111	The specified main program is selected as FG program. (On only MEM mode)
-10112	Block number isn't specified.
-10113	The block number doesn't exist in the program to be edited.
-10114	A character that isn't digit is used in block number.
-10115	Subprogram name isn't specified.
-10116	Subprogram name isn't correct.
-10117	A program that has same name as specified subprogram already exists.
-10118	The specified program number is protected.
-10119	Subprogram can't be made because the number of program is maximum.

C.7.3.2 Blank figure data input window (Fixed form and arbitrary free figure) and Free figure input window (Free figure creation window and range selection window)

(1) "ret_info" value when MANUAL GUIDE *i* window is called normally

ret_info	Meaning
-10000	G4 digit code is inserted normally.
-10001	[CANCEL] soft-key is pushed.
-10002	The mode is switched.
-10003	The mode is switched during output of program. The output is interrupted.
-10004	Path selection signal is switched.
-10005	Path selection signal is switched during output of program. The output is interrupted.
-10006	[POSITION] function key is pushed.
-10007	[PROGRAM] function key is pushed.
-10008	[OFFSET/SETTING] function key is pushed.
-10009	[SYSTEM] function key is pushed.
-10010	[MESSAGE] function key is pushed.
-10011	[GRAPH] function key is pushed.
-10012	[CUSTOM1] function key is pushed.
-10013	[COSTOM2] function key is pushed.
-10014	Unexpected error occurred on MANUAL GUIDE *i*.

(2) "ret_info" value when MANUAL GUIDE *i* window isn't called normally

ret_info	Meaning
-10101	An alarm is issued.
-10102	The emergency stop state is present.
-10103	The mode isn't correct. (Other than EDIT or MEM)
-10104	The seventh digit of "info" aren't within the range. (Other than 1-5 or 99)
-10105	The specified kind of figure isn't correct.
-10106	Specified program name to output G code is not correct.
-10107	Main program name isn't specified. (On only MEM mode)
-10108	MEM mode : Specified program doesn't exist. EDIT mode : No program is selected.
-10109	The specified main program is being operated. (On only MEM mode)
-10110	The specified main program is protected. (On only MEM mode)
-10111	The specified main program is selected as FG program. (On only MEM mode)
-10112	Block number isn't specified.
-10113	The block number doesn't exist in the program to be edited.
-10114	A character that isn't digit is used in block number.
-10115	Subprogram name isn't specified.
-10116	Subprogram name isn't correct.
-10117	A program that has same name as specified subprogram already exists.
-10118	The specified program number is protected.
-10119	Subprogram can't be made because the number of program is maximum.
-10120	The specified code doesn't exist in the specified position. (On only alteration)
-10121	The format of figure command to be altered isn't correct. (On only alteration)
-10122	The element number of arbitrary figure to be altered exceeds 90 or 300. (On only alteration)

C.7.3.3 Machining simulation window

(1) "ret_info" value when MANUAL GUIDE *i* window is called normally

ret_info	Meaning
-10001	[GRPOFF] soft-key is pushed.
-10002	The mode is switched. (other than EDIT or MEM)
-10006	[POSITION] function key is pushed.
-10007	[PROGRAM] function key is pushed.
-10008	[OFFSET/SETTING] function key is pushed.
-10009	[SYSTEM] function key is pushed.
-10010	[MESSAGE] function key is pushed.
-10011	[GRAPH] function key is pushed.
-10012	[CUSTOM1] function key is pushed.
-10013	[COSTOM2] function key is pushed.
-10014	Unexpected error occurred on MANUAL GUIDE *i*.

(2) "ret_info" value when MANUAL GUIDE *i* window isn't called normally

ret_info	Meaning
-10102	The emergency stop state is present.
-10103	The mode isn't correct. (Other than EDIT or MEM)
-10104	The seventh digit of "info" aren't within the range. (Other than 1-5 or 99)
-10123	A program to be simulated doesn't exist.
-10124	A program to be simulated already specified on other path.

C.7.3.4 NC program conversion window

(1) "ret_info" value when MANUAL GUIDE *i* window is called normally

ret_info	Meaning
-10001	[RETURN] soft-key is pushed.
-10002	The mode is switched.
-10104	The seventh digit of "info" aren't within the range. (Other than 1-5 or 99)
-10006	[POSITION] function key is pushed.
-10007	[PROGRAM] function key is pushed.
-10008	[OFFSET/SETTING] function key is pushed.
-10009	[SYSTEM] function key is pushed.
-10010	[MESSAGE] function key is pushed.
-10011	[GRAPH] function key is pushed.
-10012	[CUSTOM1] function key is pushed.
-10013	[COSTOM2] function key is pushed.
-10014	Unexpected error occurred on MANUAL GUIDE *i* .

(2) "ret_info" value when MANUAL GUIDE *i* window isn't called normally

ret_info	Meaning
-10101	An alarm is issued.
-10102	The emergency stop state is present.
-10103	The mode isn't correct. (Other than EDIT or MEM)
-10104	The seventh digit of "info" aren't within the range. (Other than 1-5 or 99)
-10123	A program to be simulated doesn't exist.
-10125	The program name of destination isn't correct.
-10126	A program that has same name as program of destination already exists.
-10127	The number of program specified as destination is protected.
-10128	A program of destination can't be made because the number of program is maximum.
-10129	NC is executing operation.

C.7.3.5 Number of string characters to be transmitted is 256 or more

"ret_info" value when MANUAL GUIDE *i* window isn't called normally

ret_info	Meaning
-10100	255 characters are given.
-10104	The seventh digit of "info" aren't within the range. (Other than 1-5 or 99)

C.7.4 Display and Operation

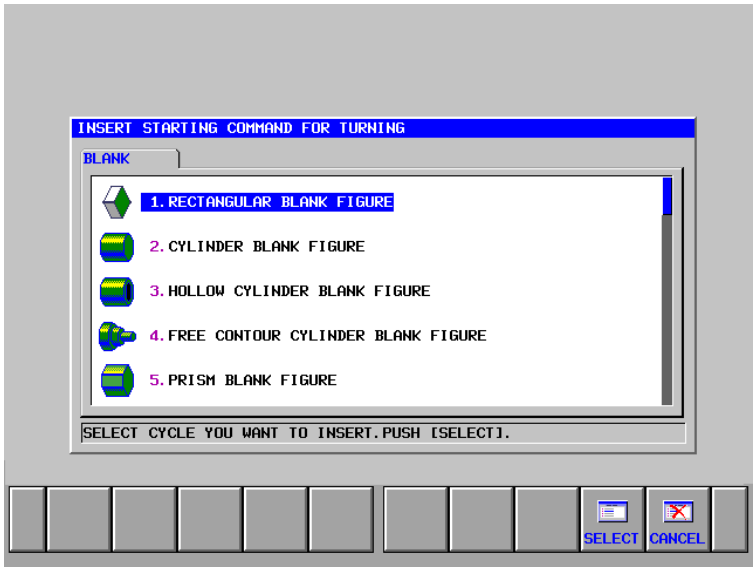
C.7.4.1 Blank figure selection window

(1) Background display (Excluding soft-keys)
The background that user application drew before calling MANUAL GUIDE *i* window is left.

(2) Operation to return screen control to user application
When one of the following operations is done, MANUAL GUIDE *i* window is closed and screen control is returned to user application.

<1> Push one of the following soft-keys after blank figure is selected, figure data is inputted and G code is inserted to program.
  (a) [CREATE] soft-key in free contour cylinder blank figure window
  (b) [OK] soft-key in "METHOD OF FREE FIGURE CREATION" window
  (c) [INSERT] soft-key in blank figure data input window for fixed form
<2> Push Soft-key [CANCEL].
<3> Change the mode.
<4> Change the path selection signal.
<5> Push one of the function keys.

(3) Example of blank figure selection window



C.7.4.2 Blank figure data input window (Fixed form and arbitrary free figure)

(1) Background display (Excluding soft-keys)
The background that user application drew before calling MANUAL GUIDE *i* window is left.

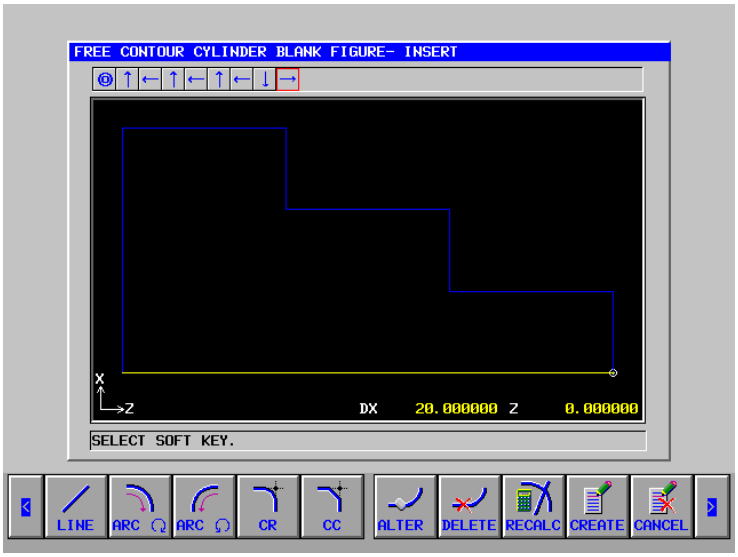
(2) Operation to return screen control to user application
When one of the following operations is done, MANUAL GUIDE *i* window is closed and screen control is returned to user application.

<1> Push one of the following soft-keys after figure data is inputted and G code is inserted to program.
  (a) [CREATE] soft-key in free contour cylinder blank figure window
  (b) [OK] soft-key in “METHOD OF FREE FIGURE CREATION” window
  (c) [INSERT] soft-key in blank figure data input window for fixed form
<2> Push soft-key [CANCEL].
<3> Change the mode.
<4> Change the path selection signal.
<5> Push one of the function keys.

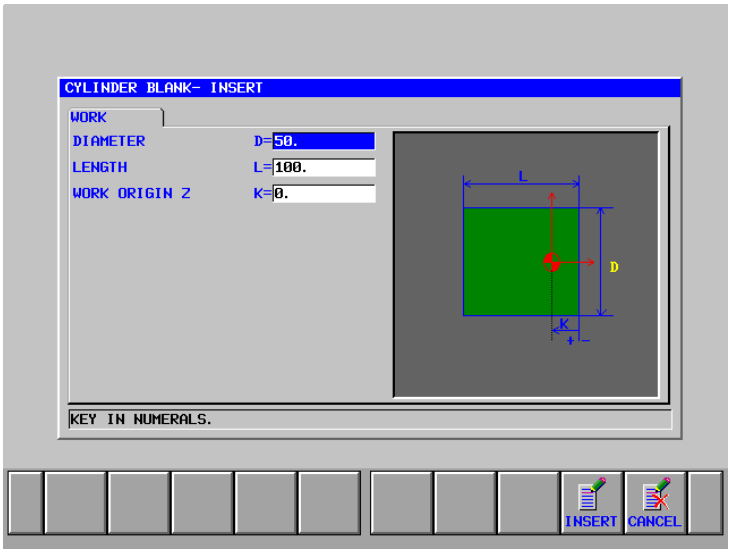
(3) Note of inserting G code
G code is not outputted to a subprogram if fixed form input screen is called.
In case of EDIT mode, G code is inserted to the current program. In case of MEM mode, G code is inserted to the selected main program.

(4) Example of blank figure selection window

<1> Example of free contour cylinder blank figure data input window



<2> Example of fixed form blank figure data input window



C.7.4.3 Free figure input window (Free figure creation window, range selection window)

(1) Background display (Excluding soft-keys)

The background that user application drew before calling MANUAL GUIDE *i* window is left.

(2) Operation to return screen control to user application

When one of the following operations is done, MANUAL GUIDE *i* window is closed and screen control is returned to user application.

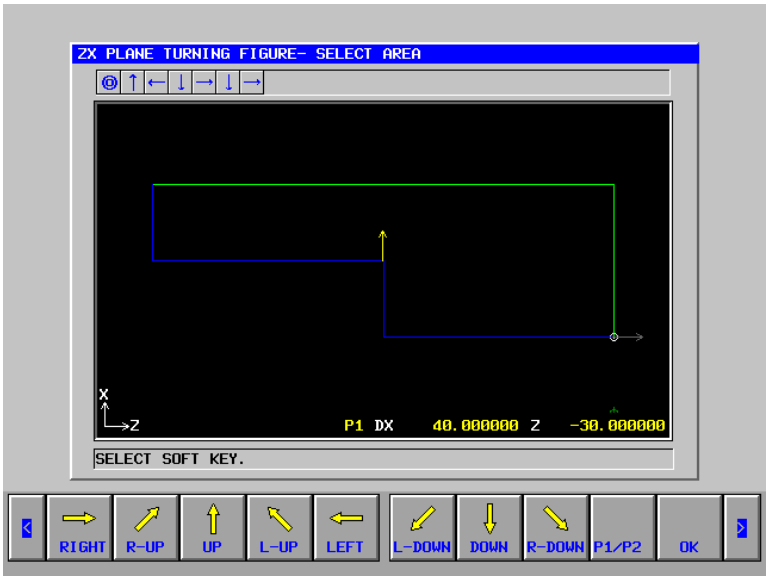
<1> Push one of the following soft-keys and G code is inputted or edited.
- (a) [CREATE] soft-key in free figure creation window
- (b) [OK] soft-key in “METHOD OF FREE FIGURE CREATION” window
- (c) [INSERT] soft-key in blank figure data input window for fixed form

<2> Push soft-key [CANCEL].

<3> Change the mode.
<4> Change the path selection signal.
<5> Push one of the function keys.

(3) Note of operating free figure creation window
Though [NXTFIG] soft-key can be used on standard screen, it can't be used when the window is called.
In case plural figures must be inserted continuously, call figure input window again after screen control is returned to user application.

(4) Example of range selection window

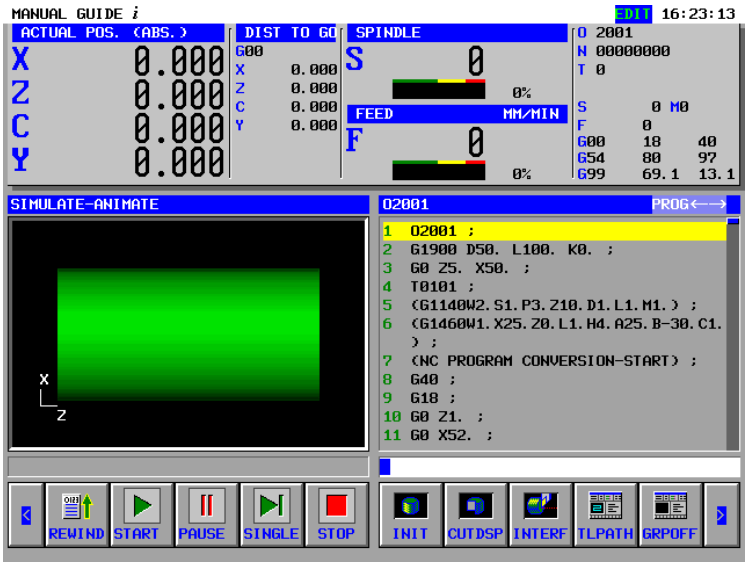


C.7.4.4 Machining simulation window

(1) Operation to return screen control to user application
When one of the following operations is done, MANUAL GUIDE *i* window is closed and screen control is returned to user application.

<1> Push soft-key [GRPOFF].
<2> Change the mode to other than EDIT or MEM.
<3> Push one of the function keys.

(2) Example of machining simulation window



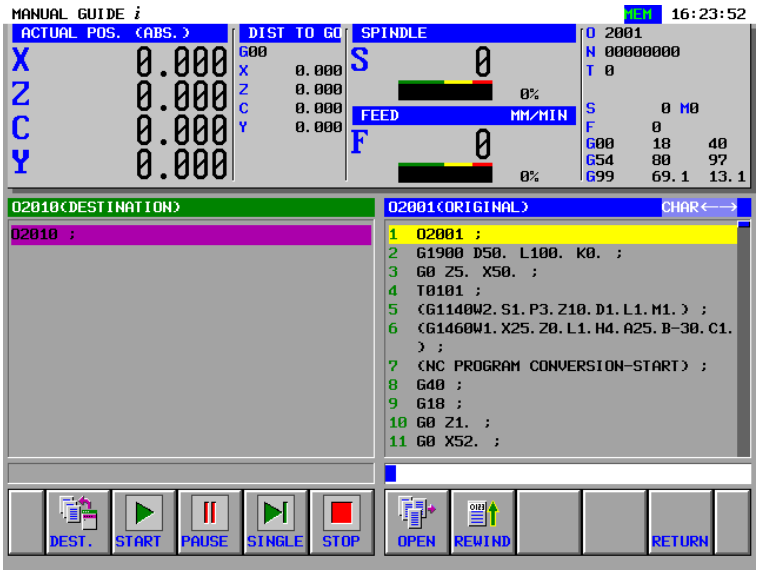
C.7.4.5 NC program conversion window

(1) Operation to return screen control to user application
When one of the following operations is done, MANUAL GUIDE *i* window is closed and screen control is returned to user application.

<1> Push soft-key [RETURN].
<2> Change the mode.
<3> Change the path selection signal.
<4> Push one of the function keys.

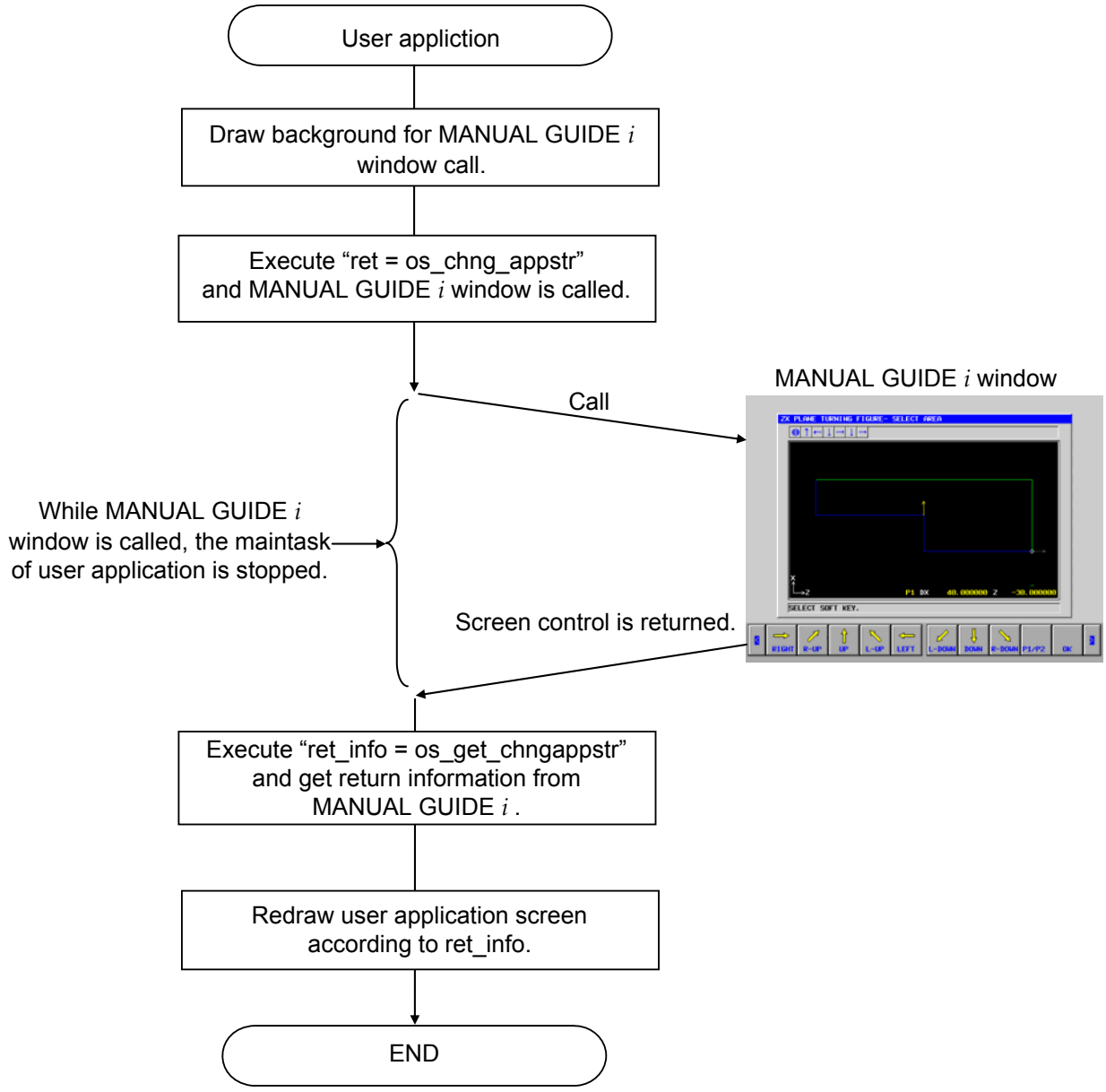
(2) Operation for displaying NC program conversion window
<1> When user application didn't specify default program name of destination
The window "CREATE PROGRAM FOR DESTINATION" is displayed. Input program name and NC program conversion window is displayed.
<2> When user application specified default program name of destination
NC program conversion window is displayed. (The window "CREATE PROGRAM FOR DESTINATION" is not displayed.)

(3) Example of NC program conversion window



C.7.5 Method of Making User Application

When this function is used, call the function “os_chng_appstr” and “os_get_chngappstr” as follows.



C.7.6 Method of Color Pallet Setting

Set graphic of user application to VGA if this function is used. And, because color pallet must be set to the same as MANUAL GUIDE *i* , set color pallet of user application as follows.

(1) When the set value of “Parameter Number” in the following table is other than 0
Set value of parameter to “Index of color pallet”.
(2) When the set value of “Parameter Number” in the following table is 0
Set “Color value” to “Index of color pallet”.

Index of color pallet	Parameter Number	When the value of parameter is 0	
		Color value	Color
16	14724	0x00003F3F	Yellow
17	14725	0x0000003F	Red
18	14726	0x00002000	Green
19	14727	0x00003F3F	Yellow
20	14728	0x003F0000	Blue
21	14729	0x002A002A	Purple
22	14730	0x003F3F3F	White
23	14731	0x003F3F3F	White
24	14732	0x00003F00	Light green
25	14733	0x00000000	Black
26	14734	0x00121212	Dark grey
27	14735	0x00303030	Bright grey
28	14736	0x003F3F00	Sky blue
29	14737	0x003C2020	Dark sky blue
30	14738	0x00181818	Grey
31	14739	0x00282828	Light grey
32	14740	0x0000003F	Red
33	14741	0x00002000	Green
34	14742	0x00003F3F	Yellow
35	14743	0x003F0000	Blue
36	14744	0x003C2020	Light purple
37	14745	0x00280030	Dark pink
38	14746	0x003F3F3F	White
39	14747	0x00102810	Light green
40	14748	0x00000000	Black
41	14749	0x003C3C00	Sky blue
42	14750	0x00303030	Bright grey
43	14751	0x00180C0C	Navy blue
44	14752	0x00000020	Dark red
45	14753	0x00101010	Dark grey
46	14754	0x00282828	Light grey
47	14755	0x00000000	Black
48	14756	0x00002000	Green
49	14757	0x003F0000	Blue
50	14758	0x003F0000	Blue
51	14759	0x003C3C00	Sky blue
52	14760	0x003C3C00	Sky blue
53	14761	0x0000003C	Red
54	14762	0x003F0000	Blue
55	14763	0x003F0000	Blue
56	14764	0x00003F3F	Yellow
57	14765	0x00303030	Bright grey

		When the value of parameter is 0	
58	14766	0x00282828	Light grey
59	14767	0x00000000	Black
60	14768	0x00000000	Black
61	14769	0x003F3F3F	White
62	14770	0x003C3C00	Sky blue
63	14771	0x00181818	Grey
64	14780	0x0000003F	Red

C.7.7 Restriction

Help window isn't displayed while a MANUAL GUIDE *i* window is being called.

C.8 DISPLAY OF SIMPLIFIED CHINESE CHARACTERS ALARM OF USER MACRO (for Series 30*i*/31*i*/32*i*-B, 0*i*-D)

Simplified chinese characters alarm message occurred in user macro can be displayed.

C.8.1 Setting for Use

(1) Enable one of the following options.
  (a) Multi-language display chinese (simplified characters) (S829)
  (b) Selection of five optional language (R521)

(2) Set the following parameters.
(When simplified chinese characters display function is used)
  (a) Parameter No.3281=15 (Language of CNC is simplified chinese characters)
  (b) Parameter No.14712=0 or 12
    0: Language of MANUAL GUIDE *i* is the same as language of CNC.
    12: Language of MANUAL GUIDE *i* is simplified chinese characters.

(When selection of five optional language function is used)
  (a) Parameter No.3281=15 (Language of CNC is simplified chinese characters)
  (b) Parameter No.3282#4=1 (Simplified chinese characters is available in selection of five optional language)
  (c) Parameter No.14712=0 or 12
    0: Language of MANUAL GUIDE *i* is the same as language of CNC.
    12: Language of MANUAL GUIDE *i* is simplified chinese characters.

(3) Define the alarm message of simplified chinese characters of user macro by using the direct language specification function of macro executor.

C.9 EXTENDED AXIS NAME FUNCTION FOR 8.4” /10.4” DISPLAY(for only Series 30*i* 31*i*/32*i*-B)

The extended axis name can be used on MANUAL GUIDE *i* screen for 8.4” /10.4” display unit.

C.9.1 Conditions to Use This Function

This function requires the following all conditions.

(1) The option “MANUAL GUIDE *i* extended axis name function”
(2) 8.4” or 10.4” display unit
(3) Extended axis name is one of following axes.
  <1> Linear axis other than basic three axis
  <2> Rotary axis of workpiece rotation
  <3> Live tool axis
(4) Extended axis name of 2 characters (Example : C2, ZS)
(5) Set-up guidance function is invalid.

C.9.2 Displaying Extended Axis Name

Extended axis name can be displayed on the following screens.

(1) Base screen
(2) Relative coordinates presetting screen
(3) Workpiece coordinate data screen
(4) Machining simulation screen

NOTE
1. In “CALCULATE OFFSET(MEASURE)” screen of workpiece coordinate data screen, extended axis name is not displayed.
2. In “CALCULATE OFFSET(+INPUT)” screen of workpiece coordinate data screen, extended axis name is not displayed.
3. The second character of axis name is not displayed but “-” (a minus sign) is displayed in “ACTUAL POS”, “DIST TO GO” and “NEXT DIST” of base screen under all the following conditions.
  <1>The axis name size is standard.
  <2>The values of “ACTUAL POS”, “DIST TO GO” and “NEXT DIST” is negative number that has 8 digits.
  (Example)
  Usual : CS 0.000
  Under the conditions : C-20000.000

C.9.3 Editing a Program Including Extended Axis Name

A program including extended axis name can be edited.

(1) Case of extended axis name + value

<1> Selecting a word by the cursor
The whole of a word is selected by the cursor.
(Example)
CS100. ;

<2> Deleting a word
The whole of a word is deleted.
(Example)
CS100. ; → ;

<3> Replacing a word
The whole of a word is replaced.
(Example)
CS100. ; → F50. ;

<4> Replacing the value of a word
Only the value of a word is replaced.
(Example)
CS100. ; → CS200. ;

(2) Case of extended axis name + ”=” + command value

<1> Selecting a word by the cursor
The whole of a word is selected by the cursor.
(Example)
C2=100. ;

<2> Deleting a word
The whole of a word is deleted.
(Example)
C2=100. ; → ;

<3> Replacing a word
The whole of a word is replaced.
(Example)
C2=100. ; → F50. ;

<4> Replacing the value of a word
Only the value of a word is replaced.
(Example)
C2=100. ; → C2=20. ;

C.10 MANUAL GUIDE *i* CUSTOMIZATION (for only Series 30*i*/31*i*/32*i*-B, 0*i*-D)

C.10.1 Automatic Operation After Changing to MANUAL GUIDE *i* Screen from User Application

(1) Explanation of function
After changing to MANUAL GUIDE *i* screen from the user application, the MDI key and the softkey can be automatically operated by using “os_chng_appstr()” function.

(2) Parameter setting for use
Parameter No.27483#0=1 is set.

(3) Specification of os_chng_appstr () function
Call “os_chng_appstr ()” function with the following arguments in the user application.

os_chng_appstr (-3, “operation command”)

- First argument : -3
- Second argument : operation command

(List of operation command)

Operation command	Explanation
0 ~ 9	Push number 0 ~ 9
-	Push sign “-“ (hyphen)
.	Push sign “.” (period)
SKEY01 ~ SKEY10	Push soft-key1~ soft-key10
SKEY-R	Push soft-key [>]
SKEY-L	Push soft-key [<]
INPUT	Push key [INPUT]
CR	Push key [→]
CL	Push key [←]
CD	Push key [↓]
CU	Push key [↑]
PAGE-D	Push key [PAGE DOWN]
PAGE-U	Push key [PAGE UP]
WAIT0.1 ~ WAIT3276.7	Waiting of 0.1(sec) ~ 3276.7(sec)

(4) Attention when operation command is specified
(a) It is possible to specify operation command up to 255 characters.
(b) When there are plural operation commands, delimit it with the space “ “.
(c) Set waiting time without putting space after "WAIT".

(5) Example
When you want to display the cylinder blank figure input window, call “os_chng_appstr ()” function as follows.

os_chng_appstr (-3, "SKEY-L SKEY01 WAIT1.5 CR CD SKEY09")
(a) (b) (c) (d) (e) (f)

(Explanation of operation command)

(a) Push key : push softkey [<]
(b) Push key : push softkey [START] of first from the left
(c) Temporary stop : wait for 1.5 seconds
(d) Push key : push softkey [→]
(e) Push key : push softkey [↓]
(f) Push key : push softkey [START] of ninth from the left

C.10.2 Change to User Application Screen After Closing MAUAL GUIDE *i* Window

(1) Explanation of function
The user application screen is changed on the following conditions.

(a) When the window of MANUAL GUIDE *i* of window ID list is closed
(b) When the following operation is done
- Push softkey [INSERT]
- Push softkey [OK]
- Push softkey [ALTER]
- Push softkey [CANCELI]
- Switch mode
- Switch path
- Push function key

(2) Parameter setting for use
Parameter No.27483#1=1 is set.

(3) Specification of function
After it changes to the user application, window ID of MANUAL GUIDE *i* that is closed at the end can be acquired in the return value of “os_get_chngappinfo ()” function.

Window ID = os_get_chngappinfo()

Window ID is expressed by the figure of six digits.
Two first digits means the method for closing the window, and four digits in the under means the input window as follows.

Window ID : - AABBBB

- AA = methods for closing window
- BBBB = input window

(List of methods for closing window)

Window ID	Method of closing window
-10	Push soft-key [INSERT] or [OK] or [ALTER]
-20	Push soft-key [CANCEL] / Switching mode or path
-30	Push function-key [SYSTEM]
-31	Push function-key [POSITION]
-32	Push function-key [PROGRAM]
-33	Push function-key [OFFSET/SETTING]
-34	Push function-key [MESSAGE]
-35	Push function-key [GRAPH]
-36	Push function-key [CUSTOM1]
-37	Push function-key [CUSTOM2]

(List of window ID for cycle input, blank input figure)

Window ID	Title	G code
-**0001	Center drilling	G1000
-**0002	Drilling	G1001
-**0003	Tapping	G1002
-**0004	Reaming	G1003
-**0005	Boring	G1004
-**0006	Fine boring	G1005
-**0007	Back boring	G1006
-**0008	Tapping	G1002
-**0010	Emboss machining roughing	G1080
-**0011	Emboss machining bottom finishing	G1081
-**0012	Emboss machining side finishing	G1082
-**0013	Emboss machining chamfering	G1083
-**0020	Facing roughing	G1020
-**0021	Facing finishing	G1021
-**0026	Pocketing roughing	G1040
-**0027	Pocketing bottom finishing	G1041
-**0028	Pocketing side finishing	G1042
-**0029	Pocketing chamfering	G1043
-**0032	Grooving roughing	G1050
-**0033	Grooving bottom finishing	G1051
-**0034	Grooving side finishing	G1052
-**0035	Grooving chamfering	G1053
-**0038	C-axis grooving	G1056
-**0039	A-axis grooving	G1058
-**0040	B-axis grooving	G1054
-**0041	Thread milling	G1010
-**0042	Outer thread milling	G1026
-**0049	Engraving	G1025
-**0050	Center drilling	G1100
-**0051	Drilling	G1101
-**0052	Tapping	G1102
-**0053	Reaming	G1103
-**0054	Boring	G1104
-**0055	Tapping	G1102

Window ID	Title	G code
-**0056	Fine boring	G1105
-**0060	Center drilling	G1110
-**0061	Drilling	G1111
-**0062	Tapping	G1112
-**0063	Reaming	G1113
-**0064	Boring	G1114
-**0065	Tapping	G1112
-**0066	Fine boring	G1115
-**0070	Outer surface roughing	G1120
-**0071	Inner surface roughing	G1121
-**0072	End face roughing	G1122
-**0073	Outer surface semi-finishing	G1123
-**0074	Inner surface semi-finishing	G1124
-**0075	End face semi-finishing	G1125
-**0076	Outer surface finishing	G1126
-**0077	Inner surface finishing	G1127
-**0078	End face finishing	G1128
-**0079	Conical boring	G1129
-**0080	Turning groove outer surface roughing	G1130
-**0081	Turning groove inner surface roughing	G1131
-**0082	Turning groove end face roughing	G1132
-**0083	Turning groove outer surface roughing and finishing	G1133
-**0084	Turning groove inner surface roughing and finishing	G1134
-**0085	Turning groove end face roughing and finishing	G1135
-**0086	Turning groove outer surface finishing	G1136
-**0087	Turning groove inner surface finishing	G1137
-**0088	Turning groove end face finishing	G1138
-**0090	Turning groove threading outer surface	G1140
-**0091	Turning groove threading inner surface	G1141
-**0094	Turning groove outer surface roughing	G1130
-**0095	Turning groove inner surface roughing	G1131
-**0096	Turning groove end face roughing	G1132
-**0097	Turning groove outer surface roughing and finishing	G1133
-**0098	Turning groove inner surface roughing and finishing	G1134
-**0099	Turning groove end face roughing and finishing	G1135
-**0100	XY plane random points	G1210
-**0101	XY plane linear points (same interval)	G1211
-**0102	XY plane linear points (different interval)	G1212
-**0103	XY plane grid points	G1213
-**0104	XY plane rectangle points	G1214
-**0105	XY plane circle points	G1215
-**0106	XY plane arc points (same interval)	G1216
-**0107	XY plane arc points (different interval)	G1217
-**0110	XY plane rectangle	G1220
-**0111	XY plane circle	G1221
-**0112	XY plane track	G1222
-**0113	XY plane radial groove	G1223
-**0114	XY plane Straight line	G1224

Window ID	Title	G code
-**0115	XY plane polygon	G1225
-**0120	YZ plane random points	G1310
-**0121	YZ plane linear points (same interval)	G1311
-**0122	YZ plane linear points (different interval)	G1312
-**0123	YZ plane grid points	G1313
-**0124	YZ plane rectangle points	G1314
-**0125	YZ plane circle points	G1315
-**0126	YZ plane arc points (same interval)	G1316
-**0127	YZ plane arc points (different interval)	G1317
-**0130	YZ plane rectangle	G1320
-**0131	YZ plane circle	G1321
-**0132	YZ plane track	G1322
-**0133	YZ plane radial groove	G1323
-**0134	YZ plane polygon	G1325
-**0135	Polygon figure	G1575
-**0150	ZX plane general-purpose thread	G1460
-**0151	ZX plane metric thread	G1461
-**0152	ZX plane unified thread	G1462
-**0153	ZX plane PT thread	G1463
-**0154	ZX plane PF thread	G1464
-**0156	ZX plane outer normal groove	G1470
-**0157	ZX plane outer trapezoidal groove	G1471
-**0158	ZX plane inner normal groove	G1472
-**0159	ZX plane inner trapezoidal groove	G1473
-**0160	ZX plane end face normal groove	G1474
-**0161	ZX plane end face trapezoidal groove	G1475
-**0169	ZC plane C-axis hole -random points- orthogonal	G1574
-**0170	XC plane rectangle	G1520
-**0171	XC plane circle	G1521
-**0172	XC plane track	G1522
-**0173	XC plane radial groove	G1523
-**0174	XC plane C axis groove	G1570
-**0175	XC plane X axis groove	G1571
-**0176	XC plane C-axis hole -circle points	G1572
-**0177	XC plane C-axis hole -random points	G1573
-**0178	XC plane polygon	G1525
-**0179	XC plane circle groove	G1526
-**0180	ZC plane C axis groove	G1670
-**0181	ZC plane X axis groove	G1671
-**0182	ZC plane C-axis hole -circle points	G1672
-**0183	ZC plane C-axis hole -random points	G1673
-**0190	ZC plane A axis groove	G1770
-**0191	ZC plane X axis groove	G1771
-**0192	ZC plane A-axis hole -circle points	G1772
-**0193	ZC plane A-axis hole -random points	G1773
-**0194	YB plane B axis groove	G1775
-**0195	YB plane X axis groove	G1776
-**0196	YB plane B-axis hole -circle points	G1777

Window ID	Title	G code
-**0197	YB plane Y B-axis hole -random points	G1778
-**0200	Contouring outer wall roughing	G1060
-**0201	Contouring outer wall bottom finishing	G1061
-**0202	Contouring outer wall side finishing	G1062
-**0203	Contouring outer wall chamfering	G1063
-**0204	Contouring Inner wall roughing	G1064
-**0205	Contouring inner wall bottom finishing	G1065
-**0206	Contouring inner wall side finishing	G1066
-**0207	Contouring inner wall chamfering	G1067
-**0208	Contouring partial roughing	G1068
-**0209	Contouring partial bottom finishing	G1069
-**0210	Contouring Partial Side finishing	G1070
-**0211	Contouring Partial Chamfering	G1071
-**0212	Outer surface residual roughing	G1160
-**0213	Inner surface residual roughing	G1161
-**0214	End face residual roughing	G1162
-**0215	Outer surface residual semi-finishing	G1163
-**0216	Inner surface residual semi-finishing	G1164
-**0217	End face residual semi-finishing	G1165
-**0218	Outer surface residual finishing	G1166
-**0219	Inner surface residual finishing	G1167
-**0220	End face residual finishing	G1168
-**0221	Polygon turning	G1145
-**0222	Tilted working plane with Euler's angle	G1850
-**0223	Tilted working plane with roll-pitch-yaw angle	G1851
-**0224	Tilted working plane with 3 point	G1852
-**0225	Tilted working plane with 2 vectors	G1853
-**0226	Tilted working plane with projection angle	G1854
-**0227	Tilted working plane with tool axis direction	G1855
-**0228	Tilted working plane cancel	G1859
-**0286	Stylus ball diameter calibration (Z axis)	G2062
-**0287	Stylus offsets calibration-A (Z axis)	G2063
-**0288	Stylus offsets calibration-B (Z axis)	G2064
-**0289	Outside diameter work-set (Z axis)	G2073
-**0290	Inside diameter work-set (Z axis)	G2074
-**0291	Outside diameter measure (Z axis)	G2083
-**0292	Inside diameter measure (Z axis)	G2084
-**0293	Stylus ball diameter calibration (X axis)	G2162
-**0294	Stylus offsets calibration-A (X axis)	G2163
-**0295	Stylus offsets calibration-B (X axis)	G2164
-**0296	Outside diameter work-set (X axis)	G2173
-**0297	Inside diameter work-set (X axis)	G2174
-**0298	Outside diameter measure (X axis)	G2183
-**0299	Inside diameter measure (X axis)	G2184
-**0300	Touch sensor position calibration (Z axis)	G2000
-**0301	Probe length calibration (Z axis)	G2001
-**0302	Stylus ball diameter calibration (Z axis)	G2002
-**0303	Stylus offsets calibration-A (Z axis)	G2003

Window ID	Title	G code
-**0304	Stylus offsets calibration-B (Z axis)	G2004
-**0305	Stylus ball diameter calibration (work rotation -Z axis)	G2005
-**0306	Stylus offsets calibration-A (work rotation -Z axis)	G2006
-**0307	Stylus offsets calibration-B (work rotation -Z axis)	G2007
-**0310	Touch sensor position calibration (X axis)	G2100
-**0311	Probe length calibration (X axis)	G2101
-**0312	Stylus ball diameter calibration (X axis)	G2102
-**0313	Stylus offsets calibration-A (X axis)	G2103
-**0314	Stylus offsets calibration-B (X axis)	G2104
-**0315	Stylus ball diameter calibration (work rotation –X axis)	G2105
-**0316	Stylus offsets calibration-A (work rotation –X axis)	G2106
-**0317	Stylus offsets calibration-B (work rotation -X axis)	G2107
-**0318	Measure for rotate-axis position calibration	G2890
-**0319	rotate-axis position calibration	G2891
-**0320	Milling tool measure (Z axis)	G2010
-**0321	Turning tool measure (Z axis)	G2011
-**0322	Milling tool measure (X axis)	G2110
-**0323	Turning tool measure (X axis)	G2111
-**0324	Milling tool measure (Z axis – non touch)	G2012
-**0325	Turning tool measure (Z axis– non touch)	G2112
-**0326	Work-piece setting error (XY plane)	G2035
-**0327	Work-piece setting error (YZ plane)	G2036
-**0328	Work-piece setting error (ZX plane)	G2037
-**0329	Work-piece setting error (set - work)	G2038
-**0330	End-face work-set Z axis direct. (Z axis)	G2022
-**0331	End-face work-set X axis direct. (Z axis)	G2020
-**0332	End-face work-set Y axis direct. (Z axis)	G2021
-**0333	Outside diameter work-set (Z axis)	G2023
-**0334	Inside diameter work-set	G2024
-**0335	Outside width work-set (Z axis)	G2025
-**0336	Inside width work-set (Z axis)	G2026
-**0337	Angle of slanted work-piece work-set	G2047
-**0341	Outside corner measure (Z axis)	G2027
-**0342	Inside corner measure (Z axis)	G2028
-**0343	Outside corner measure (X axis)	G2127
-**0344	Inside corner measure (X axis)	G2128
-**0345	C axis outside width work-set (Z axis)	G2031
-**0346	C axis inside width work-set (Z axis)	G2032
-**0347	Work-piece setting error outside-corner (Z axis)	G2033
-**0348	Work-piece setting error outside-corner (X axis)	G2133
-**0351	End-face work-set X-axis direct. (X axis)	G2120
-**0352	End-face work-set Y-axis direct. (X axis)	G2121
-**0353	Outside diameter work-set (X axis)	G2123
-**0354	Inside diameter work-set (X axis)	G2124
-**0355	Outside width work-set (X axis)	G2125
-**0356	Inside width work-set (X axis)	G2126
-**0365	C-axis Outside width work-set (X axis)	G2131
-**0366	C-axis inside width work-set (X axis)	G2132

Window ID	Title	G code
-**0367	Angle of slanted work-piece work-set (Z axis)	G2029
-**0368	Angle of slanted work-piece work-set (X axis)	G2129
-**0370	End-face measure Z-axis direct. (Z axis)	G2042
-**0371	End-face measure X-axis direct. (Z axis)	G2040
-**0372	End-face measure Y-axis direct. (Z axis)	G2041
-**0373	Outside diameter measure (Z axis)	G2043
-**0374	Inside diameter measure (Z axis)	G2044
-**0375	Outside width measure (Z axis)	G2045
-**0376	Inside width measure (Z axis)	G2046
-**0377	Outside diameter measure (work rot. - Z axis)	G2053
-**0378	Inside diameter measure (work rot. - Z axis)	G2054
-**0380	End-face measure Z-axis direct. (X axis)	G2142
-**0381	End-face measure X-axis direct. (X axis)	G2140
-**0382	End-face measure Y-axis direct. (X axis)	G2141
-**0383	Outside diameter measure (Z axis)	G2143
-**0384	Inside diameter measure (Z axis)	G2144
-**0385	Outside width measure (Z axis)	G2145
-**0386	Inside width measure (Z axis)	G2146
-**0387	Outside diameter measure (work rot. - X axis)	G2153
-**0388	Inside diameter measure (work rot.- X axis)	G2154
-**0390	Blank form Column	G1900
-**0391	Blank form Column with a hole	G1901
-**0392	Blank form Rectangular solid	G1902
-**0394	General-purpose tool	G1910
-**0395	Threading tool	G1911
-**0396	Grooving tool	G1912
-**0397	Round-nose tool	G1913
-**0398	Point nose straight tool	G1914
-**0399	Drill tool	G1921
-**0400	Counter sink tool	G1931
-**0401	Flat end mill tool	G1932
-**0402	Ball end mill tool	G1933
-**0403	Tap tool	G1922
-**0404	Reamer tool	G1923
-**0405	Boring tool	G1924
-**0406	Face mill tool	G1930
-**0407	Blank form Polygon	G1903
-**0408	Blank form polygon with a hole	G1904
-**0409	Blank form column (around X)	G1906
-**0410	Blank form column with a hole (around X)	G1907
-**0411	Blank form column (around Y)	G1756
-**0412	Blank form column with a hole (around Y)	G1757
-**0413	Blank form polygon (around Y)	G1758
-**0414	Blank form polygon with a hole (around Y)	G1759
-**0450	Work coordinate conversion direct (Y)	G1952
-**0451	Work coordinate conversion indirect (Y)	G1953
-**0452	Work coordinate conversion cancel	G1959
-**0453	Work coordinate conversion direct (Z)	G1954

Window ID	Title	G code
-**0454	Work coordinate conversion indirect (Z)	G1955
-**0460	Spindle change	G1998
-**0461	Start synchronization drawing	G1994
-**0462	Stop synchronization drawing	G1995
-**0470	Coordinate conversion (mirror image)	G1980
-**0471	Coordinate conversion cancel (mirror image)	G1981
-**0472	Coordinate conversion (rotation)	G1957
-**0473	Rotate copy	G72.1
-**0474	Scaling	G1982
-**0475	Parallel copy	G72.2
-**0476	Coordinate conversion cancel (rotation)	G1958
-**0477	Scaling cancel	G1983
-**0478	Turning outer surface roughing and finishing	G1150
-**0479	Turning inner surface roughing and finishing	G1151
-**0480	Turning end face roughing and finishing	G1152
-**0481	Blank form rectangular solid	G1902
-**0482	Cylinder blank figure (around Y)	G1756
-**0483	Hollow cylinder blank figure (around Y)	G1757
-**0484	Prism blank figure (around Y)	G1758
-**0485	Hollow prism blank figure (around Y)	G1759

(List of window ID for arbitrary figure, free contour cylinder blank figure, G contouring input)

Window ID	Title	G code
-**1100	XY plane arbitrary figure	G1200 ~ G1206
-**1200	YZ plane arbitrary figure	G1300 ~ G1306
-**1400	XC plane end face arbitrary figure	G1500 ~ G1506
-**1500	ZC plane cylindrical surface arbitrary figure	G1600 ~ G1606
-**1600	ZX plane turning figure	G1450 ~ G1456
-**1700	Free contour cylinder blank figure	G1970 ~ G1976
-**1800	XA plane cylindrical surface arbitrary figure	G1700 ~ G1706
-**1900	Free contour cylinder blank figure (around X)	G1790 ~ G1796
-**2000	Free contour cylinder blank figure (around Y)	G1780 ~ G1786
-**2100	XY plane contouring program	G1200 ~ G1206
-**2200	YZ plane contouring program	G1300 ~ G1306
-**2300	ZX plane contouring program	G1400 ~ G1406
-**2400	XC plane end face contouring program	G1500 ~ G1506
-**2500	ZC plane cylindrical surface contouring program	G1600 ~ G1606
-**5000	Milling machining start menu	-
-**5001	Milling machining menu	-
-**5002	Milling machining figure menu	-
-**5010	Turning machining start menu	-
-**5011	Turning machining menu	-
-**5012	Turning machining figure menu	-
-**5020	G contouring menu	-
-**5030	Measure menu	-

C.10.3 Unchange to Menu Window When Blank Figure Input Window is Closed by Softkey [CANCEL]

(1) Explanation of function
The base screen is changed without changing to menu window after softkey [CANCEL] is pushed on the following blank figure input windows.

Blank figure input window	G code
Cylinder blank figure	G1900
Hollow cylinder blank figure	G1901
Rectangular blank figure	G1902
Prism blank figure	G1903
Hollow prism blank figure	G1904
Cylinder blank figure (around X)	G1906
Hollow cylinder blank figure (around X)	G1907
Cylinder blank figure (around Y)	G1756
Hollow cylinder blank figure (around Y)	G1757
Prism blank figure (around Y)	G1758
Hollow prism blank figure (around Y)	G1759
Free cylinder figure	G1970 ~ G1976
Free cylinder figure (around X)	G1790 ~ G1796
Free cylinder figure (around Y)	G1780 ~ G1786

(2) Parameter setting for use
Parameter No.27483#7=1 is set.

C.11 OPERATION NAVIGATING CUSTOMIZATION FUNCTION

Create the following 3 definition files to customize operation navigating.

vgamsg1u.txt

```
…
…
20001,TURNING
20002,CREATE NEW PROG.
…
```

navilstu.txt

```
<TAB=20001>
20002,-1,1,
20003,-1,2,
<TAB=20011>
20012,-1,11,
20013,-1,12,
…
```

navidatu.txt

```
<ID=1>
<MAIN=0,SUB=-1>
<TB=-1,SK=-1>
KEY SKEY-L
KEY SKEY08
<MAIN =11,SUB=1>
<TB=-1,SK=09>
KEY SKEY-R
KEY SKEY01
KEY CR
NAVIOFF
<TB=-1,SK=10>
NAVIOFF
<ID=2>
…
…
```

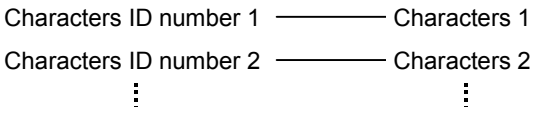
C.11.1 Window Message Definition File (vgamsg*u.txt)

This file is for defining characters displayed in title and so on of “SELECT NAVIGATION” window.
This file is unnecessary in case second method of start is used.

(1) Definition file name
vgamsg*u.txt
(“*” is language number.)

(2) Defined data
[1] Characters displayed in tab title
[2] Characters displayed in item title

(3) Data structure



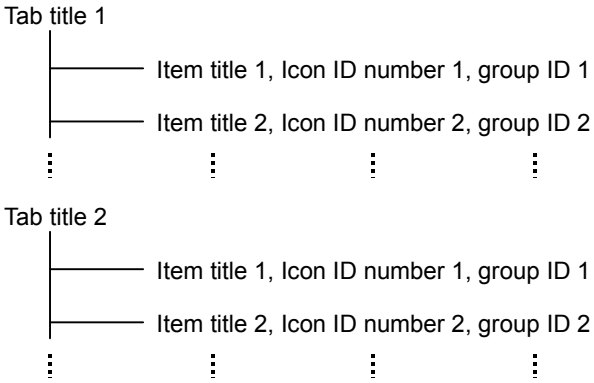
NOTE
Characters ID number machine tool builder can use are from 20000 to 29999.

C.11.2 Operation Navigating Selection Window Definition File (navilstu.txt)

This file is for defining title name and so on of “SELECT NAVIGATION” window.
This file is unnecessary in case second method of start is used.

(1) Definition file name
navilstu.txt
(2) Defined data
[1] Characters ID number of tab title
[2] Characters ID number of item title
[3] Icon ID number that is displayed next to item title on the left
[4] ID number of navigation that is executed when the item is selected (*1)
(*1) After this sentence, [4] is called “group ID”.

(3) Data structure



C.11.2.1 Specifying tab title

Specify tab title by the following form.

<TAB=Tab title ID[,SERIES=Series No.][,PARM=parameter No.,bit, 0/1]>

[Details]

(1) TAB=tab title ID
This item is for specifying tab title. Specify title name and ID number in “vgamsg*u.txt” and specify the ID number in the right side of the equation.
The number of displayed tab is same as the number of this form. But the maximum number is 5.

(2) SERIES=Series No. (This item can be omitted.)
Specify one of following numbers ID in the right side of the equation when appearing or disappearing tab should be controlled according to CNC series.

No.	CNC series to appear tab	Series ID
1	FS0i-D	09
2	FS30i-A, FS30i-B	39
3	FS0i-D, FS30i-A, FS30i-B	30

(3) PARM=parameter No.,bit, 0/1 (This item can be omitted.)
Specify this equation when appearing or disappearing tab should be controlled according to bit type parameter setting.
If “0/1” is set to 0, the tab appears when the parameter that is specified by “parameter No” and “bit” is set to 0. If “0/1” is set to 1, the tab appears when the parameter is set to 1.
But the maximum number of this equation is 8.

C.11.2.2 Specifying item title and group ID

Specify item title and group ID by the following form.

Item title ID,Icon ID,Group ID[,P=parameter No.,bit, 0/1],

[Details]

(1) Item title
This item is for specifying item title. Specify title name and ID number in “vgamsg*u.txt” and specify the ID number.
The number of displayed title is same as the number of this form. But the maximum number is 20. And the maximum number that can be displayed in a tab is 10.

(2) Icon ID
Specify ID number that is written in chapter “Available icons” when icons should be displayed next to items. Specify “-1” when icons should not be displayed.

(3) Group ID
This item is for specifying assignment of group ID to each items.
If navigation that is selected in “SELECT NAVIGATION” window starts, Navi operation codes that each group ID has are executed.
Navi operation codes are defined in “navidatu.txt”.
Specify an unique ID number to each items. Only the numbers from 1 to 127 can be used.

(4) P=parameter No.,bit, 0/1
Specify numbers only when appearing or disappearing items should be controlled according to bit type parameters.
If “0/1” is set to 0, the item appears only when the parameter that is specified by “parameter No” and “bit” is set to 0. If “0/1” is set to 1, the item appears only when the parameter is set to 1.
But the maximum number of this equation is 8.

C.11.3 Operation Navigating Execution Definition File (navidatu.txt)

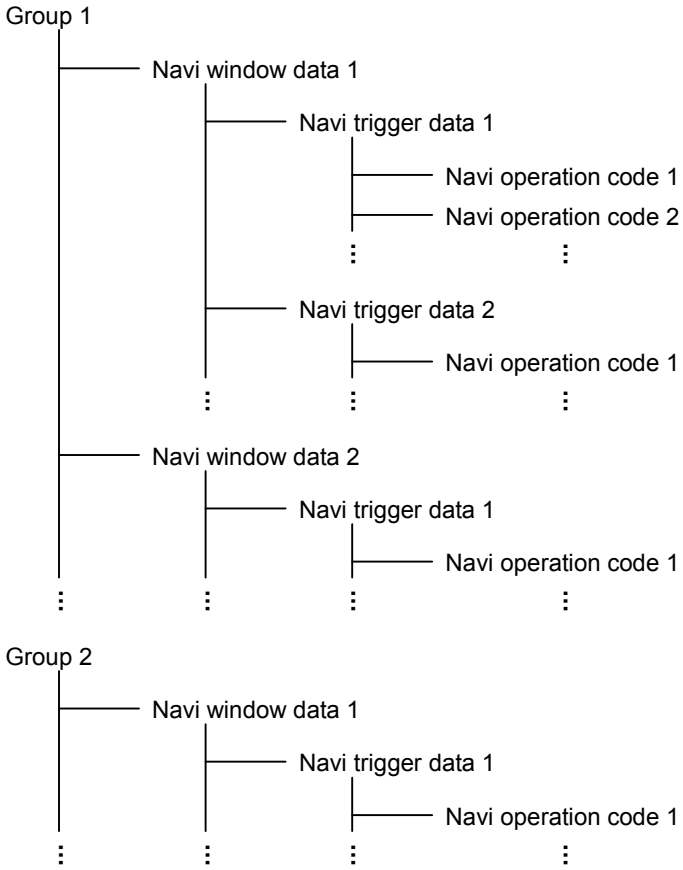
The operation navigating function is realized by the method that MANUAL GUIDE *i* automatically pushes the keys (*2) that are specified in definition files. And, the automatic key pushing is executed after a window (*3) is closed when an operator pushes a soft-key (*4).
This file is for defining the automatic key pushing and so on.

(*2) After this sentence, the key data are called “Navi operation code”.
(*3) After this sentence, the data for specifying this window are called “Navi window data”.
(*4) After this sentence, the data for specifying this soft-key are called “Navi trigger data”.

(1) Definition file name
navidatu.txt

(2) Defined data
[1] Navi window data
[2] Navi trigger data
[3] Navi operation code

(3) Data structure



C.11.3.1 Specifying group ID

Specify group ID by the following form.

<ID=group ID>

[Details]

ID=group ID

This item is for specifying group ID. The numbers from 1 to 127 can be used.
This item links group ID that is specified by “navilstu.txt” or parameter 14870.
The automatic operations after selecting group of this ID are specified in “navidatu.txt”.
The maximum number of this equation is 20.

C.11.3.2 Specifying Navi window data

Specify Navi window data by the following form.

<MAIN=main window ID,SUB=subwindow ID>

[Details]

Navigation starts if trigger soft-key is pushed while the window specified by the form is displayed.
The maximum number of this form per 1 group is 20.

(1) MAIN=main window ID
Specify one of following window ID numbers.

Main Window ID	Window	Note
0	“SELECT NAVIGATION” window just after selecting a navigation item	
1	WORK COORDINATE SYSTEM	
2	TOOL OFFSET	
4	CYCLE DATA INPUT	Specify details of window by subwindow ID.
6	INSERT MILLING CYCLE	
7	INSERT MILLING FIGURE	
8	INSERT TURNING CYCLE	
9	INSERT TURNING FIGURE	
11	CREATING NEW PROGRAM WINDOW	Specify details of window by subwindow ID.
12	INSERT M-CODE	
13	INSERT STARTING COMMAND FOR MILLING	
14	INSERT FIXED FORM SENTENCE FOR MILLING	
16	INSERT ENDING COMMAND FOR MILLING	
17	INSERT STARTING COMMAND FOR TURNING	
18	INSERT FIXED FORM SENTENCE FOR TURNING	
20	INSERT ENDING COMMAND FOR TURNING	
24	METHOD OF FREE FIGURE CREATION (This window is for selecting “INSERT IN CURRENT PROGRAM” or “CREATE AS SUB PROGRAM”.)	
40	INSERT MILLING FIGURE	
41	INSERT TURNING FIGURE	
50	MACHINING SIMULATION WINDOW	Specify details of window by subwindow ID.
66	INSERT CONTOUR PROGRAM	
76	PROGRAM LIST	Specify details of window by subwindow ID.

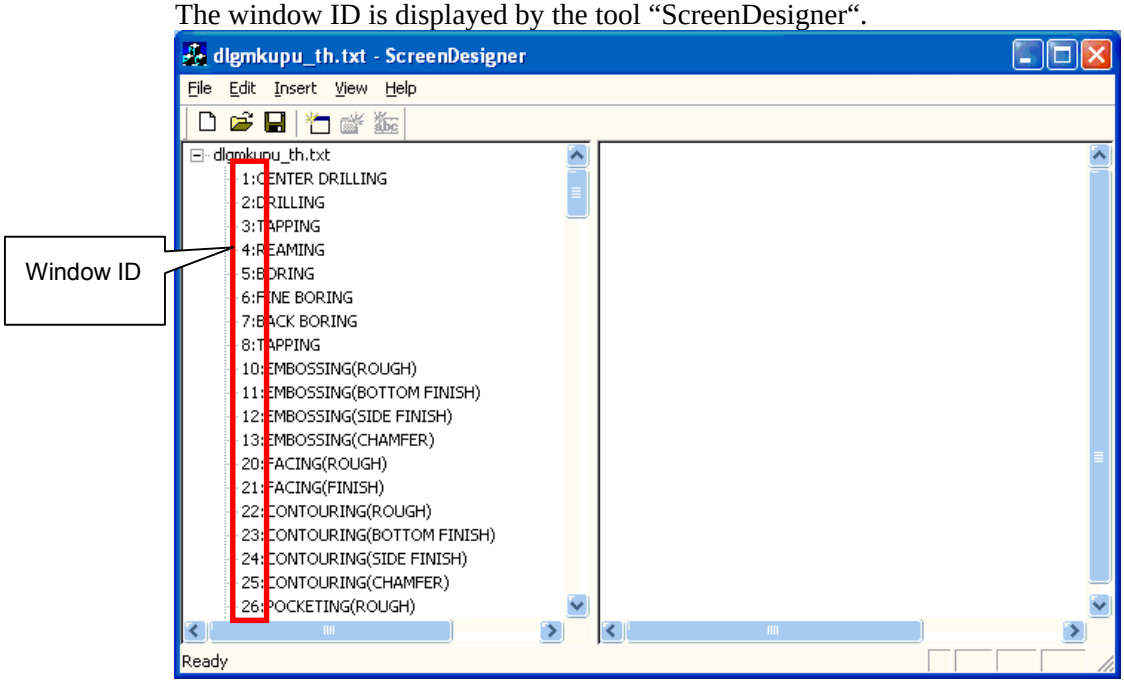
Specify window ID number “0” when second method of start is used and operation navigating starts.

(2) SUB=subwindow ID
Specify one of following subwindow ID numbers.

Main Window ID	Sub-window ID	Window	Note
4	(*)	CYCLE INPUT WINDOW	
11	1	CREATE NEW PROGRAM OR NEW FOLDER	For FS30*i*-A or FS30*i*-B
	2	CREATE NEW PROGRAM	For FS0*i*-D
24	1	METHOD OF FREE FIGURE CREATION (BLANK)	
	2	METHOD OF FREE FIGURE CREATION (TURNING)	
	3	METHOD OF FREE FIGURE CREATION (MILLING)	
	4	METHOD OF FREE FIGURE CREATION (G CONTOUR)	
50	2	TOOLPATH	
	3	ANIMATION	
76	1	OPEN PROGRAM	For FS0*i*-D
	7	PROGRAM LIST	For FS30*i*-A or FS30*i*-B
ID other than upper	-1	-	

(*) Specify window ID defined in the following definition files.
dlgmkupu.txt (User customization file)
cntmkupu.txt (User customization file)

The window ID is displayed by the tool “ScreenDesigner“.



Or following window ID is specified.

Sub-window ID	Details
-1	All window
-100	All fixed form blank data input window (For turning)
-101	All coordinate conversation window (For turning)
-102	All cutting condition data input window for turning cycle
-103	All fixed form figure data input window for turning cycle
-200	All fixed form blank data input window (For milling)
-201	All coordinate conversation window (For milling)
-202	All cutting condition data input window for milling cycle
-203	All fixed form figure data input window for milling cycle

C.11.3.3 Specifying Navi trigger data

Specify Navi trigger data by the following form.

<TB=tab number,SK=soft-key number>

[Details]

A navigation starts when specified tab is selected and specified soft-key is pushed.
The maximum number of trigger is 10 per 1 Navi window data.

(1) TB=tab number
Specify tab number that is numbered as 1, 2, 3... in order from left edge tab.
But, specify “TB=-1” in case of the followings.
- In case window that has no tab is displayed
- In case “SELECT NAVIGATION” window is displayed (Parameter No.14870 = 0)
- In case window isn’t displayed (When parameter No.14870 is set to a number more than 0 and operation navigating starts)

(2) SK=soft-key number
Specify soft-key number that is numbered as 1, 2, …, 10 in order from soft-key on left edge.
But, specify “SK=-1” in case main window ID of Navi window data is 0.
And, it is impossible to set vertical soft-keys as trigger.

C.11.3.4 Specifying Navi operation code

Specify the following codes in order of automatic key pushing and other operation.
The maximum number of Navi operation code is 20 per 1 Navi trigger data.
And incorrect Navi operation codes aren’t executed. Operation navigating is executed by a code just before the incorrect one.

Navi operation code	Detail
KEY key code	Key code for input. The details are written in (Details of key information).
WAIT waittime	Temporary stop code. Specify following figure. 0.1 - 3276.7
SIMWAIT	The code when MANUAL GUIDE i should wait by end of simulation. (Note) This code might not work if there is not a WAIT code before SIMWAIT.
NAVIOFF	Specify the code when navigation should be ended.

< Details of key information >

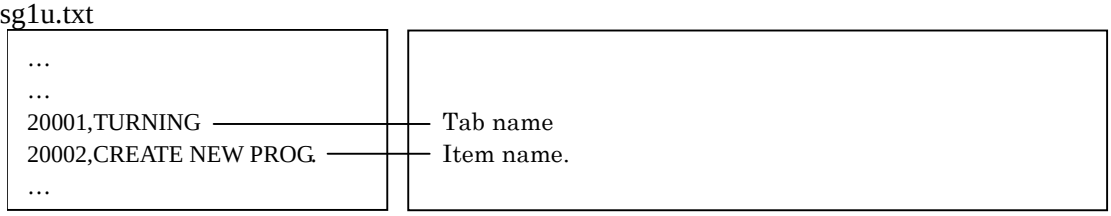
Key information	Code
0-9, A-Z	0-9, A-Z
1-10	SKEY01-SKEY10
Horizontal soft-key right	SKEY-R
Horizontal soft-key left	SKEY-L
CAN	CAN
EOB	EOB
INPUT	INPUT
Cursor right	CR
Cursor left	CL
Cursor down	CD
Cursor up	CU
PAGE down	PAGE-D
PAGE up	PAGE-U
INSERT	INSERT
DELETE	DELETE
ALTER	ALTER
HELP	HELP
SHIFT + CAN	SHIFT-CAN
SHIFT + →	SHIFT-CR
SHIFT + ←	SHIFT-CL
SHIFT + DELETE	SHIFT-DEL
SHIFT + 0	SHIFT-0
SHIFT + 1	SHIFT-1
SHIFT + 2	SHIFT-2
SHIFT + 3	SHIFT-3
SHIFT + 4	SHIFT-4
SHIFT + 5	SHIFT-5
SHIFT + 6	SHIFT-6
SHIFT + 7	SHIFT-7
SHIFT + 8	SHIFT-8
SHIFT + 9	SHIFT-9

C.11.4 Examples of Definition Files

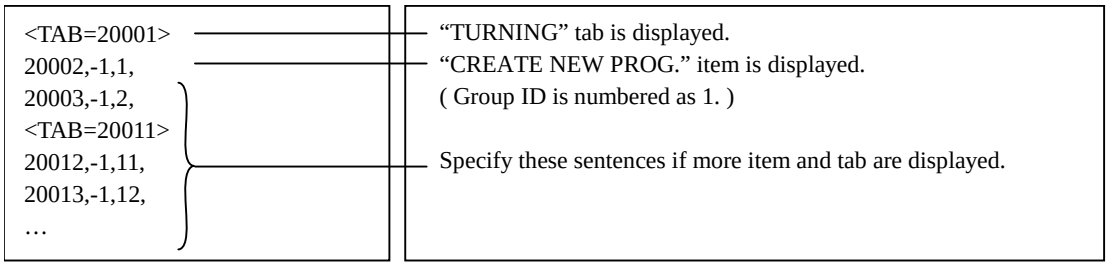
For example, the contents of definition files are shown in case the following navigation are given.

(1) “TURNING” tab is displayed in “SELECT NAVIGATION” window.
(2) “CREATE NEW PROG.” item is displayed in upper tab. No icon is displayed.
(3) After navigation of upper item starts, “CREATE NEW PROGRAM OR NEW FOLDER” window is displayed.
(4) Program name is inputted and [CREATE] soft-key is pushed, “BLANK” tab in “INSERT STARTING COMMAND FOR TURNING” window is displayed. And, operation navigating mode is released.
(5) When [CANCEL] soft-key is pushed during the state of (3), operation navigating mode is released.

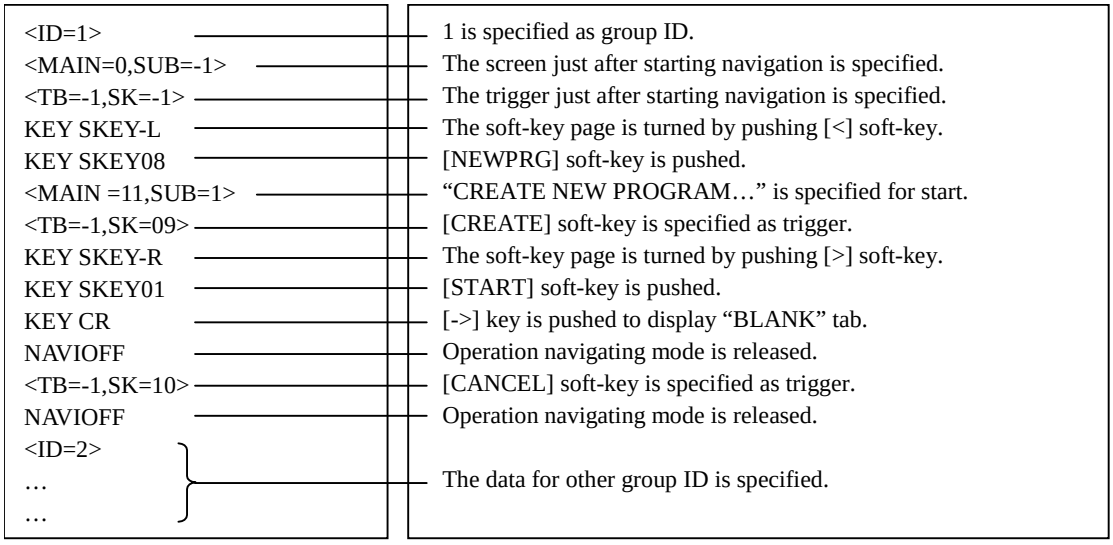
vgamsg1u.txt



navilstu.txt



navidatu.txt



C.11.5 Restriction of Customization

Customization of operation navigating function has following restrictions.

(1) Only soft-keys that are displayed in the same line of [GRPOFF] soft-key are specified as trigger while simulation window is displayed.
(2) Only soft-keys that are displayed in the same line of [CLOSE] soft-key are specified as trigger while tool offset window is displayed.

C.11.6 Available Icons

Available icons and their ID number are the followings.









C.11.7 Condition to Use Standard Operation Navigating Data

Available icons and their ID number are the followings.
The standard Operation Navigating data are available when the following conditions are met.

(1) The order of tab of “INSERT STARTING COMMAND FOR TURNING” window or “INSERT STARTING COMMAND FOR MILLING” window is “START” and “BLANK” from left edge.
(2) Parameter 14853#6 is set to 1.
(3) Parameter 14870 is set to 0.
(4) When MTB makes customization data, it is necessary to select an item that has a group ID “-1”.

D SCREEN HARD COPY

In order to make a copy to memory card of the screen of MANUAL GUIDE *i*, you need to operate as follows.

(1) Setting of parameters
In addition to the parameter for hard copy of standard CNC screen, bit 7 (HDC) of parameter No.3301= 1, setting of the parameter for C executor, bit 4 (CKM) of parameter No.8650 = 1, is necessary.
And furthermore, you need to set the parameter for memory card, No.20 = 4.

(2) Operations
Mount a memory card and display a necessary screen, then push <SHIFT> key more than 5 seconds. Release <SHIFT> key after checking the clock display on the screen stops. The clock display will move again when copying the screen completes.

(3) Created File
By the above operations, new file with name of “Hdcpy***.bmp” is created in the memory card. *** is the serial number and numbered as 001, 002. However, once the CNC power is turned off, this number is initialized to 000 from the next time of creating hard copy file.

NOTE
For an explanation of how a screen displayed by the CNC screen display function of the CNC in case of Personal computer function with WindowsCE can be copied to a memory card, refer to Section 2.10, "SCREEN COPY (*i*s SERIES)" in "FANUC OPEN CNC - CNC SCREEN DISPLAY FUNCTION OPERATOR'S MANUAL" (B-63164EN).

ADDITIONAL INFORMATION

Addition to FANUC MANUAL GUIDE *i*
Machining Center System
OPERATOR'S MANUAL

1. Type of applied technical documents

Name	FANUC MANUAL GUIDE *i* Machining Center System OPERATOR'S MANUAL
Spec. No.／Ed.	B-63874EN-2/04

2. Summary of Change

Group	Name／Outline	New, Add, Correct, Delete	Applicable Date
Basic Function	The new functions are added to PART VII.	Add	Immediately
Optional Function			
Unit			
Maintenance Parts			
Notice			
Correction			
Another			

VII. SUPPLEMENTAL INSTRUCTIONS

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2. Calculating method of measuring of setting workpiece coordinate data..... 5

1 Address R output for G02/G03 in machining cycle (for Series 30*i*/31*i*/32*i*-A/B, 0*i*-D)

1.1 Outline

In the functions on the below list, address R can be outputted for G02/G03.

(Target function)

Function
NC program conversion function
Automatic operation
Machining simulation (animated / tool path)
Input data check by simulation
Decomposed cycle display

According to the specifications of CNC, address I,J,K are outputted for G02/G03 as the below list even if address R output for G02/03 is selected.

(Output format of G02/G03 according to the center angle of arc)

CNC type	Center angle of arc		
	Center angle≦180°	180°＜Center angle ＜360°	Center angle = 360° (A round circle)
FS30i-A/FS30i-B/ FS0i-MD	By address R	By address R (Value of R is negative.)	By address I,J,K
FS0i-TD	By address R	By address I,J,K	By address I,J,K

1.2 Condition

The following parameter setting is necessary for this function.

- Parameter No.27488#0 = 1

	#7	#6	#5	#4	#3	#2	#1	#0
27488								#0

[Data type] Bit type
[Standard value] 00000000
[Path attribute] Common to paths
[Re-power on] Necessary
#0 In machining cycle,
0 : address I,J,K are outputted for G02/G03.
1 : address R are outputted for G02/G03.

1.3 Restrictions

(1) When the distance from the start point of the arc to the center point is very short, address I,J,K are outputted for G02/G03 even if address R output for G02/03 is selected.

(2) When center angle of an arc is nearly 180°, the calculated center coordinates may contain an error in the output by address R. In such a case, select the output by address I, J, K.

2 Calculating method of measuring of setting workpiece coordinate data (for Series 30*i*/31*i*/32*i*-A/B)

2.1 Outline

Calculating method of measuring of workpiece origin offset value can be selected by parameter No.27482#4 in workpiece coordinate data window.

(1) No.27482#4=0 (specifications of MANUAL GUIDE *i*)

Method of calculation	
Lathe	**Machining**
Eq.1	Eq.2

(2) No.27482#4=1 (specifications of NC screen)

Parameter	Method of calculation	
No.1015#6	**Lathe**	**Machining**
0	Eq.3	Eq.2
1	Eq.3	Eq.3

(Equation)

No.	Equation
1	(External workpiece origin offset value) Result = Current absolute coordinate value + External workpiece origin offset value - Target value of workpiece coordinate (Workpiece origin offset value) Result = Current absolute coordinate value + Workpiece origin offset value on the selected workpiece coordinate by cursor - Target value of workpiece coordinate
2	Result = Current machine coordinate value - Target value of workpiece coordinate
3	(External workpiece origin offset value) Result = Current absolute coordinate value + External workpiece origin offset value - Target value of workpiece coordinate (Workpiece origin offset value) Result = Current absolute coordinate value + Workpiece origin offset value on the modal workpiece coordinate - Target value of workpiece coordinate

2.2 Parameter setting

Please set parameter CMW(No.27482#4) to 1.

	#7	#6	#5	#4	#3	#2	#1	#0
27482				CMW				

[Data type] Bit type
[Standard value] 00000000
[System attribute] Common parameter between paths
[Re-power on] Necessary
#4 CMW Calculation methods of workpiece origin offset value
0: Specifications of MANUAL GUIDE *i* (conventional specifications).
1: The same specifications as the NC screen.

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REVISION RECORD

Edition	Date	Contents
04	Jan., 2014	• Addition of following items 2 windows machining simulation, 2 path machining center function, Simultaneous string search function of multi programs, Handle machining, C axis hole on end face (random points) - cartesian coordinates input cycle, Outer thread milling cycle, Disabling MANUAL GUIDE *i*, Extended axis name function, Dynamic axis name changing, High speed program management, Program restart in machining cycle, Usage in 8.4" LCD, Input of direction along the layout of numeric keys, Extension of fixed form sentences, Operation Navigating Function, Advanced guidance function, Calculating method of measuring of setting workpiece coordinate data, Use with peripheral axis control, MANUAL GUIDE *i* window call function, Machining simulation information setting function for each path, Improvement of icon for selected path, 4 path system, Specifying pecking clearance / return amount of drill cycle for machining center in input window, Warning message display when reset key is pushed, Display of simplified chinese characters alarm of user macro, Pocket residual machining cycle, Extended axis name function for 8.4" /10.4" display, MANUAL GUIDE *i* customization, Conversion of G41/G42 command by NC program conversion • Correction of errors
03	Nov., 2011	• Addition of explanation for Series 30*i*/31*i*/32*i*-B • Addition of following items Multi part program editing function, Machining simulation speed change function, Blank cutout display in animated simulation, G54.4 modal display, Comment display during machining, Operation restart confirmation function after editing program, Memory card input / output of offset data, Engraving cycle, Thread milling cycle, Mirror / rotation / scaling / figure copy command, Tilted working plane machining, Arc specification by central angle, Drawing of arbitrary figure on sub-program list screen, 19" display unit, Soft-key customization function, USB Memory Input/Output function • Correction of errors
02	May, 2008	• Addition of explanation for Series0*i*-MD
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